CHEMICAL REACTION ENGINEERING - II

KA GAVHANE

NIRALI PRAKASHAN
ADVANCEMENT OF KNOWLEDGE

www.pragationline.com

CONTENTS

1. INTRODUCTION TO HETEROGENEOUS REACTIONS — **1.1 TO 1.18**

- Examples of catalytic heterogeneous reactions — 1.1
- Examples of non-catalytic heterogeneous reactions — 1.1
- Rate equation for heterogeneous reactions — 1.2
- The concept of rate controlling/limiting step — 1.3
- Contacting patterns (schemes) for two-phase systems — 1.5
- Solved Examples 07 — 1.6 to 1.18

2. KINETICS OF FLUID-FLUID REACTIONS — **2.1 TO 2.42**

- Examples of fluid-fluid reactions — 2.1
- Gas absorption systems with chemical reaction — 2.2
- Rate equation — 2.2
- Rate equation for straight mass transfer — 2.4
- Rate equation for mass transfer and reaction — 2.7
- General rate expression — 2.8
- Liquid film enhancement factor — 2.9
- Instantaneous reaction with low C_B — 2.9
- Rate expression for instantaneous reaction with low C_B — 2.10
- Special case of negligible gas phase resistance — 2.12
- Instantaneous reaction with high C_B — 2.13
- Rate expression for instantaneous reaction with high C_B — 2.13
- Fast reaction with low C_B — 2.14
- Rate expression for fast reaction with low C_B — 2.14
- Fast reaction with high C_B — 2.15
- Rate expression for fast reaction with high C_B — 2.15
- Intermediate rate with respect to mass transfer — 2.16
- Slow reaction with respect to mass transfer — 2.16
- Rate expression for slow reaction — 2.17
- Infinitely slow reaction — 2.17
- Rate expression for infinitely slow reaction — 2.18
- Role of Hatta number — 2.18
- Use of solubility data to determine kinetic regime — 2.19
- Solved Examples 10 — 2.20 to 2.42

3. DESIGN OF FLUID-FLUID REACTORS — **3.1 TO 3.52**

- Tower and tank contactors for G/L reactions — 3.1
- Countercurrent flow in a tower - plug flow G + plug flow L – – straight mass transfer — 3.4
- Cocurrent flow in a tower - plug flow G + plug flow L — 3.12
- Mass transfer + not very slow reaction — 3.12
- Mass transfer + reaction in a countercurrent tower – – plug flow G + plug flow L — 3.12

- Mass transfer + reaction in a cocurrent tower –
  – plug flow G + plug flow L 3.1
- Mass transfer + reaction in an agitated tank contactors
  – mixed flow G + mixed flow L 3.1
- Mass transfer + reaction in bubble tank contactors –
  – plug flow G + mixed flow L 3.1
- Absorption + reaction in a batch agitated tank contactor
  – mixed flow G + batch of uniform L 3.1
- Solved Examples **09** 3.21 to 3.

4. KINETICS OF FLUID-PARTICLE REACTIONS 4.1 TO 4.

- Examples of fluid-solid reactions 4.1 and 4
- Selection of a kinetic model 4
- Progressive-conversion model 4
- Shrinking-core model 4
- Shrinking-core model for spherical particles of unchanging size 4
- Diffusion through gas film controls 4
- Diffusion through ash layer controls 4
- Chemical reaction controls 4.
- Rate of reaction for shrinking spherical particles 4.
- Chemical reaction controls 4.
- Diffusion through gas film controls 4.
- SCM for cylindrical particles of unchanging size 4.
- Conversion-time expressions for various shapes of particles (as per SCM) 4.
- Combination of resistances 4.
- Solved Examples **14** 4.25 to 4.

5. DESIGN OF FLUID-PARTICLE REACTORS 5.1 TO 5.

- Contacting schemes for gas-solid reactions 5
- Particles of single size - plug flow of solids + uniform gas composition 5
- Mixture of particles of different but unchanging sizes
  – plug flow of solids + uniform gas composition
- Particles of a single unchanging size
  – mixed flow of solids + uniform gas composition
- Film resistance controls
- Resistance of chemical reaction controls
- Ash resistance controls
- Mixed flow of a size mixture of particles of unchanging size + + uniform gas composition
- Fluidised bed with entrainment of solid fines 5
- Elutriation velocity constant 5
- Solved Examples **21** 5.16 to 5

6. SOLID CATALYSED REACTIONS — 6.1 TO 6.136

- Catalyst, activity and specificity of catalyst — 6.1
- Characteristics of catalysed reaction — 6.2
- Mechanism of solid catalysed reactions — 6.2
- Pore diffusion resistance combined with surface kinetics — 6.4
- Single cylindrical pore + first order reaction — 6.4
- Effectiveness factor — 6.8
- Porous catalysts — 6.12
- Proper diffusion coefficient — 6.12
- Correct measure of particle size — 6.12
- Measures of rate of reaction — 6.12
- Relationship between η and M_T for various particle shapes — 6.13
- Effectiveness factor (derivation) for a rectangular slab — 6.14
- Determining pore resistance effects from experimental data — 6.17
- Limits of pore resistance — 6.17
- Different size particles — 6.17
- Arbitrary reaction kinetics — 6.18
- Summary of diffusion resistance in pores — 6.21
- Heat effects during reaction — 6.22
- Performance equations for reactors containing porous catalysts — 6.23
- Plug flow reactor — 6.23
- Mixed flow reactor — 6.24
- Experimental methods for determining rates — 6.26
- Differential reactor — 6.26
- Integral reactor — 6.27
- Mixed flow reactor — 6.28
- Recycle reactor — 6.28
- Batch reactor — 6.29
- Film diffusion resistance — 6.30
- Pore diffusion resistance — 6.30
- Catalytic reactors — 6.133
- Fixed-bed reactors — 6.133
- Fluidised-bed reactors — 6.135
- Solved Examples 39 — 6.31 to 6.133

7. G/L REACTIONS ON SOLID CATALYSTS — 7.1 TO 7.34

- Various ways of carrying out G/L reactions catalysed by solids — 7.1
- General rate equation — 7.2
- Resistances in series in the G/L reaction on a catalyst surface — 7.3
- Performance equations for an excess of B — 7.7
- Mixed flow G + any flow L ... excess of B — 7.7
- Plug flow G + any flow L ... excess of B — 7.9

- Mixed flow G + batch L ... excess of B 7.10
- Plug flow G + batch L ... excess of B 7.13
- Special case of pure gaseous A ... excess of B 7.14
- Performance equations for an excess of A 7.15
- Solved Examples 05 7.16 to 7.34

8. DEACTIVATING CATALYSTS 8.1 TO 8.76

- Types of catalyst deactivation 8.1
- Mechanisms of catalyst deactivation 8.2
- Decay reactions 8.2
- Parallel deactivation 8.2
- Series deactivation 8.2
- Side-by-side deactivation 8.3
- Independent deactivation 8.3
- Pore diffusion 8.3
- Rate equation for main reaction and deactivation 8.3
- Rate equation using experimental data 8.5
- Determination of rate from experimental data for independent deactivation 8.6
- Batch solids + batch fluid 8.6
- Batch solids + mixed constant flow of fluid 8.9
- Batch solids + mixed changing flow of fluid 8.10
- Batch solids + plug constant flow of fluid 8.11
- Batch solids + plug changing flow of fluid 8.12
- Batch solids : determination of rate for parallel, series or side-by-side deactivation 8.13
- Flowing solids experimental reactors 8.16
- Effect of pore diffusion resistance on the kinetics of reactions with deactivating catalysts 8.16
- Solved Examples 18 8.20 to 8.76

9. ENZYME FERMENTATION 9.1 TO 9.34

- Enzyme 9.1
- Substrate 9.2
- Michaelis-Menten kinetics 9.2
- Special features of M-M equation 9.5
- Fermentors 9.5
- Batch fermentor or plug flow fermentor 9.5
- Mixed flow fermentor 9.7
- Alternate approach to evaluate k, i.e. k_3 and C_M 9.8
- Inhibition of enzyme reactions by a foreign substance 9.10
- Competitive inhibition-kinetics 9.10
- Non-competitive inhibition-kinetics 9.11
- Method to find inhibition type from experiment 9.12
- Solved Examples 08 9.13 to 9.34

Total Solved Examples = 131

Total Pages = 504

CHAPTER 1

INTRODUCTION TO HETEROGENEOUS REACTIONS

Heterogeneous reactions are those in which more than one phase is involved to proceed. Accordingly, there may be two phase and three phase reaction systems.

A heterogeneous reaction involves substances which exist in different phases and reaction usually takes place at an interface between the phases or in one of the phases.

Heterogeneous reactions may be catalytic or non-catalytic. In catalytic heterogeneous reactions usually the catalyst is a solid.

Some examples of catalytic heterogeneous reactions :

(i) **Gas-solid reactions :** Synthesis of ammonia, catalytic cracking of crude oil, oxidation of SO_2 to SO_3 using V_2O_5 catalyst in the manufacture of sulphuric acid.

(ii) **Gas-liquid-solid reactions :** Catalytic oxidation of liquid hydrocarbons (e.g. ethanol) using air or oxygen to acetic acid, catalytic hydrogenation of acetone to propanol.

A heterogeneous catalytic reaction occurs at the interface (between the phases) which is the surface of a solid catalyst immersed in a gaseous or liquid reaction mixture.

Some examples of non-catalytic heterogeneous reactions :

(i) **Gas-solid reactions :** roasting of sulphide ores, combustion of coal.

(ii) **Gas-liquid reactions :** chlorination of liquid aromatic hydrocarbons using chlorine gas, purification of gas mixtures, manufacture of ammonium nitrate by absorption of ammonia in aqueous nitric acid.

(iii) **Liquid-liquid reactions :** nitration of organic compounds with aqueous nitric acid, formation of soaps by the action of aqueous alkalies on fats or fatty acids.

(iv) **Liquid-solid reactions :** production of acetylene by the action of water on calcium carbide.

Apart from all differences, heterogeneous reactions have one thing in common : Before a chemical reaction can occur, the reactants from the bulk of one phase must be transported to the interface between the phases or into the bulk of other phase.

For heterogeneous systems we have to consider the following factors in addition to those normally considered in homogeneous systems :

(i) The modification of the rate equation resulting from the mass transfer between phases.

(ii) The contacting patterns for the reacting phases.

Rate Equation for Heterogeneous Reactions :

A heterogeneous reaction (catalytic or non-catalytic) takes place in a number of steps - one or several chemical steps (i.e., one or several chemical reactions) are always accompanied by steps that are physical in nature. These physical steps are associated with the transport of matter from one phase to the other. The concentration of matter in the bulk of one phase and at the interface is different and the concentration difference is the driving force for these physical transport processes (diffusion).

Heterogeneous reactions are treated as consisting of a series of steps or of series and parallel (mixed) steps.

In a heterogeneous system (in which the overall change is brought through various steps), the overall rate equation in general will consist mass transfer terms in addition to the usual kinetic term.

If a heterogeneous reaction consists of parallel steps, then the overall rate is equal to the sum of all the individual rates :

$$r_{overall} = \sum_{i=1}^{n} r_i$$

If, on the other hand, a heterogeneous reaction consists of several consecutive steps (i.e., if the overall change results from a number of steps that take place in series), then at steady-state conditions, the rates of the individual steps in series are all the same and equal to the overall rate of reaction :

$$r_{overall} = r_1 = r_2 = \ldots = r_n$$

Here, the rates of the individual consecutive steps adjust themselves to that of the slowest step.

When rates of the various steps are to be compared or combined, then they must be expressed identically (i.e., the rate equations of the individual steps must be written on the same basis).

Thus, when mass transfer takes, the rate is defined in terms of molar flux (moles per unit time per unit surface), the rate of chemical reaction should also be based on unit surface area instead on unit volume.

Based on unit volume of reacting fluid :

$$-r_A = -\frac{1}{V}\frac{dN_A}{dt} = \frac{\text{mol A reacted}}{\text{volume of fluid} \cdot \text{time}}$$

Based on unit mass of solid :

$$-r_A' = -\frac{1}{W}\frac{dN_A}{dt} = \frac{\text{mol A reacted}}{\text{mass of solid} \cdot \text{time}}$$

Based on unit interfacial surface / unit surface of solid :

$$-r_A'' = -\frac{1}{S}\frac{dN_A}{dt} = \frac{\text{mol A reacted}}{\text{interfacial surface} \cdot \text{time}}$$

These rates are related by

$$\frac{\text{mol A reacted}}{\text{time}} = (-r_A)\,V = (-r_A')\,W = (-r_A'')\,S$$

$$\therefore \quad -r_A = \frac{W}{V} r_A'$$

$$-r_A' = \frac{S}{W} r_A''$$

$$-r_A'' = \frac{V}{S} r_A$$

Consider steps in series. In combining rates as we usually do not know the concentration materials at intermediate positions, we have to eliminate these concentrations and express the overall rate of reaction in terms of the overall concentration difference. This is easy to do if all the steps are linear in concentration, i.e., the rate expressions for mass transfer and chemical reaction are linear in concentration - first power in concentrations or concentration differences.

CONCEPT OF RATE CONTROLLING/LIMITING STEP

We develop an expression for the overall rate of reaction and then we see how it reduces to the simplest forms if we assume one of the steps to be rate controlling.

$$A\,(g) + B\,(s) \rightarrow R\,(g)$$

A gas is transported from the bulk of gas to the solid surface, the reaction occurs at the surface and finally, product R is transported from the solid surface to the bulk gas. The reaction is first order with respect to the gaseous reactant A, and no ash is formed on the solid surface.

The overall rate can be formulated by considering only first two steps involving A.

The rate of disappearance of A can be expressed in two ways : as the rate of transport of A to the surface of B and as the rate of reaction of A at the solid surface.

The rate of mass transfer of A through the gas film by diffusion per unit surface, i.e., the flux of A, to the surface of B is

$$-r_{Al}'' = \frac{-1}{S}\frac{dN_A}{dt} = k_g\,(C_{Ag} - C_{As}) \qquad \ldots (1.1)$$

The rate of reaction $(-r''_{A2})$ based on unit surface area of B is

$$-r''_{A2} = -\frac{1}{S}\frac{dN_A}{dt} = k'' C_{As} \quad \ldots (1.2$$

These steps are in series. So at steady state, the rates of mass transfer and chemical reaction steps will be the same.

∴ Flux of A to the surface = Reaction rate at the surface

$$-r''_{A1} = -r''_{A2}$$

This equality can be used to develop the overall rate equation in terms of concentration of A in the bulk gas.

$$k_g (C_{Ag} - C_{As}) = k'' C_{As}$$

$$k_g C_{Ag} = (k'' + k_g) C_{As}$$

$$C_{As} = \frac{k_g}{k'' + k_g} C_{Ag} \quad \ldots (1.3)$$

The rate at which the overall process of mass transfer and chemical reaction proceeds can be found by substituting C_{As} from Equation (1.3) into Equation (1.2).

∴ The overall rate of reaction $(-r''_A)$ is

$$-r''_A = r''_{A2} = \frac{k'' k_g}{k'' + k_g} C_{Ag}$$

For steps in series : $r_1 = r_2 = r_{overall}$, dividing the numerator and denominator of the above equation by $k'' k_g$ gives

$$\boxed{-r''_{A1} = -r''_{A2} = -r''_A = \left(\frac{1}{\frac{1}{k_g} + \frac{1}{k''}}\right) C_{Ag}} \quad \ldots (1.4)$$

This is the expression for the overall rate in terms of the bulk reactant concentration.

By analogy with the mass transfer coefficients, the sum in the denominator of Equation (1.4) may be regarded as the sum of the resistances offered by the individual steps of a heterogeneous process. [Denominator is the sum of the reciprocal rates of the individual steps.]

The overall rate of a heterogeneous process is determined by the slowest step and hence it is called as the rate controlling step. The rate controlling step presents a maximum resistance to the overall change. Very often we find that one or other of the two steps (providing major resistance) is rate controlling or rate limiting.

If, under operating conditions, $k_g >> k''$, (mass transfer coefficient is much greater than the specific reaction rate constant) then the overall rate reduces to

$$\boxed{-r''_A = k'' C_{Ag}} \quad \ldots (1.5)$$

which implies that the chemical reaction is the rate controlling step (the chemical reaction is controlling the overall process).

[$k_g >> k''$ → so $1/k_g$ is very negligible compared to $1/k''$ and hence it can be neglected from Equation (1.4) and it then reduces to Equation (1.5)].

In this case we have rapid diffusion with respect to surface reaction - the reaction is so slow that virtually no concentration gradient exists in the gas phase and mass transfer effects are not important - when reaction is slow.

If, on the other hand, $k'' >> k_g$ (the specific reaction rate constant is much greater than the mass transfer coefficient) then the overall rate of reaction is given by

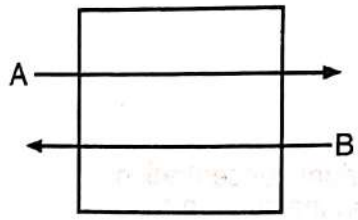
$$\boxed{r''_A = k_g C_{Ag}} \quad \text{... (1.6)}$$

[$k'' >> k_g$ ... so $1/k''$ is very small compared to $1/k_g$ and so $1/k''$ can be neglected from the overall rate expression] which implies that the mass transfer step is rate controlling. The reaction is so rapid that the rate of mass transfer to the surface limits the overall rate of reaction.

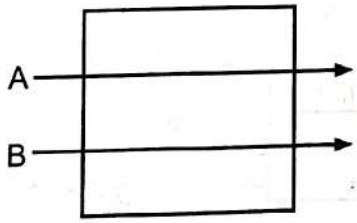
If the reaction takes place in the region intermediate between the chemically controlled region and the mass transfer controlled region, the overall rate of reaction will be composite of the two steps and is given by **Equation (1.4)**.

CONTACTING PATTERNS (SCHEMES) FOR TWO-PHASE SYSTEMS

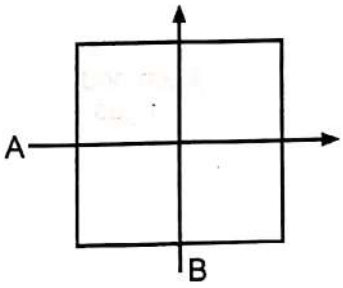
In heterogeneous reactions, contacting patterns affect both physical and chemical rate processes. There are many ways in which two phases can be contacted. Each of the contacting pattern is having a unique design equation. In ideal contacting of the phases, each phase may be in one of the two ideal flow patterns, i.e., in plug or mixed flow.



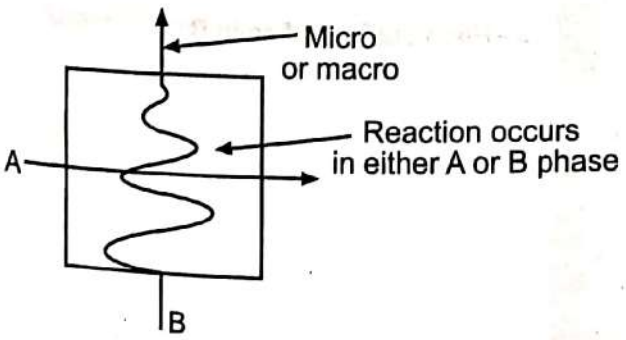
Plug A + Plug B
Countercurrent
(a)



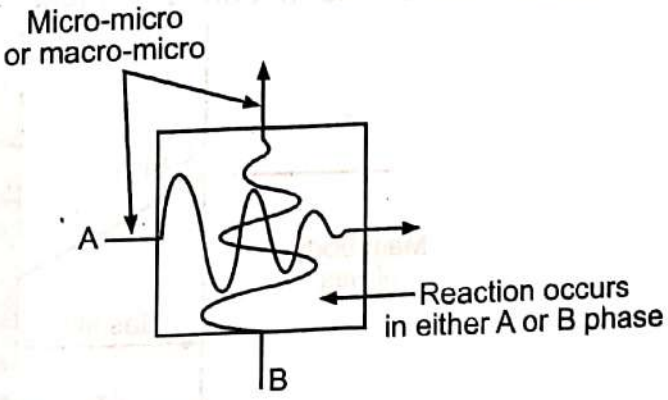
Plug A + Plug B
Cocurrent (parallel)
(b)



Plug A + Plug B
Crosscurrent (cross flow)
(c)



(d) Plug A + Mixed B



(e) Mixed A + Mixed B

Fig. 1.1 : Various ideal contacting patterns (schemes) for two flowing fluids

With ideal flow of both phases, many ways to contact the phases are possible / many combinations of contacting patterns are possible. Ideal contacting patterns for two-phase systems are shown in Fig. 1.1. If one of the phases is discontinuous (e.g., droplets or solid particles), it will be treated as a macrofluid. Distinction between macrofluids and microfluids is needed only when the phase is in mixed flow. It is possible to develop design equations for ideal flow patterns but when real flow deviates considerably from these then either we may develop models to mirror actual flow closely or may calculate the performance with ideal patterns which 'bracket' actual flow. Fortunately, many real reactors that are used for heterogeneous reaction systems can be approximated in a satisfactory manner by one of the five ideal flow patterns of Fig. 1.1 (which can be seen by comparing these flow patterns with the sketches of typical reactors for fluid-fluid and fluid-solid systems).

SOLVED EXAMPLES

Example 1.1 : *The irreversible reaction*

$$A(g) + B(s) \rightarrow R(g) \quad \text{... (first order w.r.t. A)}$$

takes place on a flat surface as shown in Fig. Ex 1.1 (a). Dilute A diffuses through a stagnant gas film onto a plain surface consisting of B. On the surface, A reacts with B to produce gaseous R which then diffuses back into the main gas stream. Develop the overall rate expression for this G/S reaction.

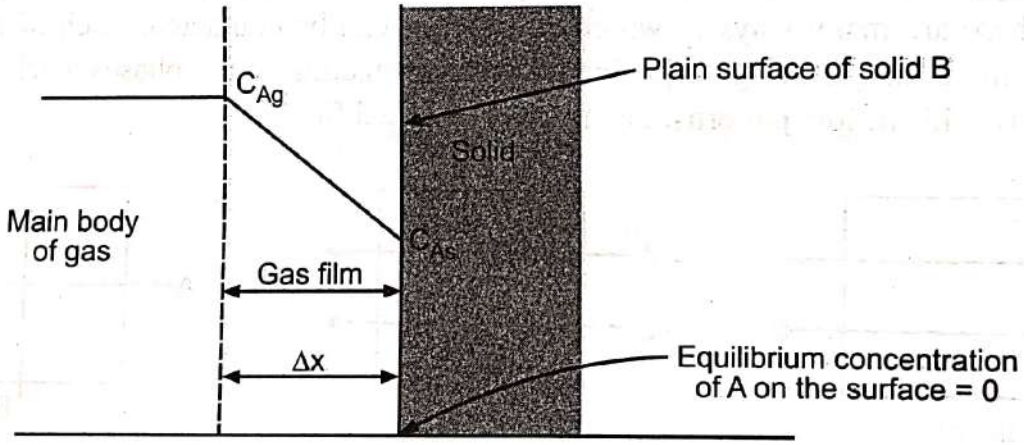


Fig. Ex 1.1 (a)

Solution : $A(g) + B(s) \rightarrow R(g)$

Given : Irreversible first order G/S reaction w.r.t. A.

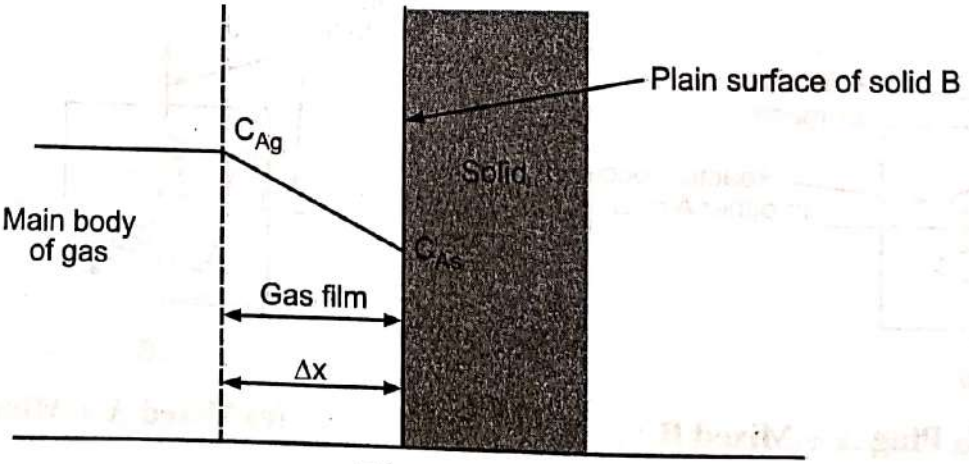


Fig. Ex 1.1 (b)

This overall process comprises of two steps in series :

(1) **Mass transfer step :** Diffusion of A through the gas film to the surface of B.

(2) **Reaction step :** The reaction of A with B on the surface of B.

(i) **Diffusion :** The mass transfer step : The flux of A (by diffusion) to the surface is given by

$$-r''_{A1} = -\frac{1}{S}\frac{dN_A}{dt} = \frac{\mathcal{D}}{\Delta x}(C_{Ag} - C_{As}) = k_g(C_{Ag} - C_{As}) \quad \ldots (1)$$

$-r''_{A1}$ is in mol/(m^2.h), C_{Ag} is in mol/m^3.

k_g is in m/s [m^3 gas/(m^2 surface · s)] – mass transfer coefficient of the gas film.

C_{Ag} is the concentration of A in the main gas stream, mol/m^3.

(ii) **Reaction step :** The reaction is first order with respect to A. The rate of reaction based on unit surface is given by

$$-r''_{A2} = -\frac{1}{S}\frac{dN_A}{dt} = k''\, C_{As} \quad \ldots (2)$$

$-r''_{A2}$ is in mol A/(m^2 surface · time), k'' is in m/s and C_{As} is in mol/m^3.

Now, we will obtain the overall rate of reaction in terms of k_{Ag}, k'' and C_{Ag} (as the concentration of A on the surface, i.e., C_{As} cannot be measured).

As both steps – diffusion and reaction are taking place in series, at steady state, these steps will proceed at the same rate. Thus

$$-r''_{A\,overall} = -r''_{A1} = -r''_{A2}$$

$$k_g(C_{Ag} - C_{As}) = k''\, C_{As}$$

$$k_g C_{Ag} = C_{As}(k_g + k'')$$

$$C_{As} = \frac{k_g}{k_g + k''} C_{Ag} \quad \ldots (3)$$

Substituting C_{As} from Equation (3) into Equation (2), we get

$$-r''_{A2} = -r''_A = \left(\frac{k''\, k_g}{k_g + k''}\right) C_{Ag}$$

$$-r''_{A1} = -r''_{A2} = -r''_A = \left(\frac{1}{\frac{1}{k_g} + \frac{1}{k''}}\right) C_{Ag}$$

Let

$$k_{overall} = \frac{1}{\frac{1}{k_g} + \frac{1}{k''}}$$

$$-r''_{A2} = -r''_{A1} = -r''_{A} = \left(\frac{1}{\frac{1}{k_g} + \frac{1}{k''}}\right) C_{Ag} = k_{overall}\, C_{Ag} \quad , \text{mol/(m}^2\cdot\text{s)}$$

This is the overall rate expression accounting for both the mass transfer and reaction steps.

Example 1.2 : *Dilute A diffuses through a stagnant liquid film onto a plane surface of solid On this plain surface, A and B react to yield liquid product R which then diffuses back throu the film into the main liquid stream. Develop the overall rate expression for the L/S reaction.*

$$A\,(l) + B\,(s) \rightarrow R\,(l) \qquad \textit{...first order w.r.t. A}$$

Solution : $A\,(l) + B\,(s) \rightarrow R\,(l)$ ... Irreversible L/S reaction ... first order w.r.t.

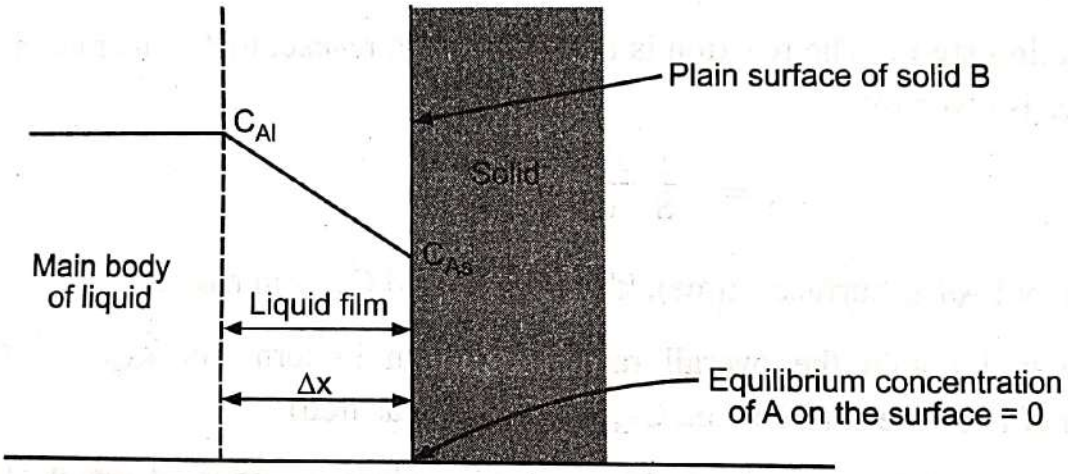


Fig. Ex 1.2

Steps in series :

(i) Diffusion of A through the liquid film to the surface.

(ii) Reaction of A with B on the surface of B.

Mass transfer step :

The molar diffusional flux of A to the surface is given by

$$-r''_{A1} = -\frac{1}{S}\frac{dN_A}{dt} = \frac{\mathscr{D}}{\Delta x}(C_{Al} - C_{As}) = k_l\,(C_{Al} - C_{As})$$

where

k_l – mass transfer coefficient of the liquid film, m/s

$-r''_{A1}$ – molar flux of A, mol/(m$^2\cdot$s)

C_{Al} – concentration of A in the bulk liquid, mol/m^3

C_{As} – concentration of A at the surface of B, mol/m^3

Reaction step :

The reaction is first order w.r.t. A. The rate of reaction based on unit surface is given by

$$-r''_{A2} = -\frac{1}{S}\frac{dN_A}{dt} = k'' C_{As} \quad \ldots (2)$$

At steady state, these steps will proceed at the same rate (steps in series). Thus,

$$-r''_{A\,overall} = -r''_{A1} = -r''_{A2}$$

$$k_l (C_{Al} - C_{As}) = k'' C_{As}$$

$$k_l C_{Al} = (k'' + k_l) C_{As}$$

$$C_{As} = \frac{k_l C_{Al}}{k'' + k_l} \quad \ldots (3)$$

Substituting the value of C_{As} from Equation (3) into Equation (2), we get

$$-r''_{A2} = \frac{k_l k'' C_{Al}}{k'' + k_l}$$

$$\boxed{-r''_{A2} = -r''_{A1} = -r''_A = \left(\frac{1}{\frac{1}{k_l} + \frac{1}{k''}}\right) C_{Al}} \text{ , mol A/(m}^2\text{·s)}$$

or $$-r''_A = k_{overall}\, C_{Al}$$

where $$k_{overall} = 1/(1/k_l + 1/k'')$$

Example 1.3 : *Repeat Example 1.1 considering the reaction step to be second order with respect to A – all else remain unchanged.*

Solution : A (g) + B (s) → R (g) … Irreversible G/S reaction with second order rate.

Diffusion-mass transfer step : Diffusion of A to the surface through a gas film.

The diffusional flux of A to the surface is given by

$$-r''_{A1} = -\frac{1}{S}\frac{dN_A}{dt} = k_g (C_{Ag} - C_{As}) \quad \ldots (1)$$

Reaction step : The reaction at the surface is second order w.r.t. A.

The rate of reaction based on unit surface is given by

$$-r''_{A2} = -\frac{1}{S}\frac{dN_A}{dt} = k'' C_{As}^2 \quad \ldots (2)$$

As the processes - diffusion and reaction at the surface occur in series, at steady state, these two steps will proceed at the same rate. Thus,

$$-r''_A = -r''_{A1} = -r''_{A2}$$

$$k_g (C_{Ag} - C_{As}) = k'' C_{As}^2$$

$$k'' C_{As}^2 + k_g C_{As} - k_g C_{Ag} = 0$$

This is a quadratic in C_{As}. The roots of this equation are :

$$C_{As} = \frac{-k_g \pm \sqrt{k_g^2 - 4k''(-k_g C_{Ag})}}{2k''}$$

The positive root is

$$C_{As} = \frac{-k_g + \sqrt{k_g^2 + 4k'' k_g C_{Ag}}}{2k''}$$

Substituting C_{As} (as it cannot be measured) from Equation (3) into Equation (1), we ge following overall rate expression :

$$-r''_{A1} = k_g\left[C_{Ag} - \left(\frac{-k_g + \sqrt{k_g^2 + 4k'' k_g C_{Ag}}}{2k''}\right)\right]$$

$$-r''_{A1} = -r''_{A2} = -r''_A = k_g\left[C_{Ag} + \frac{k_g}{2k''} - \frac{\sqrt{k_g^2 + 4k'' k_g C_{Ag}}}{2k''}\right]$$

Taking 2k'' common from the denominator gives

$$\boxed{-r''_A = \frac{k_g}{2k''}\left[2k'' C_{Ag} + k_g - \sqrt{k_g^2 + 4k'' k_g C_{Ag}}\right]}$$

Equation (4) is the desired overall rate expression that takes into account both the transfer and reaction steps.

Example 1.4 : *Gaseous reactant A diffuses through a gas film to the surface of so where it reacts with B according to a reversible first order rate,*

$$-r''_A = k''(C_{As} - C_{Ae}),\ mol/(m^2 \cdot s)$$

where C_{Ae} is the concentration of A in equilibrium with the solid surface.

Derive an expression for the rate of reaction of A that accounts for both the mass tra and reaction steps.

Solution : 1. Mass transfer step : The flux of A to the surface by diffusion (rate of transfer of A to the surface through a gas film per unit surface) is given by,

$$-r''_{A1} = -\frac{1}{S}\frac{dN_A}{dt} = k_g (C_{Ag} - C_{As})$$

C_{Ag} is the concentration of A in the main body of gas.

C_{As} is the concentration of A at the plain surface of a solid.

2. Reaction step : The rate of reaction of A at the surface based on unit surfa given by

$$-r''_{A2} = -\frac{1}{S}\frac{dN_A}{dt} = k''(C_{As} - C_{Ae})$$

At steady state the flux of A to the surface is equal to the reaction rate at the surface (steps in series so both the steps will proceed with the same rate.)

$$-r''_A = -r''_{A_1} = -r''_{A_2}$$

$$k_g (C_{Ag} - C_{As}) = k'' (C_{As} - C_{Ae})$$

$$k_g C_{Ag} + k'' C_{Ae} = C_{As} (k'' + k_g)$$

$$C_{As} = \frac{k_g C_{Ag} + k'' C_{Ae}}{k'' + k_g} \quad \ldots (3)$$

Substituting C_{As} from Equation (3) into Equation (2), thus eliminating C_{As} from the rate expression, we get

$$-r''_{A2} = k'' \left[\frac{k_g C_{Ag} + k'' C_{Ae}}{k'' + k_g} - C_{Ae} \right]$$

$$= k'' \left[\frac{k_g C_{Ag} + k'' C_{Ae} - k'' C_{Ae} - k_g C_{Ae}}{k'' + k_g} \right]$$

$$= \frac{k'' k_g}{k'' + k_g} (C_{Ag} - C_{Ae})$$

Dividing the numerator and denominator of the above equation by $k'' k_g$, we get

$$-r''_{A1} = -r''_{A2} = -r''_A = \left(\frac{1}{\frac{1}{k_g} + \frac{1}{k''}} \right) (C_{Ag} - C_{Ae})$$

$$\boxed{-r''_A = \left(\frac{1}{\frac{1}{k_g} + \frac{1}{k''}} \right) (C_{Ag} - C_{Ae})}, \text{ mol/(m}^2 \cdot \text{s)} \quad \ldots (4)$$

or $$\boxed{-r''_A = k_{overall} (C_{Ag} - C_{Ae})}$$

where $$k_{overall} = 1/(1/k_g + 1/k'')$$

Equation (4) is the desired expression for the rate of reaction of A accounting for both the mass transfer and reaction steps.

Example 1.5 : *In slurry reactors, pure reactant gas is bubbled through liquid containing suspended catalyst particles. In terms of the two-film theory, these kinetics can be viewed as shown in Fig. Ex 1.5. Hence, to reach the surface of solid, the reactant must diffuse through the liquid film into the main body of liquid and then through the film surrounding the catalyst particle. At the surface of the catalyst particle, reactant gives product according to first order kinetics. Develop an expression for the rate of reaction in terms of these resistances.*

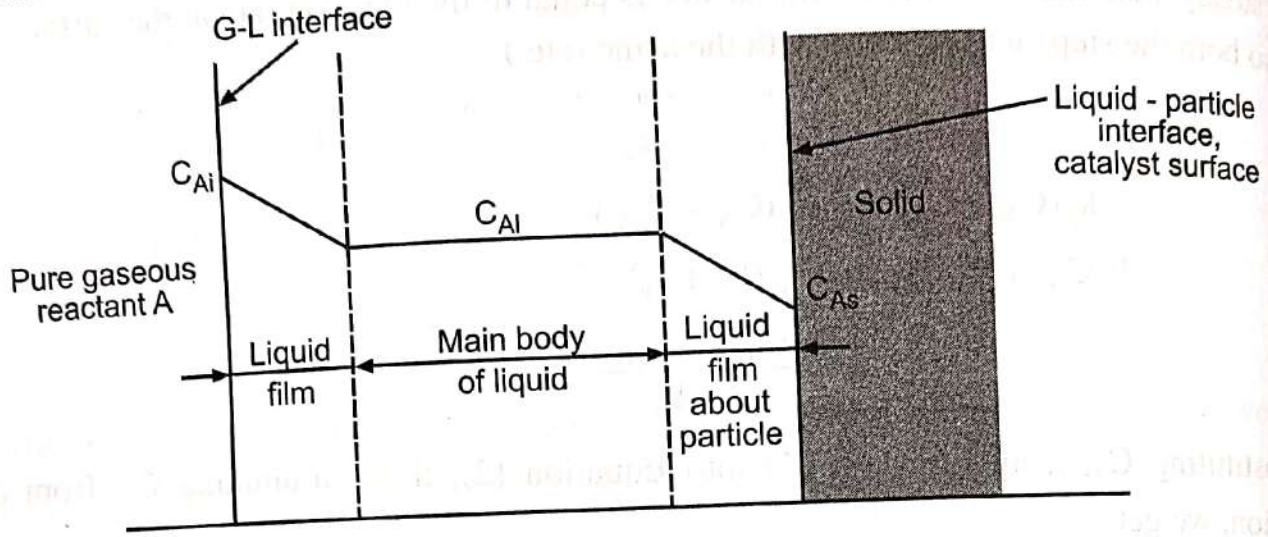


Fig. Ex 1.5

Solution : Here three resistances act in series : liquid film surrounding gas bubble, li film surrounding the catalyst particle and reaction. [As pure A is used there is no gas resistance (and for this $C_{Ag} = C_{Ai}$)].

The rate of transfer of A from gas to the surface of the catalyst particle is given by the expressions :

(i) Liquid film-surrounding gas bubble :

$$-r_A'''' = k_{Al}\, a_i\, (C_{Ai} - C_{Al})$$

where $-r_A''''$ is in mol A/(m^3 reactor·s)

$$a_i = \frac{\text{gas-liquid interface}}{\text{volume of reactor}}, \quad m^2/m^3$$

k_{Al} – liquid film mass transfer coefficient in m^3 liquid/(m^2 surface · s) i.e. in m/s.

$k_{Al}\, a_i$ is in m^3 liquid/(m^3 reactor·s)

(ii) Liquid film-surrounding the catalyst particle :

$$-r_A'''' = k_{Ac}\, a_c\, (C_{Al} - C_{As})$$

where k_{Ac} is the film coefficient about catalyst particle in m^3 liquid/(m^2 cat·s)

$$a_c = \frac{\text{external surface of particle}}{\text{volume of reactor}}, \quad m^2/m^3$$

(iii) The reaction step : The surface reaction is first order with respect to A.

The rate of reaction is given by

$$-r_A'''' = k_1\, a_c\, C_{As}$$

C_{As} is in mol/m^3 liquid, a_c is in m^2/m^3 of reactor,

k_1 is in m^3 of liquid/(m^2 cat · s).

At steady state :

$$-r_A'''' = k_{Al}\, a_i\, (C_{Ai} - C_{Al}) = k_{Ac}\, a_c\, (C_{Al} - C_{As}) = k_1\, a_c\, C_{As} \quad \ldots (4)$$

$$\frac{-r_A''''}{k_{Al}\, a_i} = C_{Ai} - C_{Al} \quad \ldots (5)$$

$$\frac{-r_A''''}{k_{Ac}\, a_c} = C_{Al} - C_{As} \quad \ldots (6)$$

$$\frac{-r_A''''}{k_1\, a_c} = C_{As} \quad \ldots (7)$$

Summing Equations (5), (6) and (7), we get

$$\frac{-r_A''''}{k_{Al}\, a_i} + \frac{-r_A''''}{k_{Ac}\, a_c} + \frac{-r_A''''}{k_1\, a_c} = C_{Ai} - C_{Al} + C_{Al} - C_{As} + C_{As}$$

$$-r_A'''' \left[\frac{1}{k_{Al}\, a_i} + \frac{1}{k_{Ac}\, a_c} + \frac{1}{k_1\, a_c}\right] = C_{Ai}$$

$\therefore$

$$\boxed{-r_A'''' = \frac{1}{\dfrac{1}{k_{Al}\, a_i} + \dfrac{1}{k_{Ac}\, a_c} + \dfrac{1}{k_1\, a_c}}\, C_{Ai}} \quad \ldots (8)$$

Equation (8) is the desired rate expression for the rate of reaction in terms of three resistances in series.

Example 1.6 : *Gas containing A contacts and reacts with a semi-infinite slab of solid B according to the reaction :*

$$A\,(g) + B\,(g) \longrightarrow R\,(g) + S\,(s)$$

As reaction progresses, a sharp reaction plane moves forward into the solid leaving behind it a product layer through which gaseous A and R must diffuse. For this situation, three resistances that act in series (– gas film, ash layer and reaction step) are as shown in Fig. Ex 1.6.

Taking into consideration that the rate of thickening of the ash layer is proportional to the rate of reaction at that instant, i.e., $dl/dt = m\,(-r_A'')$, show that the time to reach any thickness l is the sum of the time needed if each resistance acted alone, or

$$t_{actual} = t_{film\ alone} + t_{ash\ alone} + t_{reaction\ alone}$$

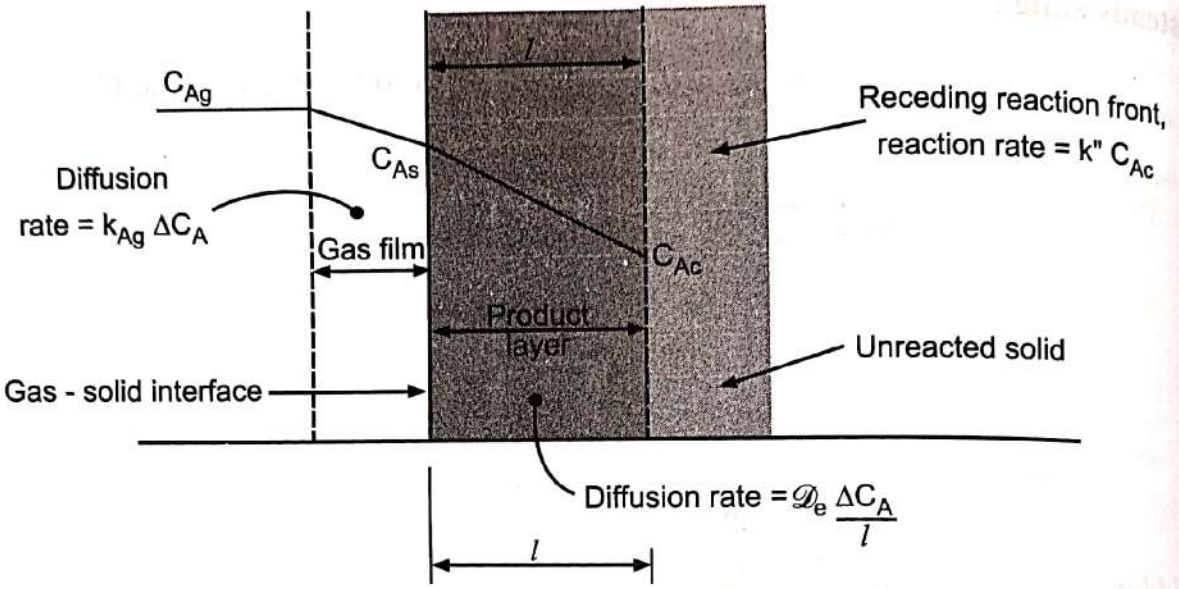


Fig. Ex 1.6

Solution : $A(g) + B(s) \rightarrow R(g) + S(s)$

Irreversible G/S reaction ... first order w.r.t. A.

Refer Fig. Ex 1.6.

Processes or steps in series :

(i) Diffusion through the gas film, (ii) diffusion through the ash layer/product layer (iii) reaction at the surface of solid.

The rate of transfer of A from gas to the surface of solid is given by the following expressions :

(i) For the gas film :

$$-r''_{A1} = -\frac{1}{S}\frac{dN_A}{dt} = k_g (C_{Ag} - C_{As})$$

$$\frac{-r''_{A1}}{k_g} = C_{Ag} - C_{As}$$

(ii) For the ash layer :

$$-r''_{A_2} = \frac{\mathscr{D}_e}{l} (C_{As} - C_{Ac})$$

$$-r''_{A2} / (\mathscr{D}_e / l) = C_{As} - C_{Ac}$$

The rate of reaction based on unit surface is

$$-r''_{A3} = k'' C_{Ac}$$

$$\frac{-r''_{A3}}{k''} = C_{Ac}$$

Adding Equations (2), (4) and (6), we get

$$\frac{-r''_{A1}}{k_g} + \frac{-r''_{A2}}{\mathcal{D}_e/l} + \frac{-r''_{A3}}{k''} = C_{Ag} - C_{As} + C_{As} - C_{Ac} + C_{Ac} = C_{Ag}$$

At steady state (since the steps are in series)

$$-r''_{A1} = -r''_{A2} = -r''_{A3} = -r''_{A} \quad \text{... steps proceed at the same rate}$$

$$\therefore \quad \frac{-r''_A}{k_g} + \frac{-r''_A}{\mathcal{D}_e/l} + \frac{-r''_A}{k''} = C_{Ag}$$

$$-r''_A \left[\frac{1}{k_g} + \frac{1}{\mathcal{D}_e/l} + \frac{1}{k''}\right] = C_{Ag}$$

$$-r''_A = \left[\frac{1}{\frac{1}{k_g} + \frac{1}{\mathcal{D}_e/l} + \frac{1}{k''}}\right] C_{Ag}$$

$$-r''_A = k_{overall}\, C_{Ag}, \quad k_{overall} = \frac{1}{\frac{1}{k_g} + \frac{1}{\mathcal{D}_e/l} + \frac{1}{k''}}$$

Given : $$\frac{dl}{dt} = m\,(-r''_A)$$

$$\therefore \quad dl = m\,(k_{overall}\, C_{Ag})\, dt$$

$$\int_0^l dl = m\,(k_{overall}\, C_{Ag}) \int_0^t dt$$

$$\therefore \quad t = \frac{l}{m\, k_{overall}\, C_{Ag}}$$

$$\therefore \quad t_{actual} = \frac{l}{m\, k_{overall}\, C_{Ag}} \quad \text{... (7)}$$

We have $$\frac{dl}{dt} = m\,(-r''_A)$$

Now we will obtain the time needed if each resistance acts alone, i.e., if a single resistance alone controls the reaction.

(i) The gas film resistance controls (if gas film resistance acts alone) : In this case, no gaseous reactant is present at the surface of solid.

$\therefore$ The concentration driving force $C_{Ag} - C_{As}$ becomes C_{Ag} and is constant at all times during reaction.

Here $$C_{As} = C_{Ac} = 0 \quad \text{... (8)}$$

$$\therefore \quad -r''_A = k_g\, C_{Ag}$$

$$\frac{dl}{dt} = m(-r_A'') = m(k_g C_{Ag})$$

$$\frac{dl}{m k_g C_{Ag}} = dt$$

$$t = t_{\text{film alone}} = \frac{l}{m C_{Ag}}\left(\frac{1}{k_g}\right)$$

(ii) The resistance of ash layer controls (diffusion through ash layer controls) : In case,

$$C_{As} = C_{Ag} \text{ and } C_{Ac} = 0$$

$\therefore$ The concentration driving force is : $C_{As} - C_{Ac} = C_{Ag}$

$\therefore$ $$-r_A'' = \frac{\mathscr{D}_e}{l}\Delta C = \frac{\mathscr{D}_e}{l} C_{Ag}$$

$$\frac{dl}{dt} = m(-r_A'') = m\left(\frac{\mathscr{D}_e}{l}\right) C_{Ag}$$

Integrating, we get

$$t = t_{\text{ash alone}} = \frac{l}{m C_{Ag}}\left(\frac{1}{\mathscr{D}_e/l}\right)$$

(iii) The resistance of surface reaction step controls :

Here $$C_{Ag} = C_{As} = C_{Ac}$$

$\therefore$ $$-r_A'' = k'' C_{Ac} = k'' C_{Ag}$$

$$\frac{dl}{dt} = m(-r_A'') = m(k'' C_{Ag})$$

Integrating, we get

$$t = t_{\text{reaction alone}} = \frac{l}{m C_{Ag}}\left(\frac{1}{k''}\right)$$

Adding Equations (9), (10) and (11), we get

$$t_{\text{film alone}} + t_{\text{ash alone}} + t_{\text{reaction alone}} = \frac{l}{m C_{Ag}}\left(\frac{1}{k_g}\right) + \frac{l}{m C_{Ag}}\left(\frac{1}{\mathscr{D}_e/l}\right) + \frac{l}{m C_{Ag}}\left(\frac{1}{k''}\right)$$

$$= \frac{l}{m C_{Ag}}\left[\frac{1}{k_g} + \frac{1}{\mathscr{D}_e/l} + \frac{1}{k''}\right]$$

Rearranging, we get

$$= \frac{l}{\left(\frac{1}{\frac{1}{k_g} + \frac{1}{\mathscr{D}_e/l} + \frac{1}{k''}}\right) m C_{Ag}}$$

but $$k_{overall} = \frac{1}{\frac{1}{k_g} + \frac{1}{\mathscr{D}_e/l} + \frac{1}{k''}}$$

$$\therefore \quad t_{film\ alone} + t_{ash\ alone} + t_{reaction\ alone} = \frac{l}{k_{overall} \cdot m\, C_{Ag}} \qquad \ldots (12)$$

Compare Equations (7) and (12). RHS of these two equations is the same. From Equations (7) and (12), therefore, we can write

$$\boxed{t_{actual} = t_{film\ alone} + t_{ash\ alone} + t_{reaction\ alone}} \qquad \ldots \textbf{Ans.}$$

Example 1.7 : *For the reaction*

A (g) + B (s) → R (g) ... A diffuses through a gas film and then reacts at the surface.

Mass transfer step : $-r_1'' = k_g (C_g - C_s)$

Reaction step : $-r_2'' = k'' C_s^2$

(i) Linearise the second order rate expression.

(ii) With this approximation, obtain a rate expression (accounting for both the mass transfer and reaction steps.)

Hint : It is decided to carry the reaction at conditions such that the concentration of gaseous reactant at the surface is somewhere in the viscinity of $C_o = 2.5$.

The Taylor expansion of any function f(C) in the viscinity of C_o is

$$f(C) = f(C_o) + \frac{C - C_o}{1!} f'(C_o) + \frac{(C - C_o)^2}{2!} f''(C_o) + \ldots\ldots$$

Solution : A (g) + B (s) ⟶ R (g)

Irreversible G/S reaction ... second order with respect to A.

1. Mass transfer step :

$$-r_1'' = k_g (C_g - C_s) \qquad \ldots (1)$$

2. Reaction step : $-r_2'' = k'' C_s^2 \qquad \ldots (2)$

$$f(C) = -r_2''$$

$$f(C_o) = k'' C_o^2$$

$$(-r_2'')' = 2k'' C_s \qquad \ldots \text{first derivative}$$

$$f'(C_o) = 2k'' C_o$$

$$(-r_2'')'' = 2k'' \qquad \ldots \text{second derivative}$$

$$f''(C_o) = 2k''$$

Now, we will make the Taylor expansion about C_o and retain the linear terms.

$$f(C) = f(C_o) + \frac{C - C_o}{1!} f'(C_o) + \frac{(C - C_o)^2}{2!} f''(C_o) + \ldots$$

$$-r_2'' = k'' C_o^2 + \frac{C_s - C_o}{1} (2k'' C_o) + \frac{(C_s - C_o)^2}{1 \times 2} (2k'') + \ldots$$

Ignoring the third term on the RHS of the above equation, we get the linear approximation of the non-linear rate expression

$$-r_2'' = k'' C_o^2 + 2k'' C_o (C_s - C_o)$$

$$= k'' C_o^2 + 2k'' C_o C_s - 2k'' C_o^2 = 2k'' C_o C_s - k'' C_o^2$$

$$-r_2'' = k'' C_o (2C_s - C_o) \quad \ldots \text{Ans.}$$

At steady state (as steps are in series) we have

$$-r_1'' = -r_2''$$

$$k_g (C_g - C_s) = k'' C_o (2C_s - C_o)$$

$$k_g C_g - k_g C_s = 2k'' C_o C_s - k'' C_o^2$$

$$C_s (k_g + 2k'' C_o) = k_g C_g + k'' C_o^2$$

$$C_s = \frac{k_g C_g + k'' C_o^2}{k_g + 2k'' C_o} \quad \ldots (3)$$

Combining Equations (1) and (3) to eliminate C_s gives

$$-r_1'' = k_g \left(C_g - \frac{(k_g C_g + k'' C_o^2)}{k_g + 2k'' C_o} \right)$$

$$= k_g \left[\frac{k_g C_g + 2k'' C_g C_o - k_g C_g - k'' C_o^2}{k_g + 2k'' C_o} \right]$$

$$= \frac{k_g k'' C_o (2C_g - C_o)}{k_g + 2k'' C_o}$$

Dividing the numerator and denominator of the above equation by $k_g k'' C_o$ gives

$$-r_1'' = \frac{2C_g - C_o}{\frac{1}{k'' C_o} + \frac{2}{k_g}}$$

$$\therefore \quad \boxed{-r_1'' = -r_2'' = -r'' = \left(\frac{1}{\frac{2}{k_g} + \frac{1}{k'' C_o}} \right) (2C_g - C_o)}$$

CHAPTER 2

KINETICS OF FLUID-FLUID REACTIONS

Heterogeneous fluid-fluid reactions are carried out to achieve one of the following objectives :

(i) to produce a particular product (fluid-fluid reactions are used on their own as processes for the manufacture of desired products)

(ii) to separate a constituent from a fluid.

Based upon fluids involved in such reactions we have :

(a) Gas-Liquid Reaction [G/L Reactions].

(b) Liquid-Liquid Reaction [L/L Reactions].

Examples of fluid-fluid reactions that are aimed at production of a desired product are :

(i) Production of chlorobenzene by reaction of gaseous chlorine with liquid benzene ... gas-liquid reaction.

(ii) Formation of soaps by action of aqueous alkalies on fatty acids or fats ... liquid-liquid reaction.

(iii) Nitration of organic compounds with aqueous nitric acid ... liquid-liquid reaction.

(iv) Production of ammonium nitrate by action of ammonia on aqueous nitric acid ..... gas-liquid reaction.

Examples of fluid-fluid reactions that are carried out to separate a constituent from a fluid are :

(i) Nitric acid by absorption of nitric oxide in water ..... gas-liquid reaction.

(ii) Removal of CO_2 from gas mixture using aqueous solutions of potassium carbonate or monoethanolamine (to scrub synthesis gas in the manufacture of ammonia).

(iii) Sulphur removal from petroleum fractions by aqueous ethanolamines ... liquid-liquid reaction.

When aim is to separate a solute A from a gas mixture, the absorption of A can be substantially accelerated by adding to the water (solvent) such a material as will react with A. For example, CO_2 may be removed from a gas mixture by absorbing it with water under pressure. When water is replaced by a solution of potassium carbonate or ethanol amine which react with CO_2, the rate of absorption can be increased enormously. Table 2.1 lists some gas absorption systems with chemical reaction of commercial importance.

Table 2.1 : Gas absorption systems with chemical reaction

Solute	Reagent (Solvent : water)
CO_2	Carbonates : Na_2CO_3, K_2CO_3
CO_2	Hydroxide : NaOH, KOH
CO_2	Ethanol amines : monoethanolamine, diethanolamine
H_2S	Ethanolamines
SO_2	$Ca(OH)_2$
SO_2	Dimethyl aniline
SO_3	H_2SO_4
NO_2	H_2O
CO	Copper ammonium salts
Cl_2	H_2O
HCN	NaOH

All these solute-reagent combinations involve absorption with a chemical reaction to effe the separation of a solute gas from a gas mixture.

Both the objectives cited above involve the reaction of materials originally present different phases. So here the reacting materials (present in the two separate phases) must fir come in contact with each other for reaction to occur. Because of this, in such reaction system both the mass transfer and chemical rates will enter into the overall rate expression.

The Rate Equation :

In the treatment that follows, we consider G/L reactions and develop the rate expressions fo gas-liquid reactions.

Here gas reactant must enter the liquid phase for reaction to occur. Thus the mass transf resistance and the resistance of chemical reaction step enter the overall rate reaction.

Our analysis consider a second order reaction in the liquid phase between reactant transferred from the gas phase, and reactant B, present in the liquid phase.

The stoichiometry of the reaction is represented by

$$A\,(g \rightarrow l) + bB\,(l) \rightarrow R \quad (s, \text{or } l \text{ or } g)$$

and the rate equation by

$$-r_A = k\,C_A\,C_B$$

Please Note that :

(i) Gaseous reactant A is soluble in the liquid. The solubility of gaseous reactant A in th liquid obeys Henry's law (the equilibrium partial pressure of the dissolved gas p_A^* above the solution is directly proportional to its concentration in the liquid phas C_{Al}). A state of equilibrium exists at the interface between the phases and so th relationship between p_A and C_A at the interface is given by

$$p_{Ai} = H_A\,C_{Ai}$$

where H_A is the Henry's law constant – the coefficient of distribution of substance between phases or phase distribution coefficient of A.

(ii) Reactant B is present in the liquid phase and is unable to enter the gas phase. B may be a liquid solvent which is reactive with the gas or it may a solute in the solvent that is reactive.

(iii) The reaction may occur in the liquid film (close to the interface between the phases), in the main body of the liquid, or in both.

(iv) The treatment given here is applicable equally well to liquid-liquid reactions.

Here the rate of reaction can be usefully defined using different bases, as follows :

$$-r_A''' = -\frac{1}{V_r}\frac{dN_A}{dt} \quad \text{... based on the volume of contactor/reactor } (V_r)$$

$$-r_{Al} = -\frac{1}{V_l}\frac{dN_A}{dt} \quad \text{... based on the volume of liquid } (V_l) \qquad \text{... (2.1)}$$

$$-r_A'' = -\frac{1}{S}\frac{dN_A}{dt} \quad \text{... based on the gas-liquid interfacial area (S)}$$

These rates are related by

$$(-r_A''')\, V_r = (-r_{Al})\, V_l = (-r_A'')\, S \qquad \text{... (2.2)}$$

Characteristics of gas-liquid reactors (contactors) :

Liquid hold-up : volume of liquid per unit volume of reactor.

$$f_l = \frac{V_l}{V_r} \quad \text{..... (L hold-up in fraction)}$$

Specific interfacial area : gas-liquid interfacial area per unit volume of reactor/reactor space.

$$a = \frac{S}{V_r}$$

Equation (2.2) can be written as

$$(-r_A''') = (-r_{Al})\frac{V_l}{V_r} = (-r_A'')\frac{S}{V_r}$$

In terms of f_l and 'a' it becomes

$$(-r_A''') = f_l\,(-r_{Al}) = a\,(-r_A'') \qquad \text{... (2.3)}$$

Here reactant A must diffuse or move from the gas phase to the liquid phase for reaction to occur. Consequently, diffusional resistances enter into the overall rate expression. Though there are several theories of mass transfer between two fluid phases, we will develop everything based on the two-film theory (Whitman's two-film theory).

According to the two-film theory of Whitman :

- Resistance to transfer in each phase is regarded as lying in two hypothetical films, one on each side of the interface.
- Equilibrium is assumed to exist at the interface - there is no resistance to transfer across the interface (at the interface itself).
- The transfer in these films is entirely by molecular diffusion.

- The concentration gradient is linear in these films and zero outside i.e. zero in the bulk fluid phases.
- The mass transfer coefficient is directly proportional to diffusivity (for solute gas A : $k_{Al} \propto \mathscr{D}_{Al}$).

[The concentration profile of A (solute gas) in the liquid film is linear for physical absorption (straight mass transfer). The concentration profile of A in the liquid film is curved for absorption with chemical reaction (i.e., when A reacts with B in the liquid phase). Here the concentration of A in the film decreases not only due to diffusion but also because of consumption of A by the reaction. Consequently, the concentration profile in the film is curved.]

The Rate Equation for Straight Mass Transfer (i.e., Physical Absorption of A) :

In physical absorption of a gaseous A in a given liquid there are two resistances in series to transfer of A from the gas to the liquid.

(i) gas-film resistance and (ii) liquid-film resistance.

There is no resistance to transfer of A across the interface and as a result the concentration p_{Ai} and C_{Ai} are equilibrium values. That is a state of equilibrium exists at the interface and for dilute concentrations of most gases the equilibrium relationship is given by Henry's law :

$$p_{Ai} = H_A C_{Ai}$$

where H_A is the Henry's law constant - the coefficient of distribution of A between the phases (Pa·m³ liquid)/mol A.

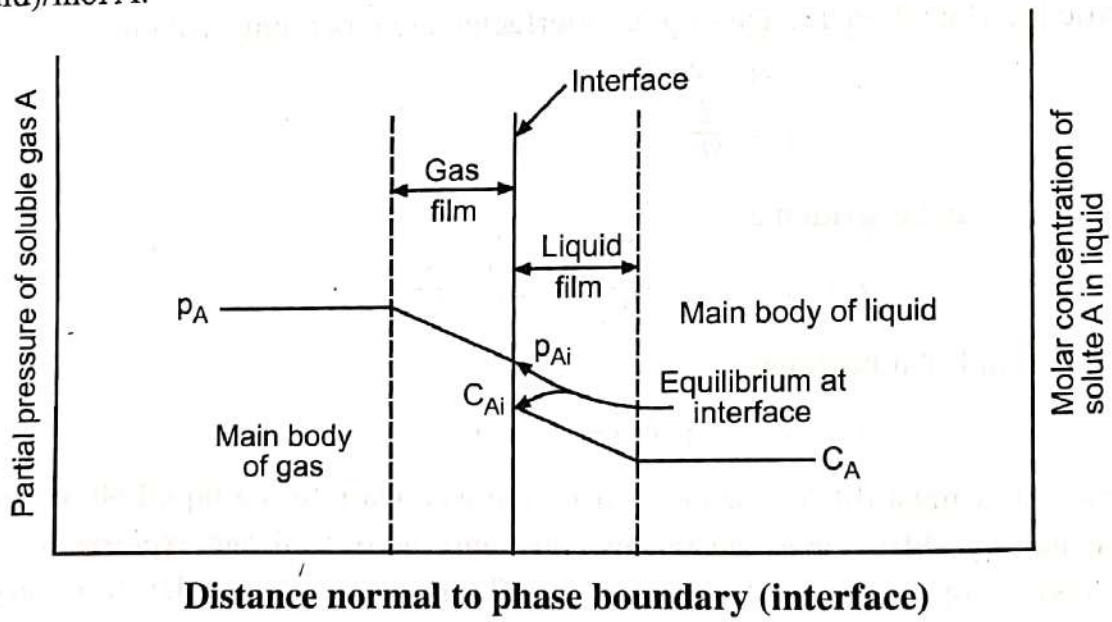


Fig. 2.1 : Establishing the rate equation for straight mass transfer (i.e. physical absorption) of A based on the two-film theory

The rate of mass transfer of A from gas to liquid is given by the following rate expressions :

(i) Through the gas film :

$$-r_A'' = k_{Ag} (p_A - p_{Ai})$$

or

$$-r_A'''' = k_{Ag}\, a\, (p_A - p_{Ai}) \quad \ldots (2.4$$

where p_A is the partial pressure of A in the bulk gas phase in Pa

p_{Ai} is the partial pressure of A at the interface.

k_{Ag} is the gas film mass transfer coefficient in mol/(m²·Pa·s).

$k_{Ag}a$ is the gas film mass transfer coefficient in mol/(m³ contactor · Pa·s)

$(-r_A'')$ is in mol/(m²·s) and $(-r_A'''')$ is in mol/(m³ contactor·s)

(ii) Through the liquid film :

$$-r_A'' = k_{Al}(C_{Ai} - C_A)$$

or
$$-r_A'''' = k_{Al}a\,(C_{Ai} - C_A) \qquad \ldots (2.5)$$

where C_{Ai} is the concentration of A at the interface

C_A is the concentration of A in the bulk liquid phase

k_{Al} is the liquid film mass transfer coefficient in m³ liquid/(m² surface·s)

$k_{Al}a$ is the liquid film mass transfer coefficient in m³ liquid/(m³ contactor·s)

Now to obtain an expression for the rate of transfer of A, accounting the resistances presented by both the gas and liquid films, rewrite Equations (2.4) and (2.5) as

$$\frac{-r_A''''}{k_{Ag}a} = p_A - p_{Ai} \qquad \ldots (2.6)$$

$$\frac{-r_A''''}{k_{Al}a} = C_{Ai} - C_A \qquad \ldots (2.7)$$

Now express C_{Ai} in terms of p_{Ai}

$$C_{Ai} = p_{Ai}/H_A \qquad \ldots \text{Henry's law} \qquad \ldots (2.8)$$

Substituting C_{Ai} from Equation (2.8) into Equation (2.7) gives

$$\frac{-r_A''''}{k_{Al}a} = \frac{p_{Ai}}{H_A} - C_A$$

$$\therefore \quad \frac{H_A}{k_{Al}a}(-r_A'''') = p_{Ai} - H_A C_A \qquad \ldots (2.9)$$

Adding Equations (2.6) and (2.9), we get

$$\frac{-r_A''''}{k_{Ag}a} + \frac{H_A}{k_{Al}a}(-r_A'''') = p_A - p_{Ai} + p_{Ai} - H_A C_A = p_A - H_A C_A$$

$$-r_A''''\left[\frac{1}{k_{Ag}a} + \frac{H_A}{k_{Al}a}\right] = p_A - H_A C_A$$

$$\boxed{-r_A'''' = \frac{1}{\dfrac{1}{k_{Ag}a} + \dfrac{H_A}{k_{Al}a}}(p_A - H_A C_A)} \qquad \ldots (2.10)$$

Equation (2.10) is the overall rate expression for straight mass transfer at any point in t contactor/absorber in terms of known parameters.

$1/k_{Ag}a$ ... gas film resistance, $(Pa \cdot s \cdot m^3)/mol$

$H_A/k_{Al}a$ ..... liquid film resistance, $(Pa \cdot s \cdot m^3)/mol$.

C_A is the concentration of A in the liquid phase, p_A^* is the concentration of A in the g phase, i.e., partial pressure of A in the gas phase that would be in equilibrium with C_A.

$$\therefore \quad p_A^* = H_A C_A \qquad \ldots (2.1$$

Equation (2.10) thus becomes

$$-r_A'''' = \frac{1}{\frac{1}{k_{Ag}a} + \frac{H_A}{k_{Al}a}} (p_A - p_A^*)$$

$$-r_A'''' = \frac{1}{\frac{1}{k_{Ag}a} + \frac{H_A}{k_{Al}a}} \Delta p_A \qquad \ldots (2.1$$

where $\Delta p_A = p_A - p_A^*$ ... overall driving force for the gas phase in terms of partial pressure.

$$\frac{1}{K_{Ag}a} = \frac{1}{k_{Ag}a} + \frac{H_A}{k_{Al}a} \qquad \ldots (2.1$$

overall /total resistance — gas-film resistance — liquid-film resistance

where $K_{Ag}a$ is the overall gas phase mass transfer coefficient in $(Pa \cdot s \cdot m^3)/mol$.

$$\therefore \quad -r_A'''' = K_{Ag}a\,(p_A - p_A^*) = K_{Ag}a\,(p_A - H_A C_A) \qquad \ldots (2.1$$

We have,

$$-r_A'''' = a\,(-r_A'')$$

$$\therefore \quad -r_A'' = \frac{-r_A''''}{a}$$

Substituting $-r_A''''$ from Equation (2.12) into the above equation gives

$$-r_A'' = -\frac{1}{S}\frac{dN_A}{dt} = \frac{1}{a}\left[\frac{1}{\frac{1}{k_{Ag}a} + \frac{H_A}{k_{Al}a}}\right] \Delta p_A$$

$$\boxed{-r_A'' = -\frac{1}{S}\frac{dN_A}{dt} = \frac{1}{\frac{1}{k_{Ag}} + \frac{H_A}{k_{Al}}} \Delta p_A} \qquad \ldots (2.1$$

$$\frac{1}{K_{Ag}} = \frac{1}{k_{Ag}} + \frac{H_A}{k_{Al}}$$

K_{Ag} is the overall gas phase mass transfer coefficient in m^3 liquid/(m^2 surface·s).

The Rate Equation for Mass Transfer and Reaction (i.e., for mass transfer + reaction) :

Here we have eight cases to consider, from infinitely fast to very slow reaction, as follows :

Case A : Instantaneous reaction with low C_B ..... reaction in the liquid film (at a plane)

Case B : Instantaneous reaction with high C_B ..... reaction in the liquid film (at the interface)

Case C : Fast reaction with low C_B ..... reaction in the liquid film (in zone in the liquid film)

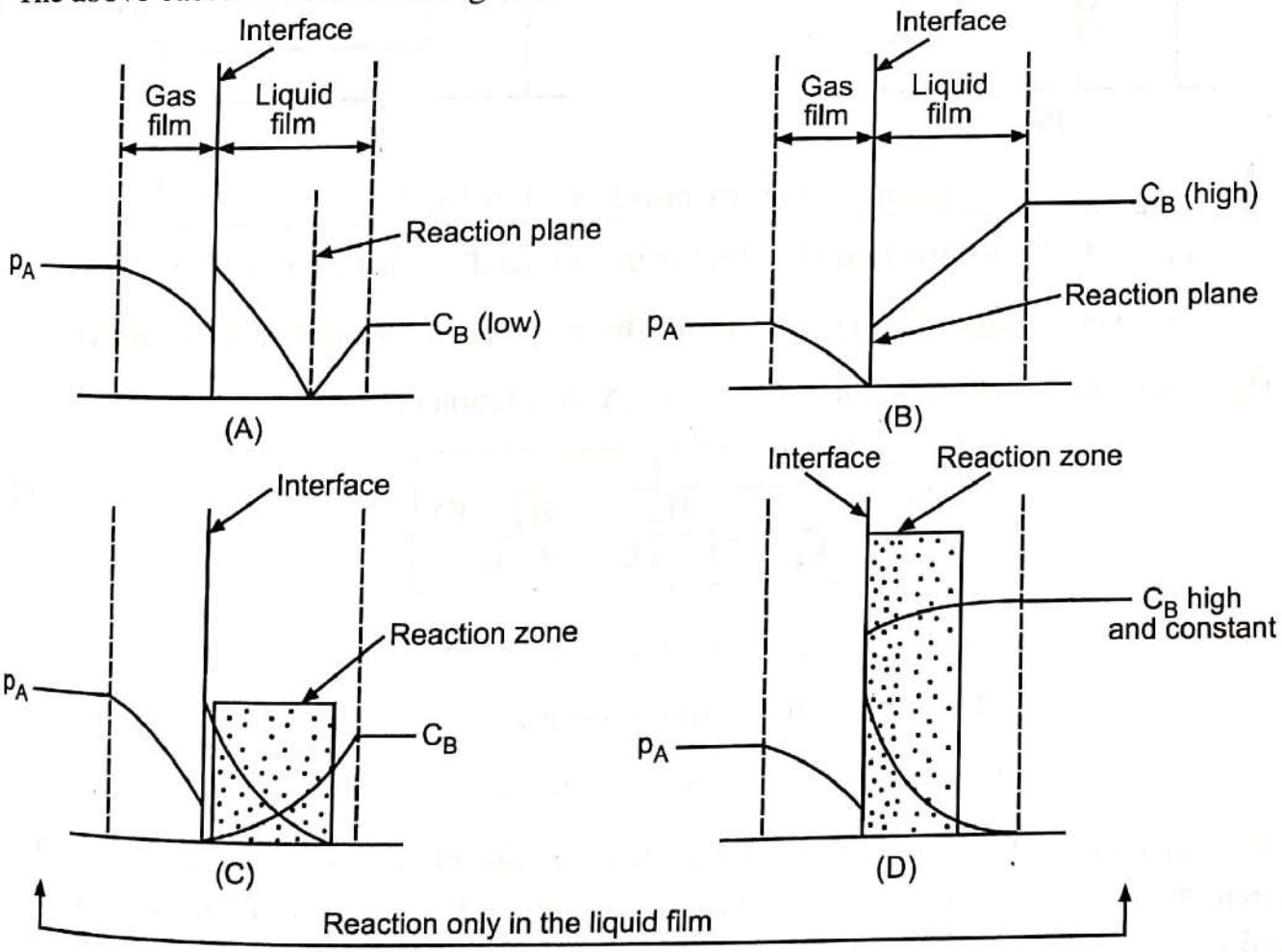
Case D : Fast reaction with high C_B ..... reaction in the liquid film in zone near the interface

Cases E and F : Intermediate rate with respect to mass transfer ..... reaction in the liquid film and in the main body of the liquid.

Case G : Slow reaction ..... reaction in the main body of the liquid with liquid film resistance.

Case H : Slow reaction ..... reaction in the main body of the liquid without any mass transfer resistance.

The above cases are shown in Fig. 2.2.



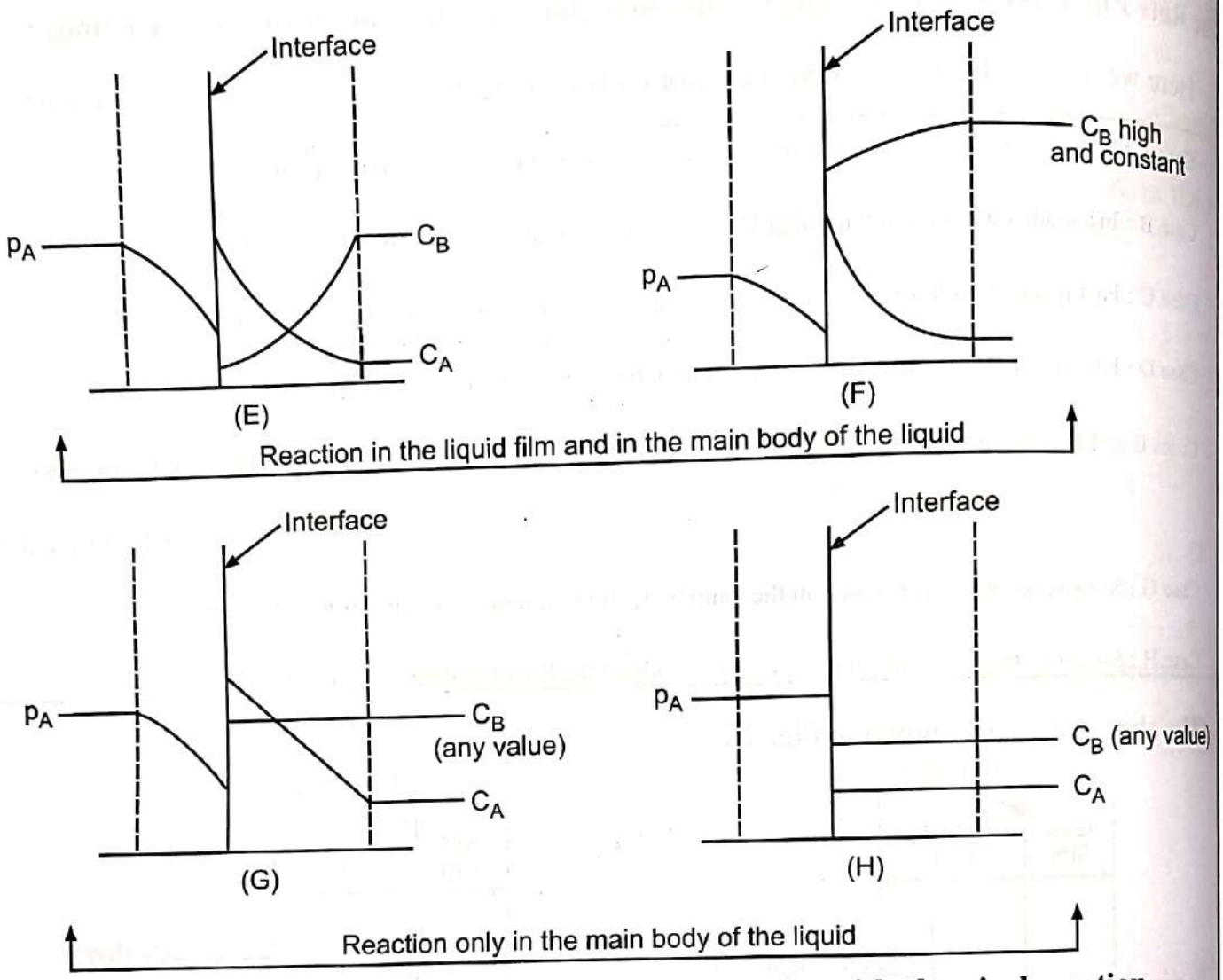


Fig. 2.2 : Concentration profiles for mass transfer with chemical reaction (covering complete range of rates of the reaction and of the mass transfer)

The general rate expression for mass transfer with reaction is

$$-r_A'''' = \frac{1}{\frac{1}{k_{Ag}\,a} + \frac{H_A}{k_{Al}\,a\,E} + \frac{H_A}{k\,C_B\,f_l}}\,p_A \qquad \ldots (2.1$$

where $1/k_{Ag}\,a$ – gas film resistance

$H_A/k_{Al}\,aE$ – liquid film resistance

$H_A/k\,C_B\,f_l$ – liquid bulk resistance

It is customary to treat gas-liquid reactions as absorption accompanied by a chemic reaction. The effect of chemical reaction is to accelerate the rate of absorption of A from g (reaction lowers the resistance of the liquid film). So, when the reaction occurs in the liquid film the absorption of A from gas is larger than for straight mass transfer (physical absorption) for t same concentrations at the two boundaries of the liquid.

While determining the liquid film enhancement factor due to chemical reaction for the same concentrations at the two boundaries of the liquid film we have to compare the rate of physical absorption with the rate of chemosorption (rate of absorption in the presence of a chemical reaction).

$$\text{Liquid film enhancement factor} = E = \frac{\text{rate of take up of A with reaction}}{\text{rate of take up of A with straight mass transfer alone}}$$

for the same C_{Ai}, C_A, C_{Bi} and C_B in the two cases.

E is always greater than or equal to one. E is a function of E_i and M_H.

E_i = enhancement factor for an infinitely fast reaction

M_H^2 = maximum possible conversion in the film compared to maximum possible diffusion through the film.

M_H = Hatta modulus.

In Fig. 2.3, the liquid film enhancement factor E is plotted against M_H (the Hatta modulus – dimensionless) with E_i as a parameter (for various values of E_i) for a second order reaction - numbers on the curves are the values of E_i (which is dimensionless).

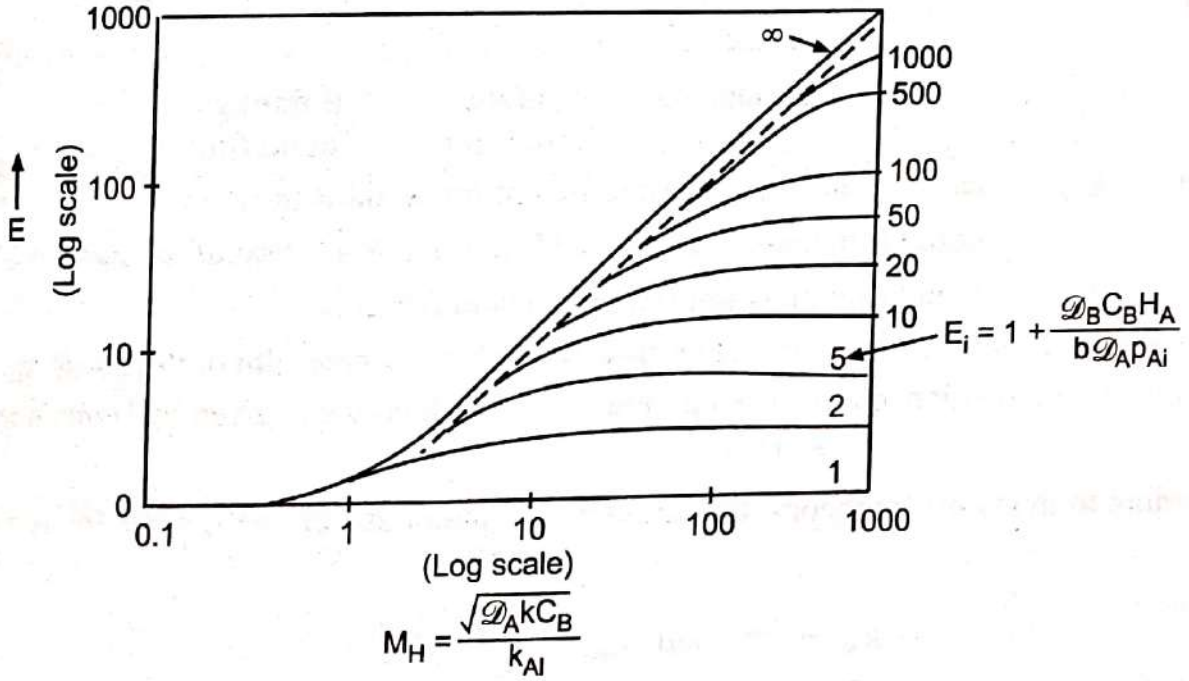


Fig. 2.3 : Enhancement factor E as a function of E_i and M_H - second order reaction

If $E_i > 5\ M_H$ ... pseudo first order reaction in the film (by the interface) ... for this case ... $E = M_H$.

If $E_i < M_H/5$ ... instantaneous reaction at a plane in the film ... for this case ... $E = E_i$.

Now we proceed to obtain the kinetic equations for some of these cases (kinetic regimes) - cited above :

A : Instantaneous Reaction (with low C_B) : For low C_B, we have the situation as shown in Fig. 2.4. Here an element of liquid contains either A or B but not both. Consequently, reaction will occur at a plane between A-containing and B-containing liquid (in the liquid film). As reactants A and B must diffuse to this reaction plane for the reaction to occur, the general rate will be decided by the rate of diffusion of A and B to this reaction plane.

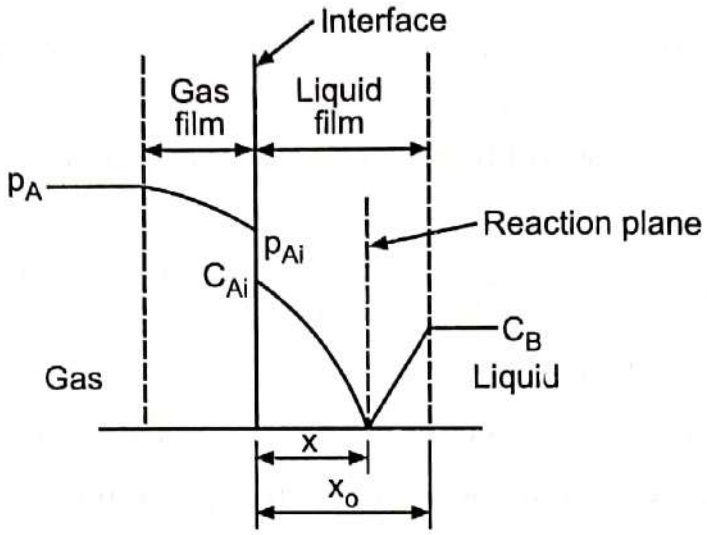


Fig. 2.4 : Concentration profiles of reactants for an infinitely fast irreversible reaction for case A – low C_B – (two-film theory)

At steady state, the flow rate of B toward the reaction zone will be b times that of A. Th the rate of disappearance of A and B based on the unit interfacial surface are given by

$$-r_A'' = \frac{-r_B''}{b} = k_{Ag}(p_A - p_{Ai}) = k_{Al}(C_{Ai} - 0)\frac{x_o}{x} = \frac{k_{Bl}}{b}(C_B - 0)\frac{x_o}{x_o - x} \quad \ldots (2.$$

A through gas film | A through liquid film | B through liquid film

where, k_{Ag} – gas film mass transfer coefficient for A, mol/(m²·Pa·s)

k_{Al} – liquid film mass transfer coefficient for A, m³ liquid/(m² surface·s)

k_{Bl} – liquid film mass transfer coefficient for B.

The liquid side coefficients are based on flow through the whole film of thickness x_o.

The equilibrium relationship between p_A and C_A at the interface is given by Henry's law

$$p_{Ai} = H_A C_{Ai} \quad \ldots (2.$$

According to the two-film theory, the liquid phase mass transfer coefficients for A and B given by

$$k_{Al} = \frac{\mathscr{D}_{Al}}{x_o} \text{ and } k_{Bl} = \frac{\mathscr{D}_{Bl}}{x_o}$$

$$\frac{k_{Al}}{k_{Bl}} = \frac{\mathscr{D}_{Al}/x_o}{\mathscr{D}_{Bl}/x_o} = \frac{\mathscr{D}_{Al}}{\mathscr{D}_{Bl}} \quad \ldots (2.$$

Now we will eliminate the unknowns x, x_o, C_{Ai}, and p_{Ai} from the above equations to obt the desired rate equation in terms of known quantities.

From Equation (2.17) we have

$$-r_A'' = k_{Ag}(p_A - p_{Ai})$$

$$\frac{-r_A''}{k_{Ag}} = p_A - p_{Ai}$$

Dividing the above equations by H_A gives

$$\frac{-r_A''}{H_A k_{Ag}} = \frac{p_A}{H_A} - \frac{p_{Ai}}{H_A} \quad \ldots (2.20)$$

$$-r_A'' = k_{Al}(C_{Ai} - 0)\frac{x_o}{x}$$

$$\frac{-r_A''}{k_{Al}} = C_{Ai}\frac{x_o}{x} \quad \ldots (2.21)$$

Also, $$k_{Al}(C_{Ai} - 0)\frac{x_o}{x} = \frac{k_{Bl}}{b}(C_B - 0)\frac{x_o}{x_o - x}$$

$\therefore$ $$\frac{C_{Ai}}{x} = \frac{k_{Bl} C_B}{b\, k_{Al}(x_o - x)}$$

$$C_{Ai} x_o - C_{Ai} x = \frac{k_{Bl}}{b\, k_{Al}} C_B x$$

$$C_{Ai} x_o = \left[\frac{k_{Bl}}{b\, k_{Al}} C_B + C_{Ai}\right] x$$

$$\frac{x_o}{x} = \frac{\left[\frac{k_{Bl}}{b\, k_{Al}} C_B + C_{Ai}\right]}{C_{Ai}} \quad \ldots (2.22)$$

Using Equation (2.22) to substitute for x_o/x, Equation (2.21) becomes

$$\frac{-r_A''}{k_{Al}} = C_{Ai}\left(\frac{\frac{k_{Bl}}{b\, k_{Al}} C_B + C_{Ai}}{C_{Ai}}\right)$$

$$\frac{-r_A''}{k_{Al}} = \frac{k_{Bl}}{b\, k_{Al}} C_B + C_{Ai} \quad \ldots (2.23)$$

Henry's law is

$$p_{Ai} = H_A C_{Ai}$$

$$C_{Ai} = \frac{p_{Ai}}{H_A} \quad \ldots (2.24)$$

Substituting C_{Ai} from Equation (2.24) into Equation (2.23) yields

$$\frac{-r_A''}{k_{Al}} = \frac{k_{Bl}}{b\, k_{Al}} C_B + \frac{p_{Ai}}{H_A} \quad \ldots (2.25)$$

Adding Equations (2.20) and (2.25) gives

$$\frac{-r_A''}{H\, k_{Ag}} + \frac{-r_A''}{k_{Al}} = \frac{p_A}{H_A} - \frac{p_{Ai}}{H_A} + \frac{k_{Bl}}{b\, k_{Al}} C_B + \frac{p_{Ai}}{H_A}$$

$$-r_A''\left[\frac{1}{H_A\, k_{Ag}} + \frac{1}{k_{Al}}\right] = \frac{p_A}{H_A} + \frac{k_{Bl}}{b\, k_{Al}} C_B \quad \ldots (2.26)$$

We have,

$$\frac{k_{Al}}{k_{Bl}} = \frac{\mathscr{D}_{Al}/x_o}{\mathscr{D}_{Bl}/x_o}$$

$$\therefore \quad \frac{k_{Bl}}{k_{Al}} = \frac{\mathscr{D}_{Bl}}{\mathscr{D}_{Al}}$$

Replacing k_{Bl}/k_{Al} by $\mathscr{D}_{Bl}/\mathscr{D}_{Al}$ in Equation (2.26) gives

$$-r''_A\left[\frac{1}{H_A k_{Ag}} + \frac{1}{k_{Al}}\right] = \frac{p_A}{H_A} + \frac{\mathscr{D}_{Bl}}{\mathscr{D}_{Al}}\frac{C_B}{b}$$

$$\boxed{-r''_A = -\frac{1}{S}\frac{dN_A}{dt} = \frac{\dfrac{\mathscr{D}_{Bl}}{\mathscr{D}_{Al}}\dfrac{C_B}{b} + \dfrac{p_A}{H_A}}{\dfrac{1}{H_A k_{Ag}} + \dfrac{1}{k_{Al}}}} \quad \ldots (2$$

..... for $k_{Ag} p_A \geq \dfrac{k_{Bl} C_B}{b}$

... reaction takes place within the liquid film ..... liquid film controls.

Special case of negligible gas phase resistance :

If pure gaseous reactant A is used, then the gas phase resistance is negligible. So for no phase resistance :

$$p_A = p_{Ai} \text{ (or } k_g \to \infty)$$

Equation (2.27) for negligible gas phase resistance, by neglecting the term $1/H_A k_{Ag}$ (a being very small), becomes

$$-r''_A = -\frac{1}{S}\frac{dN_A}{dt} = \frac{\dfrac{\mathscr{D}_{Bl} C_B}{\mathscr{D}_{Al} b} + \dfrac{p_A}{H_A}}{1/k_{Al}}$$

$$-r''_A = k_{Al}\left[\frac{p_A}{H_A} + \frac{\mathscr{D}_{Bl} C_B}{\mathscr{D}_{Al} b}\right]$$

With $p_A = p_{Ai}$, the above equation becomes

$$-r''_A = k_{Al}\left[\frac{p_{Ai}}{H_A} + \frac{\mathscr{D}_{Bl} C_B}{\mathscr{D}_{Al} b}\right]$$

We have : $p_{Ai} = H_A C_{Ai} \quad \therefore \quad p_{Ai}/H_A = C_{Ai}$

With this, above equation becomes

$$-r''_A = k_{Al}\left[C_{Ai} + \frac{\mathscr{D}_{Bl} C_B}{\mathscr{D}_{Al} b}\right]$$

Taking C_{Ai} common gives

$$-r''_A = k_{Al} C_{Ai}\left[1 + \frac{\mathscr{D}_{Bl} C_B}{\mathscr{D}_{Al} b\, C_{Ai}}\right] \quad \ldots (6$$

The maximum rate of straight mass transfer is

$$-r''_A = k_{Al} C_{Ai} \quad \ldots (6$$

Comparing Equation (2.28) with Equation (2.29) we find that the term in bracket represents the increase in rate of absorption of A due to addition of reactant B to the liquid. It is the enhancement factor E_i for infinitely fast reaction and is the ratio of rate with reaction to rate with mass transfer alone.

$$E_i = 1 + \frac{\mathscr{D}_{Bl} C_B}{b\, \mathscr{D}_{Al} C_{Ai}} = \frac{-r_A'' \text{ with reaction}}{-r_A'' \text{ with mass transfer alone}}$$

The rate expression for infinitely fast reaction and no gas phase resistance is

$$-r_A'' = k_{Al} C_{Ai} E_i \qquad \ldots (2.30)$$

B : Instantaneous Reaction (with High C_B) :

If C_B is high $\left(\text{more precisely : if } k_{Ag} p_A \leq \frac{k_{Bl} C_B}{b}\right)$, then the reaction plane moves to and stays at the gas-liquid interface. When this happens, the gas-phase resistance controls. Here $p_{Ai} = 0$ and it means that the total resistance is in the gas film and so the overall rate is controlled by diffusion of A through the gas-film. Increasing C_B further has no effect on the overall rate (the rate is not affected by raising C_B further).

Here the rate is given by the rate of diffusion of A with $p_{Ai} = 0$ ... refer Equation (2.17) :

$$\boxed{-r_A'' = k_{Ag} p_A} \qquad \ldots (2.31)$$

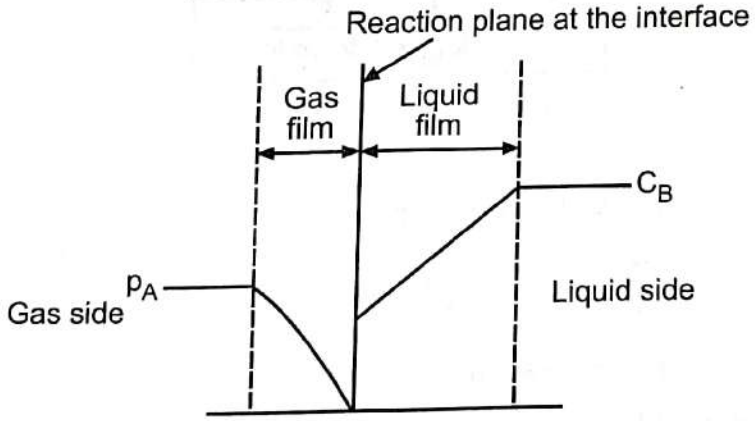


Fig. 2.5 : Concentration profiles of reactants for infinitely fast irreversible reaction – two film theory–case B – high C_B

As the rate expressions given by Equations (2.27) and (2.31) do not include the reaction rate constant, the rate of instantaneous reaction is completely mass transfer controlled.

If $k_{Ag} p_A \geq \frac{k_{Bl}}{b} C_B$, then we have Case A ... liquid film controls ... use Equation (2.27)

... reaction (at a plane) in the liquid film.

If $k_{Ag} p_A \leq \frac{k_{Bl}}{b} C_B$, then we have Case B ... gas film controls ... use Equation (2.31)

... reaction at the interface.

C : Fast Reaction (with low C_B) :

Here reaction takes place in a zone where both A and B are present (In this case the plane of reaction of Case A spreads into a zone of reaction. Refer Fig. 2.6). Since the reaction is fast enough the reaction zone remains totally within the liquid film. So, A does not enter the main body of the liquid to react there.

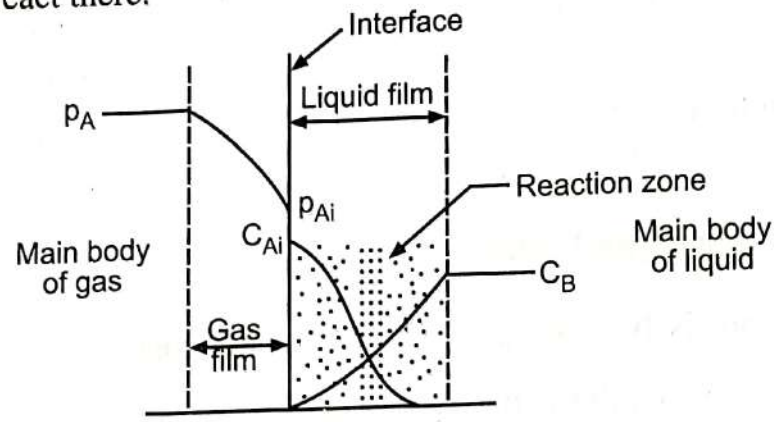


Fig. 2.6 : Fast second order reaction – Case C - low C_B – reaction zone in the liquid film

As the reaction is fast, the last resistance term in the general rate equation, Equation (2.16), is negligible (as k is large for the fast reaction). Therefore, the rate equation for this case, by neglecting the last resistance term of Equation (2.16), is

$$-r_A'''' = \frac{1}{\frac{1}{k_{Ag}\,a} + \frac{H_A}{k_{Al}\,aE}}\,p_A \qquad \ldots (2.32)$$

In this case for :

Gas film : $$-r_A'''' = k_{Ag}\,a\,(p_A - p_{Ai}) \qquad \ldots (2.33)$$

Liquid film : $$-r_A'''' = k_{Al}\,a\,C_{Ai}\,E \qquad \ldots (2.34)$$

We have : $$p_{Ai} = H_A\,C_{Ai}$$

Rearranging Equation (2.33), we get

$$\frac{(-r_A'''')}{k_{Ag}\,a} = p_A - p_{Ai} \qquad \ldots (2.35)$$

Rearranging Equation (2.34) gives

$$\frac{(-r_A'''')}{k_{Al}\,aE} = C_{Ai} = \frac{p_{Ai}}{H_A} \qquad \ldots (2.36)$$

Dividing both sides of Equation (2.35) by H_A gives

$$\frac{-r_A''''}{H_A\,k_{Ag}\,a} = \frac{p_A}{H_A} - \frac{p_{Ai}}{H_A} \qquad \ldots (2.37)$$

Adding Equations (2.36) and (2.37), we get

$$-r_A'''' \left[\frac{1}{H_A k_{Ag} a} + \frac{1}{k_{Al} aE}\right] = \frac{p_A}{H_A} - \frac{p_{Ai}}{H_A} + \frac{p_{Ai}}{H_A} = \frac{p_A}{H_A}$$

$$\therefore \quad -r_A'''' = \left[\frac{1}{\frac{1}{H_A k_{Ag} a} + \frac{1}{k_{Al} aE}}\right] \frac{p_A}{H_A}$$

Taking H_A of p_A/H_A term inside the bracket gives

$$-r_A'''' = \frac{1}{\frac{1}{k_{Ag} a} + \frac{H_A}{k_{Al} aE}} p_A \qquad \text{... [Equation (2.32)]}$$

D : Fast Reaction (with High C_B) :

When C_B is so high that it can be considered to be constant throughout the liquid film, the reaction in the film becomes pseudo first order with respect to A. Due to high C_B and fast reaction the reaction zone spreads from the interface in the liquid film but does not cross the boundary.

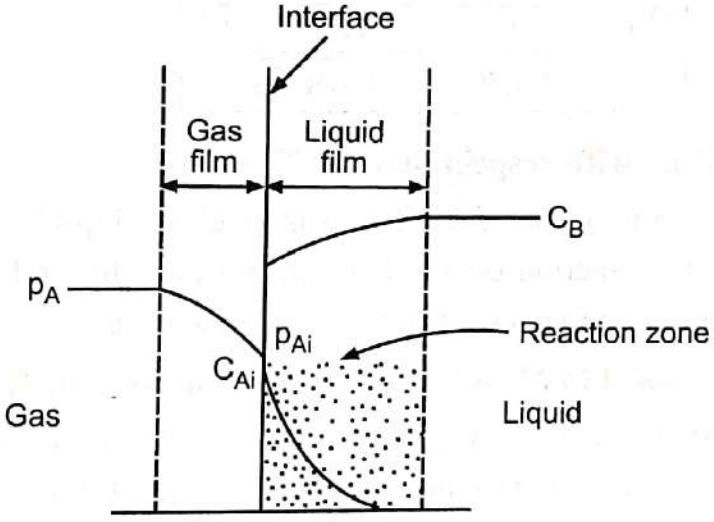


Fig. 2.7 : Fast second order reaction – Case D-high C_B - zone spread from the interface

$$-r_{Al} = k C_A C_B = (k C_B) C_A = k_1 C_A \qquad \text{... (2.38)}$$

$$-r_A'''' = k_{Ag} a (p_A - p_{Ai})$$

and

$$-r_A'''' = k_{Al} a C_{Ai} E$$

In this case, the enhancement factor is given by

$$E = M_H = \frac{\sqrt{\mathscr{D}_{Al} k C_B}}{k_{Al}} \qquad \text{... (2.39)}$$

$$\therefore \quad -r_A'''' = \frac{k_{Al} a C_{Ai} \sqrt{\mathscr{D}_{Al} k C_B}}{k_{Al}} = a \left(\sqrt{\mathscr{D}_{Al} k C_B}\right) C_{Ai} \qquad \text{... (2.40)}$$

$$\frac{(-r_A'''')}{k_{Ag} a} = p_A - p_{Ai}$$

$$\frac{-r_A''''}{H_A k_{Ag} a} = \frac{p_A}{H_A} - \frac{p_{Ai}}{H_A} \qquad \text{... (2.41)}$$

Rearranging Equation (2.40) gives

$$\frac{-r_A'''}{a\sqrt{\mathscr{D}_{Al}\,k\,C_B}} = C_{Ai} = \frac{p_{Ai}}{H_A} \qquad \ldots (2.$$

Adding Equations (2.41) and (2.42) gives

$$\frac{-r_A'''}{H_A\,k_{Ag}\,a} + \frac{-r_A'''}{a\sqrt{\mathscr{D}_{Al}\,k\,C_B}} = \frac{p_A}{H_A} - \frac{p_{Ai}}{H_A} + \frac{p_{Ai}}{H_A} = \frac{p_A}{H_A}$$

$$-r_A''' \left[\frac{1}{H_A\,k_{Ag}\,a} + \frac{1}{a\sqrt{\mathscr{D}_{Al}\,k\,C_B}}\right] = \frac{p_A}{H_A}$$

$$-r_A''' = \frac{1}{\dfrac{1}{H_A\,k_{Ag}\,a} + \dfrac{1}{a\sqrt{D_{Al}\,k\,C_B}}} \cdot \frac{p_A}{H_A}$$

$\therefore$

$$\boxed{-r_A'''' = \frac{1}{\dfrac{1}{k_{Ag}\,a} + \dfrac{H_A}{a\sqrt{\mathscr{D}_{Al}\,k\,C_B}}}\,p_A} \qquad \ldots (2.$$

E and F : Intermediate Rate with respect to Mass Transfer :

Here reaction is so slow that some of A diffuses through the liquid film into the main body the liquid. Consequently, the reaction occurs both within the film and in the main body of t liquid. In this case one has to use the general rate equation with its three resistances.

G : Slow reaction with respect to Mass Transfer (With mass transfer resistance) :

Here all of the reaction occurs in the main body of the liquid with the liquid film s offering a resistance to transfer of A into the main body of liquid. So the gas film, the liquid fi and the main body of the liquid act as resistances in series and all these resistances will enter i the overall rate expression.

For these three steps we can write :

(a) Gas film : $-r_A'''' = k_{Ag}\,a\,(p_A - p_{Ai})$

(b) Liquid film : $-r_A'''' = k_{Al}\,a\,(C_{Ai} - C_A)$

(c) Main body of liquid : $-r_{Al} = -\dfrac{1}{V_l}\dfrac{dN_A}{dt} = k\,C_A\,C_B$

but $(-r_A'''')\,V_r = (-r_{Al})\,V_l$

$$(-r_A'''') = (-r_{Al})\frac{V_l}{V_r}$$

$\therefore\ (-r_A'''') = k\,C_A\,C_B\,f_l, \qquad f_l = V_l/V_r$

$$\frac{-r_A''''}{k_{Ag}\,a} = p_A - p_{Ai} \ \ldots \text{step (a)} \ldots\ldots \text{dividing by } H_A \text{ we get}$$

$$\frac{-r_A'''}{H\,k_{Ag}\,a} = \frac{p_A}{H_A} - \frac{p_{Ai}}{H_A} = \frac{p_A}{H_A} - C_{Ai}, \quad \text{as } p_{Ai} = H_A\,C_{Ai} \qquad \ldots (2.44)$$

$$\frac{-r_A'''}{k_{Al}\,a} = C_{Ai} - C_A \qquad \ldots\ldots \text{ step (b)} \qquad \ldots (2.45)$$

$$\frac{-r_A'''}{k\,C_B\,f_l} = C_A \qquad \ldots\ldots \text{ step (c)} \qquad \ldots (2.46)$$

Adding Equations (2.44), (2.45), (2.46) gives

$$(-r_A''') \left[\frac{1}{H_A\,k_{Ag}\,a} + \frac{1}{k_{Al}\,a} + \frac{1}{k\,C_B\,f_l}\right] = \frac{p_A}{H_A} - C_{Ai} + C_{Ai} - C_{Al} + C_{Al} = \frac{p_A}{H_A}$$

$$(-r_A''') = \frac{1}{\dfrac{1}{H_A\,k_{Ag}\,a} + \dfrac{1}{k_{Al}\,a} + \dfrac{1}{k\,C_B\,f_l}} \cdot \frac{p_A}{H_A}$$

$$\boxed{-r_A''' = \frac{1}{\dfrac{1}{k_{Ag}\,a} + \dfrac{H_A}{k_{Al}\,a} + \dfrac{H_A}{k\,C_B\,f_l}}\; p_A} \qquad \ldots (2.47)$$

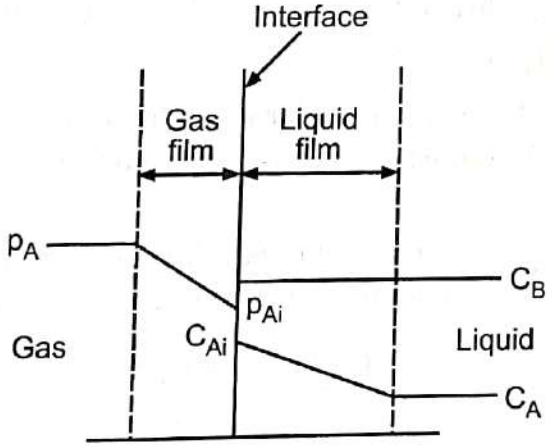


Fig. 2.8 : Slow reaction (Case G) – showing liquid-film resistance

H : Infinitely Slow Reaction :

Here the mass transfer resistance is negligible. Therefore, the concentrations of A and B are uniform throughout the liquid and the overall rate is controlled by a chemical step alone, i.e., the rate is determined by chemical kinetics alone.

The rate of reaction based on unit volume of liquid is given by

$$-r_{Al} = -\frac{1}{V_l}\frac{dN_A}{dt} = k\,C_A\,C_B$$

$(-r_A''')$ and $(-r_{Al})$ are related by

$$(-r_A''')\,V_r = (-r_{Al})\,V_l$$

$$\therefore \quad -r_A''' = (-r_{Al})\frac{V_l}{V_r}$$

$$\therefore \quad -r_A''' = k\frac{V_l}{V_r} C_A C_B$$

as $f_l = V_l/V_r$, the above equation becomes

$$\boxed{-r_A''' = k f_l C_A C_B} \quad \ldots (2.4$$

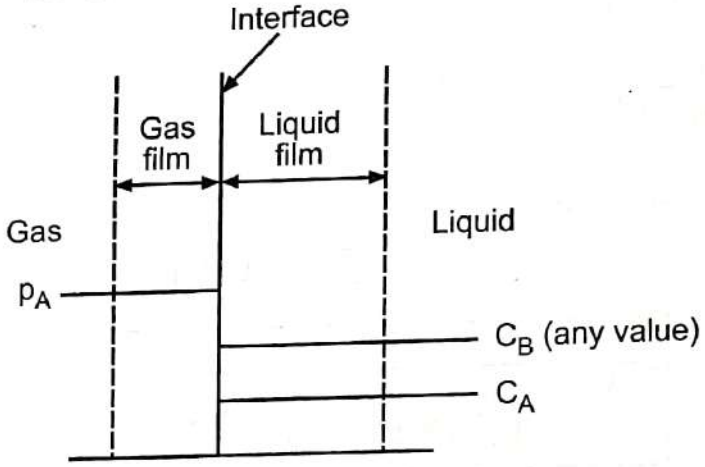


Fig. 2.9 : Infinitely slow reaction (Case H) - shows no film resistance

Role of the Hatta Number, M_H :

In order to know whether the reaction under consideration is fast or slow, we define a fi conversion factor, assuming negligible gas phase resistance and considering a unit gas-liq interfacial surface.

$$M_H^2 = \frac{\text{maximum possible conversion in the film}}{\text{maximum possible diffusion through the film}}$$

$$M_H^2 = \frac{k C_{Ai} C_B x_o}{\frac{\mathscr{D}_{Al}}{x_o} C_{Ai}} = \frac{k C_B x_o^2}{\mathscr{D}_{Al}} \quad \ldots (2.4$$

We have, $$k_{Al} = \frac{\mathscr{D}_{Al}}{x_o}$$

$$x_o = \frac{\mathscr{D}_{Al}}{k_{Al}}$$

Substituting this value of x_o into Equation (2.49) gives

$$M_H^2 = \frac{k C_B}{\mathscr{D}_{Al}} \left(\frac{\mathscr{D}_{Al}}{k_{Al}}\right)^2$$

$$M_H^2 = \frac{k C_B \mathscr{D}_{Al}}{k_{Al}^2} \quad \ldots (2.5$$

$$M_H = \frac{\sqrt{k C_B \mathscr{D}_{Al}}}{k_{Al}} \quad \ldots (2.5$$

The value of M_H provides an important indication of whether a large specific interfacial area (a) or a large liquid hold-up (in fraction - f_l) is required in the reactor to be designed for a particular reaction (with second order rate constant k).

It has been found that :

If $M_H > 2$ ... reaction occurs in the liquid film, and we have cases A, B, C and D.

If $0.02 < M_H < 2$ ... we have the cases E, F and G.

If $M_H < 0.02$ ... reaction occurs in the main body of liquid with no film resistances, and we have the Case H.

When M_H is large (fast reaction) we require a large interfacial area. Therefore for fast reactions, select a contacting device which provides a large interfacial area $\Rightarrow$ such as packed column, plate column, agitated tank (energy for agitation needs to be considered).

When M_H is very small (slow reaction) we require a large volume of liquid (i.e., large f_l). So for slow reactions select a contacting device which provides a large $f_l \Rightarrow$ such as bubble tank, bubble column, packed bubble column (bubble contactors), agitated tank [agitation needed to create large interfacial areas of contact between gas and liquid phases (by providing small bubbles) is of no benefit here].

When we require a large interfacial area as well as volume of liquid, then use agitated tanks.

Use of solubility Data to determine the Kinetic Regime :

Here we make use of solubility data expressed as the Henry's law constant [solubility data is one of the forms of equilibrium data (for absorption) that include the solubility of the solute gas in the solvent].

When the reaction occurs in the film we can predict the relative importance of both film resistances based on the numerical value of H_A (the Henry's law constant - the coefficient for the distribution of substance A between the phases–phase distribution coefficient for A.)

For doing this we consider the rate expression for straight mass transfer of A across the gas and liquid films.

$$-r_A'' = -\frac{1}{S}\frac{dN_A}{dt} = \frac{1}{\frac{1}{k_{Ag}} + \frac{H_A}{k_{Al}}}\,\Delta p_A \qquad \text{... Equation (2.15)}$$

At low values of H_A, that is, when the gas is highly soluble in the liquid : For highly soluble gases as H_A is very small, $1/k_{Ag} >> H_A/k_{Al}$, i.e., the gas film resistance is large and thus it controls the rate.

On the other hand, for slightly soluble gases as H_A is large, $H_A/k_{Al} >> 1/k_{Ag}$, i.e., the liquid film resistance is large and thus it controls the rate.

Gas film resistance controls for highly soluble gases ... e.g., ammonia in water .....

... $H_A = 0.02$ (Pa · m³)/mol at 20°C

Liquid film resistance controls for slightly soluble gases ... e.g., O_2 in water .....

... $H_A = 7400$ (Pa·m³)/mol at 20°C

A highly soluble gas is easy to absorb and has its main resistance in the gas phase. Therefore, for such gases, a chemical reaction is not beneficial.

A sparingly soluble gas is difficult to absorb and has its main resistance in the liquid phase. Therefore, for such gases, a chemical reaction that speeds-up the rate is beneficial.

SOLVED EXAMPLES

Example 2.1 : *It is proposed to remove CO_2 from air by countercurrent contact with water at 25°C.*

(i) Find the resistances of the gas and liquid films for this operation ?

(ii) Suggest the simplest form of rate equation for tower design.

(iii) For this absorption operation, would you expect reaction with absorption to be helpful ? If so, why ?

Data : For CO_2 between air and water,

$$k_g a = 0.80 \text{ mol/(h·m}^3\text{·Pa)}$$

$$k_l a = 25 \text{ h}^{-1}$$

$$H = 3000 \text{ (Pa·m}^3\text{)/mol}$$

Solution : For absorption of CO_2 (A) in water, i.e., for straight mass transfer of A, both gas and liquid films offer resistance, as the transfer of A takes place through the gas film followed by the liquid film (with interface in between).

$$k_g a = k_{Ag} a = 0.80 \text{ mol/(h·m}^3\text{·Pa)}$$

$$k_l a = k_{Al} a = 25 \text{ h}^{-1}$$

$$H = H_A = 3000 \text{ (Pa·m}^3\text{)/mol}$$

Resistance offered by the gas film $= R_g = 1/k_{Ag} a$

$= 1/0.80$

$= 1.25 \text{ (Pa·m}^3\text{·h)/mol}$

Resistance offered by the liquid film $= R_l = H_A/k_{Al} a$

$$= \frac{3000}{25} = \mathbf{120 \ (Pa \cdot m^3 \cdot h)/mol}$$ ... Ans. (i)

Total resistance to mass transfer $= R = R_g + R_l$

$= 1.25 + 120$

$= 121.25 \text{ (Pa·m}^3\text{·h)/mol}$

Resistance in the gas film $= \frac{1.25}{121.25} \times 100 = 1.03\%$

Resistance in the liquid film $= \frac{120}{121.25} \times 100 = 98.97\%$

So 1.03% of the resistance is offered by the gas film and 98.97% of the resistance is offered by the liquid film. ... Ans. (ii)

Relative resistance :

$$\text{Resistance of the liquid film relative to that of gas film} = \frac{120}{1.25} = \frac{96}{1}$$ ... Ans. (iii)

Here the gas film resistance is very very small (1.03%) in comparison with that of the liquid film (98.97%). As the main resistance is in the liquid film (and with little error ignoring the gas film resistance-being very very small), we can safely say that the absorption is liquid film controlling with rate equation :

$$\boxed{-r_A'' = k_{Al}(C_{Ai} - C_A)} \quad \text{... Ans. (ii)}$$

$$\boxed{-r_A'' = k_{Al}\left(\frac{p_A}{H_A} - C_A\right)} \quad \text{... Ans. (ii)}$$

[With negligible gas phase resistance with little error, $p_{Ai} = p_A$ and $C_{Ai} = p_{Ai}/H_A$.]

Since the major resistance resides in the liquid film, the rate of transfer of A is liquid film resistance controlled. So anything which lowers this resistance would be helpful. The reaction in the liquid phase lowers this resistance and hence can speed up the absorption of A. ... **Ans. (iii)**

Example 2.2 : *CO_2 is to be removed from air. We plan to use an aqueous NaOH solution to speed up the removal of CO_2 from air at 25 °C (instead of using pure water).*

The reaction between CO_2 and NaOH is instantaneous.

$$CO_2 + 2\,NaOH = Na_2CO_3 + H_2O$$

(i) Suggest a form of rate equation that we would use when p_{CO_2} = 1000 Pa and the solution is 2N [2 normal NaOH solution].

(ii) By what factor can absorption be speeded compared to physical absorption using pure water ?

Data :

$k_g a = 0.80$ mol/(h·m³·Pa)

$k_l a = 25\ h^{-1}$

$H = 3000$ (Pa · m³)/mol

Solution : The reaction is

$$CO_2 + 2\,NaOH = Na_2CO_3 + H_2O$$

$$A + b\,B = \text{products}, \qquad b = 2$$

We have to use 2N NaOH solution to speed up the absorption operation, i.e., for the removal of CO_2 from air.

We have,

Concentration of solution in g/*l* = Normality of solution × Equivalent weight of solute

For NaOH : valence = 1

Since valence = 1,

equivalent weight of NaOH = molecular weight of NaOH = 40

Concentration of the NaOH solution = 2 × 40 = 80 g/*l*

The concentration of NaOH solution is the amount of NaOH per litre of solution.

$$\text{Concentration of NaOH solution (in molar units)} = \frac{\text{Concentration in g/}l}{\text{Molecular weight of NaOH}}$$

$$\therefore \quad C_B = \frac{80}{40} = 2 \text{ mol/}l = 2000 \text{ mol/m}^3$$

$$\text{Given : } p_{CO_2} = p_A = 1000 \text{ Pa}$$

It is a case of instantaneous reaction with respect to mass transfer.

Now, let us have a check for instantaneous reaction with low C_B or high C_B. For this we have to estimate $k_{Ag}\, a\, p_A$ and $k_{Bl}\, a\, C_B/b$

$$k_g\, a = k_{Ag}\, a = 0.80 \text{ mol/(h·m}^3\text{·Pa)}$$

$$k_l\, a = k_{Al}\, a = 25 \text{ h}^{-1}$$

Assuming $\mathscr{D}_{Al} = \mathscr{D}_{Bl}$, we have

$$k_{Bl}\, a = k_{Al}\, a = 25 \text{ h}^{-1}$$

$$b = 2, \qquad p_A = 1000 \text{ Pa}$$

$$k_{Ag}\, a\, p_A = 0.80 \times 1000 = 800 \text{ mol/(h·m}^3\text{)}$$

$$\frac{k_{Bl}\, a\, C_B}{b} = \frac{25 \times 2000}{2} = 25000 \text{ mol/(h·m}^3\text{)}$$

$\therefore$ It is clear that : $k_{Ag}\, a\, p_A < k_{Bl}\, a\, C_B/b$

As $k_{Ag}\, a\, p_A < k_{Bl}\, a\, C_B/b$, it is a case of instantaneous reaction with high C_B.

The form of rate equation for this case is :

$$-r_A'' = -\frac{1}{S}\frac{dN_A}{dt} = k_{Ag}\, p_A \qquad \text{... Ans.}$$

$$-r_A'''' = -r_A''\, a = \text{Rate with reaction} = k_{Ag}\, a\, p_A$$

For removal of CO_2 (A) using pure water for a case of straight mass transfer of A, the rate transfer of A across the gas and liquid films is given by,

$$\text{Rate for straight mass transfer} = (-r_A'''') = \frac{p_A - H_A C_A}{\frac{1}{k_{Ag}\, a} + \frac{H_A}{k_{Al}\, a}}$$

$$\text{Maximum rate for straight mass transfer} = \frac{p_A}{\frac{1}{k_{Ag}\, a} + \frac{H_A}{k_{Al}\, a}}$$

$$\frac{\text{Rate with reaction}}{\text{Rate for straight mass transfer}} = \frac{k_{Ag}\, a\, p_A}{\left[\dfrac{p_A}{\dfrac{1}{k_{Ag}\, a} + \dfrac{H_A}{k_{Al}\, a}}\right]}$$

$$= \frac{k_{Ag}\, a}{\dfrac{1}{\dfrac{1}{k_{Ag}\, a} + \dfrac{H_A}{k_{Al}\, a}}}$$

$$= k_{Ag}\, a \left[\frac{1}{k_{Ag}\, a} + \frac{H_A}{k_{Al}\, a}\right]$$

$$= 0.80 \left[\frac{1}{0.8} + \frac{3000}{25}\right]$$

$$= 97$$

∴ By using 2N NaOH solution, the rate increases by 97 times than it for pure water.

By using 2N NaOH solution, the absorption is speeded 97 times compared to physical absorption using pure water. ... Ans. (ii)

Example 2.3 : *Repeat Example 2.2 for $p_{CO_2} = 2 \times 10^4$ Pa and for 0.2 N NaOH solution.*

Solution :

$$CO_2 + 2\,NaOH \rightarrow Na_2CO_3 + H_2O$$

$$A + bB \rightarrow \text{products},\ b = 2$$

It is given that the reaction is instantaneous.

For the instantaneous reaction, we have two cases : (i) with low C_B and (ii) with high C_B.

Now, let us check whether it is an instantaneous reaction with low C_B or with high C_B. This is required in order to find out the appropriate form of rate equation for the case under consideration.

For this we have to evaluate $k_{Ag}\, a\, p_A$ and $k_{Bl}\, a\, C_B/b$ (as was done for previous example).

$$k_{Ag}\, a = 0.80 \text{ mol/(h·m}^3\text{·Pa)}$$

$$p_A = 2 \times 10^4 \text{ Pa}$$

$$k_{Bl}\, a = k_{Al}\, a = 25 \text{ h}^{-1}$$

$$C_B = ?$$

$$b = 2$$

$$H_A = 3000 \text{ (Pa·m}^3\text{)/mol}$$

Concentration of solution in g/*l* is given by

Concentration of solution in g/*l* = Normality of solution × Equivalent weight of solute

For NaOH, valence = 1

$\therefore$ Equivalent weight of B [NaOH] = Molecular weight of B

= 40

Normality of the NaOH solution = 0.2

Concentration of the B solution = 0.2×40

= 8

$$\text{Concentration of the B solution (in molar units)} = \frac{\text{Concentration of the B solution in g/}l}{\text{Molecular weight of B}}$$

$$C_B = \frac{8}{40}$$

$$= 0.2 \text{ mol/}l$$

$$= 200 \text{ mol/m}^3$$

Alternatively :

If valence of B = 1, then

Equivalent weight of B = Molecular weight of B

$\therefore$ Molarity of solution of B = Normality of solution of B

= 0.2

But the molarity of a solution is the grammoles solute per litre of the solution. Therefore

$$C_B = 0.2 \text{ mol/}l$$

$$= 200 \text{ mol/m}^3$$

$$k_{Ag}\, a\, p_A = 0.8 \times 2 \times 10^4 = 16000 \text{ mol/(h·m}^3\text{)}$$

$$\frac{k_{Bl}\, a\, C_B}{b} = \frac{25 \times 200}{2} = 2500 \text{ mol/(h·m}^3\text{)}$$

$$\text{So } k_{Ag}\, a\, p_A > \frac{k_{Bl}\, a\, C_B}{b}$$

As $k_{Ag}\, a\, p_A > k_{Bl}\, a\, C_B/b$, we have a case of instantaneous reaction with low C_B and reaction zone is in the liquid film.

The form of rate equation to be used is :

For general case :

$$-r_A'' = -\frac{1}{S}\frac{dN_A}{dt} = \frac{\frac{\mathscr{D}_{Bl}\, C_B}{\mathscr{D}_{Al}\, b} + \frac{p_A}{H_A}}{\frac{1}{H_A\, k_{Ag}} + \frac{1}{k_{Al}}}$$

For negligible gas phase resistance :

$$-r_A'' = k_{Al}\, C_{Ai}\, E_i$$

where $$E_i = 1 + \frac{\mathscr{D}_{Bl}\, C_B}{b\, \mathscr{D}_{Al}\, C_{Ai}}$$

and $$p_{Ai} = p_A$$

Let us have a check for negligible gas phase resistance :

$$\text{Resistance offered by the gas film} = \frac{1}{k_{Ag}\, a}$$

$$= \frac{1}{0.80}$$

$$= 1.25$$

$$\text{Resistance offered by the liquid film} = \frac{H_A}{k_{Al}\, a}$$

$$= \frac{3000}{25}$$

$$= 120$$

$$\text{Gas film resistance} = \frac{1.25}{1.25 + 120} \times 100 = 1.03\ \%$$

The gas film resistance is only 1.03 % of the total resistance. We can with a little error say that it is a special case of negligible gas phase resistance.

Hence for our present case we can use the following rate equation

$$-r_A'' = k_{Al}\, C_{Ai}\, E_i$$

where $$E_i = 1 + \frac{\mathscr{D}_{Bl}\, C_B}{b\, \mathscr{D}_{Al}\, C_{Ai}}$$

$$-r_A'' = k_{Al}\, C_{Ai}\left[1 + \frac{\mathscr{D}_{Bl}\, C_B}{b\, \mathscr{D}_{Al}\, C_{Ai}}\right]$$

where

E_i = Liquid film enhancement factor for the instantaneous reaction.

$$\frac{\text{Rate with reaction}}{\text{Rate for straight mass transfer}} = 1 + \frac{\mathscr{D}_{Al}\, C_B}{b\, \mathscr{D}_{Bl}\, C_{Ai}}$$

$$= 1 + \frac{C_B}{b\, C_{Ai}}, \quad \text{assuming } \mathscr{D}_{Al} = \mathscr{D}_{Bl}$$

$$= 1 + \frac{C_B\, H_A}{b\, p_{Ai}}, \quad \text{as } p_{Ai} = H_A\, C_{Ai}$$

$$= 1 + \frac{C_B\, H_A}{b\, p_A}$$

... for negligible gas phase resistance, $p_{Ai} = p_A$

$C_B = 200$ mol/m^3, $b = 2$, $H_A = 3000$ (P·m^3)/mol, $p_A = 2 \times 10^4$ Pa

$$\frac{\text{Rate with reaction}}{\text{Rate for straight mass transfer}} = 1 + \frac{200 \times 3000}{2 \times 2 \times 10^4}$$

$$= 16$$

$\therefore$ Rate with reaction is 16 times that for the straight mass transfer. ... Ans.

With 0.2 N NaOH solution, the rate of removal of CO_2 from air increases by 16 times compared to physical absorption (with pure water). ... Ans.

Example 2.4 : *For a highly water-soluble gas such as ammonia at 10 °C*

$$H = 1.0\ (Pa \cdot m^3)/s$$

For sparingly soluble gases such as O_2, N_2, NO, CH_4, H_2, etc., at 10 °C

$$H = 10^5\ (Pa \cdot m^3)/s$$

For straight absorption of these gases in water, assuming no chemical reaction, determine:

(i) the relative resistances of the gas and liquid films.

(ii) the resistance that controls the absorption process.

(iii) the form of rate expression to be used for design in these two cases.

(iv) the effect of solubility of the slightly soluble gas affect its rate of absorption ?

(v) in which case would chemical reaction be more helpful in speeding the absorption process ? Give justification.

Data : $k_l = 10^{-5}$ m/s, $k_g RT = 0.10$ m/s

Solution :

(a) Highly soluble gas : $H = 1$ (Pa·m^3)/mol

(b) Sparingly soluble gas : $H = 10^5$ (Pa·m^3)/mol

$$k_l = 10^{-5} \text{ m/s}$$

$$k_g RT = 0.10 \text{ m/s}$$

$$R = 8.314 \text{ (kPa·m}^3\text{)/kmol·K} = 8.314 \text{ (Pa·m}^3\text{)/mol·K}$$

$$T = 10\ ^\circ\text{C} = 283 \text{ K}$$

$$k_g = \frac{0.10}{RT} = \frac{0.10}{8.314 \times 283} = 4.25 \times 10^{-5} \frac{\text{mol}}{(\text{Pa·m}^2\text{·s})}$$

It is a straight mass transfer (absorption) of A across the gas and liquid films. So there are two resistances (of gas film and liquid film) for mass transfer of A.

The rate expression for straight mass transfer of A across the gas and liquid films is :

$$-r_A'' = -\frac{1}{S}\frac{dN_A}{dt} = \frac{1}{\frac{1}{k_{Ag}} + \frac{H_A}{k_{Al}}} \Delta p_A$$

$$\text{Gas film resistance} = \frac{1}{k_{Ag}} = \frac{1}{k_g}$$

$$\text{Liquid film resistance} = \frac{H_A}{k_{Al}} = \frac{H}{k_l}$$

(a) Highly soluble gas :

$$\text{Gas film resistance} = \frac{1}{k_g}$$

$$= \frac{1}{4.25 \times 10^{-5}} = 23529.4 \text{ (Pa·m}^2\text{·s)/mol}$$

$$\text{Liquid film resistance} = \frac{H}{k_l}$$

$$= \frac{1}{10^{-5}} = 10^5 \text{ (Pa·m}^2\text{·s)/mol}$$

Relative resistance :

Liquid film resistance relative to the gas film resistance for highly soluble gas

$$= \frac{H/k_l}{1/k_g}$$

$$= \frac{10^5}{23529.4}$$

$$\boxed{= 4.25}$$ **... Ans. (i)**

(b) Sparingly (slightly) soluble gas :

$$H = 10^5 \text{ (Pa·m}^3\text{)/mol}$$

$$\text{Gas film resistance} = \frac{1}{k_g} = \frac{1}{4.25 \times 10^{-5}}$$

$$= 23529.4 \text{ (Pa·m}^2\text{·s)/mol}$$

$$\text{Liquid film resistance} = \frac{H}{k_l}$$

$$= \frac{10^5}{10^{-5}} = 10^{10} \text{ (Pa·m}^2\text{·s)/mol}$$

Relative resistance :

Liquid film resistance relative to the gas film resistance for sparingly soluble gas

$$= \frac{H/k_l}{1/k_g}$$

$$= \frac{10^{10}}{23529.4} = \boxed{4.25 \times 10^5}$$ **... Ans. (i)**

(a) Highly soluble gas :

Gas film resistance = 23529.4 (Pa·m²·s)/mol ... (19%)

Liquid film resistance = 10^5 (Pa·m²·s)/mol ... (81%)

Referring to the above values (cited in %), it is clear that we have to take into account both the resistances to find the rate of absorption. None of the resistances can be ignored. Thus we have to consider both the phases. None of the resistances separately control the absorption process. ... Ans. (ii)

(b) Sparingly soluble gas :

Gas phase resistance = 23529.4 (Pa·m² ·s)/mol ... (2.35×10^{-4} %)

Liquid phase resistance = 10^{10} (Pa·m² ·s)/mol ... (99.9997 %)

Referring to the above values (cited in %), it is clear that the gas film resistance is almost zero i.e. it does not offer resistance to mass transfer. The liquid film resistance is almost 100%. Hence, the entire resistance to mass transfer is offered by the liquid film. So for this case, **liquid phase resistance controls the absorption process. It is a liquid film controlling process.**

... Ans. (ii)

(a) Highly soluble gas :

As here we have to take into account both the resistances, the form of rate expression for straight mass transfer for this case is :

$$\boxed{-r''_A = \frac{1}{\frac{1}{k_{Ag}} + \frac{H_A}{k_{Al}}} \Delta p_A}$$ ... Ans. (iii)

$$k_g = k_{Ag} \quad \text{and} \quad k_l = k_{Al}$$

or
$$-r''_A = \frac{1}{\frac{1}{Hk_{Ag}} + \frac{1}{k_{Al}}} \Delta C_{A\,\text{overall}}$$

Δp_A or ΔC_A ... across the gas and liquid films.

(b) Sparingly soluble gas :

Here the entire resistance is the resistance offered by the liquid film. No gas phase resistance. So the form of rate expression to be used is :

$$-r''_A = k_{Al}(C_{Ai} - C_A) = k_{Al}\left[\frac{p_{Ai}}{H_A} - C_A\right]$$ ... Ans. (iv)

No gas phase resistance - $p_{Ai} = p_A$

$$\boxed{-r''_A = k_{Al}\left(\frac{p_A}{H_A} - C_A\right)}$$ ... Ans. (iv)

or
$$-r''_A = -r'' = k_l\left[\frac{p_g}{H} - C_l\right]$$

For sparingly soluble gas (slightly soluble gas) absorbed by water we have

$$-r_A'' = k_{Al}\left[\frac{p_A}{H_A} - C_A\right]$$

$$-r_A'' = \frac{k_{Al}\, p_A}{H_A}, \quad C_A \text{ very low.}$$

$$-r_A'' \alpha \frac{1}{H_A}$$

Thus the absorption is inversely proportional to H_A, i.e., absorption is proportional to the solubility of gas. **... Ans. (iv)**

In case of sparingly soluble gas/slightly soluble gas (i.e., gas difficult to absorb) since almost all resistance to mass transfer is in the liquid phase and reaction in liquid phase lowers this resistance we can definitely say that in this case the reaction in the liquid phase would be more helpful (i.e., this system would benefit greatly by a reaction in the liquid phase). **... Ans. (v)**

Example 2.5 : *Hydrogen sulfide present in a gas at 2 MPa is to be absorbed at 20 °C by a solution containing 250 mol/m³ methanolamine (MEA). The gas contains 0.10% H_2S by volume. H_2S reacts with MEA as follows :*

$$\underset{A}{H_2S} + \underset{B}{RNH_2} \rightarrow HS^- + RNH_3^+$$

This acid-base neutralisation reaction can be regarded as irreversible and instantaneous.

(i) Determine the form of rate equation which is applicable to our case.

(ii) Find the rate of absorption of H_2S in MEA solution.

(iii) Determine how much faster is absorption with MEA solution compared to physical absorption in pure water (required to find whether it is worthwhile to use MEA absorbent or not).

Data :

$$k_{Al}\, a = 0.03\ s^{-1}$$

$$k_{Ag}\, a = 6 \times 10^{-4}\ mol/(s \cdot m^3 \cdot Pa)$$

$$H_A = 10\ (Pa \cdot m^3)/mol$$

H_A is Henry's law constant for H_2S in water. The diffusivity of MEA in solution is 0.64 times that of H_2S.

Solution :

$$H_2S + RNH_2 \rightarrow HS^- + RNH_3^+$$

$$A + bB \rightarrow \text{products}, \quad b = 1$$

Given : It is an instantaneous reaction.

For the instantaneous reaction, we have two cases :

Cases :

(i) Instantaneous reaction with low C_B

(ii) Instantaneous reaction with high C_B

In order to suggest a appropriate form of rate equation for given situation we have to find whether this instantaneous reaction is with low C_B or with high C_B.

For deciding this, we have to evaluate : $k_{Ag}\, a\, p_A$ and $k_{Bl}\, a\, C_B/b$

For high C_B : $k_{Ag}\, a\, p_A \leq k_{Bl}\, a\, C_B/b$

For low C_B : $k_{Ag}\, a\, p_A \geq k_{Bl}\, a\, C_B/b$

$k_{Al}\, a = 0.03\ s^{-1}$, $k_{Ag}\, a = 6 \times 10^{-4}\ mol/(Pa \cdot m^3 \cdot s)$,

$H_A = 10\ (Pa \cdot m^3)/mol$, $b = 1$, $C_B = 250\ mol/m^3$,

$P = 2\ MPa = 2 \times 10^6\ Pa$ (gas pressure)

H_2S in the gas = 0.1 % by volume (i.e., by mole)

$$p_A = \text{mole fraction } H_2S \times P$$

$$= \left(\frac{0.1}{100}\right) \times 2 \times 10^6 = 2000\ Pa$$

Diffusivity of MEA (B) in the solution is 0.64 times that of H_2S (A).

$$\therefore \quad \mathscr{D}_{Bl} = 0.64\, \mathscr{D}_{Al}$$

$$\frac{\mathscr{D}_{Bl}}{\mathscr{D}_{Al}} = 0.64$$

As per two-film theory :

$$\frac{k_{Al}\, a}{k_{Bl}\, a} = \frac{\mathscr{D}_{Al}}{\mathscr{D}_{Bl}}$$

$$k_{Bl}\, a = k_{Al}\, a \left(\frac{\mathscr{D}_{Bl}}{\mathscr{D}_{Al}}\right)$$

$$= 0.03\ (0.64) = 0.0192\ s^{-1}$$

$$k_{Ag}\, a\, p_A = 6 \times 10^{-4} \times 2000 = 1.2\ mol/(m^3 \cdot s)$$

$$\frac{k_{Bl}\, a\, C_B}{b} = \frac{0.0192 \times 250}{1} = 4.8\ mol/(m^3 \cdot s)$$

Referring to the above two calculated values, we have

$$k_{Ag}\, a\, p_A < \frac{k_{Bl}\, a\, C_B}{b}$$

As $k_{Ag}\, a\, p_A < k_{Bl}\, a\, C_B/b$, it is an instantaneous reaction with high C_B.

For the instantaneous reaction with high C_B, the form of rate equation is

$$-r''_A = -\frac{1}{S}\frac{dN_A}{dt} = k_{Ag}\, p_A \quad \text{... Ans. (i)}$$

$$-r'''_A = (-r''_A)\, a = k_{Ag}\, a\, p_A$$

$$= 6 \times 10^{-4} \times 2000$$

$$= \mathbf{1.2\ mol/(m^3 \cdot s)} \quad \text{... Ans. (ii)}$$

$$\text{Rate for straight mass transfer} = -r''''_A = \frac{p_A - H_A C_A}{\frac{1}{k_{Ag}\, a} + \frac{H_A}{k_{Al}\, a}}$$

with $C_A = 0$

$$\text{Rate for straight mass transfer } -r''''_A = \frac{p_A}{\frac{1}{k_{Ag}\, a} + \frac{H_A}{k_{Al}\, a}}$$

$$= \frac{2000}{\frac{1}{6 \times 10^{-4}} + \frac{10}{0.03}}$$

$$= 1\ \text{mol/(m}^3 \cdot \text{s)}$$

$$\frac{\text{Rate with reaction}}{\text{Rate for straight mass transfer}} = \frac{1.2}{1} = 1.2$$

So with MEA solution, the absorption of H_2S is 20 % faster than it with pure water.

... **Ans. (iii)**

Referring to the result obtained it seems that using MEA solution instead of pure water (for absorption of H_2S) enhances absorption by 20% only.

Referring to the result obtained we can say that using MEA solution for absorption of H_2S instead of pure water is not that much worthwhile/beneficial.

Example 2.6 : *An aqueous NaOH solution is to be used for the removal of CO_2 from air at 25 °C. Assume that the reaction is instantaneous and represented by*

$$CO_2 + 2OH^- = H_2O + CO_3^{--}$$

Determine the form of rate equation to be used under the following conditions. Give necessary justification.

(i) CO_2 partial pressure 1000 Pa and NaOH solution is 1.6 N.

(ii) CO_2 partial pressure is 15000 Pa and NaOH solution is 0.4 N.

Data :

$$k_g\, a = 0.80\ \text{mol/(Pa} \cdot \text{h} \cdot \text{m}^3\text{)}$$

$$k_l\, a = 25\ \text{h}^{-1}$$

$$H = 3000\ \text{(Pa} \cdot \text{m}^3\text{)/mol}$$

Solution : $CO_2 + 2OH^- = H_2O + CO_3^{--}$

$A + bB = \text{products}, \quad b = 2$

Given : Reaction is instantaneous.

We have two cases - each having its own rate equation.

(i) Instantaneous reaction with low C_B.

(ii) Instantaneous reaction with high C_B.

To find out whether it is case (i) or (ii), we have to evaluate $k_{Ag}\, a\, p_A$ and $k_{Bl}\, a\, C_B/b$

$k_{Ag}\, a\, p_A \leq k_{Bl}\, a\, C_B/b$ ... instantaneous reaction with high

$k_{Ag}\, a\, p_A \geq k_{Bl}\, a\, C_B/b$ ... instantaneous reaction with low

$k_{Ag}\, a = k_g\, a = 0.80 \text{ mol/(Pa·h·m}^3\text{)}$

$k_{Al}\, a = k_l\, a = 25 \text{ h}^{-1}$

$H_A = 3000 \text{ (Pa·m}^3\text{)/mol}$

$b = 2$

Assuming $\mathscr{D}_{Al} = \mathscr{D}_{Bl}$, $\dfrac{k_{Al}}{k_{Bl}} = \dfrac{\mathscr{D}_{Al}}{\mathscr{D}_{Bl}}$

$\therefore \quad k_{Bl}\, a = k_{Al}\, a = 25 \text{ h}^{-1}$

(i) $p_A = 1000$ Pa and NaOH solution is 1.6 N

For NaOH : valence = 1

Since valence of NaOH = 1, Equivalent weight = Molecular weight.

Thus, Molarity = Normality = 1.6

But molarity of a solution is the gram moles solute per litre of the solution.

$\therefore \quad C_B = 1.6 \text{ mol/}l$

$= 1.6 \times 10^3 \text{ mol/m}^3$

$k_{Ag}\, a\, p_A = 0.8 \times 1000 = 800 \text{ mol/(h·m}^3\text{)}$

$\dfrac{k_{Bl}\, a\, C_B}{b} = \dfrac{25 \times 1.6 \times 10^3}{2} = 20000 \text{ mol/(h·m}^3\text{)}$

As $k_{Ag}\, a\, p_A < k_{Bl}\, a\, C_B/b$, **it is an instantaneous reaction with high C_B.**

The form of rate equation for this case (instantaneous reaction - high C_B) is :

$$\boxed{-r_A'' = -\frac{1}{S}\frac{dN_A}{dt} = k_{Ag}\, p_A}$$

... An

(ii) $p_A = 15000$ Pa and 0.4 N NaOH solution

Concentration in g/l = Normality × Equivalent weight of NaOH

$= 0.4 \times 40 = 16$

Concentration of B (molar units)

$$= \frac{\text{Concentration in g/}l\text{ of B}}{\text{Molecular weight of B}}$$

$$= \frac{16}{40} = 0.4 \text{ mol/}l$$

$\therefore$ $C_B = 0.4 \text{ mol/}l$

$= 400 \text{ mol/m}^3$

$k_{Ag}\, a\, p_A = 0.8 \times 15000 = 12000 \text{ mol/(h·m}^3\text{)}$

$$\frac{k_{Bl}\, a\, C_B}{b} = \frac{25 \times 400}{2} = 5000 \text{ mol/(h·m}^3\text{)}$$

$\therefore$ $k_{Ag}\, a\, p_A > k_{Bl}\, a\, C_B/b$

$\therefore$ It is an instantaneous reaction with low C_B. The form of rate equation for the instantaneous reaction with low C_B is :

$$\boxed{-r_A'' = -\frac{1}{S}\frac{dN_A}{dt} = \frac{\dfrac{\mathscr{D}_{Bl}\, C_B}{\mathscr{D}_{Al}\, b} + \dfrac{p_A}{H_A}}{\dfrac{1}{H_A\, k_{Ag}} + \dfrac{1}{k_{Al}}}} \quad \text{... Ans. (ii)}$$

$$\text{Gas film resistance} = \frac{1}{k_{Ag}\, a} = \frac{1}{0.8} = 1.25$$

$$\text{Liquid film resistance} = \frac{H_A}{k_{Al}\, a} = \frac{3000}{25} = 120$$

$$\text{Total resistance} = 120 + 1.25 = 121.5$$

$\therefore$ Gas film resistance = 1.03 %

$\therefore$ Liquid film resistance = 98.97 %

So with a little error, we can ignore the gas film resistance and we can safely say that it is a special case of negligible gas phase resistance and for this we can safely suggest the following form of rate equation :

$$-r_A'' = k_{Al}\, C_{Ai} \left[1 + \frac{\mathscr{D}_{Bl}\, C_B}{b\, \mathscr{D}_{Al}\, C_{Ai}}\right] \quad \text{... Ans. (ii)}$$

Example 2.7 : *Gaseous A absorbs and reacts with B in liquid according to*

$$A\,(g \rightarrow l) + B(l) \rightarrow R(l), \quad -r_{Al} = kC_A\, C_B$$

in a packed bed.

(i) Calculate the rate of reaction.

(ii) Determine the location of the major resistance (gas film, liquid film and bulk liquid) and behaviour in the liquid film (pseudo first order reaction, instantaneous etc.) at a point in the reactor where p_A = 100 Pa and C_B = 100 mol/m³ liquid.

Data :

$$k = 10^8 \text{ m}^3 \text{ liquid/(mol·h)}$$
$$H_A = 1.0 \text{ (Pa·m}^3 \text{ liquid)/mol}$$
$$k_{Ag}\,a = 0.10 \text{ mol/(h·m}^3 \text{ of reactor· Pa)}$$
$$k_{Al}\,a = 100 \text{ m}^3 \text{ liquid/(m}^3 \text{ reactor·h)}$$
$$f_l = 0.01 \text{ m}^3 \text{ liquid/m}^3 \text{ reactor}$$
$$a = 100 \text{ m}^2/\text{m}^3 \text{ reactor}$$
$$\mathscr{D}_{Al} = \mathscr{D}_{Bl} = 10^{-6} \text{ m}^2/\text{h}$$

Hint : For $E_i < M_H/5$, we have instantaneous reaction and $E \approx E_i$

Solution : $A + bB \rightarrow$ product, $b = 1$

$-r_{Al} = k\,C_A\,C_B$ with second order rate constant : $k = 10^8 \text{ m}^3/(\text{mol·h})$

$$k_{Al}\,a = 100 \text{ m}^3/(\text{m}^3\text{·h})$$
$$a = \text{m}^2/\text{m}^3$$

$\therefore$
$$k_{Al} = \frac{k_{Al}\,a}{a} = \frac{100}{100} = 1.0 \text{ m/h}$$
$$\mathscr{D}_{Al} = \mathscr{D}_{Bl} = 10^{-6} \text{ m}^2/\text{h}$$
$$p_A = p_{Ai} = 100 \text{ Pa}$$
$$C_B = 100 \text{ mol/m}^3$$

Evaluate E_i and M_H

$$M_H = \frac{\sqrt{\mathscr{D}_{Al}\,k\,C_B}}{k_{Al}}$$
$$= \frac{\sqrt{10^{-6} \times 10^8 \times 100}}{1} = 100$$
$$E_i = 1 + \frac{\mathscr{D}_{Bl}\,C_B\,H_A}{b\,\mathscr{D}_{Al}\,p_{Ai}}$$
$$= 1 + \frac{10^{-6} \times 100 \times 1}{1 \times 10^{-6} \times 100} = 2$$

From the above calculated values, we have :

$$E_i < M_H/5 \qquad \therefore \quad E = E_i \text{ (Given)}$$

Behaviour in liquid film :

Therefore, we have instantaneous second order reaction in the liquid film.

$$k_{Ag}\,a\,p_A = 0.1 \times 100 = 10 \text{ mol/(h·m}^3\text{)}$$
$$k_{Al}\,a = k_{Bl}\,a = 100 \text{ m}^3/(\text{m}^3\text{·h}) \qquad \text{as } \mathscr{D}_{Al} = \mathscr{D}_{Bl}$$
$$\frac{k_{Bl}\,a\,C_B}{b} = \frac{100 \times 100}{1} = 10^4$$
$$\therefore \quad k_{Ag}\,a\,p_A < \frac{k_{Bl}\,a\,C_B}{b}$$

$\therefore$ It is the case of instantaneous reaction with high C_B for which the suggested rate form is

$$-r_A'' = -\frac{1}{S}\frac{dN_A}{dt} = k_{Ag}\, p_A$$

$$-r_A'''' = a\,(-r_A'')$$

$$-r_A'''' = k_{Ag}\, a\, p_A$$

$$= 0.1 \times 100 = 10 \text{ mol/(h·m}^3 \text{ reactor)}$$

The general rate expression is

$$-r_A'''' = \frac{p_A}{\frac{1}{k_{Ag}\, a} + \frac{H_A}{k_{Al}\, a\, E} + \frac{H_A}{k\, C_B\, f_l}}$$

$$= \frac{100}{\frac{1}{0.10} + \frac{1}{100 \times 2} + \frac{1}{10^8 \times 100 \times 0.01}}$$

$$= \frac{100}{10 + 5 \times 10^{-3} + 1 \times 10^{-8}}$$

$$= \frac{100}{10.005} = 9.995 \approx 10 \text{ mol/(h·m}^3 \text{ reactor)}$$

Rate of reaction = 10 mol/(h·m³ reactor) ... **Ans. (i)**

Total resistance $= 10 + 5 \times 10^{-3} = 10.005$, (considering $1 \times 10^{-8} \approx 0$)

$$\text{Gas film resistance} = \frac{10}{10.005} \times 100 = 99.95\ \%$$

$$\text{Liquid film resistance} = \frac{5 \times 10^{-3}}{10.005} \times 100 = 0.0499 \approx 0.05\ \%$$

Gas film resistance = 99.95% ... **Ans. (ii)**

Major resistance is in the gas film. ... **Ans. (iii)**

Example 2.8 : *Air containing gaseous A bubbles through a tank containing aqueous B. A reacts with B in the liquid phase according to the following reaction :*

$$A\,(g \rightarrow l) + 2B\,(l) \rightarrow R\,(l), \qquad -r_A = kC_A C_B^2$$

For a point in the absorber - reactor where :

$p_A = 5000$ *Pa and* $C_B = 100$ *mol/m³ liquid*

(i) *Locate the resistance to reaction (in percent in the gas film, liquid film and main body of liquid).*

(ii) *Determine the behaviour in the liquid film (whether it is pseudo first-order reaction, instantaneous, etc.)*

(iii) *Locate the reaction zone.*

(iv) *Calculate the rate of reaction in mol/(h·m³ of reactor).*

Data :

$$k = 10^6\ m^6/(mol^2 \cdot h)$$
$$k_{Ag}\,a = 0.01\ mol/(h \cdot m^2\ of\ reactor \cdot Pa)$$
$$a = 20\ m^2/m^3\ reactor$$
$$k_{Al}\,a = 20\ m^3\ liquid/(m^3\ reactor \cdot h)$$
$$H_A = 10^5\ (Pa \cdot m^3\ liquid)/mol$$
$$f_l = 0.098\ m^3\ liquid/m^3\ reactor$$
$$\mathscr{D}_{Al} = \mathscr{D}_{Bl} = 10^{-6}\ m^2/h$$

Hint : $E_i > 5\ M_H \ldots\ E = M_H$

Solution :

$$A + 2B \rightarrow R$$
$$A + bB \rightarrow R,\quad b = 2$$

The rate is given by $-r_A = k\,C_A\,C_B^2$

This is a third order reaction and our whole treatment in this chapter is related to second order reactions. Therefore, let us replace this third order reaction with a second order approximation.

Hence, $$-r_A = k\,C_A\,C_B^2 = (kC_B)\,C_A\,C_B = k'\,C_A\,C_B$$

To find out whether it is a instantaneous, fast or slow reaction, we have to evaluate E_i and M_H.

$$M_H = \frac{(\mathscr{D}_{Al}\,k'\,C_B)^{0.5}}{k_{Al}} = \frac{(\mathscr{D}_{Al}\,k\,C_B^2)^{0.5}}{k_{Al}}$$

$$E_i = 1 + \frac{\mathscr{D}_{Bl}\,C_B\,H_A}{b\,\mathscr{D}_{Al}\,p_{Ai}}$$

For pure gaseous reactant A : $p_{Ai} = p_A$

For gaseous reactant A with air : $p_{Ai} < p_A$

Therefore, the maximum possible value of p_{Ai} is p_A.

Take $p_{Ai} = p_A = 5 \times 10^3$ Pa and evaluate E_i

We have,

$$k = 10^6\ m^6/(mol^2 \cdot h)$$
$$k_{Ag}\,a = 0.01\ mol/(h \cdot m^3 \cdot Pa)$$
$$f_l = 0.098$$
$$H_A = 10^5\ (Pa \cdot m^3)/mol$$
$$\mathscr{D}_{Al} = \mathscr{D}_{Bl} = 10^{-6}\ m^2/h$$
$$a = 20\ m^2/m^3$$
$$b = 2$$
$$k_{Al} = \frac{k_{Al}\,a}{a} = \frac{20}{20} = 1\ m/h$$
$$M_H = \frac{(10^{-6} \times 10^6 \times 100^2)^{0.5}}{1} = 100$$
$$E_i = 1 + \frac{10^{-6} \times 100 \times 10^5}{2 \times 10^{-6} \times 5000} = 10^3$$

From the above calculated values, we have :

$$E_i > 5\,M_H$$

and for any value of $p_{Ai} < 5000$ Pa (i.e. p_A) we get $E_i > 5\,M_H$

As $E_i > 5\,M_H$, we have pseudo first order reaction in the film and for this case we have

$$E = M_H = 100$$

Behaviour in the liquid film : Reaction proceeds by pseudo first order reaction of A at the interface. **... Ans. (ii)**

Reaction zone is in the liquid film. **... Ans. (iii)**

The general rate expression is

$$-r_A'''' = \frac{p_A}{\frac{1}{k_{Ag}\,a} + \frac{H_A}{k_{Al}\,a\,E} + \frac{H_A}{k\,f_l\,C_B^2}}$$

[Note C_B^2 in the third term of denominator as $-r_A = (k\,C_B)\,C_A C_B = k' C_A C_B$

$$\therefore\ H_A/k' f_l\,C_B \Rightarrow H_A/k\,f_l\,C_B^2\ ]$$

$$= \frac{5000}{\frac{1}{0.01} + \frac{10^5}{20 \times 100} + \frac{10^5}{(10^6)\,(100)^2\,(0.098)}}$$

$$= \frac{5000}{100 + 50 + 1.02 \times 10^{-4}}$$

$$= \frac{5000}{100 + 50},\ \text{as } 1.02 \times 10^{-4} \approx 0$$

$$= 33.33\ \text{mol/(h·m}^3\ \text{reactor)}$$

Rate of reaction = 33.33 mol/(h·m³ reactor) **... Ans. (iv)**

Total resistance = 100 + 50 = 150 (h·m³·Pa)/mol

Resistance in the gas film $= \frac{100}{150}$

$= 2/3^{rd}$ of total

Resistance in the gas film = 66.67 % **... Ans. (i)**

Resistance in the liquid film $= \frac{50}{150}$

$= 1/3^{rd}$ of total

$= \frac{50}{150} \times 100 = 33.33$

Resistance in the liquid film = 33.33 % **... Ans. (i)**

For fast reaction with high C_B, hence pseudo first order rate with respect to A, the rate form is :

$$-r_A'''' = \frac{p_A}{\frac{1}{k_{Ag}\,a} + \frac{H_A}{a\sqrt{\mathscr{D}_{Al}\,k\,C_B^2}}}$$

$$= \frac{5000}{\frac{1}{0.01} + \frac{10^5}{20\,(10^{-6} \times 10^6 \times 100^2)^{0.5}}}$$

$$= 33.33 \text{ mol/(h·m}^3 \text{ reactor)}$$

Example 2.9 : *Gaseous A absorbs and reacts with liquid B in liquid according to*

$$A\,(g \rightarrow l) + B\,(l) \rightarrow R\,(l), \quad -r_A = k\,C_A\,C_B$$

in a packed bed.

At a point in the reactor where p_A = 100 Pa and C_B = 1 mol/m³ liquid, calculate :

(i) the rate of reaction in mol/(h·m³ of reactor)

(ii) the resistance offered by the main body of liquid.

Data :

$k_{Ag}\,a = 0.10$ mol/(h·m³ reactor · Pa)

$f_l = 0.01$ m³ liquid/m³ reactor

$k_{Al}\,a = 100$ m³ liquid/(m³ reactor·h)

$a = 100$ m²/m³ reactor

$\mathscr{D}_{Al} = \mathscr{D}_{Bl} = 10^{-6}$ m²/h

$k = 10$ m³ liquid/(mol·h)

$H_A = 10^5$ (Pa·m³ liquid)/mol

Hint : For $M_H < 0.02$, we have infinitely slow reaction.

Solution :

$$A + bB \rightarrow R, \quad -r_A = k\,C_A\,C_B, \quad b = 1$$

First we have to determine whether it is very fast, fast, slow or very slow reaction. For this we have to evaluate M_H and E_i.

$$M_H = \frac{(\mathscr{D}_{Al}\,k\,C_B)^{1/2}}{k_{Al}}$$

$D_{Al} = 10^{-6}$ m²/h, $k = 10$ m³/(mol·h), $C_B = 1$ mol/m³

$$k_{Al} = \frac{k_{Al}\,a}{a} = \frac{100}{100} = 1.0 \text{ m/h}$$

$$M_H = \frac{(10^{-6} \times 10 \times 1)^{0.5}}{1} = 0.00316$$

$$E_i = 1 + \frac{\mathscr{D}_{Bl}\,C_B\,H_A}{b\,\mathscr{D}_{Al}\,p_{Ai}}$$

$p_{Ai} = p_A = 100$ Pa, $b = 1$, $\mathscr{D}_{Bl} = \mathscr{D}_{Al} = 10^{-6}$ m²/h, $H_A = 10^5$ (Pa·m³)/mol

$$E_i = 1 + \frac{10^{-6} \times 1 \times 10^5}{1 \times 10^{-6} \times 100} \approx 10^3$$

As $M_H < 0.02$, (M_H calculated = 0.00316), **we have infinitely slow reaction (without mass transfer resistance) occurring in the bulk of liquid.** **... Ans.**

For infinitely slow reaction, the rate is determined by kinetics alone. The form of rate expression for this case is

$$\boxed{-r_A''' = \frac{k f_l}{H_A} p_A C_B = k f_l C_A C_B} \quad \text{... Ans.}$$

We have, $f_l = 0.001$ m^3 liquid/m^3 reactor

$k = 10$ m^3 liquid/(mol·h)

$$-r_A''' = k f_l C_A C_B$$

$$= \frac{k f_l}{H_A} p_A C_B$$

$$= \frac{10 \times 0.01}{10^5} \times 100 \times 1 = 1 \times 10^{-4} \text{ mol/(m}^3 \text{ reactor·h)}$$

Rate of reaction = 1×10^{-4} mol/(m^3 reactor·h) **... Ans. (i)**

The general rate expression is

$$-r_A''' = \frac{1}{\frac{1}{k_{Ag} a} + \frac{H_A}{k_{Al} a E} + \frac{H_A}{k C_B f_l}} p_A \quad \text{... (1)}$$

In our case for infinitely slow reaction,

$$-r_A''' = \frac{k f_l}{H_A} p_A C_B = k f_l C_A C_B$$

$$= \frac{1}{\left(\frac{H_A}{k f_l C_B}\right)} p_A \quad \text{... (2)}$$

Comparing Equation (2) and with Equation (1), there is no mass transfer resistance.

$$\text{Liquid bulk resistance} = \frac{H_A}{k f_l C_B}$$

$$= \frac{10^5}{10 \times 0.01 \times 1}$$

Liquid bulk resistance = 1×10^6 (Pa·m^3·h)/mol **... Ans. (ii)**

Example 2.10 : *Gaseous A absorbs and reacts with B in liquid according to*

$$A\,(g \rightarrow l) + B\,(l) \rightarrow R\,(l), \quad -r_A = k\,C_A\,C_B$$

in a packed bed.

(i) Calculate the rate of reaction in mol/(h·m² of reactor).

(ii) Locate the resistance to reaction (what % is in gas film, in the liquid film, in the body of liquid).

(iii) Locate the reaction zone.

(iv) Determine the behaviour in the liquid film (whether pseudo first order, instantaneo etc.)

at a point in the reactor where p_A = 100 Pa and C_B = 100 mol/m³ liquid. Use the follow data :

$$k_{Ag}\,a = 0.10\ mol/h{\cdot}m^3\ of\ reactor{\cdot}Pa)$$

$$k_{Al}\,a = 100\ m^3\ liquid/(m^3\ reactor{\cdot}h)$$

$$a = 100\ m^2/m^3\ reactor$$

$$f_l = 0.01\ m^3\ liquid/m^3\ reactor$$

$$\mathscr{D}_{Al} = \mathscr{D}_{Bl} = 10^{-6}\ m^2/h$$

$$k = 10^6\ m^3\ liquid/(mol{\cdot}h)$$

$$H_A = 10^4\ (Pa{\cdot}m^3\ liquid)/mol$$

Hint : If $E_i > 5\ M_H$, we have pseudo first order reaction by the G/L interface in which $E \approx M_H$.

Solution :

$$A + bB \rightarrow R, \quad b = 1$$

$$-r_A = k\,C_A\,C_B \text{ with } k = 10^6\ m^3/(mol{\cdot}h)$$

Given :

$$k_{Ag}\,a = 0.10\ mol/(h{\cdot}m^3{\cdot}Pa)$$

$$k_{Al}\,a = 100\ m^3/(m^3{\cdot}h)$$

$$a = 100\ m^2/m^3$$

$$k_{Al} = \frac{k_{Al}\,a}{a} = \frac{100}{100} = 1.0\ m/h$$

$$H_A = 10^4\ (Pa{\cdot}m^3)/mol$$

$$\mathscr{D}_{Al} = \mathscr{D}_{Bl} = 10^{-6}\ m^2/h$$

$$p_A = 100 \text{ Pa}$$

$$C_B = 100 \text{ mol/m}^3$$

First find out whether it is instantaneous, fast, slow or infinitely slow reaction.

For fast reaction with high C_B we have pseudo first order rate with respect to A.

To find out which particular case we are having, evaluate M_H and E_i.

$$M_H = \frac{(\mathscr{D}_{Al}\, k\, C_B)^{0.5}}{k_{Al}}$$

$$E_i = 1 + \frac{\mathscr{D}_{Bl}\, C_B\, H_A}{b\, \mathscr{D}_{Al}\, p_{Ai}}$$

For pure reactant A, $p_{Ai} = p_A$

$$M_H = \frac{(10^{-6} \times 10^6 \times 100)^{0.5}}{1} = 10$$

$$E_i = 1 + \frac{10^{-6} \times 100 \times 10^4}{1 \times 10^{-6} \times 100} = 10^4$$

From the the above calculated values, we have $E_i > 5\ M_H$ and for this $E = M_H = 10$ and hence we have pseudo first order reaction, i.e., reaction proceeds by a pseudo first order reaction of A, at the interface. ... Ans. (iv)

For this case of fast reaction with high C_B hence pseudo first order rate with respect to A : the reaction zone is in the liquid film. ... Ans. (iii)

Note that $E_i > 5\ M_H$ is valid for any value of p_A less than100 Pa.

The general rate expression is

$$-r_A'''' = \frac{p_A}{\frac{1}{k_{Ag}\, a} + \frac{H_A}{k_{Al}\, a\, E} + \frac{H_A}{k\, C_B\, f_l}}$$

$$= \frac{100}{\left(\frac{1}{0.10}\right) + \left(\frac{10^4}{100 \times 10}\right) + \left(\frac{10^4}{10^6 \times 0.01 \times 100}\right)}$$

$$= \frac{100}{10 + 10 + 0.01}$$

$$= 4.9975 \approx 5.0 \text{ mol/(h·m}^3 \text{ reactor)}$$

Rate of reaction = 5.0 mol/(h·m³ reactor) **... Ans. (i)**

Resistances :

Gas film = 10 $(h \cdot m^3 \cdot Pa)/mol$

Liquid film = 10

Main body of liquid = 0.01 – negligible compared to above two val

$$\text{Gas film resistance in \%} = \frac{\text{gas film resistance}}{\text{total resistance}} \times 100$$

$$\text{Gas film resistance} = \frac{10}{(10 + 10 + 0.01)} \times 100 = 49.975\%$$

Do similar calculation for the other resistances and tabulate.

Gas film resistance = 49.975 %

Liquid film resistance = 49.975 %

Resistance of main body of liquid = 0.05 % – negligible.

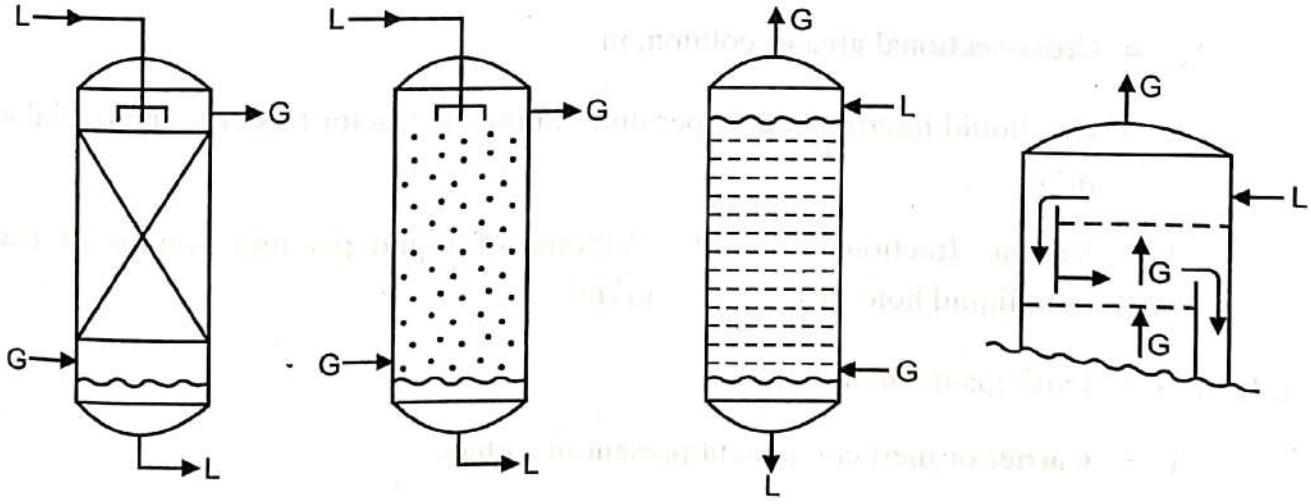
... A

CHAPTER 3

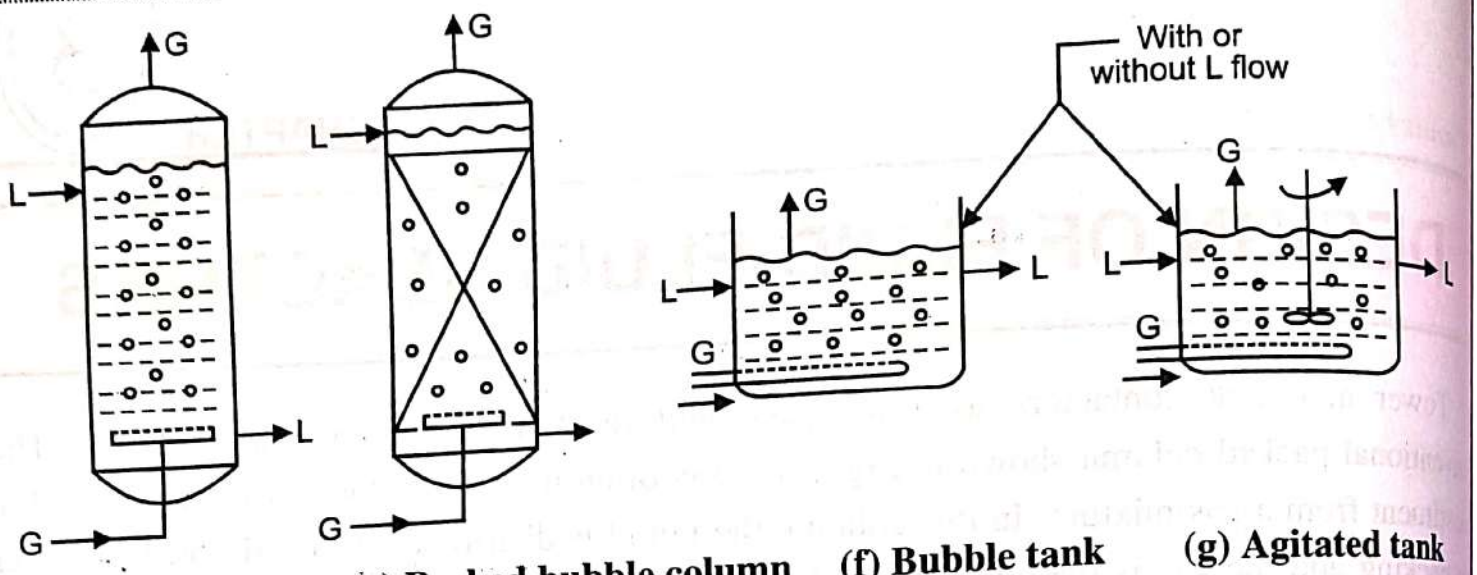
DESIGN OF FLUID-FLUID REACTORS

Tower and tank contactors used for gas-liquid reactions are shown in Fig. 3.1. The conventional packed column shown in Fig. 3.1 (a) is often used when the object is to absorb a constituent from a gas mixture. In this column, the liquid is distributed as a series of films over the packing and the gas is the continuous phase. Here the liquid hold-up is comparatively low and specific interfacial area is large. The pressure drop for the gas is relatively low, owing to which it is very suitable for handling high gas flow rates. In a spray column [Fig. 3.1 (b)], the liquid hold-up is comparatively high and the gas is the continuous phase. In a tray/plate column [Fig. 3.1 (c)], the gas is dispersed in the liquid over the tray. The tray column is suitable for stage-wise operations and for handling relatively high gas flow rates. Here the liquid hold-up is larger than with a packed column.

In situations where a high liquid hold-up is required, any one of the contactors shown in Fig. 3.1 (d) to (g) may be used. The bubble column [Fig. 3.1 (d)] is simply a vessel filled with liquid and provided with a sparger at the bottom to disperse the gas in the liquid. In such a column, there is a tendency of formation of large slugs owing to coalescence of the bubbles. With a packed bubble column [Fig. 3.1 (e)], formation of very large bubbles is avoided. It is just like an ordinary packed column operated in a flooding condition, with a sparger at the bottom for dispersing the gas.



(a) Packed column **(b) Spray column** **(c) Tray column**



(d) Bubble column **(e) Packed bubble column** **(f) Bubble tank** **(g) Agitated tank**

Fig. 3.1 : Tower and tank contactors for G/L reactions

An agitated tank contactor [Fig. 3.1 (g)] makes use of an agitator for dispersing the gas into the liquid. It provides a high interfacial area of contact between gas and liquid phases owing to the formation of small bubbles. Here the liquid hold-up is very high. The bubble tank contactor shown in Fig. 3.1 (f) provides a very low interfacial area, has a very high liquid hold-up and is cheap to build.

The idealized contacting patterns in gas-liquid contactors / reactors are shown in Fig. 3.2.

Towers approximate plug flow G + plug flow L.

Agitated tanks approximate mixed flow G + mixed flow L.

Bubble tanks approximate plug flow G + mixed flow L.

The following symbols are used in the treatment to follow :

A_{cs} = Cross-sectional area of column, m^2

a = Gas-liquid interfacial area per unit volume of reactor (specific interfacial area), m^2/m^3

f_l = Volume fraction of liquid – Volume of liquid per unit volume of reactor (i.e., liquid hold-up), V_l / V_r m^3/m^3

A, B, R, S = Participants in the reaction

U = Carrier or inert component present in a phase

T = Total moles in the reacting / liquid phase

Y_A = p_A / p_U, mol A / mol inert in the gas

$X_A = C_A / C_U$, mol A / mol inert in the liquid

F_g' = Molar flow rate of all the gas (the gas phase), mol/s.

F_l' = Molar flow rate of all the liquid (the liquid phase), mol/s.

$F_g = F_g' \frac{p_U}{\pi}$, upward molar flow rate of inerts in the gas phase, mol/s.

$F_l = F_l' \frac{C_U}{C_T}$, downward molar flow rate of inerts in the liquid phase, mol/s.

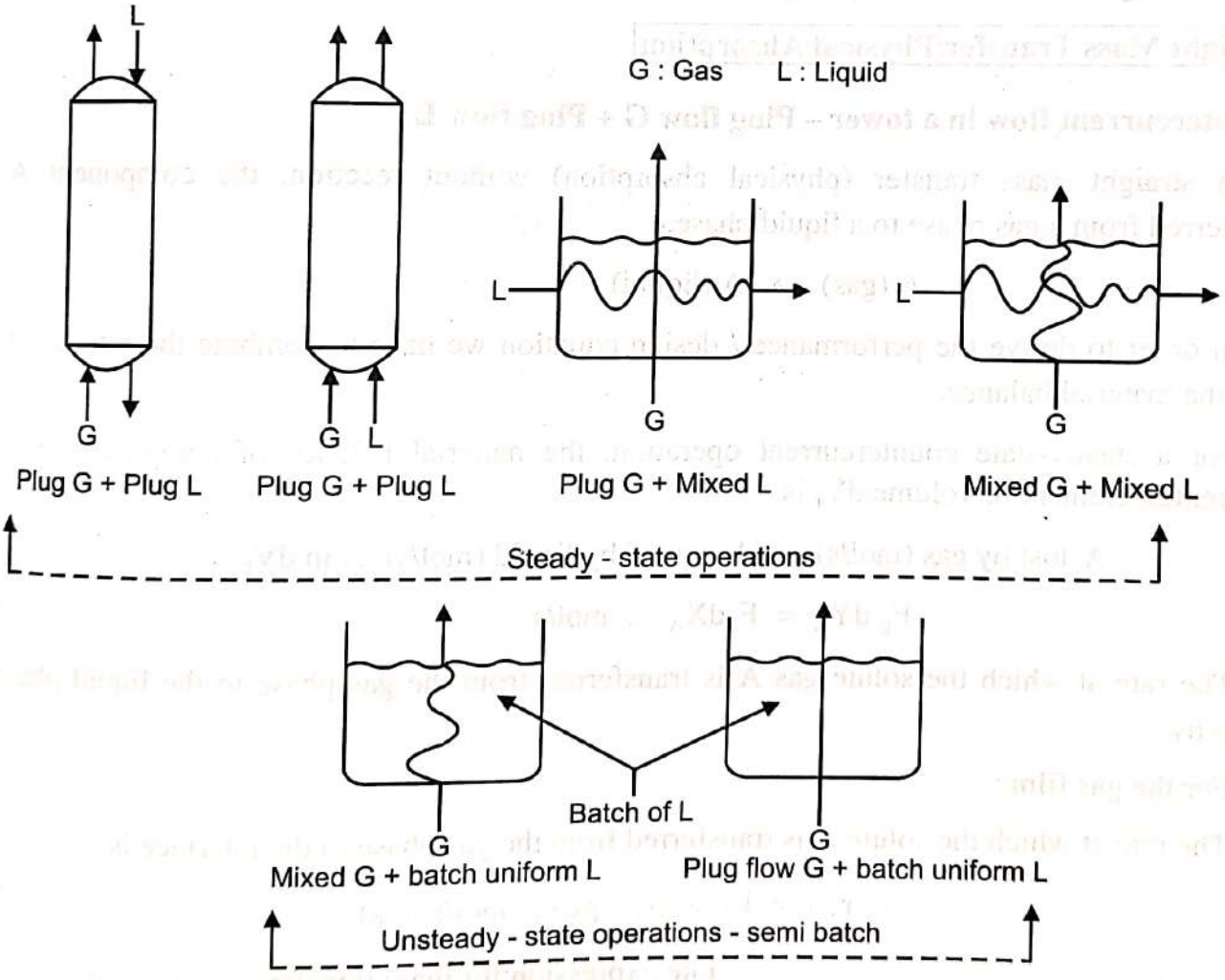


Fig. 3.2 : Various ideal contacting patterns for G/L contactors

The relationships among the various concentration measures are :

$$\pi = p_A + p_B + \ldots + p_U$$

$$C_T = C_A + C_B + \ldots + C_U$$

$$dY_A = d(p_A / p_U) = \frac{p_U\, dp_A - p_A\, dp_U}{p_U^2} \overset{\text{for dilute systems}}{\approx} \frac{dp_A}{p_U}$$

$$dX_A = d(C_A / C_U) = \frac{C_U\, dC_A - C_A\, dC_U}{C_U^2} \overset{\text{for dilute systems}}{\approx} \frac{dC_A}{C_U}$$

We will first develop the performance equations of different contactors for

A (gas) → A (liquid) ... straight mass transfer (no reaction)

and then for

A (g → *l*) + bB (*l*) → products (*l*) ... mass transfer + reaction

Straight Mass Transfer/Physical Absorption

Countercurrent flow in a tower – Plug flow G + Plug flow L

In straight mass transfer (physical absorption) without reaction, the component A is transferred from a gas phase to a liquid phase.

A (gas) → A (liquid)

In order to derive the performance / design equation we have to combine the rate equation with the material balance.

For a steady-state countercurrent operation, the material balance of component A in a differential element of volume dV_r is

'A' lost by gas (mol/s) = 'A' gained by liquid (mol/s) ... in dV_r

$$F_g\, dY_A = F_l\, dX_A \quad \text{... mol/s} \qquad \text{... (3.1)}$$

The rate at which the solute gas A is transferred from the gas phase to the liquid phase is given by

For the gas film :

The rate at which the solute A is transferred from the gas phase to the interface is

$$(-r_A''') = k_{Ag}\, a\, (p_A - p_{Ai}) \text{ ... mol/(m}^3\text{·s)} \qquad \text{... (3.2)}$$

... rate expression for mass transfer

Similarly for the liquid film :

$$(-r_A''') = k_{Al}\, a\, (C_{Ai} - C_A) \qquad \text{... (3.3)}$$

So, at steady-state :

$$(-r_A''') = k_{Ag}\, a\, (p_A - p_{Ai}) = k_{Al}\, a\, (C_{Ai} - C_A) \qquad \text{... (3.4)}$$

We can equate the rate at which the solute A is lost from the gas phase to the rate at which is transferred from the gas phase to the interface. This is also equal to the rate at which the solute A is gained by the liquid phase and the rate at which it is transferred from the interface to the liquid phase.

Thus, at steady-state for the component A in a differential element of volume dV_r (of a contactor), we can write

'A' lost from gas in mol/s = A gained by liquid in mol/s

= rate of mass transfer of A in mol/s

$$F_g\, dY_A = F_l\, dX_A = (-r_A''')\, dV_r \qquad \ldots (3.5)$$

i.e., $$F_g\, dY_A = F_l\, dX_A = k_{Ag}\, a\, (p_A - p_{Ai})\, dV_r$$

$$= k_{Al}\, a\, (C_{Ai} - C_A)\, dV_r \qquad \ldots (3.6)$$

We have, $$Y_A = p_A / p_U$$

$$dY_A = d\,(p_A / p_U)$$

$$dY_A = \frac{p_U\, dp_A - p_A\, dp_U}{p_U^2} \qquad \ldots (3.7)$$

We have : $$p_A + p_U = \pi$$

$$p_U = \pi - p_A$$

$\therefore$ $$dp_U = -\, dp_A$$

Substituting the values of p_U and dp_U in Equation (3.7), we get

$$dY_A = \frac{p_A\, dp_A - p_A\,(-\, dp_A)}{(\pi - p_A)^2} = \frac{p_U\, dp_A + p_A\, dp_A}{(\pi - p_A)^2}$$

$$dY_A = \frac{(p_U + p_A)\, dp_A}{(\pi - p_A)^2} = \frac{\pi\, dp_A}{(\pi - p_A)^2} \qquad \ldots (3.8)$$

We have $$F_g = F_g'\, p_U / \pi$$

$\therefore$ $$F_g = F_g' \frac{(\pi - p_A)}{\pi} \qquad \ldots (3.9)$$

Expressing dY_A in terms of p_A and F_g in terms of F_g' using Equations (3.8) and (3.9), we get

$$F_g\, dY_A = F_g \cdot \frac{\pi\, dp_A}{(\pi - p_A)^2} = \frac{F_g'\,(\pi - p_A)}{\pi} \cdot \frac{\pi\, dp_A}{(\pi - p_A)^2}$$

$$= \frac{F_g'\, dp_A}{(\pi - p_A)} \qquad \ldots (3.10)$$

Expressing F_g' in terms of F_g using Equation (3.9)

$$\frac{F_g' p_A}{\pi} = \frac{F_g \pi}{p_U} \cdot \frac{p_A}{\pi} = \frac{F_g p_A}{p_U}$$

$$d\left(\frac{F_g' p_A}{\pi}\right) = d\left(\frac{F_g p_A}{p_U}\right)$$

$$d\left(\frac{F_g' p_A}{\pi}\right) = F_g\, d\left(\frac{p_A}{p_U}\right)$$

But $$dY_A = d\left(\frac{p_A}{p_U}\right)$$

$$\therefore \quad d\left(\frac{F_g' p_A}{\pi}\right) = F_g\, dY_A \qquad \ldots (3.11)$$

Combining Equations (3.10) and (3.11), we get

$$F_g\, dY_A = d\left(\frac{F_g' p_A}{\pi}\right) = \frac{F_g \pi\, dp_A}{(\pi - p_A)^2} = \frac{F_g'\, dp_A}{(\pi - p_A)} \qquad \ldots (3.12)$$

We have, $$F_g\, dY_A = F_l\, dX_A = (-r_A'''')\, dV_r$$

Usually, the rate of mass transfer is defined as the flow of material per unit surface f the interface).

$$\therefore \quad (-r_A'''') = a\,(-r_A'')$$

$-r_A''$ – rate of transfer of A per unit interfacial area $= -\frac{1}{S}\frac{dN_A}{dt}$, $\frac{mol}{m^2 \cdot s}$

a = interfacial area per unit volume of reactor / contactor, m^2/m^3

$$\therefore \quad \frac{F_g}{a} \cdot \frac{dY_A}{(-r_A'')} = \frac{F_l}{a} \cdot \frac{dX_A}{(-r_A'')} = dV_r$$

Integrating the above equation for the whole tower, we get

$$\int_0^{V_r} dV_r = \int_{Y_{A1}}^{Y_{A2}} \frac{F_g}{a} \cdot \frac{dY_A}{(-r_A'')} = \int_{x_{A1}}^{x_{A2}} \frac{F_l}{a} \cdot \frac{dX_A}{(-r_A'')}$$

$$V_r = \frac{F_g}{a} \int_{Y_{A1}}^{Y_{A2}} \frac{dY_A}{(-r_A'')} = \frac{F_l}{a} \int_{x_{A1}}^{x_{A2}} \frac{dX_A}{(-r_A'')} \qquad \ldots (3.13)$$

Now we express Y_A in terms of p_A and X_A in terms C_A and obtain the expression for V_r in terms of the gas film and the liquid film driving forces.

We have, $$F_g\, dY_A = (-r_A'''')\, dV_r = (-r_A'')\, a\, dV_r = k_{Ag}\, a\, (p_A - p_{Ai})\, dV_r$$

$$F_g\, dY_A = k_{Ag}\, a\, (p_A - p_{Ai})\, dV_r \qquad \ldots (3.14)$$

Substituting $F_g\, dY_A$ from Equation (3.12), the above equation becomes

$$\frac{F_g\, \pi\, dp_A}{(\pi - p_A)^2} = \frac{F_g'\, dp_A}{(\pi - p_A)} = k_{Ag}\, a\, (p_A - p_{Ai})\, dV_r$$

Rearranging, we get

$$\frac{F_g\, \pi\, dp_A}{k_{Ag}\, a\, (\pi - p_A)^2\, (p_A - p_{Ai})} = \frac{F_g'\, dp_A}{k_{Ag}\, a\, (\pi - p_A)\, (p_A - p_{Ai})} = dV_r$$

Integrating over the whole tower gives

$$\int_0^{V_r} dV_r = F_g\, \pi \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{k_{Ag}\, a\, (\pi - p_A)^2\, (p_A - p_{Ai})}$$

$$V_r = \int_{p_{A1}}^{p_{A2}} \frac{F_g'\, dp_A}{k_{Ag}\, a\, (\pi - p_A)\, (p_A - p_{Ai})} \qquad \ldots (3.15)$$

We have $$F_l\, dX_A = (-r_A'''')\, dV_r = k_{Al}\, a\, (C_{Ai} - C_A)\, dV_r \qquad \ldots (3.16)$$

$$dX_A = d\,(C_A / C_U) = \frac{C_U\, dC}{C_U^2} \frac{- C_A\, dC_U}{} \qquad \ldots (3.17)$$

$$C_T = C_A + C_U$$

$$C_A = C_T - C_U$$

$\therefore$ $$dC_A = - dC_U$$

$$C_U = C_T - C_A$$

Substituting the values of dC_A and C_U in Equation (3.17) gives

$$dX_A = \frac{C_U\, dC_A - C_A\, (- dC_A)}{(C_T - C_A)^2}$$

$$= \frac{(C_U + C_A)\, dC_A}{(C_T - C_A)^2} = \frac{C_T\, dC_A}{(C_T - C_A)^2}$$

Using this value of dX_A, we can write

$$F_l\, dX_A = F_l \cdot \frac{C_T\, dC_A}{(C_T - C_A)^2}$$

We have $$F_l = F_l' \cdot \frac{C_U}{C_T} = \frac{F_l'(C_T - C_A)}{C_T}$$

Therefore, we can write

$$F_l\, dX_A = \frac{F_l\, C_T\, dC_A}{(C_T - C_A)^2} = \frac{F_l'(C_T - C_A)}{C_T} \cdot \frac{C_T\, dC_A}{(C_T - C_A)} = \frac{F_l'\, dC_A}{(C_T - C_A)} \quad \ldots (3.18)$$

Substituting $F_l\, dX_A$ from Equation (3.18), Equation (3.16) becomes

$$\frac{F_l\, C_T\, dC_A}{(C_T - C_A)^2} = \frac{F_l'\, dC_A}{(C_T - C_A)} = k_{Al}\, a\, (C_{Ai} - C_A)\, dV_r$$

Rearranging, we get

$$\frac{F_l\, C_T\, dC_A}{k_{Al}\, a\, (C_T - C_A)^2\, (C_{Ai} - C_A)} = \frac{F_l'\, dC_A}{k_{Al}\, a\, (C_T - C_A)\, (C_{Al} - C_A)} = dV_r$$

Integrating over the whole tower gives

$$\int_0^{V_r} dV_r = V_r = F_l\, C_T \int_{C_{A1}}^{C_{A2}} \frac{dC_A}{k_{Al}\, a\, (C_T - C_A)^2\, (C_{Ai} - C_A)}$$

$$= \int_{C_{A1}}^{C_{A2}} \frac{F_l'\, dC_A}{k_{Al}\, a\, (C_T - C_A)\, (C_{Ai} - C_A)} \quad \ldots (3.19)$$

$\therefore$ Combining Equations (3.13), (3.15) and (3.19), we get

$$V_r = \frac{F_g}{a} \int_{Y_{A1}}^{Y_{A2}} \frac{dY_A}{-r_A''} = \frac{F_l}{a} \int_{X_{A1}}^{X_{A2}} \frac{dX_A}{-r_A''}$$

$$= F_g\, \pi \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{k_{Ag}\, a\, (\pi - p_A)^2\, (p_A - p_{Ai})}$$

$$= \int_{p_{A1}}^{p_{A2}} \frac{F_g'\, dp_A}{k_{Ag}\, a\, (\pi - p_A)\, (p_A - p_{Ai})}$$

$$= F_l\, C_T \int_{C_{A1}}^{C_{A2}} \frac{dC_A}{k_{Al}\, a\, (C_T - C_A)^2\, (C_{Ai} - C_A)}$$

$$= \int_{C_{A1}}^{C_{A2}} \frac{F_l'\, dC_A}{k_{Al}\, a\, (C_T - C_A)\, (C_{Al} - C_A)} \quad \ldots (3.20)$$

The design procedure is outlined in Fig. 3.3.

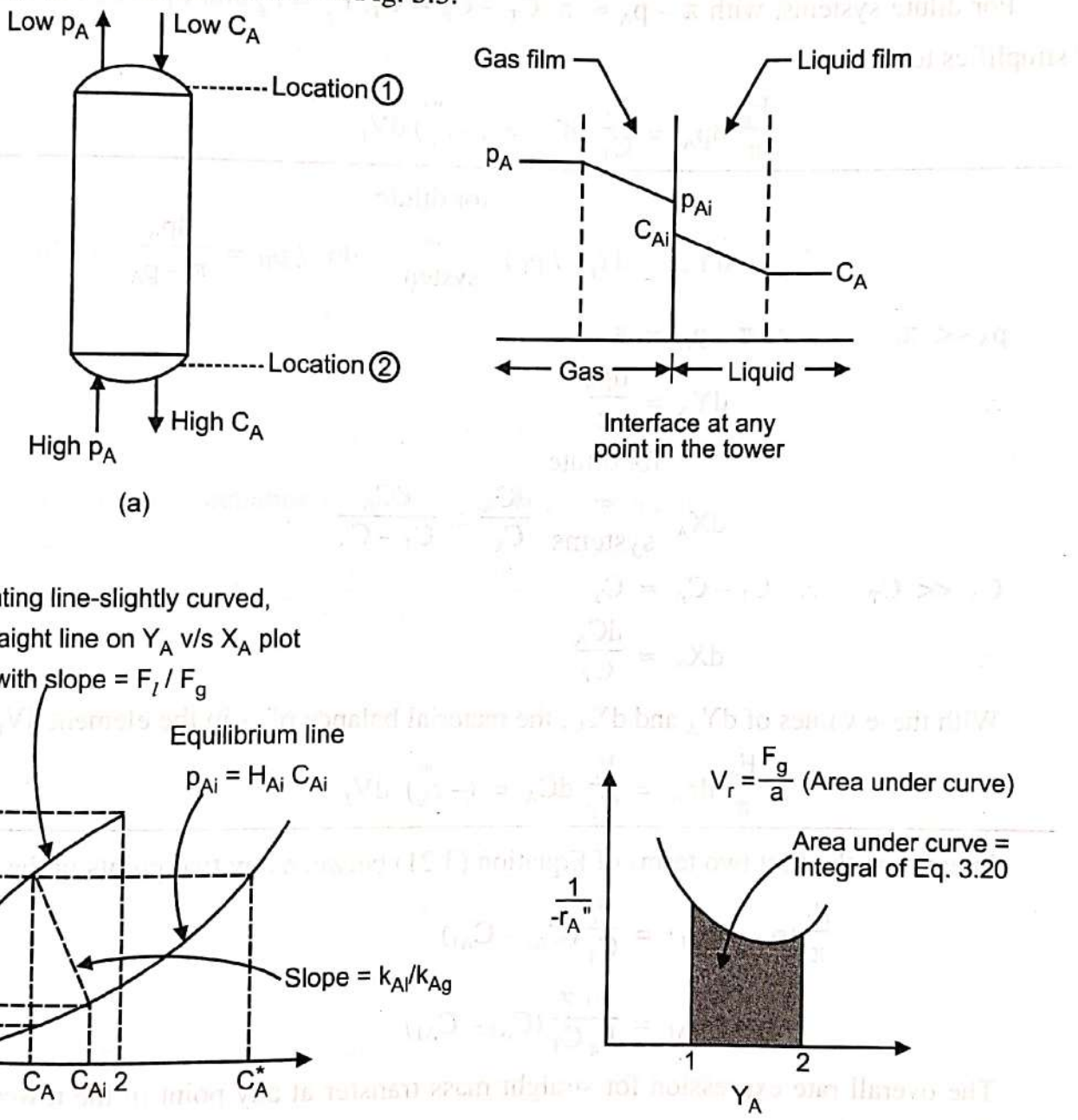


Fig. 3.3 : Design procedure for straight mass transfer - countercurrent operation

Dilute Systems :

For dilute systems, $C_A << C_T$ and $p_A << \pi$. Consequently, we have, $F_g' \approx F_g$ and $F_l' \approx F_l$.

The material balance of A in the element dV_r is

$$F_g \, dY_A = F_l \, dX_A = (-r_A'''') \, dV_r$$

In an alternate form, it can be written as (expressing dY_A in terms of dp_A and so on)

$$\frac{F_g' \, dp_A}{(\pi - p_A)} = \frac{F_l \, dC_A}{(C_T - C_A)} = (-r_A'''') \, dV_r$$

For dilute systems, with $\pi - p_A \approx \pi$, $C_T - C_A \approx C_T$, $F_g' \approx F_g$ and $F_l' \approx F_l$, the above equation simplifies to

$$\frac{F_g}{\pi} dp_A = \frac{F_l}{C_T} dC_A = (-r_A'''') dV_r \quad \ldots (3.2)$$

for dilute

$$dY_A = d(p_A / p_U) \underset{\text{systems}}{\approx} dp_A / p_U = \frac{dp_A}{\pi - p_A}$$

$p_A << \pi \quad \therefore \pi - p_A \approx \pi$

$$\therefore \quad dY_A = \frac{dp_A}{\pi}$$

for dilute

$$dX_A \underset{\text{systems}}{\approx} \frac{dC_A}{C_U} \simeq \frac{dC_A}{C_T - C_A}$$

$C_A << C_T \quad \therefore C_T - C_A \approx C_T$

$$\therefore \quad dX_A \approx \frac{dC_A}{C_T}$$

With these values of dY_A and dX_A, the material balance of A in the element dV_r becomes

$$\frac{F_g}{\pi} dp_A = \frac{F_l}{C_T} dC_A = (-r_A'''') dV_r$$

Integrating the first two terms of Equation (3.21) between any two points in the tower, we get

$$\frac{F_g}{\pi}(p_{A2} - p_{A1}) = \frac{F_l}{C_T}(C_{A2} - C_{A1})$$

$$p_{A2} - p_{A1} = \frac{F_l \pi}{F_g C_T}(C_{A2} - C_{A1}) \quad \ldots (3.2$$

The overall rate expression for straight mass transfer at any point in the tower (considering both resistances) is

$$(r_A'''') = (r_A'') a = \frac{1}{\frac{1}{k_{Ag} a} + \frac{H_A}{k_{Al} a}}(p_A - H_A C_A) \quad \ldots (3.2$$

With $p_A^* = H_A C_A$, Equation (3.23) becomes

$$(-r_A'''') = (-r_A'') a = \frac{1}{\frac{1}{k_{Ag} a} + \frac{H_A}{k_{Al} a}}(p_A - p_A^*) \quad \ldots (3.2$$

The overall mass transfer coefficients are given by

$$\frac{1}{K_{Ag} a} = \frac{1}{k_{Ag} a} + \frac{H_A}{k_{Al} a} \quad \ldots \text{gas phase} \quad \ldots (3.2$$

$$\frac{1}{K_{Al} a} = \frac{1}{H_A k_{Ag} a} + \frac{1}{k_{Al} a} \quad \ldots \text{liquid phase} \quad \ldots (3.2$$

∴ Equation (3.24) in terms of overall transfer coefficients becomes

$$(r_A'''') = (-r_A'') a = K_{Ag} a (p_A - p_A^*) = K_{Al} a (C_A^* - C_A) \quad \ldots (3.27)$$

$p_A - p_A^*$ = overall gas phase driving force

$C_A^* - C_A$ = overall liquid phase driving force

Substituting the value of $(-r_A'''')$ from Equation (3.27) in Equation (3.21) gives

$$\frac{F_g}{\pi} dp_A = \frac{F_l}{C_T} dC_A = K_{Ag} a (p_A - p_A^*) dV_r = K_{Al} a (C_A^* - C_A) dV_r \quad \ldots (3.28)$$

$$\frac{F_g}{\pi} dp_A = K_{Ag} a (p_A - p_A^*) dV_r$$

Integrating the above equation over the tower and rearranging, we get

$$V_r = \frac{F_g}{\pi K_{Ag} a} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{p_A - p_A^*} \quad \ldots (3.29)$$

$$\frac{F_l}{C_T} dC_A = K_{Al} a (C_A^* - C_A) dV_r$$

$$V_r = \frac{F_l}{C_T K_{Al} a} \int_{C_{Al}}^{C_{A2}} \frac{dC_A}{(C_A^* - C_A)} \quad \ldots (3.30)$$

$$\text{Volume of tower} = V_r = h A_{cs} \quad \ldots (3.31)$$

where h is the height of tower and A_{cs} is the cross-sectional area of tower.

For dilute systems, combining Equations (3.29), (3.30) and (3.31), we get

$$\boxed{V_r = hA_{cs} = \frac{F_g}{\pi K_{Ag} a} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{p_A - p_A^*} = \frac{F_l}{C_T K_{Al} a} \int_{C_{A1}}^{C_{A2}} \frac{dC_A}{C_A^* - C_A}} \quad \ldots (3.32)$$

$p_A^* = H_A C_A$, gas in equilibrium with liquid C_A

$C_A^* = p_A / H_A$, liquid in equilibrium with gas p_A

$K_{Al} \Rightarrow$ liquid phase overall mass transfer coefficient

$K_{Ag} \Rightarrow$ gas phase overall mass transfer coefficient

Concurrent flow in a tower : For plug flow G + plug flow L and downward flow of gas : simply replace F_g by $-F_g$ throughout in the above equations for countercurrent flow.

Mass Transfer + Not Very Slow Reaction (mass transfer accompanied by reaction) :

The design procedure of a tower for the case of mass transfer with reaction is best described by considering the reaction : A ($g \rightarrow l$) + bB (l) $\rightarrow$ products (l). Let us assume that the reaction is so fast that no unreacted A enters the main body of the liquid (i.e., the reaction is limited to the region close to the phase interface).

Mass Transfer + Reaction in a Countercurrent Tower – Plug flow G + Plug flow L :

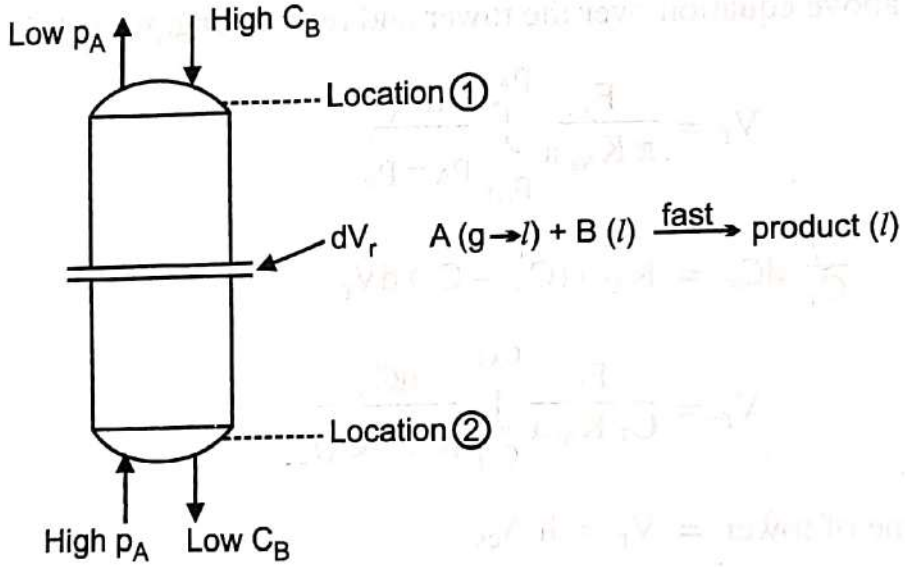


Fig. 3.4

The material balance for a differential element of tower, volume dV_r is

$$\text{'A' lost by gas} = \frac{1}{b}(\text{'B' lost by liquid})$$

$$F_g\, dY_A = -\frac{F_l\, dX_B}{b}, \frac{\text{mol}}{\text{s}} \qquad \ldots (3.33)$$

Equation (3.33) gives A reacted in the element dV_r from the material balance.

Integrating for the whole tower gives

$$F_g (Y_{A2} - Y_{A1}) = \frac{F_l}{b}(X_{B1} - X_{B2}) \qquad \ldots (3.34)$$

Compositions at any point in the tower from location 1 is given by

$$F_g (Y_A - Y_{A1}) = \frac{F_l}{b}(X_{B1} - X_B) \qquad \ldots (3.35)$$

For dilute systems (i.e., when all the participants in the reaction are dilute), $p_U \approx \pi$ and $C_U \approx C_T$

$$dY_A = d(p_A / p_U) \overset{\text{for dilute systems}}{\approx} \frac{dp_A}{p_U}$$

and $p_U \approx \pi \quad \therefore \quad dY_A = dp_A / \pi$

$$dX_A = d(C_A / C_U) \overset{\text{for dilute systems}}{\approx} \frac{dC_A}{C_U}$$

and $C_U \approx C_T \quad \therefore \quad dX_A = dC_A / C_T$

Therefore, Equation (3.33) for dilute systems simplifies to

$$\frac{F_g}{\pi} dp_A = -\frac{F_l}{bC_T} dC_A \qquad \ldots (3.35)$$

Integrating for the whole tower gives

$$\frac{F_g}{\pi}(p_{A2} - p_{A1}) = \frac{F_l}{bC_T}(C_{B1} - C_{B2}) \qquad \ldots (3.36)$$

Integrating Equation (3.35) from location 1 to any point in the tower gives

$$\frac{F_g}{\pi}(p_A - p_{A1}) = \frac{F_l}{bC_T}(C_{B1} - C_B) \qquad \ldots (3.37)$$

The volume of tower or the height of tower is found by combining the material balance and the rate equation for a differential element of tower volume dV_r.

$$\begin{pmatrix}\text{A reacted in the element} - \\ \text{from material balance}\end{pmatrix} = \begin{pmatrix}\text{A reacted in the element} - \\ \text{from rate expression}\end{pmatrix}$$

Therefore,

$$(\text{'A' lost by gas}) = \frac{1}{b}(\text{B lost by liquid}) = \begin{pmatrix}\text{disappearance of A} \\ \text{by reaction}\end{pmatrix}$$

$$F_g\, dY_A = -\frac{F_l\, dX_B}{b} = (-r_A''')\, dV_r \qquad \ldots (3.38)$$

But $\quad (-r_A'''') = (-r_A'')\, a$

$-r_A''$ is the rate based on unit interfacial area.

With $(-r_A'''') = (-r_A'')\, a$, Equation (3.38) becomes

$$F_g\, dY_A = -\frac{F_l\, dX_B}{b} = (-r_A'')\, a\, dV_r$$

Integrating the above equation and rearranging gives

$$\boxed{V_r = hA_{cs} = F_g \int_{Y_{A1}}^{Y_{A2}} \frac{dY_A}{(-r_A'')\, a} = \frac{F_l}{b} \int_{X_{B2}}^{X_{B1}} \frac{dX_B}{(-r_A'')\, a}} \qquad \ldots (3.39)$$

For dilute systems :

$$\frac{F_g}{\pi} dp_A = -\frac{F_l\, dC_B}{bC_T} = (-r_A'''')\, dV_r = (-r_A'')\, a\, dV_r \quad \ldots (3.4$$

Integrating Equation (3.40), we get

$$\boxed{V_r = hA_{cs} = \frac{F_g}{\pi}\int_{p_{A1}}^{p_{A2}} \frac{dp_A}{(-r_A'')\, a} = \frac{F_l}{bC_T}\int_{C_{B2}}^{C_{B1}} \frac{dC_B}{(-r_A'')\, a}} \quad \ldots (3.4$$

Stepwise procedure to determine V_r :

The performance equation is

$$\frac{V_r \pi}{F_g} = \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{(-r_A'')\, a}$$

1. Choose/select a few p_A values, say p_{A1}, p_{A2} and intermediate p_A values (i.e., p_A value between p_{A1} and p_{A2}) and for each p_A chosen find the corresponding C_B.

2. Evaluate the rate of reaction for each point using the rate relation :

$$(-r_A'')\, a = \frac{1}{\frac{1}{k_{Ag}\, a} + \frac{H_A}{k_{Al}\, aE} + \frac{H_A}{kC_B\, f_l}}\, p_A$$

3. Plot $1/(-r_A'')$ a v/s p_A to evaluate the integral on RHS of the performance equation [Equation (3.41)] graphically and thus to get V_r.

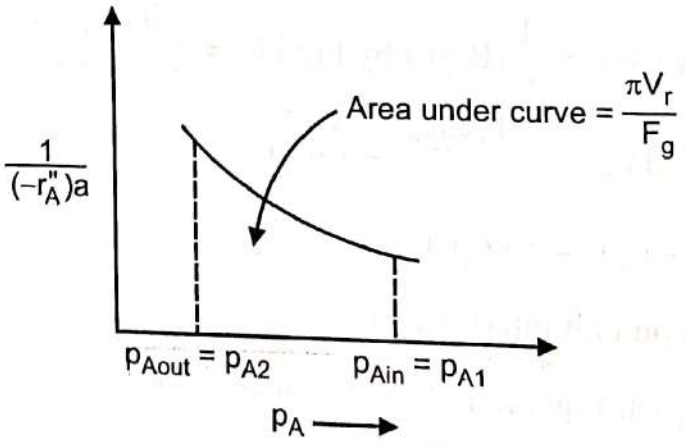


Fig. 3.5

Mass Transfer + Reaction in a Cocurrent Tower – Plug Flow G + Plug Flow L :

In case of a cocurrent tower -

(i) For downward flow of both streams simply replace F_g by $-F_g$ in the equations for countercurrent flow.

(ii) For upward flow of both streams simply replace F_l by $-F_l$ in the equations for countercurrent flow.

The rest of the procedure remains the same.

Mass Transfer + Reaction in an Agitated Tank Contactors-Mixed Flow G + Mixed Flow L :

In an agitated tank contactor, since the composition is uniform throughout the contactor, we have to make accounting of materials about the contactor as a whole. Thus,

$$\text{('A' lost by gas)} = \frac{1}{b}\text{('B' lost by liquid)} = \text{(disappearance of 'A' by reaction)}$$

or
$$F_g (Y_{A\,in} - Y_{A\,out}) = \frac{F_l}{b}(X_{B\,in} - X_{B\,out}) = (-r'''_A)\,|_{\text{at exit}} \cdot V_r \qquad \ldots (3.42)$$

Equation (3.42) for dilute systems becomes

$$\boxed{\frac{F_g}{\pi}(p_{A\,in} - p_{A\,out}) = \frac{F_l}{bC_T}(C_{B\,in} - C_{B\,out}) = (-r''''_A)\,|_{\text{at exit}} \cdot V_r} \qquad \ldots (3.43)$$

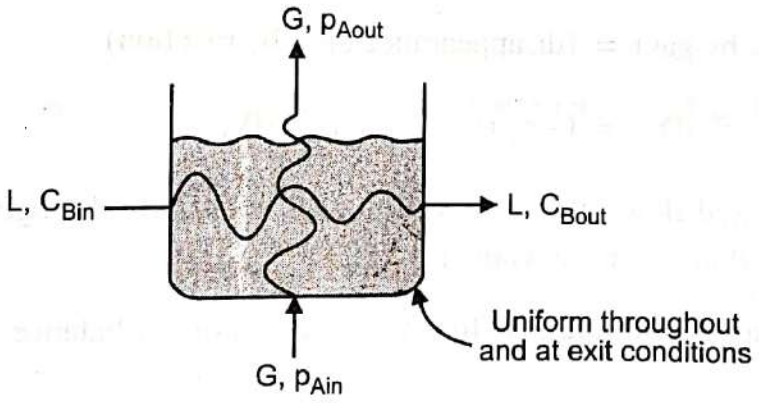


Fig. 3.6

To determine V_r :

Evaluate $(-r'''_A)$ from known stream compositions and solve Equation (3.42) or (3.43).

To find $C_{B\,out}$ or $p_{A\,out}$ for a given V_r (volume of contactor) :

(i) Guess $p_{A\,out}$, i.e., assume some $p_{A\,out}$ value and evaluate $C_{B\,out}$.

(ii) Evaluate $(-r''''_A)$ and then V_r.

(iii) Compare the calculated V_r value with the value of V_r provided. If $V_{r\,calculated}$ is different from $V_{r\,given}$, assume another [i.e., if $V_{r,\,calculate} \neq V_{r,\,given}$] $p_{A\,out}$ value and repeat the procedure till we do not get : $V_{r\,calculated} \approx V_{r\,given}$.

Mass Transfer + Reaction in Bubble Tank Contactors – Plug Flow G + Mixed Flow L :

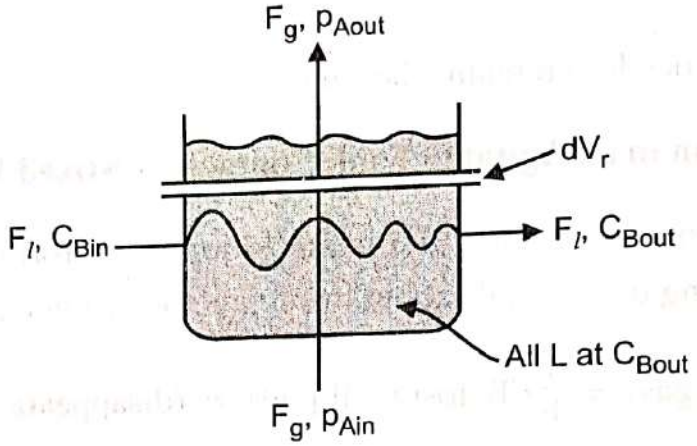


Fig. 3.7

In this case we have to make two separate material accountings, one for A and the other for B.

(i) As G is in plug flow – the composition of A is not uniform throughout - so the accounting of A must be done over a differential element of volume dV_r

Therefore, the differential balance for the loss of A from the gas is

('A' lost by gas) = (disappearance of A by reaction)

or $$F_g \, dY_A = (-r_A'''')\,|_{\text{L at exit conditions}} \cdot dV_r \qquad \ldots (3.44)$$

(ii) As L is in mixed flow - the composition of B is uniform throughout - so the accounting of B must be done over the whole reactor.

For the liquid as a whole and for the gas as whole, a balance over the whole reactor yields

(all 'A' lost by gas) $= \frac{1}{b}$ (all 'B' lost by liquid)

i.e., $$F_g \, \Delta Y_A = \frac{F_l}{b} \, \Delta X_B \qquad \ldots (3.45)$$

Integrating Equation (3.44), we get

$$\boxed{V_r = F_g \int_{Y_{A\,out}}^{Y_{A\,in}} \frac{dY_A}{(-r_A'')\,a}} \quad , \quad (-r_A'''') = (-r_A'')\,a \qquad \ldots (3.46)$$

$(-r_A'')$ is for the liquid at $C_{B\,out}$ and

$$F_g \, (Y_{A\,in} - Y_{A\,out}) = \frac{F_l}{b} \, (X_{B\,in} - X_{B\,out}) \qquad \ldots (3.47)$$

For dilute systems, Equation (3.44) becomes

$$\frac{F_g}{\pi} dp_A = (-r_A'''') dV_r = (-r_A'') a \, dV_r$$

Integrating, we get

$$\boxed{V_r = \frac{F_g}{\pi} \int_{p_{A\,out}}^{p_{A\,in}} \frac{dp_A}{(-r_A'') a}} \quad \ldots (3.48)$$

and Equation (3.47) becomes

$$\frac{F_g}{\pi} (p_{A\,in} - p_{A\,out}) = \frac{F_l}{bC_T} (C_{B\,in} - C_{B\,out}) \quad \ldots (3.49)$$

When V_r is to be determined knowing the exit conditions, then select a number of p_A values and integrate the performance equation graphically.

When $p_{A\,out}$ and $C_{B\,out}$ are to be determined for a given V_r, then guess $C_{B\,out}$, evaluate V_r and compare $V_{r\,calculated}$ with $V_{r\,given}$. See whether $V_{r\,calculated} \approx V_{r\,given\ or\ not}$ and if not, then repeat the procedure till we do not get

$$V_{r\,calculated} \approx V_{r\,given}$$

Absorption + Reaction in a Batch Agitated Tank Contactor – Mixed Flow G + Batch of uniform L :

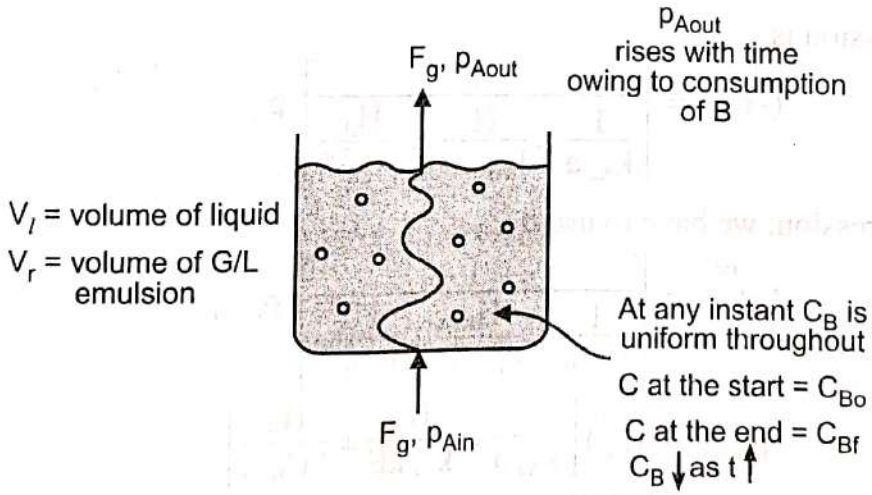


Fig. 3.8

Here C_B decreases with time because of reaction with A. However, at any instant of time, C_B is the same everywhere in the tank / uniform throughout the tank.

As it being a unsteady-state operation, composition and rates all vary with time.

At any instant of time, the material balance equates the three quantities, namely, loss of 'A' from gas, decrease of 'B' with time in L and disappearance of A or B by reaction. That is :

$$F_g (Y_{A\,in} - Y_{A\,out}) = -\frac{V_l}{b} \frac{dC_B}{dt} = (-r_A'''') V_r \quad \ldots (3.50)$$

As G is in mixed flow, use $p_{A\,out}$ in the rate expression.

For dilute systems, Equation (3.50) becomes

$$\boxed{\frac{F_g}{\pi}(p_{A\,in} - p_{A\,out}) = -\frac{V_l}{b}\frac{dC_B}{dt} = (-r_A'''')\,V_r} \quad \ldots (3.51)$$

To find the time required for a given operation :

If $p_A << \pi$ and if $E = M_H$ (then, E is independent of p_A).

Here we have to combine the first and third terms of Equation (3.50) to obtain $(-r_A'''')$.

$$Y_A = p_A / p_U = p_A / \pi - p_A$$

With this Y_A and taking the first and third terms of Equation (3.50), we get

$$F_g\left(\frac{p_{A\,in}}{\pi - p_{A\,in}} - \frac{p_{A\,out}}{\pi - p_{A\,out}}\right) = (-r_A'''')\,V_r$$

For $p_A << \pi$, it becomes

$$\frac{F_g}{\pi}(p_{A\,in} - p_{A\,out}) = (-r_A'''')\,V_r$$

Rearranging, we get

$$p_{A\,in} - p_{A\,out} = (-r_A'''')\,\frac{V_r\,\pi}{F_g} \quad \ldots (3.52)$$

The rate expression is

$$(-r_A'''') = \left[\frac{1}{\frac{1}{k_{Ag}\,a} + \frac{H}{k_{Al}\,aE} + \frac{H_A}{kC_B\,f_l}}\right] p_A$$

In the rate expression, we have to use $p_{A\,out}$.

$$\therefore \quad (-r_A'''') = \left[\frac{1}{\frac{1}{k_{Ag}\,a} + \frac{H_A}{k_{Al}\,aE} + \frac{H_A}{kC_B\,f_l}}\right] p_{A\,out}$$

$$\therefore \quad p_{A\,out} = (r_A'''')\left[\frac{1}{k_{Ag}\,a} + \frac{H_A}{k_{Al}\,aE} + \frac{H_A}{kC_B\,f_l}\right] \quad \ldots (3.53)$$

Replacing $p_{A\,out}$ in Equation (3.52) by its value given by Equation (3.53) gives

$$p_{A\,in} - (-r_A'''')\left[\frac{1}{k_{Ag}\,a} + \frac{H_A}{k_{Al}\,aE} + \frac{H_A}{kC_B\,f_l}\right] = \frac{(-r_A'''')\,\pi V_r}{F_g}$$

Taking the second term of the above equation to RHS, taking out $(-r_A'''')$ common from the terms on RHS and rearranging gives

$$\boxed{(-r_A'''') = p_{A\,in} \Big/ \left[\frac{\pi V_r}{F_g} + \frac{1}{k_{Ag}\,a} + \frac{H_A}{k_{Al}\,aE} + \frac{H_A}{kC_B\,f_l}\right]} \quad \ldots (3.54)$$

Use the above expression to evaluate $(-r_A'''')$.

Let us combine the second and third terms of Equation (3.50) to find the time required for a given operation.

$$-\frac{V_l}{b}\frac{dC_B}{dt} = (-r_A'''')\,V_r$$

$$-\frac{V_l}{V_r\,b}\frac{dC_B}{(r_A'''')} = dt$$

The limits of integration are :

At $t = 0$ (at the start) ..... $C_B = C_{Bo}$, At $t = t$ (at the end) ..... $C_B = C_{Bf}$

Integrating the above equation within the limits, we get

$$\int_0^t dt = -\frac{V_l}{V_r\,b}\int_{C_{Bo}}^{C_{Bf}}\frac{dC_B}{(-r_A'''')} = \frac{V_l}{V_r\,b}\int_{C_{Bf}}^{C_{Bo}}\frac{dC_B}{(-r_A'''')}$$

$$t = \frac{V_l}{V_r\,b}\int_{C_{Bf}}^{C_{Bo}}\frac{dC_B}{(-r_A'''')}$$

But $V_l/V_r = f_l$

$$\therefore \quad \boxed{t = \frac{f_l}{b}\int_{C_{Bf}}^{C_{Bo}}\frac{dC_B}{(-r_A'''')}} \qquad \text{... (3.55)}$$

The integral on the RHS of Equation (3.55) can be evaluated graphically by plotting $1/(-r_A'''')$ v/s C_B and measuring the area under the curve between $C_B = C_{Bo}$ and $C_B = C_{Bf}$.

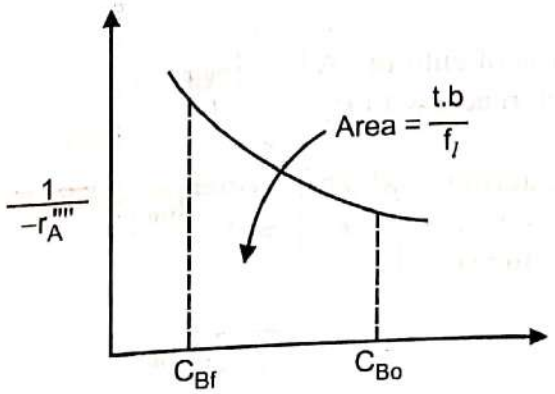


Fig. 3.9

If $p_{Aout} = 0$ at all times (all A entering reacts at all times), then the time required for processing is minimum and is given by

$$t_{min} = \frac{\frac{1}{b} V_l (C_{Bo} - C_{Bf})}{F_g \left(\frac{p_{A\,in}}{\pi - p_{A\,in}} \right)}$$

$$= \frac{\frac{1}{b} \text{(Quantity of B reacted away in the vessel)}}{\text{(Quantity of A entering the vessel in unit time)}} \quad \ldots (3.5$$

Let us obtain Equation (3.56).

We have,

$$F_g \left(\frac{p_{A\,in}}{\pi - p_{A\,in}} - \frac{p_{A\,out}}{\pi - p_{A\,out}} \right) = -\frac{V_l}{b} \left(\frac{dC_B}{dt} \right)$$

$$F_g \left(\frac{p_{A\,in}}{\pi - p_{A\,in}} - \frac{p_{A\,out}}{\pi - p_{A\,out}} \right) = -\frac{V_l}{b} \left(\frac{C_{Bf} - C_{Bo}}{t - 0} \right) = \frac{V_l}{b} \left(\frac{C_{Bo} - C_{Bf}}{t} \right)$$

With $p_{A\,out} = 0$ and $t = t_{min}$, the above equation becomes

$$F_g \left(\frac{p_{A\,in}}{\pi - p_{A\,in}} \right) = \frac{V_l}{b} \frac{(C_{Bo} - C_{Bf})}{t_{min}}$$

$\therefore$

$$t_{min} = \frac{\frac{1}{b} V_l (C_{Bo} - C_{Bf})}{F_g \left(\frac{p_{A\,in}}{\pi - p_{A\,in}} \right)}$$

The efficiency of utilisation of A is obtained by combining t and t_{min}.

$$\left[\begin{matrix} \text{Fraction of entering A} \\ \text{which reacts with B} \end{matrix} \right] = \frac{t_{min}}{t} \quad \ldots (3.2$$

$$\left[\begin{matrix} \text{Fraction of entering A which} \\ \text{remains unreacted and passes} \\ \text{through the vessel} \end{matrix} \right] = 1 - \frac{t_{min}}{t}$$

$$= \frac{t - t_{min}}{t}$$

SOLVED EXAMPLES

Example 3.1 : *The concentration of an undesirable impurity A in air (at π = 1 bar = 10^5 Pa) is to be reduced from 0.1% (or 100 Pa) to 0.02% (or 20 Pa) by absorbing the impurity A in pure water in a packed tower. Estimate the height of tower needed for countercurrent operation.*

Data :

For the packing used :

$$k_{Ag}\,a = 0.32 \text{ mol/(h.m}^3\text{.Pa)}$$

$$k_{Al}\,a = 0.10 \text{ h}^{-1}$$

The solubility of A in water is given by Henry's law constant : $H_A = 12.5 \text{ (Pa.m}^3\text{)/mol}$

The flow rates of gas and liquid per m^2 of tower cross-section are :

$$F_g/A_{cs} = 1 \times 10^5 \text{ mol/(h.m}^2\text{)}$$

$$F_l/A_{ca} = 7 \times 10^5 \text{ mol/(h.m}^2\text{)}$$

The molarity of liquid under all conditions is :

$$C_T = 56000 \text{ mol/m}^3$$

Solution :

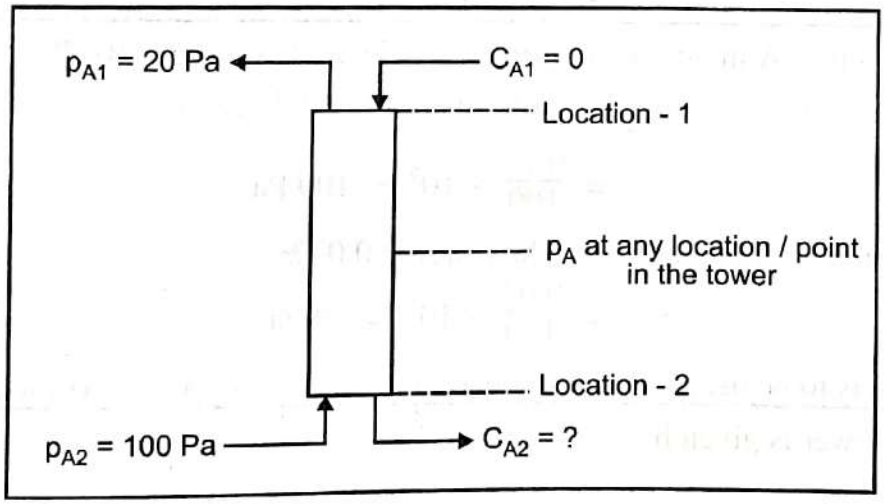


Fig. Ex 3.1 : Countercurrent tower for straight mass transfer

As impurity A is to be removed by using pure water, it is a case of :

Straight mass transfer (absorption) of A

A (gas) $\rightarrow$ A (liquid)

Also we are provided with dilute systems.

We will write first the material balance for dilute solutions to find C_{A2} in the exit water stream and then we will determine the height of tower.

For any point in the tower, p_A and C_A are related by

$$p_A - p_{A1} = \frac{(F_l/A_{cs})\,\pi}{(F_g/A_{cs})\,C_T}\,(C_A - C_{A1}) \qquad \ldots (1)$$

$p_{A1} = 20$ Pa, $C_{A1} = 0$, $\pi = 10^5$ Pa

$C_T = 56000$ mol/m^3

$F_l/A_{cs} = 7 \times 10^5$ mol/(h.m^2)

$F_g/A_{cs} = 1 \times 10^5$ mol/(h.m^2)

Substituting these values in Equation (1) gives

$$p_A - 20 = \frac{7 \times 10^5 \times 10^5}{1 \times 10^5 \times 56000} (C_A - 0)$$

$$p_A - 20 = 12.5\, C_A$$

or $$C_A = 0.08\, p_A - 1.6 \quad \ldots (1)$$

In the inlet air stream to the tower, we have :

$$p_{A2} = 100 \text{ Pa}$$

The concentration of A in the water leaving the tower is

$$C_{A2} = 0.08\, p_{A2} - 1.6$$

$$= 0.08 \times 100 - 1.6$$

$$= 6.4 \text{ mol/m}^3$$

The concentration of A in air entering the tower is 0.10% and $\pi = 10^5$ Pa

$\therefore$ p_{A2} = mole fraction × total pressure

$$= \frac{0.10}{100} \times 10^5 = 100 \text{ Pa}$$

The concentration of A in air leaving the tower is 0.02%

$\therefore$ $$p_{A1} = \frac{0.02}{100} \times 10^5 = 20 \text{ Pa}$$

This procedure is to be used when the concentration of A is given in volume or mole %.

The height of tower is given by

$$V_r = h\, A_{cs} = \frac{F_g}{\pi\, K_{Ag}\, a} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{p_A - p_A^*}$$

$$h = \frac{F_g/A_{cs}}{\pi\, K_{Ag}\, a} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{p_A - p_A^*}$$

Now we will evaluate $p_A - p_A^*$ (overall gas phase driving force)

Select a few values of p_A in the tower, usually p_{A1}, p_{A2} and one or two intermediate values are sufficient.

Evaluate : C_A corresponding to $p_A \rightarrow p_A^* \rightarrow \left(p_A - p_A^*\right)$.

C_A – concentration of A in the liquid corresponding to p_A.

p_A^* – equilibrium partial pressure of A corresponding to the liquid phase concentration C_A.

Evaluate C_A from p_A by using the relation :

$$C_A = 0.08\, p_A - 1.6$$

For $p_A = 200$ Pa, $C_A = 0$ mol/m^3

For $p_A = 100$ Pa, $C_A = 6.4$ mol/m^3

For $p_A = 60$ Pa, $C_A = 3.2$ mol/m^3

For $p_A = 80$ Pa, $C_A = 4.8$ mol/m^3

Calculate p_A^* using Henry's law :

$p_A^* = H_A C_A = 12.5\, C_A$ and tabulate.

p_A	C_A	$p_A^* = H_A C_A$	$\Delta p_A = p_A - p_A^*$, Pa
20	0	0	20
60	3.2	40	20
80	4.8	60	20
100	6.4	80	20

Here we have constant Δp_A and hence a graphical procedure is not required.

Now evaluate $K_{Ag}a$

$$\frac{1}{K_{Ag}a} = \frac{1}{k_{Ag}a} + \frac{H_A}{k_{Al}a}$$

$$= \frac{1}{0.32} + \frac{12.5}{0.10}$$

$$= 3.125 + 125 = 128.125$$

$\therefore$ Gas film resistance = (3.125/128.125) 100 = 2.5%

$\therefore$ Liquid film resistance = (125/128.125) 100 = 97.5%

$$1/K_{Ag}a = 128.125$$

$\therefore$ $$K_{Ag}a = \frac{1}{128.125} = 0.0078 \text{ mol/(h.m}^3\text{.Pa)}$$

$$h = \frac{F_g/A_{cs}}{\pi K_{Ag} a} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{p_A - p_A^*}, \quad p_{A1} = 20 \text{ Pa}, p_{A2} = 100 \text{ Pa}$$

$$= \frac{F_g/A_{cs}}{\pi K_{Ag} a \, \Delta p_A} \int_{20}^{100} dp_A$$

$$= \frac{1 \times 10^5}{10^5 \times 0.0078 \times 20} \int_{20}^{100} dp_A$$

$$= \frac{1 \times 10^5}{10^5 \times 0.0078 \times 20} (100 - 20)$$

$$= 512.82 \simeq 513 \text{ m}$$

Height of tower needed = 513 m ... Ans.

Comment : Tower is unacceptably high. Here most of the resistance (97.5%) resides in the liquid film so we can with little error consider this to be a liquid-film controlling process.

Example 3.2 : *We plan to remove 90% of the reactant A present in a gas stream by absorption in pure water in a packed tower.*

Find the volume of tower required for countercurrent operation.

Data :

For the gas stream :

$F_g = 90000 \text{ mol/h at } \pi = 10^5 \text{ Pa}$

$p_{A\,in} = 1000 \text{ Pa}, \quad p_{A\,out} = 100 \text{ Pa}$

For the liquid stream :

$F_l = 900000 \text{ mol/h}$

For the packing used :

$k_{Ag} a = 0.36 \text{ mol/(h.m}^3\text{.Pa)}$

$k_{Al} a = 72 \text{ h}^{-1}$

The molar density of liquid under all conditions is

$C_T = 55556 \text{ mol/m}^3$

and $H_A = 18 \text{ (Pa·m}^3\text{)/mol}, \quad k = 0 \quad \text{m}^3/\text{(mol.h)}$

Solution : Here we are concerned with – straight mass transfer/physical absorption of A and dilute system.

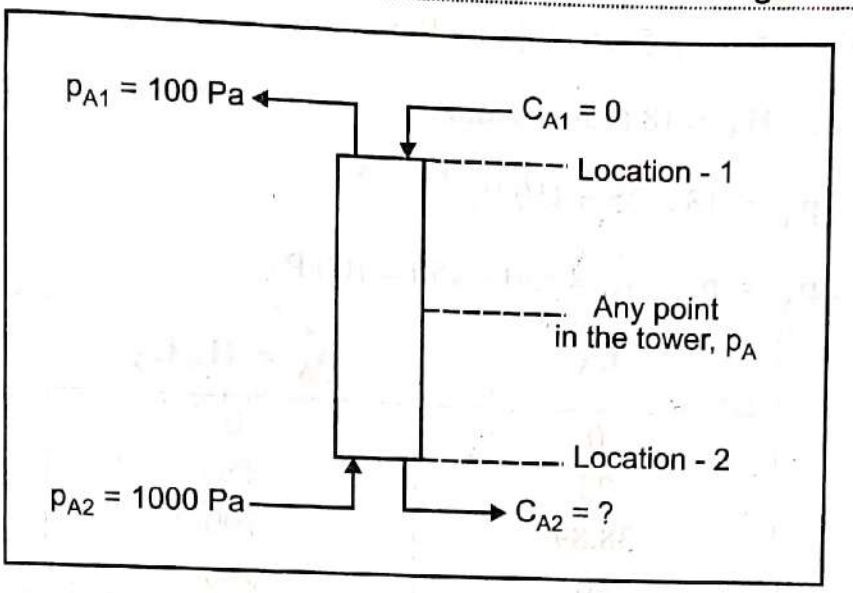


Fig. Ex 3.2

For a dilute solution for any point in the tower

p_A and C_A are related by the relation

$$p_A - p_{A1} = \frac{F_l \pi}{F_g C_T} (C_A - C_{A1})$$

$p_{A1} = 100$ Pa, $C_{A1} = 0$, $\pi = 10^5$ Pa

$C_T = 55556$ mol/m^3

$F_l = 900000$ mol/h, $F_g = 90000$ mol/h

$$p_A - 100 = \frac{900000 \times 10^5}{90000 \times 55556} (C_A - 0)$$

$$p_A - 100 = 18\, C_A$$

or $$C_A = 0.05556\, p_A - 5.5556$$

The concentration of A in water leaving the tower is given by

$$C_{A2} = 0.05556\, p_{A2} - 5.5556$$

For $p_{A2} = 1000$ Pa,

$$C_{A2} = 0.05556\,(1000) - 5.5556$$

$$= 50 \text{ mol/m}^3$$

Now select a few values of p_A in addition to p_{A1} and p_{A2} in the tower and determine $\rightarrow C_A \rightarrow p_A^* \rightarrow (p_A - p_A^*)$ and tabulate.

Take : $p_A = 550$ Pa to $p_A = 800$ Pa

For $p_A = 550$ Pa

$C_A = 0.05556\, p_A - 5.5556$

$C_A = 0.05556\,(550) - 5.5556 = 25\ mol/m^3$

$p_A^* = H_A C_A$, $H_A = 18\ (Pa \cdot m^3)/mol$

$$p_A^* = 18 \times 25 = 450\ P_a$$

$$\therefore \quad \Delta p_A = p_A - p_A^* = 550 - 450 = 100\ Pa$$

p_A	C_A	$p_A^* = H_A C_A$	$\Delta p_A = p_A - p_A^*$
100	0	0	100
550	25	450	100
800	38.89	700	100
1000	50	900	100

Here we have constant Δp_A and hence a graphical procedure is not required (as Δp_A is constant it can be taken out of the integral sign).

Next evaluate $K_{Ag}\,a$:

$$\frac{1}{K_{Ag}\,a} = \frac{1}{k_{Ag}\,a} + \frac{H_A}{k_{Al}\,a}$$

$k_{Ag}\,a = 0.36\ mol/(h.m^3.Pa)$

$k_{Al}\,a = 72\ h^{-1}$

$$= \frac{1}{0.36} + \frac{18}{72}$$

$$= 2.78 + 0.25$$

$$= 3.03$$

$$\therefore \quad K_{Ag}\,a = 1/3.03 = 0.33\ mol/(h.m^3.Pa)$$

The volume of tower is given by

$$V_r = \frac{F_g}{\pi\, K_{Ag}\,a} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{p_A - p_A^*}, \qquad \Delta p_A = p_A - p_A^*$$

$$= \frac{F_g}{\pi\, K_{Ag}\,a\, \Delta p_A} \int_{100}^{1000} dp_A$$

$$= \frac{90000}{10^5 \times 0.33 \times 100}\,(1000 - 100)$$

$$= \frac{90000}{10^5 \times 0.33 \times 100}\,(1000 - 100)$$

$$= 24.54\ m^3$$

Volume of tower required = 24.54 m³ ... Ans.

Example 3.3 : *Repeat Example 3.2 for H_A = 1.8 (Pa.m³)/mol.*

Solution : From the solution of previous example.

For any point in the tower, p_A and C_A are related by

$$C_A = 0.05556\, p_A - 5.5556$$

and $C_{A2} = 50$ mol/m³

Now select few values of p_A in addition to p_{A1} and p_{A2} and determine $\rightarrow C_A \rightarrow p_A^* \rightarrow (p_A - p_A^*)$.

Now $H_A = 1.8$ (Pa.m³)/mol

For $p_A = 550$ Pa

$$C_A = 0.05556\, p_A - 5.5556$$

$$C_A = 0.05556\,(550) - 5.5556 = 25 \text{ mol/m}^3$$

$$p_A^* = H_A C_A = 1.8 \times 25 = 45 \text{ Pa}$$

$$\Delta p_A = p_A - p_A^* = 550 - 45 = 505 \text{ Pa}$$

For $p_A = 275$ Pa

$$C_A = 0.05556\,(275) - 5.5556 = 9.7234 \text{ mol/m}^3$$

$$p_A^* = 1.8 \times 9.7234$$

$$= 17.5 \text{ Pa}$$

For $p_A = 800$ Pa

$$C_A = 0.05556\,(800) - 5.5556$$

$$= 38.89 \text{ mol/m}^3$$

$$p_A^* = 1.8 \times 38.89 = 70 \text{ Pa}$$

For $p_A = 1000$ Pa, we get $C_A = 50$ mol/m³ and $p_A^* = 90$ Pa.

p_A	C_A	$p_A^* = H_A C_A$	$\Delta p_A = p_A - p_A^*$
100	0	0	100
275	9.7234	17.5	257.5
550	25	45	505
800	38.89	70	730
1000	50	90	910

Here Δp_A is not constant.

The volume of tower is given by

$$V_r = \frac{F_g}{\pi} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{-r_A'''}$$

$$-r_A''' = K_{Ag}\, a \left(p_A - p_A^*\right)$$

From Example 3.2 : $K_{Ag}\, a = 0.33$ mol/(h.m^3.Pa)

$$\frac{\pi V_r}{F_g} = \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{-r_A'''}$$

Evaluate the integral on RHS graphically.

Plot $\frac{1}{(-r_A''')}$ v/s p_A. The area under the curve between p_{A1} and p_{A2} gives the value of integral and is equal to $\pi V_r/F_g$.

$\frac{\pi V_r}{F_g}$ = area under curve × scale x-axis × scale y-axis.

Now evaluate $-r_A'''$ for each p_A and tabulate.

For $p_A = 100$ Pa, $p_A^* = 0$ and $\Delta p_A = 100$ Pa

$$-r_A''' = K_{Ag}\, a \left(p_A - p_A^*\right) = K_{Ag}\, a\, \Delta p_A$$

$$= 0.33 \times 100 = 33 \text{ mol/(h.m}^3\text{)}$$

$$1/(-r_A''') = \frac{1}{33} = 0.0303 \text{ (h.m}^3\text{)/mol}$$

p_A	p_A^*	$\Delta p_A = p_A - p_A^*$	$-r_A'''$	$1/(-r_A''')$
100	0	100	33	0.03
275	17.5	257.5	84.975	0.012
550	45	505	166.65	0.006
1000	90	910	300.3	0.0033

Plot $\frac{1}{(-r_A''')}$ v/s p_A

We have : $p_{A1} = 100$ Pa and $p_{A2} = 1000$ Pa

Measure the area under the curve between $p_A = p_{A1} = 100$ Pa and $p_A = p_{A2} = 1000$ Pa

Refer Fig. Ex 3.3 :

Area under curve $= 2950 \text{ mm}^2 = 29.5 \text{ cm}^2$

$$\frac{\pi V_r}{F_g} = \text{Area} \times \text{scale of y-axis} \times \text{scale of x-axis}$$

$$= 29.5 \times \frac{0.005}{2} \times \frac{100}{1} = 7.375$$

$$V_r = \frac{7.375\, F_g}{\pi} = \frac{7.375 \times 90000}{10^5} = 6.64 \text{ m}^3$$

Volume of tower = 6.64 m³ **... Ans.**

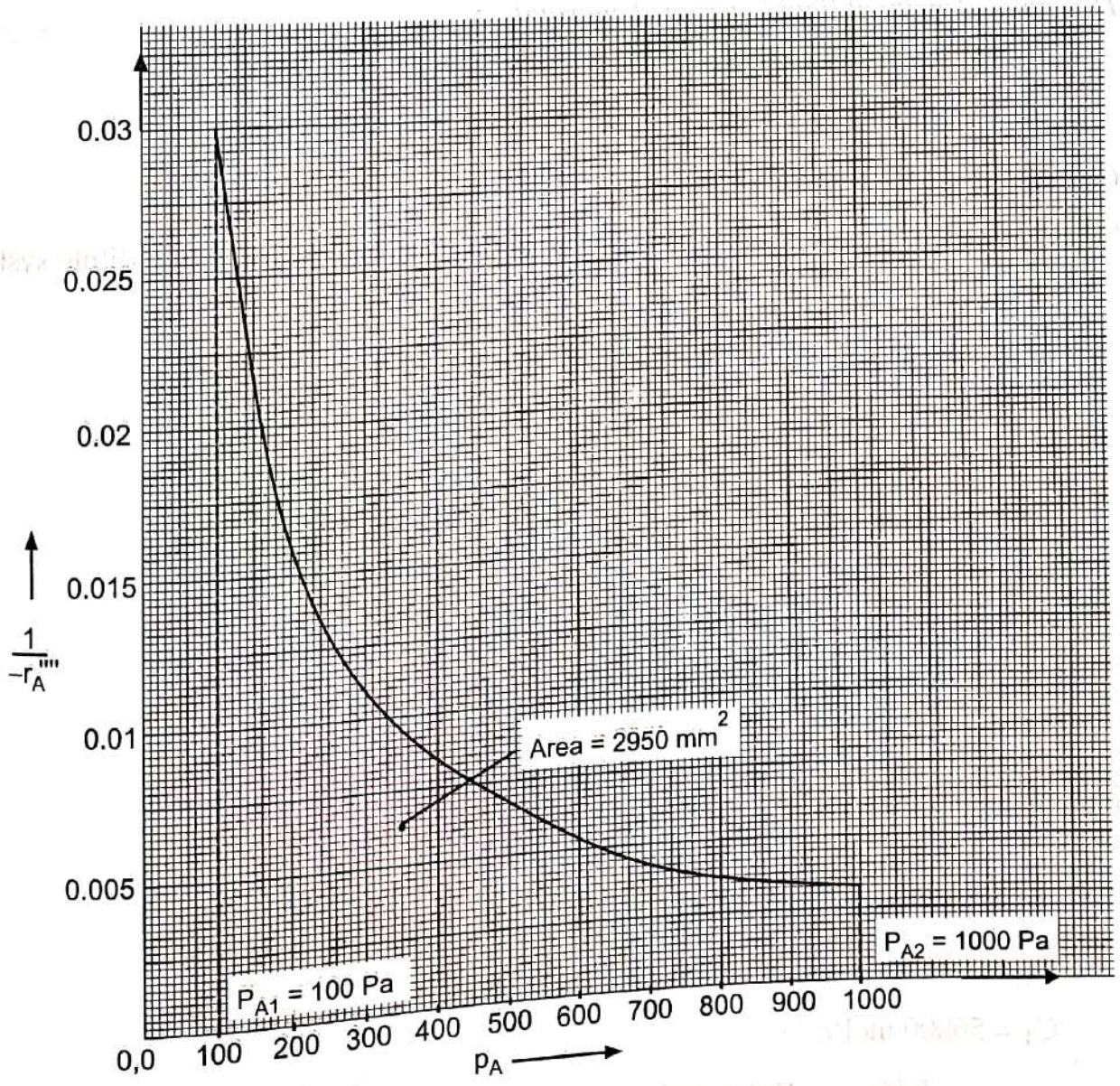


Fig. Ex 3.3

Example 3.4 : *An undesired impurity is to be removed from air by absorption in pure water in a counter current tower.*

Impurity in the inlet air = 0.5% (500 Pa)

Impurity in the outlet air = 0.1% (100 Pa)

Flow rates of gas and liquid are :

$$F_g/A_{cs} = 1 \times 10^5 \text{ mol/(h.m}^2\text{)}$$

$$F_l/A_{cs} = 10.6 \times 10^5 \text{ mol/(h.m}^2\text{)}$$

For the packing :

$$k_{Ag}\, a = 0.60 \text{ mol/(h.m}^3\text{.Pa)}$$

$$k_{Al}\, a = 0.5 \text{ h}^{-1}$$

The molar density of liquid, assumed constant, is

$$C_T = 56000 \text{ mol/m}^3$$

and $H_A = 15 \text{ (Pa.m}^3\text{)/mol}$

Calculate the height of tower required.

Solution : Here we are concerned with – straight mass transfer of A and dilute system.

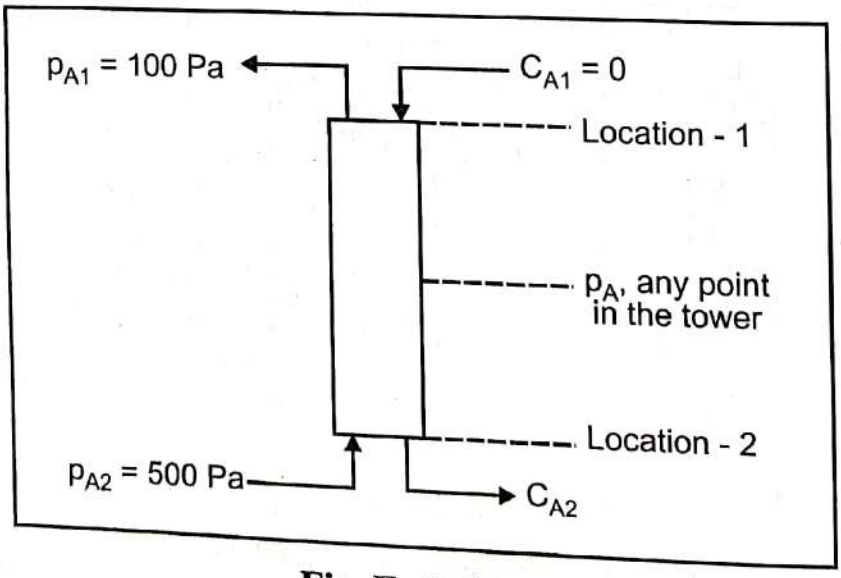


Fig. Ex 3.4 (a)

For any point in the tower p_A and C_A are related by

$$(p_A - p_{A1}) = \frac{(F_l / A_{cs})\,\pi}{(F_g / A_{cs})\,C_T}\,(C_A - C_{A1})$$

$p_{A1} = 100$ Pa, $C_{A1} = 0$, $\pi = 10^5$ Pa

$C_T = 56000$ mol/m^3

$$F_l/A_{cs} = 10.6 \times 10^5 \text{ mol/(h.m}^2\text{)}$$

$$F_g/A_{cs} = 1 \times 10^5 \text{ mol/(h.m}^2\text{)}$$

$$(p_A - 100) = \frac{10.6 \times 10^5 \times 10^5}{1 \times 10^5 \times 56000} (C_A - 0)$$

$$p_A - 100 = 18.928\, C_A$$

or $$C_A = 0.05283\, p_A - 5.283 \quad \ldots (1)$$

The concentration of the impurity (A) in water leaving the tower [C_{A2}] is obtained by using the above relation.

$$C_{A2} = 0.05283\, p_{A2} - 5.283$$

We have $p_{A2} = 500$ Pa

$\therefore$ $C_{A2} = 0.05283\,(500) - 5.283 = 21.132$ mol/m^3

$H_A = 15$ (Pa.m^3)/mol

Now select few values of p_A, determine the corresponding concentration of A (C_A) in the liquid using Equation (1), calculate the equilibrium partial pressure of A, p_A^*, corresponding to C_A and then calculate $\Delta p_A = p_A - p_A^*$, overall gas phase driving force and tabulate.

Take $p_A = 300$ Pa

$$C_A = 0.05283\, p_A - 5.283$$

$$C_A = 0.05283\,(300) - 5.283 = 10.566 \text{ mol/m}^3$$

$$p_A^* = H_A C_A$$

$$= 15 \times 10.566 = 158.49 \simeq 158.5 \text{ Pa}$$

$$\Delta p_A = p_A - p_A^* = 300 - 158.5 = 141.5 \text{ Pa}$$

p_A	C_A	$p_A^* = H_A C_A$	$p_A - p_A^* = \Delta p_A$
100	0	0	100
200	5.283	79.24	120.76
300	10.566	158.5	141.5
400	15.849	237.7	162.3
500	21.132	317	183

Here Δp_A is not constant and thus we have to adopt a graphical procedure to determine the height of tower.

We have :

$$V_r = h\, A_{cs} = \frac{F_g}{\pi} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{-r_A}$$

$$\frac{\pi h}{(F_g/A_{cs})} = \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{-r_A}$$

We have :

$$k_{Ag}\,a = 0.60 \text{ mol/(h.m}^3\text{.Pa)}$$

$$k_{Al}\,a = 0.5 \text{ h}^{-1}$$

$$H_A = 15 \text{ (Pa.m}^3\text{)/mol}$$

We know : $$\frac{1}{K_{Ag}\,a} = \frac{1}{k_{Ag}\,a} + \frac{H_A}{k_{Al}\,a}$$

$$= \frac{1}{0.6} + \frac{15}{0.5}$$

$$= 1.667 + 30 = 31.667$$

$$\therefore \quad K_{Ag}\,a = \frac{1}{31.667} = 0.0316 \text{ mol/(h.m}^3\text{.Pa)}$$

Gas film resistance = (1.667/31.667) 100 = 5.26%

Liquid film resistance = (30/31.667) 100 = 94.74%

So it is a liquid film controlling process (as the major resistance is offered by the liquid film).

$$-r_A'''' = K_{Ag}\,a\left(p_A - p_A^*\right)$$

Now evaluate $-r_A''''$ for each value of p_A and tabulate.

p_A	$p_A - p_A^*$	$-r_A''''$	$1/\left(-r_A''''\right)$
100	100	3.16	0.316
200	120.76	3.82	0.262
300	141.5	4.47	0.224
400	162.3	5.13	0.195
500	183	5.78	0.173

$$\frac{\pi h}{(F_g/A_{cs})} = \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{-r_A''''}$$

$$= \int_{100}^{500} \frac{dp_A}{-r_A''''}$$

= Area under the curve × Scale of y-axis × Scale of x-axis

Plot $\frac{1}{\left(-r_A'''\right)}$ v/s p_A and measure area under curve between $p_{A1} = 100$ Pa and $p_{A2} = 500$ Pa.

Refer Fig. Ex 3.4 (b) :

Area under the curve = 1829 mm^2 = 18.29 cm^2

$$\frac{\pi h}{F_g/A_{cs}} = \text{Area} \times \text{Scale of y-axis} \times \text{Scale of x-axis}$$

$$\frac{\pi h}{(F_g/A_{cs})} = 18.29 \times \frac{0.1}{2} \times \frac{100}{1}$$

$$= 91.45$$

$$h = \frac{91.45\,(F_g/A_{cs})}{\pi}$$

$$= \frac{91.45 \times 1 \times 10^5}{10^5}$$

$$= 91.45 \text{ m}$$

Height of tower = 91.45 m **... Ans.**

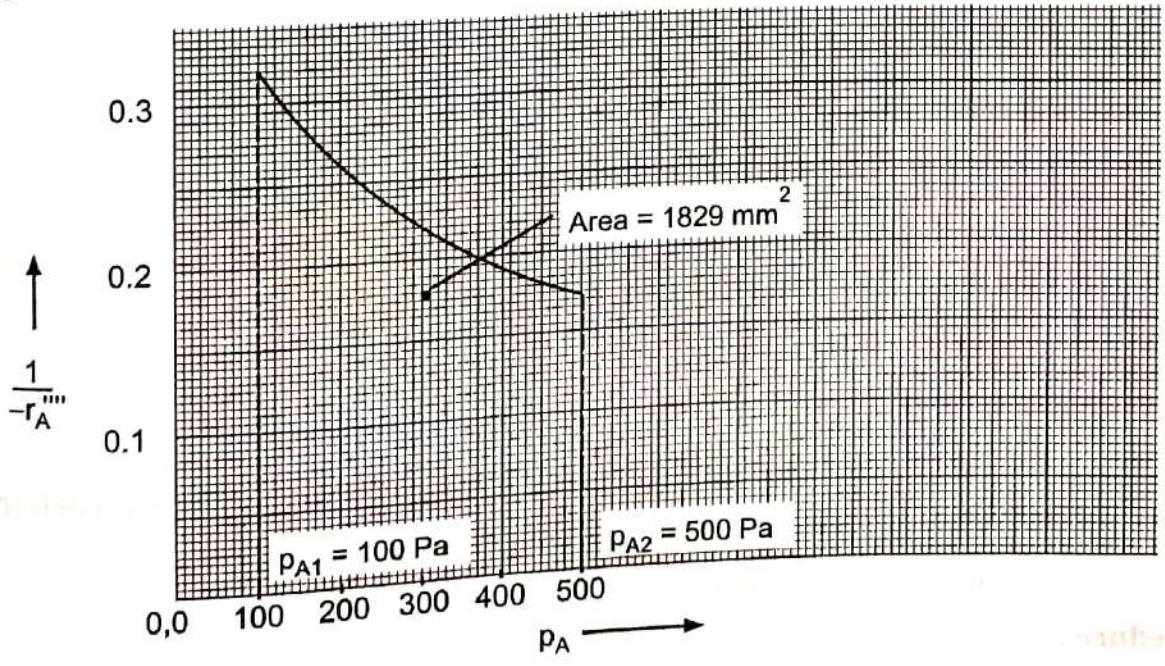


Fig. Ex 3.4 (b)

Example 3.5 : *An undesired impurity (A) is to be removed from air by absorption in water which contains reactant B using a packed tower. Chemicals A and B react in the liquid as follows :*

$$A\,(g \rightarrow l) + B(l) \rightarrow R(l)$$

The reaction is extremely rapid so $k = \infty$.

A in air entering the tower = 0.5% (500 Pa)

A in air leaving the tower = 0.1% (100 Pa)

The air flows through the tower at $\pi = 10^5$ *Pa*

For the packing used :

$$k_{Ag}\,a = 0.20\ mol/(h.m^3.Pa)$$

$$k_{Al}\,a = 0.5\ h^{-1}$$

The solubility of A in water is given by Henry's law constant :

$$H_A = 15\ (Pa.m^3)/mol$$

The molar density of liquid under all conditions is

$$C_T = 56000\ mol/m^3$$

The flow rate per m^2 *of tower cross-section are :*

$$F_g/A_{cs} = 1 \times 10^5\ mol/(h.m^2)$$

$$F_l/A_{cs} = 8.7 \times 10^5\ mol/(h.m^2)$$

The concentration of B in water entering the tower = $C_B = 400\ mol/m^3$

Assume that the diffusivities of A and B in water are the same.

Calculate the height of tower required for countercurrent operation.

Solution :

Here we are concerned with mass transfer plus chemical reaction :

$$A\,(g \rightarrow l) + B(l) \rightarrow R(l)\ ,\ k = \infty$$

Here : $b = 1$

As $k = \infty$, it is an instantaneous reaction (extremely rapid reaction)

Given : Diffusitivities of A and B in water are the same and thus

$$k_{Al}\,a = k_{Bl}\,a = 0.5\ h^{-1}$$

Procedure :

1. Find C_{B2} in the exit stream with the help of material balance.
2. Find the form of rate equation (to be used).
3. Find the tower height.

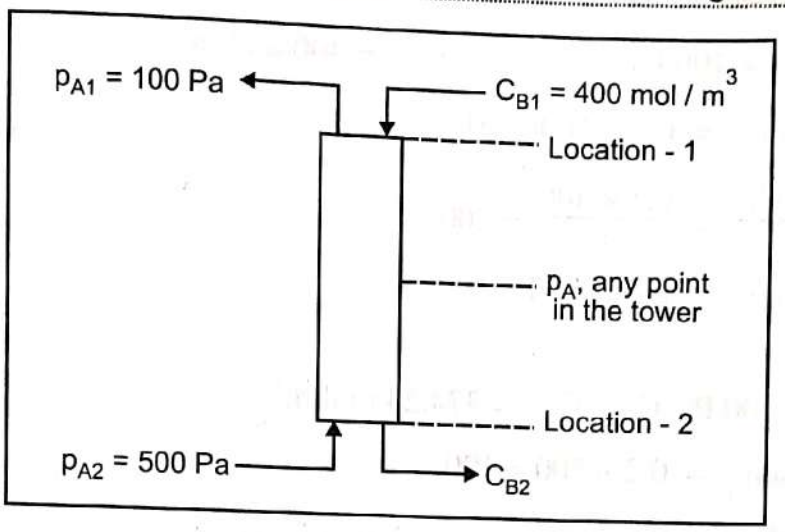


Fig. Ex 3.5

For dilute solutions with extremely rapid reaction at any point in the tower, p_A and C_B are related by the relation

$$(p_A - p_{A1}) = \frac{(F_l/A_{cs})}{(F_g/A_{cs})\, b\, C_T} (C_{B1} - C_B) \qquad \ldots (1)$$

We have : $F_l/A_{cs} = 8.7 \times 10^5$ mol/(h.m^2), $F_g/A_{cs} = 1 \times 10^5$ mol/(h.m^2)

$b = 1$, $C_T = 56000$ mol/m^3

$\pi = 10^5$ Pa, $p_{A1} = 100$ Pa, $C_{B1} = 400$ mol/m^3

Substituting the values of the parameters in Equation (1) gives

$$(p_A - 100) = \frac{8.7 \times 10^5 \times 10^5}{1 \times 10^5 \times 1 \times 56000} (400 - C_B)$$

$$p_A - 100 = 6214.4 - 15.536\ C_B$$

$$p_A = 6314.4 - 15.536\ C_B$$

or $$C_B = 406.444 - 0.0644\ p_A$$

Let us calculate C_{B2} using the above relation.

At the bottom of the tower : $p_A = p_{A2} = 500$ Pa

$\therefore$ $$C_{B2} = 406.44 - 0.0544\ p_{A2}$$

$$C_{B2} = 406.44 - 0.0544\ (500)$$

$$= 374.24 \text{ mol/m}^3$$

To find the appropriate form of rate equation, let us have check at both the ends of the tower : whether $k_{Ag}\, a\, p_A$ is greater than or less than $\frac{k_{Bl}\, a\, C_B}{b}$ (to find whether it is an instantaneous reaction with low C_B or high C_B).

We have : $k_{Ag}\, a = 0.2$ mol/(h.m^3.Pa)

$k_{Bl}a = k_{Al}a = 0.5$ h^{-1}

At top : $p_A = p_{A1} = 100$ Pa, $C_B = C_{B1} = 400$ mol/m^3

$$k_{Ag} a\, p_A = 0.2 \times 100 = 20$$

$$\frac{k_{Bl}\, a\, C_B}{b} = \frac{0.5 \times 400}{1} = 200$$

$$\therefore \quad k_{Ag}\, a\, p_A < k_{Bl}\, a\, C_B/b$$

At bottom :

$p_A = p_{A2} = 500$ Pa, $C_A = C_{A2} = 374.24$ mol/m^3

$$k_{Ag}\, a\, p_A = 0.2 \times 500 = 100$$

$$\frac{k_{Bl}\, a\, C_B}{b} = 0.5 \times 374.24 = 187.12$$

$$k_{Ag}\, a\, p_A < k_{Bl}\, a\, C_B/b$$

As we have $k_{Ag}\, a\, p_A < k_{Bl}\, a\, C_B / b$ at both ends of tower we have :

Instantaneous reaction with high C_B. Reaction zone is at the interface. The resistance of the controls and we have a pseudo first order reaction and thus the rate given by

$$-r_A''' = k_{Ag}\, a\, p_A$$

For instantaneous (extremely rapid) reaction with high C_B the rate form is

$$-r_A'' = k_{Ag}\, p_A$$

or $$-r_A''' = \left(-r_A''\right) a = k_{Ag}\, a\, p_A$$

We have, $k_{Ag}\, a = 0.2$ mol/(h.m^3.Pa)

$$\therefore \quad -r_A''' = 0.20\, p_A$$

The height of tower is given by

$$h = \frac{F_g/A_{cs}}{\pi} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{-r_A'''}$$

$$h = \frac{F_g/A_{cs}}{\pi} \int_{100}^{500} \frac{dp_A}{0.20\, p_A}$$

$$= \frac{1 \times 10^5}{10^5 \times 0.20} \left[ln\, p_A\right]_{100}^{500}$$

$$= 5[ln\, 500 - ln\, 100] = 5\, ln\, (500/100)$$

$$= 8.05 \text{ m}$$

The height of tower required for countercurrent operation is 8.05 m ... Ans.

Example 3.6 : *We plan to remove about 80% of the A present in a gas stream by absorption in water containing reactant B in a packed tower. Material B reacts with A as per the following reaction :* $A(g \rightarrow l) + B(l) \rightarrow R(l)$

The reaction is extremely rapid. Therefore, $k = \infty$.

Determine the height of tower for countercurrent operation using the following data.

Data : The gas and liquid flow rates are :

$$F_g/A_{cs} = 1 \times 10^5 \text{ mol/(h.m}^2\text{) at } \pi = 10^5 \text{ Pa}$$

$$F_l/A_{cs} = 7 \times 10^5 \text{ mol/(h.m}^2\text{)}$$

$$H_A = 12.5 \text{ (Pa.m}^3\text{)/mol}$$

C_T (molar density of liquid) = 56000 mol/m^3

$$p_{A\,in} = 100 \text{ Pa}$$

For the packed bed :

$$k_{Ag}\,a = 0.32 \text{ mol/(h.m}^3\text{.Pa)}$$

$$k_{Al}\,a = 0.10 \text{ h}^{-1}$$

$C_{B\,in}$ = concentration of reactant B in water entering the tower = 800 mol/m^3.

Assume that the diffusivities of A and B in water are the same.

Solution : $A(g \rightarrow l) + B(l) \rightarrow R(l)$, $k = \infty$

It is an extremely rapid reaction (an instantaneous reaction)

Here $b = 1$ (stoichiometric coefficient of B)

$$p_{A\,in} = p_{A2} = 100 \text{ Pa}$$

80% of A is removed from air

$\therefore$

$$p_{A\,out} = p_{A1} = 100\,(1 - 0.80) = 20 \text{ Pa}$$

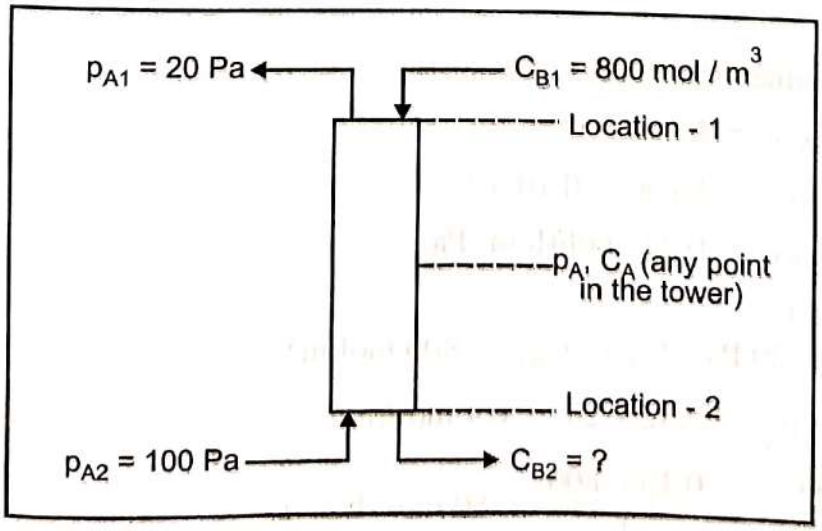


Fig. Ex 3.6

First find C_{B2} with the help of material balance.

C_{B2} is the concentration of B in the exit stream from the tower.

For dilute solutions, the simplified material balance yields

$$(p_A - p_{A1}) = \frac{(F_l/A_{cs})\,\pi}{(F_g/A_{cs})\,b\,C_T}(C_{B1} - C_B) \quad \ldots (1)$$

$p_{A1} = 20$ Pa, $\pi = 10^5$ Pa, $b = 1$, $C_T = 56000$ mol/m^3, $C_{B1} = 800$ mol/m^3

$$F_l/A_{cs} = 7 \times 10^5 \text{ mol/(h.m}^2\text{)}$$

$$F_g/A_{cs} = 1 \times 10^5 \text{ mol/(h.m}^2\text{)}$$

Substituting the above values, Equation (1) becomes

$$p_A - 20 = \frac{7 \times 10^5 \times 10^5}{1 \times 10^5 \times 1 \times 56000}(800 - C_B)$$

$$p_A = 10020 - 12.5\,C_B$$

or $$C_B = 801.6 - 0.08\,p_A$$

The above equation relates C_B to p_A at any point in the tower.

At the bottom of the tower :

$$p_A = p_{A2} = 100 \text{ Pa}$$

We have : $$C_{B2} = 801.6 - 0.08\,p_{A2}$$

$\therefore$ $$C_{B2} = 801.6 - 0.08 \times 100 = 793.6 \text{ mol/m}^3$$

It is given that the reaction is extremely rapid (instantaneous).

To find the appropriate form of the rate equation, we have to determine whether it is an instantaneous reaction with low or high C_B and for this estimate $k_{Ag}\,a\,p_A$ and $k_{Bl}\,a\,C_B/b$ at both ends of the tower.

As $\mathscr{D}_{Bl} = \mathscr{D}_{Al}$ (same diffusivities in water)

$$k_{Al}\,a = k_{Bl}\,a$$

We have $$k_{Al}\,a = k_{Bl}\,a = 0.10 \text{ h}^{-1}$$

$$k_{Ag}\,a = 0.32 \text{ mol/(h.m}^3\text{.Pa)}$$

At top of the tower :

$$p_A = p_{A1} = 20 \text{ Pa}, \quad C_B = C_{B1} = 800 \text{ mol/m}^3$$

$$k_{Ag}\,a\,p_A = 0.32 \times 20 = 6.4 \text{ mol/(h.m}^3\text{)}$$

$$\frac{k_{Bl}\,a\,C_B}{b} = \frac{0.10 \times 800}{1} = 80 \text{ mol/(h.m}^3\text{)}$$

$\therefore$ $$k_{Ag}\,a\,p_A < k_{Bl}\,aC_B/b$$

At bottom of the tower :

$$p_A = p_{A2} = 100 \text{ Pa}, \quad C_B = C_{B2} = 793.6 \text{ mol/m}^3$$

$$k_{Ag}\, a\, p_A = 0.32 \times 100 = 32 \text{ mol/(h.m}^3\text{)}$$

$$\frac{k_{Bl}\, a\, C_B}{b} = \frac{0.1 \times 793.16}{1} = 79.32 \text{ mol/(h.m}^3\text{)}$$

At both ends of tower, $k_{Ag}\, a\, p_A < k_{Bl}\, a\, C_B/b$

As $k_{Ag}\, a\, p_A < k_{Bl}\, a\, C_B/b$, we have :

Instantaneous reaction with high C_B. Gas phase resistance controls. Reaction zone is at the interface and so the form of rate equation is

$$-r_A''' = \left(-r_A''\right) a = k_{Ag}\, a\, p_A$$

$$-r_A''' = 0.32\, p_A$$

The height of tower is given by

$$h = \frac{(F_g/A_{cs})}{\pi} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{-r_A'''}$$

$$= \frac{(F_g/A_{cs})}{\pi\,(0.32)} \int_{20}^{100} \frac{dp_A}{p_A}$$

$$= \frac{1 \times 10^5}{1 \times 10^5 \times 0.32} \ln (100/20)$$

$$= 5.03 \text{ m}$$

Height of the tower for countercurrent operation = 5.03 m **... Ans.**

Example 3.7 : *Repeat Example 3.6 for $C_{Bl} = C_{Bin} = 32$ mol/m^3 (Rest of the data remain the same).*

Solution : $A(g \rightarrow l) + B(l) \rightarrow R(l), \quad k = \infty$

$p_{A\,in} = 100$ Pa, $p_{A\,out} = 20$ Pa

$C_{Bl} = 32$ mol/m^3, $C_{B2} = ?$

Refer to the solution of Example 3.6.

For any point in the tower, we have :

$$p_A - p_{Al} = \frac{(F_l/A_{cs})\pi}{(F_g/A_{cs})\, b\, C_T} (C_{Bl} - C_B)$$

$$p_A - 20 = \frac{7 \times 10^5 \times 10^5}{1 \times 10^5 \times 1 \times 56000} (32 - C_B)$$

$$p_A = 420 - 12.5\, C_B$$

or $$C_B = 33.6 - 0.08\, p_A$$

At the bottom :

$$p_{A2} = 100 \text{ Pa}$$

$$C_{B2} = 33.6 - 0.08\, p_{A2}$$

$$C_{B2} = 33.6 - 0.08 \times 100 = 25.6 \text{ mol/m}^3$$

To find the form of rate equation to be used, estimate $k_{Ag}\, a\, p_A$ and $k_{Bl}\, a\, C_B/b$ at both ends of the tower.

At the top :

$p_A = p_{A1} = 20$ Pa, $C_B = C_{B1} = 32$ mol/m^3

$k_{Ag}\, a = 0.32$ mol/(h.m^3.Pa), $k_{Bl}\, a = 0.10$ h^{-1}

$[k_{Bl}\, a = k_{Al}\, a$ as $\mathscr{D}_{Al} = \mathscr{D}_{Bl}]$

$k_{Ag}\, a\, p_A = 0.32 \times 20 = 6.4$ mol/(h.m^3)

$$\frac{k_{Bl}\, a\, C_B}{b} = \frac{0.10 \times 32}{1} = 3.2 \text{ mol/(h.m}^3\text{)}$$

At the bottom :

$p_A = p_{A2} = 100$ Pa, $C_B = C_{B2} = 25.6$ mol/m^3

$k_{Ag}\, a\, p_A = 0.32 \times 100 = 32$ mol/(h.m^3)

$$\frac{k_{Bl}\, a\, C_B}{b} = \frac{0.1 \times 25.6}{1} = 2.56 \text{ mol/(h.m}^3\text{)}$$

At both ends of the tower, $k_{Ag}\, a\, p_A > k_{Bl}\, a\, C_B/b$

As $k_{Ag}\, a\, p_A > k_{Bl}\, a\, C_B/b$, we have :

Instantaneous reaction with low C_B.

Reaction takes place within the liquid film and so the rate equation to be used is

$$-r_A''' = \left(-r_A''\right) a = \frac{\dfrac{\mathscr{D}_{Bl}\, C_B}{\mathscr{D}_{Al}\, b} + \dfrac{p_A}{H_A}}{\dfrac{1}{H_A\, k_{Ag} a} + \dfrac{1}{k_{Al}\, a}}$$

As $b = 1$, $\mathscr{D}_{Al} = \mathscr{D}_{Bl}$, the above equation reduces to

$$-r_A''' = \frac{C_B + \dfrac{p_A}{H_A}}{\dfrac{1}{H_A k_{Ag} a} + \dfrac{1}{k_{Al}\, a}}$$

$$-r_A''' = \frac{H_A\, C_B + p_A}{\dfrac{1}{k_{Ag}\, a} + \dfrac{H_A}{k_{Al}\, a}}$$

$$\frac{1}{k_{Ag}\,a} + \frac{H_A}{k_{Al}\,a} = \frac{1}{0.32} + \frac{12.5}{0.10} = 128.125$$

$$\therefore \quad -r_A''' = \frac{1}{128.125}\,(p_A + H_A\,C_B) = 0.0078\,(p_A + H_A\,C_B)$$

Now evaluate $(p_A + H_A\,C_B)$ at different locations in the tower. Take p_{A1}, p_{A2} and one intermediate value of p_A.

We have :

$$C_B = 33.6 - 0.08\,p_A$$

Take $p_A = p_{A1} = 20$ Pa

$$C_B = 33.6 - 0.08 \times 20 = 32 \text{ mol/m}^3 \quad \ldots (C_B = C_{B1})$$

$$H_A\,C_B = 12.5 \times 32 = 400$$

$$\therefore \quad p_A + H_A\,C_B = 20 + 400 = 420 \text{ Pa}$$

p_A	C_B	$H_A\,C_B$	$p_A + H_A\,C_B$
20	32	400	420
40	30.4	380	420
50	29.6	370	420
100	25.6	320	420

As $p_A + H_A\,C_B$ is constant throughout the tower and as $-r_A''' = 0.0078\,(p_A + H_A\,C_B)$, $-r_A'''$ can be taken out of the integral sign of design equation (as it is being constant).

$$-r_A''' = 0.0078\,(p_A + H_A\,C_B) = 0.0078 \times 420 = 3.276 \text{ mol/(h.m}^3\text{)}$$

The height of tower is given by

$$h = \frac{(F_g/A_{cs})}{\pi} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{-r_A'''}$$

$$= \frac{(F_g/A_{cs})}{\pi\,(-r_A''')} \int_{20}^{100} dp_A$$

$$= \frac{1 \times 10^5}{10^5 \times 3.276}\,(100 - 20)$$

$$= 24.42 \text{ m}$$

Height of the tower for countercurrent operation = 24.42 m ... Ans.

Example 3.8 : *We plan to remove 90% of an undesirable impurity (A) present in a gas stream by absorption in water containing reactive B in a packed tower.*

A and B react in the liquid as follows :

$$A(g \rightarrow l) + B(l) \rightarrow R(l), \quad -r_A = k\, C_A\, C_B$$

Determine the volume of tower needed for countercurrent operation using the following data.

Data :

$F_g = 90000$ mol/h at $\pi = 10^5$ Pa

$p_{A\,in} = 1000$ Pa, $p_{A\,out} = 100$ Pa

$F_l = 900000$ mol/h, $C_{B\,in} = 55.56$ mol/m^3

$k_{Ag}\, a = 0.36$ mol/(h.m^3.Pa)

$k_{Al}\, a = 72\ h^{-1}$, $a = 100$ m^2/m^3

$f_l = \dfrac{V_l}{V} = 0.08$

$\mathscr{D} = 3.6 \times 10^{-6}$ m^2/h

$C_U = 55556$ mol H_2O/m^3 liquid, at all C_B

$H_A = 10^5$ (Pa.m^3)/mol

and $k = 2.6 \times 10^7$ m^3/(mol. h)

Solution : $A\,(g \rightarrow l) + B(l) \rightarrow R(l)$, $b = 1$

It is a second-order reaction with $k = 2.6 \times 10^7$ m^3/(mol.h)

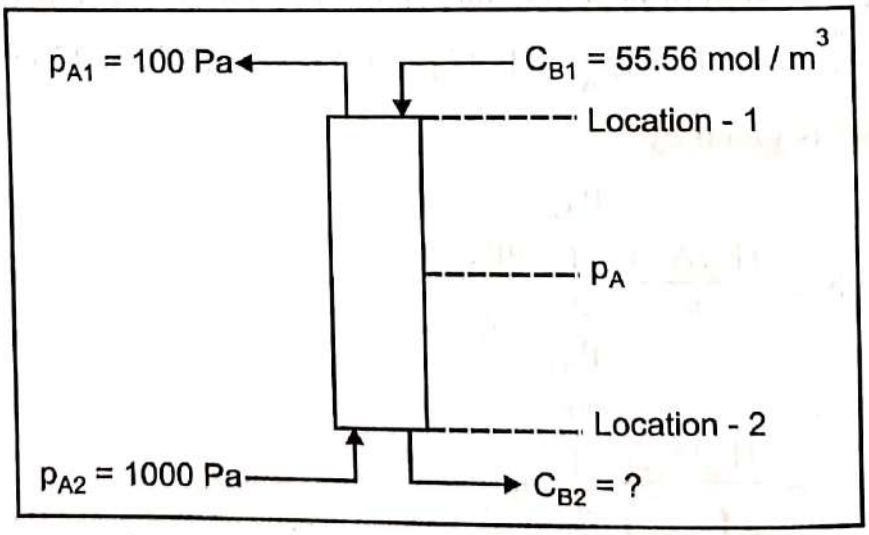


Fig. Ex 3.8 (a)

Procedure : find C_{B2} → find the form of rate equation → find the volume of tower

At any point in the tower, for dilute solutions material balance yields :

$$p_A - p_{A1} = \frac{F_l\, \pi}{F_g\, b\, C_T}\,(C_{B1} - C_B)$$

$F_g = 90000$ mol/h, $F_l = 900000$ mol/h.

$b = 1$, $p_{A1} = 100$ Pa, $C_{B1} = 55.56$ mol/m^3

$C_T \simeq C_U = 55556$ mol/m^3, $\pi = 10^5$ Pa

$$p_A - 100 = \frac{900000 \times 10^5}{90000 \times 1 \times 55556} (55.56 - C_B)$$

$$p_A = 1100 - 18\, C_B$$

or $$C_B = 61.11 - 0.0555\, p_A$$

Now we will find C_{B2} in the exit stream.

$$p_A = p_{A2} = 1000 \text{ Pa}$$

and $$C_{B2} = 61.11 - 0.0555\, p_{A2}$$

$\therefore$ $$C_{B2} = 61.11 - 0.0555\,(1000) = 5.61 \text{ mol/m}^3$$

Let us now find the form of rate equation to be used :

$$k_{Ag}\, a = 0.36 \text{ mol/(h.m}^3\text{.Pa)}$$

$$k_{Al}\, a = 72 \text{ h}^{-1}, \quad a = 100 \text{ m}^2/\text{m}^3$$

$\therefore$ $$k_{Al} = \frac{k_{Al}\, a}{a} = \frac{72}{100} = 0.72 \text{ m/h}$$

$$\mathscr{D} = \mathscr{D}_{Bl} = \mathscr{D}_{Al} = 3.6 \times 10^{-6} \text{ m}^2/\text{h}$$

$$k = 2.6 \times 10^7 \text{ m}^3/\text{(mol.h)}$$

$$H_A = 10^5 \text{ (Pa.m}^3\text{)/mol}$$

At the top of the tower :

$$p_A = p_{A1} = 100 \text{ Pa and } C_B = C_{B1} = 55.56 \text{ mol/m}^3$$

$$M_H = \frac{(\mathscr{D}_{Al}\, C_B\, k)^{1/2}}{k_{Al}}$$

$$= \frac{(3.6 \times 10^{-6} \times 55.56 \times 2.6 \times 10^7)^{1/2}}{0.72} = 100$$

$$E_i = 1 + \frac{\mathscr{D}_{Bl}\, C_B\, H_A}{\mathscr{D}_{Al}\, p_{Ai}} = 1 + \frac{C_B\, H_A}{p_{Ai}}$$

For $p_{Ai} = 100$ Pa – liquid-film controls

For no gas-phase resistance : $p_{Ai} = p_A = 100$ Pa and in this case

$$E_i = 1 + \frac{55.56 \times 10^5}{100} = 55560$$

Here $E_i > 5\, M_H$ and for any other small value of p_{Ai} : $E_i > 5\, M_H$

At the bottom of the tower :

$$p_A = p_{A2} = 1000 \text{ Pa}, \quad C_B = C_{B2} = 5.61 \text{ mol/m}^3$$

$$M_H = \frac{(3.6 \times 10^{-6} \times 5.61 \times 2.6 \times 10^7)^{1/2}}{0.72} = 31.83$$

Assume $p_{Ai} = p_A = 1000$ Pa

$$E_i = 1 + \frac{C_B H_A}{p_{Ai}} = 1 + \frac{5.61 \times 10^5}{1000} = 561$$

Here also $E_i > 5\ M_H$ and for any other small value of p_{Ai} : $E_i > 5\ M_H$.

$\therefore$ At both ends of the tower, we have : $E_i > 5\ M_H$.

As we have $E_i > 5\ M_H$ at both ends of the tower, we have a pseudo first-order reaction by gas-liquid interface and for this $E = M_H$.

It is a case of fast reaction with high C_B, the reaction zone is (near) by the interface in the liquid film.

For this case, the form of rate equation to be used is

$$\left(-r_A''\right) a = -r_A'''' = \frac{1}{\frac{1}{k_{Ag}\, a} + \frac{H_A}{a.\sqrt{\mathscr{D}_{Al}\, k\, C_B}}}\, p_A$$

or

$$-r_A'''' = \frac{1}{\frac{1}{k_{Ag}\, a} + \frac{H_A}{k_{Al}\, aE}} \text{ with } E = M_H = \frac{\sqrt{\mathscr{D}_{Al}\, k\, C_B}}{k_{Al}}$$

As M_H is different at the top and the bottom, the rate also varies and so we have to adopt a graphical procedure determine the tower height.

For graphical procedure, in addition to p_{A1} and p_{A2} take two additional p_A values and evaluate C_B at these values and then the rate and tabulate.

We have

$$C_B = 61.11 - 0.0555\, p_A$$

Use the above equation to evaluate C_B

For $p_A = 300$ Pa , $C_B = 44.46$ mol/m^3

For $p_A = 500$ Pa , $C_B = 33.61$ mol/m^3

For $p_A = 750$ Pa , $C_B = 19.48$ mol/m^3

Now we will estimate the rate :

For $p_A = 100$ Pa, $C_B = 55.56$ mol/m^3

$\mathscr{D}_{Al} = 3.6 \times 10^{-6}$ m^2/h, $k = 2.6 \times 10^7$ m^3/(mol. h)

$H_A = 10^5$ (Pa.m^3)/mol, $a = 100$ m^2/m^3

$k_{Ag}\, a = 0.36$ mol/(h.m^3.Pa)

$$-r_A'''' = \left(-r_A''\right) a = \frac{100}{\frac{1}{0.36} + \frac{10^5}{100\sqrt{3.6 \times 10^{-6} \times 2.6 \times 10^7 \times 55.56}}}$$

$$= 6 \text{ mol/(h.m}^3\text{)}$$

Calculate the rate for other p_A values and tabulate :

p_A	C_B	$1/k_{Ag}a$	$H_A/a\,(\mathscr{D}_{Al}\,k\,C_B)^{1/2}$	$-r_A''''$	$\frac{1}{\left(-r_A''''\right)}$
100	55.56	2.78	13.87	6	0.167
300	44.46	2.78	15.50	16.41	0.061
500	33.61	2.78	17.83	24.26	0.041
750	19.48	2.78	23.41	28.64	0.035
1000	5.61	2.78	43.64	21.54	0.046
200	50.61	2.78	14.62	11.5	0.087

The volume of tower (for dilute systems) is given by

$$V_r = \frac{F_g}{\pi} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{\left(-r_A''\right) a} = \frac{F_g}{\pi} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{-r_A''''}$$

$$V_r = \frac{F_g}{\pi} \int_{100}^{1000} \frac{dp_A}{-r_A''''}$$

To evaluate the integral on the RHS of the above equation, plot $\frac{1}{\left(-r_A''''\right)}$ i.e., $\frac{1}{\left(-r_A''\right) a}$ v/s p_A and measure the area under the curve between $p_A = p_{A1} = 100$ Pa and $p_A = p_{A2} = 1000$ Pa.

From the plot [Fig. Ex 3.8 (b)] :

Area under the curve = 1880 mm^2 = 18.80 cm^2.

$\therefore$ $V_r = \frac{F_g}{\pi} \times$ Area under the curve $\times$ scale of x-axis $\times$ scale of y-axis

$$= \frac{90000}{10^5} \times 18.80 \times \frac{100}{1} \times \frac{0.05}{2}$$

$$= 42.3 \text{ m}^3$$

Volume of the tower required for the countercurrent operation = 42.3 m^3 **... Ans.**

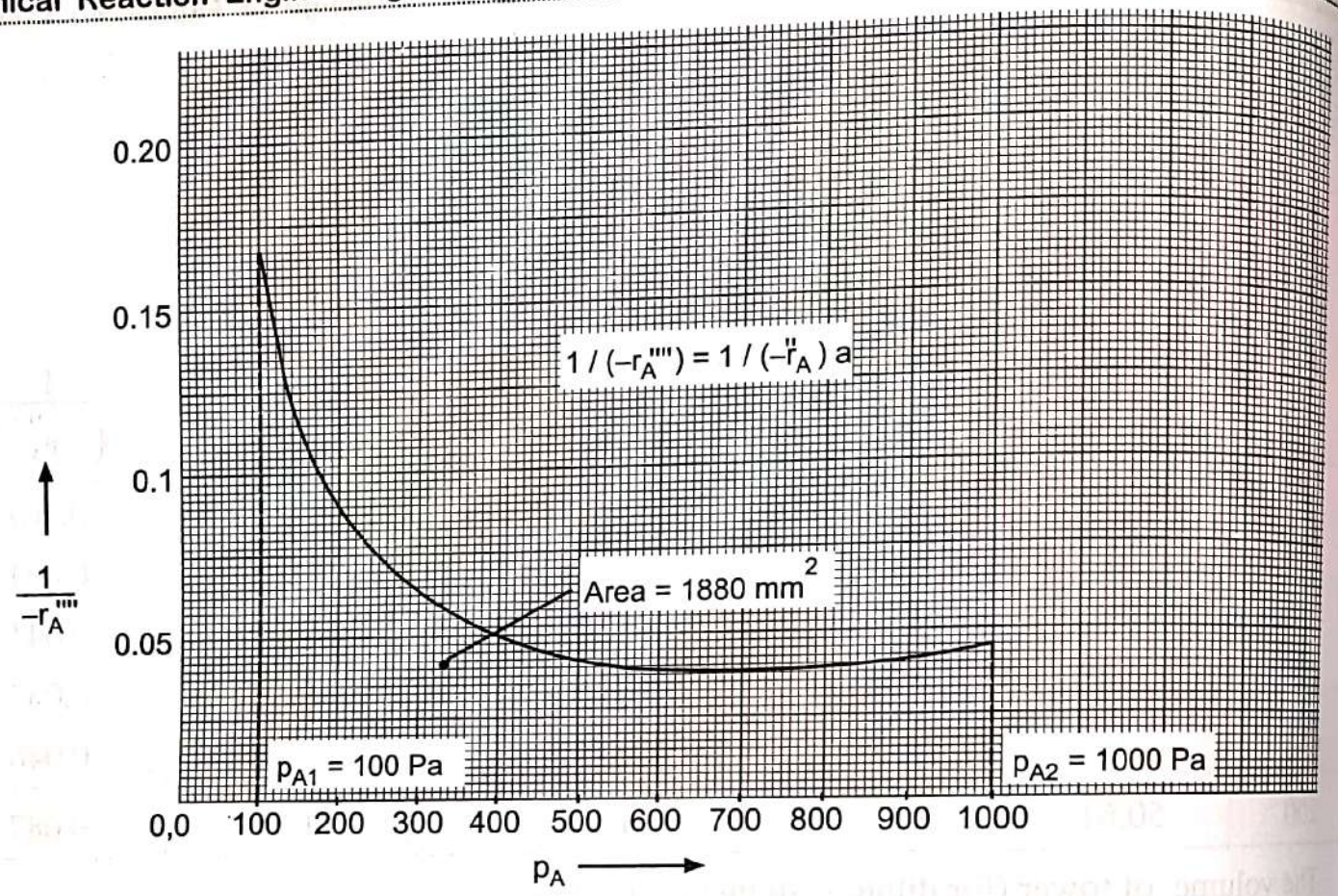


Fig. Ex 3.8 (b)

Example 3.9 : *The concentration of an undesirable impurity (A) in air (at 1 bar = 10^5 Pa) is to be reduced from 0.10% (or 100 Pa) to 0.02% (or 20 Pa) by absorption using a dilute acid of concentration $C_B = 3.2$ mol/m^3.*

B reacts with A extremely rapidly :

$A\,(g \rightarrow l) + B(l) \rightarrow R(l)\,,\ k = \infty.$

Calculate the height of tower required for countercurrent operation.

Data : The flow rates per m^2 of tower cross-section are

$$F_g/A_{cs} = 1 \times 10^5 \text{ mol/(h.m}^2\text{)}$$

$$F_l/A_{cs} = 7 \times 10^5 \text{ mol/(h.m}^2\text{)}$$

Molar density of liquid under all conditions is

$$C_T = 56000 \text{ mol/m}^3$$

The solubility of A is given by Henry's law constant

$$H_A = 12.5 \text{ (Pa.m}^3\text{)/mol.}$$

Diffusivities of A and B in water are the same.

$k_{Ag}\,a = 0.32$ mol/(h.m^3.Pa)

$k_{Al}\,a = 0.10\ h^{-1}$

Solution : $A(g \rightarrow l) + B(l) \rightarrow R(l)$, $k = \infty$

$\therefore \quad b = 1$

The reaction is extremely rapid (instantaneous) and occurs in liquid phase.

As C_{B1} is very low and reaction is rapid we see that B will get completely converted in the tower.

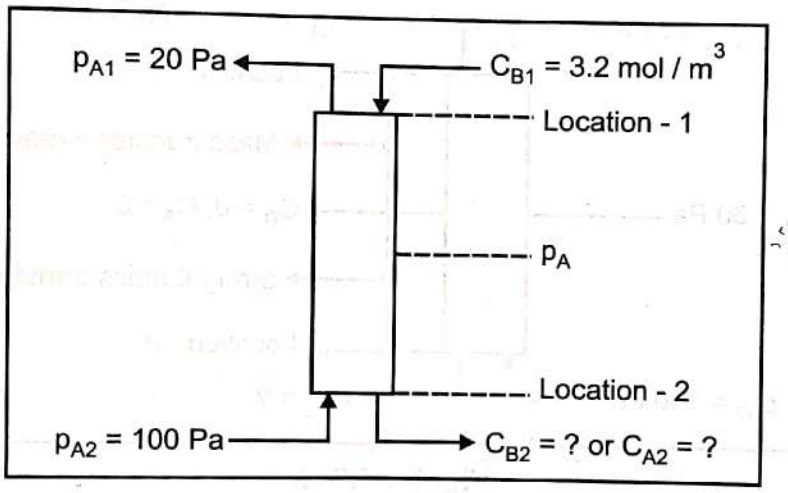


Fig. Ex 3.9 (a)

We have : $F_g/A_{cs} = 1 \times 10^5$ mol/(h.m^2) at $\pi = 10^5$ Pa

$F_l/A_{cs} = 7 \times 10^5$ mol/(h.m^2)

$C_T = 56000$ mol/m^3, $\quad b = 1$

For dilute solutions material balance for any two points in the tower yields :

$$p_A - p_{A1} = \frac{(F_l/A_{cs})\,\pi}{(F_l/A_{cs})\,b\,C_T}\,(C_{B1} - C_B)$$

$$p_A - 20 = \frac{7 \times 10^5 \times 10^5}{1 \times 10^5 \times 1 \times 56000}\,(3.2 - C_B)$$

$$p_A = 60 - 12.5\,C_B$$

or $$C_B = 4.8 - 0.08\,p_A$$

At the bottom of the tower :

$$C_{B2} = 4.8 - 0.08\,p_{A2} \quad , \quad p_{A2} = 100 \text{ Pa}$$

$$C_{B2} = 4.8 - 0.08\,(100) = -3.2$$

– ve concentration of B (not realistic) indicates that B is completely converted somewhere in the tower. Now let us find a point in the tower where it occurs.

As at both extremes of the upper part :

$k_{Ag}\, a\, p_A > k_{Bl}\, a\, C_B / b$. So we have :

Instantaneous reaction with low C_B

Reaction plane is in the liquid film and the form of the rate equation is

$$-r_A''' = \left(-r_A''\right) a = \frac{\dfrac{\mathscr{D}_{Bl}\, C_B}{\mathscr{D}_{Al}\, b} + \dfrac{p_A}{H_A}}{\dfrac{1}{H_A\, k_{Ag}\, a} + \dfrac{1}{k_{Al}\, a}}$$

With $\mathscr{D}_{Al} = \mathscr{D}_{Bl}$ and $b = 1$ above equation becomes

$$-r_A''' = \frac{H_A C_B + p_A}{\dfrac{1}{k_{Ag}\, a} + \dfrac{H_A}{k_{Al}\, a}}$$

$$\therefore \quad -r_A''' = \frac{p_A + H_A C_B}{\dfrac{1}{0.32} + \dfrac{12.5}{0.10}} = 0.0078\left(p_A + H_A\, C_B\right)$$

Now take a few values of p_A and estimate $-r_A'''$ and see whether it is constant or changing with p_A.

For the upper part of tower :

at one end : $p_A = 20$ Pa, $C_B = 3.2$ mol/m^3

$$\therefore \quad p_A + H_A\, C_B = 20 + 12.5 \times 3.2 = 60 \text{ Pa}$$

and the other end : $p_A = 60$ Pa and $C_B = 0$ mol/m^3

$$\therefore \quad p_A + H_A\, C_B = 60 + 12.5 \times 0 = 60 \text{ Pa}$$

As $p_A + H_A\, C_B$ is the same at both points it will be constant throughout this part and so the rate is constant and will be taken out of the integral sign.

For $p_A + H_A\, C_B$ to be constant let us have check by considering one more point in this part.

For this upper part we have

$$C_B = 4.8 - 0.08\, p_A$$

Take $p_A = 40$ Pa

$$C_B = 4.8 - 0.08 \times 40 = 1.6 \text{ mol/m}^3$$

$$\therefore \quad p_A + H_A\, C_B = 40 + 12.5 \times 1.6 = 60 \text{ Pa}$$

So $p_A + H_A\, C_B = 60$ Pa and hence $-r_A'''$ is constant throughout this section.

$$-r_A''' = 0.0078\,(p_A + H_A\, C_B)$$

$$= 0.0078 \times 60 = 0.468 \text{ mol/(m}^3\text{.h)}$$

$$h_1 = \frac{(F_g/A_{cs})}{\pi\left(-r_A'''\right)} \int_{20}^{60} dp_A$$

$$= \frac{1 \times 10^5}{10^5 \times 0.468} (60 - 20) = 85.47 \text{ m}$$

Height of the upper part of the tower = 85.47 m

Now let us find the height of the lower part of the tower where there is straight mass transfer.

$$h_2 = \frac{F_g/A_{cs}}{\pi} \int_{p_{A1}}^{p_{A2}} \frac{dp_A}{-r_A'''}$$

Limits of integration for this lower part of tower are :

$$p_{A1} = 60 \text{ Pa} \quad \text{and} \quad p_{A2} = 100 \text{ Pa}$$

For straight mass transfer the rate is given by

$$-r_A''' = \left(-r_A'''\right) a = K_{Ag} a \left(p_A - p_A^*\right)$$

$$h_2 = \frac{(F_g/A_{cs})}{\pi K_{Ag} a} \int_{60}^{100} \frac{dp_A}{p_A - p_A^*}$$

Let us have check whether $p_A - p_A^*$ is constant throughout this part or not.

At the start of this part of tower :

$p_A = 60$ Pa and $C_A = 0$

$\therefore \quad p_A - p_A^* = p_A - H_A C_A = 60 - 12.5 \times 0 = 60$ Pa

At the end of this part of tower i.e. at lower end of the tower.

$p_A = p_{A2} = 100$ Pa and $C_A = C_{A2} = 3.2$ mol/m^3

$\therefore \quad p_A - p_A^* = 100 - 12.5 \times 3.2 = 60$ Pa

Take one more point :

Say $p_A = 80$ Pa

for this part we have,

$$C_A = 0.08\, p_A - 4.8$$

$$C_A = 0.08 \times 80 - 4.8 = 1.6 \text{ mol/m}^3$$

$\therefore \quad p_A - p_A^* = p_A - H_A C_A = 80 - 12.5 \times 1.6 = 60$ Pa

So $p_A - p_A^*$ is constant and thus can be taken out of the integral sign.

$$\therefore \quad p_A - p_A^* = \Delta p_A$$

$$h = \frac{(F_g/A_{cs})}{\pi K_{Ag}a\ \Delta p_A} \int_{60}^{100} dp_A$$

$$= \frac{(F_g/A_{cs})}{\pi K_{Ag}a\ \Delta p_A} (100 - 60)$$

Evaluate $K_{Ag}a$

$$\frac{1}{K_{Ag}a} = \frac{1}{k_{Ag}a} + \frac{H_A}{k_{Al}a}$$

$$= \frac{1}{0.32} + \frac{12}{0.} \frac{5}{)} = 128.125$$

$$K_{Ag}a = 0.0078 \text{ m}^{\cdot} \text{/(h.m}^3\text{.Pa)}$$

$$h_2 = \frac{1 >}{10^5 \times 0.(} \quad \overline{8 \times 60} \ (100 - 60)$$

$$= 85.47 \text{ m}$$

Height of the lower part of tower = 85.47 m

The height of tower is

$$h = h_1 + h_2$$

$$= 85.47 + 85.47 = 170.94 \text{ m} \simeq 171 \text{ m}$$

The height of tower required for the countercurrent operation is 171 m ... Ans

CHAPTER 4

KINETICS OF FLUID-PARTICLE REACTIONS

This chapter deals with the kinetics of fluid-particle reactions - non-catalytic heterogeneous reactions in which a gas or liquid is brought in contact with a solid and reacts with the solid to transform (convert) it into a product. Fluid-particle reactions yield fluid and/or solid products and thus may be represented by

$$A\,(\text{fluid}) + bB\,(\text{solid}) \rightarrow \text{Fluid products} \quad \ldots (4.1)$$

$$\rightarrow \text{Solid products} \quad \ldots (4.2)$$

$$\rightarrow \text{Fluid and solid products} \quad \ldots (4.3)$$

In case of such reactions we have two different cases to consider :

1. The particle size remains unchanged during reaction. In fluid-solid reactions this will happen when -

(a) solid particles contain large amount of inert material which remain as a non-flaking ash.

or

(b) a firm product material is formed by the reactions of Equation (4.2) or Equation (4.3).

Examples of fluid-solid reactions in which the solid does not appreciably change in size during reaction are :

(i) The roasting of sulphide ores in air to yield the metal oxides. For example, zinc sulphide is roasted to produce zinc oxide.

$$ZnS\,(s) + \frac{1}{2}\,O_2\,(g) \rightarrow ZnO\,(s) + SO_2\,(g)$$

(ii) The reduction of metal oxides to the corresponding metals. For example, reduction of magnetite, Fe_3O_4, by hydrogen occurs according to the following reaction to produce iron.

$$Fe_3O_4\,(s) + 4H_2\,(g) \rightarrow 3Fe\,(s) + 4H_2O\,(g)$$

2. The particle size changes with the progress of reaction (the particle decreases in size, i.e., shrinks during the course of reaction until it disappears).

In fluid-particle reactions this will happen when -

(a) a flaking ash or product material is formed or

(b) pure B is used in the reaction of Equation (4.1)

Examples of fluid-solid reactions in which the size of solid changes are :

(i) The production of carbon disulphide from the elements :

$$C(s) + 2S(g) \rightarrow CS_2(g)$$

(ii) The production of water gas by the action of steam on coal (with low ash content) :

$$C(s) + H_2O(g) \rightarrow CO(g) + 2H_2(g)$$

(ii) The burning of pure carbon in air :

$$C(s) + O_2(g) \rightarrow CO_2(g)$$

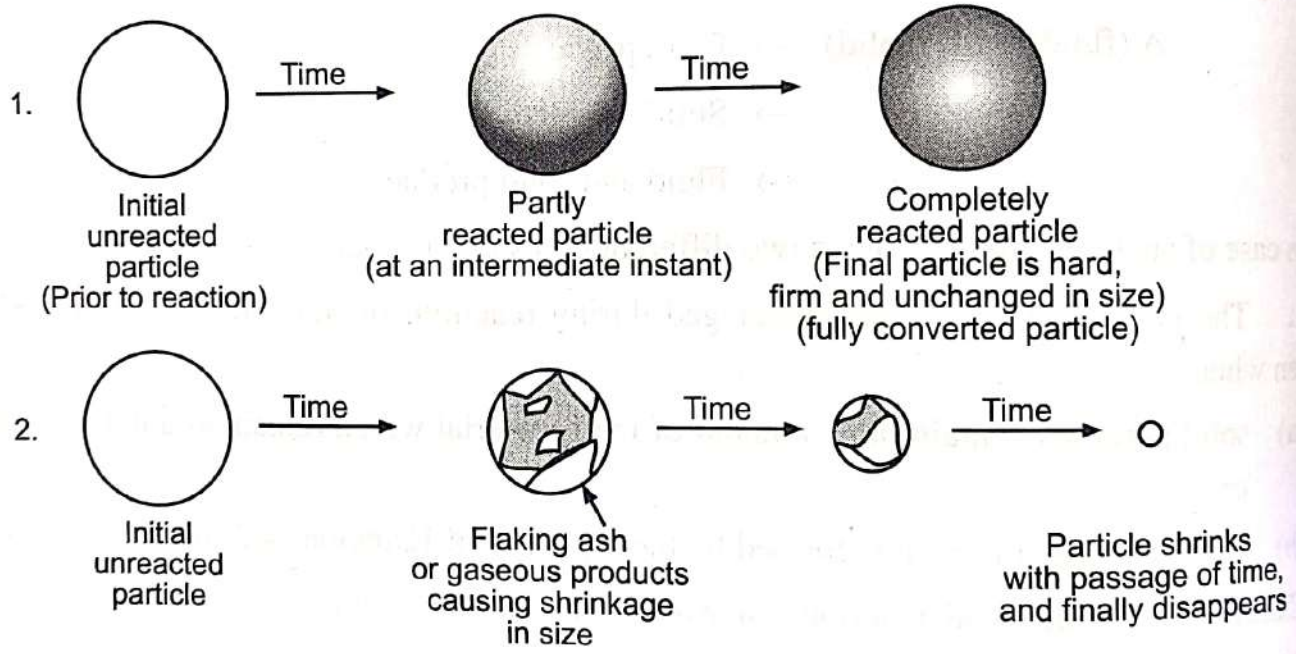


Fig. 4.1 : Different sorts of behaviour of reacting solid particles in fluid-solid reactions

Selection of a kinetic model :

Every conceptual picture or model describing the progress of reaction has its own mathematical representation, its rate equation. Therefore, if we select a model we have to accept its rate equation and vice versa. If a model matches closely with what really happens, then its rate equation will closely predict and describe the actual kinetics of the reaction, whereas if a model differs widely from what really happens, then its rate equation will be useless. It is worthless to select a model which involves too many mathematical complexities even though it very closely reflects/represents reality. One has to select a model which is the closest representation of reality and at the same time it should be easy to treat mathematically (i.e., the kinetic model selected can be described with relatively simple equations).

Many kinetic models have been developed for the progress of the reaction within a single solid particle. To describe the progress of the non-catalytic reaction of particles with surrounding fluid, we consider two simple models regarded as perfect in reality, namely the progressive - conversion model and the shrinking unreacted - core model.

Progressive - Conversion Model (PCM) :

This model assumes that reactant gas enters and reacts throughout the particle at all times (reaction proceeds continuously throughout the particle). The rate of reaction is different at different locations within the particle.

Hence, the whole reactant particle is continuously and progressively converted into product as shown in Fig. 4.2 (which also shows how the concentration of the solid reactant changes throughout the solid particle at different instants after the onset of the reaction). [If the product is itself is a firm solid, the particle size remains unchanged, but if the product is volatile or flaking, the particle size decreases until it disappears.]

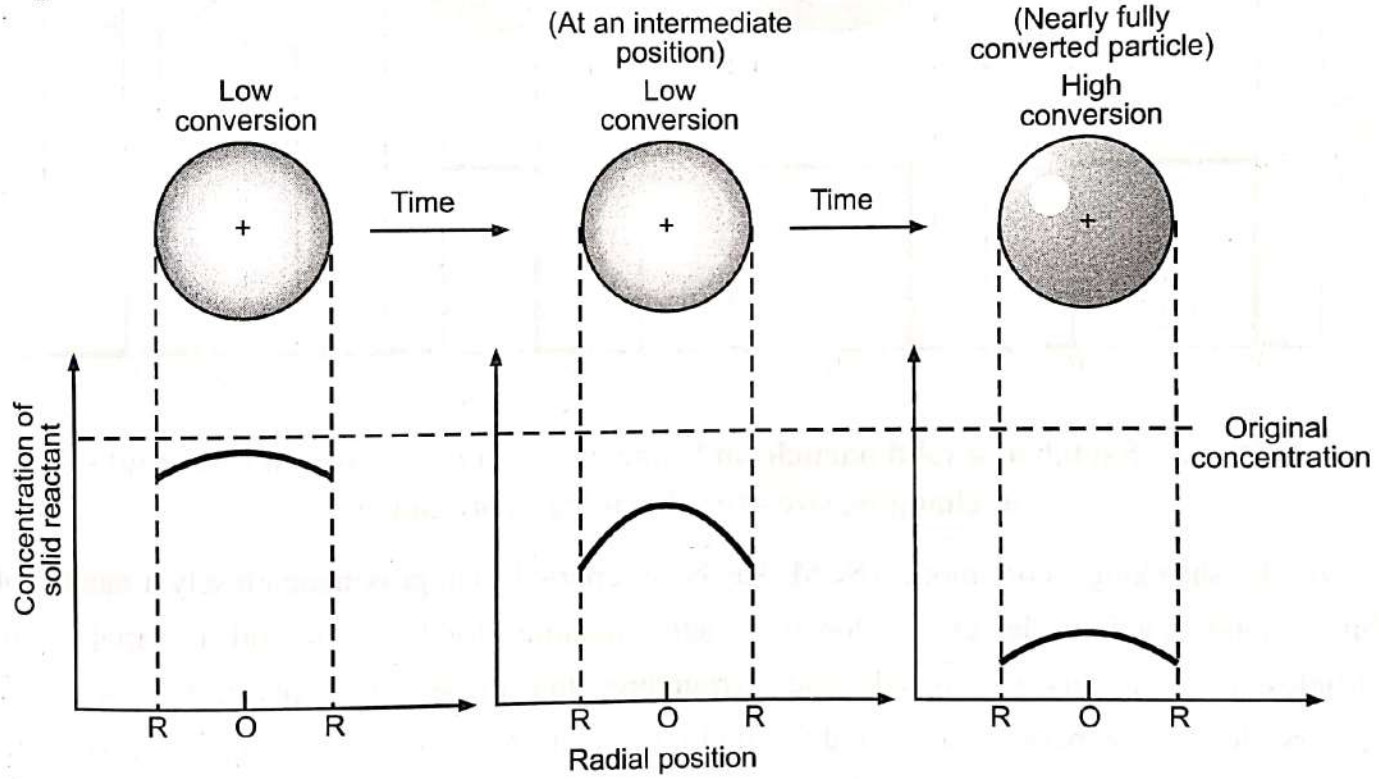


Fig. 4.2 : Sketch of a solid particle and concentration profile of solid reactant (B) with progress of reaction - unchanging size - progressive - conversion model

Shrinking - Core Model (SCM) :

This model assumes that reaction begins first at the outer surface of the solid particle and the deeper layers do not take part in the reaction until all the outer layer has transformed into solid or gaseous product. Gradually, the reaction zone then moves inward (into the solid), constantly reducing the size of core of the unreacted solid and leaving behind completely converted solid (solid product) and inert material (inert constituent of the solid reactant).

We define converted solid (i.e., solid product) and inert material as **ash**. An unreacted core of solid continuously decreases in size / shrinks in size during reaction until it is entirely consumed (Here the amount of solid material to be consumed by reaction is shrinking with time.)

At any given instant, the solid particle comprises of a core surrounded by an envelope. The core is the unreacted reactant (that shrinks with time) so this model is referred to as shrinking unreacted-core model. The envelope consists of a solid product and inert material.

The shrinking of size of unreacted core of solid with reaction time is shown in Fig. 4.3.

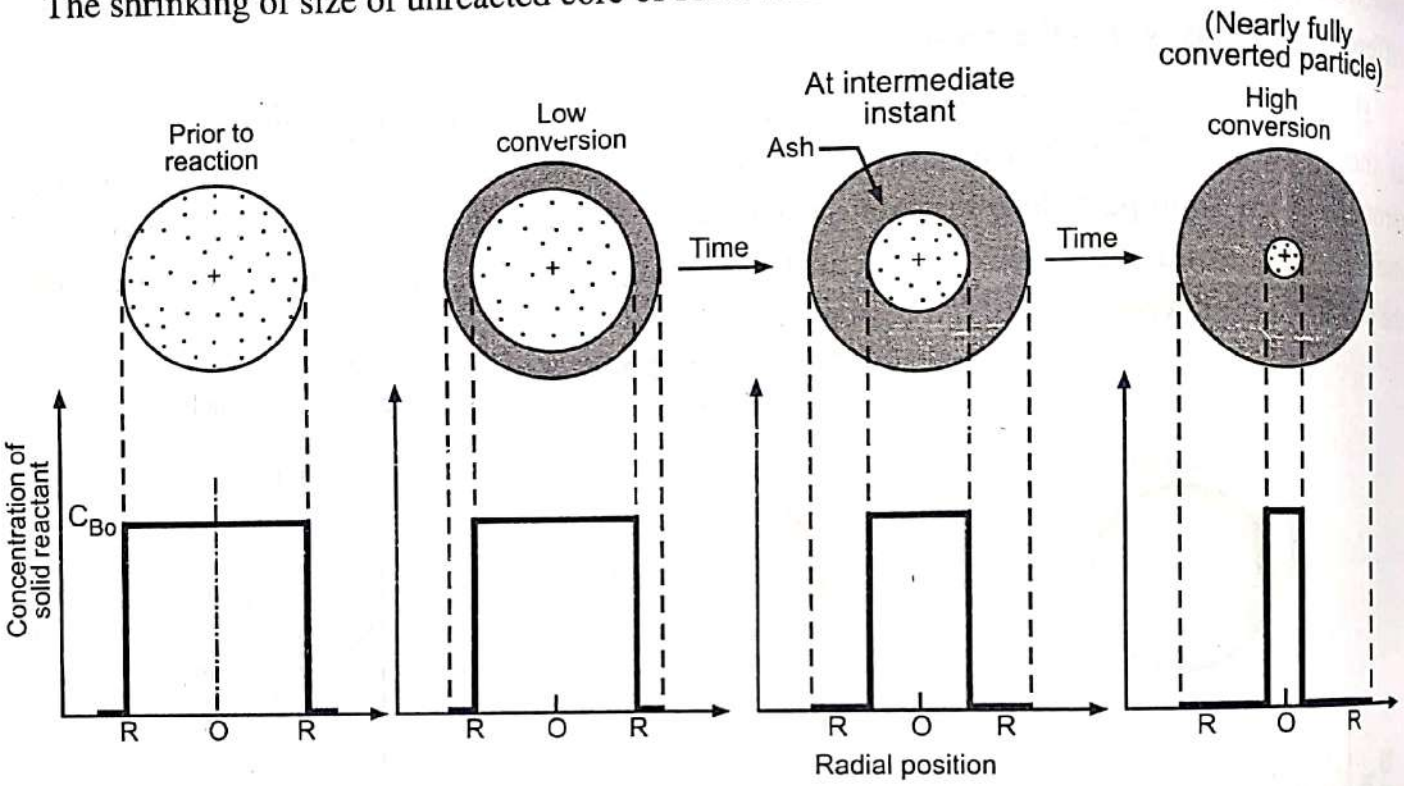


Fig. 4.3 : Sketch of a solid particle and concentration profile of solid reactant - unchanging size - the shrinking - core model

As the shrinking - core model (SCM) has been reported to approximate closely a number of fluid - solid reactions, let us develop its kinetic equations for fixed as well as variable size particles. In doing this, we consider the surrounding fluid to be a gas, spherical particles and irreversible reaction between a gas and a solid [any one of the reactions given by Equations (4.1), (4.2) and (4.3)]. Please note that in the analysis to follow we consider the surrounding fluid to be a gas but keep in mind that the analysis given below is applicable equally well to liquids.

Shrinking - Core Model (SCM) for Spherical Particles of Unchanging Size / Fixed Size :

According to this model, the following five steps occur in succession during reaction :

(i) Diffusion of gaseous reactant A through the gas film surrounding the particle to the surface of the solid particle ... reactant A diffuses to the external surface of the solid particle from the bulk gas phase through the gas film ..... external diffusion.

(ii) Penetration of A by diffusion through pores and cracks in the blanket (layer) of ash to the surface of the unreacted core ... pore or internal diffusion.

(iii) Reaction of gaseous A with solid B at the surface of the unreacted core (reaction surface).

(iv) Diffusion of gaseous products back to the exterior surface of the solid through the ash (ash layer) ... reverse internal diffusion.

(v) Diffusion of gaseous products through the gas film back into the main body of gas ... reverse external diffusion.

Fig. 4.4 shows the concentration profile of a gaseous reactant A reacting with a solid particle – the concentration of reactant A at different points of the reaction space.

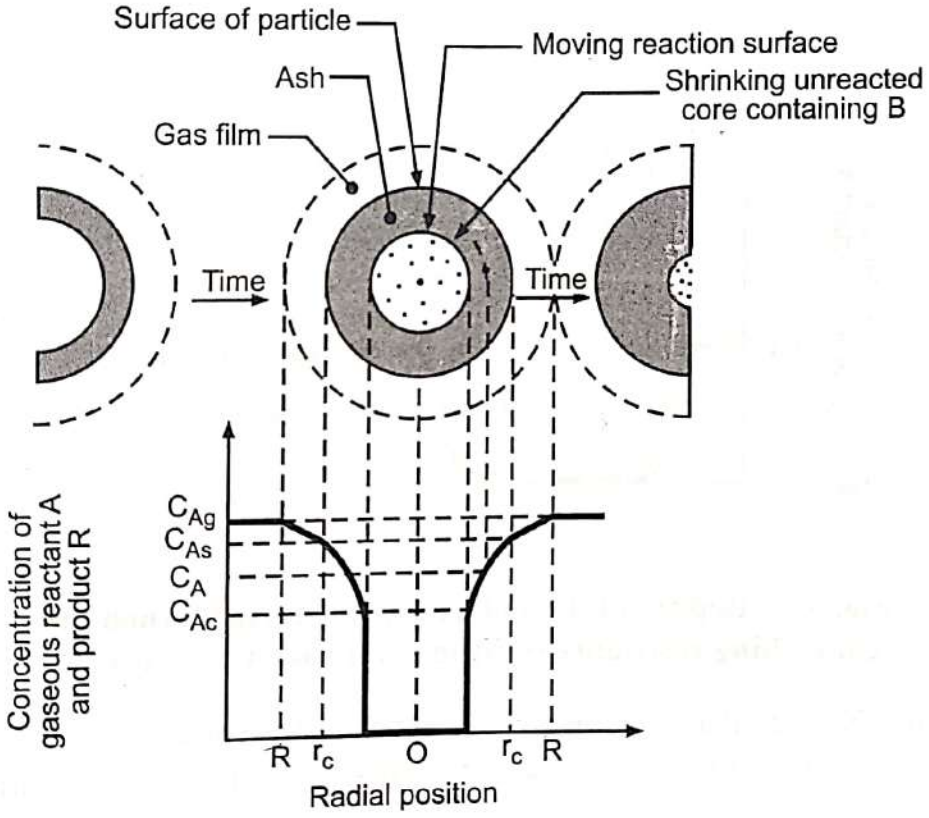


Fig. 4.4 : Concentration profile of a gaseous reactant A reacting with a solid particle : A(g) + bB(s) → solid product - particle of unchanging size - shrinking-core model

In all gas-solid reactions, all these steps may not exist. If no gaseous products are formed, steps (iv) and (v) do not exist and thus do not contribute to the resistance to reaction. All these steps offer resistance to reaction and the magnitude of the resistances of these steps usually differ greatly from each other. In such cases, the step with the highest resistance is considered as the rate - controlling / rate - determining for the overall reaction process.

One of the following three steps may control the overall reaction rate. Therefore, we will develop the conversion equations for spherical particles by considering each of the following steps separately to be rate-controlling.

(i) Diffusion through the gas film.

(ii) Diffusion through the ash (converted solid and inert material) / ash layer.

(iii) Chemical reaction.

Diffusion through Gas Film Controls (Controlling resistance-gas film diffusion) :

Whenever the resistance of the gas film controls the overall reaction rate, the concentration profile for gaseous reactant A will be as shown in Fig. 4.5.

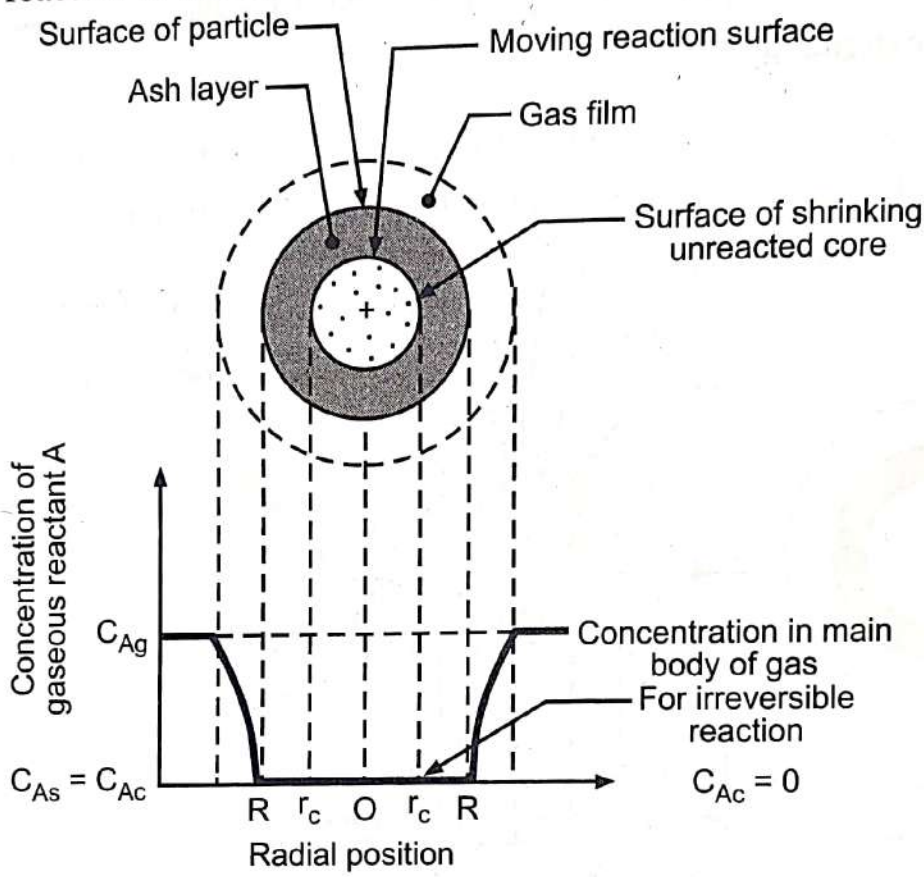


Fig. 4.5 : Representation of a reacting particle when the controlling resistance is diffusion through the gas film

It is clear from Fig. 4.5 that no gaseous reactant A is present at the surface of particle and thus, the concentration driving force, $C_{Ag} - C_{As}$, becomes C_{Ag} and is constant during the course of reaction.

As the rate of mass transfer is defined as the rate of flow of material normal to a unit surface, we base the rate on the unchanging exterior surface of a particle. The rate of reaction of A per unit surface is equal to the molar flux of A from the main body of gas (the bulk gas phase) to the surface of the particle.

$$-\frac{1}{S_{ex}}\frac{dN_A}{dt} = -\frac{1}{4\pi R^2}\frac{dN_A}{dt} = k_g (C_{Ag} - C_{As}) = k_g C_{Ag} = \text{constant} \quad \ldots (4.4)$$

[as C_{Ag} and k_g both are constant for the case under consideration]

$$A\,(g) + bB\,(s) \rightarrow \text{products}$$

From the stoichiometry of the reaction :

$$-\frac{dN_A}{dt} = -\left(\frac{1}{b}\right)\frac{dN_B}{dt} = -\frac{1}{b}\frac{dN_B}{dt}$$

$$-\frac{dN_B}{dt} = -b\frac{dN_A}{dt} \qquad \ldots (4.5)$$

Combining Equations (4.4) and (4.5), we get

$$-\frac{1}{S_{ex}}\frac{dN_B}{dt} = -\frac{1}{4\pi R^2}\frac{dN_B}{dt} = -\frac{b}{4\pi R^2}\frac{dN_A}{dt} = bk_g C_{Ag} = \text{constant} \qquad \ldots (4.6)$$

The amount of B present in a particle is given by

$$N_B = \rho_B V \qquad \ldots (4.7)$$

where ρ_B is the molar density of B in the solid in moles/m^3 solid and V is the volume of particle (m^3 solid).

The decrease in volume or radius of unreacted core as a result of the disappearance of dN_B moles of solid reactant is given by

$$-dN_B = -b\,dN_A = -d(\rho_B V) = -\rho_B\,dV = -\rho_B\,d\left(\frac{4}{3}\pi r_c^3\right)$$

$$= -4\pi\rho_B r_c^2\,dr_c \qquad \ldots (4.8)$$

Substituting the value of $-dN_B$ from Equation (4.8) into Equation (4.6), we get an expression for the rate of reaction in terms of the shrinking radius of unreacted core :

$$-\frac{1}{S_{ex}}\frac{dN_B}{dt} = -\frac{4\pi\rho_B r_c^2}{4\pi R^2}\frac{dr_c}{dt} = -\frac{\rho_B r_c^2}{R^2}\frac{dr_c}{dt} = bk_g C_{Ag} \qquad \ldots (4.9)$$

where k_g is the mass transfer coefficient between fluid and particle, $m^3/(m^2 \cdot s)$.

Rearranging Equation (4.9) gives

$$-\frac{\rho_B}{R^2} r_c^2\,dr_c = bk_g C_{Ag}\,dt$$

Integrating the above equation between the limits : At $t = 0$ ... $r_c = R$, At $t = t$, ... $r_c = r_c$, we get

$$-\frac{\rho_B}{R^2}\int_R^{r_c} r_c^2\,dr_c = bk_g\int_0^t dt$$

$$-\frac{\rho_B}{R^2}\left[\frac{r_c^3}{3}-\frac{R^3}{3}\right] = \frac{\rho_B R}{3R^3}\left[R^3 - r_c^3\right]$$

$$= \frac{\rho_B R}{3}\left[1-\left(\frac{r_c}{R}\right)^3\right] = bk_g C_{Ag} t$$

$$\therefore \quad \boxed{t = \frac{\rho_B R}{3bk_g C_{Ag}}\left[1-\left(\frac{r_c}{R}\right)^3\right]} \quad \ldots (4.10)$$

Equation (4.10) gives us the idea regarding how the unreacted core shrinks with time.

Let the time for complete conversion of a particle be τ. At complete conversion : $r_c = 0$. Therefore, by putting $r_c = 0$ in Equation (4.10), we obtain an expression for τ (for this, replace t by τ).

$$\tau = \frac{\rho_B R}{3bk_g C_{Ag}}\left[1-\left(\frac{0}{R}\right)^3\right]$$

$$\therefore \quad \boxed{\tau = \frac{\rho_B R}{3bk_g C_{Ag}}} \quad \ldots (4.11)$$

Dividing Equation (4.10) by Equation (4.11), we get

$$\frac{t}{\tau} = 1-\left(\frac{r_c}{R}\right)^3 \quad \ldots (4.12)$$

Equation (4.12) expresses the radius of unreacted core in terms of fractional time for complete conversion.

The radius of unreacted core in terms of fractional conversion is given by

$$1 - X_B = \text{fraction of B unreacted}$$

where X_B is the fractional conversion of B.

$$= \frac{\text{Volume of unreacted core}}{\text{Total volume of particle}}$$

$$= \frac{\frac{4}{3}\pi r_c^3}{\frac{4}{3}\pi R^3} = \left(\frac{r_c}{R}\right)^3 \quad \ldots (4.13)$$

Combining Equations (4.12) and (4.13), we get

$$\boxed{\frac{t}{\tau} = 1-\left(\frac{r_c}{R}\right)^3 = X_B} \quad \ldots (4.14)$$

Equation (4.14) gives the relationship of time with the radius of unreacted core and fractional conversion of B.

Diffusion through Ash Layer Controls (Controlling resistance-ash layer diffusion) :

Whenever the resistance of the ash layer controls the overall rate of reaction, the concentration profile of the gaseous reactant A will be as shown in Fig. 4.6.

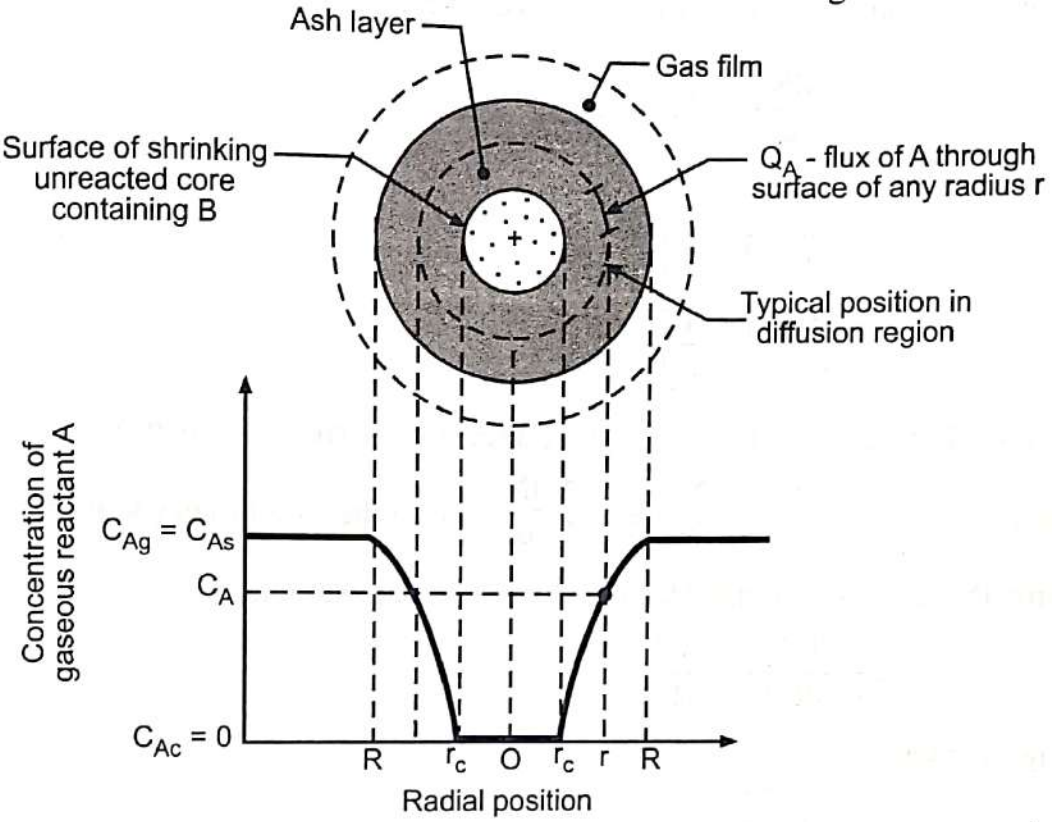


Fig. 4.6 : Representation of a reacting particle when the controlling resistance is diffusion through the ash layer

Under this condition, the instantaneous rate of reaction of A at any time is given by the rate of diffusion of A to the reaction surface, that is,

$$-\frac{dN_A}{dt} = 4\pi r^2 Q_A = 4\pi R^2 Q_{As} = 4\pi r_c^2 Q_{Ac} = \text{constant} \qquad \ldots (4.14)$$

where Q_A is the flux of A through a spherical surface of any radius r in the ash layer, Q_{As} is the flux of A through the exterior surface of particle and Q_{Ac} is the flux of A to the reaction surface.

The flux of A through the ash layer may be given by Fick's law :

$$Q_A = \mathscr{D}_e \frac{dC_A}{dr} \qquad \ldots (4.15)$$

(Please note both Q_A and dC_A/dr are positive.)

where $\mathscr{D}_e$ is the effective diffusion coefficient of gaseous reactant A in the ash layer.

Combining Equations (4.14) and (4.15), we get for any r

$$-\frac{dN_A}{dt} = 4\pi r^2 \mathscr{D}_e \frac{dC_A}{dr} = \text{constant} \qquad \ldots (4.16)$$

The limits of integration are :

At $r = R$ ... $C_A = C_{As} = C_{Ag}$ and at $r = r_c$ ... $C_A = C_{Ac} = 0$

Integrating Equation (4.16) across the ash layer, we get

$$-\frac{dN_A}{dt}\int_R^{r_c}\frac{dr}{r^2} = 4\pi\mathscr{D}_e\int_{C_{Ag}}^{0} dC_A$$

$$-\frac{dN_A}{dt}\left[\frac{1}{R}-\frac{1}{r_c}\right] = -4\pi\mathscr{D}_e C_{Ag}$$

$$-\frac{dN_A}{dt}\left[\frac{1}{r_c}-\frac{1}{R}\right] = 4\pi\mathscr{D}_e C_{Ag} \qquad \text{... (4.17)}$$

Equation (4.17) represents the condition of a reacting particle at any time.

We have : $$-\frac{dN_A}{dt} = -\frac{1}{b}\frac{dN_B}{dt}$$ (from the stoichiometry of reaction)

Replacing dN_A/dt by $-1/b\ (dN_B/dt)$ in Equation (4.17), we obtain

$$-\frac{1}{b}\frac{dN_B}{dt}\left[\frac{1}{r_c}-\frac{1}{R}\right] = 4\pi\mathscr{D}_e C_{Ag}$$

Rearranging gives

$$-dN_B\left[\frac{1}{r_c}-\frac{1}{R}\right] = 4\pi b\mathscr{D}_e C_{Ag}\,dt \qquad \text{... (4.18)}$$

We have : $$-dN_B = -\rho_B\,dV = -\rho_B d\left(\frac{4}{3}\pi r_c^3\right) = -4\pi\rho_B r_c^2\,dr_c$$

Replacing $-dN_B$ in Equation (4.18) by its value given above, we get

$$-4\pi\rho_B r_c^2\,dr_c\left[\frac{1}{r_c}-\frac{1}{R}\right] = 4\pi b\mathscr{D}_e C_{Ag}\,dt \qquad \text{... (4.19)}$$

$$-\rho_B\int\left[\frac{1}{r_c}-\frac{1}{R}\right] r_c^2\,dr_c = b\mathscr{D}_e C_{Ag}\int dt$$

$$-\rho_B\int\left[r_c-\frac{r_c^2}{R}\right] dr_c = b\mathscr{D}_e C_{Ag}\int dt$$

The limits of integration are : $t = 0 \rightarrow r_c = R$ and $t = t \rightarrow r_c = r_c$. Thus, integration gives

$$-\rho_B\int_R^{r_c}\left[r_c-\frac{r_c^2}{R}\right] dr_c = b\mathscr{D}_e C_{Ag}\int_0^t dt$$

$$-\rho_B\left[\frac{r_c^2}{2}-\frac{r_c^3}{3R}\right]_R^{r_c} = b\mathscr{D}_e C_{Ag}\,t$$

$$-\rho_B\left[\frac{r_c^2}{2}-\frac{R^2}{2}-\frac{r_c^3}{3R}+\frac{R^3}{3R}\right] = -\rho_B\left[-\frac{R^2}{6}+\frac{r_c^2}{2}-\frac{r_c^3}{3R}\right] = b\mathscr{D}_e C_{Ag} t$$

$$\therefore \quad \rho_B\left[\frac{R^2}{6}-\frac{r_c^2}{2}+\frac{r_c^2}{3R}\right] = b\mathscr{D}_e C_{Ag} t$$

$$t = \frac{\rho_B}{b\mathscr{D}_e C_{Ag}}\left[\frac{R^2}{6}-\frac{r_c^2}{2}+\frac{r_c^3}{3R}\right]$$

Taking $R^2/6$ common from the RHS of the above equation, we get

$$\boxed{t = \frac{\rho_B R^2}{6b\mathscr{D}_e C_{Ag}}\left[1-3\left(\frac{r_c}{R}\right)^2+2\left(\frac{r_c}{R}\right)^3\right]} \quad \ldots (4.20)$$

Equation (4.20) relates the radius of a unreacted core with time when the diffusion through the ash layer controls the overall reaction rate.

The time required for the complete conversion of a particle (τ) is obtained by putting $r_c = 0$ in Equation (4.20) as for the complete conversion of the particle : $r_c = 0$. Therefore, putting $r_c = 0$ and replacing t by τ gives

$$\tau = \frac{\rho_B R^2}{6b\mathscr{D}_e C_{Ag}}\left[1-3\left(\frac{0}{R}\right)^2+2\left(\frac{0}{R}\right)^3\right]$$

$$\boxed{\tau = \frac{\rho_B R^2}{6b\mathscr{D}_e C_{Ag}}} \quad \ldots (4.21)$$

Dividing Equation (4.20) by Equation (4.21), we get

$$\boxed{\frac{t}{\tau} = 1-3\left(\frac{r_c}{R}\right)^2+2\left(\frac{r_c}{R}\right)^3} \quad \ldots (4.22)$$

Equation (4.22) gives us idea regarding the progress of reaction in terms of time required for the complete conversion.

We have, $$1 - X_B = \left(\frac{r_c}{R}\right)^3$$

$$\therefore \quad (1-X_B)^{1/3} = \frac{r_c}{R}$$

$\therefore$ Equation (4.22) in terms of fractional conversion by making use of the above relation becomes

$$\boxed{\frac{t}{\tau} = 1-3(1-X_B)^{2/3}+2(1-X_B)} \quad \ldots (4.23)$$

Chemical Reaction Controls (Controlling resistance-chemical reaction step) :

Whenever the resistance of chemical reaction step controls the overall rate of reaction, the concentration profile for gaseous reactant A will be as shown in Fig. 4.7. In this case, diffusion steps - diffusion of A through the gas film and the ash layer - are rapid with respect to the surface reaction (slow reaction) and thus virtually no concentration gradients exist in the gas film and in the ash layer, i.e., as neither the gas film nor the ash layer offers any significant resistance to

mass transfer, the concentration of A at the reaction surface at the radial position r_c will be C_{Ag} the same as in the bulk of the gas.

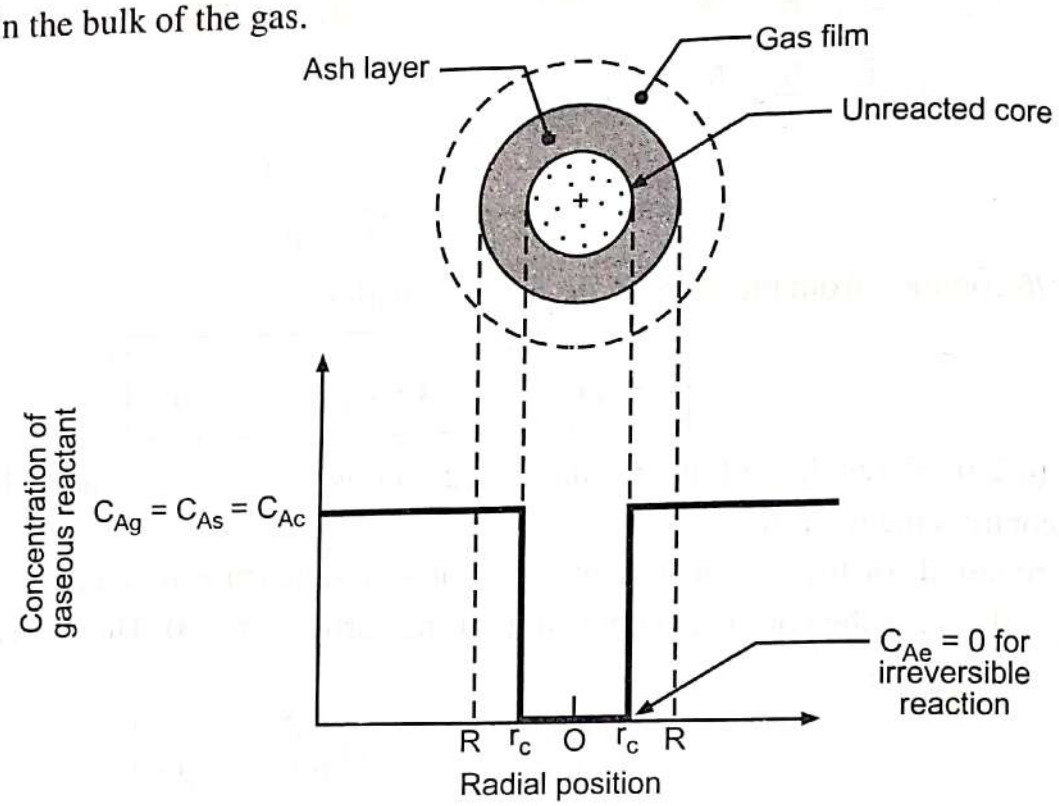


Fig. 4.7 : Representation of a reacting particle when controlling resistance is chemical reaction – A(g) + b B(s) → products

Here the overall rate of reaction is proportional to the available surface of unreacted core. Thus, taking a unit surface of unreacted core of radius r_c as the basis for the reaction rate (i.e., to define the rate), the rate of reaction of A, i.e., the rate at which moles of A are consumed in the reaction per unit surface is given by

$$-\frac{1}{4\pi r_c^2}\frac{dN_A}{dt} = k'' C_{Ag} \qquad \ldots (4.24)$$

From the stoichiometry of reactions represented by Equations (4.1), (4.2) and (4.3),

$$-\frac{dN_A}{dt} = -\frac{1}{b}\frac{dN_B}{dt} \qquad [A\,(g) + bB\,(s) \rightarrow \text{Products}] \qquad \ldots (4.25)$$

Combining Equations (4.24) and (4.25) gives

$$-\frac{1}{4\pi r_c^2}\frac{dN_B}{dt} = -\frac{b}{4\pi r_c^2}\frac{dN_A}{dt} = bk'' C_{Ag} \qquad \ldots (4.26)$$

where k'' is the first order rate constant for the surface reaction, m/s.

Also, we have

$N_B = \rho_B V$ ... amount of B present in a particle

$$dN_B = \rho_B\, dV = \rho_B\, d\left(\frac{4}{3}\pi r_c^3\right) = 4\pi\, \rho_B\, r_c^2\, dr_c$$

$$\therefore \quad \frac{dN_B}{dt} = 4\pi\rho_B\, r_c^2 \frac{dr_c}{dt} \qquad \ldots (4.27)$$

Replacing dN_B/dt in Equation (4.26) by its value in terms of the shrinking radius given by Equation (4.27), we get

$$-\frac{1}{4\pi r_c^2}\, 4\pi\rho_B\, r_c^2 \frac{dr_c}{dt} = -\rho_B \frac{dr_c}{dt} = bk''\, C_{Ag} \qquad \ldots (4.28)$$

Integrating the above equation within the limits :

at $t = 0$ ... $r_c = R$ and at $t = t$ ... $r_c = r_c$ gives

$$-\rho_B \int_R^{r_c} dr_c = bk''\, C_{Ag} \int_0^t dt$$

$$-\rho_B\,(r_c - R) = \rho_B\,(R - r_c) = bk''\, C_{Ag}\, t$$

$$\therefore \quad \boxed{t = \frac{\rho_B}{bk''C_{Ag}}(R - r_c)} \qquad \ldots (4.29)$$

The time τ required for complete conversion corresponds to $r_c = 0$ (zero radius of the unreacted core) and is obtained by replacing t by τ and putting $r_c = 0$ in Equation (4.29). Therefore, doing this, we get

$$\boxed{\tau = \frac{\rho_B R}{bk''C_{Ag}}} \qquad \ldots (4.30)$$

Dividing Equation (4.29) by Equation (4.30), we get

$$\frac{t}{\tau} = \frac{R - r_c}{R} = 1 - \frac{r_c}{R} \qquad \ldots (4.31)$$

We have $$1 - X_B = \left(\frac{r_c}{R}\right)^3$$

$$\therefore \quad (1 - X_B)^{1/3} = \frac{r_c}{R} \qquad \ldots (4.32)$$

Combining Equations (4.31) and (4.32),

$$\boxed{\frac{t}{\tau} = 1 - \frac{r_c}{R} = 1 - (1 - X_B)^{1/3}} \qquad \ldots (4.33)$$

Equation (4.33) gives the time t at which the core has a radius r_c and the fractional conversion X_B for the particle in terms of τ.

Rate of Reaction for Shrinking Spherical Particles (Variable / Changing Particle Size) :

When no ash (converted solids and inert material) forms (no ash layer cover the unreacted core as the reaction proceeds), the reacting particle continuously shrinks in size with time during reaction until it disappears. This process is illustrated in Fig. 4.8.

In case of such reactions as no ash layer is present, the following three steps occur in succession/series :

(i) Diffusion of gaseous reactant A through the gas film to the surface of the solid - from the main body of gas.

(ii) Reaction on the surface between gaseous reactant A and solid reactant B.

(iii) Diffusion of reaction products from the surface of the solid back into the main body of gas through the gas film.

Here the rate controlling step may be gas film diffusion or chemical reaction.

Chemical Reaction Controls (Controlling resistance-reaction step) :

The expressions derived for t, τ and X_B–t [Equations (4.29), (4.30) and (4.33)] for single particles of unchanging size are directly applicable to the shrinking spherical particles (spherical particles of changing size) when chemical reaction step controls the overall rate of reaction.

Diffusion through Gas Film Controls (Controlling resistance-film diffusion) :

When diffusion through the gas film controls, the situation becomes more complicated because of the changing particle size in turn changing the film resistance. Film resistance at the surface of a particle depends upon the following parameters :

(i) Size of particle,

(ii) Properties of fluid,

(iii) Relative velocity between particle and solid.

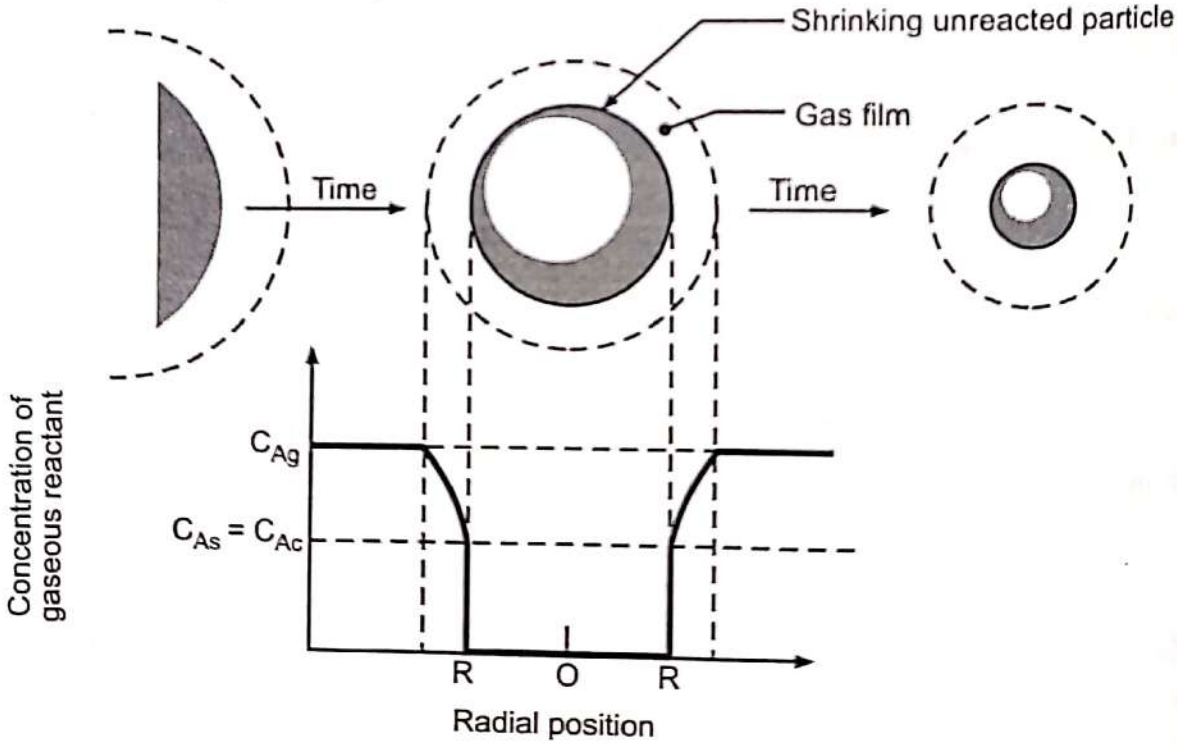


Fig. 4.8 : Concentration profile of a gaseous reactant A for the reaction between a shrinking solid particle and gaseous reactant A as per the stoichiometry :

$$\mathbf{A\ (g) + bB\ (s) \rightarrow rR\ (g)}$$

These parameters have been correlated for various ways of contacting fluid with solid. The correlation for mass transfer of a component with mole fraction y in a fluid to free-falling solids is :

$$\frac{k_g\, dp\, y}{\mathscr{D}} = 2 + 0.6 N_{Re}^{1/2}\, N_{Sc}^{1/3} \qquad \ldots (4.34)$$

$$N_{Sc} = \frac{\mu}{\rho \mathscr{D}} \qquad \ldots \text{ Schmidt number}$$

$$N_{Re} = \frac{dp\, u\rho}{\mu} \qquad \ldots \text{ Reynolds number}$$

$$\frac{k_g\, dp\, y}{\mathscr{D}} = 2 + 0.6 \left(\frac{\mu}{\rho \mathscr{D}}\right)^{1/3} \left(\frac{dp\, u\, \rho}{\mu}\right)^{1/2} \qquad \ldots (4.35)$$

This relationship is referred to as the Froessling correlation.

Since a particle changes in size with time, k_g varies during reaction. In general, k_g increases with increase in gas velocity and with smaller particles.

$$k_g \propto \frac{1}{dp} \qquad \ldots \text{ for small dp and low u} \qquad \ldots (4.36)$$

$$k_g \propto \frac{u^{1/2}}{d_p^{1/2}} \qquad \ldots \text{ for large dp and high u} \qquad \ldots (4.37)$$

Equation (4.36) represents particles in the Stokes law regime $\left[\frac{dp\, u\, \rho}{\mu} < 0.3\right]$

Let us derive the conversion-time relationship for such particles.

Stokes Regime – Small Particles :

At the time when a particle of the original size R_o has shrunk to a size R, we can write

$$dN_B = d(\rho_B V) = \rho_B\, dV$$

$$= \rho_B\, d\left(\frac{4}{3}\pi R^3\right) = 4\pi\rho_B R^2\, dR \qquad \ldots (4.38)$$

The rate of reaction in terms of size R of the shrinking unreacted particle is given by an expression similar to Equation (4.24).

$$-\frac{1}{S_{ex}}\frac{dN_B}{dt} = b k_g C_{Ag} \qquad \ldots (4.39)$$

Replacing dN_B in Equation (4.39) by its value given by Equation (4.38) and replacing S_{ex} by $4\pi R^2$ (the surface area of shrinking unreacted particle) in Equation (4.39), we get

$$-\frac{1}{S_{ex}}\frac{dN_B}{dt} = -\left(\frac{1}{4\pi R^2}\right) 4\pi\rho_B R^2 \frac{dR}{dt}$$

$$= -\rho_B \frac{dR}{dt} = b k_g C_{Ag} \qquad \ldots (4.40)$$

In the Stokes regime, Equation (4.35) reduces to

$$k_g = \frac{2\mathscr{D}}{dp} = \frac{\mathscr{D}}{R} \quad \ldots (4.41)$$

Therefore, Equation (4.40) becomes

$$-\rho_B \frac{dR}{dt} = \frac{b\mathscr{D}C_{Ag}}{R} \quad \ldots (4.42)$$

Rearranging, we get

$$\frac{dR}{dt} = -\frac{bC_{Ag}\mathscr{D}}{\rho_B R}$$

$$\int R\,dR = -\frac{bC_{Ag}\mathscr{D}}{\rho_B}\int dt$$

The limits of integration are :

At $t = 0$ ... $R = R_o$ and at $t = t$ ... $R = R$. Integrating gives

$$\therefore \int_{R_o}^{R} R\,dR = -\frac{bC_{Ag}\mathscr{D}}{\rho_B}\int_0^t dt$$

$$\left[\frac{R^2}{2} - \frac{R_o^2}{2}\right] = -\frac{bC_{Ag}\mathscr{D}t}{\rho_B}$$

Rearranging gives

$$t = -\frac{\rho_B}{bC_{Ag}\mathscr{D}}\left[\frac{R^2}{2} - \frac{R_o^2}{2}\right]$$

$$t = \frac{\rho_B}{2bC_{Ag}\mathscr{D}}\left(R_o^2 - R^2\right)$$

$$\boxed{t = \frac{\rho_B R_o^2}{2bC_{Ag}\mathscr{D}}\left[1 - \left(\frac{R}{R_o}\right)^2\right]} \quad \ldots (4.43)$$

The time for complete disappearance of a particle [i.e., for complete conversion] (τ) is obtained by replacing t by τ and putting $R = 0$ in Equation (4.43). Therefore,

$$\boxed{\tau = \frac{\rho_B R_o^2}{2bC_{Ag}\mathscr{D}}} \quad \ldots (4.44)$$

Dividing Equation (4.43) by Equation (4.44), we get

$$\frac{t}{\tau} = 1 - \left(\frac{R}{R_o}\right)^2 \quad \ldots (4.45)$$

Equation (4.45) can be written in terms of fractional conversion by noting that

$$1 - X_B = \left(\frac{R}{R_o}\right)^3$$

$$\therefore (1 - X_B)^{1/3} = \frac{R}{R_o} \quad \ldots (4.46)$$

Therefore, with this Equation (4.45) becomes

$$\boxed{\frac{t}{\tau} = 1 - \left(\frac{R}{R_o}\right)^2 = 1 - (1 - X_B)^{2/3}} \quad \ldots (4.47)$$

Equation (4.47) expresses the relationship between size and time for shrinking particles in the Stokes regime and well represents small burning solid particles as well as small burning liquid droplets.

SCM for Cylindrical Particles (radius R and length L) of Unchanging Size/Fixed Size) :

1. Diffusion Through Gas Film Controls :

Rate of reaction is given by

$$-\frac{1}{S_{ex}}\frac{dN_B}{dt} = \frac{b}{S_{ex}}\frac{dN_A}{dt} = bk_g C_{Ag} = \text{constant}$$

S_{ex} = unchanging exterior surface of the cylindrical particle

$$= \pi DL = 2\pi RL$$

$$-dN_B = -\rho_B\, dV = -\rho_B\, d(\pi r_c^2 L) = -2\pi\rho_B L r_c\, dr_c$$

$$-\frac{1}{S_{ex}}\frac{dN_B}{dt} = bk_g C_{Ag}$$

Substituting for S_{ex} and dN_B, we get

$$-\frac{2\pi\rho_B L r_c}{2\pi RL}\frac{dr_c}{dt} = -\rho_B \frac{r_c}{R}\frac{dr_c}{dt} = bk_g C_{Ag}$$

$$-\frac{\rho_B}{R}\int_R^{r_c} r_c\, dr_c = bk_g C_{Ag}\int_0^t dt$$

$$-\frac{\rho_B}{R}\left[\frac{r_c^2}{2} - \frac{R^2}{2}\right] = bk_g C_{Ag}\, t$$

$$\frac{\rho_B}{R}\left[\frac{R^2}{2} - \frac{r_c^2}{2}\right] = bk_g C_{Ag}\, t$$

$$t = \frac{\rho_B}{bk_g C_{Ag} R}\left[\frac{R^2}{2} - \frac{r_c^2}{2}\right]$$

$$\boxed{t = \frac{\rho_B R}{2bk_g C_{Ag}}\left[1 - \left(\frac{r_c}{R}\right)^2\right]} \quad \ldots (4.48)$$

τ is obtained by putting $r_c = 0$. Replacing for t by τ and putting $r_c = 0$ in the above equation, we get

$$\boxed{\tau = \frac{\rho_B R}{2bk_g C_{Ag}}} \quad \ldots (4.49)$$

Dividing Equation (4.48) by Equation (4.49) gives

$$\frac{t}{\tau} = 1 - \left(\frac{r_c}{R}\right)^2 \quad \ldots (4.50)$$

This can be written in terms of fractional conversion by noting that

$$1 - X_B = \frac{\text{Volume of unreacted core}}{\text{Total volume of particle}}$$

For cylindrical particles, it becomes

$$1 - X_B = \frac{\pi r_c^2 L}{\pi R^2 L} = \left(\frac{r_c}{R}\right)^2 \quad \ldots (4.51\,a)$$

$$\therefore \quad 1 - \left(\frac{r_c}{R}\right)^2 = X_B \quad \ldots (4.51\,b)$$

Combining Equations (4.50) and (4.51 b), we get

$$\boxed{\frac{t}{\tau} = 1 - \left(\frac{r_c}{R}\right)^2 = X_B} \quad \ldots (4.52)$$

2. Chemical Reaction Controls :

Based on the surface of unreacted core r_c, we have

$$-\frac{1}{S_{ex,\,c}}\frac{dN_B}{dt} = -\frac{b}{S_{ex,\,c}}\frac{dN_A}{dt} = -\frac{b}{S_{ex,\,c}}\frac{dN_A}{dt} = bk''C_{Ag}$$

$$-\frac{1}{2\pi r_c L}\frac{dN_B}{dt} = -\frac{b}{2\pi r_c L}\frac{dN_A}{dt} = bk''C_{Ag}$$

$$-\frac{1}{2\pi r_c L}\frac{dN_B}{dt} = bk''C_{Ag}$$

$$-dN_B = -\rho_B\, dV = -\rho_B\, d(\pi r_c^2 L)$$

$$= -2\pi\rho_B L r_c\, dr_c$$

$$\therefore \quad -\frac{2\pi\rho_B L}{2\pi r_c L}\, r_c \frac{dr_c}{dt} = bk''C_{Ag}$$

$$-\rho_B\, dr_c = bk''C_{Ag}\, dt$$

$$-\rho_B \int_R^{r_c} dr_c = bk''C_{Ag} \int_0^t dt$$

$$-\rho_B [r_c - R] = \rho_B (R - r_c) = bk''C_{Ag}\, t$$

$$\therefore \quad t = \frac{\rho_B}{bk''C_{Ag}}(R - r_c)$$

$$\therefore \quad \boxed{t = \frac{\rho_B R}{bk''C_{Ag}}\left[1 - \left(\frac{r_c}{R}\right)\right]} \quad \ldots (4.53)$$

The time τ required for complete conversion is obtained by putting $r_c = 0$ and $t = \tau$ in the above equation.

$$\boxed{\tau = \frac{\rho_B R}{b k'' C_{Ag}}} \quad \ldots (4.54)$$

Dividing Equation (4.53) by Equation (4.54), we get

$$\frac{t}{\tau} = 1 - \left(\frac{r_c}{R}\right) \quad \ldots (4.55)$$

We have for cylindrical particles :

$$1 - X_B = \left(\frac{r_c}{R}\right)^2$$

$$(1 - X_B)^{1/2} = \frac{r_c}{R} \quad \ldots (4.56)$$

Combining Equations (4.55) and (4.56), we get

$$\boxed{\frac{t}{\tau} = 1 - \left(\frac{r_c}{R}\right) = 1 - (1 - X_B)^{1/2}} \quad \ldots (4.57)$$

3. Diffusion through ash layer controls :

We have : $$-\frac{dN_A}{dt} = 2\pi r L Q_A$$

and $$Q_A = \mathscr{D}_e \frac{dC_A}{dr}$$

$$-\frac{dN_A}{dt} = 2\pi r L \mathscr{D}_e \frac{dC_A}{dr}$$

$$-\frac{dN_A}{dt} \int_R^{r_c} \frac{dr}{r} = 2\pi L \mathscr{D}_e \int_{C_{Ag}}^{0} dC_A$$

$$-\frac{dN_A}{dt} [\ln r_c - \ln R] = -2\pi L \mathscr{D}_e C_{Ag}$$

$$-\frac{dN_A}{dt} [\ln R - \ln r_c] = 2\pi L \mathscr{D}_e C_{Ag}$$

$$-\frac{dN_A}{dt} = \frac{2\pi L \mathscr{D}_e C_{Ag}}{\ln R - \ln r_c}$$

We have : $$-dN_B = -\rho_B dV = -\rho_B (\pi r_c^2 L) = -2\pi\rho_B L r_c dr_c$$

We have : $$-\frac{dN_A}{dt} = -\frac{1}{b}\frac{dN_B}{dt}$$

Substituting for $-\frac{dN_A}{dt}$, we get

$$-\frac{1}{b}\frac{dN_B}{dt} [\ln R - \ln r_c] = 2\pi L \mathscr{D}_e C_{Ag}$$

Substituting for dN_B in the above equation gives

$$-\frac{2\pi L\rho_B r_c}{b}\frac{dr_c}{dt}[\ln R - \ln r_c] = 2\pi L \mathscr{D}_e C_{Ag}$$

$$-\frac{\rho_B}{b}\int_R^{r_c}[\ln R - \ln r_c]\, r_c\, dr_c = \mathscr{D}_e C_{Ag}\int_0^t dt$$

$$-\int_R^{r_c}[\ln R - \ln r_c]\, r_c\, dr_c = \frac{b\mathscr{D}_e C_{Ag} t}{\rho_B}$$

$$-\int_R^{r_c} r_c[\ln R - \ln r_c]\, dr_c = -\int_R^{r_c}[r_c \ln R]\, dr_c + \int_R^{r_c} r_c \ln r_c\, dr_c$$

Integrating the second term by parts, we get

$$= -\left[\ln R \cdot \frac{r_c^2}{2}\right]_R^{r_c} + \left[\frac{r_c^2}{2}\ln r_c - \frac{r_c^2}{4}\right]_R^{r_c}$$

$$= -\left[\ln R \cdot \frac{r_c^2}{2} - \frac{R^2}{2}\ln R\right] + \left[\frac{r_c^2}{2}\ln r_c - \frac{r_c^2}{4} - \frac{R^2}{2}\ln R + \frac{R^2}{4}\right]$$

$$= -\frac{r_c^2}{2}\ln R + \frac{R^2}{2}\ln R + \frac{r_c^2}{2}\ln r_c - \frac{r_c^2}{4} - \frac{R^2}{2}\ln R + \frac{R^2}{4}$$

$$= \frac{R^2}{4} - \frac{r_c^2}{4} + \frac{r_c^2}{2}\ln r_c - \frac{r_c^2}{2}\ln R$$

$$= \frac{R^2}{4} - \frac{r_c^2}{4} + \frac{r_c^2}{2}[\ln r_c - \ln R]$$

$$= \frac{R^2}{4} - \frac{r_c^2}{4} + \frac{r_c^2}{2}\ln\left(\frac{r_c}{R}\right)$$

$$= \frac{R^2}{4}\left[1 - \left(\frac{r_c}{R}\right)^2 + 2\left(\frac{r_c}{R}\right)^2 \ln\left(\frac{r_c}{R}\right)\right]$$

$$\therefore \frac{R^2}{4}\left[1 - \left(\frac{r_c}{R}\right)^2 + 2\left(\frac{r_c}{R}\right)^2 \ln\left(\frac{r_c}{R}\right)\right] = \frac{b\mathscr{D}_e C_{Ag} t}{\rho_B}$$

$$\therefore \quad t = \frac{\rho_B R^2}{4b\mathscr{D}_e C_{Ag}}\left[1 - \left(\frac{r_c}{R}\right)^2 + 2\left(\frac{r_c}{R}\right)^2 \ln\left(\frac{r_c}{R}\right)\right] \quad \ldots (4.58)$$

When $r_c = 0$, t becomes τ ... time for complete conversion and therefore the above equation reduces to

$$\boxed{\tau = \frac{\rho_B R^2}{4b\mathscr{D}_e C_{Ag}}} \quad \ldots (4.59)$$

Dividing Equation (4.58) by Equation (4.59), we get

$$\frac{t}{\tau} = 1 - \left(\frac{r_c}{R}\right)^2 + 2\left(\frac{r_c}{R}\right)^2 \ln\left(\frac{r_c}{R}\right) \quad \ldots (4.60)$$

We have : $1 - X_B = \left(\frac{r_c}{R}\right)^2$

$\therefore$ $(1 - X_B)^{1/2} = \frac{r_c}{R}$

Therefore, Equation (4.60) in terms of fractional conversion, by replacing r_c/R by its value cited above, becomes

$$\frac{t}{\tau} = 1 - \left[(1 - X_B)^{1/2}\right]^2 + 2\left[(1 - X_B)^{1/2}\right]^2 \ln (1 - X_B)^{1/2}$$

$$\frac{t}{\tau} = 1 - (1 - X_B) + 2(1 - X_B) \ln (1 - X_B)^{1/2}$$

$$\frac{t}{\tau} = 1 - 1 + X_B + 2(1 - X_B) \times \frac{1}{2} \ln (1 - X_B)$$

$$\boxed{\frac{t}{\tau} = X_B + (1 - X_B) \ln (1 - X_B)} \quad \ldots (4.61)$$

Table 4.1 lists the conversion-time expressions for various shapes of particles (as per SCM).

Table 4.1

Controlling Resistance	Constant size particles		
	Flat plate	**Cylinder**	**Sphere**
	$X_B = 1 - \frac{l}{L}$ $L = \frac{\text{Thickness}}{2}$	$X_B = 1 - \left(\frac{r_c}{R}\right)^2$	$X_B = 1 - \left(\frac{r_c}{R}\right)^3$
Film diffusion (diffusion through the gas film)	$\frac{t}{\tau} = X_B$ $\tau = \frac{\rho_B L}{b k_g C_{Ag}}$	$\frac{t}{\tau} = X_B$ $\tau = \frac{\rho_B R}{2b k_g C_{Ag}}$	$\frac{t}{\tau} = X_B$ $\tau = \frac{\rho_B R}{3b k_g C_{Ag}}$
Ash diffusion (diffusion through the ash layer)	$\frac{t}{\tau} = X_B^2$ $\tau = \frac{\rho_B L^2}{2b \mathscr{D}_e C_{Ag}}$	$\frac{t}{\tau} = X_B + (1 - X_B) \ln (1 - X_B)$ $\tau = \frac{\rho_B R^2}{4b \mathscr{D}_e C_{Ag}}$	$\frac{t}{\tau} = 1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)$ $\tau = \frac{\rho_B R^2}{6b \mathscr{D}_e C_{Ag}}$
Chemical reaction	$\frac{t}{\tau} = X_B$ $\tau = \frac{\rho_B L}{b k'' C_{Ag}}$	$\frac{t}{\tau} = 1 - (1 - X_B)^{1/2}$ $\tau = \frac{\rho_B R}{b k'' C_{Ag}}$	$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$ $\tau = \frac{\rho_B R}{b k'' C_{Ag}}$

Table 4.1 continued.....

Controlling Resistance	Shrinking spheres	
	Small particles stokes regime	Large particle (u = constant)
Film diffusion (Diffusion through the gas film)	$\frac{t}{\tau} = 1-(1-X_B)^{2/3}$ $\tau = \frac{\rho_B R_o^2}{2b\mathscr{D}C_{Ag}}$	$\frac{t}{\tau} = 1-(1-X_B)^{1/2}$ $\tau = (\text{constant})\frac{R_o^{3/2}}{C_{Ag}}$
Chemical reaction	$\frac{t}{\tau} = 1-(1-X_B)^{1/3}$ $\tau = \frac{\rho_B R_o}{bk''C_{Ag}}$	$\frac{t}{\tau} = 1-(1-X_B)^{1/3}$ $\tau = \frac{\rho_B R}{bk''C_{Ag}}$

Combination of Resistances-Diffusion through-gas film + ash layer + Reaction step :

We have derived the above conversion - time expressions assuming that a single resistance controls throughout reaction of the solid particle with gas. However, the magnitude of these individual resistances will vary with conversion of the particle. For example, for a fixed / unchanging size particle the resistance of the gas film remains unchanged, the resistance to reaction increases as the conversion increases owing to a continuous decrease in the surface of unreacted core, while the ash layer resistance (which is absent at the start) increases as the ash layer builds up with the conversion of particle. Thus, in general, it seems that it is not reasonable to consider that just a single resistance (i.e., one step) controls throughout reaction of the particle.

Accounting of the simultaneous action of these three resistances is easy/simple as they act in series and are all linear in concentration. Thus combining Equations (4.9), (4.17) and (4.28) with their individual driving forces and eliminating intermediate concentrations we can prove that the time required to achieve a specific level of conversion is the sum of the times needed if each resistance acted alone or mathematically,

$$\boxed{t_{total} = t_{film\ alone} + t_{ash\ alone} + t_{reaction\ alone}} \quad \ldots (4.62)$$

Similarly, τ_{total} for complete conversion is given as

$$\boxed{\tau_{total} = \tau_{film\ alone} + \tau_{ash\ alone} + \tau_{reaction\ alone}} \quad \ldots (4.63)$$

On combining directly the individual resistances, the overall rate of reaction at any particular level of conversion is given by

$$-\frac{1}{S_{ex}}\frac{dN_B}{dt} = \frac{bC_A}{\underbrace{\frac{1}{k_g}}_{\text{film resistance}} + \underbrace{\frac{R(R-r_c)}{r_c\mathscr{D}_e}}_{\text{ash resistance}} + \underbrace{\frac{R^2}{r_c^2 k''}}_{\text{reaction resistance}}} \quad \ldots (4.63\,a)$$

or
$$-\frac{dr_c}{dt} = \frac{bC_A/\rho_B}{\frac{r_c^2}{R^2 k_g} + \frac{r_c(R-r_c)}{R\mathcal{D}_e} + \frac{1}{k''}} \qquad \text{... (4.63 b)}$$

gas film ash layer reaction step

Let us derive Equation (4.63 a).

For film diffusion controls :
$$-\frac{1}{S_{ex}}\frac{dN_B}{dt} = bk_g(C_{Ag} - C_{As}) = bk_g(C_A - C_{As}) \qquad \text{... (4.64)}$$
$$\frac{1}{k_g} = \frac{b}{\left(-\frac{1}{S_{ex}}\cdot\frac{dN_B}{dt}\right)}(C_A - C_{As}) \qquad \text{... (4.65)}$$

For ash diffusion controls :
$$-\frac{dN_A}{dt}\left[\frac{1}{r_c} - \frac{1}{R}\right] = 4\pi\mathcal{D}_e(C_{As} - C_{Ac})$$

We have,
$$-dN_A = \left(-\frac{1}{b}\right)dN_B$$

Combining the above two equations, we get

$\therefore$
$$-\frac{1}{b}\frac{dN_B}{dt}\left(\frac{1}{r_c} - \frac{1}{R}\right) = 4\pi\mathcal{D}_e(C_{As} - C_{Ac})$$
$$-\frac{dN_B}{dt}\left(\frac{R-r_c}{Rr_c}\right) = 4\pi b\mathcal{D}_e(C_{As} - C_{Ac})$$

Dividing both sides by S_{ex} gives
$$-\frac{1}{S_{ex}}\frac{dN_B}{dt}\left(\frac{R-r_c}{Rr_c}\right) = \frac{4\pi b\mathcal{D}_e(C_{As} - C_{Ac})}{S_{ex}}$$
$$-\frac{1}{S_{ex}}\frac{dN_B}{dt}\left(\frac{R-r_c}{Rr_c}\right) = \frac{4\pi b\mathcal{D}_e}{4\pi R^2}(C_{As} - C_{Ac}), \text{ since } S_{ex} = 4\pi R^2$$

Rearranging gives
$$\frac{R^2}{\mathcal{D}_e}\left(\frac{R-r_c}{Rr_c}\right) = \frac{b}{\left(-\frac{1}{S_{ex}}\cdot\frac{dN_B}{dt}\right)}(C_{As} - C_{Ac})$$
$$\frac{R(R-r_c)}{\mathcal{D}_e r_c} = \frac{b}{\left(-\frac{1}{S_{ex}}\cdot\frac{dN_B}{dt}\right)}(C_{As} - C_{Ac}) \qquad \text{... (4.66)}$$

For chemical reaction controls :
$$\frac{1}{4\pi r_c^2}\frac{dN_B}{dt} = bk''C_{Ac}$$
$$[C_{Ac} - C_{Ae} = C_{Ac} \text{ as } C_{Ae} = 0 \text{ for an irreversible reaction}]$$

Dividing both sides of the above equation by S_{ex} and rearranging, we get

$$-\frac{1}{S_{ex}}\frac{dN_B}{dt} = \frac{4\pi b r_c^2 k'' C_{Ac}}{S_{ex}}$$

$$-\frac{1}{S_{ex}}\frac{dN_B}{dt} = \frac{4\pi b k'' r_c^2 C_{Ac}}{4\pi R^2} = \frac{b k'' r_c^2 C_{Ac}}{R^2}$$

$$\frac{1}{r_c^2 k''/R^2} = \frac{R^2}{r_c^2 k''} = \frac{b}{\left(-\frac{1}{S_{ex}}\cdot\frac{dN_B}{dt}\right)} C_{Ac} \quad \ldots (4.67)$$

Adding Equations (4.65), (4.66) and (4.67), we get

$$\frac{1}{k_g} + \frac{R(R-r_c)}{\mathscr{D}_e r_c} + \frac{R^2}{r_c^2 k''} = \frac{b(C_A - C_{As})}{\left(-\frac{1}{S_{ex}}\cdot\frac{dN_B}{dt}\right)} + \frac{b(C_{As} - C_{Ac})}{\left(-\frac{1}{S_{ex}}\cdot\frac{dN_B}{dt}\right)} + \frac{bC_{Ac}}{\left(-\frac{1}{S_{ex}}\cdot\frac{dN_B}{dt}\right)}$$

$$\frac{1}{k_g} + \frac{R(R-r_c)}{\mathscr{D}_e r_c} + \frac{R^2}{r_c^2 k''} = \frac{b}{\left(-\frac{1}{S_{ex}}\cdot\frac{dN_B}{dt}\right)} (C_A - C_{As} + C_{As} - C_{Ac} + C_{Ac})$$

$$\frac{1}{k_g} + \frac{R(R-r_c)}{\mathscr{D}_e r_c} + \frac{R^2}{r_c^2 k''} = \frac{bC_A}{-\frac{1}{S_{ex}}\cdot\frac{dN_B}{dt}}$$

Rearranging the above equation, we get

$$\mathbf{-\frac{1}{S_{ex}}\cdot\frac{dN_B}{dt} = \frac{bC_A}{\frac{1}{k_g} + \frac{R(R-r_c)}{\mathscr{D}_e r_c} + \frac{R^2}{r_c^2 k''}}} \quad \ldots \text{(Equation (4.63 a)}$$

Taking into account the whole progress of reaction right from fresh to completely converted solid for a constant / fixed size particle, the relative contributions of these three resistances on the average is given by

$$\mathbf{-\frac{1}{S_{ex}}\frac{d\bar{N}_A}{dt} = \bar{k}'' C_A = \frac{1}{\frac{1}{k_g} + \frac{R}{2\mathscr{D}_e} + \frac{3}{k''}} C_A} \quad \ldots (4.68)$$

For ash-free particles which shrink with the progress of reaction there are only two resistances acting in series, namely, gas film and surface reaction. Taking these resistances into account, the overall rate of reaction based on the changing exterior surface of a particle is given by

$$-\frac{1}{S_{ex}}\frac{dN_A}{dt} = \frac{1}{\frac{1}{k_g} + \frac{1}{k''}} C_A \quad \ldots (4.69)$$

The time required to achieve the same fractional conversion for particles of different but fixed/constant/unchanging sizes is given by

$t \propto R^{1.5 \text{ to } 2.0}$ ... for film diffusion as the rate controlling step ... (4.70)

$t \propto R^2$ ... for ash diffusion as the rate controlling step ... (4.71)

$t \propto R$ ... for chemical reaction as the rate controlling step ... (4.72)

SOLVED EXAMPLES

Example 4.1 : *A batch of spherical solids (of single size, i.e., of uniform size) is treated by gas in a uniform environment. Solid is converted to a firm non-flaking product according to the shrinking core model (SCM). The conversion is 87.5% in a reaction time of 1 h and it is complete (100%) in 2 h. Determine the rate controlling mechanism.*

Solution : We get non-flaking product – it means particles remain unchanged in size after reaction.

For constant size spherical particles, the time-conversion expressions for different mechanisms are :

1. $\frac{t}{\tau} = X_B$ ..... film diffusion controls.

2. $\frac{t}{\tau} = 1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)$ ..... ash diffusion controls.

3. $\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$ ..... chemical reaction controls.

For any mechanism to be rate controlling the corresponding expression should be satisfied, i.e., data should fit the expression. Now let us check which of these mechanisms controls the rate.

Given : t = 1 h ..... $X_B = \frac{87.5}{100} = 0.875$ and $\tau = 2$ h.

(A) When film diffusion (i.e., resistance to diffusion through gas film) controls the rate ...

$$\frac{t}{\tau} = X_B \text{ ..... the date should satisfy this expression}$$

LHS of time-conversion expression :

$$\frac{t}{\tau} = \frac{1}{2} = 0.5$$

RHS of time-conversion expression :

$$X_B = 0.875$$

$$\therefore \quad \frac{t}{\tau} \neq X_B, \quad (0.5 \neq 0.875)$$

As the data provided do not fit the time-conversion expression ..... film diffusion (i.e., resistance to diffusion through gas film) does not control the rate ..... this mechanism is not rate controlling.

(B) When ash diffusion controls ..... time-conversion expression is

$$\frac{t}{\tau} = 1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)$$

For this to be rate controlling the above expression should fit the data provided.

LHS : $\frac{t}{\tau} = \frac{1}{2} = 0.5$

RHS : $1 - 3(1 - X_B)^{2/3} + 2(1 - X_B) = 1 - 3(1 - 0.875)^{2/3} + 2(1 - 0.875)$

$= 0.5$

As LHS = RHS .....

..... ash diffusion controls, i.e., resistance to diffusion through ash layer controls the overall rate.

Let us have check for the third mechanism.

(C) For chemical reaction controlling i.e. chemical step controlling the overall rate, we have

$$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$$

and for it to be rate controlling, the data should satisfy the above expression.

LHS : $\frac{t}{\tau} = \frac{1}{2} = 0.5$

RHS : $1 - (1 - X_B)^{1/3} = 1 - (1 - 0.875)^{1/3} = 0.5$

$\therefore$ LHS = RHS

So the data provided fit the time-conversion expression for this mechanism too.

So here we have a coincidence where both the ash diffusion and chemical reaction-mechanisms fit the data ... Ans.

By referring to the plot of $(1 - X_B)$ v/s $\frac{t}{\tau}$, we see that the data provided corresponds to precise point where the ash diffusion curve crosses that of chemical reaction controlling.

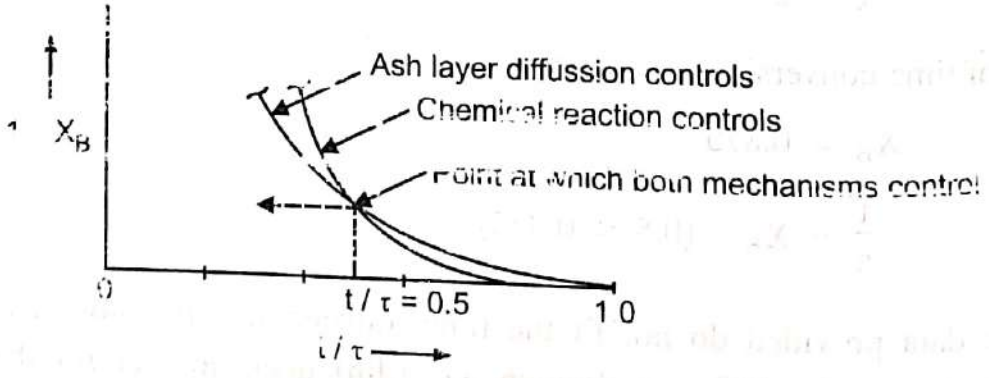


Fig. Ex 4.1

Example 4.2 : *Calculate the time required for complete burning of particles of graphite (size : R_0 = 5 mm, density : ρ_B = 2.2 g/cm^3) in an 8% oxygen stream at 900 °C and 1 atm.*

For the high gas velocity used assume that film diffusion does not offer any resistance to transfer and reaction.

Data : Rate constant = k'' = 20 cm/s

Solution : In burning of particles of graphite (pure C) there is no ash formation. The reacting particle shrinks in size as reaction progresses and finally disappears (on complete combustion).

As the ash layer is absent there are only two possible resistances :

(i) resistance offered by film diffusion.

(ii) resistance offered by chemical reaction.

But we are provided with : film diffusion does not offer any resistance.

So in our case we have only one resistance : resistance offered by chemical reaction, i.e., in our present case chemical reaction controls the rate.

For small spherical particles when chemical reaction controls the rate, the time required for complete conversion (i.e., the time required for complete burning of particle of graphite) is given by

$$\tau = \frac{\rho_B R_o}{bk'' C_{Ag}}$$

Combustion Reaction :

$C(s) + O_2(g) \rightarrow CO_2(g)$ ..... compare ..... $A(g) + bB(s) \rightarrow$ fluid product.

$\therefore$ b = 1 (stoichiometric coefficient of C : Carbon)

R_o = 5 mm = 0.5 cm

density of graphite = ρ_B = 2.2 g/cm^3

rate constant = k'' = 20 cm/s

Evaluate C_{Ag} : bulk gas phase concentration of A.

Here A is O_2.

$$C_{Ag} = \frac{p_{Ag}}{RT}$$

A in the gas stream : 8% by volume, total pressure = 1 atm.

For an ideal gas : volume % A = pressure % A

$$\text{pressure \% A} = 8 = \frac{p_{Ag}}{P} \times 100$$

$$\therefore \quad p_{Ag} = \frac{8 \times 1}{100} = 0.08 \text{ atm.}$$

$$T = 900\,^{\circ}C = 1173 \text{ K}$$

$$R = 82.06 \text{ (cm}^3\text{·atm)/mol.K}$$

$$C_{Ag} = \frac{0.08}{82.06 \times 1173} = 8.31 \times 10^{-7} \text{ mol/cm}^3$$

ρ_B = 2.2 g/cm^3 convert this mass density of carbon into molar density mol/cm^3

$$\rho_B = \frac{2.2}{12} = 0.183 \text{ mol/cm}^3, \text{ (At. Wt. C = 12)}$$

Substituting the values of R_o, ρ_B, etc. in Equation (1), we get

$$\tau = \frac{0.183 \times 0.50}{1 \times 20 \times 8.31 \times 10^{-7}} = 5505.4 \text{ s}$$

Time required for the complete burning of particles of graphite = 5505.4 s or ≃ 1.53 h ... Ans.

Example 4.3 : *Two small samples of solids are kept in a constant environment oven for period of 1 h. Under the conditions prevailing in the oven, the 4 mm particles are 57.8% converted, the 2 mm particles are 87.5% converted :*

(a) Find the rate controlling mechanism for the conversion of solids.

(b) Find the time required for complete conversion of 1 mm particle in this oven.

Solution : For constant size sphericle particles the possible mechanisms controlling the rate are :

(i) film diffusion

(ii) ash diffusion and

(iii) chemical reaction.

Data given :

R, mm	t, h	X_B
$R_1 = \frac{4}{2} = 2$	1	$X_{B1} = 0.578$
$R_2 = \frac{2}{2} = 1$	1	$X_{B2} = 0.875$

(i) For film diffusion controlling we have for constant size spherical particles :

$$\frac{t}{\tau} = X_B$$

and $$\tau \propto R$$

$$\frac{t}{\tau_1} = X_{B1} \quad \ldots\ldots\ldots (1) \ldots\ldots\ldots \text{for 4 mm particle}$$

$$\frac{t}{\tau_2} = X_{B2} \quad \ldots\ldots\ldots (2) \ldots\ldots\ldots \text{for 2 mm particle}$$

$$\tau_1 \propto R_1 \qquad \therefore \quad \tau_1 = c\,R_1, \qquad c = \text{constant}$$

$$\tau_2 \propto R_2 \qquad \therefore \quad \tau_2 = c\,R_2$$

From Equations (1) and (2), we get

$$t = X_{B1}\,\tau_1 = X_{B2}\,\tau_2$$

$$\therefore \quad X_{B1}\,c\,R_1 = X_{B2}\,c\,R_2$$

$$X_{B1}\,R_1 = X_{B2}\,R_2 \quad \ldots\ldots\ldots \text{this should be satisfied}$$

$$X_{B1}\,R_1 = 0.578 \times 2 = 1.158$$

$$X_{B2}\,R_2 = 0.875 \times 1 = 0.875$$

$$\therefore \quad X_{B1}\,R_1 \neq X_{B2}\,R_2 \ldots\ldots\ldots$$

..... so film diffusion does not control the rate.

OR :

We have :

$$\frac{t}{\tau_1} = X_{B1}$$

$$\frac{t}{\tau_2} = X_{B2}$$

$$\frac{\frac{t}{\tau_1}}{\frac{t}{\tau_2}} = \frac{X_{B1}}{X_{B2}}$$

$$\frac{\frac{t}{\tau_1}}{\frac{t}{\tau_2}} = \frac{\frac{t}{c\,R_1}}{\frac{t}{c\,R_2}} = \frac{R_2}{R_1} = \frac{X_{B1}}{X_{B2}}$$

$$\frac{R_2}{R_1} = \frac{1}{2} = 0.5$$

$$\frac{X_{B1}}{X_{B2}} = \frac{0.578}{0.875} = 0.66$$

$$\therefore \quad \frac{R_2}{R_1} \neq \frac{X_{B1}}{X_{B2}} \ldots\ldots\ldots$$

..... so film diffusion does not control the rate.

(B) For Ash diffusion controlling, we have

$$\frac{t}{\tau} = 1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)$$

and $\tau \propto R^2$

$$\therefore \quad \frac{t}{\tau_1} = 1 - 2(1 - X_{B1})^{2/3} + 2(1 - X_{B1}) \ldots\ldots\ldots \text{4 mm particle}$$

$$\tau_1 \propto R_1^2 \quad \therefore \ \tau_1 = c R_1^2$$

$$\frac{t}{\tau_2} = 1 - 3(1 - X_{B2})^{2/3} + 2(1 - X_{B2}) \ldots\ldots\ldots \text{2 mm particle}$$

$$\tau_2 \propto R_2^2 \quad \therefore \ \tau_2 = c R_2^2$$

$$\frac{\frac{t}{\tau_1}}{\frac{t}{\tau_2}} = \frac{\frac{t}{c R_1^2}}{\frac{t}{c R_2^2}} = \frac{R_2^2}{R_1^2} = \frac{1 - 3(1 - X_{B1})^{2/3} + 2(1 - X_{B1})}{1 - 3(1 - X_{B2})^{2/3} + 2(1 - X_{B2})}$$

$$\frac{R_2^2}{R_1^2} = \frac{(1)^2}{(2)^2} = \frac{1}{4} = 0.25$$

$$\frac{1 - 3(1 - X_{B1})^{2/3} + 2(1 - X_{B1})}{1 - 3(1 - X_{B2})^{2/3} + 2(1 - X_{B2})} = \frac{1 - 3(1 - 0.578)^{2/3} + 2(1 - 0.578)}{1 - 3(1 - 0.875)^{2/3} + 2(1 - 0.875)}$$

$$= 0.312$$

$$0.25 \neq 0.312$$

$$\therefore \quad \frac{\frac{t}{\tau_1}}{\frac{t}{\tau_2}} = \frac{R_2^2}{R_1^2} \neq \frac{1 - 3(1 - X_{B1})^{2/3} + 2(1 - X_{B1})}{1 - 3(1 - X_{B2})^{2/3} + 2(1 - X_{B2})}$$

..... so ash diffusion does not control the rate.

(C) For chemical reaction controlling, we have

$$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$$

and $\tau \propto R$

$$\frac{t}{\tau_1} = 1 - (1 - X_{B1})^{1/3} \ldots\ldots (1) \ldots\ldots \text{4 mm particles}$$

$$\tau_1 \propto R_1 \quad \therefore \ \tau_1 = c R_1$$

$$\frac{t}{\tau_2} = 1 - (1 - X_{B2})^{1/3} \ldots\ldots (2) \ldots\ldots \text{2 mm particles}$$

$$\tau_2 \propto R_2 \quad \therefore \ \tau_2 = c R_2$$

From Equations (1) and (2), we get

$$t = \tau_1 \left[1 - (1 - X_{B1})^{1/3}\right] = \tau_2 \left[1 - (1 - X_{B2})^{1/3}\right]$$

$$c\,R_1 \left[1 - (1 - X_{B1})^{1/3}\right] = c\,R_2 \left[1 - (1 - X_{B2})^{1/3}\right]$$

$$R_1 \left[1 - (1 - X_{B1})^{1/3}\right] = R_2 \left[1 - (1 - X_{B2})^{1/3}\right]$$

$$\text{LHS} : R_1 \left[1 - (1 - X_{B1})^{1/3}\right] = 2 \left[1 - (1 - 0.578)^{1/3}\right]$$

$$= 0.50$$

$$\text{RHS} : R_2 \left[1 - (1 - X_{B2})^{1/3}\right] = 1 \left[1 - (1 - 0.875)^{1/3}\right]$$

$$= 0.50$$

$\therefore$ LHS = RHS

As LHS = RHS ..... Chemical reaction controls the rate **... Ans. (a)**

Rate of controlling mechanism : Chemical reaction **... Ans. (a)**

OR

$$\frac{\frac{t}{\tau_1}}{\frac{t}{\tau_2}} = \frac{\frac{t}{c\,R_1}}{\frac{t}{c\,R_2}} = \frac{R_2}{R_1} = \frac{1 - (1 - X_{B1})^{1/3}}{1 - (1 - X_{B2})^{1/3}}$$

$$\text{LHS} : \frac{R_2}{R_1} = \frac{1}{2} = 0.50$$

$$\text{RHS} : \frac{1 - (1 - X_{B1})^{1/3}}{1 - (1 - X_{B2})^{1/3}} = \frac{1 - (1 - 0.578)^{1/3}}{1 - (1 - 0.875)^{1/3}}$$

$$= \frac{0.25}{0.5} = 0.50$$

$\therefore$ LHS = RHS

So, chemical reaction controls the rate **... Ans. (a)**

We have for 2 mm particle :

$t = 1$ h, $X_B = 0.875$

For chemical reaction controlling, we have

$$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$$

$$\frac{1}{\tau} = 1 - (1 - 0.875)^{1/3}$$

$\therefore$ $\tau = 2$ h

Time required for 2 mm particle (R = 1 mm) for complete conversion is 2 h.

Now we have to find out the time required for complete conversion of 1 mm particle.

For chemical reaction controlling, we have

$$\tau \propto R$$

Let τ_1..... for 2 mm particle ($R_1 = 1$ mm)

τ_2..... for 1 mm particle ($R_2 = 0.5$ mm)

$$\tau_1 \propto R_1$$

$$\tau_2 \propto R_2$$

$$\therefore \quad \frac{\tau_1}{\tau_2} = \frac{R_1}{R_2}$$

$$\tau_2 = \tau_1 \frac{R_2}{R_1} = 2 \times \frac{0.5}{1} = 1 \text{ h}$$

∴ Time required for complete conversion for 1 mm particle = 1 h ... Ans. (b)

Example 4.4 : *Spherical solid particles containing B are roasted at constant temperature in an oven by gas of constant composition. Solids are converted to give a firm non-flaking product according to the shrinking-core model (SCM). From the following conversion data determine the rate controlling mechanism for the transformation of solid.*

Data :

dp, mm	X_B	t, s
2	0.875	1
1	1	1

Solution : Here as the solids are converted to a firm non-flaking product, the final solids after reaction remain unchanged in size.

So we have constant size spherical particles.

Now we will determine the mechanism that controls the rate of the transformation of solid.

1. **Film diffusion controls :** For diffusion through gas film controls we have the following conversion-time expression.

$$\frac{t}{\tau} = X_B$$

and $$\tau \propto R$$

Data : $dp_1 = 2$ mm, $t_1 = 1$ s, $X_{B1} = 0.875$

and $dp_2 = 1$ mm, $t_2 = 1$ s, $X_{B2} = 1.0$

For 1 mm particle : $t_2 = \tau_2 = 1$ s as conversion is complete ($X_{B2} = 1$)

$$\left(\frac{t}{\tau}\right) 1 = X_{B1}$$

$$\frac{t_1}{\tau_1} = X_{B1}$$

$$\tau_1 = \frac{t_1}{X_{B1}}$$

$$= \frac{1}{0.875} = 1.143$$

$$\tau \propto R$$

$$\tau_1 \propto R_1 \quad \text{and} \quad \tau_2 \propto R_2$$

$$\therefore \quad \frac{\tau_1}{\tau_2} = \frac{R_1}{R_2} = \frac{dp_1}{dp_2}$$

LHS : $$\frac{\tau_1}{\tau_2} = \frac{1.143}{1} = 1.143$$

RHS : $$\frac{R_1}{R_2} = \frac{dp_1}{dp_2} = \frac{2}{1} = 2$$

$$\therefore \quad \text{LHS} \neq \text{RHS}$$

$$\frac{\tau_1}{\tau_2} \neq \frac{dp_1}{dp_2}$$

So film diffusion does not control the rate.

or $$\frac{\frac{t}{\tau_1}}{\frac{t}{\tau_2}} = \frac{\frac{t_1}{c\,R_1}}{\frac{t_2}{c\,R_2}} = \frac{X_{B1}}{X_{B2}}$$

$$\frac{t_1 R_2}{t_2 R_1} = \frac{t_1\, dp_2}{t_2\, dp_1} = \frac{X_{B1}}{X_{B2}}$$

LHS : $$\frac{t_1\, dp_2}{t_2\, dp_1} = \frac{1 \times 1}{1 \times 2} = 0.5$$

RHS : $$\frac{X_{B1}}{X_{B2}} = \frac{0.875}{1} = 0.875$$

$$\text{LHS} \neq \text{RHS}$$

$\therefore$ Film diffusion does not control the rate.

2. **Ash diffusion controls :** For this mechanism, the conversion-time expression for constant size spherical particle is

$$\frac{t}{\tau} = 1 - 3\,(1 - X_B)^{2/3} + 2\,(1 - X_B)$$

and $\tau \propto R^2 \qquad \tau = c R^2$

$$\frac{t_1}{\tau_1} = \frac{t_1}{c R_1^2} = 1 - 3 (1 - X_{B1})^{2/3} + 2 (1 - X_{B1})$$

$$\frac{t_2}{\tau_2} = \frac{t_2}{c R_2^2} = 1 - 3 (1 - X_{B2})^{2/3} + 2 (1 - X_{B2})$$

$$\frac{\frac{t_1}{\tau_1}}{\frac{t_2}{\tau_2}} = \frac{t_1 R_2^2}{t_2 R_1^2} = \frac{1 - 3 (1 - X_{B1})^{2/3} + 2 (1 - X_{B1})}{1 - 3 (1 - X_{B2})^{2/3} + 2 (1 - X_{B2})}$$

LHS : $$\frac{t_1 R_2^2}{t_2 R_1^2} = \frac{t_1 (dp_2)^2}{t_2 (dp_1)^2} = \frac{1 \times (1)^2}{1 \times (2)^2} = 0.25$$

RHS : $$\frac{1 - 3 (1 - X_{B1})^{2/3} + 2 (1 - X_{B1})}{1 - 3 (1 - X_{B2})^{2/3} + 2 (1 - X_{B2})} = \frac{1 - 3 (1 - 0.875)^{2/3} + 2 (1 - 0.875)}{1 - 3 (1 - 1)^{2/3} + 2 (1 - 1)}$$

$$= 0.5$$

LHS $\neq$ RHS

$\therefore$ Ash diffusion does not control the rate.

3. Chemical reaction controls :

For this we have

$$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$$

and $\tau \propto R$

$$\frac{t_1}{\tau_1} = 1 - (1 - X_{B1})^{1/3}$$

For $dp_1 = 2$ mm, $t_1 = 1$ s, $X_{B1} = 0.875$

$$\frac{1}{\tau_1} = 1 - (1 - 0.875)^{1/3}$$

$$\tau_1 = 2 \text{ s}$$

For $dp_2 = 1$ mm, $\tau_2 = t_2 = 1$ (as conversion is complete)

$\tau_1 \propto R_1$ and $\tau_2 \propto R_2$

$$\frac{\tau_1}{\tau_2} = \frac{R_1}{R_2} = \frac{dp_1}{dp_2}$$

LHS : $$\frac{\tau_1}{\tau_2} = \frac{2}{1} = 2$$

RHS : $$\frac{R_1}{R_2} = \frac{dp_1}{dp_2} = \frac{2}{1} = 2$$

LHS = RHS

∴ **Chemical reaction controls the rate**
i.e., this transformation of solid is chemical reaction controlling **... Ans.**

OR : $$\frac{t_1}{\tau_1} = 1 - (1 - X_{B1})^{1/3}$$

$$\frac{t_2}{\tau_2} = 1 - (1 - X_{B2})^{1/3}$$

$$\frac{\frac{t_1}{\tau_1}}{\frac{t_2}{\tau_2}} = \frac{\frac{t_1}{c\,R_1}}{\frac{t_2}{c\,R_2}} = \frac{1 - (1 - X_{B1})^{1/3}}{1 - (1 - X_{B2})^{1/3}}$$

$$\frac{t_1 R_2}{t_2 R_1} = \frac{t_1 dp_2}{t_2 dp_1} = \frac{1 - (1 - X_{B1})^{1/3}}{1 - (1 - X_{B2})^{1/3}}$$

$dp_1 = 2$ mm, $dp_2 = 1$ mm, $t_1 = 1$s, $t_2 = 1$s

LHS : $$\frac{t_1 R_2}{t_2 R_1} = \frac{t_1 dp_2}{t_2 dp_1} = \frac{1 \times 1}{1 \times 2} = 0.5$$

$X_{B1} = 0.875$ and $X_{B2} = 1$

RHS : $$\frac{1 - (1 - X_{B1})^{1/3}}{1 - (1 - X_{B2})^{1/3}} = \frac{1 - (1 - 0.875)^{1/3}}{1 - (1 - 1)^{1/3}} = 0.5$$

LHS = RHS

So chemical reaction controls the rate of transformation of solid **... Ans.**

OR

Rate controlling mechanism for the transformation of solid : chemical reaction ... Ans.

Example 4.5 : *Repeat Example 4.4 for the following conversion-time data.*

Data :

dp, mm	X_B	t, s
1	1	200
1.5	1	450

Solution : Here conversion in case of both the spherical particles is complete ($X_B = 1$) therefore, the time given is τ.

For $dp_1 = 1$ mm $\quad \tau_1 = t_1 = 200$ s

For $dp_2 = 1.5$ mm $\quad \tau_2 = t_2 = 450$ s

1. Film diffusion controls :

We have $\tau \propto R$

$\tau_1 \propto R_1$ and $\tau_2 \propto R_2$

$$\therefore \quad \frac{\tau_1}{\tau_2} = \frac{R_1}{R_2} = \frac{dp_1}{dp_2}$$

LHS : $$\frac{\tau_1}{\tau_2} = \frac{200}{450} = 0.444$$

RHS : $$\frac{R_1}{R_2} = \frac{dp_1}{dp_2} = \frac{1}{1.5} = 0.667$$

$$\text{LHS} \neq \text{RHS}$$

$\therefore$ Film diffusion does not control the rate.

2. Ash diffusion controls :

We have $\tau \propto R^2$

$\tau_1 \propto R_1^2$ and $\tau_2 \propto R_2^2$

$$\therefore \quad \frac{\tau_1}{\tau_2} = \frac{R_1^2}{R_2^2} = \frac{(dp_1)^2}{(dp_2)^2}$$

LHS : $$\frac{\tau_1}{\tau_2} = \frac{200}{450} = 0.444$$

RHS : $$\frac{R_1^2}{R_2^2} = \frac{(dp_1)^2}{(dp_2)^2} = \frac{(1)^2}{(1.5)^2} = 0.444$$

$$\text{LHS} = \text{RHS}$$

$\therefore$ **Ash diffusion controls the rate of the transformation of solid** ... Ans.

OR

Rate controlling mechanism for the transformation of solid : Ash diffusion ... Ans.

3. Chemical reaction controls : We have again $\tau \propto R$ as that with the film diffusion. Therefore, we obtain the same results as that for the film diffusion.

So the chemical reaction does not control the rate.

Example 4.6 : *Repeat Example 4.4 for the following conversion-time data.*

Data :

dp, mm	X_B	t, s
1	0.3	2
1	0.75	5

Solution : Here the data are provided for a single size spherical particle.

$dp_1 = dp_2 = 1$ mm

$t_1 = 2$ s ..... $X_{B1} = 0.30$

$t_2 = 5$ s ..... $X_{B2} = 0.75$

1. Film diffusion controls :

For this we have :

$$\frac{t}{\tau} = X_B$$

$$\frac{t_1}{\tau_1} = X_{B1}$$

$$\frac{t_1}{\tau_2} = X_{B2}$$

$$\frac{\frac{t_1}{\tau_1}}{\frac{t_2}{\tau_2}} = \frac{X_{B1}}{X_{B2}}$$

but $\tau_1 = \tau_2 = \tau$ for a single size particle

$$\frac{t_1}{t_2} = \frac{X_{B1}}{X_{B2}}$$

LHS : $\frac{t_1}{t_2} = \frac{2}{5} = 0.40$

RHS : $\frac{X_{B1}}{X_{B2}} = \frac{0.30}{0.75} = 0.40$

LHS = RHS

$\therefore$ **Film diffusion controls the rate** **... Ans.**

OR

$$\frac{t_1}{\tau_1} = X_{B1}$$

$$\tau_1 = \frac{t_1}{X_{B1}} = \frac{2}{0.3} = 6.67 \text{ s}$$

$$\frac{t_2}{\tau_2} = X_{B2}$$

$$\tau_2 = \frac{t_2}{X_{B2}}$$

$$\tau_2 = \frac{t_2}{X_{B2}}$$

$$= \frac{5}{0.75} = 6.67 \text{ s}$$

Therefore, the transformation of solid is film diffusion controlling. **... Ans.**

2. Chemical reaction controls :

We have $$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$$

$$\tau \propto R$$

$$\frac{\frac{t_1}{\tau_1}}{\frac{t_2}{\tau_2}} = \frac{\frac{t_1}{c\,R_1}}{\frac{t_2}{c\,R_2}} = \frac{\frac{t_1}{c\,dp_1}}{\frac{t_2}{c\,dp_2}} = \frac{1 - (1 - X_{B1})^{1/3}}{1 - (1 - X_{B2})^{1/3}}$$

We have $dp_1 = dp_2 = 1$ mm

$$\therefore \quad \frac{t_1}{t_2} = \frac{1 - (1 - X_{B1})^{1/3}}{1 - (1 - X_{B2})^{1/3}}$$

LHS : $$\frac{t_1}{t_2} = \frac{2}{5} = 0.40$$

RHS : $$\frac{1 - (1 - X_{B1})^{1/3}}{1 - (1 - X_{B2})^{1/3}} = \frac{1 - (1 - 0.30)^{1/3}}{1 - (1 - 0.75)^{1/3}} = 0.303$$

$$\text{LHS} \neq \text{RHS}$$

$\therefore$ **Chemical reaction does not control the rate** ... Ans.

Example 4.7 : *Spherical particles of zinc sulphide of size R = 1 mm are roasted in an 8% oxygen stream at 900 °C and 1 atm.*

The reaction that proceeds according to the shrinking-core model (SCM) is as follows :

$$2\,ZnS + 2\,O_2 \rightarrow 2\,ZnO + 2\,SO_2$$

Using the following data, calculate :

(i) the time required for complete conversion of a particle.

(ii) the relative resistance of ash layer diffusion during the roasting operation.

Data : $\rho_B = 4.13$ g/cm^3 $= 0.0425$ mol/cm^3

$k'' = 2.2$ cm^2/s

$\mathcal{D}_e = 0.08$ cm^2/s

Film resistance can safely be neglected as long as a growing ash layer is present.

Solution : $2\,ZnS + 3\,O_2 \rightarrow 2\,ZnO + 2\,SO_2$

Specific : $\frac{2}{3}\,ZnS + O_2 \rightarrow \frac{2}{3}\,ZnO + \frac{2}{3}\,SO_2$

General : A (fluid) + b B (solid) $\rightarrow$ Fluid and solid products

Comparing the above two reactions,

$b = \frac{2}{3}$

$R = 1 \text{ mm} = 0.10 \text{ cm}$

$\rho_B = 4.13 \text{ g/cm}^3$

$T = 900\,^{\circ}\text{C} = 1173 \text{ K}, \quad P = 1 \text{ atm}$

$k'' = 2.2 \text{ cm/s}, \quad \mathscr{D}_e = 0.08 \text{ cm}^2/\text{s}$

In our case : $A = O_2$

Volume % of O_2 in the gas stream = 8

Mole % O_2 = Volume % O_2 (for an ideal gas)

Therefore, Mole % $O_2 = 8$

$\therefore$ Mole fraction $O_2 = \frac{8}{100} = 0.08$

$P = 1 \text{ atm}$

$\therefore \qquad p_{Ag} = x_A \cdot P = 0.08 \times 1 = 0.08 \text{ atm}$

C_{Ag} = concentration of A in the bulk gas phase

The concentration of A is given by

$$C_{Ag} = \frac{p_{Ag}}{RT}$$

where $R = 82.06 \text{ (cm}^3\text{.atm.)/mol. K}$

$$\therefore \qquad C_{Ag} = \frac{0.08}{82.06 \times 1173} = 8.31 \times 10^{-7} \text{ mol/cm}^3$$

Given : Neglect the gas film resistance, i.e., film diffusion is not to be considered.

We have to find out the time required for complete conversion of a particle.

Here we have to consider the following resistances :

(i) resistance offered by ash layer.

(ii) resistance offered by chemical reaction (chemical step).

Time required for complete conversion is given by

$$\tau_{total} = \tau_{ash\ alone} + \tau_{reaction\ alone}$$

Now calculate $\tau_{ash\ alone}$ and $\tau_{reaction\ alone}$

1. For ash diffusion alone :

For constant size spherical particle, τ_{ash} is given by

$$\tau_{ash} = \frac{\rho_B R^2}{6\, b\, \mathscr{D}_e C_{Ag}}$$

$$= \frac{0.0425 \times (0.1)^2}{6\left(\frac{2}{3}\right)(0.08)\,(8.31 \times 10^{-7})} = 1598 \text{ s}$$

2. For chemical reaction alone :

Time for complete conversion is given by

$$\tau_{reaction} = \frac{\rho_B R}{b\, k''\, C_{Ag}}, \quad \rho_B = 0.0425 \text{ mol/m}^3$$

$$= \frac{0.0425 \times 0.10}{\frac{2}{3} \times 2 \times 8.31 \times 10^{-7}}$$

$$= 3836 \text{ s}$$

$$\tau_{total} = 1598 + 3836 = 5434 \text{ s}$$

Time required for the complete conversion of the particle = 5434 s ... Ans. (i)

Resistance of ash layer $= \frac{1598}{5434} \times 100 = 29.4\%$

Resistance of ash layer = 29.4% ... Ans. (ii)

Example 4.8 : *Repeat Example 4.7 for particles of size R = 0.05 mm.*

Solution : From the solution of Example 4.7, we have

$$\tau_{ash\ alone} = 1598 \text{ s}$$

$$\tau_{reaction\ alone} = 3836 \text{ s}$$

Now we will calculate τ_{ash} and $\tau_{reaction}$ for spherical particle of size R = 0.05 mm by making use of results for particle of size R = 1 mm.

For particle of size R = 0.05 mm

Particle size	$\tau_{ash\ alone}$	$\tau_{reaction\ alone}$
$R_1 = 1$ mm	$\tau_{a1} = 1598$ s	$\tau_{r1} = 3836$ s
$R_2 = 0.05$ mm	$\tau_{a2} = ?$	$\tau_{r2} = ?$

1. Ash diffusion step alone :

Here we have $\tau \propto R^2$

$$\tau_{a1} \propto R_1^2, \quad \tau_{a2} \propto R_2^2$$

$$\therefore \quad \frac{\tau_{a2}}{\tau_{a1}} = \frac{R_2^2}{R_1^2}$$

$$\tau_{a2} = \tau_{ash} = \frac{\tau_{a1} \cdot R_2^2}{R_1^2} = \frac{1598\ (0.05)^2}{(1)^2} = 3.995 \approx 4s$$

$\tau_{ash\ alone}$ for particle of size R (= 0.05 mm) = 4s.

2. Chemical reaction step alone :

Here we have $\tau \propto R$

$$\tau_{r1} \propto R \quad \text{and} \quad \tau_{r2} \propto R_2$$

$$\frac{\tau_{r2}}{\tau_{r1}} = \frac{R_2}{R_1}$$

$$\tau_{r2} = \tau_{r1} \frac{R_2}{R_1}$$

$$\tau_{reaction} = \tau_{r2} = \frac{3836 \times 0.05}{1} = 191.8 \approx 192 \text{ s}$$

$\tau_{reaction\ alone}$ for the particle of size R = 0.05 mm = 192 s

The total time required for complete conversion of the spherical particle of size R = 0.05 mm is

$$\tau_{total} = \tau_{ash} + \tau_{reaction}$$
$$= 4 + 192 = 196 \text{ s}$$

Time required for the complete conversion = 196 s **... Ans. (i)**

$$\text{Ash resistance} = \frac{4}{196} \times 100 = 2.04\%$$

Resistance of ash layer = 2.04% **... Ans. (ii)**

Example 4.9 : *On doubling the particle size from R to 2R the time needed for complete conversion of a particle triples.*

Determine the contribution of ash diffusion to the overall resistance for particles of size (i) R and (ii) 2R. Neglect gas film resistance.

Solution : (i) Particles of size R :

The time required for complete conversion of particles of size R, neglecting film resistance, is :

$$\tau_{total} = \tau_{ash\ alone} + \tau_{reaction\ alone}$$

$$z_1 = x + y \quad \text{... for particles of size R}$$

$$\tau_{ash\ alone} \propto R^2$$

$$\tau_{reaction\ alone} \propto R$$

(ii) For particles of size 2 R :

$$\tau_{ash\ alone}\ (\text{for } 2R) \propto (2R)^2$$

$$\tau_{ash\ alone}\ (\text{for } R) \propto R^2$$

Therefore, by taking the ratio

$$\tau_{ash}\,(\text{for } 2R) = \frac{\tau_{ash}\,(\text{for } R)\,(2R)^2}{R^2}$$

$$= x \cdot 4 = 4x$$

$$\tau_{reaction}\,(\text{for } 2R) \propto 2R$$

$$\tau_{reaction}\,(\text{for } R) \propto R$$

Therefore, $\tau_{reaction}\,(\text{for } 2R) = \tau_{reaction}\,(\text{for } R) \cdot \frac{2R}{R}$

$$= y \cdot 2 = 2y$$

The time required for complete conversion of particles of size 2R is

$$\tau_{total} = z_2 = \tau_{ash\ alone} + \tau_{reaction\ alone}$$

$$z_2 = 4x + 2y$$

We have :

$$\tau_{total} = z_1 = x + y \quad \ldots\ldots \text{ for particles of size R}$$

$$\tau_{total} = z_2 = 4x + 2y \quad \ldots\ldots \text{ for particles of size 2R}$$

Given : On doubling the particles size the time for complete conversion triples. That is,

$$z_2 = 3\,z_1$$

Substituting the values of z_1 and z_2 gives

$$4x + 2y = 3\,(x + y)$$

$$4x + 2y = 3x + 3y$$

$$\therefore \quad x = y$$

$$\text{Resistance of ash diffusion for particles of size R} = \left(\frac{x}{x+y}\right) 100$$

$$= \left(\frac{x}{2x}\right) 100\,, \text{ as } x = y$$

$$= 50\%$$

Contribution of the ash diffusion to the overall resistance for the particle of size R = 50% ... Ans.

$$\text{Resistance of ash diffusion for particles of size 2R} = \left(\frac{4x}{4x+2y}\right) 100$$

$$= \left(\frac{4x}{6x}\right) 100, \text{ as } x = y$$

$$= 66.67\%$$

Contribution of ash diffusion to the overall resistance for the particle of size 2R = 66.67% ... Ans.

Example 4.10 : *The reduction of iron ore of size R = 5 mm and density ρ_B = 4.6 g/cm³ by hydrogen gas can be represented well by the shrinking-core model (SCM). With no water vapour present, the reaction stoichiometry is :* $4H_2 + Fe_3O_4 \rightarrow 4H_2O + 3Fe$.

The rate is approximately proportional to the concentration of hydrogen gas.

The first order rate constant is given by

$$k'' = 1.93 \times 10^5 \; e^{-24000/RT}, \; cm/s$$

Take $\mathscr{D}_e$ = 0.03 cm²/s, the diffusion coefficient for hydrogen penetration in the product layer. Calculate the time needed for complete conversion of a particle of oxide to metal at a temperature of 600 °C and a pressure of 1 atm.

At. Wt. Fe = 56, O = 16.

Film resistance can safely be neglected in the presence of ash layer.

Solution : $4H_2(g) + Fe_3O_4(s) \rightarrow 3Fe(s) + 4H_2O(g)$

It is a fluid-solid reaction in which the solid does not appreciably change in size. So we have constant size particles.

Given : neglect film resistance.

The above reaction can be written as

Specific : $H_2(g) + \frac{1}{4} Fe_3O_4(s) \rightarrow \frac{3}{4} Fe(s) + H_2O(g)$

General : A(g) + b B (s) → Fluid and solid products

Comparing the above two reactions, we have

We have : $b = \frac{1}{4} = 0.25$.

R = 5 mm = 0.5 cm

ρ_B = 4.6 g/cm³, $\mathscr{D}_e$ = 0.03 cm²/s

Mol. Wt. of Fe_3O_4 = 3 × 56 + 4 × 16 = 232 g/mol.

Molar density of solid = $\rho_B = \frac{4.6}{232} = 0.0198$ mol/cm³

Temperature = 600 °C = 873 K

$$k'' = 1.93 \times 10^5 \; e^{-24000/RT}, \quad R = 1.987 \text{ cal/(mol.K)}$$

$$= 1.93 \times 10^5 \; e^{-24000/1.987 \times 873}$$

$$= 0.189 \text{ cm/s}$$

Since pure hydrogen is used, we have : $p_{Ag} = P = 1$ atm.

$$C_{Ag} = \text{concentration of hydrogen in the gas phase}$$

$$= \frac{p_{Ag}}{RT}$$

$$= \frac{1}{82.06 \times 873} = 1.396 \times 10^{-5} \text{ mol/cm}^3$$

The time required for complete conversion of a particle (from oxide to metal) is

$$\tau_{total} = \tau_{ash\ alone} + \tau_{reaction\ alone}$$

1. For ash diffusion alone :

For constant size spherical particle, τ is given by

$$\tau_{ash} = \frac{\rho_B R^2}{6b\ \mathscr{D}_e C_{Ag}}$$

$$= \frac{0.0198 \times (0.5)^2}{6 \times 0.25 \times 0.03 \times 1.396 \times 10^{-5}}$$

$$= 7879.66 \simeq 7880 \text{ s}$$

2. For chemical reaction alone :

$$\tau_{reaction} = \frac{\rho_B R}{b\ k'' C_{Ag}}$$

$$= \frac{0.0198 \times (0.50)}{0.25 \times 0.189 \times 1.396 \times 10^{-5}}$$

$$= 15009 \text{ s}$$

$$\tau_{total} = \tau_{ash\ alone} + \tau_{reaction\ alone}$$

$$= 7880 + 15009$$

$$= 22889 \text{ s}$$

Time required for complete conversion of the particle of size R (= 5 mm) = 22889 s ... Ans.

$$\text{Resistance of ash layer} = \frac{7880}{22889} \times 100 = 34.43\%$$

$$\text{Resistance of chemical step} = \frac{15009}{22889} \times 100 = 65.57\%$$

Example 4.11 : *Calculate the time required to burn to completion spherical particles of graphite (radius 12 mm, bulk density 2.4 g/cm³) in a 12% oxygen stream at 900 °C and 1 atm. Assume gas film resistance to be negligible.*

Data : Surface reaction rate constant = k'' = 25 cm/s.

Solution : Here the reacting particle shrinks during reaction and finally disappears.

$C(s) + O_2(g) \rightarrow CO_2(g)$

From the reaction, b = 1

Given : R = 12 mm = 1.2 cm

$\rho_B = 2.4$ g/cm^3

Molar density of graphite (C) = $\rho_B = \frac{2.4}{12} = 0.2$ mol/cm^3

Given : T = 900 °C = 1173 K, P = 1 atm.

O_2 in the gas stream = 12% by volume

Specific : $C(s) + O_2(g) \rightarrow CO_2(g)$

General : $b\,B(s) + A(g) \rightarrow$ fluid product

Comparing the above two equations, we get b = 1 and A = O_2

C_{Ag} - concentration of O_2 in the bulk gas phase

For an ideal gas : mole % = volume %

$\therefore$ Mole % O_2 in the gas stream = 12

We know :

$$p_{Ag} = x_A.P = \frac{12}{100} \times 1 = 0.12 \text{ atm.}$$

The concentration of O_2 is calculated using the relation :

$$C_{Ag} = \frac{p_{Ag}}{RT}, \quad R = 82.06 \text{ (cm}^3\text{·atm)/mol.K}$$

$$= \frac{0.12}{82.06 \times 1173}$$

$$= 1.25 \times 10^{-6} \text{ mol/cm}^3$$

Given : $k'' = 25$ cm/s

Given : Negligible gas film resistance.

Therefore, for the shrinking spherical particle, in our case, chemical reaction controls the rate.

For the situation where chemical reaction controls, we have for shrinking sphere :

$$\tau = \frac{\rho_B R}{b\, k''\, C_{Ag}} = \frac{0.2 \times 12}{1 \times 25 \times 1.25 \times 10^{-6}}$$

$$= 7680 \text{ s}$$

Time required for the complete conversion of graphite particles of size R = 12 mm = 7680 s ... **Ans.**

Example 4.12 : *Iron ore of density $\rho_B = 4.6$ g/cm³ and size $R = 5$ mm is reduced as per the following reaction by pure hydrogen gas :*

$$4\,H_2 + Fe_3O_4 \rightarrow 4\,H_2O + 3\,Fe$$

Estimate the time required for 98% conversion of a particle from oxide to metal in a hydrogen environment at constant hydrogen pressure of 1 atm and at 600 °C.

The rate is proportional to the concentration of hydrogen gas.

The first-order reaction rate constant is given by

$$k'' = 1.93 \times 10^5\, e^{-12000/T},\ cm/s$$

Hydrogen diffusion coefficient, $\mathscr{D}_e = 0.03$ cm²/s

Film diffusion coefficient, $k_g = 10$ cm/s. At. wt. : Fe = 56, O = 16

Solution : Here the size of the particle does not change appreciably during the reaction. Therefore, so we are concerned with a constant size spherical particle.

$$4\,H_2 + Fe_3O_4 \rightarrow 3\,Fe(s) + 4\,H_2O$$

$$H_2(g) + \frac{1}{4}\,Fe_3O_4\,(s) \rightarrow \frac{3}{4}\,Fe + H_2O\,(g)$$

$$A(g) + b\,B(s) \rightarrow \text{Fluid and solid product}$$

Comparing the above two reactions, we have

$$b = \frac{1}{4} = 0.25$$

R = 5 mm = 0.5 cm, ... (particle size)

Mol. Wt. of Fe_3O_4 = 232 g/mol $(= 3 \times 56 + 4 \times 16)$

$$\rho_B = 4.6\ g/cm^3$$

Molar density of B = $\rho_B = \frac{4.6}{232} = 0.0198$ g/mol

T = 600 °C = 873 K, R = 82.06 (cm³.atm) / mol.K ... (gas constant)

$$C_{Ag} = \frac{p_{Ag}}{RT},\ p_{Ag} = P = 1 \text{ atm - being pure hydrogen}$$

$$= \frac{1}{82.06 \times 873} = 1.396 \times 10^{-5}\ mol/cm^3$$

$$\mathscr{D}_e = 0.03\ cm^2/s, \qquad k_g = 10\ cm/s$$

$$k'' = 1.93 \times 10^5\, e^{-12000/T}$$

$$k'' = 1.93 \times 10^5\, e^{-12000/873} = 0.207\ cm/s$$

We have to consider the resistances of gas film, ash layer and chemical reaction.

Time needed for any stage of conversion is

$$t_{total} = t_{film\ alone} + t_{ash\ alone} + t_{reaction\ alone}$$

First we calculate τ and then t for each of these three resistances.

1. Film diffusion controls (film diffusion alone) :

τ for constant size spherical particle is given by

$$\tau = \frac{\rho_B R}{3\, b\, k_g\, C_{Ag}}$$

$$= \frac{0.0198 \times 0.50}{3 \times 0.25 \times 10 \times 1.396 \times 10^{-5}}$$

$$= 94.55 \text{ s}$$

Conversion to be achieved = $X_B = 98\% = 0.98$

We have

$$\frac{t}{\tau} = X_B$$

$$t = \tau X_B$$

$$= 94.55 \times 0.98 = 92.66 \text{ s}$$

$$t_{film\ alone} = 92.66 \text{ s}$$

2. Ash diffusion controls (ash diffusion alone) :

$$\tau = \frac{\rho_B R^2}{6\, b\, \mathscr{D}_e\, C_{Ag}}$$

$$= \frac{0.0198 \times (0.50)^2}{6 \times 0.25 \times 0.03 \times 1.396 \times 10^{-5}}$$

$$= 7880 \text{ s}$$

We have

$$\frac{t}{\tau} = 1 - 3\,(1 - X_B)^{2/3} + 2\,(1 - X_B), \quad X_B = 0.98$$

$$\frac{t}{\tau} = 1 - 3\,(1 - 0.98)^{2/3} + 2\,(1 - 0.98)$$

$$= 0.819$$

$$t_{ash\ alone} = 0.819\,\tau$$

$$= 0.819 \times 7880 = 6453 \text{ s}$$

3. Chemical reaction controls (chemical reaction alone) :

$$\tau = \frac{\rho_B R}{b\, k'' C_{Ag}}$$

$$= \frac{0.0198 \times 0.50}{0.25 \times 0.207 \times 1.396 \times 10^{-5}} = 13703 \text{ s}$$

We have

$$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$$

$$= 1 - (1 - 0.98)^{1/3}$$

$$= 0.728$$

$$\frac{t}{\tau} = 0.728$$

$$t_{\text{reaction alone}} = 0.728\,\tau$$

$$= 0.728 \times 13703 = 9976 \text{ s}$$

Total time required for 98% conversion of the particle is calculated by using the relation

$$t_{\text{total}} = t_{\text{film alone}} + t_{\text{ash alone}} + t_{\text{reaction alone}}$$

$$t_{\text{total}} = 92.66 + 6453 + 9976$$

$$= 16521.66 \text{ s}$$

$$= 16522 \text{ s}$$

Total time required for 98% conversion of the particle from oxide to metal = 16522 s or 4.59 h ... Ans.

Example 4.13 : *Spherical particles of an ore of size R = 5 mm are roasted isothermally in an air stream.*

The time-conversion data obtained are as follows :

t, min	*15*	*30*	*60*
X_B	*0.334*	*0.584*	*0.88*

Assume that the reaction follows the SCM.

(i) Find the rate controlling mechanism.

(ii) Determine the time needed for complete conversion of spherical particles of size R = 0.5 mm, all other parameters remain unchanged.

Solution : Here we have fluid-solid reaction in which the solid does appreciable change in size during the reaction.

So we have constant size spherical particles. We have three rate controlling mechanisms :

1. diffusion through gas film
2. diffusion through ash layer
3. chemical reaction at the surface.

We will check which of these fits the data.

1. **Film diffusion controls/resistance of gas film controls :** In this case, the time-conversion expression is

$$\frac{t}{\tau} = X_B$$

$$\tau = \frac{t}{X_B}$$

Data are provided for a single size particle. τ calculated from the data should be constant for this mechanism to be rate controlling.

t, min	15	30	60
X_B	0.334	0.584	0.88
$\tau = t/X_B$, min	44.19	51.37	68.18

As τ is not constant (i.e., as τ is not the same throughout) .....

..... film diffusion does not control the rate.

2. **Chemical reaction controls :**

For this mechanism, the time-conversion expression for a constant size spherical particle is

$$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$$

$$\tau = \frac{t}{\left[1 - (1 - X_B)^{1/3}\right]}$$

If we get almost same value of τ for all the data points, then this mechanism controls the rate.

t	X_B	$\tau = \frac{t}{\left[1 - (1 - X_B)^{1/3}\right]}$
15	0.334	118.38
30	0.584	118.34
60	0.88	118.40

Here τ is almost constant .....

So chemical reaction controls the rate.

Rate controlling mechanism : chemical reaction **... Ans. (i)**

3. **Ash diffusion controls :**

For this case, we have

$$\frac{t}{\tau} = 1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)$$

$$\tau = \frac{t}{1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)}$$

t	X_B	$\tau = \dfrac{t}{1-3(1-X_B)^{2/3}+2(1-X_B)}$
15	0.334	340.15
30	0.584	187.26
60	0.88	117.61

Large variation in τ values for the given data points. $\tau \neq$ constant .....

..... so ash diffusion does not control the rate.

(ii) For chemical reaction controlling the rate, we have :

$$\frac{t}{\tau} = 1-(1-X_B)^{1/3}$$

For R = 2 mm we have

t = 30 min $X_B = 0.584$

First we will determine the time needed for complete conversion of a particle of size R = 2 mm.

$$\tau = \frac{t}{\left[1-(1-X_B)^{1/3}\right]}$$

$$= \frac{30}{1-(1-0.584)^{1/3}} = 118.34 \text{ min}$$

Now we will calculate the time required for complete conversion of a particle of size R = 0.5 mm.

For chemical reaction controlling, we have

$$\tau \propto R$$

$$\tau_1 \propto R_1 \quad \text{and} \quad \tau_2 \propto R_2$$

$$\therefore \quad \tau_2 = \frac{\tau_1 R_2}{R_1}$$

For $R_1 = 2$ mm , $\tau_1 = 118.34$ mm (calculated above)

For $R_2 = 0.5$ mm , $\tau_2 = ?$

$$\tau_2 = \frac{118.34 \times 0.5}{2} = 29.585 \approx 29.6 \text{ min.}$$

Time required for complete conversion of the particles of size R = 0.5 mm = 29.5 min

... Ans. (ii)

Example 4.14 : *The reduction of uniform-sized spherical particles of UO_3 to UO_2 in a uniform environment yielded the following results :*

t, h	*0.180*	*0.347*	*0.453*	*0.567*	*0.733*
X_B	*0.45*	*0.68*	*0.80*	*0.95*	*0.98*

Assuming that the reaction proceeds follows the shrinking-core model, determine the rate-controlling mechanism.

Solution : The mechanism which will best fits the data will be the rate controlling.

Now we will check the mechanisms to fit the data.

1. **Film diffusion controls/diffusion through gas film controls :**

For this, the time-conversion expression for constant size spherical particle is

$$\frac{t}{\tau} = X_B$$

$$\tau = \frac{t}{X_B} \quad \text{..... this should be satisfied.}$$

For single size particle τ calculated from each conversion-time should be the same if at all film diffusion controls.

t	0.180	0.347	0.453	0.567	0.733
X_B	0.45	0.68	0.80	0.95	0.98
$\tau = \frac{t}{X_B}$	0.40	0.50	0.57	0.597	0.748

We see from the table that there is a variation in τ values.

As τ is not constant ..... film diffusion does not control the rate.

2. **Chemical reaction controls :**

In this case, the time-conversion expression for constant size spherical particle is :

$$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$$

$$\tau = \frac{t}{\left[1 - (1 - X_B)^{1/3}\right]}$$

The data should fit the above expression for this mechanism to be rate controlling.

t	X_B	$\tau = \frac{t}{\left[1 - (1 - X_B)^{1/3}\right]}$
0.18	0.45	$0.996 \approx 1.0$
0.347	0.68	1.098
0.453	0.80	1.091
0.567	0.95	0.898
0.733	0.98	1.006

We see from the table that τ calculated for the majority of $t - X_B$ data is almost close to 1 (τ is almost same) .....

..... so chemical reaction controls the rate. .

As the $t - X_B$ expression reasonably fits the data :

Rate controlling mechanism : Chemical reaction ... Ans.

Let us also have check for :

3. Ash diffusion controls :

In this case, the time-conversion expression for constant size spherical particle is

$$\frac{t}{\tau} = 1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)$$

$$\tau = \frac{t}{\left[1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)\right]}$$

t	X_B	$\tau = \frac{t}{\left[1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)\right]}$
0.18	0.45	2.09
0.347	0.68	1.467
0.453	0.80	1.21
0.567	0.95	0.818
0.733	0.98	0.895

We see that τ is not constant .....

..... so, ash diffusion does not control the rate.

We see that the $t - X_B$ data best fit the time-conversion expression for the mechanism no. 2.

Rate controlling mechanism : chemical reaction ... Ans.

□□□

CHAPTER 5

DESIGN OF FLUID-PARTICLE REACTORS

This chapter treats the design of reactors for non-catalytic fluid-solid reactions. In the treatment that follows, we consider the simplified idealised systems in which the reactor kinetics, flow patterns of solid and fluid in the reactor and size distribution of solid particles (to be treated) are known.

Contacting Schemes / Patterns :

Let us discuss briefly the various types of contacting schemes used for gas-solid reactions shown in Fig. 5.1.

Solids and Gas both in Plug Flow [Plug flow S + Plug flow G] :

When solids and gas pass through the reactor in plug flow, the compositions of both solids and gas will change along their flow paths. Also, under such conditions reactor operations are non-isothermal.

The plug flow contacting of phases may be achieved by one of the following ways :

(i) Countercurrent flow of phases : In blast furnaces and cement kilns (rotary kilns), the flow of gas is countercurrent to the flow of solids.

(ii) Cocurrent flow of phases : In rotary dryers used for heat sensitive materials (polymer dryers), the flow of gas is in cocurrent with solids.

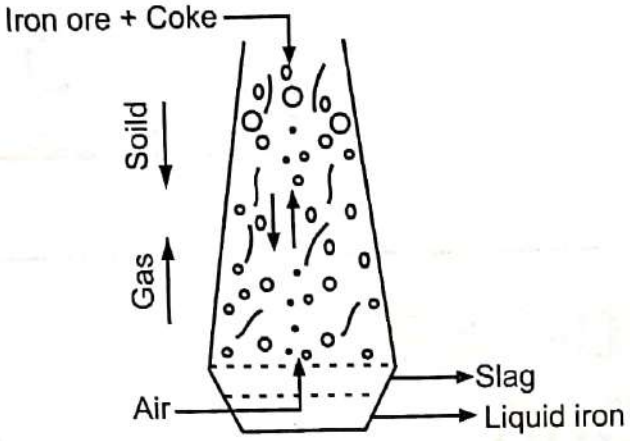
(iii) Cross flow of phases : In moving belt feeders for furnaces (combustion of coal) the flow of gas is normal (crosscurrent) to the flow path of solids.

Solids in Mixed Flow :

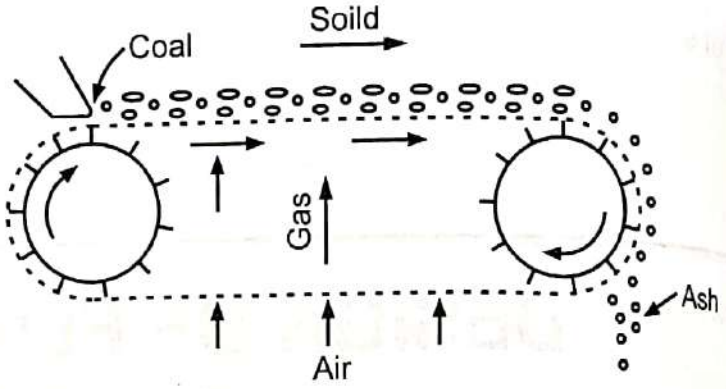
In fluidised-bed reactors, the solids are in mixed flow. The gas flow in such a reactor is complex, not well known, but it is certainly far from plug flow with considerable bypassing. These reactors can operate under isothermal conditions.

Semi-batch Operations (Plug flow G + Stationary S) :

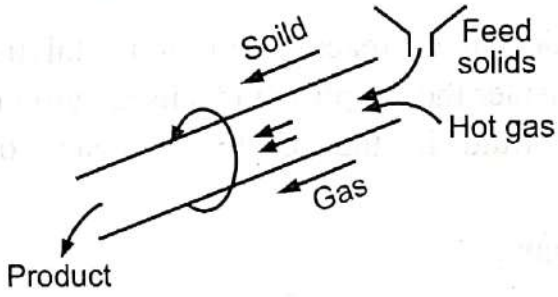
In semi-batch reactors, the gas passes through a stationary bed of solids, charged and removed batchwise. Here the flow of gas closely approximates plug flow. It is used for small-scale operations. The ion-exchange column shown in Fig. 5.1 (e) is an example of semi-batch operation.



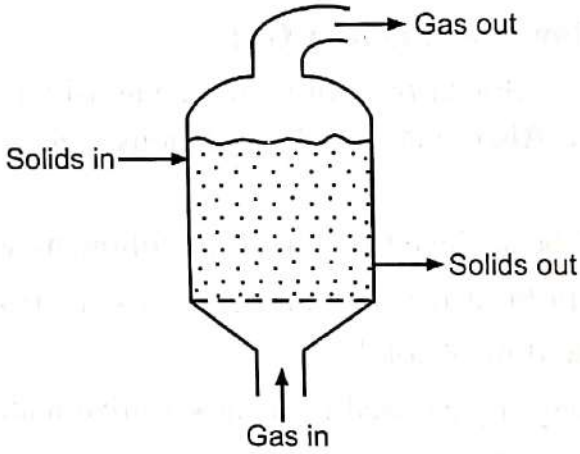
(a) Blast furnace



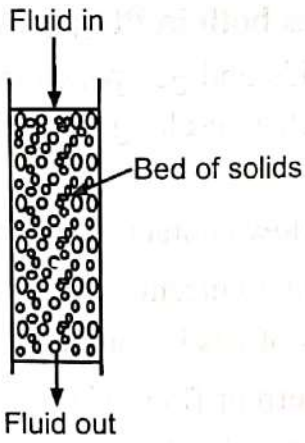
(b) Moving feeder for coal furnaces



(c) Rotary dryer



(d) Fluidised-bed reactor



(e) Ion-exchange bed

Fig. 5.1 : Various contacting patterns (schemes) in fluid-solid reactors : (a) countercurrent plug flow, (b) crosscurrent plug flow, (c) cocurrent plug flow ... (of G and S), (d) intermediate gas flow and mixed solid flow, (e) semi-batch operations (plug flow F + stationary S)

Analysis and design of fluid-solid systems become highly simplified if we assume uniform composition of the fluid throughout the reactor. This assumption is fairly correct in situations where fractional conversion of fluid-phase reactants is not too high (valid when a sufficient excess of the fluid phase reactants is used) and where fluid back mixing is considerable (true in fluidised beds). Therefore, we can make use of this assumption frequently without much deviation from reality.

Let us now develop the performance equations for various contacting patterns, using the assumption of uniform gas composition within the reactor in every case.

Particles of Single Size – Plug flow of solids + Uniform Gas Composition :

Here the contact time or reaction time required to achieve a specified conversion of solid reactant is obtained by using the equations given in Table 4.1 (Chapter 4).

Mixture of Particles of Different but Unchanging sizes (Fixed sizes) – Plug flow of solids + Uniform Gas Composition :

Consider a solid feed comprising of a mixture of different-size particles. The size distribution of this feed may be given by screen analysis.

The size distribution of feed can be represented as a continuous distribution or as a discrete distribution. Here we will use the latter representation as screen analysis (sieve analysis), our way of measuring size distributions, which yields/results in measurements in a discrete form : the mass fraction of particle with a mean size R_i is $\psi(R_i)$; with fraction of particles with a mean size R_{i+1} is $\psi(R_i + 1)$, etc. $R_1, \ldots, R_i, \ldots, R_m$ are the mean sizes of the particles left on the sieve of a given mesh size [The mean size of the particles remaining on the (say) fine sieve is the arithmetic mean of the two mesh sizes (mesh size of a fine sieve and mesh size of the next coarser sieve)].

The discrete size distribution of particles is presented as a bar diagram [Refer Fig. 5.2]. Here instead of using mass fraction we use the mass feed rate of particles. The feed rate of particles of a mean size R_i is $F(R_i)$.

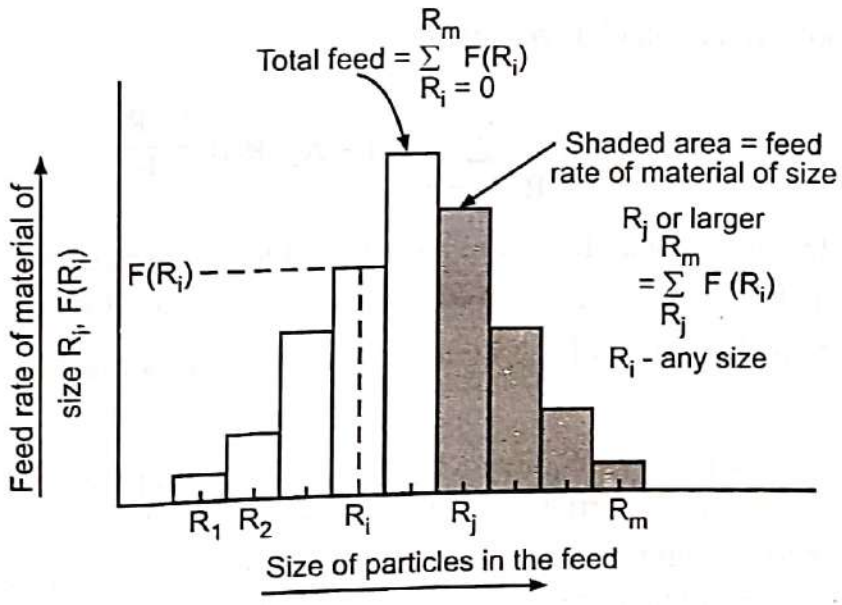


Fig. 5.2 : Bar diagram showing particle size distribution of the feed rate of a mixture of particles

Let F be the quantity of solid being treated per unit time. As the density of solid may change during reaction, in the general case we treat F as the volumetric feed rate of solid. But when the density change of solid is negligible, we treat F as the mass feed rate of solid.

Let F (R_i) be the quantity of material of size about R_i (mean size) and R_m is the largest particle size in the feed to the reactor.

Then, for particles of unchanging size, we can write

$$F = \sum_{R_i = 0}^{R_m} F(R_i), \quad m^3/s \text{ or } kg/s$$

As solids are in plug flow, they all will stay in the reactor for the same length of time t_p (residence time of all solids).

Let X_B (R_i) be the fractional conversion of reactant B for any size of particle R_i. It can be obtained from t_p and kinetics for any particular controlling resistance.

Fraction of the reactant B converted in particles of size $R_i = X_B(R_i)$

Fraction of the reactant B unconverted in particles of size $R_i = 1 - X_B(R_i)$.

Fraction of the feed which is of size $R_i = \frac{F(R_i)}{F}$.

The mean conversion $\bar{X}_B$ of the solids leaving the reactor can be obtained by summing the contribution to conversion of all sizes of particles.

$$\therefore \begin{pmatrix} \text{Mean value for} \\ \text{the fraction of} \\ \text{reactant B unconverted} \end{pmatrix} = \sum_{\text{all sizes}} \begin{pmatrix} \text{Fraction of reactant} \\ \text{B unconverted in} \\ \text{particles of size } R_i \end{pmatrix} \begin{pmatrix} \text{fraction of feed} \\ \text{which is of} \\ \text{size } R_i \end{pmatrix} \quad \ldots (5.1)$$

In the form symbols, Equation (5.1) becomes

$$1 - \bar{X}_B = \sum_{R(t_p = \tau)}^{R_m} [1 - X_B(R_i)] \frac{F(R_i)}{F} \quad \ldots (5.2)$$

Note : As particles smaller than R ($t_p = \tau$) are completely converted they do not contribute to the fraction unconverted and thus are not included in the summation of Equation (5.2).

Where R ($t_p = \tau$) is the radius of the largest particle which is completely converted in the reactor.

We know that a smaller particle requires a shorter time for complete conversion compared to a large one. Therefore, the feed particles of sizes smaller than R ($t_p = \tau$) will be completely converted during their passage through the reactor. If we consider such particles [that are smaller than R ($t_p = \tau$)] while calculating the fraction unconverted using Equation (5.2), we will get X_B values greater than 1, which is meaningless. Thus, this condition is incorporated into the

lower limit of the summation of Equation (5.2) and it indicates that particles smaller than R ($t_p = \tau$) are completely converted and do not contribute to the fraction unconverted, $1 - \bar{X}_B$.

Particles of a Single Unchanging Size (Fixed Size) – Mixed Flow of Solids + Uniform Gas Composition :

With mixed flow of solids and uniform gas composition we have a fluidised bed reactor.

Consider a fluidised bed reactor with constant flow rates of both solids and gas into and out of the reactor. In this case there is no elutriation of fine particles.

The conversion of reactant in a single particle depends on its length of stay in the fluidised bed. Its proper controlling resistance is given by Equations (4.14), (4.23) or (4.33). Here as the length of stay is not the same for all the particles in the reactor we must calculate a mean conversion $\bar{X}_B$ of solid material. For this we consider that the solid behaves as a macrofluid.

For a macrofluid from the topic : Basic aspects of non-ideal flow [in CRE-I], the mean conversion in the exit stream is given by

$$\bar{X} = \int_0^\infty X\, E\, dt$$

$$\therefore \quad 1 - \bar{X} = \int_0^\infty (1 - X)\, E\, dt$$

Hence, for the solids leaving the reactor, the mean value of the fraction unconverted is given by

$$\begin{pmatrix}\text{Mean value for the fraction of reactant B unconverted}\end{pmatrix} = \sum_{\substack{\text{particles} \\ \text{of all ages}}} \begin{pmatrix}\text{fraction of reactant B unconverted for particles staying for time between t and t + dt in the reactor}\end{pmatrix} \begin{pmatrix}\text{fraction of exit stream that has stayed for time t and t + dt in the reactor}\end{pmatrix}$$

... (5.3)

Using symbols for the terms involved, Equation (5.3) becomes

$$1 - \bar{X}_B = \int_0^\infty (1 - X_B)\, E\, dt, \quad X_B \le 1$$

or

$$1 - \bar{X}_B = \int_0^\tau (1 - X_B)\, E\, dt \qquad \ldots (5.4)$$

where E is the exit age distribution of the solids in the reactor. The exit age distribution function E is such that the fraction of the exit stream with residence times between t and t + dt is given by

Edt. In Equation (5.4), Edt is the fraction of the exit stream that has stayed in the reactor for times between t and t + dt and in which $1 - X_B$ is still unconverted.

[E for a mixed flow reactor is given by : $E = \frac{e^{-t/\tau}}{\tau}$]

$\therefore$ For mixed flow of solids with mean residence time $\bar{t}$ in the reactor, E is given by

$$E = \frac{e^{-t/\bar{t}}}{\bar{t}} \qquad \ldots (5.5)$$

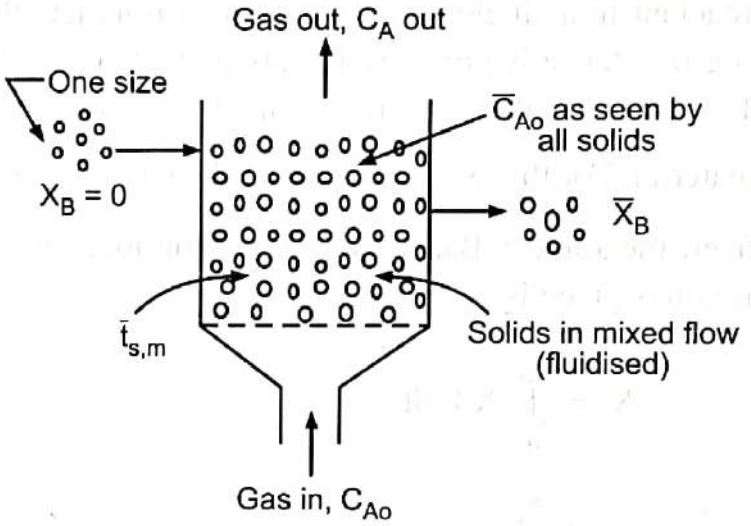


Fig. 5.3 : Conversion of solids of one size in mixed flow

Hence, by combining Equations (5.4) and (5.5), for mixed flow of the single size of solid which is completely converted in time τ, we obtain

$$1 - \bar{X}_B = \int_0^{\tau} (1 - X_B)_{\text{individual particle}} \cdot \frac{e^{-t/\bar{t}}}{\bar{t}}\, dt \qquad \ldots (5.6)$$

Now we will integrate Equation (5.6) for the various controlling resistances.

1. Film Resistance Controls (Controlling resistance – film diffusion) :

For diffusion through gas film controls, the time-conversion relation is

$$\frac{t}{\tau} = X_B$$

$\therefore$
$$1 - X_B = 1 - \frac{t}{\tau} \qquad \ldots (5.7)$$

Combining Equations (5.6) and (5.7), we get

$$1 - \bar{X}_B = \int_0^{\tau} \left(1 - \frac{t}{\tau}\right) \frac{e^{-t/\bar{t}}}{\bar{t}}\, dt \qquad \ldots (5.8)$$

Integrating Equation (5.8), we get

$$\bar{X}_B = \frac{\bar{t}}{\tau}\left(1 - e^{-\tau/\bar{t}}\right) \qquad \ldots (5.9)$$

We can write Equation (5.9) in an equivalent expanded form as

$$\boxed{1 - \bar{X}_B = \frac{1}{2}\frac{\tau}{\bar{t}} - \frac{1}{3!}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{1}{4!}\left(\frac{\tau}{\bar{t}}\right)^3 - \ldots\ldots} \qquad \ldots (5.10)$$

Equation (5.10) is useful for large $\bar{t}/\tau$ and thus for very high conversion.

Let us obtain Equation (5.9).

We have,

$$1 - \bar{X}_B = \int_0^{\tau}\left(1 - \frac{t}{\tau}\right)\frac{e^{-t/\bar{t}}}{\bar{t}}\,dt$$

Let, $t/\bar{t} = x \quad \therefore\ t = \bar{t}x \quad dt = \bar{t}\,dx$

At $t = 0 \ldots \quad x = 0 \quad$ At $t = \tau \ldots\ldots \quad x = \tau/\bar{t}$

$$1 - \bar{X}_B = \int_0^{\tau/\bar{t}}\left(1 - \frac{\bar{t}x}{\tau}\right)\frac{e^{-x}}{\bar{t}}\,\bar{t}\,dx$$

$$= \int\left(1 - \frac{\bar{t}}{\tau}x\right)e^{-x}\,dx$$

$$= \int e^{-x}\,dx - \frac{\bar{t}}{\tau}\int x\,e^{-x}\,dx$$

$$= \left[-e^{-x} - \frac{\bar{t}}{\tau}\left(-x\,e^{-x} - e^{-x}\right)\right]_0^{\tau/\bar{t}}$$

$$= \left[-e^{-x} + \frac{\bar{t}}{\tau}x\,e^{-x} + \frac{\bar{t}}{\tau}e^{-x}\right]_0^{\tau/\bar{t}}$$

$$= \left\{e^{-x}\left[\frac{\bar{t}}{\tau}x + \frac{\bar{t}}{\tau} - 1\right]\right\}_0^{\tau/\bar{t}}$$

$$\int u\,dv = uv - \int v\,du$$

$u = x \rightarrow du = dx$

$v = e^{-x} \rightarrow dv = -e^{-x}\,dx$

$$-\int x\,e^{-x}\,dx = x\,e^{-x} - \int e^{-x}\,dx$$

$$-\int x\,e^{-x}\,dx = x\,e^{-x} + e^{-x}$$

$$\therefore \int x\,e^{-x}\,dx = -x\,e^{-x} - e^{-x}$$

$$= e^{-\tau/\bar{t}}\left(\frac{\bar{t}}{\tau}\cdot\frac{\tau}{\bar{t}}+\frac{\bar{t}}{\tau}-1\right)-e^{-0}\left(\frac{\bar{t}}{\tau}(0)+\frac{\bar{t}}{\tau}-1\right)$$

$$1-\bar{X}_B = e^{-\tau/\bar{t}}\left(\frac{\bar{t}}{\tau}\right)-\frac{\bar{t}}{\tau}+1, \qquad \because\ e^{-0}=1$$

$$\therefore\quad -\bar{X}_B = \frac{\bar{t}}{\tau}e^{-\tau/\bar{t}}-\frac{\bar{t}}{\tau}$$

$$\bar{X}_B = -\frac{\bar{t}}{\tau}e^{-\tau/\bar{t}}+\frac{\bar{t}}{\tau}$$

$$\therefore\quad \bar{X}_B = \frac{\bar{t}}{\tau}\left(1-e^{-\tau/\bar{t}}\right) \quad \ldots \text{ which is Equation (5.9).}$$

Let us obtain Equation (5.10).

$$\bar{X}_B = \frac{\bar{t}}{\tau}\left(1-e^{-\tau/\bar{t}}\right)$$

Let $\tau/\bar{t} = y$

$$\therefore\quad \bar{X}_B = \frac{1}{y}(1-e^{-y})$$

Since, $$e^{-y} = 1-y+\frac{y^2}{2!}-\frac{y^3}{3!}+\frac{y^4}{4!}-\ldots\ldots$$

$$\therefore\quad \bar{X}_B = \frac{1}{y}\left[1-1+y-\frac{y^2}{2!}+\frac{y^3}{3!}-\frac{y^4}{4!}+\ldots\ldots\right]$$

$$\bar{X}_B = 1-\frac{y}{2!}+\frac{y^2}{3!}-\frac{y^3}{4!}+\ldots\ldots$$

$$\therefore\quad 1-\bar{X}_B = \frac{y}{2!}-\frac{y^2}{3!}+\frac{y^3}{4!}+\ldots\ldots$$

$$\therefore\quad 1-\bar{X}_B = \frac{1}{2!}\left(\frac{\tau}{\bar{t}}\right)-\frac{1}{3!}\left(\frac{\tau}{4!}\right)^2+\frac{1}{4!}\left(\frac{\tau}{\bar{t}}\right)^3-\ldots\ldots$$

With $2! = 1\times 2 = 2$, we get

$$1-\bar{X}_B = \frac{1}{2}\left(\frac{\tau}{\bar{t}}\right)-\frac{1}{3!}\left(\frac{\tau}{\bar{t}}\right)^2+\frac{1}{4!}\left(\frac{\tau}{\bar{t}}\right)^3-\ldots\ldots \quad \text{Equation (5.10)}$$

2. Resistance of chemical reaction controls (Controlling resistance - reaction step) :

For chemical reaction controls, the time-conversion relation is :

$$\frac{t}{\tau} = 1-(1-X_B)^{1/3}$$

$$\therefore\quad (1-X_B)^{1/3} = \left(1-\frac{t}{\tau}\right)$$

$$(1-X_B) = \left(1-\frac{t}{\tau}\right)^3 \qquad \ldots (5.11)$$

Combining Equations (5.6) and (5.11) gives

$$1 - \bar{X}_B = \int_0^{\tau} \left(1 - \frac{t}{\tau}\right)^3 \cdot \frac{e^{-t/\bar{t}}}{\bar{t}} \, dt \qquad \ldots (5.12)$$

Equation (5.12) on integration gives

$$\boxed{\bar{X}_B = 3\frac{\bar{t}}{\tau} - 6\left(\frac{\bar{t}}{\tau}\right)^2 + 6\left(\frac{\bar{t}}{\tau}\right)^3 (1 - e^{-\tau/\bar{t}})} \qquad \ldots (5.13)$$

For large $\bar{t}/\tau$ thus for very high conversion, Equation (5.13) in an equivalent expanded form is

$$\boxed{1 - \bar{X}_B = \frac{1}{4}\left(\frac{\tau}{\bar{t}}\right) - \frac{1}{20}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau}{\bar{t}}\right)^3 - \ldots} \qquad \ldots (5.14)$$

3. Ash resistance controls (Controlling resistance - ash diffusion) : For diffusion through the ash layer controlling the overall rate, the mean of fraction of B unconverted is given by

$$\boxed{1 - \bar{X}_B = \frac{1}{5}\left(\frac{\tau}{\bar{t}}\right) - \frac{19}{420}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{41}{4620}\left(\frac{\tau}{\bar{t}}\right)^3 - \ldots\ldots} \qquad \ldots (5.15)$$

Mixed Flow of a Size Mixture of Particles of Unchanging/Fixed Size + Uniform Gas Composition :

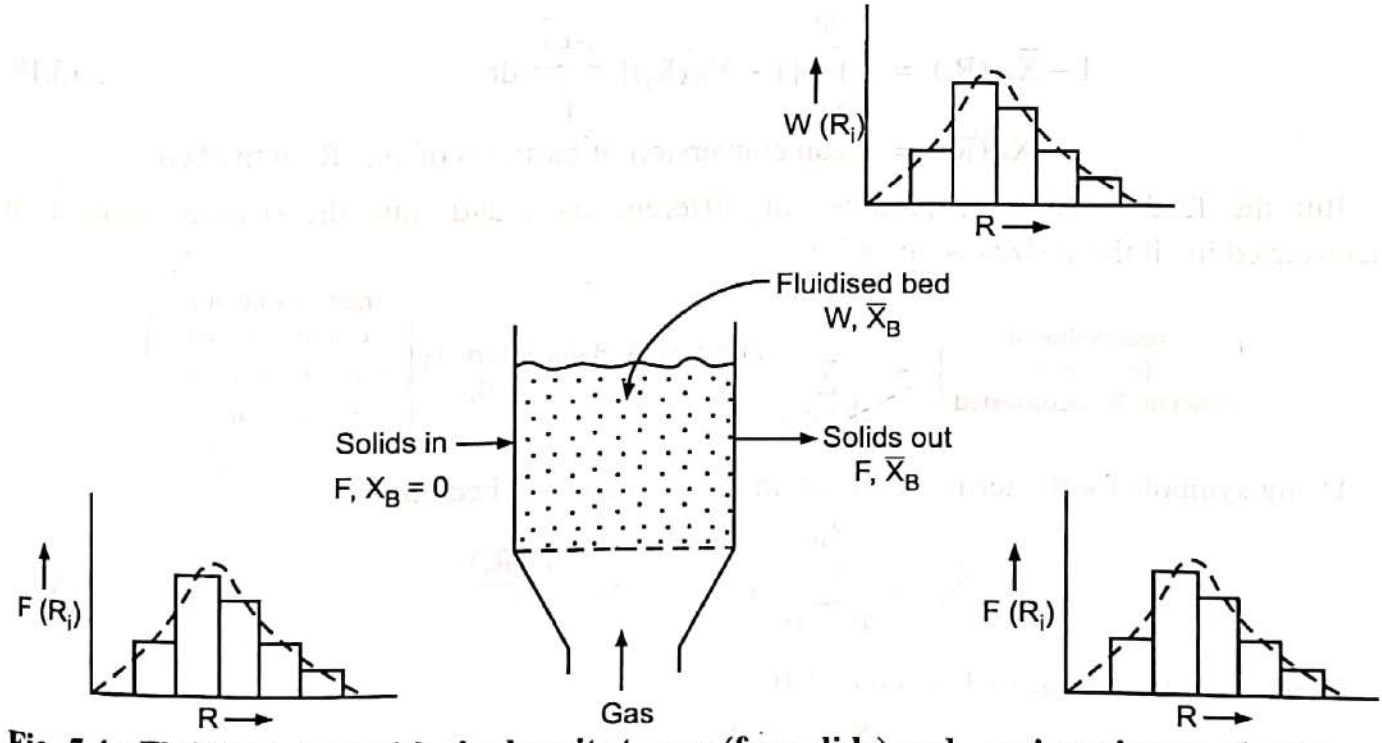


Fig. 5.4 : Fluidised bed with single exit stream (for solids) and – a size mixture of solids as feed. The size distribution of the feed stream, the exit stream and the bed are all the same

Consider the reactor of Fig. 5.4 with a feed consisting of a mixture of different size particles. In this case the solids are in mixed flow, the composition of gas is uniform and there is no elutriation of the fines, i.e., no carry over (a cyclone separator is used to separate and return to the bed any solids that may be entrained by the gas). Therefore, we have a single exit stream for all solids. As the exit stream represents the bed conditions (as the solids are in mixed flow), we have the same size distribution of the bed, feed and exit stream.

$$\therefore \quad \frac{F(R_i)}{F} = \frac{W(R_i)}{W} \qquad \ldots (5.16)$$

where,

F = feed rate of solids

$F(R_i)$ = quantity of material of size R_i in the feed

W = quantity of material in the reactor

$W(R_i)$ = quantity of material of size R_i in the reactor.

For this flow, also we have :

$$\bar{t} = \bar{t}(R_i) = \frac{W}{F} = \frac{\text{weight of all solids in the reactor (i.e. in the bed)}}{\text{feed rate of all solids to the reactor}} \qquad \ldots (5.17)$$

$\bar{t}$ = mean residence time of all solids in the bed

$\bar{t}(R_i)$ = mean residence time of material of any size R_i in the bed.

Let $\bar{X}_B(R_i)$ be the mean conversion of particles of size R_i in the bed.

$\therefore$ For mixed flow of material (particles) of size R_i which gets completely converted in time τ, the mean of fraction of B unconverted in particles of size R_i in the bed is given by

$$1 - \bar{X}_B(R_i) = \int_0^{\tau(R_i)} [1 - X_B(R_i)] \frac{e^{-t/\bar{t}}}{\bar{t}}\, dt \qquad \ldots (5.18)$$

$X_B(R_i)$ = mean conversion of particles of size R_i in the bed.

But the feed consists of particles of different sizes and thus the overall mean of B unconverted in all these sizes is given by

$$\begin{pmatrix}\text{mean value of}\\ \text{fraction of}\\ \text{reactant B unconverted}\end{pmatrix} = \sum_{\text{all sizes}} \begin{pmatrix}\text{fraction of B unconverted}\\ \text{in particles of size } R_i\end{pmatrix} \begin{pmatrix}\text{fraction of exit or}\\ \text{entering stream}\\ \text{which is of size } R_i,\\ \text{i.e., consisting of}\\ \text{particles of size } R_i\end{pmatrix}$$

Using symbols for the terms involved, the above equation becomes

$$1 - \bar{X}_B = \sum_{R_i = 0}^{R_m} [1 - \bar{X}_B(R_i)] \frac{F(R_i)}{F} \qquad \ldots (5.19)$$

Combining Equations (5.18) and (5.19) gives

$$1 - \bar{X}_B = \sum^{R_m} \int_0^{\tau(R_i)} \left[[1 - X_B(R_i)] \frac{e^{-t/\bar{t}}}{\bar{t}}\, dt \right] \frac{F(R_i)}{F} \qquad \ldots (5.20)$$

Replacing the first term expression of Equation (5.20) with the expression for a particular controlling resistance for each size of particle we can obtain final expressions for the various controlling resistances.

1. **For Diffusion through Gas Film Controlling (controlling resistance - film diffusion)**, i.e., when resistance of the gas film controls ... for high conversion ... the mean value of the fraction of B unconverted is given by

$$1 - \bar{X}_B = \sum^{R_m} \left\{ \frac{1}{2!}\left[\frac{\tau(R_i)}{\bar{t}}\right] - \frac{1}{3!}\left[\frac{\tau(R_i)}{\bar{t}}\right]^2 + \frac{1}{4!}\left[\frac{\tau(R_i)}{\bar{t}}\right]^3 - \ldots \right\} \frac{F(R_i)}{F} \qquad \ldots (5.21)$$

Let us obtain Equation (5.21).

We have : $$1 - \bar{X}_B = \sum^{R_m} \int_0^{\tau(R_i)} \left[[1 - X_B(R_i)] \frac{e^{-t/\bar{t}}}{\bar{t}} dt \right] \frac{F(R_i)}{F} \qquad \ldots (A)$$

For film diffusion controlling, the time-conversion (i.e., $t - X_B$) relationship is

$$\frac{t}{\tau} = X_B$$

For particles of size R_i

$$\frac{t}{\tau(R_i)} = X_B(R_i)$$

$$1 - X_B(R_i) = \left(1 - \frac{t}{\tau(R_i)}\right) \qquad \ldots (B)$$

Combining Equations (A) and (B) gives

$$1 - \bar{X}_B = \sum^{R_m} \left[\int_0^{\tau(R_i)} \left(1 - \frac{t}{\tau(R_i)}\right) \frac{e^{-t/\bar{t}}}{\bar{t}} dt \right] \frac{F(R_i)}{F} \qquad \ldots (C)$$

For the integration of the bracketed term of Equation (C) please refer to the integration procedure given to derive Equation (5.9).

Therefore, on the same line as Equation (5.9), we can write

$$\bar{X}_B = \sum^{R_m} \left\{ \frac{\bar{t}}{\tau(R_i)} \left(1 - e^{-\tau(R_i)/\bar{t}}\right) \right\} \frac{F(R_i)}{F} \qquad \ldots (D)$$

and in an equivalent expanded form for high conversion [refer to the procedure given to derive the Equation (5.10)].

$$1 - \bar{X}_B = \sum^{R_m} \left\{ \frac{1}{2!}\left[\frac{\tau(R_i)}{\bar{t}}\right] - \frac{1}{3!}\left[\frac{\tau(R_i)}{\bar{t}}\right]^2 + \frac{1}{4!}\left[\frac{\tau(R_i)}{\bar{t}}\right]^3 - \ldots\ldots \right\} \frac{F(R_i)}{F} \qquad \ldots (5.21)$$

2. **For chemical reaction step controlling** ... when the resistance of a chemical step controls, the mean value of fraction of B unconverted, for high conversion is given by

$$1 - \bar{X}_B = \sum^{R_m} \left\{ \frac{1}{4}\left[\frac{\tau(R_i)}{\bar{t}}\right] - \frac{1}{20}\left[\frac{\tau(R_i)}{\bar{t}}\right]^2 + \frac{1}{120}\left[\frac{\tau(R_i)}{\bar{t}}\right]^3 - \ldots \right\} \frac{F(R_i)}{F} \qquad \ldots (5.22)$$

3. **For diffusion through ash layer controlling (controlling resistance - ash diffusion),** the mean value of fraction of B unconverted, for high conversion, is given by

$$1 - \bar{X}_B = \sum^{R_m} \left\{ \frac{1}{5}\left[\frac{\tau(R_i)}{\bar{t}}\right] - \frac{19}{420}\left[\frac{\tau(R_i)}{\bar{t}}\right]^2 + \frac{41}{4620}\left[\frac{\tau(R_i)}{\bar{t}}\right]^3 - \cdots \right\} \frac{F(R_i)}{F} \quad \ldots (5.23)$$

where $\tau(R_i)$ is the time needed for complete reaction, conversion, of particles of size R_i.

Fluidised Bed with Entrainment of Solid Fines/Solid carry over in Fluidised Bed :

When the feed to a fluidised bed consists of a wide size distribution of solids, carryover of fines may occur. Where no cyclone separator is used, the solids entrained by the fluidising gas leave the system from the top as carryover and thus in such cases for solids we have one input stream (feed stream) and two exit streams.

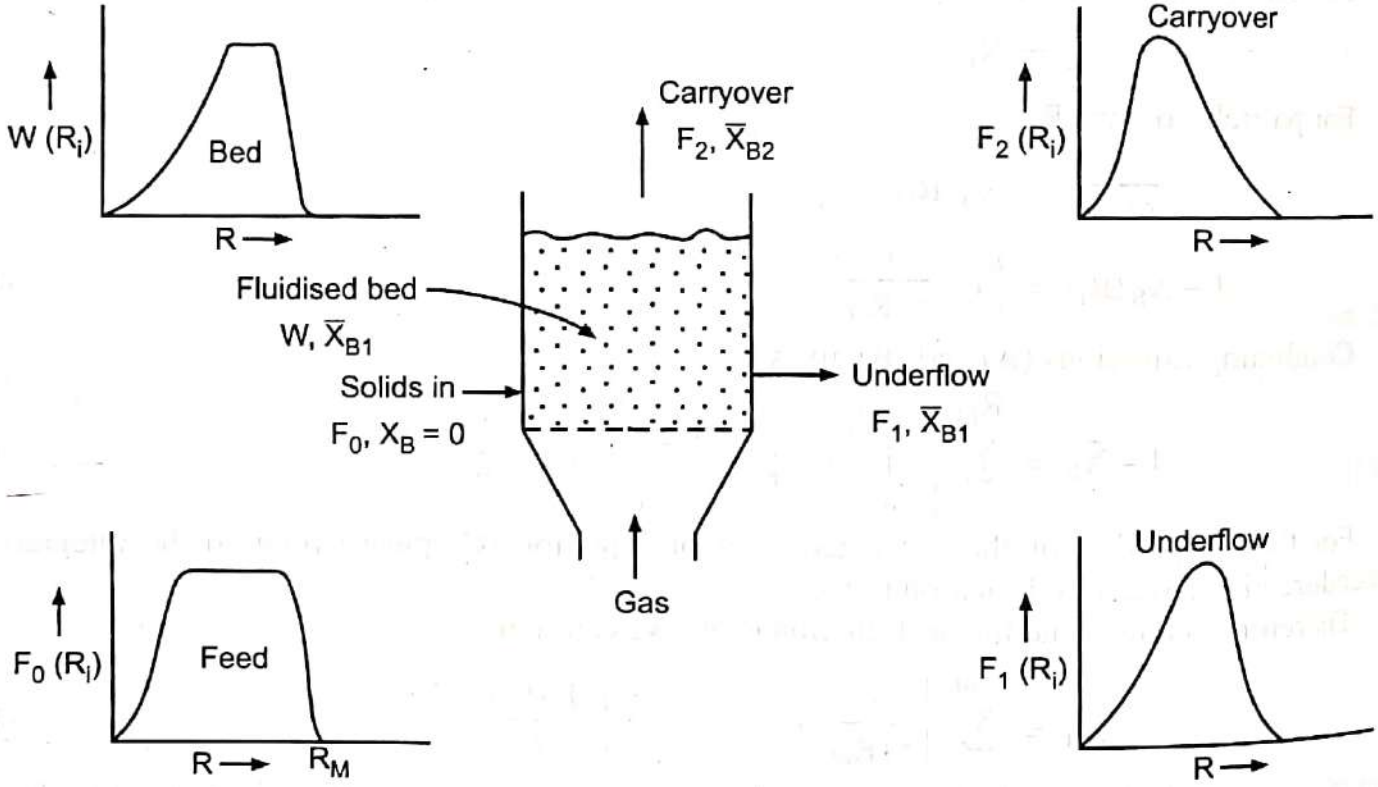


Fig. 5.5 : Fluidised bed with carryover of fines, showing size distributions of all streams and the bed

Let F_0 = mass flow rate of feed stream to the unit

F_1 = mass flow rate of underflow stream from the unit

F_2 = mass flow rate of carryover stream from the unit.

Then, the overall solid balance is

$$F_0 = F_1 + F_2 \quad \ldots (5.24)$$

The balance for particles of size R_i is

$$F_0(R_i) = F_1(R_i) + F_2(R_i) \quad \ldots (5.25)$$

As the solids are in mixed flow, the composition of the underflow stream is the same as the composition of solids within the bed. That is :

$$\frac{F_1(R_i)}{F_1} = \frac{W(R_i)}{W} \quad \ldots (5.26)$$

The mean residence time of particles of different sizes is not the same as small particles that are more likely to be blown out of the bed will stay in the bed for a shorter time than the larger particles.

For particles of size R_i, the mean residence time $\bar{t}(R_i)$ is given by

$$\bar{t}(R_i) = \frac{\text{weight of particles of size } R_i \text{ in the bed}}{\text{flow rate of particles of size } R_i \text{ in or out of the bed}}$$

$$= \frac{W(R_i)}{F_0(R_i)} \quad \ldots (5.27)$$

Using $F_0(R_i)$ from Equation (5.25), Equation (5.27) becomes

$$\bar{t}(R_i) = \frac{W(R_i)}{F_1(R_i) + F_2(R_i)} \quad \ldots (5.28)$$

From Equation (5.26), we get

$$F_1(R_i) = \frac{W(R_i)\, F_1}{W} \quad \ldots (5.29)$$

Combining Equations (5.28) and (5.29), we get

$$\bar{t}(R_i) = \frac{W(R_i)}{\frac{W(R_i)\, F_1}{W} + F_2(R_i)}$$

Dividing both numerator and denominator of the above equation by $W(R_i)$, we get

$$\bar{t}(R_i) = \frac{1}{\frac{F_1}{W} + \frac{F_2(R_i)}{W(R_i)}}$$

$$\therefore \quad \boxed{\bar{t}(R_i) = \frac{W(R_i)}{F_0(R_i)} = \frac{W(R_i)}{F_1(R_i) + F_2(R_i)} = \frac{1}{\frac{F_1}{W} + \frac{F_2(R_i)}{W(R_i)}}} \quad \ldots (5.30)$$

Therefore, the mean residence time, hence conversion, of particles of any given size is the same in the underflow and carryover stream.

The mean conversion of particles of size R_i is given by [Refer Equation (5.6)].

$$1 - \bar{X}_B = \int_0^{\tau(R_i)} [1 - X_B(R_i)] \frac{e^{-t/\bar{t}(R_i)}}{\bar{t}(R_i)}\, dt \quad \ldots (5.31)$$

For a feed consisting of a mixture of particles of different sizes, we have

$$\begin{pmatrix}\text{Mean value for fraction of reactant B unconverted}\end{pmatrix} = \sum_{\text{all sizes}} \begin{pmatrix}\text{fraction of B unconverted in particles of size } R_i\end{pmatrix} \begin{pmatrix}\text{fraction of feed consisting of particles of size } R_i \text{ i.e. fraction of feed which is of size } R_i\end{pmatrix}$$

or in terms of symbols

$$1 - \bar{X}_B = \sum_{R_m} [1 - \bar{X}_B (R_i)] \frac{F_0 (R_i)}{F_0} \quad \ldots (5.32)$$

Combining Equations (5.31) and (5.32) with the time-conversion relationships for various controlling resistances, we get

1. **For film diffusion controlling (controlling resistance - film diffusion) :** The mean value of B unconverted is given by (for high conversion)

$$\boxed{1 - \bar{X}_B = \sum_{R_m} \left\{ \frac{1}{2!} \left[\frac{\tau (R_i)}{\bar{t} (R_i)} \right] - \frac{1}{3!} \left[\frac{\tau (R_i)}{\bar{t} (R_i)} \right]^2 + \frac{1}{4!} \left[\frac{\tau (R_i)}{\bar{t} (R_i)} \right]^3 - \ldots \right\} \frac{F_0 (R_i)}{F_0}} \quad \ldots (5.33)$$

2. **For chemical reaction controlling (controlling resistance - reaction step) :**

$$\boxed{1 - \bar{X}_B = \sum_{R_m} \left\{ \frac{1}{4} \left[\frac{\tau (R_i)}{\bar{t} (R_i)} \right] - \frac{1}{20} \left[\frac{\tau (R_i)}{\bar{t} (R_i)} \right]^2 + \frac{1}{120} \left[\frac{\tau (R_i)}{\bar{t} (R_i)} \right]^3 - \ldots \right\} \frac{F_0 (R_i)}{F_0}} \quad \ldots (5.34)$$

3. **For ash diffusion controlling (controlling resistance - ash layer diffusion) :**

$$\boxed{1 - \bar{X}_B = \sum_{R_m} \left\{ \frac{1}{5} \left[\frac{\tau (R_i)}{\bar{t} (R_i)} \right] - \frac{19}{420} \left[\frac{\tau (R_i)}{\bar{t} (R_i)} \right]^2 + \frac{41}{4620} \left[\frac{\tau (R_i)}{\bar{t} (R_i)} \right]^3 - \ldots \right\} \frac{F_0 (R_i)}{F_0}} \quad \ldots (5.35)$$

In the above three expressions, $\bar{t} (R_i)$ is an unknown quantity. To use these expressions to find the conversion, $\bar{t} (R_i)$ must be known. Therefore, we present here a procedure to obtain $\bar{t} (R_i)$.

Elutriation experiments show that the rate at which particles of a given size blown out of a fluidised bed is proportional to the weight of such particles present in the fluidised bed.

Rate of carryover $\propto$ hold up in the bed ... for particles of any single size in the bed.

At steady-state, for particles of size R_i in the bed

$$\begin{pmatrix} \text{Rate of carryover of} \\ \text{particles of size } R_i \end{pmatrix} \propto (\text{weight of particles of size } R_i \text{ present in the bed})$$

$$\text{Rate of carryover of particles of size } R_i = \kappa \, (\text{weight of particles of size } R_i \text{ present in the bed})$$

or in terms of symbols,

$$F_2 (R_i) = \kappa (R_i) \, W (R_i)$$

$$\therefore \quad \kappa (R_i) = \frac{F_2 (R_i)}{W (R_i)} \quad \ldots (5.36)$$

where $\kappa (R_i)$ is called as the elutriation velocity constant for particles of size R_i. It is a function of the properties of system. It has the units of reciprocal time.

$\kappa (R_i)$ data are available from independent elutriation experiments.

Combining Equations (5.30) and (5.36) gives

$$\bar{t}\,(R_i) = \frac{W\,(R_i)}{F_0\,(R_i)} = \frac{1}{\frac{F_1}{W} + \frac{F_2\,(R_i)}{W\,(R_i)}}$$

$$\bar{t}\,(R_i) = \frac{W\,(R_i)}{F_0\,(R_i)} = \frac{1}{F_1/W + \kappa\,(R_i)} \qquad \ldots (5.37)$$

Combine Equations (5.37) and (5.26) to obtain an expression for F_1.

$$\bar{t}\,(R_i) = \frac{W\,(R_i)}{F_0\,(R_i)} = \frac{1}{F_1/W + \kappa\,(R_i)}$$

$$\therefore \quad W\,(R_i) = \frac{F_0\,(R_i)}{F_1/W + \kappa\,(R_i)}$$

$$\frac{F_1\,(R_i)}{F_1} = \frac{W\,(R_i)}{W}$$

$$\frac{F_1\,(R_i)}{F_1} = \frac{F_0\,(R_i)}{W\,[F_1/W + \kappa\,(R_i)]}$$

$$F_1\,(R_i) = \frac{F_0\,(R_i)\,F_1}{W\,[F_1/W + \kappa\,(R_i)]} = \frac{F_0\,(R_i)}{\frac{W}{F_1}\,[F_1/W + \kappa\,(R_i)]}$$

$$F_1\,(R_i) = \frac{F_0\,(R_i)}{1 + (W/F_1)\,K\,(R_i)} \qquad \ldots (5.38)$$

Summing over all particle sizes gives

$$F_1 = F_1\,(R_1) + F_1\,(R_2) + \ldots + F_1\,(R_m)$$

$$\boxed{F_1 = \sum_{}^{R_m} \frac{F_0\,(R_i)}{1 + (W/F_1)\,\kappa\,(R_i)}} \qquad \ldots (5.39)$$

Equation (5.39) is the desired expression for F_1. Knowing the size distribution of the feed, the size of bed and κ for all particle sizes, we can evaluate F_1 by using Equation (5.39) and adopting a trial and error procedure.

Procedure :

(i) Select a value of F_1 (Guess F_1).

(ii) Evaluate $F_1\,(R_i)$ using Equation (5.38).

(iii) Sum up all the $F_1\,(R_i)$ to get $F_{1\ \text{calculated}}$ [as in Equation (5.39)].

(iv) Check for $F_{1\ \text{calculated}} \approx F_{1\ \text{selected}}$.

If $F_{1\ \text{calculated}} \neq F_{1\ \text{selected}}$ then select/guess another value for F_1 and repeat the procedure till we do not get $F_{1\ \text{calculated}} \approx F_{1\ \text{selected}}$.

(v) Using Equation (5.38), calculate $\bar{t}\,(R_i)$ – the mean residence times of the various sizes of particles.

(vi) Use an appropriate expression depending upon the controlling resistance [Equations (5.33), (5.34) or (5.35)] to obtain the conversion (X_B).

SOLVED EXAMPLES

Example 5.1 : *Solid particles of uniform size are 80% converted, according to the shrinking core model (SCM) with ash diffusion step as the rate controlling, on passing through a reactor with uniform gas environment. What would be the conversion of solids if the reactor is made twice as large - all else remaining unchanged ?*

(i) Case - I : The solids are in plug flow.

(ii) Case - II : The solids are in mixed flow.

Solution :

Case - I : The solids are in plug flow.

Given : Ash diffusion step is rate controlling.

For ash diffusion controlling, particles of single unchanging (uniform) size and plug flow of solids, the conversion-time expression is

$$\frac{t_p}{\tau} = 1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)$$

$$\left[\frac{t}{\tau} = 1 - 3(1 - X_B)^{2/3} + 2(1 - X_B) \ldots\ldots \text{ash diffusion controls}\right]$$

Given : $X_B = \frac{80}{100} = 0.80$

$$\frac{t_p}{\tau} = 1 - 3(1 - 0.80)^{2/3} + 2(1 - 0.80)$$

$$\Rightarrow \quad \frac{t_p}{\tau} = 0.374 \ldots\ldots\ldots \text{for original reactor}$$

We have to determine the conversion of solids in the new reactor.

The size of the new reactor is twice that of the original reactor. This is the only difference between these reactors.

$\therefore$ For the new reactor with all other parameters except size remaining unchanged

$$\left(\frac{t_p}{\tau}\right)_{new} = 2\left(\frac{t_p}{\tau}\right)_{original}$$

$$= 2 \times 0.374 = 0.748$$

Conversion in the new reactor :

$$\frac{t_p}{\tau} = 1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)$$

$$0.748 = 1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)$$

Find X_B by a trial and error procedure.

$$\text{LHS} = 0.748$$

$$\text{RHS} = 1 - 3\,(1 - X_B)^{2/3} + 2\,(1 - X_B)$$

For $X_B = 0.94$, RHS = 0.66, $X_B = 0.96$, RHS = 0.729

For $X_B = 0.965$, RHS = 0.749

For $X_B = 0.9648$, RHS = 0.7482 $\simeq$ LHS

$\therefore$ $X_B = 0.9648$ as for this value LHS $\simeq$ RHS

Conversion of solids in the new reactor of size twice as large for the solids in plug flow = 0.9648 or 96.48% **... Ans. (i)**

Case - II : The solids are in mixed flow :

For the original reactor : $\bar{X}_B = 0.80$

Given : ash diffusion step is rate controlling. For mixed flow of particles of single unchanging size, uniform gas composition and ash resistance controlling we have :

$$1 - \bar{X}_B = \frac{1}{5}\frac{\tau}{\bar{t}} - \frac{19}{420}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{41}{4620}\left(\frac{\tau}{\bar{t}}\right)^3 - \ldots\ldots$$

$$1 - 0.8 = 0.2 = \frac{1}{5}\frac{\tau}{\bar{t}} - \frac{19}{420}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{41}{4620}\left(\frac{\tau}{\bar{t}}\right)^3$$

$$\text{LHS} = 0.20$$

$$\text{RHS} = \frac{1}{5}\frac{\tau}{\bar{t}} - \frac{19}{420}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{41}{4620}\left(\frac{\tau}{\bar{t}}\right)^3$$

Let $\frac{\tau}{\bar{t}} = x$

$$\text{RHS} = 0.2\,x - 0.04524\,x^2 + 8.87 \times 10^{-3}\,x^3$$

For $x = 1$ ... RHS = 0.1636, $x = 1.3$ ... RHS = 0.203

$x = 1.5$ ... RHS = 0.228, $x = 1.29$ ... RHS = 0.2017

$x = 1.4$ ... RHS = 0.2157, $x = 1.28$ ... RHS = 0.20 = LHS

$$\text{LHS} = \text{RHS for } x = \frac{\tau}{\bar{t}} = 1.28$$

$$x = \frac{\tau}{\bar{t}} = 1.28 \ldots\ldots \text{ for original reactor}$$

$\therefore$ $$\frac{\bar{t}}{\tau} = \frac{1}{1.28} = 0.78125 \ldots\ldots \text{ for original reactor.}$$

For the new reactor of size twice as large

$$W_{new} = 2\,W_{original}$$

and $$\bar{t} = \frac{W}{F}, \quad F \text{ is the same in both the cases}$$

$$\therefore \quad \left(\frac{\bar{t}}{\tau}\right)_{new} = 2\left(\frac{\bar{t}}{\tau}\right)_{original}$$

$$= 2 \times 0.78125 = 1.5625$$

$$\frac{\tau}{\bar{t}} \text{ for the new reactor} = \frac{1}{1.5625} = 0.64$$

Conversion in the new reactor (for ash resistance controlling) :

$$1 - \bar{X}_B = \frac{1}{5}\frac{\tau}{\bar{t}} - \frac{19}{420}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{41}{4620}\left(\frac{\tau}{\bar{t}}\right)^3$$

$$1 - \bar{X}_B = \frac{1}{5}(0.64) - \frac{19}{420}(0.64)^2 + \frac{41}{4620}(0.64)^3$$

$$\bar{X}_B = 0.9362$$

Mean conversion of solids in the reactor of size twice as large, for mixed flow of solids = 0.9362 or 93.62% ... Ans. (ii)

Example 5.2 : *Particles of uniform size are 60% converted in a single fluidised bed as per the shrinking-core model (SCM) with reaction controlling. Find the conversion if the reactor is made twice as large but containing the same amount of solids with the same gas environment.*

Solution : In a fluidised bed reactor for chemical reaction controlling, we have :

$$\bar{X}_B = 3\left(\frac{\bar{t}}{\tau}\right) - 6\left(\frac{\bar{t}}{\tau}\right)^2 + 6\left(\frac{\bar{t}}{\tau}\right)^3 \left(1 - e^{-\tau/\bar{t}}\right)$$

For the original reactor, conversion is 60%. Find $\frac{\bar{t}}{\tau}$ by a trial and error procedure.

$$\bar{X}_B = 0.60$$

$$0.60 = 3\left(\frac{\bar{t}}{\tau}\right) - 6\left(\frac{\bar{t}}{\tau}\right)^2 + 6\left(\frac{\bar{t}}{\tau}\right)^3 \left(1 - e^{-\tau/\bar{t}}\right)$$

Take $\frac{\bar{t}}{\tau} = 0.6$ ..... first trial

RHS = 0.6912 not matching

For $\frac{\bar{t}}{\tau} = 0.5$ ... RHS = 0.648

$\frac{\bar{t}}{\tau} = 0.40$ ... RHS = 0.5924

$\frac{\bar{t}}{\tau} = 0.41$ ... RHS = 0.5988

$$\therefore \quad \frac{\bar{t}}{\tau} = 0.415 \quad \cdots \quad \text{RHS} = 0.602$$

$$\therefore \quad \text{LHS} \simeq \text{RHS} \quad \text{for} \quad \frac{\bar{t}}{\tau} = 0.415$$

For the new reactor : It is two times the original reactor in size. Therefore,

$$\frac{\bar{t}}{\tau} = 2\,(\bar{t}/\tau \text{ for } R-1) = 2 \times 0.415 = 0.83$$

Now we will find the conversion in the new reactor.

$$\bar{X}_B = 3\left(\frac{\bar{t}}{\tau}\right) - 6\left(\frac{\bar{t}}{\tau}\right)^2 + 6\left(\frac{\bar{t}}{\tau}\right)^3 \left(1 - e^{-\tau/\bar{t}}\right)$$

$$= 2\,(0.83) - 6\,(0.83)^2 + 6\,(0.83)^3 \left(1 - e^{-1/0.83}\right)$$

$$= 0.7589 \simeq 0.759$$

Conversion in the new reactor of size double that of the original = 75.9% **... Ans.**

Example 5.3 : *In a fluidised bed reactor, particles of iron sulphide of uniform size are to be roasted. The time required for complete conversion is 20 min. The mean residence time of particles in the fluidised bed is 60 min. The particles remained as hard solids during reaction. τ and R relationship found to be $\tau \propto R^{1.5}$.*

Calculate the fraction of the original iron sulphide that remains unreacted.

Solution : As a hard material is formed during the reaction we can exclude film diffusion as a possibility of the controlling resistance.

With this we are left with :

(i) ash layer diffusion, and

(ii) chemical reaction – as the controlling resistance.

Given : τ and R are related by $\tau \propto R^{1.5}$

1. For chemical reaction controlling, τ and R are related by $\tau \propto R$.

2. For ash layer diffusion controlling, τ and R are related by $\tau \propto R^2$.

As the given relationship between τ and R lies between 1 and 2, both these mechanisms offer resistance to conversion.

So we will first calculate the conversion by considering :

(i) Chemical reaction as the controlling resistance.

(ii) Ash layer diffusion as the controlling resistance and then we will find the average of the conversion values obtained for (i) and (ii) to get the result.

Given : time required for complete conversion $= \tau = 20$ min.

mean residence time of particles in the bed $= \bar{t} = 60$ min.

$$\frac{\tau}{\bar{t}} = \frac{20}{60} = 0.333 \quad \text{and} \quad \frac{\bar{t}}{\tau} = \frac{60}{20} = 3 \text{ (large value)}$$

(i) Chemical reaction controlling :

$$1 - \bar{X}_B = \frac{1}{4}\left(\frac{\tau}{\bar{t}}\right) - \frac{1}{20}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau}{\bar{t}}\right)^3 - \cdots\cdots$$

$$= \frac{1}{4}(0.333) - \frac{1}{20}(0.333)^2 + \frac{1}{120}(0.333)^3$$

$$= 0.078 \text{ (i.e., 7.8\%)}$$

(ii) Ash layer diffusion controlling :

$$1 - \bar{X}_B = \frac{1}{5}\left(\frac{\tau}{\bar{t}}\right) - \frac{19}{420}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{41}{4620}\left(\frac{\tau}{\bar{t}}\right)^3 - \cdots\cdots$$

$$= \frac{1}{5}(0.333) - \frac{19}{420}(0.333)^2 + \frac{41}{4620}(0.333)^3$$

$$= 0.0619 \simeq 0.062 \text{ (i.e., 6.2\%)}$$

The fraction of iron sulphide remaining unconverted is

(i) Chemical reaction controlling : $1 - \bar{X}_B = 0.078$

(ii) Ash layer diffusion controlling : $1 - \bar{X}_B = 0.062$

Lower limit to the expected conversion is given by (i) and

Upper limit to the expected conversion is given by (ii)

The fraction of iron sulphide that remains unconverted is between 0.062 (i.e., 6.2%) and 0.078 (i.e., 07.8%)

Average fraction that remains unconverted $= \dfrac{0.062 + 0.078}{2} = 0.07$

$\therefore$ **Fraction of the original sulphide that remains unreacted $= 1 - \bar{X}_B = 0.07$ (i.e., 7%)** ... Ans.

Example 5.4 : *A steady flow fluidised bed reactor of size 20 cm ID × 2 m long operates with a solid feed consisting of*

30% of 50 μm radius particles

40% of 100 μm radius particles

30% of 200 μm radius particles

The fluidising gas is the gas phase reactant and has uniform composition. The time required for complete conversion is 5, 10 and 20 minutes for the three sizes of feed at the planned operating conditions. The feed rate to the reactor is 1 kg solid/min and the fluidised bed contains 10 kg of solids.

The solids are hard and remain unchanged in size and weight during reaction. The change in gas phase composition in the bed is negligible.

Calculate the conversion of solids in the fluidised bed reactor.

Solution : Here we have mixed flow of a size mixture of unchanging size and uniform gas composition.

For mixed flow of solids, the mean residence time $\bar{t}\ (R_i)$ of material of any size R_i is equal to the mean residence time of all solids in the bed.

Given : W = weight of all solids in the reactor = 10 kg

F = feed rate of all solids to the reactor = 1 kg/min

$$\bar{t} = \bar{t}\ (R_i) = \frac{W}{F} = \frac{10}{1} = 10 \text{ min}$$

$$F(R_i) = \text{Wt. fraction of particles of size } R_i = \frac{\text{Wt. \%}}{100} \times F$$

$\tau(R_i)$ is the time required for complete conversion of particles of size R_i

$F(50\ \mu m) = 0.30 \times 1 = 0.3$ kg/min, $\tau(50\ \mu m) = 5$ min

[Since the feed consists of 30% of 50 μm - radius particles]

Similarly,

$F(100\ \mu m) = 0.40 \times 1 = 0.4$ kg/min, $\tau(100\ \mu m) = 10$ min

$F(200\ \mu m) = 0.3 \times 1 = 0.3$ kg/min, $\tau(200\ \mu m) = 20$ min

For 50 μm radius particles ..... $\tau = 5$ min

For 100 μm radius particles ..... $\tau = 10$ min

For 200 μm radius particles ..... $\tau = 20$ min

We see from the above cited values of R and τ that

$$\tau \propto R$$

$$\frac{\tau_1}{\tau_2} = \frac{\tau_2}{\tau_3} = \frac{R_1}{R_2} = \frac{R_2}{R_3}$$

$$\frac{5}{10} = \frac{10}{20} = \frac{50}{100} = \frac{100}{200} = 0.50$$

Since $\tau \propto R$, chemical reaction step is the rate controlling, i.e., chemical reaction controls the rate.

Thus, for the chemical reaction controlling in a fluidised bed reactor with no elutriation of fine particles, the mean value for fraction of B unconverted is given by

$$1-\bar{X}_B = \sum_{R_m} \left\{\frac{1}{4}\frac{\tau(R_i)}{\bar{t}} - \frac{1}{20}\left(\frac{\tau(R_i)}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau(R_i)}{\bar{t}}\right)^3\right\} \frac{F(R_i)}{F}$$

$$= \left[\frac{1}{4}\frac{\tau(50)}{\bar{t}} - \frac{1}{20}\left(\frac{\tau(50)}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau(50)}{\bar{t}}\right)^3\right] \frac{F(50)}{F}$$

$$+ \left[\frac{1}{4}\frac{\tau(100)}{\bar{t}} - \frac{1}{20}\left(\frac{\tau(100)}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau(100)}{\bar{t}}\right)^3\right] \frac{F(100)}{F}$$

$$+ \left[\frac{1}{4}\frac{\tau(200)}{\bar{t}} - \frac{1}{20}\left(\frac{\tau(200)}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau(200)}{\bar{t}}\right)^3\right] \frac{F(200)}{F}$$

$$1-\bar{X}_B = \left[\frac{1}{4}\times\frac{5}{10} - \frac{1}{20}\left(\frac{5}{10}\right)^2 + \frac{1}{120}\left(\frac{5}{10}\right)^3\right] \frac{0.3}{1}$$

$$+ \left[\frac{1}{4}\times\frac{10}{10} - \frac{1}{20}\left(\frac{10}{10}\right)^2 + \frac{1}{120}\left(\frac{10}{10}\right)^3\right] \frac{0.40}{1}$$

$$+ \left[\frac{1}{4}\times\frac{20}{10} - \frac{1}{20}\left(\frac{20}{10}\right)^2 + \frac{1}{120}\left(\frac{20}{10}\right)^3\right] \frac{0.30}{1}$$

$$1-\bar{X}_B = 0.03406 + 0.0833 + 0.11 = 0.22736 \approx 0.2274$$

$$\bar{X}_B = 0.7726$$

Mean conversion of solids $= 0.7726 \times 100 = 77.26\%$

$\therefore$ **Mean conversion of solids in the fluidised bed reactor = 77.26%** ... Ans.

Example 5.5 : *An ore of uniform size particles is to be roasted in a fluidised bed reactor. The time required for complete conversion of solid particles is 20 min and the mean residence time of particles in the bed is 48 min. The solids remain unchanged in size during reaction. Calculate the fraction of the original ore remaining unconverted assuming :*

(i) the chemical reaction step to be rate controlling.

(ii) the ash diffusion step to be rate controlling.

Solution : Time required for complete conversion $= \tau = 20$ min

Mean residence time of solids in the bed $= \bar{t} = 48$ min

(i) Chemical reaction step is rate controlling.

For particles of single unchanging size in a fluidised bed with chemical reaction controlling, we have :

$$1-\bar{X}_B = \frac{1}{4}\left(\frac{\bar{t}}{\tau}\right) - \frac{1}{20}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau}{\bar{t}}\right)^3 - \ldots\ldots$$

$$\tau = 20 \text{ min}, \ \bar{t} = 48 \text{ min}, \ \therefore \ \frac{\tau}{\bar{t}} = \frac{20}{48} = 0.4167$$

$$1 - \bar{X}_B = \frac{1}{4}(0.4167) - \frac{1}{20}(0.4167)^2 + \frac{1}{120}(0.4167)^3$$

$$= 0.096$$

The fraction of the original ore remaining unconverted $(1 - \bar{X}_B)$ is 0.096 or 9.6% ... Ans. (i)

(ii) Ash diffusion step is rate controlling :

For this mechanism the mean value for the fraction of solids unconverted is given by

$$1 - \bar{X}_B = \frac{1}{5}\left(\frac{\tau}{\bar{t}}\right) - \frac{19}{420}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{41}{4620}\left(\frac{\tau}{\bar{t}}\right)^3 - \ldots\ldots$$

where $\frac{\tau}{\bar{t}} = 0.4167$

$$1 - \bar{X}_B = \frac{1}{5}(0.4167) - \frac{19}{420}(0.4167)^2 + \frac{41}{4620}(0.4167)^3$$

$$= 0.076$$

The fraction of the original ore remaining unconverted $(1 - \bar{X}_B)$ is 0.076 or 7.6% ... Ans. (ii)

Example 5.6 : *A feed consisting of 20% of 1 mm particles, 30% of 2 mm particles and 50% of 4 mm particles passes through a rotating tubular reactor (resembling a cement kiln) where it reacts with gas to yield a hard non-friable solid product according to the shrinking-core model (SCM) with chemical reaction controlling. The time required for complete conversion of 4 mm particles (τ) is 4 h.*

Determine the mean conversion of the solids for a residence time of 15 min.

Solution : As the solids are continuously passing through the rotating tubular reactor, they are considered to be in plug flow.

Time required for complete conversion of 4 mm particles (τ) = 4 h.

Now we will calculate τ for 1 mm and 2 mm size particles.

Given : Chemical reaction step is the rate controlling.

For chemical reaction controlling, we have

$$\tau \propto R$$

$\tau_1 \propto R_1$ $\quad \tau_2 \propto R_2$ and $\tau_4 \propto R_4$

$R_1 = 1$ mm $\quad \tau_1 = ?$

$R_2 = 2$ mm $\quad \tau_2 = ?$

$R_4 = 4$ mm $\quad \tau_4 = 4$ h

Therefore,

$$\frac{\tau_1}{\tau_4} = \frac{R_1}{R_4}$$

$$\therefore \quad \tau_1 = \frac{\tau_4 R_1}{R_4} = \frac{4 \times 1}{4} = 1 \text{ h}$$

and

$$\tau_2 = \frac{\tau_4 R_2}{R_4} = \frac{4 \times 2}{4} = 2 \text{ h}$$

Given : Residence time of 15 min in the reactor

$$\bar{t} = t_p = 15 \text{ min} = 0.25 \text{ h}$$

Now we will calculate the mean conversion of solids $\left(\bar{X}_B\right)$ for $t_p = 0.25$ h

For a mixture of particles of different but unchanging sizes and plug flow of solids, the fraction of B unconverted is given by

$$1 - \bar{X}_B = \sum_{}^{R_m} \left[1 - X_B (R_i)\right] \frac{F(R_i)}{F} \quad \ldots (1)$$

For chemical reaction controlling, the conversion-time relationship is

$$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$$

In the present case of mixture of particles in plug flow of each size, the above relation becomes

$$\frac{t_p}{\tau(R_i)} = 1 - \left[1 - X_B(R_i)\right]^{1/3}$$

$$\left[1 - X_B(R_i)\right] = \left(1 - \frac{t_p}{\tau(R_i)}\right)^3 \quad \ldots (2)$$

Combining Equations (2) and (3) for three sizes of particles, we can write

$$1 - \bar{X}_B = \left(1 - \frac{t_p}{\tau_1}\right)^3 \frac{F(1)}{F} + \left(1 - \frac{t_p}{\tau_2}\right)^3 \frac{F(2)}{F} + \left(1 - \frac{t_p}{\tau_4}\right)^3 \frac{F(4)}{F} \quad \ldots (3)$$

For 1 mm particles : $\quad \frac{F(1)}{F} = \frac{0.20\,F}{F} = 0.20, \ \tau_1 = 1$ h

For 2 mm particles : $\quad \frac{F(2)}{F} = \frac{0.30\,F}{F} = 0.30, \ \tau_2 = 2$ h

For 4 mm particles : $\frac{F(4)}{F} = \frac{0.50\,F}{F} = 0.50, \qquad \tau_4 = 4\text{ h}$

and $t_p = 0.25\text{ h}$

Substituting the values of the terms involved, Equation (3) becomes

$$1 - \bar{X}_B = \left(1 - \frac{0.25}{1}\right)^3 \times 0.20 + \left(1 - \frac{0.25}{2}\right)^3 \times 0.3 + \left(1 - \frac{0.25}{4}\right)^3 \times 0.5$$

$$= 0.697$$

$$\bar{X}_B = 1 - 0.697 = 0.303, \text{ i.e., } 30.3\%$$

Mean conversion of solids for residence time of 15 min = 30.3% **... Ans.**

Extra :

The mean conversion of solids for the residence time of 30 min (= 0.5 h) is

$$1 - \bar{X}_B = \left(1 - \frac{0.5}{1}\right)^3 \times 0.2 \div \left(1 - \frac{0.50}{2}\right)^3 \times 0.3 + \left(1 - \frac{0.5}{4}\right)^3 \times 0.5$$

$$1 - \bar{X}_B = 0.486$$

$$\bar{X}_B = 1 - 0.486 = 0.514, \quad \text{i.e., } 51.4\%$$ **... Ans.**

Example 5.7 : *A moving grate is continuously fed with a feed consisting of 30% of 50 μm radius particles, 40% of 100 μm radius particles and 30% of 200 μm radius particles. The feed is fed in the form of thin layer and moves on the grate in a crosscurrent fashion to the flow of reactant gas. The time required for complete conversion is*

5 min for 50 μm particles

10 min for 100 μm particles

and 20 min for 200 μm particles.

Calculate the conversion of solids for a residence time of 8 min over the moving grate.

Solution : It is clear from the statement of the problem that the solids are moving in plug flow over the grate. The residence time available for reaction over the grate (t_p) is 8 min.

For a mixed feed with unchanging size, the mean conversion $\bar{X}_B$ of the solids is given by

$$1 - \bar{X}_B = \sum_{R_i = 0}^{R_m} [1 - X_B(R_i)] \frac{F(R_i)}{F}$$

For our three sizes of particles, the above equation becomes

$$1 - \bar{X}_B = [1 - X_B(50\,\mu m)] \frac{F(50\,\mu m)}{F} + [1 - X_B(100\,\mu m)] \frac{F(100\,\mu m)}{F}$$

$$+ [1 - X_B(200\,\mu m)] \frac{F(200\,\mu m)}{F} \qquad \ldots (1)$$

For 50 μm radius particles : $\frac{F(50\mu)}{F} = \frac{0.30\,F}{F} = 0.30$, $\tau\,(50\ \mu m) = 5$ min

For 100 μm radius particles : $\frac{F(100\ \mu m)}{F} = 0.40$, $\tau\,(100\ \mu m) = 10$ min

F contains 40% of 100 μm radius particles.

$\therefore$ $F\,(100\ \mu m) = 0.40\,F$

$$\frac{F\,(100\ \mu m)}{F} = \frac{0.40\,F}{F} = 0.40$$

For 200 μm radius particles : $\frac{F\,(200\ \mu m)}{F} = 0.30$, $\tau\,(200\ \mu m) = 20$ min

For three size fractions of particles :

$R_1 = 50\ \mu m$ $\tau_1 = 5$ min

$R_2 = 100\ \mu m$ $\tau_2 = 10$ min

$R_3 = 200\ \mu m$ $\tau_3 = 20$ min

$$\frac{R_2}{R_1} = \frac{\tau_2}{\tau_1} \qquad \frac{100}{50} = 2 = \frac{10}{5} = 2$$

$$\frac{R_3}{R_2} = \frac{\tau_3}{\tau_2} \qquad \frac{200}{100} = 2 = \frac{20}{10} = 2$$

$\therefore$ $R_1 : R_2 : R_3 = \tau_1 : \tau_2 : \tau_3$

Hence, $\tau \propto R$

$\tau_1 \propto R_1$ $\tau_2 \propto R_2$

and we get $\frac{\tau_2}{\tau_1} = \frac{R_2}{R_1}$

Since $\tau \propto R$, chemical reaction controls the rate, i.e., the chemical reaction step is the rate controlling.

For chemical reaction controlling, the conversion-time relation for each size is given by

$$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$$

In our present case, the above equation becomes

$$\frac{t_p}{\tau(R_i)} = 1 - \left[1 - \bar{X}_B\,(R_i)\right]^{1/3} \quad \text{... for any size } R_i$$

$$\left[1 - \bar{X}_B\,(R_i)\right] = \left[1 - \frac{t_p}{\tau(R_i)}\right]^3 \qquad \text{... (2)}$$

Given : $t_p = 8$ min

So the particles having τ value less than $t_p = 8$ min are completely converted on the grate as the residence time is greater than the time required for complete conversion.

In our case, the particles of size 50 μm with $\tau = 5$ min (less than $t_p = 8$ min) are completely converted and hence do not contribute to the fraction of solids unconverted, $1 - \bar{X}_B$.

So for calculating the fraction of solids unconverted we have to consider only 100 μm and 200 μm particles.

Combining Equations (1) and (2) and setting the term for 50 μm particles equal to zero, we get

$$1 - \bar{X}_B = \left[1 - \frac{t_p}{\tau\,(50\ \mu m)}\right]^3 \times 0.3 + \left[1 - \frac{t_p}{\tau\,(100\ \mu m)}\right]^3 \times 0.40 + \left[1 - \frac{t_p}{\tau\,(200\ \mu m)}\right]^3 \times 0.30$$

$$1 - \bar{X}_B = 0 + \left[1 - \frac{8}{10}\right]^3 \times 0.40 + \left[1 - \frac{8}{20}\right]^3 \times 0.30$$

$$1 - \bar{X}_B = 0.068 \text{ (i.e. } 6.8\%)$$

$$\therefore \quad \bar{X}_B = 1 - 0.068 = 0.932 \text{ or } 93.2\%$$

Fraction of solid converted (conversion of solids) = 93.2% ... Ans.

Example 5.8 : *A feed comprising of 20% of 1 mm particles and smaller, 30% of 2 mm particles and 50% of 4 mm particles passes through a rotating tubular reactor where it reacts with gas to yield a hard nonfriable solid product according to the shrinking - core model (SCM) with the reaction step as rate controlling. The time required for complete conversion of 4 mm particles (τ) is 4 h.*

(a) Determine the residence time needed in the tubular reactor for :

(i) 100% conversion of solids

(ii) 75% conversion of solids

(iii) 95% conversion of solids.

Solution : Here we consider that the solids are in a plug flow.

Time required for complete conversion of 4 mm particles (τ) = 4 h.

Now we will evaluate τ for 1 mm and 2 mm size particles.

Given : Chemical reaction step is rate controlling.

For chemical reaction controlling, we have :

$$\tau \propto R$$

$$\tau_1 \propto R_1 \quad \text{and} \quad \tau \propto R_2 \qquad \tau_4 \propto R_4$$

$$\tau_1 = \frac{\tau_4 R_1}{R_4}, \qquad \tau_2 = \frac{\tau_4 R_2}{R_4}$$

$R_1 = 1$ mm $\quad \tau_1 = ?$

$R_2 = 2$ mm $\quad \tau_2 = ?$

$R_4 = 4$ mm $\quad \tau_4 = 4$ h

For 1 mm particles :

$$\tau_1 = \frac{\tau_4 R_1}{R_4} = \frac{4 \times 1}{4} = 1 \text{ h}$$

For 2 mm particles :

$$\tau_2 = \frac{\tau_4 R_2}{R_4} = \frac{4 \times 2}{4} = 2 \text{ h}$$

Now we will find $\bar{t} = t_p$ (here) for $\bar{X}_B = 1.0$ (100% conversion)

For mixture of particles of different but unchanging sizes and plug flow of solids, the fraction of B unconverted is given by

$$1 - \bar{X}_B = \sum_{}^{R_m} [1 - X_B (R_i)] \frac{F(R_i)}{F} \qquad \ldots (1)$$

For chemical reaction controlling, the conversion-time relationship is

$$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$$

In present case of mixture of particles in plug flow for each size, the above relation becomes

$$\frac{t_p}{\tau(R_i)} = 1 - [1 - X_B(R_i)]^{1/3}$$

$$\therefore \qquad \left[1 - \bar{X}_B (R_i)\right] = \left(1 - \frac{t_p}{\tau(R_i)}\right)^3 \qquad \ldots (2)$$

Combining Equations (1) and (2) for three sizes of particles, we can write :

$$1 - \bar{X}_B = \left(1 - \frac{t_p}{\tau_1}\right)^3 \frac{F(1)}{F} + \left(1 - \frac{t_p}{\tau_2}\right)^3 \frac{F(2)}{F} + \left(1 - \frac{t_p}{\tau_4}\right) \frac{F(4)}{F}$$

For 1 mm particles :

$$\frac{F(1)}{F} = \frac{0.20\,F}{F} = 0.20 \quad , \quad \tau_1 = 1 \text{ h}$$

For 2 mm particles :

$$\frac{F(2)}{F} = \frac{0.30\,F}{F} = 0.30 \quad , \quad \tau_2 = 2 \text{ h}$$

For 4 mm particles :

$$\frac{F(4)}{F} = \frac{0.50\,F}{F} = 0.50 \quad , \quad \tau_4 = 4\,h$$

$$\bar{X}_B = 1.0$$

$$\therefore \quad 1 - 1 = \left(1 - \frac{t_p}{1}\right)^3 (0.20) + \left(1 - \frac{t_p}{2}\right)^3 (0.3) + \left(1 - \frac{t_p}{4}\right)^3 (0.5)$$

$$0 = 0.2\,(1 - t_p)^3 + 0.3\left(1 - \frac{t_p}{2}\right)^3 + 0.5\left(1 - \frac{t_p}{4}\right)^3 \qquad \ldots (3)$$

Adopt a trial and error procedure to get t_p :

If we assume $t_p = 1$, the first term on the RHS of Equation (3) becomes zero but other than the terms yield + value > 0.

If we take $t_p < 1$, all of the three terms will be there and RHS is > 0.

If we take $t_p = 2$ min, the first two terms will be zero (as $\tau_1 < t_p$) but with the third term, RHS is > 0.

For RHS to be zero, we have to take $t_p = 4$ h so that contribution of the first two terms will be zero (as $\tau_1 = 1$ h and $\tau_2 = 2$ h – less than $t_p = 4$h) and the third term also become zero.

So $t_p = 4\,h$

With $t_p = 4$ h, the third term of Equation (3) becomes zero and the first two terms of Equation (3) will not contribute to the fraction unconverted as :

$\tau_1 = 1$ h, $\tau_2 = 2$ h both are less than 4 h.

$\therefore$ **Residence time needed for 100% conversion of solids = 4 h** **... Ans. (i)**

(i) Residence time needed for 75% conversion of solids, i.e., for $\bar{X}_B = 0.75$.

We have : $1 - \bar{X}_B = 0.2\,(1 - t_p)^3 + 0.3\left(1 - \frac{t_p}{2}\right)^3 + 0.5\left(1 - \frac{t_p}{4}\right)^3$

$$1 - 0.75 = 0.25 = 0.2\,(1 - t_p)^3 + 0.3\left(1 - \frac{t_p}{2}\right)^3 + 0.5\left(1 - \frac{t_p}{4}\right)^3$$

LHS = 0.25

For $t_p = 1$ h

RHS = 0 + 0.0375 + 0.211 = 0.285

For $t_p = 0.98$ h RHS = 0.255

For $t_p = 0.99$ h RHS = 0.2577

For $t_p = 0.995$ h RHS = 0.25

So for $t_p = 0.995$ h LHS = RHS = 0.25

Residence time needed for 75% conversion of solids = 0.995 h = 59.7 min **... Ans. (ii)**

(ii) Residence time needed for 95% conversion of solids, i.e., for $\bar{X}_B = 0.95$

$$1 - 0.95 = 0.05 = 0.2\,(1 - t_p)^3 + 0.3\left(1 - \frac{t_p}{2}\right)^3 + 0.5\left(1 - \frac{t_p}{4}\right)^3$$

LHS = 0.05

For $t_p = 2$ h, the first term will not contribute as $\tau_2 = 1\text{ h} < t_p$.

RHS = 0 + 0 + 0.0625

For $t_p = 2.15$ h, the first two terms will not contribute to the fraction unconverted as $\tau_1 = 1$ h and $\tau_2 = 2$ h – both are $< t_p = 2.15$ h.

RHS = 0 + 0 + 0.0495

For $t_p > 2$ h, the first two terms will not contribute to the fraction of solid unconverted as $\tau_1 = 1$ h, $\tau_2 = 2$ h will be less than t_p.

$$\therefore \quad 0.05 = 0 + 0 + 0.5\left(1 - \frac{t_p}{4}\right)^3$$

$$\therefore \quad \left(\frac{0.05}{0.5}\right)^{1/3} = 1 - \frac{t_p}{4}$$

$$t_p = 2.143\text{ h}$$

Residence time needed for 95% conversion of solids = 2.143 h (128.6 min) ... Ans. (iii)

Example 5.9 : *Solids of unchanging size (R = 0.3 mm) are reacted with gas in a steady flow laboratory-scale fluidised bed reactor according to the shrinking-core model (SCM) with the reaction steps to be rate controlling. Following result is obtained in the laboratory scale reactor.*

For F_o = 10 g/s, W = 1000 g : $\bar{X}_B$ = 0.75.

Design a commercial scale fluidised bed reactor (i.e., determine W) to treat 4t/h of solids of size R = 0.3 mm to 98% conversion. [1 t = 1000 kg].

Solution : In the laboratory scale fluidised bed reactor :

Feed rate = F_o = 10 g/s, Mean conversion = $\bar{X}_B$ = 0.75

Weight of solids in the reactor = W = 1000 g

$$\bar{t} = \text{mean residence time} = \frac{W}{F_o} = \frac{1000}{10} = 100\text{ s}$$

Given : Chemical reaction step is rate controlling and single unchanging size.

For single unchanging size solids in a fluidised bed reactor and for chemical reaction controlling – the mean value of the fraction of solid (B) unconverted is given by $\left(\text{for large } \frac{\bar{t}}{\tau}\right)$:

$$1 - \bar{X}_B = \frac{1}{4}\left(\frac{\tau}{\bar{t}}\right) - \frac{1}{20}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau}{\bar{t}}\right)^3$$

Let $\frac{\tau}{\bar{t}} = y$

$$1 - \bar{X}_B = \frac{1}{4}y - \frac{1}{20}y^2 + \frac{1}{120}y^3$$

$$1 - 0.75 = 0.25 = 0.25\,y - 0.05\,y^2 + 8.33 \times 10^{-3}\,y^3$$

$$\text{LHS} = 0.25$$

$$\text{RHS} = 0.25\,y - 0.05\,y^2 + 8.33 \times 10^{-3}\,y^3$$

Let $y = 1$ ... RHS = 0.208

$y = 1.5$ ... RHS = 0.2906

$y = 1.2$ ... RHS = 0.2424

$y = 1.25$ ... RHS = 0.2506

$y = 1.245$ ... RHS = $0.2498 \simeq 0.25$

$\text{LHS} \simeq \text{RHS}$ for $y = 1.245$

$\therefore$ $y = 1.245$

$$y = \frac{\tau}{\bar{t}} = 1.245$$

$$\tau = 1.245\,\bar{t}$$

$$= 1.245 \times 100 = 124.5 \text{ s}$$

Time required for complete conversion of solid = τ = 124.5 s.

We have to make use of this τ to design a commercial reactor.

(i) Commercial scale fluidised bed reactor :

For chemical reaction controlling, τ = 124.5 s

Given : Conversion = 98% $\quad \therefore \bar{X}_B = 0.98$

We will calculate $\bar{t}$ and then W. For chemical reaction controlling, the fraction of solid remain unconverted is given by (for high conversion) :

$$1 - \bar{X}_B = \frac{1}{4}\left(\frac{\tau}{\bar{t}}\right) - \frac{1}{20}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau}{\bar{t}}\right)^3$$

Let $\frac{\tau}{\bar{t}} = y$

$$1 - 0.98 = \frac{1}{4}(y) - \frac{1}{20}y^2 + \frac{1}{120}y^3$$

$$0.02 = 0.25\,y - 0.05\,y^2 + 8.33 \times 10^{-3}\,y^3$$

LHS = 0.02

RHS = $0.25\,y - 0.05\,y^2 + 8.33 \times 10^{-3}\,y^3$

$y = 0.1$ ... RHS = 0.0245

$y = 0.08$ ... RHS = 0.0197

$y = 0.085$ ... RHS = 0.0209

$y = 0.082$ ... RHS = 0.02017

$y = 0.0815$ ... RHS = 0.02005 $\simeq$ 0.02

LHS $\simeq$ RHS for $y = 0.0815$

$\therefore$ $y = \frac{\tau}{\bar{t}} = 0.0815$

$$\bar{t} = \frac{\tau}{0.0815}$$

$$= \frac{124.5}{0.0815} = 1527.6 \text{ s}$$

Feed rate = F = 4 t/h = 4000 kg/h = 1.111 kg/s

$$\bar{t} = \frac{W}{F}$$

$$W = \bar{t}\,F$$

$$= 1527.6 \times 1.11$$

$$= 1697.2 \text{ kg} = 1.697 \text{ t} \simeq 1.7 \text{ t}$$

W = Weight of solids in the reactor = 1.7 t ... Ans.

OR

For chemical reaction controlling, the mean conversion $\bar{X}_B$ is given by

$$\bar{X}_B = 3\frac{\bar{t}}{\tau} - 6\left(\frac{\bar{t}}{\tau}\right)^2 + 6\left(\frac{\bar{t}}{\tau}\right)^3 \left(1 - e^{-\tau/\bar{t}}\right)$$

Let $\frac{\bar{t}}{\tau} = x$

$$\bar{X}_B = 3x - 6x^2 + 6x^3\left(1 - e^{-1/x}\right)$$

$$0.75 = 3x - 6x^2 + 6x^2\left(1 - e^{-1/x}\right)$$

$$\text{LHS} = 0.75$$

$$\text{RHS} = 3x - 6x^2 + 6x^3\left(1 - e^{-1/x}\right)$$

Let $x = 1$ ... RHS = 0.7917

$x = 0.9$ ... RHS = 0.774

$x = 0.8$ ... RHS = 0.7518

$x = 0.79$ ... RHS = 0.7493

$x = 0.795$ ... RHS = 0.7506

$x = 0.794$ ... RHS = 0.7504

$x = 0.793$ ... RHS = 0.7501 $\simeq$ 0.75

For $x = 0.793$, LHS $\simeq$ RHS

$\therefore$ $x = 0.793$

$$x = \frac{\bar{t}}{\tau} = 0.793$$

$$\tau = \frac{\bar{t}}{0.793}$$

$$= \frac{100}{0.793} = 126.1$$

In further calculations we will use this value of τ, i.e., $\tau = 126.1$ s.

For commercial fluidised bed reactor :

$$F = 4 \text{ t/h} = 4000 \text{ kg/h} = 1.111 \text{ kg/s}$$

$$\bar{X}_B = \frac{98}{100} = 0.98$$

For chemical reaction controlling :

$$\tau = 126.1 \text{ s}$$

We will first calculate $\bar{t}$ using $\tau = 126.1$ s and $\bar{X}_B = 0.98$ and then W using $\bar{t}$ calculated and given F.

For chemical reaction controlling for unchanging particles in a fluidised bed, the mean conversion $\bar{X}_B$ is given by

$$\bar{X}_B = 3\frac{\bar{t}}{\tau} - 6\left(\frac{\bar{t}}{\tau}\right)^2 + 6\left(\frac{\bar{t}}{\tau}\right)^3\left(1 - e^{-\tau/\bar{t}}\right)$$

Let $\frac{\bar{t}}{\tau} = x$

$$\bar{X}_B = 3x - 6x^2 + 6x^3\left(1 - e^{-1/x}\right)$$

$$0.98 = 3x - 6x^2 + 6x^3\left(1 - e^{-1/x}\right)$$

LHS = 0.98

$$RHS = 3x - 6x^2 + 6x^3\left(1 - e^{-1/x}\right)$$

$x = 1$	RHS = 0.7927				
$x = 3$	RHS = 0.922	$x = 5$	RHS = 0.952		
$x = 7$	RHS = 0.9653	$x = 9$	RHS = 0.973		
$x = 11$	RHS = 0.9777	$x = 12$	RHS = 0.9795		
$x = 12.5$	RHS = 0.9803	$x = 12.3$	RHS = 0.98		

For $x = 12.3$ LHS = RHS

$\therefore$ $x = \frac{\bar{t}}{\tau} = 12.3$

$\bar{t} = 12.3\,\tau = 12.3 \times 126.1 = 1551$ s

$\bar{t} = \frac{W}{F}$

$\therefore$ $W = \bar{t}\,F = 1551 \times 1.111 = 1723$ kg $\simeq$ **1.723 t** ... Ans.

% error by two different approaches $= \frac{1723 - 1697}{1697} \times 100 = 1.53$ (acceptable)

Example 5.10 : *In a uniform gas environment, 4 mm solid particles are 87.5% converted to product in 5 min according to the shrinking core model. The chemical reaction step is rate controlling. The solids remain unchanged in size during reaction. Find the mean residence time of solids needed to achieve the same mean conversion of solids in a fluidised bed reactor operating with the same gas environment using 1000 kg/h of a feed consisting of equal quantities of 2 mm and 1 mm particles. Also find the solid hold-up in the bed.*

Solution : Given : For R = 4 mm, $X_B = 0.875$ and t = 5 min

For chemical reaction controlling and for solids of unchanging size, we have :

$$\frac{t}{\tau} = 1 - (1 - X_B)^{1/3}$$

$$\frac{5}{\tau} = 1 - (1 - 0.875)^{1/3}$$

$\therefore$ $\tau = 10$ min

For chemical reaction controlling, τ and R are related by

$$\tau \propto R$$

$$\tau_1 \propto R_1 \qquad \text{and} \qquad \tau_4 \propto R_4$$

$$\frac{\tau_1}{\tau_4} = \frac{R_1}{R_4}$$

Now we will calculate τ for particles of size R = 2 mm and 1 mm.

τ_4 = 10 min ..... R_4 = 4 mm

τ_2 = ? ..... R_2 = 2 mm

$$\tau \text{ for 2 mm size particle} = \tau_1 = \frac{\tau_4 R_2}{R_4}$$

$$= \frac{10 \times 2}{4} = 5 \text{ min}$$

τ_4 = 10 min ..... R_4 = 4 mm

τ_1 = ? ..... R_1 = 1 mm

$$\tau \text{ for 1 mm size particle} = \tau_2 = \frac{\tau_4 R_1}{R_4}$$

$$= \frac{10 \times 1}{4} = 2.5 \text{ min}$$

For fluidised bed reactor with a mixture of particles of unchanging size :

For chemical reaction controlling, the mean value for the fraction of B unconverted is given by

$$1 - \bar{X}_B = \sum_{R_m} \left\{ \frac{1}{4} \frac{\tau(R_i)}{\bar{t}} - \frac{1}{20} \left[\frac{\tau(R_i)}{\bar{t}} \right]^2 + \ldots\ldots \right\} \frac{F(R_i)}{F}$$

We have particles of size : R_1 and R_2

For R_1 = 1 mm : τ_1 = 2.5 min

For R_2 = 2 mm : τ_2 = 5 min

F = 1000 kg/h

$F(R_1) + F(R_2) = F$

Given : equal quantities of particles of size R_1 and R_2

$$\therefore \quad F(R_1) = \frac{F}{2} = \frac{1000}{2} = 500 \text{ kg/h}, \quad F(R_2) = 500 \text{ kg/h}$$

$$\bar{t}(R_1) = \bar{t}(R_2) = \bar{t}$$

$$1 - \bar{X}_B = \left[\frac{1}{4} \frac{\tau_1}{\bar{t}} - \frac{1}{20} \left(\frac{\tau_1}{\bar{t}} \right)^2 \right] \frac{F(R_1)}{F} + \left[\frac{1}{4} \frac{\tau_2}{\bar{t}} - \frac{1}{20} \left(\frac{\tau_2}{\bar{t}} \right)^2 \right] \frac{F(R_2)}{F}$$

$$1 - \bar{X}_B = \left[\frac{1}{4}\left(\frac{2.5}{\bar{t}}\right) - \frac{1}{20}\left(\frac{2.5}{\bar{t}}\right)^2\right]\frac{500}{1000} + \left[\frac{1}{4}\left(\frac{5}{\bar{t}}\right) - \frac{1}{20}\left(\frac{5}{\bar{t}}\right)^2\right]\frac{500}{1000}$$

$$1 - \bar{X}_B = \left[\frac{0.625}{\bar{t}} - \frac{0.3125}{\bar{t}^2}\right] \times 0.5 + \left[\frac{1.25}{\bar{t}} - \frac{1.25}{\bar{t}^2}\right] \times 0.5$$

Conversion is same in both the cases. Therefore, $\bar{X}_B = 0.875$

$$1 - 0.875 = \frac{0.3125}{\bar{t}} - \frac{0.15625}{\bar{t}^2} + \frac{0.625}{\bar{t}} - \frac{0.625}{\bar{t}^2}$$

$$0.125 = \frac{0.9375}{\bar{t}} - \frac{0.78125}{\bar{t}^2}$$

$$0.125\,\bar{t}^2 = 0.9375\,\bar{t} - 0.78125$$

$$0.125\,\bar{t}^2 - 0.9375\,\bar{t} + 0.78125 = 0$$

$$\therefore \quad \bar{t} = \frac{-(-0.9375) \pm \left[(0.9375)^2 - 4(0.125 \times 0.78125)\right]^{1/2}}{2 \times 0.125}$$

$$= 6.545 \text{ min (positive root)}$$

Mean residence time of solid particles in the bed = 6.545 min ... Ans.

$$\bar{t} = \frac{W}{F}$$

$$F = 1000 \text{ kg/h} = 16.67 \text{ kg/min}$$

$$W = \bar{t}\,F$$

$$= 6.545 \times 16.67$$

$$= 109.1 \text{ kg}$$

Weight of all solids in the reactor [solid hold-up in the bed] = 109.1 kg ... Ans.

Example 5.11 : *It is planned to use a fluidised-bed reactor for continuous conversion of solid B to solid R. In order to find the mean residence time in this flow reactor, the following data are obtained using a batch fluidised bed unit.*

At one minute interval, solids are taken out from the batch reactor and analysed for B and R. At exactly 50% conversion of B to R, the following results are found :

Size of particles in the batch reactor	*4 mm*	*12 mm*
Temperature of run	*550 °C*	*590 °C*
Time required for 50% conversion of B	*15 min*	*120 min*

Calculate the mean residence time needed to achieve 98% conversion of B in the flow reactor operated at 550°C with a feed of 2 mm solid particles. The particles are unchanged in size during reaction.

Solution : Here we will make use of experimental runs to find out the rate controlling mechanism.

Since the solids are unchanged in size during reaction, we can safely ignore the gas film resistance.

So the possibility : film diffusion controls is ruled out.

So we are left with :

(i) Chemical reaction controlling

(ii) Ash layer diffusion controlling

Time required to achieve the same fractional conversion for particles of different but unchanging size is given by :

$t \propto R^2$ ..... ash diffusion controlling

$t \propto R$ ..... chemical reaction controlling

(i) Chemical reaction controlling :

For 50% conversion of B at 550°C

$$t \propto R$$

$$t_4 = R_4 \quad \text{and} \quad t_{12} \propto R_{12}$$

$$\therefore \quad \frac{t_{12}}{t_4} = \frac{R_{12}}{R_4} \qquad t_4 = 15 \text{ min},\ R_4 = 4 \text{ min},\ R_{12} = 12 \text{ min}$$

$$t_{12} = \frac{t_4 R_{12}}{R_4} = \frac{15 \times 12}{4} = 45 \text{ min}$$

(ii) Ash diffusion controlling :

For 50% conversion of B at 550°C

$$t \propto R^2$$

$$t_{12} \propto R_{12}^2, \quad t_4 \propto R_4^2$$

$$t_{12} = t_4 \left(\frac{R_{12}}{R_4}\right)^2 = 15 \left(\frac{12}{4}\right)^2 = 135 \text{ min, i.e., 2 h and 15 min}$$

Time required for 50% conversion of 12 mm particles at 550 °C ..... chemical reaction controlling ..... is 45 min.

Time required for 50% conversion at 550°C of 12 mm particles ..... ash diffusion controlling ..... is 135 min.

For 12 mm particles :

Mechanism	Time for 50% conversion at 550 °C
Chemical reaction controlling	45 min
Ash diffusion controlling	135 min (2 h, 15 min)
Actual experiment shows (for 12 mm particle)	Time for 50% conversion at 590°C is 2h.

(i) With chemical reaction controlling, time for 50% conversion at 550 °C is 45 min. The actual experiment time for 50% conversion at 590 °C is 2h.

We expect time to reduce further at 590 °C but at 590 °C, t is 2h. This behaviour is opposite to our expectation. Hence, the chemical reaction does not control the rate, i.e., chemical reaction is not controlling mechanism.

(ii) With ash diffusion controlling, time required for 50% conversion at 550 °C is 136 min. (2h, 15 min), Time (t) is expected to reduce at 590 °C and 2h < 2 h, 15 min and hence ash diffusion is the controlling mechanism.

Ash diffusion is the controlling mechanism.

Now we will calculate the mean residence time in the fluidised bed reactor for feed of 2 mm solid particles at 550°C.

First we will find τ for 4 mm and then for 2 mm solid particles.

At 550 °C : for $R_4 = 4$ mm, t = 15 min for 50% conversion.

For ash diffusion controlling, we have :

$$\frac{t}{\tau} = 1 - 2(1 - X_B)^{2/3} + 2(1 - X_B)$$

$$t = 15 \text{ min for 4 mm particles, } X_B = 0.50$$

$$\frac{15}{\tau} = 1 - 3(1 - 0.50)^{2/3} + 2(1 - 0.5)$$

$$\tau = 136.22 \text{ min}$$

τ for 4 mm particles = 136.22 min.

For ash diffusion controlling : $\tau \propto R^2$

$$\tau_2 = R_2^2 \qquad \tau_4 \propto R_4^2$$

$$\tau_2 = \tau_4 \left(\frac{R_2}{R_4}\right)^2$$

$$= 136.22 \times \left(\frac{2}{4}\right)^2$$

$$= 34.055 \text{ min}$$

$$\simeq 34.06 \text{ min}$$

τ for 2 mm solid particles = 34.06 min

We will make use of this τ value to calculate $\bar{t}$.

Given : $\bar{X}_B = \frac{98}{100} = 0.98$

For ash diffusion controlling, for a fluidised bed reactor, mean of fraction of solid unconverted for single unchanging size particles is given by

$$1 - \bar{X}_B = \frac{1}{5}\frac{\tau}{\bar{t}} - \frac{19}{120}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{41}{4620}\left(\frac{\tau}{\bar{t}}\right)^3 - \ldots\ldots$$

Let $\frac{\tau}{\bar{t}} = y$

$$1 - 0.98 = 0.02 = 0.2\,y - 0.0452\,y^2 + 8.87 \times 10^{-3}\,y^3$$

$$LHS = 0.02$$

$$RHS = 0.2\,y - 0.0452\,y^2 + 8.87 \times 10^{-3}\,y^3$$

$y = 0.10$ RHS = 0.01956, $y = 0.104$ RHS = 0.0203

$y = 0.11$ RHS = 0.02146, $y = 0.102$ RHS = 0.01994 $\simeq$ LHS

For $y = 0.102$ LHS $\simeq$ RHS

$\therefore \quad y = \frac{\tau}{\bar{t}} = 0.102$

We have $\tau = 34.06$ min

$\therefore \quad \bar{t} = \frac{\tau}{0.102} = \frac{34.06}{0.102} = 333.92 \simeq 334$ min.

Mean residence time needed to achieve 98% conversion = $\bar{t}$ = 334 min in the fluidised bed reactor **... Ans.**

Example 5.12 : *In a uniform gas environment, 4 mm solid particles are 87.5% converted to product in 5 min according to the SCM with the ash diffusion step as rate controlling. The solids remain unchanged in size during reaction. Find the mean conversion that can be obtained in a fluidised bed reactor operating with the same environment as before, using a feed consisting of equal amounts of 2 mm and 1 mm particles. The mean residence time of solids in the fluidised bed reactor is 30 min.*

Solution : For R = 4 mm, $X_B = 0.875$ and t = 5 min

For solids of unchanging size and for ash diffusion controlling, we have

$$\frac{t}{\tau} = 1 - 3(1 - X_B)^{2/3} + 2(1 - X_B)$$

$$\frac{5}{\tau} = 1 - 3(1 - 0.875)^{2/3} + 2(1 - 0.875)$$

$\therefore \quad \tau = 10$ min

For ash diffusion controlling, τ and R are related by

$$\tau = R^2$$

$$\tau_1 = R_1^2 \qquad \tau_2 \propto R_2^2 \qquad T_4 \propto R_4^2$$

We have $\tau_4 = 10$ min for $R_4 = 4$ mm

Now we will calculate τ for 2 mm and 1 mm particles.

$R_1 = 1$ mm, $\tau_1 = ?$ $R_2 = 2$ mm $\tau_2 = ?$

$$\tau \text{ for 1 mm particles} = \tau_1 = \tau_4 \left(\frac{R_1}{R_4}\right)^2$$

$$= 10 \left(\frac{1}{4}\right)^2 = 0.625 \text{ min}$$

$$\tau \text{ for 2 mm particles} = \tau_2 = \tau_4 \left(\frac{R_2}{R_4}\right)^2$$

$$= 10 \left(\frac{2}{4}\right)^2 = 2.5 \text{ min}$$

We will make use of the τ values obtained to find out the mean conversion in the fluidised bed reactor.

Mean residence time = 30 min

$R_1 = 1$ mm ..... $\tau_1 = 0.625$ min

$R_2 = 2$ mm ..... $\tau_2 = 2.5$ min

$R_4 = 4$ mm ..... $\tau_4 = 10$ min

The mean value of the fraction of unconverted solids for mixture of particles of different but unchanging size for ash diffusion controlling is given by

$$1 - \bar{X}_B = \sum^{R_m} \left\{ \frac{1}{5} \frac{\tau(R_i)}{\bar{t}} - \frac{19}{420} \left(\frac{\tau(R_i)}{\bar{t}}\right)^2 + \frac{41}{4620} \left(\frac{\tau(R_i)}{\bar{t}}\right)^3 \right\} \frac{F(R_i)}{F} \qquad \ldots (1)$$

We have to find out the mean conversion in the reactor with the feed consisting of equa weights of 2 mm and 1 mm particles.

$$F = F(R_1) + F(R_2) \text{ but } F(R_1) = F(R_2)$$

$$\therefore \quad F(R_1) = F(R_2) = 0.5\,F$$

For 1 mm particles : $\dfrac{F(R_i)}{F} = \dfrac{F(R_1)}{F} = \dfrac{0.5\,F}{F} = 0.5$

and $\tau(R_1) = 0.625$ min

For 2 mm particles : $\dfrac{F(R_2)}{F} = \dfrac{0.5\,F}{F} = 0.50$

and $\tau(R_2) = 2.5$ min

Putting the values in Equation (1) [$\bar{t}$ = 30 min, τ_1 = 0.625 min, etc.] :

$$1 - \bar{X}_B = \left[\frac{1}{5}\left(\frac{0.625}{30}\right) - \frac{19}{420}\left(\frac{0.625}{30}\right)^2 + \frac{41}{4620}\left(\frac{0.625}{30}\right)^3\right] \times 0.5$$

$$+\left[\frac{1}{5}\left(\frac{2.5}{30}\right) - \frac{19}{420}\left(\frac{2.5}{30}\right)^2 + \frac{41}{4620}\left(\frac{2.5}{30}\right)^3\right] \times 0.5$$

$$1 - \bar{X}_B = 0.00207 + 0.00818$$

$$= 0.01025$$

$$\bar{X}_B = 1 - 0.01025 = 0.98975 = 0.9897$$

Mean conversion obtainable in the fluidised bed reactor = $\bar{X}_B$ = 0.9897 or 98.97% ... Ans.

Example 5.13 : *In a gas phase environment, C_{Ao}, particles of B are converted to solid product according to the shrinking core model with reaction step as rate controlling as per the following reaction :*

$$A(g) + B(s) \rightarrow S(s) + R(g)$$

The time required for complete conversion of particles is 1h.

It is required to design a fluidised bed reactor to treat 1t/h of solids to achieve 90% conversion using a stoichiometric feed rate of A, fed at C_{Ao}. Determine the weight of solids in the reactor assuming gas to be in mixed flow. Please note that the gas in the reactor is not at C_{Ao}.

Solution : A(g) + B(s) → S(s) + R(g) ..... with chemical reaction controlling

Given : SCM with chemical reaction controlling. For chemical reaction controlling, we have

$$\tau = \frac{\rho_B R}{bk'' C_{Ag}}$$

with b = 1 (stoichiometric coefficient of B)

$$\tau = \frac{\rho_B R}{k'' C_{Ag}}$$

In a C_{Ao} environment :

$$\tau = \frac{\rho_B R}{k'' C_{Ao}} = 1 \text{ h} \quad (\text{Given : } \tau = 1 \text{ h})$$

In any other environment :

$$\tau \propto \frac{1}{C_{Ag}}$$

Given : Stoichiometric feed rate of A, i.e., the feed rate of A required by the stoichiometry of the reaction corresponding to that of B.

From stoichiometry of the reaction, one mole A should enter for every mole of B (as the stoichiometric coefficient of both A and B is 1)

Suppose that 2 mol of A per 2 mol of B are entering in the reactor.

Let X_A be the conversion of A and $\bar{X}_B$ be the conversion of B. Then

$$\text{Amount of A reacted} = 2 X_A \quad \ldots (1)$$

$$\text{Amount of B reacted} = 2 \bar{X}_B$$

$$A(g) + B(s) \rightarrow \text{product, } a = b = 1 \text{ (stoichiometric coefficients)}$$

From the reaction :

$$\text{Amount of A reacted corresponding to B reacted} = \frac{1}{1}\left(2\bar{X}_B\right) = 2 \bar{X}_B \quad \ldots (2)$$

Equating (1) and (2),

$$2 X_A = 2 \bar{X}_B$$

$$\therefore \quad X_A = \bar{X}_B$$

$$\text{Conversion of solids} = 90\%$$

$$\therefore \quad \bar{X}_B = \frac{90}{100} = 0.90$$

and $$C_A = C_{Ao}(1 - X_A)$$

but $$X_A = X_B = 0.90$$

$$C_A = C_{Ao}(1 - 0.90) = 0.10\, C_{Ao}$$

In the mixed flow reactor, the concentration of A is C_A. So the particles are exposed to C_A in the reactor.

We have : $$\tau \propto \frac{1}{C_{Ag}} \quad \ldots\ldots \text{ chemical reaction controlling}$$

$$\tau(C_{Ao}) \propto \frac{1}{C_{Ao}}$$

$$\tau(C_A) \propto \frac{1}{C_A}$$

$$\therefore \quad \frac{\tau(C_A)}{\tau(C_{Ao})} = \frac{C_{Ao}}{C_A}$$

$$\tau(C_{Ao}) = 1 \text{ h}$$

$$C_A = 0.1\, C_{Ao}$$

$$\tau(C_A) = \tau(C_{Ao}) \cdot \frac{C_{Ao}}{C_A}$$

$$= \frac{1 \times C_{Ao}}{0.10\, C_{Ao}}$$

$$= 10 \text{ h}$$

$\therefore$ $\tau = 10$ h

The mean value for the fraction of solids unconverted in a fluidised bed reactor for single size solids for chemical reaction controlling is given by

$$1 - \bar{X}_B = \frac{1}{4}\left(\frac{\tau}{\bar{t}}\right) - \frac{1}{20}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau}{\bar{t}}\right)^3$$

$\bar{X}_B = 0.90, \quad \tau = 10$ h

Let $\frac{\tau}{\bar{t}} = y$

$1 - 0.90 = 0.10 = 0.25\, y - 0.05\, y^2 + 8.33 \times 10^{-3}\, y^3$

LHS = 0.10

RHS = $0.25\, y - 0.05\, y^2 + 8.33 \times 10^{-3}\, y^3$

$y = 0.40$	RHS = 0.092	$y = 0.44$	RHS = 0.101
$y = 0.42$	RHS = 0.0968	$y = 0.437$	RHS = 0.100 = LHS
$y = 0.43$	RHS = 0.0989		

For $y = 0.437$ LHS = RHS

$\therefore$ $y = \frac{\tau}{\bar{t}} = 0.437$

$\bar{t} = \frac{\tau}{0.437} = \frac{10}{0.437}$

$= 22.88$ h

For a fluidised bed reactor :

$$\bar{t} = \frac{W}{F}$$

Given : $F = 1$ t/h

$W = \bar{t}\, F$

$= 22.88 \times 1$

$= 22.88$ t $= 22880$ kg $(1$ t $= 1000$ kg$)$

Weight of solids in the reactor = 22.88 t or 22880 kg ... Ans.

Example 5.14 : *Repeat Example 5.13 assuming that gas passes through the reactor in plug flow.*

Solution : For plug flow gas τ is evaluated at C_{Ao}.

In a C_{Ao} environment :

$$\tau = 1 \text{ h}$$

Here we have :

$$1 - \bar{X}_B = \frac{1}{4}\left(\frac{\tau}{\bar{t}}\right) - \frac{1}{20}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau}{\bar{t}}\right)^3$$

Let $\frac{\tau}{\bar{t}} = y, \qquad \bar{X}_B = 0.90$

$$1 - 0.9 = 0.1 = 0.25\, y - 0.05\, y^2 + 8.33 \times 10^{-3}\, y^3$$

$$\text{LHS} = 0.1$$

$$\text{RHS} = 0.25\, y - 0.05\, y^2 + 8.33 \times 10^{-3}\, y^3$$

For $y = 0.437$ LHS = RHS (See Example 5.13)

$$y = \frac{\tau}{\bar{t}} = 0.437$$

$$\bar{t} = \frac{\tau}{0.437}$$

$$\tau = 1 \text{ h}$$

$$\bar{t} = \frac{1}{0.437} = 2.288 \text{ h}$$

In the fluidised bed reactor :

$$\bar{t} = \frac{W}{F}$$

$$W = \bar{t}\, F$$

$$F = 1 \text{ t/h}$$

$\therefore$ $$W = 2.288 \times 1 = 2.288 \text{ t}$$

Weight of solids in the reactor = 2.288 t or 2288 kg ... Ans.

Example 5.15 : *Repeat Example 5.13 for twice the stoichiometric ratio of A to B fed to the reactor all else remaining unchanged.*

Solution : $A(g) + B(s) \rightarrow S(s) + R(g)$

Stoichiometric ratio of A to B (as per reaction stoichiometry) is $\frac{1}{1} = 1$.

A and B are fed to the reactor such that ratio of feed rate of A to B on mole basis is 2 times the stoichiometric ratio of A to B.

With this, 2 mol of A enters the reactor per mol of B.

If X_A is the conversion of A and $\bar{X}_B$ is the conversion of B for 2 mol of A per 1 mol of B :

Amount of A reacted $= 2 \times X_A = 2\,X_A$ ... (1)

Amount of B reacted $= 1 \times \bar{X}_B = \bar{X}_B$

From reaction :

Amount of A reacted corresponding to the amount of B $= \frac{1}{1} \times \bar{X}_B = \bar{X}_B$ ... (2)

Equating (1) and (2)

Amount of A reacted $= 2\,X_A = \bar{X}_B$

$\therefore \quad 2\,X_A = \bar{X}_B$

$\therefore \quad X_A = \frac{\bar{X}_B}{2}$

i.e. conversion of A is 50% of that of B.

Given : $\bar{X}_B = 0.90$

Therefore, $X_A = \frac{0.90}{2} = 0.45$

$C_A = C_{Ao}(1 - X_A)$

$= C_{Ao}(1 - 0.45) = 0.55\,C_{Ao}$

In a mixed flow reactor, gas is at C_A.

For chemical reaction controlling :

$$\tau = \frac{\rho_B R}{bk''C_{Ag}}$$

with b = 1 $$\tau = \frac{\rho_B R}{k''C_{Ag}}$$

In a C_{Ao} environment $$\tau = \frac{\rho_B R}{k''C_{Ao}} = 1\ \text{h}$$

In any other environment

$$\tau \propto \frac{1}{C_{Ag}}$$

$\therefore \quad \tau(C_{Ao}) \propto \frac{1}{C_{Ao}}$

and $$\tau\,(C_A) \propto \frac{1}{C_A}$$

$$\frac{\tau(C_A)}{\tau(C_{Ao})} = \frac{C_{Ao}}{C_A}$$

$$\tau(C_A) = \frac{\tau(C_{Ao}) \cdot C_{Ao}}{C_A}$$

$$\tau(C_{Ao}) = 1 \text{ h}, \qquad C_A = 0.55\, C_{Ao}$$

$\therefore$ $$\tau(C_{Ao}) = \frac{1 \times C_{Ao}}{0.55\, C_{Ao}}$$

$$= 1.818 \text{ h}$$

We have for chemical reaction controlling (See Example 5.13)

$$1 - X_B = \frac{1}{4}\left(\frac{\tau}{\bar{t}}\right) - \frac{1}{20}\left(\frac{\tau}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau}{\bar{t}}\right)^2$$

$$y = \frac{\tau}{\bar{t}}, \qquad \bar{X}_B = 0.90$$

$$1 - 0.9 = 0.1 = 0.25\, y - 0.05\, y^2 + 8.33 \times 10^{-3}\, y^3$$

$$\text{LHS} = 0.10$$

$$\text{RHS} = 0.25\, y - 0.05\, y^2 + 8.33 \times 10^{-3}\, y^3$$

For $$y = 0.437, \quad \text{LHS} = \text{RHS}$$

$\therefore$ $$y = \frac{\tau}{\bar{t}} = 0.437$$

$$\bar{t} = \frac{\tau}{0.437}$$

$$\tau = 1.818 \text{ h}$$

$$\bar{t} = \frac{1.818}{0.437} = 4.16 \text{ h}$$

$$\bar{t} = \frac{W}{F}$$

$$W = \bar{t}\, F$$

Given : $$F = 1 \text{ t/h}$$

$\therefore$ $$W = 4.16 \times 1 = 4.16 \text{ t}$$

Weight of solids in the reactor = 4.16 t (4160 kg) ... Ans.

Example 5.16 : *A fluidised bed is charged with particles of single size at a rate of 2 kg/min. The bed contains 60 kg of solids. Find the mean residence time of solid for no carry over of particles.*

For the same solid feed rate, gas flow rate is increased and this increase in the gas flow rate results in expansion of the bed so that the bed contains only 50 kg of solids and 0.86 kg solid per minute are carried out of the bed by gas. Find the mean residence time of solids under these conditions.

Solution : (i) Without carry over of particles :

Feed rate of solids to the bed = F = 2 kg/min

Weight of solids in the bed = W = 60 kg

$$\text{Mean residence time of solids} = \bar{t} = \frac{W}{F}$$

$$= \frac{60}{2} = 30 \text{ min}$$

Mean residence time of solids for no carry over = 30 min **... Ans. (i)**

(ii) With carry over of particles by the gas :

$W = 50$ kg

$F_0 = 2$ kg/min. (same feed rate)

F_1 = underflow stream from reactor

F_2 = carryover stream from reactor

$F_0 = F_1 + F_2$ (by material balance around reactor)

$F_2 = 0.86$ kg/min.

$F_1 = F_0 - F_2$

$= 2 - 0.86 = 1.14$ kg/min.

In case of a fluidised bed with entrainment of solid $\bar{t}$ is given by

$$\bar{t}(R_i) = \frac{1}{\frac{F_1}{W} + \frac{F_2(R_i)}{W(R_i)}}$$

$$= \frac{1}{\frac{1.14}{50} + \frac{0.86}{50}}$$

$$= 25 \text{ min.}$$

Mean residence time of solids in the reactor (with carryover of solid particles) = 25 min **... Ans. (ii)**

Example 5.17 : *A fluidised fed reactor is fed with 1.5 kg/min feed consisting of 25% 50 μm particles, 40% 100 μm radius particles and 35% 200 μm radius particles. The bed contains 15 kg of solids. The fluidising gas is the gas phase reactant. τ (100 μm) = 10 min. Ash diffusion step controls rate. The solids are hard and remain unchanged in size during reaction.*

The solids are entrained by the fluidising gas. The change in gas phase composition is negligible. The elutriation velocity constant is given by $k = \frac{(500\ \mu m^2/min)}{R^2}$ *where R is particle radius in μm.*

Find the conversion of solids in the reactor.

Solution : Calculate : $\tau \rightarrow k \rightarrow F_1 \rightarrow \bar{t}\ (R_i) \rightarrow \bar{X}_B$ for given mechanism.

Given : Ash diffusion step is the rate controlling mechanism.

For ash diffusion controlling :

$$\tau \propto R^2$$

$$\tau_2 = \tau_1 \left(\frac{R_2}{R_1}\right)^2$$

$$\tau\ (100\ \mu m) = 10\ min$$

$$\therefore \quad \tau\ (50\ \mu m) = \tau\ (100\ \mu m) \left(\frac{50}{100}\right)^2$$

$$= 10 \left(\frac{50}{100}\right)^2 = 2.5\ min$$

$$\tau\ (200\ \mu m) = \tau\ (100\ \mu m) \left(\frac{200}{100}\right)^2$$

$$= 10 \left(\frac{200}{100}\right)^2 = 40\ min$$

$$R\ (50\ \mu m) \ldots\ldots \tau\ (50\ \mu m) = 2.5\ min$$

$$R\ (100\ \mu m) \ldots\ldots \tau\ (100\ \mu m) = 10\ min$$

$$R\ (200\ \mu m) \ldots\ldots \tau\ (20\ \mu m) = 40\ min$$

The elutriation velocity constant is given by

$$k = \frac{500\ \mu m^2/min}{R^2}$$

$$\text{Feed rate} = F_o = 1.5\ kg/min$$

$$F_o\ (R_i) = \frac{\text{Wt. \% of particles of size } R_i}{100} \times F_o$$

Solids in the bed = W = 15 kg

$$k\,(50\ \mu m) = \frac{500}{(50)^2} = 0.2\ min^{-1}, \qquad F_o\,(50\ \mu m) = \frac{25}{100} \times 1.5 = 0.375\ kg/min$$

$$k\,(100\ \mu m) = \frac{500}{(100)^2} = 0.05\ min^{-1}, \qquad F_o\,(100\ \mu m) = \frac{40}{100} \times 1.5 = 0.60\ kg/min$$

$$k\,(200\ \mu m) = \frac{500}{(200)^2} = 0.0125\ min^{-1}, \qquad F_o\,(200\ \mu m) = \frac{35}{100} \times 1.5 = 0.525\ kg/min$$

The underflow stream (F_1) is given by

$$F_1 = \sum^{R_m} \frac{F_o R_i}{1 + \left(\frac{W}{F_1}\right) k\,(R_i)}$$

In our case

$$F_1 = \frac{F_o\,(50\ \mu m)}{1 + \left(\frac{W}{F_1}\right) k\,(50\ \mu m)} + \frac{F_o\,(100\ \mu m)}{1 + \left(\frac{W}{F_1}\right) k\,(100\ \mu m)} + \frac{F_o\,(200\ \mu m)}{1 + \left(\frac{W}{F_1}\right) k\,(200\ \mu m)}$$

$$F_1 = \frac{0.375}{1 + \frac{15}{F_1} \times 0.2} + \frac{0.60}{1 + \frac{15}{F_1} \times 0.05} + \frac{0.525}{1 + \frac{15}{F_1} \times 0.0125}$$

$$F_1 = \frac{0.375}{1 + \frac{3}{F_1}} + \frac{0.60}{1 + \frac{0.75}{F_1}} + \frac{0.525}{1 + \frac{0.1875}{F_1}}$$

Find F_1 by a trial and error procedure.

F_1 = 1 kg/min ..... LHS = 1 ..... RHS = 0.878

F_1 = 0.8 kg/min ..... LHS = 0.8 ..... RHS = 0.814

F_1 = 0.82 kg/min ..... LHS = 0.82 ..... RHS = 0.82117

F_1 = 0.822 kg/min ..... LHS = 0.822 ..... RHS = 0.82188 $\simeq$ 0.822

$\therefore$ LHS $\simeq$ RHS for F_1 = 0.822 kg/min.

Now we will evaluate $\bar{t}\ (R_i)$ for particles of three sizes

$$\bar{t}\,(R_i) = \frac{1}{\frac{F_1}{W} + k\,(R_i)}$$

$$\bar{t}\,(50\ \mu m) = \frac{1}{\left(\frac{0.822}{15}\right) + 0.20} = 3.92\ min$$

$$\bar{t}\,(100\ \mu m) = \frac{1}{\left(\frac{0.822}{15}\right) + 0.05} = 9.54 \text{ min}$$

$$\bar{t}\,(200\ \mu m) = \frac{1}{\left(\frac{0.822}{15}\right) + 0.0125} = 14.85 \text{ min}$$

For ash diffusion controlling, the mean value of fraction of solids unconverted is given by

$$1 - \bar{X}_B = \sum_{R_m} \left\{ \frac{1}{5} \frac{\tau\,(R_i)}{\bar{t}\,(R_i)} - \frac{19}{420} \left(\frac{\tau\,(R_i)}{\bar{t}\,(R_i)} \right)^2 + \frac{41}{4620} \left(\frac{\tau\,(R_i)}{\bar{t}\,(R_i)} \right)^3 \right\} \frac{F_o\,(R_i)}{F_o}$$

$$1 - \bar{X}_B = \left[\frac{1}{5}\left(\frac{2.5}{3.92}\right) - \frac{19}{420}\left(\frac{2.5}{3.92}\right)^2 + \frac{41}{4620}\left(\frac{2.5}{3.92}\right)^3 \right] \frac{0.375}{1.5}$$

$$+ \left[\frac{1}{5}\left(\frac{10}{9.54}\right) - \frac{19}{420}\left(\frac{10}{9.54}\right)^2 + \frac{41}{4620}\left(\frac{10}{9.54}\right)^3 \right] \frac{0.60}{1.5}$$

$$+ \left[\frac{1}{5}\left(\frac{40}{14.85}\right) - \frac{19}{420}\left(\frac{40}{14.85}\right)^2 + \frac{41}{4620}\left(\frac{40}{14.85}\right)^3 \right] \frac{0.525}{1.5}$$

$$1 - \bar{X}_B = 0.0278 + 0.068 + 0.1343 = 0.2301$$

$$\bar{X}_B = 1 - 0.2301 = 0.7699 \simeq 0.77$$

Mean conversion of solids in the reactor is $0.77 \times 100 = 77\%$ ... Ans.

Extra : We have $F_1 = 0.822$ kg/min and $F_o = 1.5$ kg/min

$$F_o = F_1 + F_2$$

$$F_2 = F_o - F_1 = 1.5 - 0.822 = 0.678 \text{ kg/min.}$$

Carry over of the solids by the fluidising gas = $F_2 = 0.678$ kg/min.

$$F_1\,(R_i) = \frac{F_o\,(R_i)}{1 + \left(\frac{W}{F_1}\right) k\,(R_i)}$$

$$F_1\,(50\ \mu m) = \frac{0.375}{1 + \left(\frac{15}{0.822}\right) \times 0.20} = 0.0806 \text{ kg/min.}$$

$$F_1\,(100\ \mu m) = \frac{0.60}{1 + \left(\frac{15}{0.822}\right) \times 0.05} = 0.3137 \text{ kg/min.}$$

$$F_1\,(200\ \mu m) = \frac{0.525}{1 + \left(\frac{15}{0.822} \times 0.0125\right)} = 0.4275 \text{ kg/min.}$$

F_2 (50 μm) = F_o (50 μm) – F_1 (50 μ)

= 0.375 – 0.0806 = 0.2944 kg/min.

F_2 (100 μm) = 0.60 – 0.3137 = 0.2863 kg/min.

F_2 (200 μm) = 0.525 – 0.4275 = 0.0975 kg/min.

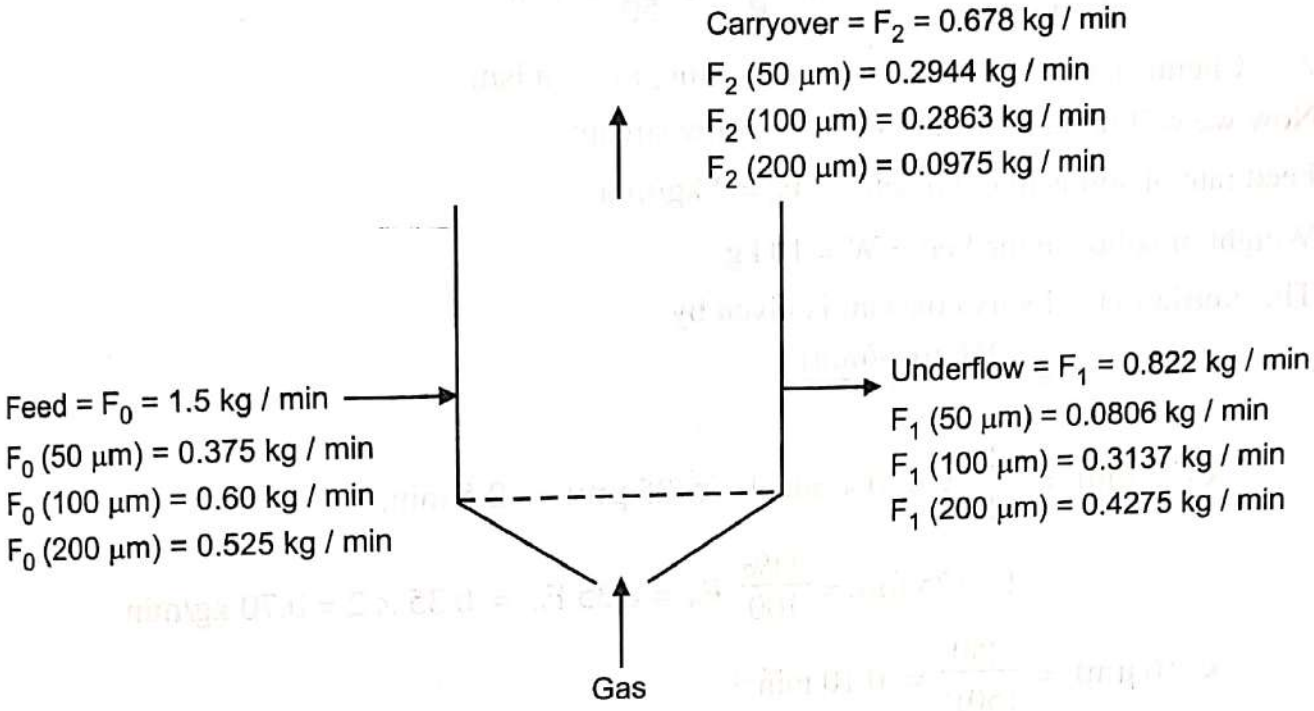


Fig. Ex 5.17

Example 5.18 : *A solid feed consisting of 35% of 25 μm radius particles, 35% of 50 μm radius particles and 30% of 100 μm radius particles is to be reacted in a steady state fluidised bed reactor. The fluidising gas is the gas phase reactant. Time required for complete conversion of these three particle sizes of feed is 2.5, 5 and 10 min respectively. The reactor is fed at a rate of 2 kg solids/min and the bed contains 16 kg solids.*

These solids are hard and remain unchanged in size during reaction. There is negligible change in the gas-phase composition. The solids are entrained by the fluidising gas.

The elutriation velocity constant is given by

$$\kappa = (250\ \mu m^2/min)/R^2$$

Find the conversion of solids in the reactor.

Solution : Procedure : Find ... mechanism ... κ values ... F_1 ... $\bar{t}$ (R_i) values ... $\bar{X}_B$.

First we will find rate controlling mechanism whether ash diffusion or chemical reaction.

R = 25 μm ..... τ = 2.5 min

R = 50 μm ..... τ = 5 min

R = 100 μm ..... τ = 10 min

From the above values of R and τ, it is clear that

$$\tau \propto R$$

$$\therefore \quad \frac{\tau_1}{\tau_2} = \frac{R_1}{R_2}$$

$$\frac{\tau\,(25\ \mu m)}{\tau\,(50\ \mu m)} = \frac{2.5}{5} = 0.5 = \frac{R_{25}}{R_{50}} = \frac{25}{50} = 0.50$$

$\therefore$ Chemical reaction is the rate controlling mechanism.

Now we will proceed to find F_1 (underflow stream)

Feed rate of solids to the reactor = $F_o = 2$ kg/min

Weight of solids in the bed = W = 10 kg

The elutriation velocity constant is given by

$$\kappa = \frac{(250\ \mu m^2/min)}{R^2}$$

$$\kappa\,(25\ \mu m) = \frac{250}{(25)^2} = 0.4\ min^{-1},\quad \tau\,(25\ \mu m) = 2.5\ min,$$

$$F_o\ (25\ \mu m) = \frac{wt\%}{100}\ F_o = 0.35\ F_o = 0.35 \times 2 = 0.70\ kg/min$$

$$\kappa\,(50\ \mu m) = \frac{250}{(50)^2} = 0.10\ min^{-1},$$

$$\tau\,(50\ \mu m) = 5\ min,\ F_o\ (50\ \mu m) = 0.35 \times 2 = 0.7\ kg/min$$

$$\kappa\,(100\ \mu m) = \frac{250}{(100)^2} = 0.025\ min^{-1},$$

$$\tau\,(100\ \mu m) = 10\ min,\ F_o\ (100\ \mu m) = 0.30 \times 2 = 0.6\ kg/min$$

The underflow stream (F_1) is given by

$$F_1 = \sum^{R_m} \frac{F_o\,(R_i)}{1 + \left(\frac{W}{F_1}\right) k\,(R_i)}$$

In our case, it becomes

$$F_1 = \frac{F_o\,(25\ \mu m)}{1 + \left(\frac{W}{F_1}\right) k\,(25\ \mu m)} + \frac{F_o\,(50\ \mu m)}{1 + \left(\frac{W}{F_1}\right) k\,(50\ \mu m)} + \frac{F_o\,(100\ \mu m)}{1 + \left(\frac{W}{F_1}\right) k\,(100\ \mu m)}$$

$$F_1 = \frac{0.70}{1 + \frac{16}{F_1}(0.4)} + \frac{0.70}{1 + \frac{16}{F_1}(0.1)} + \frac{0.60}{1 + \frac{16}{F_1}(0.025)}$$

$$F_1 = \frac{0.70}{1 + \frac{6.4}{F_1}} + \frac{0.70}{1 + \frac{1.6}{F_1}} + \frac{0.60}{1 + \frac{0.4}{F_1}}$$

Find F_1 by a trial and error procedure.

F_1 = 1 kg/min ..... LHS = 1 ..... RHS = 0.79

F_1 = 1.5 kg/min ..... LHS = 1.5 ..... RHS = 0.88

F_1 = 0.8 kg/min ..... LHS = 0.8 ..... RHS = 0.71

F_1 = 0.6 kg/min ..... LHS = 0.6 ..... RHS = 0.6145

F_1 = 0.62 kg/min ..... LHS = 0.62 ..... RHS = 0.622

F_1 = 0.625 kg/min ..... LHS = 0.625 ..... RHS = 0.6247

$\therefore$ LHS = RHS for F_1 = 0.625 kg/min

$$F_1(R_i) = \frac{F_o(R_i)}{1 + \left(\frac{W}{F_1}\right) k(R_i)}$$

$$F_1(25\ \mu m) = \frac{F_o(25\ \mu m)}{1 + \left(\frac{W}{F_1}\right) k(25\ \mu m)} = \frac{0.70}{1 + \frac{16}{0.625} \times 0.4} = 0.6023 \text{ kg/min}$$

Similarly,

$$F_1(50\ \mu m) = \frac{0.70}{1 + \frac{16}{0.625} \times 0.10} = 0.1966 \text{ kg/min}$$

$$F_1(100\ \mu m) = \frac{0.60}{1 + \frac{16}{0.625} \times 0.025} = 0.3658 \text{ kg/min}$$

The mean residence times of three sizes of particles are :

$$\bar{t}(R_i) = \frac{1}{\frac{F_1}{W} + k(R_i)}$$

F_1 = 0.625 kg/min, W = 16 kg

$$\bar{t}(25\ \mu m) = \frac{1}{\left(\frac{0.625}{16}\right) + 0.40} = 2.277 \simeq 2.78 \text{ min}$$

$$\bar{t}(50\ \mu m) = \frac{1}{\left(\frac{0.625}{16}\right) + 0.10} = 7.19 \text{ min}$$

$$\bar{t}(100\ \mu m) = \frac{1}{\left(\frac{0.625}{16}\right) + 0.025} = 15.60 \text{ min}$$

Mean value for the fraction of B unconverted for the case of chemical reaction controlling is given by

$$1 - \bar{X}_B = \sum^{R_m} \left\{ \frac{1}{4} \frac{\tau(R_i)}{\bar{t}(R_i)} - \frac{1}{20} \left(\frac{\tau(R_i)}{\bar{t}(R_i)} \right)^2 + \frac{1}{120} \left(\frac{\tau(R_i)}{\bar{t}(R_i)} \right)^3 \right\} \frac{F_o(R_i)}{F_o}$$

$$1 - \bar{X}_B = \left\{ \frac{1}{4} \frac{\tau(25\ \mu m)}{\bar{t}(25\ \mu m)} - \frac{1}{20} \left(\frac{\tau(25\ \mu m)}{\bar{t}(25\ \mu m)} \right)^2 + \frac{1}{120} \left(\frac{\tau(25\ \mu m)}{\bar{t}(25\ \mu m)} \right)^3 \right\} \frac{F_o(25\ \mu m)}{F_o}$$

+ same terms for 50 μm + same terms for 100 μm

$$1 - \bar{X}_B = \left[\frac{1}{4} \left(\frac{2.5}{2.78} \right) - \frac{1}{20} \left(\frac{2.5}{2.78} \right)^2 + \frac{1}{120} \left(\frac{2.5}{2.78} \right)^3 \right] \frac{0.70}{2.0}$$

$$+ \left[\frac{1}{4} \left(\frac{5}{7.19} \right) - \frac{1}{20} \left(\frac{5}{7.19} \right)^2 + \frac{1}{120} \left(\frac{5}{7.19} \right)^3 \right] \frac{0.70}{2.0}$$

$$+ \left[\frac{1}{4} \left(\frac{10}{15.60} \right) - \frac{1}{20} \left(\frac{10}{15.60} \right)^2 + \frac{1}{120} \left(\frac{10}{15.60} \right)^3 \right] \frac{0.60}{2.0}$$

$$1 - \bar{X}_B = 0.0667 + 0.0534 + 0.0426 = 0.1627$$

$$\bar{X}_B = 1 - 0.1627 = 0.8373$$

Mean conversion of solids in the reactor is 0.8373 or 83.73% ... Ans

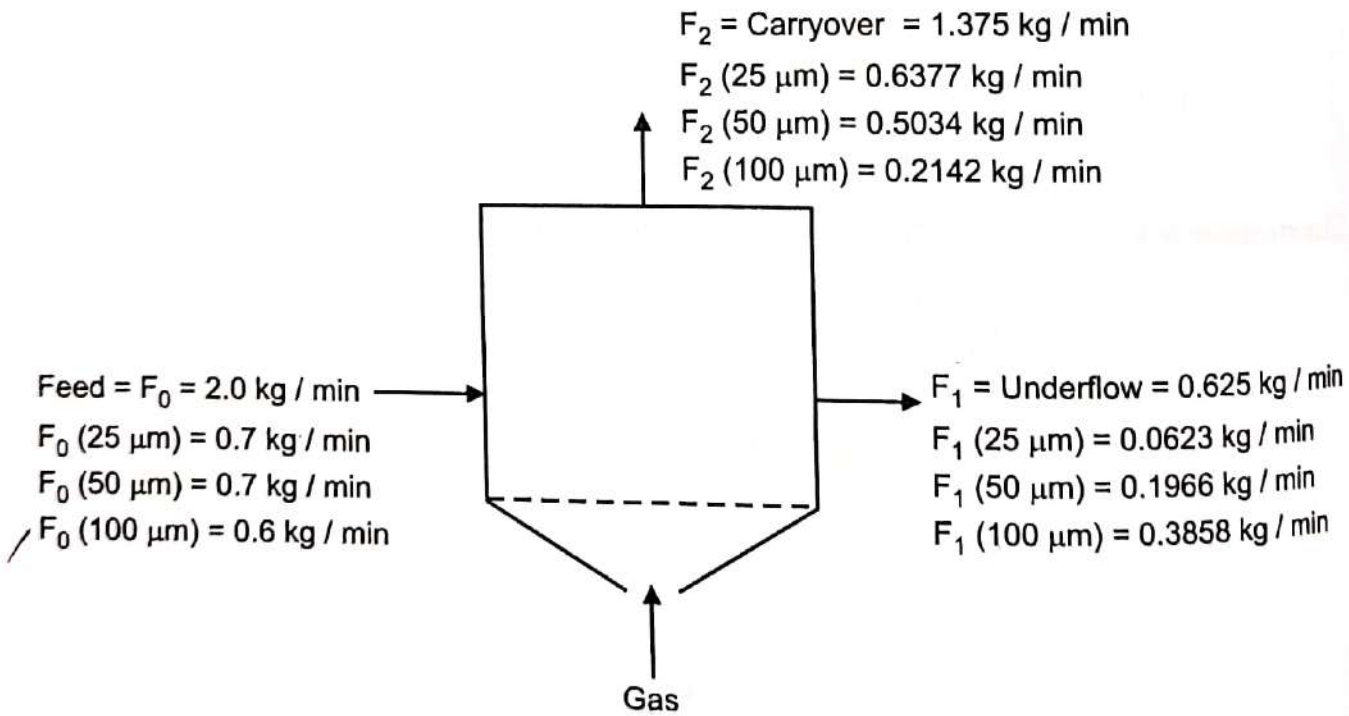


Fig. Ex 5.18

$$F_o = F_1 + F_2$$

$$2 = 0.625 + F_2$$

$$\therefore \quad F_2 = 1.375 \text{ kg/min}$$

Example 5.19 : *An experimental set-up consisting of a fluidised bed followed by a cyclone (which separates and returns all entrained solids to the bed) is used to conduct experiments on the elutriation rate of solids.*

In a typical experiment, a batch of solids consisting of 400 g of 25 μm radius particles, 600 g of 50 μm radius particles and 1000 g of 150 μm radius particles is fluidised by a gas at constant velocity. At steady state, the carryover rate is 22 g/min composed of 27.3% of 50 μm radius particle and 72.7% of 25 μm radius particles. Neglect the amount of solid in the recycle system as it being small (10 g) at any instant.

Find the elutriation velocity as a function of radius of particles in centimeters.

Assume the relationship of the type :

$$\kappa(R) = \alpha R^{\beta}$$

Solution : Particle sizes : 25 μm, 50 μm and 150 μm.

W (25 μm) = 400 g, W (50 μm) = 600 g, W (150 μm) = 1000 g

Carryover rate = F_2 = 22 g/min

It contains 27.3% of 50 μm particles and 72.7% of 25 μm particles.

$$F_2\,(50\ \mu m) = \frac{\text{Wt. \% of 25 μm particle}}{100} \times F_2$$

$$= \frac{27.3}{100} \times 22 = 6.006 \text{ m/min.}$$

$$F_2\,(25\ \mu m) = \frac{72.7}{100} \times 22 = 15.994 \text{ g/min}$$

For particles of size R_i in the bed at steady state,

$$\kappa\,(R_i) = \frac{F_2\,(R_i)}{W(R_i)}$$

$$\kappa\,(50\ \mu m) = \frac{F_2\,(50\ \mu m)}{W\,(50\ \mu m)} = \frac{6.006}{600} = 0.01001 \approx 0.01 \text{ min}^{-1}$$

$$\kappa\,(25\ \mu m) = \frac{F_2\,(25\ \mu m)}{W\,(25\ \mu m)} = \frac{15.994}{400} = 0.039985 \approx 0.04 \text{ min}^{-1}$$

The relationship between $\kappa\,(R_i)$ and R_i is

$$\kappa\,(R_i) = \alpha\,(R_i)^{\beta}$$

$$\kappa\,(25\ \mu m) = \alpha\,(25\ \mu m)^{\beta}$$

$$\kappa\,(50\ \mu m) = \alpha\,(50\ \mu m)^{\beta}$$

$$\frac{\kappa\,(25\ \mu m)}{\kappa\,(50\ \mu m)} = \frac{\alpha\,(25\ \mu m)^{\beta}}{\alpha\,(50\ \mu m)^{\beta}}$$

$$\frac{0.04}{0.01} = \left(\frac{25}{50}\right)^{\beta}$$

$$4 = (0.5)^{\beta}$$

$$\ln(4) = \beta \ln(0.5) \qquad 1.3863 = \beta(-0.6931)$$

$$\beta = -2.0$$

or $\quad LHS = 4 \qquad RHS = (0.5)^{\beta}$

for $\quad \beta = -2 \qquad LHS = RHS$

$\therefore \quad \beta = -2$

Now we will evaluate α :

$$\kappa(50\ \mu m) = \alpha(50\ \mu m)^{\beta}$$

$$0.01 = \alpha(50)^{-2}$$

$\therefore \quad \alpha = 25\ \mu m^2/min$

$$k = \alpha R^{-2}$$

$\therefore \quad \boxed{\kappa = \dfrac{(25\ \mu m^2/min)}{R^2}}$, where R – particle radius in μm

$$1\ \mu m = 10^{-6}\ m = 10^{-6} \times 100\ cm = 10^{-4}\ cm$$

$\therefore \quad \kappa = \dfrac{25\ (10^{-4}\ cm)^2/min}{R^2}$

$$\boxed{\kappa = \frac{(25 \times 10^{-8}\ cm^2/\ min)}{R^2}\ \text{, where R is the particle radius in cm}}$$ ... Ans.

Example 5.20 : *A feed consisting of 40% of 10 μm radius particles and 60% of 40 μm radius particles is fed at a rate of 35.5 g/min to an experimental fluidised bed reactor.*

Data :

$\kappa(10\ \mu m) = 0.40\ min^{-1}$, $\kappa(40\ \mu m) = 0.01\ min^{-1}$

$W = 1100\ g$

$\tau(10\ \mu m) = 2.5\ min$, $\quad \tau(40\ \mu m) = 10\ min$

(a) Find the conversion of solids in the experimental reactor using the above data.

(b) It is proposed to use a cyclone to separate and return to the reactor all entrained solids as a possible means to increase the conversion. Determine the effect of such a device on the conversion of solids.

Solution : $\tau(10\ \mu m) = 2.5\ min$

$\tau(40\ \mu m) = 10\ min$

Let us find the rate controlling mechanism.

For chemical reaction as the rate controlling mechanism :

$$\tau \propto R$$

$$\therefore \quad \frac{\tau_1}{\tau_2} = \frac{R_1}{R_2}$$

$$\therefore \quad \frac{\tau\,(10\ \mu m)}{\tau\,(40\ \mu m)} = \frac{10\ \mu m}{40\ \mu m}$$

$$\frac{2.5}{10} = 0.25 = \frac{10}{40} = 0.25$$

$$\therefore \quad \text{LHS} = \text{RHS}$$

$\therefore$ Chemical reaction is the rate controlling mechanism.

$F_o = 35.5$ g/min, $W = 1100$ g

$k\,(10\ \mu m) = 0.40\ min^{-1}$, $k\,(40\ \mu m) = 0.01\ min^{-1}$

(a) Find the conversion with entrainment of solids (no cyclone is used) :

$$F_o\,(10\ \mu m) = \frac{wt\%}{100} F_o = 0.40 \times 35.5 = 14.2\ \text{g/min}$$

$$F_o\,(40\ \mu m) = 0.60 \times 35.5 = 21.3\ \text{g/min}$$

Evaluate F_1(underflow stream) :

F_1 is given by

$$F_1 = \sum_{}^{R_m} \frac{F_o\,(R_i)}{1 + \left(\frac{W}{F_1}\right) k\,(R_i)}$$

$$F_1 = \frac{F_o\,(10\ \mu m)}{1 + \left(\frac{W}{F_1}\right) k\,(10\ \mu m)} + \frac{F_o\,(40\ \mu m)}{1 + \left(\frac{W}{F_1}\right) k\,(40\ \mu m)}$$

$$F_1 = \frac{14.2}{1 + \frac{1100 \times 0.40}{F_1}} + \frac{21.3}{1 + \frac{1100 \times 0.01}{F_1}}$$

$$F_1 = \frac{14.2}{1 + \frac{440}{F_1}} + \frac{21.3}{1 + \frac{11}{F_1}}$$

Find F_1 by a trial and error procedure.

$F_1 = 10$ g/min ... LHS = 10 ... RHS = 10.46

$F_1 = 11$ g/min ... LHS = 11 ... RHS = 10.996 $\simeq$ 11

$\therefore$ LHS $\simeq$ RHS for $F_1 = 11$ g/min

Evaluate $\bar{t}\,(R_1)$:

$F_1 = 11$ g/min

$$\bar{t}\,(R_i) = \frac{1}{\left(\frac{F_1}{W}\right) + k\,(R_i)}$$

$$\bar{t}\,(10\,\mu m) = \frac{1}{\frac{11}{1100} + 0.40} = 2.439 \simeq 2.44 \text{ min}$$

$$\bar{t}\,(40\,\mu m) = \frac{1}{\frac{11}{1100} + 0.01} = 50 \text{ min}$$

Evaluate mean overall conversion of solids, $\bar{X}_B$:

For chemical reaction controlling we have :

$$1 - \bar{X}_B = \sum \left\{ \frac{1}{4}\left(\frac{\tau(R_i)}{\bar{t}\,(R_i)}\right) - \frac{1}{20}\left(\frac{\tau(R_i)}{\bar{t}\,(R_i)}\right)^2 + \frac{1}{120}\left(\frac{\tau(R_i)}{\bar{t}\,(R_i)}\right)^3 \right\} \frac{F_o(R_i)}{F_o}$$

$\tau\,(10\,\mu m) = 2.5$ min, $\tau\,(40\,\mu m) = 10$ min

$F_o\,(10\,\mu m) = 14.2$ g/min, $F_o\,(40\,\mu m) = 21.3$ g/min

$F_o = 35.5$ g/min

$$1 - \bar{X}_B = \left[\frac{1}{4}\left(\frac{2.5}{2.44}\right) - \frac{1}{20}\left(\frac{2.5}{2.44}\right)^2 + \frac{1}{120}\left(\frac{2.5}{2.44}\right)^3\right]\frac{14.2}{35.5} + \left[\frac{1}{4}\left(\frac{10}{50}\right) - \frac{1}{20}\left(\frac{10}{50}\right)^2 + \frac{1}{120}\left(\frac{10}{50}\right)^3\right]\frac{21.3}{35.5}$$

$$1 - \bar{X}_B = 0.085 + 0.0288 = 0.1138$$

$$\bar{X}_B = 1 - 0.1138 = 0.8862$$

$\therefore$ With elutriation : Mean conversion of solids is 0.8862 or 88.62% ... Ans. (a

(b) With a cyclone to separate and return the entrained solids :

$$F_2 = 0$$

$$\bar{t} = \bar{t}\,(R_i)$$

$$\bar{t} = \bar{t}\,(10\,\mu m) = \bar{t}\,(40\,\mu m) = \frac{W}{F}$$

$W = 1100$ g and $F = 35.5$ g/min

$$\bar{t} = \frac{1100}{35.5} = 30.986 \text{ min} \approx 31 \text{ min}$$

$$F = 35.5 \text{ g/min}$$

$$F(R_i) = \frac{\text{Wt. \% of particles of size } R_i}{100} \times F$$

$$F(10\,\mu m) = \frac{40}{100} \times 35.5 = 14.2 \text{ g/min}$$

$$F(40\,\mu m) = \frac{60}{100} \times 35.5 = 21.3 \text{ g/min}$$

For chemical reaction controlling :

$$1 - \bar{X}_B = \sum_{R_m} \left\{ \frac{1}{4}\frac{\tau(R_i)}{\bar{t}} - \frac{1}{20}\left(\frac{\tau(R_i)}{\bar{t}}\right)^2 + \frac{1}{120}\left(\frac{\tau(R_i)}{\bar{t}}\right)^3 \right\} \frac{F(R_i)}{F}$$

$$= \left[\frac{1}{4}\left(\frac{2.5}{31}\right) - \frac{1}{20}\left(\frac{2.5}{31}\right)^2 + \frac{1}{120}\left(\frac{2.5}{31}\right)^3\right]\frac{14.2}{35.5}$$

$$+\left[\frac{1}{4}\left(\frac{10}{31}\right) - \frac{1}{20}\left(\frac{10}{31}\right)^2 + \frac{1}{120}\left(\frac{10}{31}\right)^3\right]\frac{21.3}{35.5}$$

$$1 - \bar{X}_B = 0.00794 + 0.0454 = 0.05334$$

$$\therefore \quad \bar{X}_B = 1 - 0.05334 = 0.94666 \simeq 0.9467$$

Without elutriation : The mean conversion of solids is 0.9467 or 94.67%

... Ans. (b)

Thus, use of the cyclone increases the conversion in the reactor.

Conversion with a cyclone (no entrained solids) = $\left(\bar{X}_B\right)_1 = 94.67\%$

Conversion without a cyclone (with entrained solids) = $\left(\bar{X}_B\right)_2 = 88.62\%$

Increase in the conversion by using cyclone $= \dfrac{94.67 - 88.62}{88.62} \times 100$

$= 6.827\%$

Increase in the conversion by using cyclone = 6.827%

... Ans. (b)

Example 5.21 : *A batch of solids containing 20 kg of A, 20 kg B and 50 kg C is fluidised at high air velocity. The entrained solids are removed, collected and analysed. After 10 minutes of operation, an analysis of the entrained solids shows 16 kg A, 10 kg B and 2 kg C.*

(i) Determine the elutriation constants for these solids.

(ii) Find the entrainment rate of solids if the entrained solids are collected in a cyclone and immediately returned to the bed.

Solution : For elutriation experiment, we have

ln (fraction of marked particles remaining in the bed) = – kt

where κ is the elutriation velocity constant.

For A :

Initial solids = 20 kg, Solids entrained = 16 kg

Solids in the bed after time t = 20 – 16 = 4 kg

t = 10 min

Fraction of marked particles remaining in the bed = 4/20

$$\therefore \quad \ln(4/20) = -\kappa_A \times 10$$

$$\therefore \quad \kappa_A = \frac{-\ln(4/20)}{10} = \frac{-(-1.609)}{10} = 0.1609 \approx 0.161 \text{ min}^{-1}$$

Elutriation constant for A = 0.161 min^{-1}

... Ans. (i)

For B :

Initial solids = 20 kg, Solids entrained = 10 kg

Solids in bed after time t = 20 – 10 = 10 kg

t = 10 min

Fraction of marked particles remaining in the bed = 10/20

$$\therefore \quad \ln (10/20) = -\kappa_B \times 10$$

$$\therefore \quad \kappa_B = \frac{-(-0.693)}{10} = 0.0693 \text{ min}^{-1}$$

Elutriation constant for B = 0.0693 min^{-1} ... Ans. (ii)

For C :

Initial solids = 50 kg, Solids entrained = 2 kg

Solids in the bed after time t = 50 – 2 = 48 kg

t = 10 min

Fraction of marked particles remaining in the bed = 48/50

$$\therefore \quad \ln (48/50) = -\kappa_C \times 10$$

$$\therefore \quad \kappa_C = \frac{-(-0.041)}{10} = 0.0041 \text{ min}^{-1}$$

Elutriation constant for C = 0.0041 min^{-1} ... Ans. (i)

Part (ii) :

For particles of size R_i in the bed and at steady state,

$$\kappa (R_i) = \frac{\text{Rate of carryover of particles of size } R_i}{\text{Weight of such particles present in the bed}}$$

$$= \frac{F_2 (R_i)}{W (R_i)}$$

For A :

$$\kappa_A = \frac{F_{2A}}{W_A} \quad , \quad W_A = 20 \text{ kg ... initial solids}$$

$$F_{2A} = \kappa_A W_A$$

$$= 0.161 \times 20 = 3.22 \text{ kg/min}$$

For B :

$$\kappa_B = \frac{F_{2B}}{W_B}$$

$$F_{2B} = \kappa_B W_B = 0.0693 \times 20 = 1.386 \text{ kg/min}$$

For C :

$$F_{2C} = \kappa_C W_C = 0.0041 \times 50 = 0.205 \text{ kg/min}$$

$\therefore$ Total entrainment rate of solids = 3.22 + 1.386 + 0.205 = 4.811 kg/min

Total elutriation of solids = 4.811 kg/min ... Ans. (ii)

CHAPTER 6

SOLID CATALYSED REACTIONS

Solid catalysed fluid-phase reactions are the most important and commonly encountered types in industrial processes : production of sulphuric acid, nitric acid, ethylene oxide, methanol, etc. In the production of petrochemicals, solid catalysts play a key role in the reactions such as cracking, isomerisation, alkylation, etc. Heterogeneous catalytic reactions occur at the fluid-solid interface (which is the surface of a solid catalyst immersed in a gaseous or liquid reaction mixture). A major advantage of these reactions is ease in complete separation of the fluid product mixture from the catalyst (for reuse).

A substance that enhances the rate of a chemical reaction and is recovered at the end of the reaction without change in its mass and chemical composition is called a **catalyst** (Note : normally we talk about a catalyst that speeds up a reaction). A catalyst is a substance that speeds-up a chemical reaction.

A catalytic reaction occurs at the fluid-solid interface. Therefore, a large interfacial area is essential to attain a significant reaction rate. To provide this area, solid catalysts are normally porous (catalyst contains many fine pores and the surface of these pores supplies the area required for the high rate of reaction).

A catalyst which has a large area resulting from pores is called as a porous catalyst. For example, nickel catalyst used in the hydrogenation of vegetable oils. Most catalyst solids have surface areas in 5 to 1000 m^2/g range.

Two important characteristics of solid catalysts are : (i) activity and (ii) selectivity.

Activity : It is a measure of the accelerating effect that a catalyst has on the rate of reaction.

Specificity (or selectivity) : It refers to the ability of a catalyst to speed up the main reaction in the presence of several side reactions.

A catalyst does enter into a chemical reaction but does not appear in the overall reaction. It usually speeds up a reaction by providing an alternate path (by promoting a different reaction mechanism) for formation of products that requires a lower activation energy than that for the uncatalysed reaction. In presence of a catalyst, a reaction proceeds along a pathway which is energetically more favourable, i.e., the reaction proceeds by such a mechanism that the activation energy of each catalytic step of the mechanism is less than that for the uncatalysed reaction (so the reaction is carried out at a higher rate).

Characteristics of catalysed reaction :

1. A catalyst remains unchanged in its amount and chemical composition at the end of the reaction.
2. A small amount of catalyst is often sufficient to catalyse (speed up) a reaction.
3. A catalyst does not change the state/position of chemical equilibrium of a reaction (since it does not depend on the pathway of the reaction).
4. A catalyst is usually specific in nature.
5. A catalysed reaction generally has a lower activation energy than the non-catalysed reaction. Consequently, the catalysed reaction proceeds at a faster rate.

A catalyst is able to speed up only those reactions that are thermodynamically feasible. It lowers the activation energy of the reaction (i.e., it reduces the potential energy barrier over which the reactants must pass to form products) by providing a different path to products and thus accelerates the reaction. The lowering of energy barrier/activation energy is shown in Fig. 6.1.

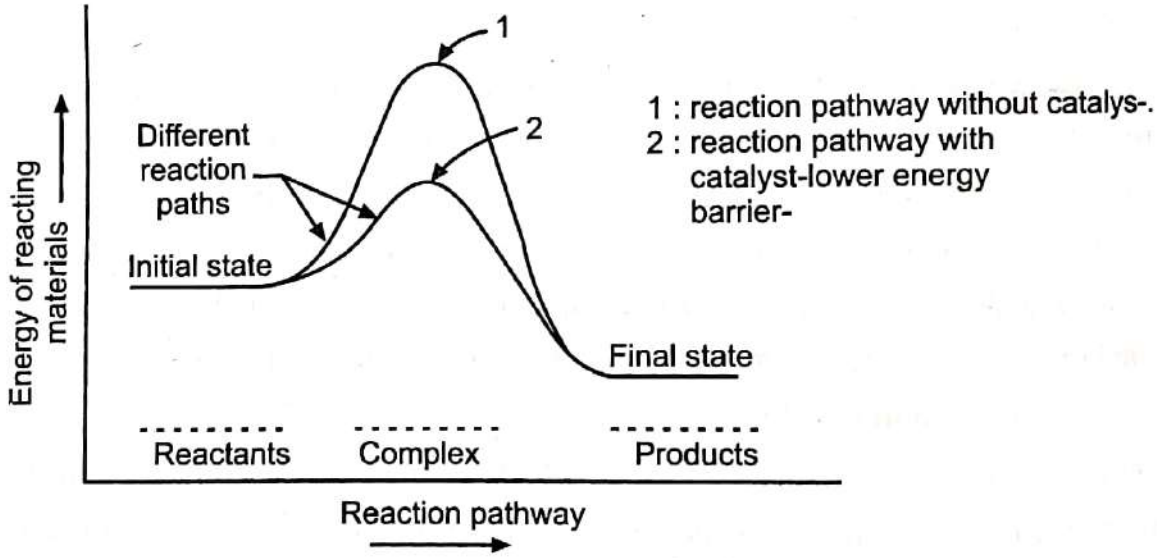


Fig. 6.1 : Energy diagram of uncatalysed and catalysed reaction – effect of use of a catalyst

Since the internal surface of a commercial catalyst is many times greater than its external surface, a catalytic reaction takes place mainly at the surface of the pores in the catalyst grain.

Mechanism of solid catalysed reactions :

The steps involved in solid catalysed fluid-phase reactions (which may cause resistance to reaction) are as follows :

1. Diffusion of the reactant(s) (e.g., species A) from the bulk fluid phase to the external surface of the catalyst through the fluid film ... external or surface diffusion step. It may involve the adsorption of reactant on the catalyst surface reaction on the surface or description of the product.

2. Diffusion of the reactant from the pore mouth into the catalyst pores.

   The greater proportion of reactant molecules diffuse inside the pores of the catalyst ..... the pore diffusion step.

3. Adsorption of the reactant on the catalyst surface (external and pore surfaces).

4. Chemical reaction of the adsorbed reactant on the surface of the catalyst (e.g., $A \rightarrow R$) to form the products ..... surface reaction.

5. Desorption of the products (e.g., R) from the surface (to the fluid phase near the surfaces).

6. Diffusion of the desorbed products from the interior of the catalyst (pores) to the pore mouth at the external surface, i.e., to the external surface ..... reverse pore diffusion.

7. Diffusion of the products from the external surface into the bulk fluid phase through the fluid film surrounding the catalyst ..... reverse external-diffusion.

Steps 1, 2, 6 and 7 ..... diffusion steps.

Steps 3, 4 and 5 ..... reaction steps.

Collectively, steps 3, 4 and 5 may be regarded as surface chemical reaction.

The overall rate of reaction is equal to the rate of the slowest step in the mechanism. With a porous catalyst, diffusion within catalyst pores may limit the rate of reaction.

When a porous catalyst particle is immersed/cover completely by reactant A, the rate of reaction of A for the whole particle may depend on one or more of the following factors :

1. Surface kinetics.

2. Pore (diffusion) resistance. Because of this resistance, the interior of the particle will be starved for reactant. We can reduce the pore diffusion resistance by reducing the catalyst particle size, which thus shortens the diffusion path.

3. Particle ΔT – temperature gradient within the catalyst particle (internal temperature gradient). This results from evolution or absorption of a large amount of heat during reaction. It causes the reaction rate to vary inside the catalyst (i.e., with position).

4. Film ΔT – temperature difference between the outer surface of the catalyst particle and the bulk gas stream (external temperature gradient) - resistance to heat transfer is within the gas film enveloping the catalyst particle.

5. Film diffusion resistance – external diffusion resistance - concentration gradient across the gas film surrounding the catalyst particle.

In the majority of cases with porous catalyst particles, the rate is influenced by surface reaction and pore diffusion. Therefore, here we will consider surface kinetics and pore diffusion resistance. Also, we will use a simple empirical rate expression – first-order or n^{th} order rate expression, which adequately fits the data, to represent the surface reaction.

Pore Diffusion Resistance Combined with Surface Kinetics (Pore Diffusion + Surface Reaction)

Single Cylindrical Pore + First-Order Reaction : Consider a single cylindrical catalyst pore of length L as shown in Fig. 6.2. Further consider that reactant A diffuses into the pore, reacts on the surface (the walls of the pore) by a first order reaction and product diffuses out of the pore.

The reaction and rate based on unit surface of solid are given by

$A \rightarrow$ product, $\qquad -r_A'' = -\frac{1}{S}\frac{dN_A}{dt} = k''C_A$

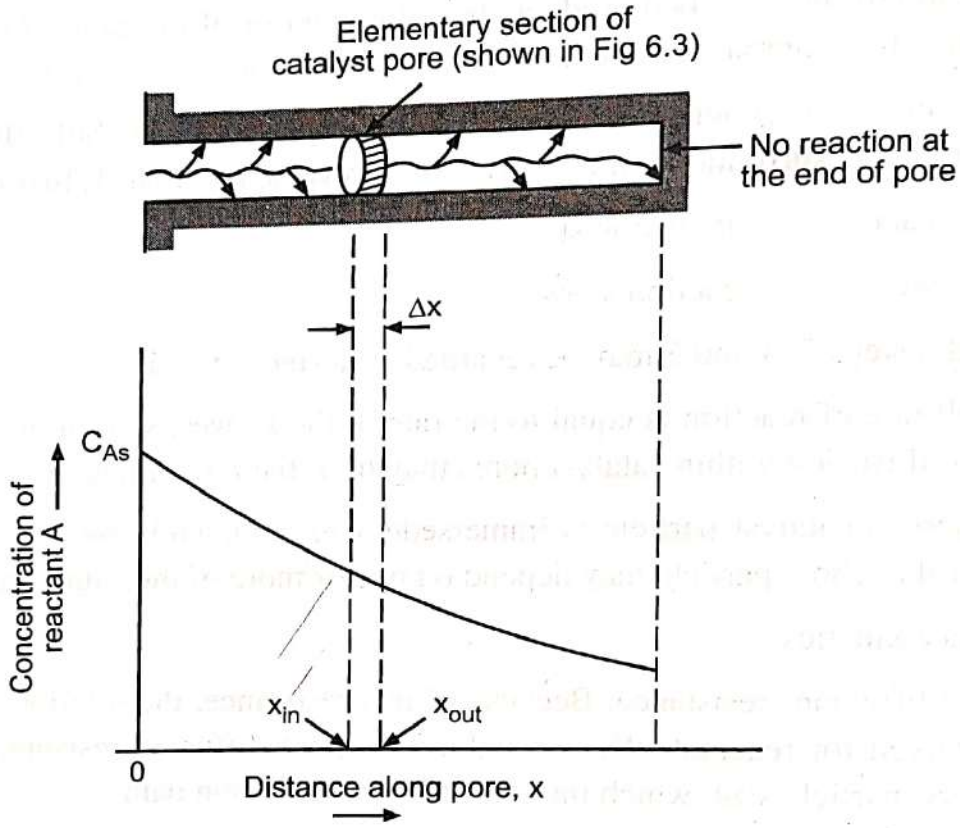


Fig. 6.2 : A straight cylindrical catalyst pore and reactant concentration distribution along pore length

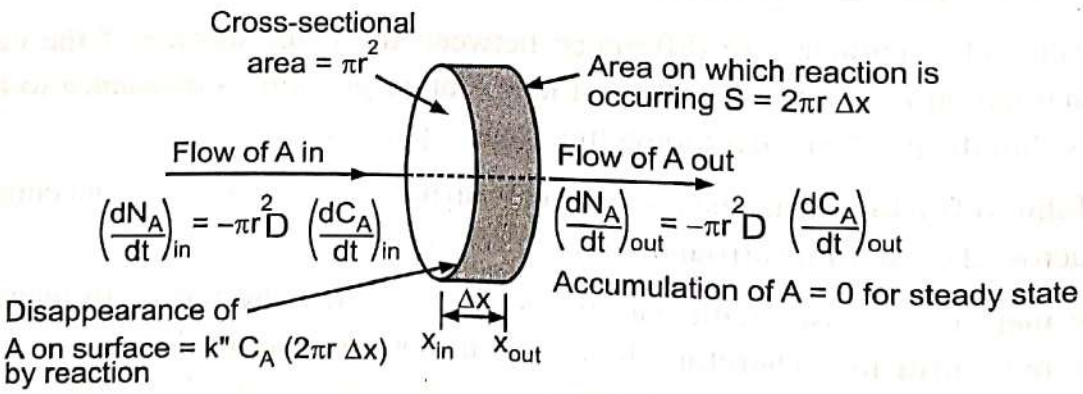


Fig. 6.3 : Elementary slice of catalyst pore for material balance

Fig. 6.3 shows the elementary section of the catalyst pore of length Δx. The steady state material balance for reactant A across this elementary section is

input of A = output of A + disappearance of A by reaction

output of A – input of A + disappearance of A by reaction = 0 ... (6.1)

The individual terms in mol A/s are as follows :

Reactant A entering by diffusion (at x_{in}, i.e., at x) :

$$\left(\frac{dN_A}{dt}\right)_{in} = \pi r^2 \mathscr{D} \left(\frac{dC_A}{dx}\right)_{in}, \quad \pi r^2 = \text{cross-sectional area of the section}$$

Reactant A leaving by diffusion (at x_{out}, i.e., at $x + \Delta x$) :

$$\left(\frac{dN_A}{dt}\right)_{out} = -\pi r^2 \mathscr{D} \left(\frac{dC_A}{dt}\right)_{out}$$

Disappearance of reactant A on the surface by reaction :

$$\text{Disappearance of A by reaction} = \left(\frac{\text{rate of disappearance of A}}{\text{unit surface}}\right) \text{(surface)}$$

$$= \left(-\frac{1}{S}\frac{dN_A}{dt}\right) \text{(surface)}$$

$$= k'' C_A (2\pi r \Delta x)$$

The surface on which reaction is occurring = $S = \pi D \cdot \Delta x = 2\pi r \Delta x$.

Substituting for all these terms, Equation (6.1) becomes

$$-\pi r^2 \mathscr{D} \left(\frac{dC_A}{dx}\right)_{out} + \pi r^2 \mathscr{D} \left(\frac{dC_A}{dx}\right)_{in} + k'' C_A (2\pi r \Delta x) = 0$$

Changing sign and dividing by $\pi r^2 \mathscr{D} \Delta x$ throughout gives

$$\frac{\left(\frac{dC_A}{dx}\right)_{out} - \left(\frac{dC_A}{dx}\right)_{in}}{\Delta x} - \frac{2k''}{\mathscr{D} r} C_A = 0 \quad \text{... (6.2)}$$

[For any quantity Q which is a smooth continuous function of x

$$\lim_{x_2 \to x_1} \frac{Q_2 - Q_1}{x_2 - x_1} = \lim_{\Delta x \to 0} \frac{\Delta Q}{\Delta x} = \frac{dQ}{dx} \ ]$$

So taking the limit as $\Delta x \to 0$, we obtain the following second order differential equation

$$\frac{d^2 C_A}{dx^2} - \frac{2k''}{\mathscr{D} r} C_A = 0 \quad \text{... (6.3)}$$

Here as the rate is based on unit surface area of the wall of the catalyst pore k" has the units of length per time (m/s).

The rates on different bases are related by

$$(-r_A) V = (-r_A') W = (-r_A'') S \quad \text{... (6.4)}$$

In terms of rate constant it becomes

$$kV = k'W = k''S \quad \ldots (6.5)$$

Therefore for cylindrical catalyst pore,

$$k = k'' \frac{S}{V} = k'' \left(\frac{\text{surface area}}{\text{volume}}\right) = k'' \left(\frac{2\pi rL}{\pi r^2 L}\right) = \frac{2k''}{r} \quad \ldots (6.6)$$

So in terms of volumetric units, Equation (6.3) becomes

[substituting k for 2k''/r in Equation (6.3), we get]

$$\frac{d^2C_A}{dx^2} - \frac{k}{\mathscr{D}} C_A = 0 \quad \ldots (6.7)$$

Let $k/D = m^2$

With this Equation (6.3) becomes

$$\frac{d^2C_A}{dx^2} - m^2 C_A = 0 \quad \ldots (6.8)$$

We seek the particular solutions of the above equation in the form :

$$C_A = e^{Ax}$$

then $$\frac{dC_A}{dx} = A e^{Ax} \text{ and } \frac{d^2C_A}{dx^2} = A^2 e^{Ax}$$

Therefore, Equation (6.8) becomes

$$A^2 e^{Ax} - m^2 e^{Ax} = 0$$

Hence, $$A = \pm m$$

Thus, the particular solutions of Equation (6.8) are :

$$C_A = e^{mx} \text{ and } C_A = e^{-mx}$$

General solution of Equation (6.8) is the linear combination of the particular solutions.

$$C_A = M_1 e^{mx} + M_2 e^{-mx}$$

(where M_1 and M_2 are constants of integration.)

With this the general solution of the linear differential Equation [Equation (6.7)] is

$$C_A = M_1 e^{mx} + M_2 e^{-mx} \quad \ldots (6.9)$$

where $$m = \sqrt{k/\mathscr{D}} = \sqrt{2k''/\mathscr{D} r}$$

and where M_1 and M_2 are arbitrary constants of integration.

Let us evaluate M_1 and M_2 using the boundary conditions.

The boundary conditions are :

(i) At the pore entrance, the concentration is C_{As} :

$$C_A = C_{As} \text{ at } x = 0 \quad \ldots (6.10\,a)$$

(ii) There is no flux or movement of material through the interior end of the pore. Therefore,

$$\frac{dC_A}{dx} = 0 \quad \text{at} \quad x = L \qquad \ldots(6.10\ b)$$

With $C_A = C_{As}$ at $x = 0$ Equation (6.9) becomes

$$C_{As} = M_1 + M_2 \qquad \ldots (6.11)$$

We have : $C_A = M_1 e^{mx} + M_2 e^{-mx}$

$$\frac{dC_A}{dx} = m M_1 e^{mx} - m M_2 e^{-mx}$$

With $dC_A/dx = 0$ at $x = L$, the above becomes

$$0 = m M_1 e^{mL} - m M_2 e^{-mL}$$

$$M_1 e^{mL} - M_2 e^{-mL} = 0 \qquad \ldots (6.12)$$

Combining Equations (6.11) and (6.12) gives

$$M_1 e^{mL} - (C_{As} - M_1)\, e^{-mL} = 0$$

$$M_1 (e^{mL} + e^{-mL}) - C_{As} e^{-mL} = 0$$

$$\therefore \quad M_1 = \frac{C_{As} e^{-mL}}{e^{mL} + e^{-mL}} \qquad \ldots (6.13\ a)$$

Similarly, $$M_2 = \frac{C_{As} e^{mL}}{e^{mL} + e^{-mL}} \qquad \ldots (6.13\ b)$$

Substituting M_1 and M_2 from Equation (6.13) into Equation (6.9) gives

$$C_A = \frac{C_{As} e^{-mL} \cdot e^{mx}}{e^{mL} + e^{-mL}} + \frac{C_{As} e^{mL} \cdot e^{-mx}}{e^{mL} + e^{-mL}}$$

$$C_A = C_{As}\left[\frac{e^{-m(L-x)} + e^{m(L-x)}}{e^{mL} + e^{-mL}}\right]$$

The dimensionless concentration profile of reactant A within the pore is

$$\frac{C_A}{C_{As}} = \frac{e^{m(L-x)} + e^{-m(L-x)}}{e^{mL} + e^{-mL}} = \frac{\cosh m(L-x)}{\cosh mL} \qquad \ldots (6.14)$$

[As $$\cosh mL = \frac{e^{mL} + e^{-mL}}{2}$$

$$\cosh m(L-x) = \frac{e^{m(L-x)} + e^{-m(L-x)}}{2}$$

$$\cosh x = \frac{e^{x} + e^{-x}}{2}\ ]$$

Fig. 6.4 shows the concentration profile of reactant within the pore and it is seen to be dependent on the dimensionless quantity mL, or M_T. The dimensional quantity mL, or M_T, is called the Thiele modulus.

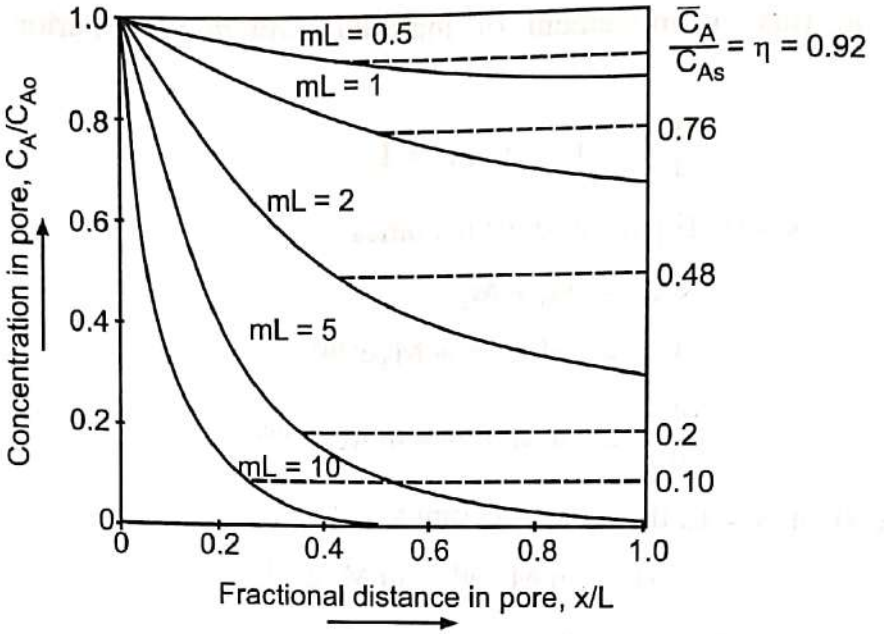


Fig. 6.4 : Distribution and average concentration of reactant in a catalyst pore as a function of mL (Thiele modulus) for a first order surface reaction

Now, we define the quantity η called the effectiveness factor in order to measure how much the reaction rate is lowered due to the resistance of pore diffusion.

It is defined as the *ratio of the actual average reaction rate within the pore to the reaction rate in the absence of pore diffusion resistance.*

$$\eta = \frac{\text{actual average reaction rate within pore}}{\text{reaction rate in the absence of pore diffusion resistance}}$$

$$= \frac{\text{actual average reaction rate within pore}}{\text{reaction rate that would result if entire pore surface were exposed to conditions at the external surface/pore mouth } C_{As},\ T_S}$$

The rate of reaction in the absence of pore diffusion resistance is the maximum rate of reaction. The rate of reaction will be maximum when the concentration of reactant along the pore is the same as that at the external surface, C_{As}.

$$\eta = \frac{-\bar{r}_{A \text{ with diffusion}}}{-r_{A \text{ without diffusion resistance}}} \qquad \ldots(6.15)$$

The actual rate is also referred to as the observed rate of reaction, $(-r_A)_{obs}$.

The effectiveness factor ranges from 0 to 1.

For first order reactions, since the rate is proportional to the concentration, the effectiveness factor is given by

$$\eta_{\text{first order}} = \bar{C}_A/C_{As}$$

Evaluating the average rate in the pore from Equation (6.14) yields the following relationship

$$\eta_{\text{first order}} = \frac{\bar{C}_A}{C_{As}} = \frac{\tanh mL}{mL} \quad \ldots (6.15)$$

Let us obtain the relationship given by Equation (6.15).

For first-order reactions :

$$\eta_{\text{first order}} = \frac{k\bar{C}_A}{kC_{As}} = \frac{\bar{C}_A}{C_{As}} \quad \ldots (6.16)$$

$\bar{C}_A$ is given by $$\bar{C}_A = \frac{\int_0^L C_A\, dx}{\int_0^L dx} = \frac{1}{L}\int_0^L C_A\, dx$$

With this $\bar{C}_A$, Equation (6.16) becomes

$$\eta_{\text{first order}} = \frac{1}{L\, C_{As}} \int_0^L C_A\, dx \quad \ldots (6.17)$$

We have, $$\frac{C_A}{C_{As}} = \frac{\cosh m(L - x)}{\cosh mL}$$

$$\therefore \quad C_A = C_{As}\, \frac{\cosh m(L - x)}{\cosh mL}$$

Substituting C_A into Equation (6.17) in terms of C_{As} gives

$$\eta_{\text{first order}} = \frac{1}{L\, C_{As}} \int_0^L \frac{C_{As} \cosh m(L - x)\, dx}{\cosh mL}$$

$$= \frac{C_{As}}{L\, C_{As} \cosh mL} \int_0^L \cosh m(L - x)\, dx$$

$$= \frac{1}{L \cosh mL} \int_0^L \cosh m(L - x)\, dx$$

$$= \frac{1}{L \cosh mL} \left[\frac{-1}{m} \sinh m(L - x)\right]_0^L$$

$$= \frac{-1}{mL \cosh mL} [0 - \sinh mL]$$

$$= \frac{\sinh mL}{mL \cosh mL}$$

$$\therefore \quad \eta_{\text{first order}} = \frac{\tanh mL}{mL} \quad \ldots (6.18)$$

Alternately (for first order reactions) :

The rate of reaction if there were no slow down effect from pore diffusion is the maximum rate of reaction. This is the rate when the reactant concentration along the pore is the same as that at the external surface (pore mouth), C_{As}.

$$\left(-\frac{1}{S}\frac{dN_A}{dt}\right)_{max} = k'' C_{As}$$

$$\therefore \quad \left(-\frac{dN_A}{dt}\right)_{max} = k'' C_{As} S, \quad \text{mol A/s}$$

where S is the internal surface area of a cylindrical pore

$$S = 2\pi rL$$

The maximum rate (i.e., rate in the absence of pore diffusion) in mol A/s is :

$$\left(\frac{-dN_A}{dt}\right)_{max} = 2\pi rL\, k'' C_{As} \qquad \ldots (6.19)$$

The actual mean rate of reaction is the rate when slow down is caused by pore diffusion.

The actual rate of reaction of A is the rate with pore diffusion affects (i.e., when slow down is caused by pore diffusion) and it cannot exceed the rate at which reactant A enters the mouth of the pore due to molecular diffusion.

At steady state :

The actual rate of reaction of A (i.e., rate of consumption of A by reaction)

= rate of mass transfer of A at the pore entrance

$$\therefore \quad \left(\frac{-dN_A}{dt}\right)_{avg} = \left[-\mathscr{D}\frac{dC_A}{dx}\right]_{x=0} \cdot S_1, \quad \text{mol/s}$$

rate of reaction in mol/s — molar flux of A · area

where S_1 is the surface area of pore mouth

$$S_1 = \pi r^2$$

Actual rate (mol A/s) :

$$\left(\frac{-dN_A}{dt}\right)_{avg} = \pi r^2 \left[-\mathscr{D}\frac{dC_A}{dx}\right]_{x=0} \qquad \ldots (6.20)$$

$$\eta = \frac{\text{actual}}{\text{rate in absence of pore diffusion}}$$

$$\eta = \frac{-\mathscr{D}\pi r^2 [dC_A/dx]_{x=0}}{2\pi rL\, k'' C_{As}} \qquad \ldots (6.21)$$

Now we will evaluate $[dC_A/dx]_{x=0}$: $-dC_A/dx$ at $x = 0$, i.e., at the mouth of the pore.

We have, $$\frac{C_A}{C_{As}} = \frac{\cosh m(L-x)}{\cosh mL}$$

$$C_A = C_{As}\frac{\cosh m(L-x)}{\cosh mL}$$

$$\frac{dC_A}{dx} = \frac{C_{As}(-m)\sinh m(L-x)}{\cosh mL}$$

$$\left[\frac{dC_A}{dx}\right]_{x=0} = \frac{-m\,C_{As}\sinh mL}{\cosh mL} = -m\,C_{As}\tanh mL \quad \text{(by putting } x = 0\text{)}$$

Substituting this value of $[dC_A/dx]_{x=0}$ into Equation (6.21) gives

$$\therefore \quad \eta = \frac{-\mathscr{D}\pi r^2[-m\,C_{As}\tanh mL]}{2\pi rL\,k''C_{As}}$$

$$= \frac{m\,\mathscr{D}\,r\tanh mL}{L\,2k''}$$

$$\eta = \frac{m\tanh mL}{L(2k''/\mathscr{D}r)} \quad \ldots (6.22)$$

We have, $$m = \sqrt{k/D} = \sqrt{2k''/\mathscr{D}r}$$

$$m^2 = 2k''/\mathscr{D}r \quad \ldots (6.23)$$

Combining Equations (6.22) and (6.23) gives

$$\eta = \frac{m\tanh mL}{Lm^2}$$

$$\eta = \frac{\tanh mL}{mL} \quad \ldots (6.24)$$

A plot of effectiveness factor (η) as a function of the Thiele modulus (mL) is shown in Fig. 6.5.

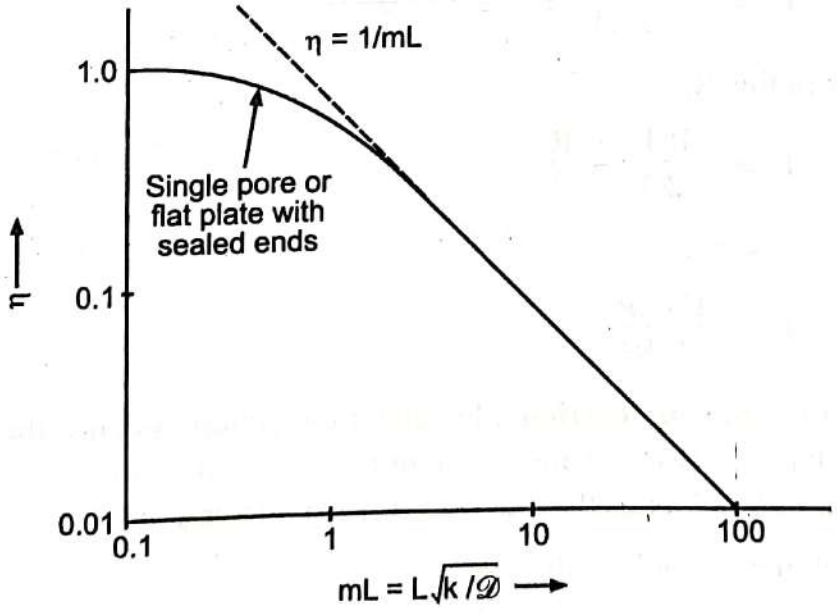


Fig. 6.5 : The effectiveness factor as a function of the Thiele modulus (mL or M_T)

For small values of the Thiele modulus (mL < 0.4), tanh mL ≈ mL. Consequently, the effectiveness factor at low values of mL is practically equal to unity ($\eta \approx 1$). This is a case when the catalyst has a short, wide pores and the reaction rate constant (k) is small in comparison with the diffusion coefficient. Under these conditions the concentration of reactant does not drop appreciably within the pore (Refer Fig. 6.4) and hence pore diffusion offers negligible resistance (reaction is not limited by diffusion). This can also be confirmed by noting that a small value for $mL = L\,(k/\mathscr{D})^{1/2}$ means either a short pore (small L), slow reaction (small k), or fast diffusion (large $\mathscr{D}$), all these factors tending to lower the resistance to pore diffusion.

For large values of the Thiele modulus (mL > 4), tanh mL ≈ 1 and so $\eta = 1/mL$. Under these conditions the concentration of reactant drops rapidly to zero on moving into the pore (Refer Fig. 6.4) and hence pore diffusion strongly influences the rate of reaction (the reaction is diffusion limited – pore diffusion limits the rate of reaction). We call this the regime of strong resistance to pore diffusion or the regime of strong pore resistance.

Porous Catalysts :

The results for a single pore can be applied for porous catalyst particles of various shapes (spheres, cylinders, etc.).

(i) Proper diffusion coefficient : Here the molecular diffusion coefficient $\mathscr{D}$ is to be replaced by the effective diffusion coefficient of fluid in the porous solids $\mathscr{D}_e$, $m^3/(h \cdot m\ cat)$.

(ii) Correct measure of particle size : A characteristic size of particle of any shape is defined as

$$L = \frac{\text{volume of particle}}{\text{external surface available for reactant penetration}} \quad \ldots (6.25)$$

For flat plates without end effects (of thickness c) :

$$L = \frac{a \times b \times c}{2 \times a \times b} = \frac{c}{2} = \frac{\text{thickness}}{2}$$

For cylinders (of radius R) :

$$L = \frac{\pi R^2 L}{2\pi R L} = \frac{R}{2}$$

For spheres (of radius R) :

$$L = \frac{4/3\ \pi R^3}{4\pi R^2} = \frac{R}{3}$$

(iii) Measures of rates of reaction : In catalytic reaction systems, the rate of reaction can be expressed in a number of ways (using different bases – void volume, weight of catalyst, etc.). For example, for first order kinetics the rate of reaction in its various forms is :

Based on unit volume of voids in the reactor :

$$-r_A = -\frac{1}{V}\frac{dN_A}{dt} = kC_A, \text{ mol A reacted/(m}^3 \text{ voids} \cdot \text{s)} \quad \ldots (6.26)$$

Based on unit weight (mass) of catalyst pellets :

$$-r_A' = -\frac{1}{W}\frac{dN_A}{dt} = k'C_A, \text{ mol A reacted/(kg cat·s)} \quad \ldots(6.27)$$

Based on unit catalyst surface :

$$-r_A'' = -\frac{1}{S}\frac{dN_A}{dt} = k''C_A, \text{ mol A reacted/(m}^2\text{ cat·s)} \quad \ldots(6.28)$$

Based on unit volume of catalyst pellets :

$$-r_A''' = -\frac{1}{V_p}\frac{dN_A}{dt} = k'''C_A, \text{ mol A reacted/(m}^3\text{ solid·s)} \quad \ldots(6.29)$$

[Here use V_p or $V_s \rightarrow$ m^3 cat or m^3 solid]

Based on unit volume of reactor :

$$-r_A'''' = -\frac{1}{V_r}\frac{dN_A}{dt} = k''''C_A, \text{ mol A reacted/(m}^3\text{ reactor·s)} \quad \ldots(6.30)$$

These rates are related by

$$(-r_A)A = (-r_A')W = (-r_A'')S = (-r_A''')V_s = (-r_A'''')V_r \quad \ldots(6.31)$$

For porous catalyst particles, rates based on unit mass and unit volume of particles are the useful measures. Therefore, for n^{th} order reactions :

$$-r_A' = k'C_A^n, \text{ mol A/(kg cat·s)} \quad \ldots(6.32)$$

The rate of constant, k', has the units of

$$k' \Rightarrow [(m^3\text{ gas})^n/[(\text{mol A})^{n-1}\cdot \text{kg cat·s}]]$$

$$-r_A''' = k'''C_A^n, \text{ mol A/(m}^3\text{ cat·s)} \quad \ldots(6.33)$$

The rate constant, k''', has the units of

$$k''' \Rightarrow [(m^3\text{ gas})^n/[(\text{mol A})^{n-1}\cdot m^3\text{ cat}\cdot s]]$$

(iv) The relationship between η (effectiveness factor) and **M_T** (Thiele modulus) for the various particle shapes are given below.

For a first-order isothermal surface reaction of the type

$A \rightarrow R$, $\quad -r_A''' = k'''C_{As}\eta$

the relationship between η and M_T are :

1. $\eta = \frac{1}{M_T}\tanh M_T$ ... for flat plates/slabs with sealed edges ... (6.34)

2. $\eta = \frac{1}{M_T}\cdot\frac{I_1(2M_T)}{I_0(2M_T)}$ ... for cylinders with sealed ends (cylindrical catalyst pellets) ...(6.35)

where I_0 and I_1 represent zero and first order modified Bessel functions of the first kind.

3. $\eta = \frac{1}{M_T}\left(\frac{1}{\tanh 3M_T} - \frac{1}{3M_T}\right)$ ... for spheres (spherical catalyst pellets) ... (6.36)

where $M_T = L\sqrt{k'''/\mathscr{D}_e}$ ... (6.37)

k''' is expressed in m^3 gas/(m^3 cat·s).

The above relationships are shown in Fig. 6.6.

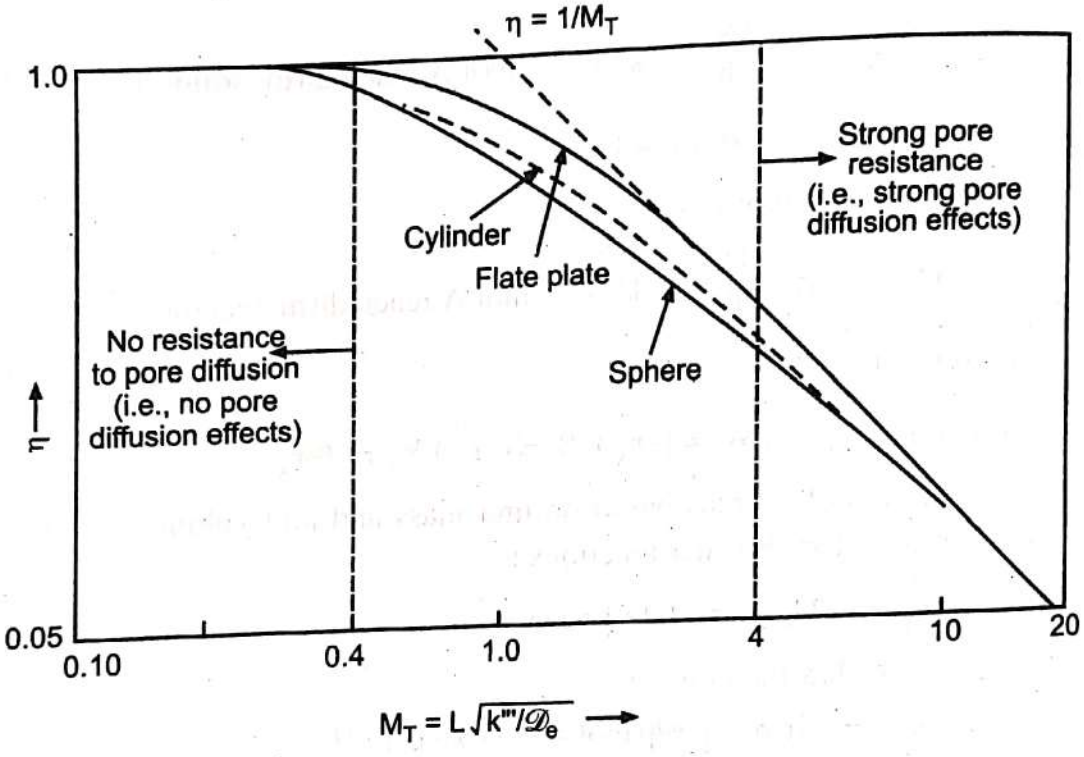


Fig. 6.6 : Effectiveness factor as a function of M_T for porous particles of various shapes

Let us derive the expression for the effectiveness factor of a rectangular slab (flat plate) of porous catalyst. The edges of slab are sealed so that diffusion occurs in one dimension only.

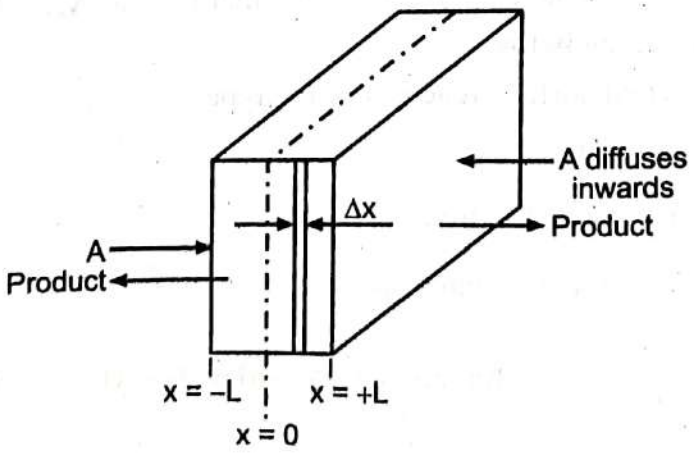


Fig. 6.7 : Geometry of slab type catalyst pellets

The steady state material balance of reactant A is

output – input + disappearance = 0

$$-A\,\mathscr{D}_e\left(\frac{dC_A}{dx}\right)_{out} + A\,\mathscr{D}_e\left(\frac{dC_A}{dx}\right)_{in} + k''C_A\,A\,\Delta x = 0 \qquad \ldots (6.38)$$

Changing sign throughout gives

$$A\,\mathscr{D}_e\left(\frac{dC_A}{dx}\right)_{out} - A\,\mathscr{D}_e\left(\frac{dC_A}{dx}\right)_{in} - k'''\,C_A\,A\,dx = 0$$

Dividing by $A\,\mathscr{D}_e\,\Delta x$ gives

$$\frac{\left(\frac{dC_A}{dx}\right)_{out} - \left(\frac{dC_A}{dx}\right)_{in}}{\Delta x} - \frac{k'''}{\mathscr{D}_e}\,C_A = 0$$

As $\Delta x \to 0$

$$\frac{d^2C_A}{dx^2} - \frac{k'''}{\mathscr{D}_e}\,C_A = 0 \qquad \ldots (6.39)$$

Let $$\frac{k'''}{\mathscr{D}_e} = m^2$$

$$\frac{d^2C_A}{dx^2} - m^2C_A = 0 \qquad \ldots (6.40)$$

The general solution of this equation is

$$C_A = M_1\,e^{mx} + M_2\,e^{-mx} \qquad \ldots (6.41)$$

The boundary conditions are

(i) $C_A = C_{As}$ at $x = \pm L$ … at the exterior surface of the slab.

(ii) $\frac{dC_A}{dx} = 0$ at $x = 0$ … at the centre of the slab

(the net flux through the plane at $x = 0$ is zero).

$$C_A = M_1\,e^{mx} + M_2\,e^{-mx}$$

$$C_{As} = M_1\,e^{mL} + M_2\,e^{-mL} \quad \ldots \text{ with the boundary condition (i)}$$

$$C_A = M_1\,e^{mx} + M_2\,e^{-mx}$$

$$\frac{dC_A}{dx} = m\,M_1\,e^{mx} - m\,M_2\,e^{-mx}$$

$$0 = m\,M_1\,e^{m(0)} - m\,M_2\,e^{-m(0)} \ldots \text{ with the boundary condition (ii)}$$

$\therefore$ $$M_1 = M_2$$

$$C_{As} = M_1\,e^{mL} + M_2\,e^{-mL}$$

$$C_{As} = M_1\,e^{mL} + M_1\,e^{-mL} = M_1\,(e^{mL} + e^{-mL}), \quad \text{as } M_1 = M_2$$

$$M_1 = M_2 = \frac{C_{As}}{e^{mL} + e^{-mL}}$$

$$\therefore \quad C_A = \frac{C_{As}\, e^{mx}}{e^{mL} + e^{-mL}} + \frac{C_{As}\, e^{-mx}}{e^{mL} + e^{-mL}} = C_{As}\left(\frac{e^{mx} + e^{-mx}}{e^{mL} + e^{-mL}}\right)$$

$$C_A = C_{As}\frac{\cosh mx}{\cosh mL}, \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

$$\frac{dC_A}{dx} = \frac{m\, C_{As} \sinh mx}{\cosh mL}$$

$$\left[\frac{dC_A}{dx}\right]_{x=L} = \frac{m\, C_{As} \sinh mL}{\cosh mL} = m\, C_{As} \tanh mL$$

At steady state, the actual rate of reaction of A is equal to the total flux of A at the external surfaces.

$$\text{Actual rate of reaction in mol/s} = -2A\left[-\mathscr{D}_e\left[\frac{dC_A}{dx}\right]_{x=L}\right]$$

$$= 2A\, \mathscr{D}_e\left[\frac{dC_A}{dx}\right]_{x=L}$$

$$= 2A\, \mathscr{D}_e\, m\, C_{As} \tanh mL$$

Rate in the absence of pore diffusion in mol/s = $k'''C_{As} \cdot 2AL$

$$\therefore \quad \eta = \frac{\text{actual rate}}{\text{rate in absence of diffusion}}$$

$$\eta = \frac{2A\, \mathscr{D}_e\, [dC_A/dx]_{x=L}}{2AL\, k'''C_{As}}$$

$$= \frac{2A\, \mathscr{D}_e\, m\, C_{As} \tanh mL}{2AL\, k'''C_{As}}$$

$$= \frac{m\, \mathscr{D}_e \tanh mL}{L\, k'''}$$

$$= \frac{m \tanh mL}{L\,(k'''/\mathscr{D}_e)}, \quad \text{we have, } m = (k'''/\mathscr{D}_e)^{1/2}$$

$$= \frac{m \tanh mL}{m^2 L}$$

$$= \frac{\tanh mL}{mL}, \quad m = (k'''/\mathscr{D}_e)^{1/2}$$

$$= \frac{\tanh (k'''/\mathscr{D}_e)^{1/2} L}{L\,(k'''/\mathscr{D}_e)^{1/2}}$$

With M_T = Thiele modulus = $L\,(k'''/\mathscr{D}_e)^{1/2}$ the above equation becomes

$$\eta = \frac{\tanh M_T}{M_T} \quad \ldots \text{ desired expression}$$

(v) Determining pore resistance effects from experimental data : To tell whether pore resistance is or is not interferring to slow the rate of reaction from the observed rate measurement for a given catalyst pallets in a known environment, we define the Wagner modulus (M_W).

$$M_W = M_T^2 \eta = \frac{\left(-r_A'''/C_A\right)_{obs}}{\mathscr{D}_e} L^2 \qquad \ldots (6.42)$$

All the terms in the above equation are either measured or known.

(vi) Limits of pore resistance : When reactant completely penetrates the particles (catalyst pellet) and saturates thouroughly all its surfaces, i.e., when reactant is uniformly distributed throughout the particles (almost the same reactant concentration throughout), then the particle is in the diffusion resistance free regime/diffusion free regime. This happens when $M_T < 0.4$ or $M_W < 0.15$.

When reactant is found close to the exterior surface of the particle (penetrates to a small depth into the pores) and thereby the centre of the particle is lacked for reactant (i.e., there is no reactant) and remains unused, then the particle is in the strong pore diffusion resistance regime/strong pore resistance regime. This happens when $M_T > 4$ or $M_W > 4$. (Here the reaction is diffusion limited in the pellet).

These limits are shown in Fig. 6.8.

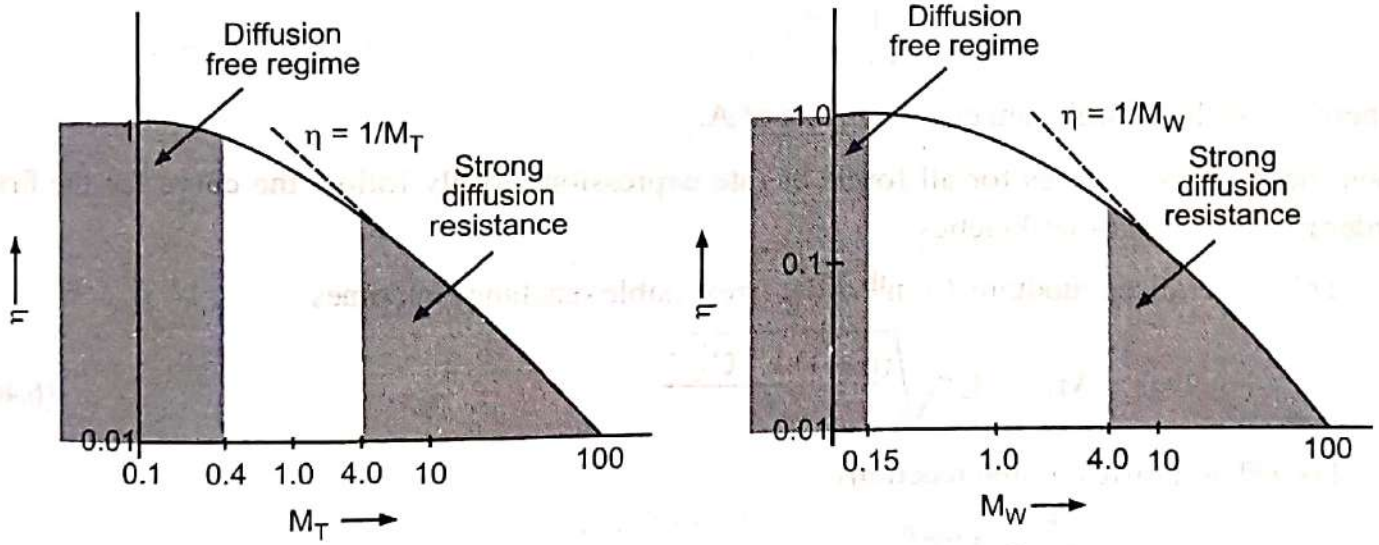


Fig. 6.8 : Limits for negligible and for strong pore diffusion resistance

(vii) Different size particles : The effectiveness factor accounts for pore diffusion resistance. Hence, based on unit mass of catalyst, considering pore diffusion resistance, the rate is given by

$$-r_A' = k'C_A\eta$$

[considering the gas film resistance to be absent : $C_{As} = C_A$ – concentration in the bulk gas stream]

$$-r'_{A_1} = k'C_A\eta_1 \quad \text{... for particles of size } R_1$$

$$-r'_{A_2} = k'C_A\eta_2 \quad \text{... for particles of size } R_2$$

In the diffusion-free regime, $\eta = 1$ and hence for the two sizes of particles, we have

$$\frac{-r'_{A_1}}{-r'_{A_2}} = \frac{k'C_A\eta_1}{k'C_A\eta_2} = \frac{\eta_1}{\eta_2} = 1, \quad [\text{as } \eta_1 = \eta_2 = 1] \qquad \ldots (6.43)$$

In the strong pore diffusion resistance regime, $\eta = 1/M_T$ and hence for the two sizes of particles, we have

$$\frac{-r'_{A_1}}{-r'_{A_2}} = \frac{k'C_A\eta_1}{k'C_A\eta_2} = \frac{\eta_1}{\eta_2} = \frac{1/M_{T1}}{1/M_{T2}} = \frac{M_{T2}}{M_{T1}} = \frac{R_2}{R_1} \quad [\text{as } M_T \propto R] \qquad \ldots (6.44)$$

Thus, the rate of reaction is inversely proportional to the particle size.

Arbitrary Reaction Kinetics :

If we generalise the Thiele modulus as

$$M_T = \frac{(-r'''_{As})\, L}{\left[2\,\mathscr{D}_e \int_{C_{Ae}}^{C_{As}} (-r'''_A)\right]^{1/2}} \qquad \ldots (6.45)$$

where C_{Ae} is the equilibrium concentration of A.

then the η v/s M_T curves for all forms of rate expression closely follow the curve for the first-order reaction/first order kinetics.

This generalised modulus for n^{th} order irreversible reactance becomes

$$M_T = L\sqrt{\frac{(n+1)\,k'''\,C_{As}^{n-1}}{2\mathscr{D}_e}} \qquad \ldots (6.46)$$

For n^{th} order irreversible reactions :

$$-r'''_{As} = k'''C_{As}^n$$

$$-r'''_A = k'''C_A^n$$

For irreversible reactions, $C_{Ae} = 0$

$$\therefore \quad M_T = \frac{k'''C_{As}^n\, L}{\left[2\,\mathscr{D}_e \int_0^{C_{As}} k'''C_A^n\, d\,C_A\right]^{1/2}}$$

$$= \frac{k''' C_{As}^{n} L}{\left[\frac{2\mathscr{D}_e\, k'''[C_A^{n+1}]_0^{C_{As}}}{(n+1)}\right]^{1/2}} = \frac{k''' C_{As}^{n} L}{\left[\frac{2\mathscr{D}_e\, k''' C_{As}^{n+1}}{n+1}\right]^{1/2}}$$

$$= L\left[\frac{(n+1)\, k'''}{2\,\mathscr{D}_e}\right]^{1/2} \left[C_{As}^{n} \cdot C_{As}^{-(n+1)/2}\right]$$

$$= L\left[\frac{(n+1)\, k'''}{2\,\mathscr{D}_e}\right]^{1/2} \left[C_{As}^{(2n-n-1)/2}\right]$$

$$M_T = L\left[\frac{(n+1)\, k'''}{2\,\mathscr{D}_e}\right]^{1/2} \cdot C_{As}^{(n-1)/2} = L\sqrt{\frac{(n+1)\, k''' C_{As}^{n-1}}{2\mathscr{D}_e}} \qquad \ldots (6.47)$$

For n^{th} order reactions,

$$-r_A''' = k''' C_{As}^{n}\, \eta \qquad \ldots (6.48)$$

We have, $\eta = 1/M_T$

$$\therefore \quad -r_A''' = k''' C_{As}^{n} \cdot \frac{1}{M_T} \qquad \ldots (6.49)$$

Substituting M_T from Equation (6.47) into the above equation gives

$$-r_A''' = k''' C_{As}^{n} \cdot \frac{1}{L\left[(n+1)\, k''' C_{As}^{(n-1)/2} / 2\mathscr{D}_e\right]^{1/2}}$$

$$= k''' C_{As}^{n} \cdot \frac{1}{L}\left(\frac{2\,\mathscr{D}_e}{(n+1)\, k''' C_{As}^{n-1}}\right)^{1/2},$$

$$C_{As}^{n} \cdot C_{As}^{-(n-1)/2} = C_{As}^{(2n-n+1)/2} = C_{As}^{(n+1)/2}$$

$$-r_A''' = \left[\frac{2}{(n+1)} \cdot \frac{k'''\mathscr{D}_e}{L^2}\right]^{1/2} C_{As}^{(n+1)/2} \qquad \ldots (6.50)$$

The rate depends on $C_{As}^{(n+1)/2} \rightarrow C_{As}^{n'}$

Therefore, in the regime of strong pore diffusion, the measured/observed reaction order n' is related to the true reaction order n as follows :

$$n' = \frac{n+1}{2}$$

true order → 0 order ... becomes 1/2 order ← observed order/apparent order

1st order ... remains 1st order

2nd order ... becomes 1.5 order

The strong pore resistance also affects the temperature dependency of reactions. For n^{th} order reactions, the observed rate constant is

$$k'''_{obs} = \left[\frac{2}{n+1} \cdot \frac{k''' \mathscr{D}_e}{L^2}\right]^{1/2}$$

Taking the natural logarithms of both sides gives

$$\ln k'''_{obs} = \frac{1}{2} \ln \left(\frac{2}{n+1} \cdot \frac{k''' \mathscr{D}_e}{L^2}\right)$$

$$\ln k'''_{obs} = \frac{1}{2} \ln \left(\frac{2}{(n+1)\, L^2}\right) + \frac{1}{2} \ln k''' + \frac{1}{2} \ln D_e$$

Differentiating with respect to temperature, we get

$$\frac{d\,(\ln k'''_{obs})}{dT} = \frac{1}{2} \frac{d \ln k'''}{dT} + \frac{1}{2} \frac{d \ln \mathscr{D}_e}{dT} = \frac{1}{2} \left[\frac{d \ln k'''}{dT} + \frac{d \ln \mathscr{D}_e}{dT}\right] \quad \ldots (6.51)$$

[the diffusional process is also to some extent temperature dependent]

Temperature dependency from Arrhenius' law :

For reaction : $k''' = k'''_o \, e^{-E_{true}/RT}$

For diffusion : $\mathscr{D}_e = \mathscr{D}_{eo} \, e^{-E_{diff}/RT}$

$$\frac{d \ln k'''}{dT} = \frac{d \ln (k'''_o \, e^{-E_{true}/RT})}{dT} = \frac{d \ln k'''_o}{dT} + \frac{d \ln (e^{-E_{true}/RT})}{dT} = \frac{E_{true}}{RT^2}$$

Adopting this procedure to the other terms of Equation (6.51) gives

$$\frac{E_{obs}}{RT^2} = \frac{1}{2} \left[\frac{E_{true}}{RT^2} + \frac{E_{diff}}{RT^2}\right]$$

$$\therefore \quad E_{obs} = \frac{E_{true} + E_{diff}}{2} \quad \ldots (6.52)$$

The activation energy for gas phase reactions is very very high in comparison with that for diffusion.

Since $E_{true} >> E_{diff}$, we can write roughly

$$E_{obs} \approx \frac{E_{true}}{2} \quad \ldots (6.53)$$

The observed activation energy for reactions influenced by strong pore diffusion resistance (when reactions occur in the regime of strong pore diffusion) is approximately one-half the true value.

Summary of Diffusion Resistance in Pores :

Here we give a brief account of our finding for first order isothermal surface reaction as follows :

$-r_A''' = k''' C_{As} \eta$ ... rate accounting pore diffusion resistance (actual rate)

$-r_A''' = k''' C_{As}$ ... rate without diffusion effects

η = ... effectiveness factor, it ranges from 0 to 1 and accounts for the resistance to pore diffusion. Indicates the utilisation of the internal surface of a catalyst

$$\eta = \frac{\text{actual rate of reaction}}{\text{rate that would result if the entire surface were exposed to external surface concentration, } C_{As}}$$

$\eta = 1$ ... for no pore diffusion resistance

$\eta = \dfrac{\tanh M_T}{M_T}$ ... expression for flat plate, good approximation for all particle shapes.

$M_T = L\sqrt{k'''/\mathscr{D}_e}$ ... Thiele modulus - useful to predict reactor behaviour from known k'''.

L ... characteristic size of the particle ... characteristic length

$\mathscr{D}_e$ ... effective diffusion coefficient in porous solids, m^3 gas/(m^3 solid·s)

$M_W = M_T^2 \eta = \dfrac{L(-r_A'''/C_A)_{obs}}{\mathscr{D}_e}$ ... Wagner modulus, useful to interpret experiments.

$M_T < 0.4$ and $\eta = 1$ ... reactant is evenly distributed throughout the pellet, no pore diffusion resistance and surface reaction controls.

$M_T > 4$ and $\eta < 1$ ... strong pore diffusion resistance, reaction is diffusion limited in the pellet, i.e., pore diffusion controls.

Fine solids are free of pore diffusion resistance (very small L → low M_T → $\eta = 1$ → no diffusion resistance) but are difficult to use in a packed bed because of plugging and very high pressure drop. On the other hand, a catalyst bed of large particles offer low pressure drop but are susceptible to be in the regime of strong pore diffusion where much of the interior of pellets remains unused. Therefore, for effective operations use the largest particles size which is free of pore diffusion resistance – the particles size corresponding to $M_T = 0.4$ or $M_W = 0.15$.

Heat Effects (Exothermicity/Endothermicity) During Reaction :

In very fast reactions, the heat released (or absorbed) in the pellet can not be removed rapidly /quickly. Thus it is not possible to keep the pellet close to the temperature of the surrounding fluid and this consequently results in non-isothermal conditions. In such cases, the following types of temperature effects may be encountered :

(i) **Particle ΔT :** A temperature gradient may exist within the pellet – temperature varies within the catalyst pellet.

(ii) **Film ΔT :** A temperature gradient may exist across the fluid film (between the outer surface of pellet and main gas stream). The whole pellet may be hotter (or colder) than the surrounding fluid, though uniform in temperature.

In case of exothermic reactions, heat is generated and particles are hotter than the surrounding fluid. Consequently, the non-isothermal rate (of reaction) is higher than the isothermal rate which is based on the bulk stream conditions. On the other hand, in case of endothermic reactions, heat is absorbed and the particles are cooler than the surrounding fluid. Consequently, the non-isothermal rate is lower than the isothermal rate.

Thus we would promote nonisothermal behaviour in exothermic reactions provided no harmful effects of thermal shock, or sintering of the catalyst, or drop in selectivity occur with hot particles. On the other hand, we would depress nonisothermal behaviour in endothermic reactions.

The following simple calculations will give us idea regarding the form of non-isothermal effect, if any, that may be present.

(i) **Film ΔT :** To find the maximum ΔT across the film, the rate of heat generation by reaction within the pellet is set equal to the rate of heat removal through the film.

Rate of heat generation by reaction within the pellet =

$$= Q_{generated} = V_{pellet}\left(-r'''_{A\,obs}\right) \cdot (-\Delta H_r)$$

$$[V_{pellet} \cdot (-r'''_{Aobs}) \cdot (-\Delta H_r)] \Rightarrow (m^3\,cat)\left(\frac{mol\ A}{(m^3\,cat \cdot s)}\right)\left(\frac{kJ}{mol\ A}\right) = kJ/s$$

where $\left(-r'''_{Aobs}\right)$ = observed reaction rate

$(-\Delta H_r)$ = heat of reaction/enthalpy change resulting from reaction

V_{pellet} = volume of pellet = $S_{pellet} \times L$ (as $L = V_p/S_p$)

L = characteristic size of the pellet

$$Q_{generated} = S_{pellet} \cdot L \cdot \left(-r'_{Aobs}\right) \cdot (-\Delta H_r)$$

Rate of heat removal through the film = $Q_{removed} = h \cdot S_{pellet} (T_g - T_s)$

where h = film heat transfer coefficient, kJ/(m²·s·K)

T_g = temperature of main gas stream/bulk gas, K

T_s = temperature at the outer surface of the pellet, K

$$Q_{removed} = Q_{generated}$$

$$h \cdot S_{pellet} (T_g - T_s) = S_{pellet} \cdot L \left(-r'''_{Aobs}\right) (-\Delta H_r)$$

$$\boxed{\Delta T_{film} = (T_g - T_s) = \frac{L \left(-r'''_{Aobs}\right) (-\Delta H_r)}{h}, \text{ K}} \quad \ldots (6.54)$$

(ii) Particle ΔT : In steady state, the heat flux within an elementary thickness of the particle is balanced by the heat generated by chemical reaction. At any point in the pellet,

$$-k_{eff} \frac{dT}{dx} = \mathscr{D}_e \frac{dC_A}{dx} (-\Delta H_r) \quad \ldots (6.55)$$

For the pellet as a whole, integrating the above equation from the exterior surface where $T = T_s$ and $C_A = C_{As}$ to the centre of the particle where say $T = T_{centre}$ and $C_A = C_{A\ centre}$, we obtain

$$\boxed{\Delta T_{particle} = (T_{centre} - T_s) = \frac{\mathscr{D}_e (C_{As} - C_{A\ centre}) (-\Delta H_r)}{k_{eff}}} \quad \ldots (6.56)$$

where k_{eff} is the effective thermal conductivity of the particle, J/(s·m cat·K).

When Thiele modulus is large, $C_{Acentre}$ is zero and consequently the maximum temperature difference between the centre and exterior surface of the particle is $\mathscr{D}_e (-\Delta H_r) C_{As}/k_{eff}$.

Performance Equations for Reactors Containing Porous Catalysts :

Plug Flow Reactor :

Consider a thin slice of the solid catalysed plug flow reactor as shown in Fig. 6.9.

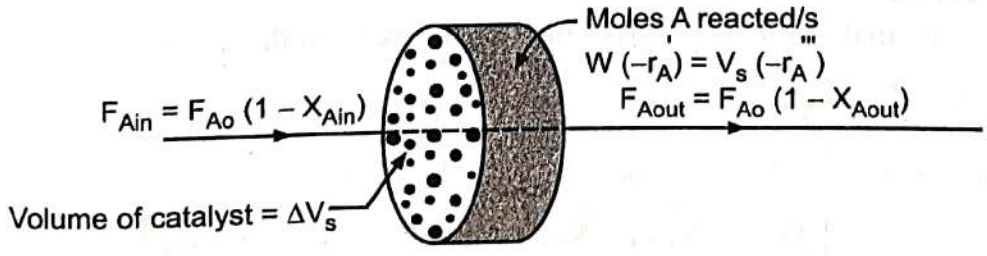


Fig. 6.9 : Elementary slice of PFR containing solid catalyst particles

The steady state material balance for reactant A is

$$\text{input of A} = \text{output of A} + \text{disappearance of A by reaction}$$

in symbols

$$F_{Ao}(1 - X_{A\,in}) = F_{Ao}(1 - X_{A\,out}) + (-r'_A)\,\Delta W, \text{ mol A/s}$$

The above material balance in differential form is written as :

$$F_{Ao}\,dX_A = (-r'_A)\,dW$$

We have, $(-r'_A)\,W = (-r'''_A)\,V_S$, V_S – volume of solids

$$\therefore \quad F_{Ao}\,dX_A = (-r'_A)\,dW = (-r'''_A)\,dV_S \qquad \ldots (6.57)$$

For the whole reactor the above equation must be integrated.

Integrating over the reactor as a whole, we get

$$\frac{dW}{F_{Ao}} = \frac{dX_A}{-r'_A} \quad \text{or} \quad \frac{dV_S}{F_{Ao}} = \frac{dX_A}{-r'''_A}$$

$$\therefore \quad \boxed{\frac{W}{F_{Ao}} = \int_0^{X_{A\,out}} \frac{dX_A}{-r'_A} \quad \text{or} \quad \frac{V_S}{F_{Ao}} = \int_0^{X_{A\,out}} \frac{dX_A}{-r'''_A}} \qquad \ldots (6.58)$$

Equation (6.58) is analogous (similar) to the plug flow performance equation for homogeneous reactions. To bring this analogy closer let

$$\frac{W\,C_{Ao}}{F_{Ao}} = \tau', \text{ (kg cat·s)/m}^3 \qquad \ldots (6.59)$$

and

$$\frac{V_S\,C_{Ao}}{F_{Ao}} = \tau''', \quad (\text{m}^3 \text{ cat·s})/\text{m}^3 \qquad \ldots (6.60)$$

By analogy to the space time for homogeneous systems, we may call τ' as the weight-time and τ''' as the volume-time. Therefore, the integrated plug flow performance equation for first order catalytic reactions with $\varepsilon_A \neq 0$ analogous to that for homogeneous reactions is

$$\boxed{k'\tau' = k'''\tau''' = (1 + \varepsilon_A)\ln\left[\frac{1}{1 - X_{Aout}}\right] - \varepsilon_A X_{Aout}} \qquad \ldots (6.61)$$

Mixed Flow Reactor :

The steady state material balance over the reactor as a whole is

$$F_{Ao}(1 - X_{A\,in}) = F_{Ao}(1 - X_{A\,out}) + (-r'_{Aout})\,W$$

$$F_{Ao}(X_{A\,out} - X_{A\,in}) = (-r'_{Aout})\,W = (-r'''_{Aout})\,V_S$$

$$\therefore \quad \boxed{\frac{W}{F_{Ao}} = \frac{X_{A\,out} - X_{A\,in}}{r'_{A\,out}} \quad \text{or} \quad \frac{V_S}{F_{Ao}} = \frac{X_{A\,out} - X_{A\,in}}{(-r'''_{A\,out})}} \qquad \ldots (6.62)$$

For first order catalytic reactions with $C_{A\,in} = C_{Ao}$ and $\varepsilon_A \neq 0$, the performance equation given by Equation (6.62) becomes

$$\boxed{k'\tau' = k'''\tau''' = \frac{X_{A\,out}(1 + \varepsilon_A X_{A\,out})}{(1 - X_{A\,out})}} \quad \ldots (6.63)$$

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out} - X_{A\,in}}{-r'_{A\,out}}$$

With $C_{A\,in} = C_{Ao}$, $X_{A\,in} = 0$

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out}}{-r'_{A\,out}}$$

We have : $-r'_{A\,out} = k' C_{A\,out} = k' C_{Ao}(1 - X_{A\,out})/(1 + \varepsilon_A X_{A\,out})$

$\therefore$ $$\frac{W}{F_{Ao}} = \frac{X_{A\,out}(1 + \varepsilon_A X_{A\,out})}{k' C_{Ao}(1 - X_{A\,out})}$$

$$\frac{k' C_{Ao} W}{F_{Ao}} = k'\tau' = \frac{X_{A\,out}(1 + \varepsilon_A X_{A\,out})}{(1 - X_{A\,out})}$$

Reactor containing a Batch of Catalyst and a Batch of Gas :

The material balance of A is

input of A = output of A + rate of disappearance of A by reaction + rate of accumulation of A within the reactor.

Here input of A = output of A = 0

$\therefore$ (rate of disappearance of A by reaction) = – (rate of accumulation of A in the reactor)

$$-r_A V = -\frac{dN_A}{dt} = \frac{d[N_{Ao}(1 - X_A)]}{dt} = N_{Ao}\frac{dX_A}{dt}$$

$$\int_0^t dt = N_{Ao}\int_0^{X_A} \frac{dX_A}{(-r_A)\,V}$$

$$t = C_{Ao}\int_0^{X_A} \frac{dX_A}{(-r_A)}$$

$$\frac{t}{C_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r_A}$$

We have : $(-r_A)\,V = (-r'_A)\,W = (-r'''_A)\,V_s$

$\therefore$ $$\frac{t}{C_{Ao}} = \int_0^{X_A} \frac{dX_A}{(-r'_A)\,W/V} \text{ or } \frac{t}{C_{Ao}} = \int_0^{X_A} \frac{dX_A}{(-r'''_A)\,V_s/V}$$

$$\boxed{\frac{t}{C_{Ao}} = \frac{V}{W}\int_0^{X_A} \frac{dX_A}{-r'_A} \text{ or } \frac{t}{C_{Ao}} = \frac{V}{V_s}\int_0^{X_A} \frac{dX_A}{-r'''_A}, \ (m^3 \cdot s)/mol} \quad \ldots (6.64)$$

Experimental Methods for Determining Rates :

The following experimental reactors are used to obtain kinetic data of catalytic reactions. This type of data are required to determine the rate law parameters, reaction order and rate constant, i.e., to obtain the complete rate equation for a given catalytic reaction under consideration. Here the experimental strategy is to measure the extent of conversion of gas passing in steady flow through a batch of solid catalyst particles. In these reactions only one fluid phase in present and hence the rates can be found by a way similar to that adopted for homogeneous reactions.

Now we will deal with the following experimental reactors.

(i) Differential (flow) reactor.

(ii) Integral (plug flow) reactor.

(iii) Mixed flow reactor.

(iv) Batch reactor – batch of catalyst and batch of fluid.

In these reactions, only one fluid phase is present.

Differential Reactor :

A differential reactor is normally used to determine the rate of reaction as a function of initial or entering concentration of reactant. It consists of a tube containing a very small amount of catalyst, in the form of thin disk. Here the change in reactant concentration through the bed of catalyst is extremely small and the reaction rate is considered to be constant at all points within the reactor.

The performance equation for a differential reactor is

$$\boxed{\frac{W}{F_{Ao}} = \frac{X_{A\,out} - X_{A\,in}}{(-r_A')_{avg}}} \qquad \ldots(6.65)$$

from which the average rate is

$$(-r_A') = \frac{X_{A\,out} - X_{in}}{W/F_{Ao}} = \frac{F_{Ao}(X_{A\,out} - X_{A\,in})}{W} = \frac{F_{A\,in} - F_{A\,out}}{W} \qquad \ldots(6.66)$$

The average rate for each run - the rate at the average concentration in the reactor is obtained by using Equation (6.66). A series of runs gives a set of rate-concentration data which can then be analysed to obtain the rate law/rate equation.

The average concentration of reactant A within the reactor is given by

$$C_{A\,avg.} = \frac{C_{A\,in} + C_{A\,out}}{2}$$

Step-wise procedure to obtain a rate equation :

(i) Conduct a series of runs using different $C_{A\,in}$.

(ii) Treat the highest $C_{A\,in}$ as C_{Ao} and use it as the basis for calculating F_{Ao}, $X_{A\,in}$ and $X_{A\,out}$.

(iii) Calculate F_{Ao}, W, $X_{A\,in}$, $X_{A\,out}$ and $C_{A\,avg.}$ for each run.

(iv) Using Equation (6.66) calculate the reaction rate for each run.

(v) Tabulate rate $(-r'_A)$ – concentration $(C_{A\,avg.})$ data.

(vi) Apply the differential method of analysis of rate data to obtain a rate equation (i.e., to obtain kinetic/rate law parameters - order of reaction and rate constant).

Integral (Packed-bed) Reactor :

Here the variation in reaction rate within a reactor is large owing to a significant change in the concentration of reactant along a flow path through the reactor. In this case we may adopt either integral or differential-method of analysis :

Integral analysis : This method is suited for testing simpler rate expressions. Here we guess a rate equation and put it to test by integrating the plug flow performance equation.

(i) Conduct a series of runs in in this experimental factor with fixed feed (C_{Ao}) but with changing W and/or F_{Ao} so as to cover a wide range of W/F_{Ao} and $X_{A\,out}$.

(ii) Guess a rate equation to be tested and integrate the plug flow performance equation after inserting the rate expression in it.

$$\boxed{\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r'_A}} \qquad \ldots (6.67)$$

(iii) For each run evaluate numerically both sides of the above equation.

(iv) Plot LHS v/s RHS of the above equation and test for linearity.

Differential analysis :

It is used in more complicated situations, i.e., when rate expressions are complicated.

By differentiating Equation (6.67), we get

$$-r'_A = \frac{dX_A}{dW/F_{Ao}} = \frac{dX_A}{d(W/F_{Ao})} \qquad \ldots (6.68)$$

Step-wise procedure is as follows :

(i) Take a series of runs in an experimental packed bed reactor with fixed feed (C_{Ao}) but with varying W and/or F_{Ao}.

(ii) Plot $X_{A\,out}$ v/s W/F_{Ao} for all runs.

(iii) Draw a smooth curve passing through the origin.

(iv) From Equation (6.68) it is clear that the rate of reaction at any value X_A is simply the slope of the tangent to this curve at X_A. Therefore, find the slope of this curve (i.e., the rate of reaction) by drawing tangents to this curve at a number of X_A values and also find C_A corresponding to each X_A.

(v) Tabulate rate - concentration data.

(vi) Test n^{th}-order rate form ($-r'_A = k' C_A^n$) by taking logarithms of the rate equation to obtain the rate equation parameters k and n (i.e., to obtain the rate equation to represent the rate concentration data).

Mixed Flow Reactor :

In a mixed flow reactor, the composition of fluid is uniform throughout. With gas-solid systems this ideal is closely approached in a basket-type mixed flow reactor, which consists of four rapidly spinning wire baskets containing catalyst pellets.

The performance equation for a mixed flow reactor is

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out}}{-r'_{A\,out}} \qquad \ldots (6.69)$$

Rearranging gives

$$-r'_{A\,out} = \frac{F_{Ao} X_{A\,out}}{W} \qquad \ldots (6.70)$$

For each run calculate the rate at the exit fluid composition (C_{Aout}) by using Equation (6.70).

Tabulate rate-concentration data and test directly for n^{th}-order kinetics. For this take the logarithm of both sides of $-r'_A = k' C_A^n$ and construct a plot of log $(-r'_A)$ v/s log C_A, i.e., log $C_{A\,out}$.

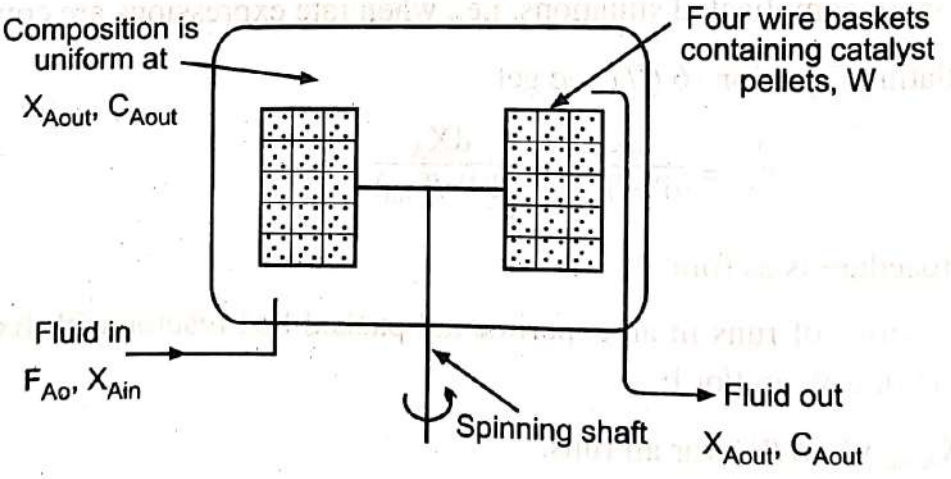


Fig. 6.10 : (Carberry) Basket-type experimental mixed flow reactor

Recycle Reactor :

When recycle ratio is very large, mixed flow is approached. So in such cases, the methods of the mixed flow reactor-direct evaluation of rate for each run - can be used.

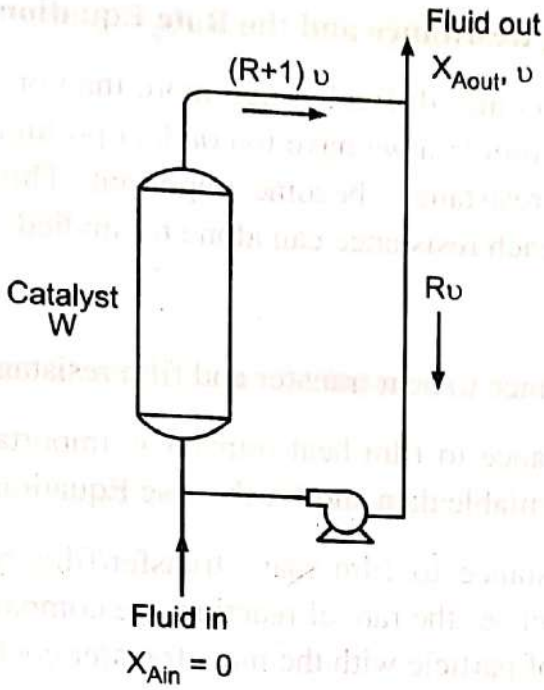


Fig. 6.11 : Experimental recycle reactor

Batch Reactor :

Fig. 6.12 shows an experimental batch reactor. This experimental reactor uses a batch of catalyst and a batch of fluid. In this reactor we have to measure the composition of fluid at different times during the course of reaction and interpret the data collected using the performance equation of a batch reactor.

$$\frac{t}{C_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r_A} = \frac{V}{W} \int_0^{X_A} \frac{dX_A}{-r_A'} \qquad \ldots (6.71)$$

where V is the volume of gas and $(-r_A)\ V = (-r_A')\ W$.

Here the procedure is the same as that used with the homogeneous batch reactor. In order to obtain meaningful results, the composition of fluid must be uniform throughout the system at any instant of time and this necessitates a small per pass conversion across the catalyst bed.

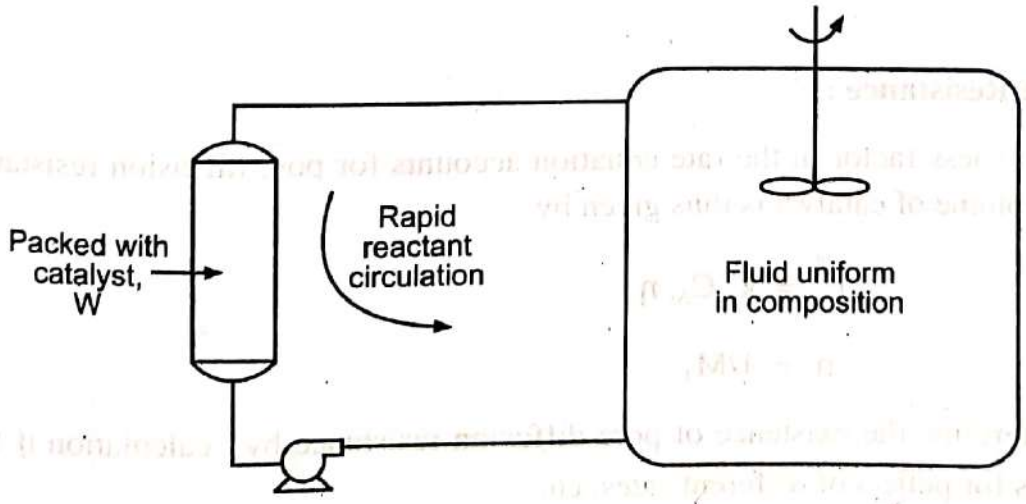


Fig. 6.12 : Batch reactor for catalytic reactions

Determination of Controlling Resistance and the Rate Equation :

Analysis of kinetic data becomes difficult when more than one resistance affects the rate of reaction. In order to avoid this problem we have to conduct preliminary runs to find the limits of operations where the various resistances become important. This will permit us to select the operating conditions in which each resistance can alone be studied.

Film Diffusion Resistance :

Here we have : film resistance to heat transfer and film resistance to mass transfer.

To find whether the resistance to film heat transfer is important, i.e., whether it influences the rate of reaction from the available data and for this use Equation (6.54).

To find whether the resistance to film mass transfer/film resistance to mass transfer is important, i.e., whether it influences the rate of reaction, we compare the observed first order rate constant based on the volume of particle with the mass transfer coefficient.

Using data available, calculate : $k'''_{obs} V_p$ ... observed rate, mole A/s

: $k_g S_{ex}$ ... rate if film resistance controls (film mass transfer rate), mol A/s

For spherical particle : $V_p = 4/3\ \pi R^3 = \pi/6\ d_p^3$, $S_{ex} = 4\pi R^2 = \pi d_p^2$

If $k'''_{obs} V_p$ is much smaller than $k_g S_{ex}$ then the resistance to film mass transfer does not affect/influence the rate of reaction. Consequently, we ignore the resistance to mass transfer through the gás film.

If the above two terms are of the same order of magnitude we may presume that the film resistance affects the rate of reaction.

With porous catalysts, the film mass transfer resistance does not influence the rate of reaction.

Pore Diffusion Resistance :

The effectiveness factor in the rate equation accounts for pore diffusion resistance. The rate based on unit volume of catalyst is thus given by

$$-r'''_A = k''' C_{As} \eta$$

where $\eta = 1/M_T$

We can determine the existence of pore diffusion resistance by : calculation if $\mathscr{D}_e$ is known, comparing rates for pellets of different sizes, etc.

SOLVED EXAMPLES

Example 6.1 : *A solid catalysed first order reaction A → R takes place with 55% conversion in a basket type mixed reactor. Find the conversion if the reactor size is trebled-all else, i.e., temperature, composition, etc. remains unchanged.*

Solution : $A \rightarrow R, \; -r'_A = k'C_A$

The performance equation for basket type mixed flow reactor is

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out}}{-r'_{A\,out}} = \frac{X_{A\,out}}{k'C_{A\,out}}$$

For $A \rightarrow R : \varepsilon_A = 0$

$$\therefore \quad C_{A\,out} = C_{Ao}(1 - X_{A\,out})$$

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out}}{k'C_{Ao}(1 - X_{A\,out})}$$

Case – I : For original reactor, let $W = W_1$

$$X_A = X_{A\,out} = \frac{55}{100} = 0.55$$

$$\frac{W_1}{F_{Ao}} = \frac{0.55}{k'C_{Ao}(1 - 0.55)} = \frac{1.22}{k'C_{Ao}}$$

$$W_1 = \frac{1.22\,F_{Ao}}{k'C_{Ao}}$$

Case-II : For new reactor :

Now we have to find out the conversion in the **new reactor.**

Given : The reactor size is trebled (three times as much as) - all else remains unchanged.

Size of new reactor = 3 × size of the original reactor

$$W = 3\,W_1$$

So for new reactor :

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out}}{k'\,C_{Ao}(1 - X_{A\,out})}$$

$$W = 3\,W_1$$

$$\frac{3W_1}{F_{Ao}} = \frac{X_{A\,out}}{k'C_{Ao}(1 - X_{A\,out})}$$

We have, $W_1 = \dfrac{1.22\, F_{Ao}}{k' C_{Ao}}$

$\therefore \quad \dfrac{3 \times 1.22\, F_{Ao}}{k' C_{Ao} \cdot F_{Ao}} = \dfrac{X_{A\,out}}{k' C_{Ao}\,(1 - X_{A\,out})}$

$\therefore \quad 3.66 = \dfrac{X_{A\,out}}{1 - X_{A\,out}}$

$3.66\,(1 - X_{A\,out}) = X_{A\,out}$

$\therefore \quad X_{A\,out} = 0.7854$ or 78.54%

$\therefore$ **Conversion of A in the reactor of size three times as large (New reactor) = 0.7854 or 78.54%** ... Ans.

Example 6.2 : *The results of the kinetic runs on the reaction A → R made in an experimental packed bed reactor using a fixed feed rate F_{Ao} = 10 kmol/h are as follows :*

W, kg catalyst	*1*	*2*	*3*	*4*	*5*	*6*	*7*
X_A	*0.12*	*0.20*	*0.27*	*0.33*	*0.37*	*0.41*	*0.44*

(i) Find the reaction rate at 40% conversion.

(ii) For a feed rate of 400 kmol/h to large scale packed bed reactor, find the amount of catalyst needed for 40% conversion.

(iii) Find the amount of catalyst that would be required in part (ii) if the reactor employed a very large recycle of product stream.

Solution : $A \rightarrow R$

(i) Finding the reaction rate at 40% conversion in an experimental packed bed reactor :

The kinetic data provided at fixed feed rate (F_{Ao} = 10 kmol/h) are :

The conversion as a function of the amount of catalyst used.

For a packed bed reactor, the performance equation is

$$\frac{W}{F_{Ao}} = \int \frac{dX_A}{-r_A'}$$

In the differential form, it becomes

$$\frac{dW}{dF_{Ao}} = \frac{dX_A}{-r_A'}$$

$$d\left(\frac{W}{F_{Ao}}\right) = \frac{dX_A}{-r_A'}$$

$$\therefore \quad -r_A' = \frac{dX_A}{d\left(\dfrac{W}{F_{Ao}}\right)}$$

If we construct a plot of X_A v/s $\frac{W}{F_{Ao}}$ from the given kinetic data, then,

$$-r_A' = \frac{dX_A}{d\left(\frac{W}{F_{Ao}}\right)}$$ is the slope of the curve

Therefore, plot X_A v/s $\frac{W}{F_{Ao}}$ and find the slope of the curve at 40% conversion to get the reaction rate $\left(-r_A'\right)$ at 40% conversion.

For $F_{Ao} = 10$ kmol/h

W	1	2	3	4	5	6	7
$\frac{W}{F_{Ao}}$	0.10	0.20	0.30	0.40	0.50	0.60	0.70
X_A	0.12	0.20	0.27	0.33	0.37	0.41	0.44

Draw a tangent to the curve at point P, i.e., at $X_A = 0.40$ and measure the slope.

To have precision in plotting use enlarged scale for y-coordinate

From Fig. Ex 6.2 (b) :

$$-r_A' = \text{slope at point P} = 0.38 \frac{\text{kmol converted}}{\text{kg catalyst·h}}$$

Reaction rate $= -r_A'$ at 40% conversion $= 0.38 \frac{\text{kmol (converted)}}{\text{kg catalyst·h}}$ **... Ans. (i)**

(ii) Determination of the amount of catalyst needed in the packed bed reactor for $F_{Ao} = 400$ kmol/h to achieve 40% conversion :

Refer Fig. Ex 6.2 (b) :

Read x-coordinate of point P, i.e., $\frac{W}{F_{Ao}}$ for $X_A = 0.40$ [40% conversion]

$$\frac{W}{F_{Ao}} = 0.57 \text{ at 40\% conversion}$$

$$F_{Ao} = 400 \text{ kmol/h}$$

$$W = 0.57\, F_{Ao}$$

$$= 0.57 \times 400 = 228 \text{ kg}$$

∴ **Amount of catalyst needed in the large-scale packed bed reactor = 228 kg**

... Ans. (ii)

(iii) Finding the amount of catalyst required in the packed bed reactor employing a very large recycle of product stream :

With a very large recycle of product stream (i.e., large recycle ratio) mixed flow is approached. Therefore, here we will use the performance equation for mixed flow.

$$\therefore \quad \frac{W}{F_{Ao}} = \frac{X_A}{-r'_{A\,out}}$$

$$-r'_{A\,out}\ (\text{at } X_A = 0.40) = 0.38\ \frac{\text{kmol}}{\text{kg catalyst}\cdot\text{h}}$$

$$X_A = 0.40$$

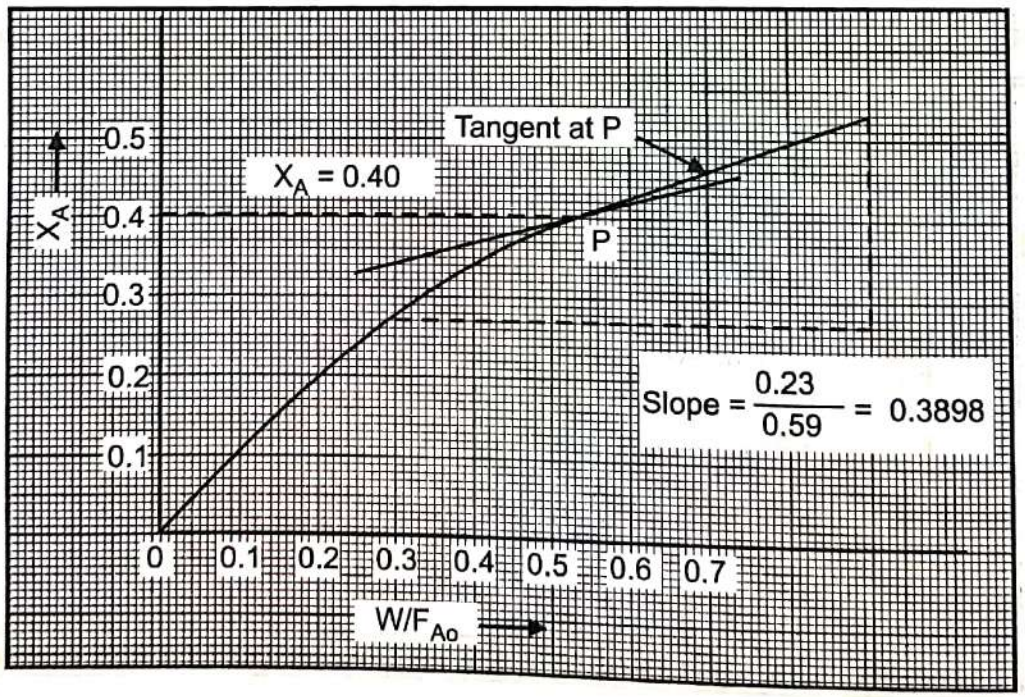
$$F_{Ao} = 400\ \text{kmol/h}$$

$$W = \frac{F_{Ao}\cdot X_A}{-r'_{A\,out}}$$

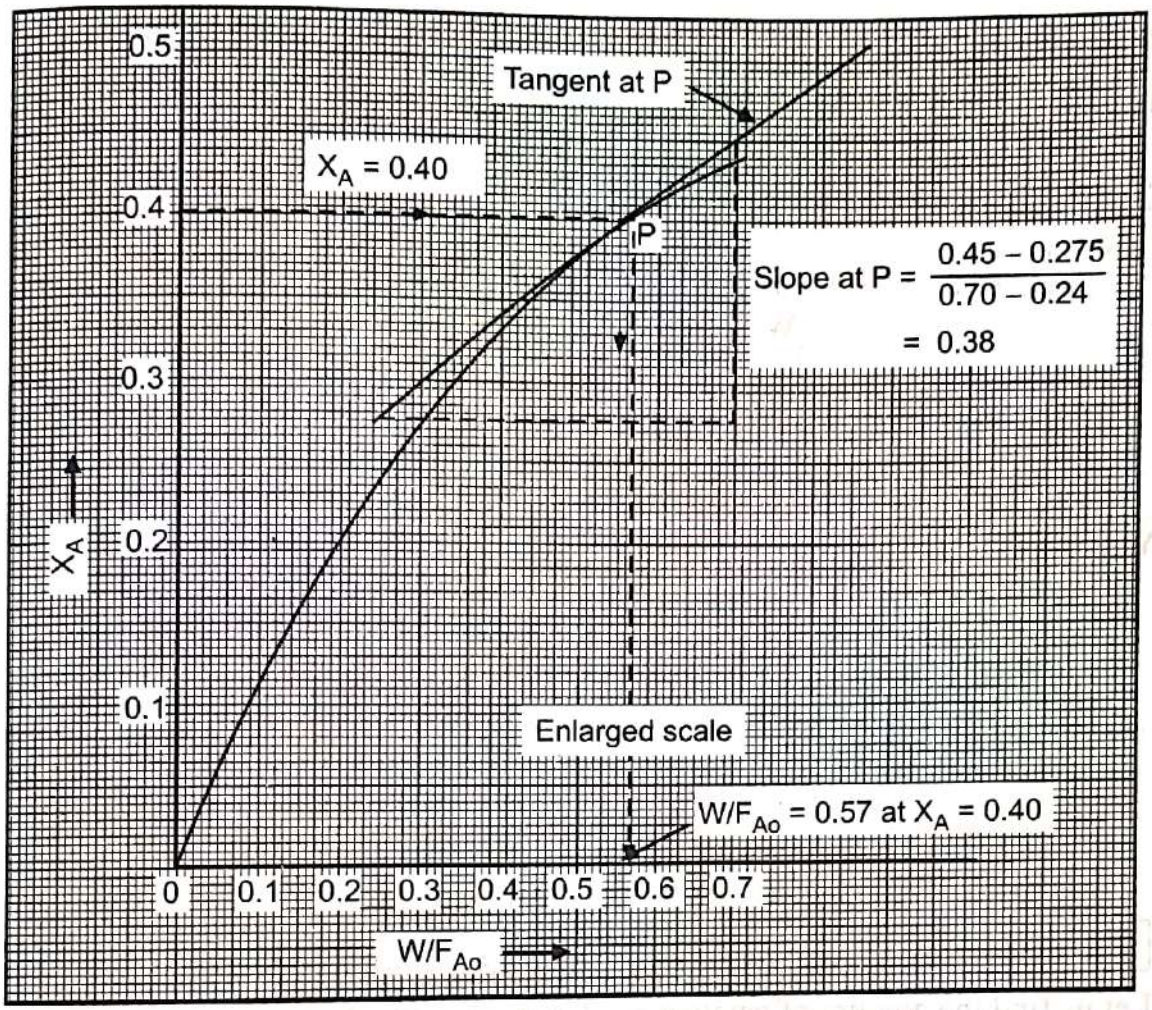
$$= \frac{400 \times 0.40}{0.38}$$

$$= 421\ \text{kg}$$

Amount of catalyst required in the packed bed reactor employing a very large recycle of product stream = 421 kg ... Ans. (iii)



(a)



(b)

Fig. Ex 6.2 (a) and (b)

Example 6.3 : *The second order reaction A → R is studied in an experimental recycle reactor with very large recycle ratio.*

The data recorded are as follows :

Void volume of reactor = 1 l

Amount of catalyst used = 3 g

Feed to the reactor : v_o = 1 l/h with C_{Ao} = 2 mol/l

Concentration of A in the exit stream from the recycle system ($C_{A\ out}$) = 0.5 mol/l

(i) Find the rate constant for this reaction.

(ii) Calculate the amount of catalyst needed in a packed bed reactor to achieve 80% conversion for feed to the reactor :
v_o = 1000 l/h and C_{Ao} = 1 mol/l

(iii) Repeat part (ii) if the reactor is packed with 1 part catalyst to 3 parts inert solid.

Solution : (i) Let us find the rate constant for the second order reaction A → R

In case of recycle reactor using very large recycle ratio, mixed flow is approached. Therefore, here we will make use of the performance equation of a mixed flow reactor.

$$\frac{W}{F_{Ao}} = \frac{X_{A\ out}}{-r'_{A\ out}}$$

$A \rightarrow R, \quad \varepsilon_A = 0$

$$\therefore \quad C_{A\,out} = C_{Ao}(1 - X_{A\,out})$$

$$\therefore \quad X_{A\,out} = \frac{C_{Ao} - C_{A\,out}}{C_{Ao}}$$

$$-r'_{A\,out} = k' C^2_{A\,out} \quad \ldots\ldots \text{ second order reaction}$$

$$\frac{W}{F_{Ao}} = \frac{(C_{Ao} - C_{A\,out})}{C_{Ao} \cdot k' C^2_{A\,out}}$$

$$k' = \frac{(C_{Ao} - C_{A\,out})\, F_{Ao}}{C_{Ao} \cdot C^2_{A\,out} \cdot W}$$

$W = 3$ g, $C_{Ao} = 2$ mol/l

$$C_{A\,out} = 0.5 \text{ mol/}l$$

$$v_o = 1 \text{ } l\text{/h}$$

$$F_{Ao} = C_{Ao} \cdot v_o = 2 \times 1 = 2 \text{ mol/h}$$

$$k' = \frac{(2 - 0.50) \times 2}{2 \times (0.5)^2 \times 3} \; \frac{(\text{mol/}l)\,(\text{mol/h})}{(\text{mol/}l)\,(\text{mol/}l)^2\,(\text{g cat})}$$

$$= 2\,(\text{mol/h})\,(\text{mol/h})^{-2}\,(\text{g cat})^{-1}$$

$$= 2\ l^2 / (\text{g cat. mol. h})$$

The second order rate constant is 2 l^2 / (g cat. mol. h) ... Ans. (i)

(ii) Let us find the amount of catalyst needed in a packed bed reactor :

$v_o = 1000$ l/h, $C_{Ao} = 1$ mol/h, $X_A = \frac{80}{100} = 0.80$

$F_{Ao} = \dot{v}_o \,.\, C_{Ao} = 1000 \times 1 = 1000$ mol/h.

For a packed bed reactor, the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{\left(-r'_A\right)} = \int_0^{X_A} \frac{dX_A}{k' C^2_{Ao}\,(1 - X_A)^2}$$

$$\frac{W}{F_{Ao}} = \frac{1}{k' C^2_{Ao}} \left[\frac{X_A}{1 - X_A}\right]$$

$$W = \frac{F_{Ao}}{k' C^2_{Ao}} \left[\frac{X_A}{1 - X_A}\right]$$

$$W = \frac{1000 \times 0.80}{2 \times (1)^2\,(1 - 0.80)} = 2000 \text{ g} = 2 \text{ kg}$$

Amount of catalyst needed = 2 kg ... Ans. (ii)

(iii) Let us find the amount of catalytic material needed in the packed bed which contains 1 part catalyst with 3 parts inert solid.

For every 1 kg of catalyst the total weight of material (i.e., catalyst + inert solid) in the packed bed reactor is 4 kg (1 kg + 3 kg).

Amount of catalyst from part (ii) = 2 kg.

∴ Total weight of the material containing catalyst plus inert solid $= \frac{4}{1} \times 2 = 8$ kg.

Total weight of the material containing catalyst plus inert solid = 8 kg **... Ans. (iii)**

Example 6.4 : *1 m^3/h of a gas containing A (C_{Ao} = 2 mol/m^3) is fed to a plug flow reactor with recycle loop (0.02 m^3 loop volume, 3 kg of catalyst) and the composition of the exit stream from the reactor system is 0.5 mol A/m^3.*

Find the rate equation for the decomposition of A for :

1. *Very large recycle, A → 3R, n = 2 and feed contains 50% A and 50% inerts.*
2. *Very large recycle, A → 3R, n = 1, feed contains 50% A and 50% inerts.*
3. *No recycle, A → 3R, n = 2, feed contains 25% A and 75% inerts.*
4. *Very large recycle, A → R, $n = \frac{1}{2}$*

Solution : 1. Very large recycle, A → 3R, n = 2 and feed contains 50% A and 50% inerts.

Given : v_o = 1 m^3/h, C_{Ao} = 2 mol/m^3, W = 3 kg, C_A at exit = 0.5 mol/m^3, A → 3R.

With 50% and 50% inerts, two volumes of reactant mixture on complete conversion yield four volumes (3 moles of R and 1 mole of inert) of product mixture. The fractional change in volume (ε_A) is

$$\varepsilon_A = \frac{4-2}{2} = 1$$

OR : Basis : 100 mol feed.

It contains 50 mol A and 50 mol inerts.

Moles at no conversion = initial moles = 100 mol

Moles of R produced = (3/1) × 50 = 150 mol

Moles of complete conversion = 0 mol A + 150 mol R + 50 mol inerts

= 200 mol

$$\therefore \quad \varepsilon_A = \frac{200-100}{100} = 1$$

For changing density systems, C_A and X_A are related by

$$\frac{C_A}{C_{Ao}} = \frac{1-X_A}{1+\varepsilon_A X_A} = \frac{1-X_A}{1+X_A}, \text{ with } \varepsilon_A = 1$$

or

$$X_A = \frac{1-\frac{C_A}{C_{Ao}}}{1+\varepsilon_A \frac{C_A}{C_{Ao}}}$$

with $\varepsilon_A = 1$

$$X_A = \frac{1-\frac{C_A}{C_{Ao}}}{1+\frac{C_A}{C_{Ao}}}$$

$F_{Ao} = v_o.C_{Ao} = 1 \times 2 = 2$ mol/h

$C_{A\,out} = 0.5$ mol/m^3

With very large recycle, mixed flow is approached. Therefore, the performance equation of a mixed flow reactor is to be used.

$$\frac{W}{F_{Ao}} = \frac{X_{out}}{-r'_{A\,out}}$$

$-r'_{A\,out} = k' C^2_{A\,out}$ ..... second order decomposition

$$\frac{W}{F_{Ao}} = \left(\frac{1-\frac{C_{A\,out}}{C_{Ao}}}{1+\frac{C_{A\,out}}{C_{Ao}}}\right) \times \frac{1}{k' C^2_{A\,out}}$$

$$k' = \left(\frac{1-\frac{C_{A\,out}}{C_{Ao}}}{1+\frac{C_{A\,out}}{C_{Ao}}}\right) \cdot \frac{F_{Ao}}{W \times C^2_{A\,out}}$$

$$= \left(\frac{1-\frac{0.5}{2}}{1+\frac{0.5}{2}}\right) \times \frac{(2\ \text{mol/h})}{(3\ \text{kg cat})\ (0.5\ \text{mol/m}^3)^2}$$

$$= 1.6\ (\text{mol/h})\ (\text{kg cat.})^{-1}\ (\text{mol/m}^3)^{-2}$$

$$= 1.6\ \text{m}^6/(\text{mol.h.kg cat})$$

The rate equation for second order decomposition of A is

$$-r'_A = \left(1.6 \frac{\text{m}^6}{\text{mol.h.kg cat}}\right) \left(C_A \frac{\text{mol}}{\text{m}^3}\right)^2$$ ... Ans. (1)

Units of $-r'_A$ are $\Rightarrow \frac{\text{m}^6}{\text{mol.h.kg cat}} \times \frac{\text{mol}^2}{\text{m}^6}$

$\Rightarrow$ mol/(h. kg cat.)

$$-r'_A = -\frac{1}{W}\frac{dN_A}{dt} \Rightarrow \left(\frac{1}{\text{kg cat.}}\right)\left(\frac{\text{mol}}{\text{h}}\right) \Rightarrow \frac{\text{mol}}{(\text{h. kg cat})}$$

2. Very large recycle, $A \rightarrow 3R$, $n = 1$ and feed contains 50% A and 50% inerts. Here $\varepsilon_A = 1$ (obtained previously for same feed), $F_{Ao} = 2$ mol/h, $W = 3$ kg, $C_{A\,out} = 0.5$ mol/m^3, $C_{Ao} = 2$ mol/m^3

With large recycle, use performance equation of a mixed flow reactor.

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out}}{-r'_{A\,out}}, \quad -r'_{A\,out} = k' C_{A\,out} \text{ ... first order decomposition of A}$$

$$\frac{W}{F_{Ao}} = \left(\frac{1-\frac{C_{A\,out}}{C_{Ao}}}{1+\frac{C_{A\,out}}{C_{Ao}}}\right) \times \frac{1}{k' C_{A\,out}}$$

$$k' = \left(\frac{1-\frac{C_{A\,out}}{C_{Ao}}}{1+\frac{C_{A\,out}}{C_{Ao}}}\right)\frac{F_{Ao}}{W\,C_{Aout}} = \left(\frac{1-\frac{0.5}{2}}{1+\frac{0.5}{2}}\right)\times\frac{2\ (mol/h)}{0.5\ (mol/m^3)\times 3\ kg\ cat}$$

$$k' = 0.8\ m^3/(h.kg\ cat)$$

$$\therefore \quad \boxed{-r_A' = -\frac{1}{W}\frac{dN_A}{dt} = \left(0.8\frac{m^3}{h.\ kg\ cat}\right)\left(C_A\frac{mol}{m^3}\right)} \quad \text{... Ans. (2)}$$

3. No recycle, $A \rightarrow 3R$, $n = 2$, feed contains 25% A and 75% inerts.
With 25% A and 75% inerts, 4 volumes, i.e., 4 moles of feed mixture on complete conversion yield 6 volumes, i.e., 6 moles of product mixture.

Therefore, $\varepsilon_A = \frac{6-4}{4} = 0.50$

OR : Basis : 100 mol feed.

It contains 25 mol A + 75 mol inerts.

Initial moles (moles at no conversion) = 25 mol A + 75 mol inerts = 100 moi

Moles of R produced = (3/1) × 25 = 75 mol

Moles at complete conversion = 0 mol A + 75 mol R = 75 mol inerts = 150 mol

Therefore, $\varepsilon_A = \frac{150-100}{100} = 0.50$

For changing density systems, C_A and X_A are related by

$$C_A = C_{Ao}\left(\frac{1-X_A}{1+\varepsilon_A X_A}\right)$$

$$C_A = C_{Ao}\left(\frac{1-X_A}{1+0.5\,X_A}\right) \quad \text{... with } \varepsilon_A = 0.50$$

With no recycle we have a plug flow reactor.

The performance equation for a plug flow reactor is

$$\frac{W}{F_{Ao}} = \int_0^{X_A}\frac{dX_A}{-r_A'}$$

$$-r_A' = k'C_A^2 = k'C^2\left(\frac{1-X_A}{1+0.5\,X_A}\right)^2$$

$$\frac{W}{F_{Ao}} = \frac{1}{k'C_{Ao}^2}\int_0^{X_A}\frac{dX_A}{\left(\frac{1-X_A}{1+0.5\,X_A}\right)^2} = \frac{1}{k'C_{Ao}^2}\int_0^{X_A}\frac{(1+0.5\,X_A)^2}{(1-X_A)^2}\,dX_A$$

$$\int_0^{X_A} \frac{(1+0.5\,X_A)^2}{(1-X_A)^2}\,dX_A = \int_0^{X_A} \frac{\left(1+X_A+0.25\,X_A^2\right)}{(1-X_A)^2}\,dX_A$$

$$= \int_0^{X_A} \frac{dX_A}{(1-X_A)^2} + \int_0^{X_A} \frac{X_A\,dX_A}{(1-X_A)^2} + 0.25\int_0^{X_A} \frac{X_A^2\,dX_A}{(1-X_A)^2}$$

(I) (II) (III)

$$I = \int_0^{X_A} \frac{dX_A}{(1-X_A)^2} = \frac{X_A}{(1-X_A)}$$

$$II = \int_0^{X_A} \frac{X_A\,dX_A}{(1-X_A)^2} = \frac{X_A}{1-X_A} + ln\,(1-X_A)$$

$$III = 0.25\int_0^{X_A} \frac{X_A^2\,dX_A}{(1-X_A)^2} = \frac{0.25\,X_A}{(1-X_A)} + 0.50\,ln\,(1-X_A) + 0.25\,X_A$$

$$I + II + III = \frac{2.25\,X_A}{(1-X_A)} + 1.50\,ln\,(1-X_A) + 0.25\,X_A$$

$$\therefore \quad \frac{W}{F_{Ao}} = \frac{1}{k'C_{Ao}^2}\left[\frac{2.25\,X_A}{(1-X_A)} + 1.50\,ln\,(1-X_A) + 0.25\,X_A\right]$$

We have : $C_{Ao} = 2$ mol/m^3, $C_A = 0.5$ mol/m^3

$$X_A = \frac{1-\frac{C_A}{C_{Ao}}}{1+\varepsilon_A\frac{C_A}{C_{Ao}}}, \quad \varepsilon_A = 0.5 = \frac{1-\left(\frac{0.5}{2}\right)}{1+\left(\frac{0.5\times 0.5}{2}\right)} = 0.6666 \simeq 0.67$$

$$\therefore \quad k' = \left[\frac{2.25\,X_A}{(1-X_A)} + 1.5\,ln\,(1-X_A) + 0.25\,X_A\right] \times \frac{F_{Ao}}{C_{Ao}^2 \times W}$$

$$k' = \left[\frac{2.25\times 0.67}{1-0.67} + 1.5\,ln\,(1-0.67) + 0.25\times 0.67\right] \times$$

$$\frac{2\ (mol/h)}{(2)^2\ (mol/m^3)^2 \times 3\ (kg\ cat)}$$

$$= 0.512\ (mol/h)\ (mol/m^3)^{-2}\ (kg\ cat)^{-1}$$

$$= 0.512\ m^6/(mol.h.kg\ cat)$$

Therefore, the rate equation for decomposition of A is

$$\boxed{-r'_A = -\frac{1}{W}\frac{dN_A}{dt} = \left[0.512 \frac{m^6}{(mol.h.kg\ cat)}\right]\left(C_A \frac{mol}{m^3}\right)^2} \quad \text{... Ans. (3)}$$

4. Very large recycle, $A \rightarrow R$, $n = \frac{1}{2}$

For $A \rightarrow R$, $\varepsilon_A = 0$

Here also mixed flow is approached

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out}}{-r'_{A\,out}}$$

$$X_{A\,out} = \frac{C_{Ao} - C_{A\,out}}{C_{Ao}}, \text{ for } \varepsilon_A = 0$$

$$-r'_{A\,out} = k' C_{A\,out}^{1/2} \quad \ldots\ldots\ n = \frac{1}{2} \quad \ldots\ldots \text{ half order decomposition}$$

$$\frac{W}{F_{Ao}} = \left(\frac{C_{Ao} - C_{A\,out}}{C_{Ao}}\right) \times \frac{1}{k' C_{A\,out}^{1/2}}$$

$$k' = \left(\frac{C_{Ao} - C_{A\,out}}{C_{Ao}}\right) \frac{F_{Ao}}{(C_{A\,out})^{1/2} \times W}$$

$C_{Ao} = 2\ mol/m^3$, $C_{A\,out} = 0.50\ mol/m^3$

$F_{Ao} = 2\ mol/h$, $W = 3\ kg$

$$k' = \left(\frac{2 - 0.5}{2}\right) \frac{(mol/m^3)}{(mol/m^3)} \times \frac{2\ (mol/h)}{(0.5)^{1/2}\ (mol/m^3)^{1/2} \times 3\ (kg\ cat)}$$

$$= 0.707\ (mol/h)\ (mol/m^3)^{-1/2}\ (kg\ cat)^{-1}$$

$$= 0.707\ (m^3)^{1/2}\ (mol)^{1/2}/(h.\ kg\ cat)$$

The rate equation is

$$\boxed{-r'_A = -\frac{1}{W}\frac{dN_A}{dt} = \left(0.707 \frac{(m^3.mol)^{1/2}}{(h.\ kg\ cat)}\right)\left(C_A \frac{mol}{m^3}\right)^{1/2}} \quad \text{... Ans. (4)}$$

Example 6.5 : *Calculate the amount of catalyst needed in a packed bed reactor to achieve 90% conversion of 1000 m^3/h of pure gaseous A (C_{Ao} = 100 mol/m^3) for :*

(i) $A \rightarrow R$, $\quad -r'_A = \dfrac{50\ C_A}{1 + 0.02\ C_A} \dfrac{mol}{(h.kg\ cat)}$

(ii) $A \rightarrow R$, $\quad -r'_A = 8\ C_A^2 \dfrac{mol}{(h.kg\ cat)}$

Solution : (i) $A \rightarrow R$ with $-r_A' = \dfrac{50\, C_A}{1 + 0.02\, C_A}\ \dfrac{mol}{h.kg\ cat}$

$C_{Ao} = 100\ mol/m^3,\ X_A = \dfrac{80}{100} = 0.80$

$\nu_o = 1000\ m^3/h$

$F_{Ao} = \nu_o\, C_{Ao} = 1000 \times 100 = 10^5\ mol/h$

For a packed bed reactor (assuming plug flow), the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r_A'}$$

For $A \rightarrow R$, $\varepsilon_A = 0$

The concentration and conversion are related by

$$C_A = C_{Ao}\ (1 - X_A)$$

$$-r_A' = \frac{50\, C_A}{1 + 0.02\, C_A} = \frac{50\, C_{Ao}\,(1 - X_A)}{1 + 0.02\, C_{Ao}\,(1 - X_A)}$$

$$= \frac{50 \times 100\,(1 - X_A)}{1 + 0.02 \times 100\,(1 - X_A)}$$

$$= \frac{5000\,(1 - X_A)}{1 + 2\,(1 - X_A)}$$

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{\dfrac{5000\,(1 - X_A)}{1 + 2\,(1 - X_A)}} = \int_0^{X_A} \frac{1 + 2\,(1 - X_A)\,dX_A}{5000\,[1 - X_A]}$$

$$\int_0^{X_A} \frac{[1 + 2\,(1 - X_A)]\,dX_A}{5000\,(1 - X_A)}$$

$$= \frac{1}{5000}\left[\int_0^{X_A} \frac{dX_A}{1 - X_A} + 2\int_0^{X_A} \frac{(1 - X_A)\,dX_A}{(1 - X_A)}\right]$$

$$= \frac{1}{5000}\left[ln\left(\frac{1}{1 - X_A}\right) + 2\,X_A\right]$$

$$\frac{W}{F_{Ao}} = \frac{1}{5000}\left[ln\left(\frac{1}{1 - X_A}\right) + 2\,X_A\right]$$

$$W = \frac{F_{Ao}}{5000}\left[ln\left(\frac{1}{1-X_A}\right) + 2\,X_A\right]$$

$$= \frac{10^5}{5000}\left[ln\left(\frac{1}{1-0.8}\right) + 2\times 0.80\right]$$

$$= 64.188 \simeq 64.2 \text{ kg}$$

Amount of catalyst needed = 64.2 kg **... Ans. (i)**

(ii) $A \rightarrow R,\ -r'_A = 8\,C_A^2 \frac{mol}{(h.kg\ cat)}$

With $\varepsilon_A = 0 : C_A = C_{Ao}(1 - X_A)$

Assuming plug flow, we have

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r'_A}$$

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{8\,C_A^2} = \frac{1}{8}\int_0^{X_A} \frac{dX_A}{C_{Ao}^2(1-X_A)^2}$$

$$\frac{W}{F_{Ao}} = \frac{1}{8\,C_{Ao}^2}\left[\frac{X_A}{1-X_A}\right]$$

$$W = \frac{F_{Ao}}{8\,C_{Ao}^2}\left[\frac{X_A}{1-X_A}\right]$$

$C_{Ao} = 100$ mol/m^3, $F_{Ao} = 10^5$ mol/h, $X_A = 0.80$

$$W = \frac{10^5}{8\,(100)^2}\left[\frac{0.80}{1-0.80}\right]$$

$$= 5 \text{ kg}$$

Amount of catalyst needed = 5 kg **... Ans. (ii)**

Example 6.6 : *10 m^3/h of a gaseous feed containing A and B passes through an experimental reactor packed with 4 kg of catalyst.*

The stoichiometry and rate are given by:

$$A + B \rightarrow R + S,\ -r'_A = 0.6\,C_A\,C_B \frac{mol}{kg.h}$$

Find the conversion of reactants if the feed to the reactors contains 0.1 mol A/m^3 and 10 mol B/m^3.

Solution : The stoichiometry and rate are given by

$$A + B \rightarrow R + S$$

$$-r'_A = 0.6\, C_A\, C_B$$

Given : $C_{Ao} = 0.1$ mol/m^3, $C_{Bo} = 10$ mol/m^3

$\therefore$ $C_{Ao} \neq C_{Bo}$

Given : ν_o = volumetric feed rate = 10 m^3/h

and W = amount of catalyst = 4 kg

Therefore, $F_{Ao} = \nu_o\, C_{Ao} = 10 \times 0.1 = 1$ mol/h

The amounts of A and B reacted are equal and are given by

$$\frac{C_{Ao}\, X_A}{a} = \frac{C_{Bo}\, X_B}{b}$$

Here $a = b = 1$

$\therefore$ $C_{Ao}\, X_A = C_{Bo}\, X_B$

For $\varepsilon_A = 0$, we have : $C_A = C_{Ao}\,(1 - X_A)$

and $C_B = C_{Bo}\,(1 - X_B)$

$= C_{Bo} - C_{Bo}\, X_B = C_{Bo} - C_{Ao}\, X_A$

Let $M = \dfrac{C_{Bo}}{C_{Ao}}$

then $C_B = C_{Ao}\,(M - X_A)$

Given : $-r'_A = 0.6\, C_A\, C_B$

Therefore, $-r'_A = 0.6\, C_{Ao}\,(1 - X_A)\, C_{Ao}\,(M - X_A)$

$= 0.6\, C_{Ao}^2\,(1 - X_A)\,(M - X_A)$

For a packed bed assuming plug flow, the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r'_A} = \int_0^{X_A} \frac{dX_A}{0.6\, C_{Ao}^2\,(1 - X_A)\,(M - X_A)}$$

$$\frac{0.6\, W\, C_{Ao}^2}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{(1 - X_A)\,(M - X_A)}$$

$$\int_0^{X_A} \frac{dX_A}{(1-X_A)(M-X_A)} = \frac{1}{(M-1)}\left[\int_0^{X_A} \frac{dX_A}{1-X_A} - \int_0^{X_A} \frac{dX_A}{M-X_A}\right]$$

$$= \frac{1}{M-1}\left[ln\left(\frac{1}{1-X_A}\right) + ln\left(\frac{M-X_A}{M}\right)\right]$$

$$\frac{0.60\, W\, C_{Ao}^2}{F_{Ao}} = \frac{1}{M-1}\left[ln\left(\frac{1}{1-X_A}\right) + ln\left(\frac{M-X_A}{M}\right)\right]$$

$$M = \frac{C_{Bo}}{C_{Ao}} = \frac{10}{0.10} = 100$$

$W = 4$ kg, $C_{Ao} = 0.1$ mol/m^3, $F_{Ao} = 1$ mol/h

$$\frac{0.60 \times 4 \times (0.1)^2}{1} = \frac{1}{(100-1)}\left[ln\left(\frac{1}{1-X_A}\right) + ln\left(\frac{100-X_A}{100}\right)\right]$$

$$2.376 = ln\left(\frac{1}{1-X_A}\right) + ln\left(\frac{100-X_A}{100}\right)$$

Use a trial and error procedure to find X_A.

LHS = 2.376

For $X_A = 0.70 \rightarrow$ RHS = 1.19 $X_A = 0.90 \rightarrow$ RHS = 2.29

$X_A = 0.50 \rightarrow$ RHS = 0.688 $X_A = 0.91 \rightarrow$ RHS = 2.398

$X_A = 0.95 \rightarrow$ RHS = 2.986 $X_A = 0.905 \rightarrow$ RHS = 2.345

$X_A = 0.907 \rightarrow$ RHS = 2.366

$X_A = 0.9075 \rightarrow$ RHS = 2.3714

$X_A = 0.908 \rightarrow$ RHS = 2.3768

$X_A = 0.9079 \rightarrow$ RHS = 2.3757 $\simeq$ 2.376 = LHS

LHS = RHS for $X_A = 0.9079$

Conversion of reactant A = 0.9079 × 100 = 90.79% **... Ans.**

We have :

$$C_{Ao}\, X_A = C_{Bo}\, X_B$$

$$0.10\,(0.9079) = 10\, X_B$$

$$X_B = 9.079 \times 10^{-3}$$

Conversion of reactant B = $9.079 \times 10^{-3} \times 100 = 0.9079 \simeq 0.908\%$

Conversion of reactant B = 0.908% **... Ans.**

Example 6.7 : *The following kinetic data on the solid catalysed reaction $A \rightarrow 3R$ are obtained in a basket type of mixed flow reactor 960 cm^3 in volume and containing 1 g of catalyst, by conducting runs at 8 atm and 700 °C using pure A feed.*

Kinetic data : *The partial pressure of A in the exit stream at various feed rates :*

Feed rate, l/h	*100*	*22*	*4*	*1*	*0.60*
$p_{A\,out}/p_{A\,in}$	*0.8*	*0.5*	*0.2*	*0.1*	*0.05*

Find the rate equation for this reaction.

Solution : $A \rightarrow 3R$

Feed to the reactor : pure A

Runs are conducted at 8 atm. and 700 °C

Catalyst used = W = 1 g

$A \rightarrow 3R$

For pure A, 1 volume of feed on complete conversion yields 3 volumes of product mixture. Therefore,

$$\varepsilon_A = \frac{3-1}{1} = 2$$

OR : Basis : 100 mol feed and pure A

Moles at no conversion = 100 mol A = 100 mol

Moles of R produced = $(3/1) \times 100 = 300$ mol

Moles at complete conversion = 0 mol A + 300 mol R = 300 mol

Therefore, $$\varepsilon_A = \frac{300-100}{100} = 2$$

[ε_A – fractional change in volume]

For changing density systems, C_A and X_A are related by

$$X_A = \frac{1 - \frac{C_A}{C_{Ao}}}{1 + \frac{\varepsilon_A C_A}{C_{Ao}}}$$

For a basket type mixed flow reactor, the performance equation is

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out}}{-r'_{A\,out}}$$

$$-r'_{A\,out} = \frac{X_{A\,out} F_{Ao}}{W}$$

$$X_{A\,out} = \frac{1 - \frac{C_{A\,out}}{C_{Ao}}}{1 + \varepsilon_A \frac{C_{A\,out}}{C_{Ao}}} = \frac{1 - \frac{C_{A\,out}}{C_{Ao}}}{1 + 2\frac{C_{A\,out}}{C_{Ao}}}, \quad \text{as } \varepsilon_A = 2$$

Find C_{Ao} : $C_{Ao} = \frac{p_{Ao}}{RT} = \frac{P}{RT}$ (p_{Ao} = P as feed pure A)

$= \frac{8}{0.082096 \times 973}$, $p_{Ao} = 8$ atm., $T = 700 + 273 = 973$ K

$= 0.10$ mol/l. $R = 0.08206$ l.atm/(mol.K)

Convert the data given in a form suitable to use :

Since $C_A \propto p_A$, we get :

$$\frac{p_{A\,out}}{p_{A\,in}} = \frac{C_{A\,out}}{C_{A\,in}} = \frac{C_{A\,out}}{C_{Ao}}, \quad C_{A\,in} = C_{Ao}$$

We know C_{Ao}. Therefore, for each run obtain $C_{A\,out}$.

Sample calculations for Run no. 1 :

$$\frac{p_{A\,out}}{p_{A\,in}} = 0.80 \quad \text{and} \quad \upsilon_o = 100\ l/\text{h}$$

$$\therefore \quad \frac{C_{A\,out}}{C_{Ao}} = \frac{p_{A\,out}}{p_{A\,in}} = 0.80$$

$$C_{A\,out} = 0.8\, C_{Ao} = 0.8 \times 0.1 = 0.08\ \text{mol}/l$$

$$F_{Ao} = \nu_o\, C_{Ao} = 100 \times 0.10 = 10\ \text{mol/h}$$

$$X_{A\,out} = \frac{1 - \frac{C_{A\,out}}{C_{Ao}}}{1 + 2\frac{C_{A\,out}}{C_{Ao}}} = \frac{1 - 0.80}{1 + 2 \times 0.80} = 0.0769$$

$$-r'_{A\,out} = \frac{X_{A\,out}\, F_{Ao}}{W} = \frac{0.0769 \times 10}{1} = 0.769\ \text{mol/(h. g cat)}$$

In this way, convert the data and tabulate :

ν_o	100	22	4	1	0.60
$\frac{p_{A\,out}}{p_{A\,in}} = \frac{C_{A\,out}}{C_{Ao}}$	0.80	0.5	0.2	0.1	0.05
$C_{A\,out} = C_A$	0.08	0.05	0.02	0.01	0.005
$X_{A\,out} = \frac{1 - \frac{C_{A\,out}}{C_{Ao}}}{1 + 2\frac{C_{A\,out}}{C_{Ao}}}$	0.0769	0.25	0.571	0.75	0.864
$F_{Ao} = \nu_o\, C_{Ao}$	10	2.2	0.40	0.10	0.06
$-r'_{A\,out} = -r'_A = \frac{X_{A\,out}\, F_{Ao}}{W}$	0.769	0.55	0.23	0.075	0.052

Now we will find out the rate equation that fits the data provided.

Let us start with the simplest of rate forms, i.e., first order rate equation.

$$-r'_{A\,out} = k' C_{A\,out}$$

i.e., $$-r'_A = k' C_A$$

Plot $-r'_A$ v/s C_A. If we get a straight line passing through the origin, then we say that the data fits the rate equation guessed (i.e., the decomposition of A is first order)

Refer Fig. Ex 6.7 (a).

Plot of $-r'_A$ v/s C_A yields a good straight line through the origin, indicating that the reaction is first order.

Slope = k' = 10 *l*/(h.g cat)

The rate in terms of mol A reacted/(h.g cat) is given by

$$\boxed{-r'_A = -\frac{1}{W}\frac{dN_A}{dt}\left(10\,\frac{l}{\text{h.g cat}}\right)\left(C_A\,\frac{\text{mol}}{l}\right)}$$ ... Ans.

OR :

We can use general n^{th} order rate equation and find k' and n.

$$-r'_A = k' C_A^n$$

$$\ln\left(-r'_A\right) = \ln k' + n \ln C_A$$

C_A	0.08	0.05	0.02	0.01	0.005
$\ln C_A$	– 2.526	– 2.996	– 3.912	– 4.605	– 5.298
$\left(-r'_A\right)$	0.769	0.55	0.23	0.075	0.052
$\ln\left(-r'_A\right)$	– 0.263	– 0.598	– 1.469	– 2.59	– 2.956

Plot $\ln\left(-r'_A\right)$ v/s $\ln C_A$

slope = n and intercept = $\ln k'$

Refer Fig. Ex 6.7 (b) :

slope = n = 1

Intercept = $\ln k'$ = 2.3

$\therefore$ k' = 9.974 *l*/(h.g cat) $\simeq$ 10 (previously obtained)

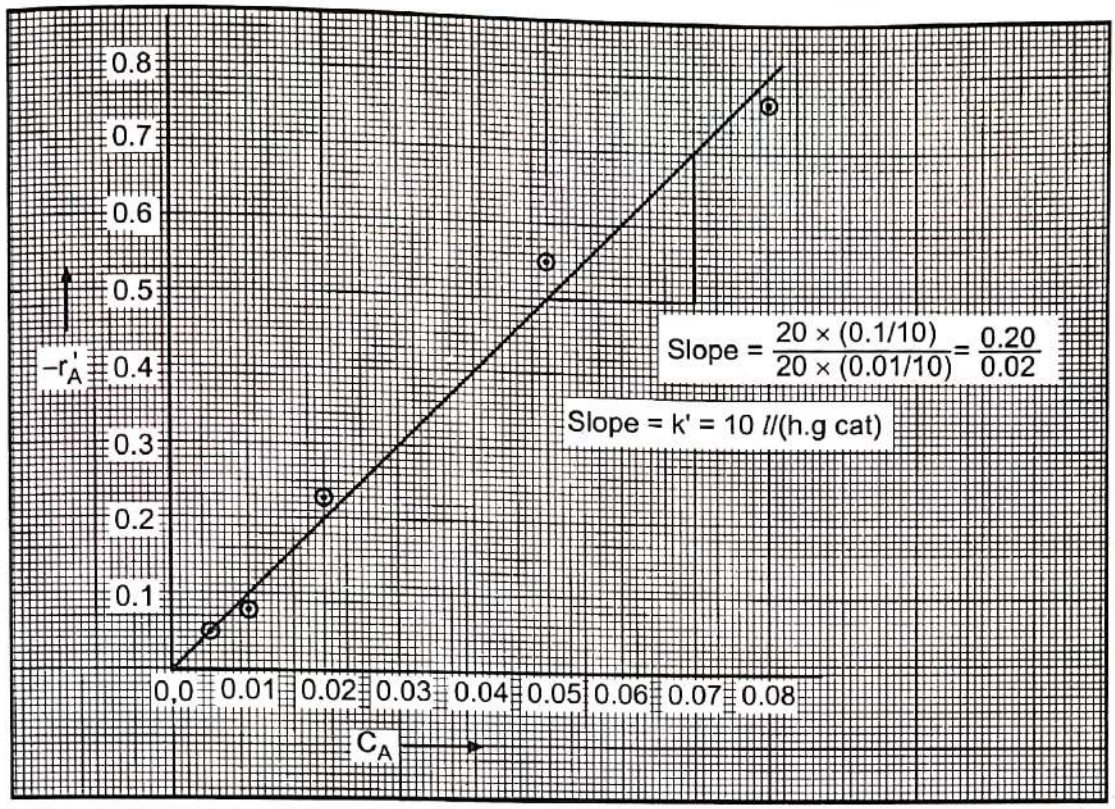
Therefore, we obtain almost same results.

$\therefore$ The reaction is first order with

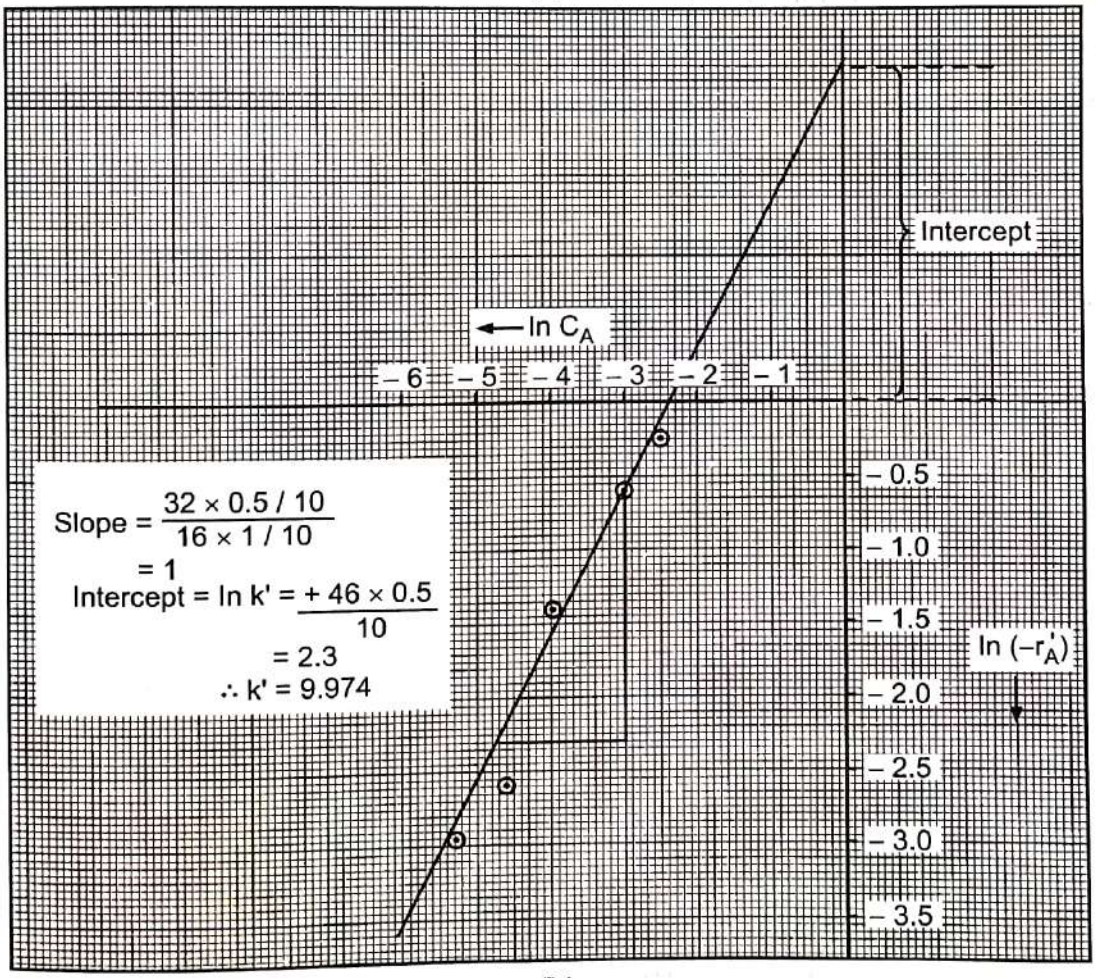
k = 10 *l*/(h.g cat)

The rate equation is

$$\boxed{-r'_A = -\frac{1}{W}\frac{dN_A}{dt} = \left(10\,\frac{l}{\text{h.g cat}}\right)\left(C_A\,\frac{\text{mol}}{l}\right)}$$ ... Ans.



(a)



(b)

Fig. Ex 6.7 (a) and (b)

Example 6.8 : *The following conversion data on the gaseous reaction A → R are obtained in an experimental reactor. Find the rate equation for this reaction.*

1. *Mixed flow, 4 g of catalyst, $C_{Ao} = 10$ mol/m³*

v_o m³/h	3	2	1.2
X_A	0.20	0.30	0.50

2. *Plug flow, $C_{Ao} = 60$ mol/m³, $v_o = 3$ l/min.*

W, g	0.50	1.0	2.5
C_A, mol/m³	30	20	10

Solution :

1. Mixed flow behaviour :

A → R, gas phase reaction

The fractional change in volume is

$$\varepsilon_A = \frac{1-1}{1} = 0$$

One mol A on complete conversion gives 1 mol R. Hence there is no change in volume on reaction and we have a constant density system.

$W = 4$ g, $C_{Ao} = 10$ mol/m³

For a mixed flow reactor, the performance equation is

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out}}{-r'_{A\,out}}$$

Calculate F_{Ao}, $-r'_{A\,out}$ and $C_{A\,out}$ for each run.

Sample calculations for Run no. 1 :

$v_o = 3$ m³/h, $X_A = 0.20$

$$F_{Ao} = v_o\, C_{Ao}$$
$$= 3 \times 10 = 30 \text{ mol/h}$$
$$-r'_{A\,out} = X_{A\,out} \frac{F_{Ao}}{W}$$
$$= 0.20 \times \frac{30}{4}$$
$$= 1.5 \text{ mol/(h.g cat)}$$

For $\varepsilon_A = 0$, C_A and X_A are related by

$$C_A = C_{Ao}(1 - X_A)$$
$$C_{A\,out} = C_{Ao}(1 - X_{A\,out})$$
$$= 10(1 - 0.20)$$
$$= 8 \text{ mol/m}^3$$

Similarly convert the remaining data and tabulate.

ν_o	3	2	1.2
F_{Ao}	30	20	12
$X_A = X_{A\,out}$	0.20	0.30	0.50
$C_A = C_{A\,out}$	8	7	5
$-r'_A = -r'_{A\,out}$	1.5	1.5	1.5

As the rate of reaction $\left(-r'_A\right)$ is constant for all C_A values (i.e., it is independent of concentration of A) the reaction is zero order with respect to A.

$$-r'_A = k' C_A^o = k' = 1.5 \text{ mol/(h.g cat)}$$

The rate equation is :

$$\boxed{-r'_A = \left(1.5 \frac{\text{mol}}{\text{h.g cat}}\right)\left(C_A \frac{\text{mol}}{\text{m}^3}\right)^o}$$ **... Ans. (1)**

2. Plug flow behaviour :

$$C_{Ao} = 60 \text{ mol/m}^3$$

$$\nu_o = 3 \text{ } l\text{/min} = 3 \times 10^{-3} \text{ m}^3\text{/min}$$

$$F_{Ao} = \nu_o C_{Ao}$$

$$= 3 \times 10^{-3} \times 60 = 0.18 \text{ mol/min}$$

For a plug flow reactor, the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r'_A}$$

For $A \rightarrow R$, $\varepsilon_A = 0$

$$C_A = C_{Ao}(1 - X_A) \text{ or } X_A = \frac{C_{Ao} - C_A}{C_{Ao}}$$

Calculate X_A for each run.

Run 1 : $X_A = \dfrac{60 - 30}{60} = 0.50,$ $\dfrac{W}{F_{Ao}} = \dfrac{0.50}{0.18} = 2.78$

Run 2 : $X_A = \dfrac{60 - 20}{60} = 0.67,$ $\dfrac{W}{F_{Ao}} = \dfrac{1}{0.18} = 5.56$

Run 3 : $X_A = \dfrac{60 - 10}{60} = 0.833,$ $\dfrac{W}{F_{Ao}} = \dfrac{2.5}{0.18} = 13.89$

Let us find the rate equation :

Guess-1 : first order reaction

$$-r'_A = k' C_A = k' C_{Ao} (1 - X_A)$$

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r'_A} = \int_0^{X_A} \frac{dX_A}{k' C_{Ao} (1 - X_A)} = \frac{1}{k' C_{Ao}} ln \left(\frac{1}{1 - X_A}\right)$$

$$\frac{W}{F_{Ao}} = \frac{1}{k' C_{Ao}} \left[ln \left(\frac{1}{1 - X_A}\right)\right] \ldots\ldots\ldots y = mx$$

If a plot of $\frac{W}{F_{Ao}}$ v/s $ln \left(\frac{1}{1 - X_A}\right)$ yields a straight line then only we say that our guess is correct, i.e., the reaction is first order.

Run	1	2	3
$\frac{W}{F_{Ao}}$	2.78	5.56	13.89
$ln \left(\frac{1}{1 - X_A}\right)$	0.693	1.108	1.789

Refer Fig. Ex 6.8 (a) :

As we are not getting a straight line passing through the origin, our guess regarding the order is incorrect, i.e., the reaction is not first order (first order rate equation does not fit the data).

Guess : Second order reaction

$$-r'_A = k' C_A^2 = k' C_{Ao}^2 (1 - X_A)$$

$$\frac{W}{F_{Ao}} = \frac{1}{k' C_{Ao}^2} \int_0^{X_A} \frac{dX_A}{(1 - X_A)^2}$$

$$\frac{W}{F_{Ao}} = \frac{1}{k' C_{Ao}^2} \left[\frac{X_A}{1 - X_A}\right] \ldots\ldots\ldots y = mx$$

Plot of $\frac{W}{F_{Ao}}$ v/s $\frac{X_A}{(1 - X_A)}$ should yield a straight line through origin for our guess to be correct.

Run	1	2	3
$\frac{W}{F_{Ao}}$	2.78	5.56	13.89
$\frac{X_A}{1 - X_A}$	1.0	2.03	4.988

Refer to Fig. Ex 6.8 (b) :

As we obtained a straight line passing through the origin, the reaction is second order with spect to A (that is, our guess is correct).

$$\text{slope} = \frac{1}{k' C_{Ao}^2} = 2.773$$

$$k' = \frac{1}{2.773\, C_{Ao}^2}, \quad C_{Ao} = 60 \text{ mol/m}^3$$

$$k' = \frac{1}{2.773 \times (60)^2}$$

$$= 1 \times 10^{-4} \text{ m}^6/(\text{mol. min. g cat})$$

The rate equation is :

$$\boxed{-r_A' = \left(1 \times 10^{-4} \frac{\text{m}^6}{\text{mol.min.g cat}}\right) \left(C_A \frac{\text{mol}}{\text{m}^3}\right)^2}$$ **... Ans. (2)**

Alternatively :

$$\frac{W}{F_{Ao}} = \frac{1}{k' C_{Ao}^2} \left[\frac{X_A}{1 - X_A}\right]$$

$$k' = \frac{F_{Ao}}{W C_{Ao}^2} \left[\frac{X_A}{1 - X_A}\right]$$

Calculate k' for the data of each run.

Run 1 : $k' = \frac{0.18}{0.5 \times (60)^2} \left[\frac{0.50}{1 - 0.50}\right] = 1 \times 10^{-4}$ m^6/(mol. min. g cat)

Run 2 : $k' = \frac{0.18}{1 \times (60)^2} \left[\frac{0.67}{1 - 0.67}\right] = 1.015 \times 10^{-4}$ m^6/(mol. min. g cat)

Run 3 : $k' = \frac{0.18}{2.5\ (60)^2} \left[\frac{0.833}{1 - 0.833}\right] = 9.976 \times 10^{-5}$ m^6/(mol. min. g cat)

As k' values obtained are almost same our guess regarding the order is correct. So A → R is a second order reaction.

$$\text{Avg. } k' = \frac{1 \times 10^{-4} + 1.015 \times 10^{-4} + 9.976 \times 10^{-5}}{3}$$

$$= 1.0042 \times 10^{-4} \approx 1 \times 10^{-4} \text{ m}^6/(\text{mol. min. g cat})$$

k' by both the methods is the same.

The rate equation representing the reaction A → R is

$$\boxed{-r_A' = \left(1 \times 10^{-4} \frac{\text{m}^6}{\text{mol. min. g cat}}\right) \left(C_A \frac{\text{mol}}{\text{m}^3}\right)^2}$$ **... Ans. (2)**

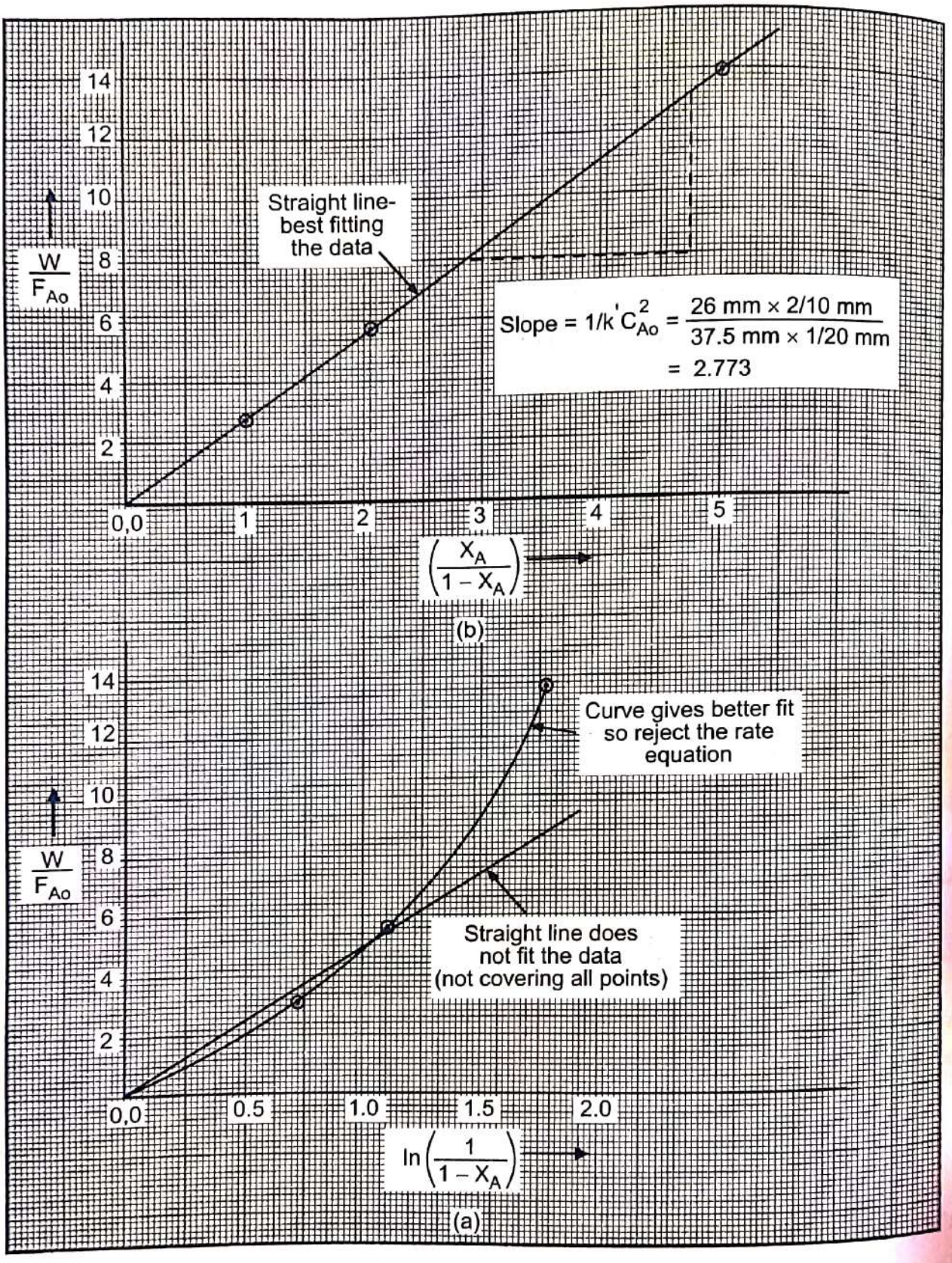


Fig. Ex 6.8 (a) and (b)

Example 6.9 : *At 700 °C, the rate of decomposition of A as per the reaction A → 3R using specific catalyst of size dp = 3 mm is found to be*

$$-r'_A = -\frac{1}{W}\frac{dN_A}{dt} = 10\ [l/(\text{mol. g cat})]\ C_A$$

A tubular reactor of 20 mm ID to be used on a pilot plant scale is packed with – 25% of these active catalyst pellets evenly mixed with 75% of inert pellets. The added inerts ensure isothermal operations.

For a flow rate of 400 mol/h of feed containing 50% A and 50% inert gas at 8 atm and 700 °C, what must be the length of reactor so that the ratio of partial pressure of A in the exit stream to partial pressure of A in the inlet (i.e., $p_{A\,out}/p_{A\,in}$) is 0.111.

Data : Catalyst and inert pellets are porous and of the same size.
Particle density $\rho_s = 2\ g/cm^3$. Bulk voidage of packed bed = 50%.

Solution : A → 3 R ..... first order catalytic gas phase decomposition of A.

$$-r'_A = 10\ l/(\text{mol. g cat})\ C_A$$

Feed to the reactor contains 50% A and 50% inert gas.

Feed flow rate (total) = $F_o = 400$ mol/h

$$\therefore \quad F_{Ao} = \frac{50}{100} \times F_o = 0.50 \times 400 = 200 \text{ mol/h}$$

With 50% A and 50% inert gas at the start, two volumes of reactant mixture on complete conversion yield four volumes (3 volumes of R + 1 volume of inert gas) of product mixture.

$$\therefore \quad \varepsilon_A = \frac{4-2}{2} = 1$$

OR : Basis : 1 mol of feed

Therefore,

A in feed = 0.5 mol, Inert gas in feed = 0.5 mol

Moles at no conversion = 1 mol

Moles on complete conversion of A :

Product mixture contains 0.5 mol of inert gas and $\frac{3}{1} \times 5 = 1.5$ mol of R

Moles of product mixture = 0.50 + 1.5 = 2 mol

Moles on complete conversion = 2 mol

$$\varepsilon_A = \frac{\text{moles on complete conversion} - \text{initial moles}}{\text{initial moles}}$$

$$\therefore \quad \varepsilon_A = \frac{2-1}{1} = 1$$

For changing density systems, C_A and X_A are related by

$$X_A = \frac{1 - \frac{C_A}{C_{Ao}}}{1 + \frac{\varepsilon_A C_A}{C_{Ao}}}$$

with $\varepsilon_A = 1$

$$X_A = \frac{1 - \frac{C_A}{C_{Ao}}}{1 + \frac{C_A}{C_{Ao}}}$$

Given : We have to determine length of reactor for

(i) $F_{Ao} = 200$ mol/h (total flow = 400 mol/h)

(ii) $p_{A\,out} / p_{A\,in} = 0.111$

Using the ratio of partial pressures of A exit to inlet, we will find X_A.

For gases

$$C_A \propto p_A$$

$$\therefore \quad \frac{C_A}{C_{Ao}} = \frac{p_{A\,out}}{p_{A\,in}} = \frac{p_A}{p_{Ao}} = 0.111$$

$$\therefore \quad X_A = \frac{1 - \frac{p_{A\,out}}{p_{A\,in}}}{1 + \frac{p_{A\,out}}{p_{A\,in}}}$$

$$= \frac{1 - 0.111}{1 + 0.111} = 0.80$$

Given : Conversion of A is 80%.

i.e., we have to determine the length of reactor for 80% conversion.

Now we will evaluate W (amount of active catalyst) needed for 80% conversion.

For a packed bed reactor assuming plug flow, the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r_A'}$$

Given : $\quad -r_A' = 10\, C_A$

We have for a changing density system :

$$C_A = C_{Ao}\left(\frac{1 - X_A}{1 + \varepsilon_A X_A}\right)$$

With $\varepsilon_A = 1$, this becomes

$$C_A = C_{Ao}\left[\frac{1 - X_A}{1 + X_A}\right]$$

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{10\, C_{Ao}\left(\frac{1 - X_A}{1 + X_A}\right)}$$

For gaseous A :

$$C_{Ao} = \frac{p_{Ao}}{RT}$$

$p_{Ao} = x_A P = 0.5 \times 8 = 4$ atm, P = 8 atm, T = 700 °C = 973 K, R = 0.08206 *l*.atm/(mol.K)

$$C_{Ao} = \frac{4}{0.08206 \times 973} = 0.05 \text{ mol/}l$$

$$\frac{W}{F_{Ao}} = \frac{1}{10\, C_{Ao}} \int_0^{X_A} \frac{(1 + X_A)\, dX_A}{(1 - X_A)} \leftarrow \text{put } 1 - X_A = y \text{ and integrate}$$

$$\frac{W}{F_{Ao}} = \frac{1}{10\, C_{Ao}} [-2\, ln\, (1 - X_A) - X_A]$$

$$W = \frac{F_{Ao}}{10\, C_{Ao}} [-2\, ln\, (1 - X_A) - X_A]$$

$$= \frac{200}{10 \times 0.05} [-2\, ln\, (1 - 0.80) - 0.80]$$

$$= 967.55 \text{ g}$$

Packed bed consists of 25% of active catalyst pellets + 75% of inert pellets.

Total weight of the bed = Weight of active catalyst + Weight of inert solid

$$= 967.55 + \frac{75}{25} \times 967.55$$

$$= 3870.2 \text{ g}$$

OR : Since the bed contains 25% of active catalyst pellets,

$$\text{Weight of the bed} = \frac{\text{Weight of active catalyst}}{\frac{25}{100}}$$

$$= \frac{967.55}{0.25} = 3870.2 \text{ g}$$

Particle density ρ_s = 2 g/cm^3 ..... density of all solid material in the bed.

$$\text{Volume of catalyst + inert particles} = \frac{3870.2}{\rho_s}$$

$$= \frac{3870.2}{2} = 1935.1 \text{ cm}^3$$

Now we will make use of bulk voidage of the bed to find the reactor volume.

Bulk voidage of packed bed = 50%

$$\text{Fractional voidage of bed } \epsilon = \frac{50}{100} = 0.5$$

$$\text{Fractional voidage of packed bed} = \frac{\text{Empty volume in the bed}}{\text{Total bed volume}}$$

$$\epsilon = \frac{\text{Empty volume in the bed}}{\text{Total bed volume}} = \frac{\text{Volume of fluid}}{\text{Internal volume of reactor}}$$

$$\therefore \quad 1 - \epsilon = 1 - \frac{\text{Empty volume in bed}}{\text{Total bed volume}}$$

$$1 - \epsilon = \frac{\text{Total bed volume} - \text{Empty volume in bed}}{\text{Total bed volume}}$$

Total bed volume – Empty volume in the bed = Volume of the catalyst + inert pellets

$$= 1935.1 \text{ cm}^3$$

$$\therefore \quad 1 - 0.5 = \frac{1935.1}{\text{Total bed volume}}$$

$$\text{Total bed volume} = 3870.2 \text{ cm}^3$$

But Total bed volume = Internal volume of the reactor

∴ Volume of the reactor = V = 3870.2 cm^3

$$V = \frac{\pi}{4} d^2 L = 3870.2, \text{ Given } d = 2 \text{ cm}$$

$$\frac{\pi}{4} (2)^2 L = 3870.2 \qquad \therefore L = 1231.92 \approx 1232 \text{ cm} = 1.232 \text{ m}$$

Length of the reactor that must be needed for $\frac{p_{A\,out}}{p_{A\,in}} = 0.111$, i.e., for 80% conversion = 1232 cm

Length of the reactor needed = 1.232 m ... Ans.

Example 6.10 : *Gaseous reactant A decomposes on a solid catalyst as per the following reaction :* $A \rightarrow R$ *with* $-r_A = k C_A^2$

A pilot plant scale tubular reactor packed with 2 l of catalyst is fed with 2 m^3/h of pure A at 300 °C and 20 atm. 65% of A is converted.

It is desired to treat 100 m^3/h of feed gas at 40 atm and 300 °C consisting of 60% A and 40% diluent (mole basis) in a larger plant to achieve 85% conversion of A. Find the internal volume of the reactor needed.

Solution : $A \rightarrow R$, $-r_A = k C_A^2$ ..... second order decomposition

(i) Pilot plant scale tubular reactor :

v_{01} = 2 m^3/h, pure A feed at 300 °C and 20 atm.

V_1 = 2*l* ..... as packed with 2*l* of catalyst

$$X_{A1} = \frac{65}{100} = 0.65$$

(ii) Large plant reactor :

v_{02} = 100 m³/h, feed containing 60% A + 40% diluent – 300 °C and 40 atm.

$$X_{A2} = \frac{85}{100} = 0.85$$

V_2 = ?

$A \rightarrow R$ ..... Gas phase reaction

$$\varepsilon_A = \frac{1-1}{1} = 0$$

with $\varepsilon_A = 0$ we have :

$\therefore \quad C_A = C_{Ao}(1 - X_A)$

Assuming plug flow in both the cases, the performance equation based on the internal volume of reactor is

$$\frac{V}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r_A}$$

$$\frac{V}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{k C_{Ao}^2 (1 - X_A)^2} = \frac{1}{k C_{Ao}^2}\left[\frac{X_A}{1 - X_A}\right]$$

For the pilot plant reactor :

$$\frac{V_1}{F_{Ao1}} = \frac{1}{kC_{Ao1}^2}\left[\frac{X_{A1}}{1 - X_{A1}}\right] \quad \ldots (1)$$

For large the plant reactor :

$$\frac{V_2}{F_{Ao2}} = \frac{1}{kC_{Ao2}^2}\left[\frac{X_{A2}}{1 - X_{A2}}\right] \quad \ldots (2)$$

$F_{Ao1} = v_{01}\ C_{Ao1}$

$$C_{Ao1} = \frac{p_{Ao1}}{RT} = \frac{P_1}{RT} \quad \ldots\ldots \text{pure A}$$

$$= \frac{20}{0.08206 \times 573} = 0.4253 \text{ mol/}l$$

v_{01} = 2 m³/h = 100×10^3 l/h

Feed contains 60 mole % A.

Mole fraction of A in feed = $\frac{60}{100} = 0.60$

We know that :

$p_A = x_A P$

$$C_{Ao2} = \frac{p_{Ao2}}{RT} \qquad p_{Ao2} = x_A \cdot P = 0.60 \times 40 \text{ atm.}$$

$$= \frac{0.60 \times 40}{0.08206 \times 573}$$

$$= 0.5104 \text{ mol/}l$$

$$F_{Ao2} = \nu_{o2}.C_{Ao2}$$

$$= 100 \times 10^3 \times 0.5104 = 51040 \text{ mol/h}$$

Dividing Equation (2) by Equation (1)

$$\frac{V_2 F_{Ao1}}{F_{Ao2}.V_1} = \frac{\frac{1}{k C_{Ao2}^2}\left[\frac{X_{A2}}{1-X_{A2}}\right]}{\frac{1}{k C_{Ao1}^2}\left[\frac{X_{A1}}{1-X_{A1}}\right]}$$

$$V_2 = \frac{F_{Ao2} V_1}{F_{Ao1}} \frac{C_{Ao1}^2}{C_{Ao2}^2}\left[\frac{X_{A2}}{1-X_{A2}}\right]\left[\frac{1-X_{A1}}{X_{A1}}\right]$$

$$V_2 = \frac{51040 \times 2 \times (0.4253)^2}{850.6 \times (0.5104)^2}\left[\frac{0.85}{1-0.85}\right]\left[\frac{1-0.65}{0.65}\right]$$

$$= 254.25l \simeq 254\, l$$

Internal volume of the larger plant reactor = 254 *l* ... Ans.

OR : Calculate k using Equation (1) and make use of this k in Equation (2) to calculate V_2.

The reactor is packed with 2*l* of catalyst, i.e., the catalyst material when filled in the reactor occupies volume of 2*l* (including void space in the bed) so this equal to the internal volume of the reactor.

Example 6.11 : *A particular first-order gas phase reactor, in absence of pore diffusion resistance, proceeds as reported below :*

$$-r_A''' = 10^{-6} \text{ mol/(s. cm}^3 \text{ cat)}$$

at $C_A = 10^{-5}$ *mol/cm*^{3} *at 1 atm and 400 °C*

Determine the size of catalyst pellets that should be used to insure that pore resistance effects do not intrude to slow the rate of reaction.

Data : The effective diffusivity of A in pellet = $\mathscr{D}_e = 10^{-3}$ *cm*^{2}*/s.*

Solution : The pore resistance effects do not intrude to slow the rate of reaction ..... when there is a negligible resistance to pore diffusion.

For no pore resistance,

$$M_w = \frac{\left(-r_A'''\right)_{obs} L^2}{\mathscr{D}_e C_{Ag}} < 0.15$$

where M_w = Wagner modulus

For largest particle size free of pore diffusion resistance

$$\frac{\left(-r_A'''\right)_{obs} L^2}{\mathscr{D}_e C_{Ag}} = 0.15$$

$$L^2 = \frac{0.15\, \mathscr{D}_e C_{Ag}}{\left(-r_A'''\right)_{obs}}$$

For pure A at 1 atm and 400 °C, C_{Ag} is calculated by using the relation :

$$C_{Ag} = \frac{p_{Ao}}{RT}$$

$p_{Ag} = P = 1$ atm, $T = 400\,^{\circ}C = 673$ K

$R = 0.08206$ *l*.atm/(mol. K) $= 82.06$ cm^3.atm/(mol. K)

$$C_{Ag} = \frac{1}{82.06 \times 673}$$

$$= 1.81 \times 10^{-5} \text{ mol/cm}^3$$

$$\mathscr{D}_e = 10^{-3} \text{ cm}^2/\text{s}$$

$$\left(-r_A'''\right)_{obs} = 10^{-6} \text{ mol/(s. cm}^3 \text{ cat)}$$

$$L^2 = \frac{0.15 \times 10^{-3} \times 1.81 \times 10^{-5}}{10^{-6}}$$

$$= 2.715 \times 10^{-3}$$

$L = 0.052$ cm ..... characteristic size of catalyst particle

For spherical pellets :

$$L = \frac{R}{3}$$

$$\frac{R}{3} = 0.052$$

$$R = 3 \times 0.052 = 0.156 \text{ cm}$$

$$= 1.56 \text{ mm}$$

Diameter of catalyst pellets

$$= 2R$$

$$= 2 \times 1.56$$

$$= 3.12 \text{ mm}$$

Size of catalyst pellets – diameter = 3.12 nm ... Ans.

Example 6.12 : *A packed bed reactor filled with catalyst – C1 runs the gas phase reaction* $A \rightarrow R$ *at 10 atm. and 336 °C and gives 90% conversion of a pure A feed.*

The same reaction with a new porous catalyst – C2 [$\mathscr{D}_e = 2 \times 10^{-6}$ $m^3/(m\ cat.\ s)$], *in the absence of any pore diffusion resistance, is found to proceed with a rate given by.*

$$-r_A''' = 0.88\, C_A \ mol/(m^3\ cat{\cdot}s)$$

It is decided to use the new catalyst - C2 [which can do much better job than the catalyst - C1] as a next catalyst charge when the next replacement comes. What diameter of catalyst balls should we order ?

Solution : $A \rightarrow R$ ..... gas phase reaction ..... pure A feed ..... at 10 atm and 336 °C.

$$X_A = \frac{90}{100} = 0.90 \quad \text{..... with catalyst – C1}$$

$p_{Ag} = P = 10$ atm ... pure A feed, $T = 336\ ^\circ C = 609$ K

We have :
$$C_{Ag} = \frac{p_{Ag}}{RT}$$
$$= \frac{10}{0.08206 \times 609}$$
$$= 0.20 \text{ mol/}l$$
$$= 200 \text{ mol/m}^3$$
$$\varepsilon_A = \frac{1-1}{1} = 0$$

$\therefore$
$$C_A = C_{Ao}\ (1 - X_A)$$
$$= 200\ (1 - 0.90)$$
$$= 20 \text{ mol/m}^3 \quad \text{..... use it for catalyst - C2}$$

For new catalyst – C2, in the absence of pure diffusion resistance :

$$-r_A''' = 0.88\, C_A$$
$$= 0.88 \times 20$$
$$= 17.6 \text{ mol/(m}^3 \text{ cat·s)}$$

For no pore resistance, we have :

$$\frac{\left(\frac{-r_A'''}{C_A}\right)_{obs} L^2}{\mathscr{D}_e} \leq 0.15$$

$$\frac{\left(-r_A'''\right)_{obs} L^2}{\mathscr{D}_e C_{Ag}} = 0.15 \qquad (C_A)_{obs} = C_{Ag}$$

$$L^2 = \frac{0.15\, \mathscr{D}_e\, C_{Ag}}{\left(-r'''_A\right)_{obs}}, \qquad \mathscr{D}_e = 2 \times 10^{-6}\ m^3/(m\ cat.\ s)$$

$$= \frac{0.15 \times 2 \times 10^{-6} \times 200}{17.6}$$

$$= 3.409 \times 10^{-6}$$

$$L = 1.846 \times 10^{-3}\ m \ldots\ldots \text{characteristic size of particle}$$

For spherical pellets :

$$L = \frac{R}{3}$$

$$R = 3\,L$$

$$R = 3 \times 1.846 \times 10^{-3}\ m$$

$$= 5.5 \times 10^{-3}\ m$$

$$= 5.5\ mm$$

$$dp = 2R$$

$$= 2 \times 5.5$$

$$= 11\ mm \simeq 11\ mm$$

Diameter of the catalyst balls to be ordered = 11 mm. **... Ans.**

Example 6.13 : *A solid catalysed gas phase reaction A → 4R is conducted at 3.2 atm. and 117 °C in a plug flow reactor containing 10 g of catalyst and using a feed consisting of the partially converted product of 20 l/h of pure unreacted A. The following results are obtained :*

Run	*1*	*2*	*3*	*4*
$C_{A\ in}$	*0.10*	*0.08*	*0.06*	*0.04*
$C_{A\ out}$	*0.084*	*0.07*	*0.055*	*0.038*

Find the rate equation for this reaction.

Solution : A → 4R ..... gas phase reaction.

Examine the data given. For run-1 there is a maximum variation between $C_{A\ in}$ and $C_{A\ out}$.

$C_{A\ in} = 0.1$ mol/*l*, $C_{A\ out} = 0.084$ mol/*l*

$$C_{A\ mean} = \frac{0.1 + 0.084}{2} = 0.092\ \text{mol/}l$$

Maximum variation about the mean concentration is 8% (0.1 – 0.092 = 0.08 or 8%) of $C_{A\ in}$.

0.1 – 0.092 = 0.08, i.e., 8%

As the maximum variation about the mean concentration is 8% (not large) we may treat this to be differential reactor.

Consider pure A at 3.2 atm. and 117 °C as the basis for calculating conversion for all runs.

Let us obtain C_{Ao} :

$$C_{Ao} = \frac{p_{Ao}}{RT}$$

$p_{Ao} = P = 3.2$ atm ..... since pure A

$R = 0.08206$ *l*.atm/(mol.K)

$T = 117\,^{\circ}C = 390$ K

$$C_{Ao} = \frac{3.2}{0.08206 \times 390}$$

$= 0.10$ mol/*l*.

Molal flow rate of A.

$F_{Ao} = C_{Ao}\, v$, $v = 20$ *l*/h of pure unreacted A

$= 0.1 \times 20 = 2$ mol/h

For variable density systems, concentration and conversion are related by

$$\frac{C_A}{C_{Ao}} = \frac{1 - X_A}{1 + \varepsilon_A X_A}$$

or

$$X_A = \frac{1 - \dfrac{C_A}{C_{Ao}}}{1 + \dfrac{\varepsilon_A C_A}{C_{Ao}}}$$

For $A \rightarrow 4R$.

For pure A : Basis : 1 mol feed

Moles at no conversion = 1 mol A = 1 mol

Moles of R produced = $(4/1) \times 1 = 4$ mol

Moles on complete conversion = 0 mol A + 4 mol R = 4 mol

The fractional change in volume of the reaction system is

$$\varepsilon_A = \frac{4 - 1}{1} = 3 \text{ ..... for pure A (basis selected)}$$

For a differentiated flow reactor, the performance equation is

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out} - X_{A\,in}}{\left(-r'_A\right)_{avg}}$$

$$\left(-r'_A\right) = \frac{(X_{A\,out} - X_{A\,in})\, F_{Ao}}{W}$$

Now we will calculate $X_{A\,in}$, $X_{A\,out}$, $C_{A\,avg}$ and $\left(-r'_A\right)$ for each run.

Run 1 : $C_{A\,in} = 0.1$ mol/l $C_{A\,out} = 0.084$ mol/l $C_{Ao} = 0.1$ mol/l

$$C_{A\,avg} = \frac{0.1 + 0.084}{2} = 0.092 \text{ mol/}l$$

$$X_{A\,in} = \frac{1 - \dfrac{C_{A\,in}}{C_{Ao}}}{1 + \varepsilon_A \dfrac{C_{A\,in}}{C_{Ao}}} = \frac{1 - \left(\dfrac{0.1}{0.1}\right)}{1 + 3\left(\dfrac{0.1}{0.1}\right)} = 0$$

$$X_{A\,out} = \frac{1 - \dfrac{C_{A\,out}}{C_{Ao}}}{1 + \varepsilon_A \dfrac{C_{A\,out}}{C_{Ao}}} = \frac{1 - \left(\dfrac{0.084}{0.10}\right)}{1 + 3\left(\dfrac{0.084}{0.10}\right)} = 0.04545 \simeq 0.0455$$

$$\left(-r_A'\right)_{avg} = -r_A' = \frac{(X_{A\,out} - X_{A\,in})\, F_{Ao}}{W}$$

$W = 10$ g $= 0.01$ kg, $F_{Ao} = 2$ mol/h

$$-r_A' = \frac{(0.0455 - 0) \times 2}{0.01} = 9.1 \frac{\text{mol}}{\text{(h. kg cat)}}$$

Do similar calculations for the remaining runs and tabulate.

Run	1	2	3	4
$C_{A\,in}$	0.10	0.08	0.06	0.04
$C_{A\,out}$	0.084	0.07	0.055	0.038
$X_{A\,in} = \dfrac{1 - \dfrac{C_{A\,in}}{C_{Ao}}}{1 + 3\dfrac{C_{A\,in}}{C_{Ao}}}$	0	0.0588	0.143	0.273
$X_{A\,out} = \dfrac{1 - \dfrac{C_{A\,out}}{C_{Ao}}}{1 + 3\dfrac{C_{A\,out}}{C_{Ao}}}$	0.0455	0.0968	0.170	0.290
$\Delta X_A = X_{A\,out} - X_{A\,in}$	0.0455	0.038	0.027	0.017
$C_{A\,avg} = \dfrac{C_{A\,in} + C_{A\,out}}{2}$	0.092	0.075	0.0575	0.039
$-r_A' = \dfrac{\Delta X_A \cdot F_{Ao}}{W}$	9.1	7.6	5.4	3.4
$W = 0.01$ kg, $C_{Ao} = 0.10$ mol/l, $F_{Ao} = 2$ mol/h.				

Now we have : rate $\left(-r_A'\right)$ – concentration ($C_{A\,avg}$) data. Using this data we can find the rate equation for the reaction $A \rightarrow 3R$.

Guess : First order reaction - decomposition of A to be first order, i.e., as a first guess try a first order rate.

$\therefore$ $-r_A' = k' C_A$

For our guess to be correct, plot of $-r_A'$ v/s C_A should yield a straight line through the origin.

Plot $-r_A'$ v/s C_A

Refer Fig. Ex 6.13 :

Plot of $-r_A'$ v/s C_A results in a straight line passing through the origin.

So the reaction is first order.

slope = k' = 96.55 *l*/(h. kg cat)

The rate equation is :

$$\boxed{-r_A' = -\frac{1}{W}\frac{dN_A}{dt} = \left(96.55\ \frac{l}{\text{(h. kg cat)}}\right)\left(C_A\ \frac{\text{mol}}{l}\right)}$$

... Ans.

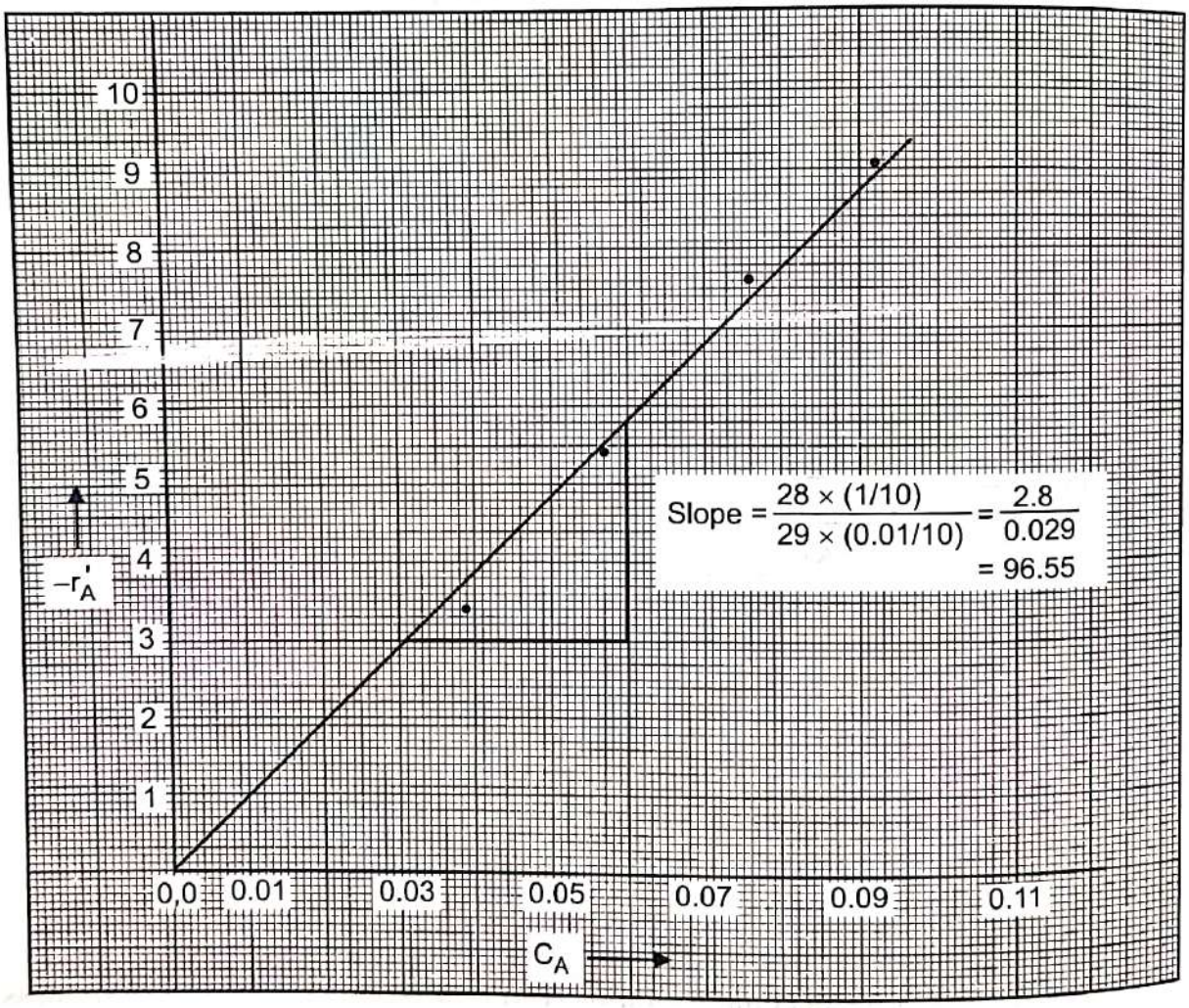


Fig. Ex 6.13

Example 6.14 : *How much catalyst is needed in a packed bed reactor (assume plug flow) for 35% conversion of A to R for a feed of 2000 mol/h of pure gaseous A at 3.2 atm and 117 °C if the stoichiometry and rate are given by*

$$A \rightarrow 4R, \quad -r'_A = 96.55 \left(\frac{l}{h.\ kg\ cat}\right) C_A$$

Solution : $A \rightarrow 4R$ ..... gas phase reaction.

For pure A, 1 volume of feed on complete conversion yields 4 volumes of product mixture. Therefore, the fractional change in volume of the reaction system is

$$\varepsilon_A = \frac{4-1}{1} = 3$$

For a plug flow behaviour, the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r'_A}$$

$$-r'_A = 96.55\, C_A = k'C_A, \quad k' = 96.55\ l\ /\ (\text{h. kg cat})$$

For a variable density system, the concentration and conversion are related by

$$C_A = \left(\frac{1-X_A}{1+\varepsilon_A X_A}\right) C_{Ao}$$

For pure A at 3.2 atm and 117 °C,

$$C_{Ao} = \frac{p_{Ao}}{RT}$$

p_{Ao} = total pressure = 3.2 atm ..... for pure A feed

$R = 0.08206\ l.\text{atm/(mol. K)}$

$T = 117\ ^\circ\text{C} = 390\ \text{K}$

$$C_{Ao} = \frac{3.2}{0.08206 \times 390} = 0.1\ \text{mol}/l$$

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r'_A} = \int_0^{X_A} \frac{dX_A}{k'C_A}$$

$$= \frac{1}{k'C_{Ao}} \int_0^{X_A} \left(\frac{1+\varepsilon_A X_A}{1-X_A}\right) dX_A$$

Integrating the above equation, we get

$$\frac{W}{F_{Ao}} = \frac{1}{k'C_{Ao}} \left[(1+\varepsilon_A)\, ln \left(\frac{1}{1-X_A}\right) - \varepsilon_A X_A\right]$$

$$W = \frac{F_{Ao}}{k'C_{Ao}} \left[(1+\varepsilon_A)\, ln \left(\frac{1}{1-X_A}\right) - \varepsilon_A X_A\right]$$

where : $k' = 96.55\ l/(h.\ kg\ cat)$

$C_{Ao} = 0.10\ mol/l,\ X_A = \frac{35}{100} = 0.35$

$\varepsilon_A = 3,\ F_{Ao} = 2000\ mol/h$

$$W = \frac{2000}{96.55 \times 0.10}\left[(1+3)\ ln\left(\frac{1}{1-0.35}\right) - 3 \times 0.35\right]$$

$$= 207.146\ [1.723 - 1.05]$$

$$= 139.41\ kg$$

Amount of catalyst needed in the packed bed reactor = 139.41 kg ... Ans.

Example 6.15 : *How much catalyst is needed in a packed bed reactor with very large recycle rate (assume mixed flow) for 35% conversion of 2000 mol/h of pure gaseous A at 3.2 atm and 117 °C if the stoichiometry and rate are given by :*

$$A \rightarrow 4R,\ -r_A' = 96.55\ l(h.\ kg\ cat)\ C_A$$

Solution : $A \rightarrow 4R$

$$-r_A' = 96.55\left(\frac{l}{h.\ kg\ cat}\right) C_A = k'\ C_A$$

$$k' = 96.55\ l/(h.\ kg\ cat)$$

For pure gaseous A at 3.2 atm. and 117 °C

$$C_{Ao} = \frac{p_{Ao}}{RT}$$

$p_{Ao} = 3.2$ atm. ..... pure A

$R = 0.08206\ l.atm/(mol.\ K)$

$T = 117\ °C = 390\ K$

$$C_{Ao} = \frac{3.2}{0.08206 \times 390} = 0.10\ mol/l$$

$F_{Ao} = 2000\ mol/h$

$A \rightarrow 4R$ ..... gas phase reaction

For this reaction and pure A, the fractional change in volume of the reaction system is

$$\varepsilon_A = \frac{4-1}{1} = 3$$

For a packed bed reactor with very large recycle rate mixed flow is approached. Therefore the performance equation is

$$\frac{W}{F_{Ao}} = \frac{X_{A\ out}}{-r_A'} = \frac{X_{A\ out}}{k'\ C_{A\ out}}$$

For a variable density system, X_A and C_A are related by

$$C_A = C_{Ao}\left(\frac{1 - X_A}{1 + \varepsilon_A X_A}\right)$$

$$C_{A\,out} = C_{Ao}\left(\frac{1 - X_{A\,out}}{1 + \varepsilon_A X_{A\,out}}\right), \text{ with this the performance equation becomes}$$

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out}(1 + \varepsilon_A X_{A\,out})}{k'(1 - X_{A\,out})\,C_{Ao}}$$

$$W = \frac{F_{Ao} X_{A\,out}(1 + \varepsilon_A X_{A\,out})}{k'(1 - X_{A\,out})\,C_{Ao}}$$

$k' = 96.55\ l/(\text{h. kg cat})$, $X_{A\,out} = \frac{35}{100} = 0.35$

$C_{Ao} = 0.10\ \text{mol}/l$, $F_{Ao} = 2000\ \text{mol/h}$, $\varepsilon_A = 3$

$$W = \frac{2000 \times 0.35\,(1 + 3 \times 0.35)}{96.55\,(1 - 0.35) \times 0.10}$$

$$= 228.66\ \text{kg} \simeq 229\ \text{kg}$$

Amount of catalyst needed = 229 kg **... Ans.**

Example 6.16 : *The first-order decomposition reaction $A \rightarrow R$ is run in an experimental mixed flow reactor. From the following data determine whether the runs were conducted under diffusion-free, strong pore resistance or intermediate conditions.*

dp	*W*	C_{Ao}	*v*	X_A
3	*1*	*100*	*9*	*0.40*
12	*4*	*300*	*8*	*0.80*

Solution : $A \rightarrow R$ ..... first order decomposition

For mixed flow reactor :

$$\frac{W}{F_{Ao}} = \frac{X_{A\,out}}{-r'_{A\,out}}$$

$$-r'_{A\,out} = \frac{X_{A\,out} F_{Ao}}{W}$$

$$-r'_{A\,out} = \frac{X_{A\,out} C_{Ao} \cdot v}{W}$$

$$-r'_{A} = \frac{X_A C_{Ao} v}{W}$$

Using the above equation and data provided we will check in which regime a given reaction is conducted.

In the diffusion-free regime for particles of size R_1 and R_2 :

$$\frac{-r'_{A1}}{-r'_{A2}} = 1$$

$$\therefore \quad -r'_{A1} = -r'_{A2}$$

Let us check whether the data satisfies this condition or not.

For particle of size $dp_1 = 3 : R_1 = \frac{3}{2}$, $W_1 = 1$, $C_{Ao1} = 100$

$$\nu_1 = 9 \text{ and } X_{A1} = 0.40$$

$$\therefore \quad -r'_{A1} = \frac{X_{A1} C_{Ao1} \nu_1}{W_1}$$

$$= \frac{0.4 \times 100 \times 9}{1} = 360$$

For particle of size $dp_2 = 12$:

$R_1 = \frac{12}{2}$, $W_2 = 4$, $C_{Ao2} = 300$

$\nu_2 = 8$, $X_{A2} = 0.60$

$$-r'_{A2} = \frac{X_{A2} C_{Ao2} \nu_2}{W_2}$$

$$= \frac{0.60 \times 300 \times 8}{4} = 360$$

Therefore, $r'_{A1} = r'_{A2} = 360$

The data satisfies the condition $\Rightarrow \frac{-r'_{A1}}{-r'_{A2}} = 1$

Since $-r'_{A1} = -r'_{A2}$ $\left(\text{i.e.}, -r'_{A1} / -r'_{A2} = 1\right)$ the runs were conducted under diffusion-free condition.

Reaction is run in the diffusion-free regime ... Ans.

Example 6.17 : *The following kinetic data are obtained in a basket type mixed flow reactor. The catalyst is porous. Assuming isothermal behaviour interpret the data in terms of controlling resistances.*

Data :

Pellet diameter	Concentration of reactant in the exit stream	Spinning rate of baskets	Measured reaction rate, $-r'_A$
1	1	high	3
3	1	low	1
3	1	high	1

Solution :

Run	dp	C_A	RPM	Rate
1	1	1	high	3
2	3	1	low	1
3	3	1	high	1

Examine the data for its interpretation in terms of controlling resistances.

Look at run-2 and run-3 (for large particles) :

These two runs use different spinning rates but have the same reaction rate.

For diffusion to control, higher the spinning rate the higher will be the k_g and hence the rate. But for run-3, though the spinning rate is higher compared to run-2 the measured reaction rate is the same as that for run-2 (not higher than that for run-2). Therefore, film diffusion is not controlling resistance for large particles.

Since the film diffusion is not controlling for large pellets, it will not be controlling in case of small pellets (run-1).

First conclusion : Film diffusion resistance is not controlling, i.e., film diffusion resistance does not influence the rate.

Now we will check for pore diffusion resistance (to be controlling resistance).

In the regime of strong pore diffusion resistance .....

$$-r'_A \propto \frac{1}{R}$$

i.e., $$\frac{-r'_{A1}}{-r'_{A2}} = \frac{R_2}{R_1}$$ ..... should be satisfied for particles of sizes R_1 and R_2.

Compare run-1 with either run-2 or -3 :

Run-1 and -2 :

$$-r'_{A1} = 3 \qquad R_1 = \frac{1}{2}$$

$$-r'_{A2} = 1 \qquad R_2 = \frac{3}{2}$$

$$\frac{-r'_{A1}}{-r'_{A2}} = \frac{3}{1} = 3$$

and $$\frac{R_2}{R_1} = \frac{\frac{3}{2}}{\frac{1}{2}} = 3$$

$$\therefore \quad \frac{-r'_{A1}}{-r'_{A2}} = \frac{R_2}{R_1} \quad \ldots\ldots\ldots \text{ is satisfied}$$

Same result is obtained by comparing run-1 with run-3.

$$\frac{-r'_{A1}}{-r'_{A3}} = \frac{3}{1} = 3 \quad \text{and} \quad \frac{R_3}{R_1} = \frac{\frac{3}{2}}{\frac{1}{2}} = 3$$

As the rate is inversely proportional to particle size, the pore diffusion resistance is controlling the rate. Therefore, the runs are conducted in the regime of strong pore resistance.

Final conclusion/final interpretation of data :

- negligible film diffusion resistance.
- strong pore diffusion resistance.

Pore diffusion resistance is controlling the rate of reaction ... Ans.

Example 6.18 : *Experiments are conducted on the decomposition of A with a particular catalyst. Is the reaction influenced by pore diffusion ?*

Data : For spherical particle :

$R = 1.2\ mm$

$$\text{Effective mass diffusivity} = 5 \times 10^{-5}\ m^3/(m\ cat.\ h)$$

$$\text{Observed reaction rate} = 10^5\ mol/(h.\ m^3\ cat)$$

$$C_{Ag} = 20\ mol/m^3\ (at\ 1\ atm.\ and\ 336\ ^\circ C)$$

Solution :

R = 1.2 mm ..... spherical particle.

For spheres, the characteristic size of particle is

$$L = \frac{R}{3} = \frac{1.2}{3} = 0.4\ mm = 4 \times 10^{-4}\ m$$

To determine whether the reaction is influenced by pore diffusion or not we have to calculate the Wagner modulus (M_W).

Based upon the numerical value of the Wagner modulus :

If $M_W < 0.15$ ..... diffusion-free regime (no pore resistance)

If $M_W > 4$ ..... strong pore diffusion resistance regime (strong pore diffusion effects)

$$M_W = \frac{L^2 \left(\frac{-r'''_A}{C_A}\right)_{obs}}{\mathcal{D}_e}$$

where $L = 4 \times 10^{-4}$ m, $\mathscr{D}_e = 5 \times 10^{-5}$ m^3/(h. m^3 cat)

$$\left(-r_A'''\right)_{obs} = 10^5 \text{ mol/(h. m}^3 \text{ cat)}$$

$$C_{A\,obs} = C_{Ag} = 20 \text{ mol/m}^3$$

$$M_W = \frac{(4 \times 10^{-4})^2 \times 10^5}{20 \times 5 \times 10^{-5}}$$

$$= 16$$

$\therefore$ $M_W > 4$

As M_W is greater than 4 ... pore diffusion is influencing and slowing the rate of reaction **... Ans.**

Example 6.19 : *The catalytic reaction*

$$R \rightarrow 3.5\,S$$

is carried out in a plug flow reactor, containing 0.015 kg of catalyst at 3 atm. and 125 °C. Feed to the reactor is partially converted and contains 28 l/h of pure unreacted R. The following results are obtained by conducting the runs.

Run	*1*	*2*	*3*	*4*
$C_{R\,in}$, *mol/l*	*0.092*	*0.08*	*0.06*	*0.05*
$C_{R\,out}$, *mol/l*	*0.077*	*0.07*	*0.055*	*0.047*

Find the rate equation for this reaction.

Solution : Maximum difference (variation) in the concentration of R (refer Run - 1) :

$C_{R\,in} = 0.092$ mol/*l* and $C_{R\,out} = 0.077$ mol/*l* is 0.015

$$\text{Mean concentration} = C_{R\,mean} = \frac{0.092 + 0.077}{2} = 0.0845 \text{ mol/}l$$

Maximum variation about the mean concentration is 8.1522% of C_{Rin}. Therefore, we may consider this to be a differential reactor (... not large variation ... for small conversions).

Consider pure R at 3 atm and 125 °C as the basis for calculating conversion for all runs :

$$C_{Ro} = \frac{p_{Ro}}{RT}, \qquad p_{Ro} = 3 \text{ atm. ..... pure R}$$

$T = 125\ ^\circ\text{C} = 398$ K

$R = 0.08206$ *l*.atm/(mol. K)

$$C_{Ro} = \frac{3}{0.08206 \times 398}$$

$$= 0.09186 \approx 0.092 \text{ mol/}l$$

Molar flow rate of R :

$$F_{Ro} = C_{Ro} \cdot \nu, \qquad \nu = 28\ l/h$$
$$= 0.092 \times 28 = 2.576\ \text{mol/h}$$

The fractional change in volume of the reaction system between no conversion and complete conversion is

For the reaction : $R \rightarrow 3.5\ S$

$$\varepsilon_R = \frac{3.5 - 1}{1} = 2.5 \quad \ldots\ldots \text{for pure R (basis selected)}$$

Thus, it is a changing density reaction system.

For a variable density system, concentration and conversion are related by

$$\frac{C_R}{C_{Ro}} = \frac{1 - X_R}{1 + \varepsilon_R X_R}$$

or

$$X_R = \frac{1 - \dfrac{C_R}{C_{Ro}}}{1 + \varepsilon_R \dfrac{C_R}{C_{Ro}}}$$

For differential flow reactor, we have

$$\frac{W}{F_{Ro}} = \frac{X_{R\,out} - X_{R\,in}}{\left(-r_R'\right)_{avg}}$$

$$\left(-r_R'\right)_{avg} = -r_R' = \frac{(X_{Rout} - X_{Ain})\, F_{Ro}}{W}$$

$$C_{R\,avg} = \frac{C_{R\,out} + C_{R\,in}}{2}$$

Now we will calculate $X_{R\,in}$, $X_{R\,out}$, $C_{R\,avg}$ and $-r_R'$ for each run.

Run 1 : $C_{R\,in} = 0.092\ \text{mol}/l$, $C_{R\,out} = 0.077\ \text{mol}/l$

$$C_{R\,avg} = \frac{0.092 + 0.077}{2} = 0.0845\ \text{mol}/l$$

$$X_{R\,in} = \frac{1 - \dfrac{C_{R\,in}}{C_{R\,o}}}{1 + \varepsilon_R \dfrac{C_{R\,in}}{C_{R\,o}}} = \frac{1 - \dfrac{0.092}{0.092}}{1 + 2.5 \times \dfrac{0.092}{0.092}} = 0$$

$$X_{R\,out} = \frac{1 - \dfrac{C_{R\,out}}{C_{R\,o}}}{1 + \varepsilon_R \dfrac{C_{R\,out}}{C_{R\,o}}}$$

$$= \frac{1 - \left(\dfrac{0.077}{0.092}\right)}{1 + \left(2.5 \times \dfrac{0.077}{0.092}\right)} = 0.0527$$

$$\left(-r_R'\right)_{avg} = -r_R' = \frac{(X_{R\,out} - X_{R\,in})\,F_{Ro}}{W}, \quad W = 0.015 \text{ kg}$$

$$= \frac{(0.0527 - 0) \times 2.576}{0.015} = 9.05 \text{ mol/(h. kg cat)}$$

Do similar calculations for the remaining runs and tabulate.

Run	1	2	3	4
$C_{R\,in}$	0.092	0.08	0.06	0.05
$C_{R\,out}$	0.077	0.07	0.055	0.047
$C_{R\,avg}$	0.0845	0.075	0.0575	0.0485
$X_{R\,in}$	0	0.041	0.132	0.193
$X_{R\,out}$	0.0527	0.0824	0.161	0.215
ΔX_R	0.0527	0.0414	0.029	0.022
$-r_R'$	9.05	7.11	4.98	3.78

We have : rate $\left(-r_R'\right)$ – concentration $C_{R\,avg}$ data. With the help of this data we can find the rate equation.

For n^{th} order reaction we have :

$$-r_R' = k' C_R^n, \quad C_R = C_{R\,avg}$$

$$ln\left(-r_R'\right) = ln\, k' + n\, ln\, C_R$$

Plot $ln\left(-r_R'\right)$ v/s $ln\, C_R$, the slope of this plot gives the value of n and the intercept gives the value of k.

$-r_R'$	9.05	7.11	4.98	3.78
$ln\left(-r_R'\right)$	2.20	1.96	1.60	1.33
C_R	0.0845	0.075	0.0575	0.0485
$ln\, C_R$	– 2.47	– 2.59	– 2.856	– 3.03

From the plot, refer Fig. Ex 6.19 :

$n = 1.48$ and $k' = 338.7\ (l/mol)^{0.48} / (h.kg\ cat)$

Thus, the rate equation is :

$$\boxed{-r_R' = \left[338.7 \left(\frac{(l/mol)^{0.48}}{(h.\ k\ cat)}\right)\right] \left(C_R \frac{mol}{l}\right)^{1.48}, \frac{mol}{(h.\ kg\ cat)}}$$

... Ans.

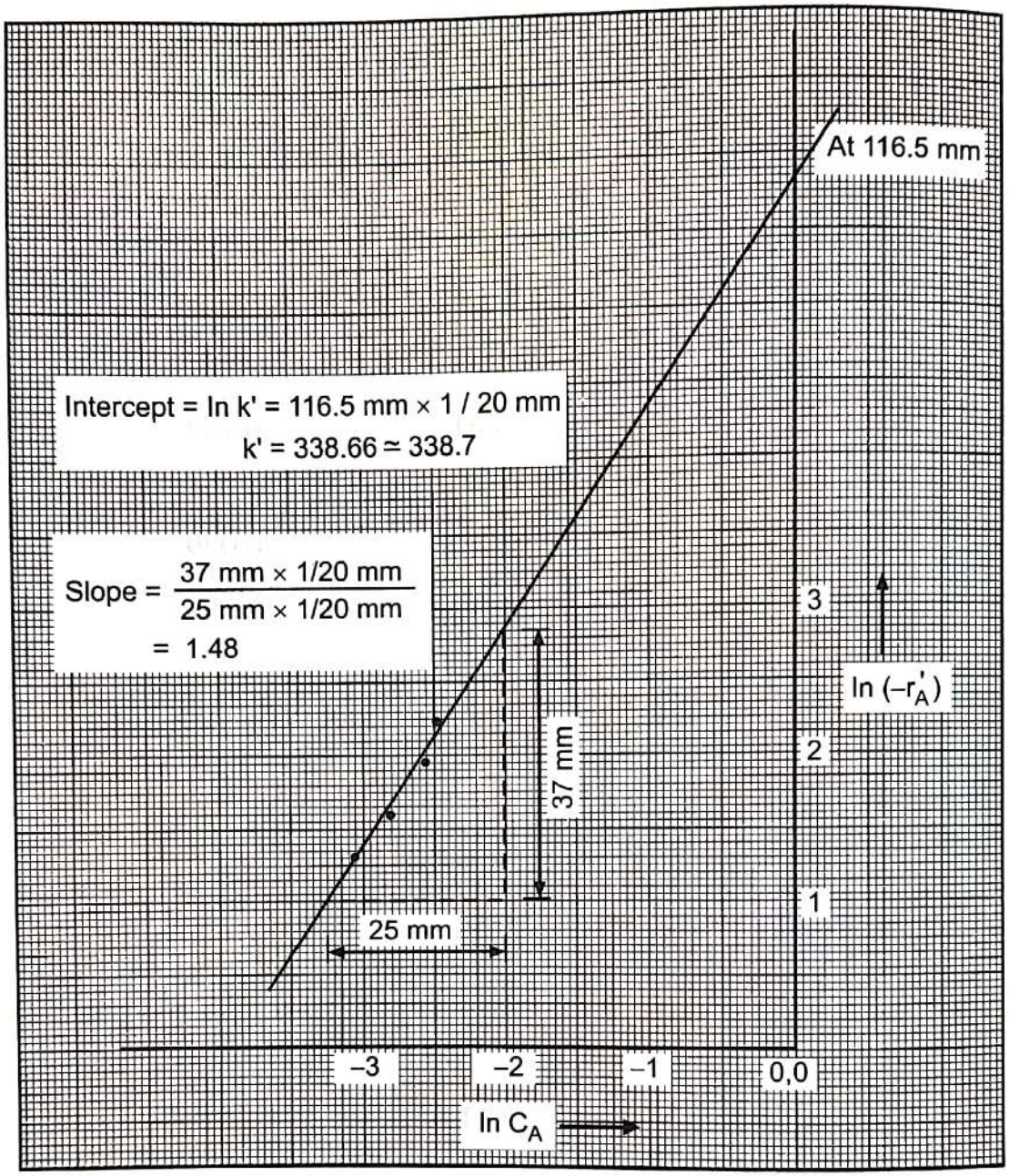


Fig. Ex 6.19

Example 6.20 : *The catalytic reaction $A \rightarrow 4R$ is carried out in a plug flow reactor with $C_{Ao} = 0.12$ mol/l. The packed bed contains 0.125 kg catalyst. The feed consists of 45 l/h of pure A.*

The data collected in different runs are

Run	***1***	***2***	***3***	***4***
$C_{A\ in}$, mol/l	0.12	0.096	0.072	0.048
$C_{A\ out}$, mol/l	0.1008	0.084	0.066	0.0456

Solution : Examine the data given :

After examining the data we come to know that for Run-1 the difference between $C_{A\ in}$ and $C_{A\ out}$ is maximum (among all the runs).

Refer Run-1.

Mean concentration of A = $C_{A\ mean} = \dfrac{0.12 + 0.1008}{2} = 0.01104$

Maximum variation about the mean concentration of $C_{A\ in}$ and $C_{A\ out}$ is

$$\left[\frac{(0.12 - 0.1104)}{0.12}\right] \times 100 = 8\% \text{ of } C_{A\ in}$$

As the maximum variation about the mean concentration is 8% we may consider this to be a differential reactor and apply the performance equation of a differential flow reactor to find the reaction rate.

For $A \rightarrow 4R$

$$\varepsilon_A = \frac{4-1}{1} = 3 \quad \ldots\ldots \text{ for pure A}$$

$$C_{Ao} = 0.12 \text{ mol/}l \quad \ldots\ldots \text{ given}$$

Molal flow rate of A :

$$F_{Ao} = C_{Ao} \cdot v \ , \quad v = 45\ l\text{/h}$$

$$= 0.12 \times 45 = 5.4 \text{ mol/h}$$

$$W = 0.125 \text{ kg}$$

Use pure A at $C_{Ao} = 0.12$ mol/l as a basis for calculation for conversion for all runs.

For variable density system, we have

$$\frac{C_A}{C_{Ao}} = \frac{1 - X_A}{1 + \varepsilon_A X_A}$$

or

$$X_A = \frac{1 - \dfrac{C_A}{C_{Ao}}}{1 + \varepsilon_A \dfrac{C_A}{C_{Ao}}}$$

For differential flow reactor,

$$\frac{W}{F_{Ao}} = \frac{X_{A\ out} - X_{A\ in}}{\left(-r'_A\right)_{avg}}$$

$$-r'_{A\ avg} = \frac{(X_{A\ out} - X_{A\ in})\ F_{Ao}}{W}$$

$$C_{A\ avg} = \frac{C_{A\ out} + C_{A\ in}}{2}$$

Calculate $X_{A\ in}$, $X_{A\ out}$, $C_{A\ avg}$ and $\left(-r'_A\right)_{avg}$ for each run and tabulate.

Run - 1 :

$C_{A\,in} = 0.12$ mol/l, $C_{A\,out} = 0.1008$ mol/l

$$C_{A\,avg} = \frac{0.12 + 0.1008}{2} = 0.1104 \text{ mol/}l$$

$$X_{A\,in} = \frac{1 - \frac{C_{A\,in}}{C_{Ao}}}{1 + \varepsilon_A \frac{C_{A\,in}}{C_{Ao}}}$$

$$= \frac{1 - \left(\frac{0.12}{0.12}\right)}{1 + \left(\frac{3 \times 0.12}{0.12}\right)} = 0$$

$$X_{A\,out} = \frac{1 - \frac{C_{A\,out}}{C_{Ao}}}{1 + \varepsilon_A \frac{C_{A\,out}}{C_{Ao}}}$$

$$= \frac{1 - \left(\frac{0.1008}{0.12}\right)}{1 + 3\left(\frac{0.1008}{0.12}\right)}$$

$$= 0.04545$$

$$\left(-r_A'\right)_{avg} = -r_A' = \frac{(0.04545 - 0) \times 5.4}{0.125} = 1.963 \text{ mol/(h. kg cat)}$$

Run	1	2	3	4
$C_{A\,in}$	0.12	0.096	0.072	0.048
$C_{A\,out}$	0.1008	0.084	0.066	0.0456
$C_{A\,avg}$	0.1104	0.09	0.069	0.0468
$X_{A\,in}$	0	0.0588	0.143	0.273
$X_{A\,out}$	0.04545	0.0968	0.170	0.290
ΔX_A	0.04545	0.038	0.027	0.017
$-r_A' = \Delta X_A F_{Ao}/W$	1.963	1.642	1.17	0.734

Now we have : rate $\left(-r_A'\right)$ – concentration ($C_{A\ avg}$) data. Using this data we can find the rate equation.

Guess : Assume that the decomposition of A ($A \rightarrow 4R$) is a first order reaction.

$$\therefore \quad -r_A' = k' C_A, \qquad C_A = C_{A\ avg}$$

$$\therefore \quad k' = \frac{-r_A'}{C_A}$$

For decomposition to be first order, a plot of $-r_A'$ v/s C_A should yield a straight line passing through the origin or k' calculated for all runs using $k' = \frac{-r_A'}{C_A}$ should be almost same.

Adopt a graphical procedure. Refer Fig. Ex 6.20 :

We are getting straight line passing through the origin. Therefore, our guess regarding the reaction order is correct.

$\therefore$ The decomposition of A is a first order reaction.

From Fig. Ex 6.19,

$$\text{slope} = k' = 17.15 \ \frac{l}{\text{h. kg cat}}$$

The rate equation is :

$$\boxed{-r_A' = -\frac{1}{W}\frac{dN_A}{dt} = \left(17.15 \frac{l}{\text{h. kg cat}}\right)\left(C_A \frac{\text{mol}}{l}\right)} \quad \text{... Ans.}$$

Extra :

$$k' = \frac{-r_A'}{C_A}$$

For run 1 :

$$k' = \frac{1.963}{0.1104} = 17.78 \ \frac{l}{\text{(h. kg cat)}}$$

For run 2 :

$$k' = \frac{1.642}{0.09} = 18.24 \ \frac{l}{\text{(h. kg cat)}}$$

For run 3 :

$$k' = \frac{1.17}{0.069} = 16.96 \frac{l}{(\text{h. kg cat})}$$

For run 4 :

$$k' = \frac{0.734}{0.0468} = 15.69 \frac{l}{(\text{h. kg cat})}$$

..... k' values are almost same

$$\text{Average value of } k' = \frac{17.78 + 18.24 + 16.96 + 15.69}{4}$$

$$= 17.1675$$

$$\simeq 17.17 \frac{l}{(\text{h. kg cat})}$$

k' calculated and k' obtained graphically are almost same.

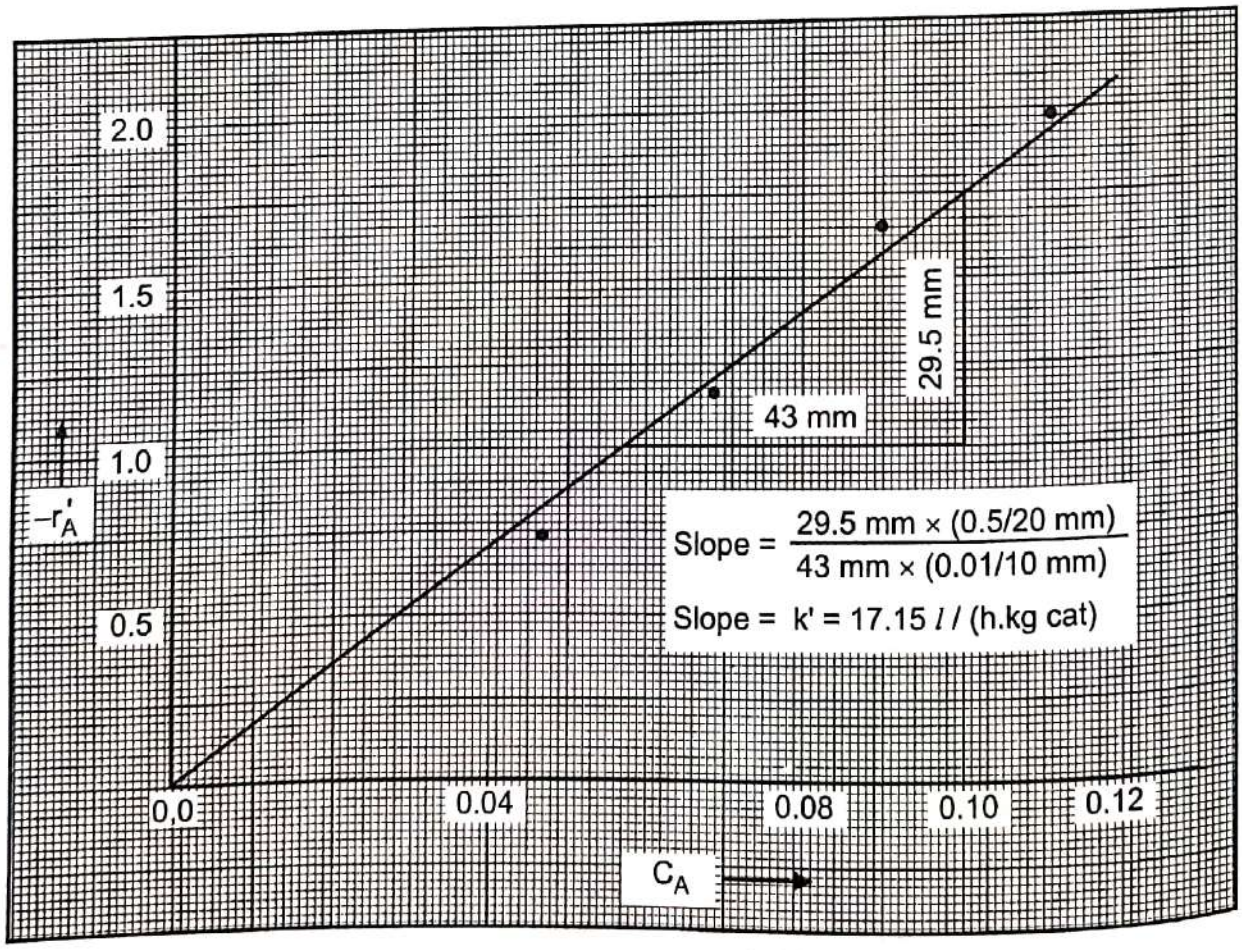


Fig. Ex 6.20

Example 6.21 : *A reaction A → 3 R is carried at 800 °C in a catalytic tubular reactor acked with 30% active catalyst pellets and 70% inerts to maintain isothermal conditions.*

The rate equation is :

$$-r'_A = -\frac{1}{W}\frac{dN_A}{dt} = \left(10\frac{l}{h.\ g\ cat}\right) C_A,\ \frac{mol}{h.\ g\ cat}$$

For 500 mol/h of a feed consisting of 60% A and 40% inert gas at 8.5 atm and 800 °C, what ust be the length of reactor so that $\frac{p_{A\ out}}{p_{A\ in}} = 0.15$ *?*

Data :

(i) Catalyst and inert pellets are porous (dp = 3 mm)

(ii) Particle density $\rho_s = 2\ g/cm^3$

(iii) Bulk voidage of packed bed, $\varepsilon = 50\%$

(iv) Internal diameter of tubular reactor = 2 cm

Solution : A → 3R

Basis : 1 mol of feed. It contains 0.60 mol A and 0.40 mol inerts.

For feed consisting of 60% A and 40% inert gas :

A	inert gas	R	
0.6	0.4	0	... at zero conversion
0	0.4	0.6 × 3 = 1.8	... at 100% conversion

Moles at no conversion = 1 mol

Moles on complete conversion = 1.8 + 0.40 = 2.2 mol

$$\varepsilon_A = \frac{2.2 - 1}{1} = 1.2$$

Thus, it is a changing density system.

Molar flow rate of total feed = 500 mol/h

Feed contains 60% A

Mole fraction of A in the feed = x_A = 60/100 = 0.60

P = 8.5 atm

Molar flow rate of A :

$$F_{Ao} = 0.60 \times 500 = 300\ mol/h$$

Feed is at 8.5 atm and 800 °C

C_{Ao} can be calculated by

$$C_{Ao} = \frac{p_{Ao}}{RT}$$

We have : $p_A = x_A P$. Therefore,

$$p_{Ao} = 0.60 \times 8.5 = 5.1\ atm$$

$$T = 800\ °C = 1073\ K$$

$$R = 0.08206\ l.atm/(mol.\ K)$$

$$C_{Ao} = \frac{5.1}{0.08206 \times 1073} = 0.0579\ mol/l$$

We have to determine the length of reactor for $\frac{p_{A\,out}}{p_{A\,in}}$ to be 0.15. We will make use of this pressure ratio to find out the conversion of A.

For an ideal gas :

$$C_A = \frac{p_A}{RT} \text{ , i.e., } C_A \propto p_A$$

For variable density system, conversion and concentration are related by

$$X_A = \frac{1 - \frac{C_A}{C_{Ao}}}{1 + \varepsilon_A \frac{C_A}{C_{Ao}}} = \frac{1 - \frac{C_{A\,out}}{C_{A\,in}}}{1 + \varepsilon_A \frac{C_{A\,out}}{C_{A\,in}}}$$

$[\text{as } C_A = C_{A\,out} \text{ and } C_{Ao} = C_{A\,in}]$

Since $C_A \propto p_A$, the above expression becomes

$$X_A = \frac{1 - \frac{p_{A\,out}}{p_{A\,in}}}{1 + \varepsilon_A \frac{p_{A\,out}}{p_{A\,in}}}$$

$$\varepsilon_A = 1.2 \quad , \quad \frac{p_{A\,out}}{p_{A\,in}} = 0.15$$

$$X_A = \frac{1 - 0.15}{1 + (1.2 \times 0.15)} = 0.72$$

Using X_A we will find the weight of active catalyst pellets.

For plug flow, the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r_A'}$$

Given : $-r_A' = 10\, C_A = k'\, C_A$ ..... first order reaction

where $k' = 10\ l/(\text{h. g cat})$

$$C_A = \frac{C_{Ao}(1 - X_A)}{(1 + \varepsilon_A X_A)}$$

$$\frac{W}{F_{Ao}} = \frac{1}{k' C_{Ao}} \int_0^{X_A} \left(\frac{1 + \varepsilon_A X_A}{1 - X_A}\right) dX_A$$

Integrating and rearranging, we get

$$W = \frac{F_{Ao}}{k'C_{Ao}} [-(1+\varepsilon_A) \ln (1-X_A) - \varepsilon_A X_A]$$

$\varepsilon_A = 1.2$, $X_A = 0.72$, $F_{Ao} = 300$ mol/h

$k' = 10$ l/(h. g cat), $C_{Ao} = 0.0579$ mol/l

$$W = \frac{300}{10 \times 0.0579} [-(1+1.2) \ln (1-0.72) - 1.2 \times 0.72]$$

$$= 1003.4 \text{ g}$$

Amount (Weight) of active catalyst = 1003.4 g

The reactor is packed with 30% active catalyst pellets and 70% inerts.

$\therefore$ Weight of packed bed (catalyst + inerts)

$$= \frac{1003}{0.30}$$

$$= 3344.67 \text{ g}$$

Given : Particle density

$$\rho_s = 2 \text{ g/cm}^3$$

$$\text{Volume of catalyst + inert particles} = \frac{3344.67}{2}$$

$$= 1672.335 \text{ cm}^3$$

Given : Bulk voidage = 50%

We have :

$$V = \epsilon V_r$$

$$\epsilon = \frac{V}{V_r} = \frac{\text{Volume of reacting fluid}}{\text{Internal volume of reactor}}$$

$$= \frac{\text{Empty volume in bed}}{\text{Total bed volume}}$$

$$1 - \epsilon = 1 - \frac{\text{Empty volume in bed}}{\text{Total bed volume}}$$

$$1 - \epsilon = \frac{\text{Total bed volume} - \text{Empty volume of bed}}{\text{Total bed volume}}$$

But Total bed volume – Empty volume in bed = Volume of particles in the bed

Volume of particles in the bed = Volume of catalyst + Inerts

$\therefore$

$$1 - \epsilon = \frac{\text{Volume of catalyst + Inerts}}{\text{Total bed volume}}$$

Bulk voidage = 50% $\therefore \epsilon = 0.50$

$$1 - 0.50 = \frac{1672.335}{\text{Total bed volume}}$$

$$\text{Total bed volume} = \frac{1672.335}{(1 - 0.50)}$$

$$= 3344.67 \text{ cm}^3$$

Total bed volume = Internal volume of reactor = 3344.67 cm^3

$$\text{Internal volume of reactor} = \frac{\pi}{4} d^2 L = 3344.67$$

Given $d = 2$ cm

$$\frac{\pi}{4} (2)^2 L = 3344.67$$

$$\therefore \quad L = 1064.64 \approx 1065 \text{ cm}$$

Length of tubular reactor = 1065 cm ... Ans.

Extra : Empty volume in the bed is the void volume, i.e., the volume available for reacting fluid.

Volume of particles is the bed volume actually occupied by the active catalyst and inert pellets.

Total bed volume (internal volume of reactor) is the sum of the volume occupied by particles (active catalyst + inerts) and void volume, i.e., empty volume.

Example 6.22 : *The Thiele modulus for a first order isothermal reaction for a flat plate geometry catalyst is found to be 2. Calculate the catalyst effectiveness factor.*

Solution : For a flat plate, the relationship between effectiveness factor η and Thiele modulus (M_T) is

$$\eta = \frac{\tanh M_T}{M_T}$$

$$\text{Thiele modulus} = M_T = 2$$

$$\eta = \frac{\tanh (2)}{2}$$

$$= \frac{0.964}{2} = 0.482$$

Effectiveness factor = 0.482 ... Ans.

Example 6.23 : *The observed rate using a cube of 2 cm catalyst pellet is 10^{-3} mol/(s.cm^3 cat). Calculate the value of first order reaction rate constant. The effective diffusivity is 0.01 cm^2/s and concentration of reactant at the exterior surface of pellet is 2×10^{-4} mol/cm^3.*

Solution : The characteristic size of cube of side 2 cm is

$$L = \frac{\text{Volume of particle pellet}}{\text{Exterior surface available for reactant penetration}}$$

$$= \frac{a^3}{6a} = \frac{a}{6}, \quad a - \text{side of cube}$$

Given : Side of cube = a = 2 cm

$$\therefore \quad L = \frac{2}{6} = 0.333 \text{ cm}$$

For first order reaction $A \rightarrow R$

$$-r_A''' = k''' C_{As} \eta$$

where $$\eta = \frac{\tanh M_T}{M_T}$$

$$-r_A''' = k''' C_{As} \frac{\tanh M_T}{M_T}$$

We have $$M_T = L\left(\frac{k'''}{\mathscr{D}_e}\right)^{0.5} \quad \text{... first order reaction}$$

$$\therefore \quad -r_A''' = \frac{k''' C_{As} \tanh\left[L\left(\frac{k'''}{\mathscr{D}_e}\right)^{0.5}\right]}{L\left(\frac{k'''}{\mathscr{D}_e}\right)^{0.5}}$$

Given : $-r_A''' = 10^{-3}$ mol/(s. cm^3 cat)

$L = 0.333$ cm, $\mathscr{D}_e = 0.01$ cm^2/s

$C_{As} = 2 \times 10^{-4}$ mol/cm^3

$$10^{-3} = \frac{2 \times 10^{-4}\ k''' \tanh\left[0.333\ \left(\frac{k'''}{0.01}\right)^{0.5}\right]}{0.333\ \left(\frac{k'''}{0.01}\right)^{0.5}}$$

$$5 = \frac{k''' \tanh\left[3.33\ (k''')^{0.5}\right]}{3.33\ (k''')^{0.5}}$$

$$(k''')^{0.5} \tanh\left[3.33\ (k''')^{0.5}\right] = 16.65$$

We can obtain the value of k''' by a trial and error procedure. Assume k''' and chec LHS = RHS or not.

$k''' = 200$	LHS = 14.14	RHS = 16.65
$k''' = 270$	LHS = 16.43	RHS = 16.65
$k''' = 278$	LHS = 16.67	RHS = 16.65
$k''' = 277.2$	LHS = 16.649	RHS = 16.65

LHS $\simeq$ RHS for $k''' = 277.2\ s^{-1}$

First order rate constant $= k''' = 277.2\ s^{-1}$ ... A

Units of $k''' \Rightarrow$ cm^3 gas/(cm^3 cat. s) $= s^{-1}$

As tanh $[3.33\ (k''')^{0.5}]$ for $k''' > 2.3$ is 1, we can use easy approach (for getting k''') as gi below :

$A \rightarrow R$ ..... first order reaction

$$-r_A''' = k''' C_{As} \eta$$

Assume strong pore resistance

For strong pore resistance :

$$\eta = \frac{1}{M_T}$$

$$-r_A''' = \frac{k''' C_{As}}{M_T}$$

$$M_T = L\left(\frac{k'''}{\mathscr{D}_e}\right)^{0.50}$$

$$-r_A''' = \frac{k''' C_{As}}{L\left(\frac{k'''}{\mathscr{D}_e}\right)^{0.50}}$$

$$(k''')^{0.50} = \frac{\left(-r_A'''\right) L}{C_{As}\, \mathscr{D}_e^{0.50}}$$

$$k''' = \left[\frac{-r_A''' L}{C_{As}}\right]^2 \times \frac{1}{\mathscr{D}_e}$$

$$= \left[\frac{10^{-3} \times 0.333}{2 \times 10^{-4}}\right]^2 \times \frac{1}{0.01}$$

$$= 277.2\ s^{-1}$$

First order rate constant $= k''' = 277.2\ s^{-1}$

Units of $k''' \Rightarrow$ cm^3 gas/(cm^3 cat. s)

Example 6.24 : *To determine effect of pore diffusion, experiments are carried out on different sizes of crushed catalyst for first order irreversible reaction. The surface concentration of reactant A was $C_{As} = 2 \times 10^{-4}$ mol/cm³. Determine the true rate constant and effective diffusivity. Consider cylindrical pore model and spherical particles.*

Data :

diameter of particle, cm	*0.20*	*0.02*
observed reaction rate, mol/(h. cm³ cat)	*0.12*	*1.03*

Solution : For first order reaction in cylindrical pore :

$$\eta = \frac{\tanh mL}{mL} = \frac{\tanh M_T}{M_T}$$

$$\frac{\eta_1}{\eta_2} = \frac{(M_T)_2 \tanh (M_T)_1}{(M_T)_1 \tanh (M_T)_2}$$

$$M_T = L\left(\frac{k'''}{\mathscr{D}_e}\right)^{0.50}, \quad L = \frac{R}{3} \quad \text{..... for spherical particle}$$

$$\frac{(M_T)_1}{(M_T)_2} = \frac{L_1}{L_2} = \frac{R_1}{R_2}$$

$$\therefore \quad \frac{\eta_1}{\eta_2} = \frac{\left(-r_A'''\right)_1}{\left(-r_A'''\right)_2} = \frac{R_2}{R_1} \frac{\tanh (M_T)_1}{\tanh (M_T)_2}$$

$$\frac{\tanh (M_T)_1}{\tanh (M_T)_2} = \frac{\left(-r_A'''\right)_1 R_1}{\left(-r_A'''\right)_2 R_2}$$

$$R_1 = \frac{0.20}{2} = 0.1 \text{ cm} \quad , \quad R_2 = \frac{0.02}{2} = 0.01 \text{ cm}$$

$$\left(-r_A'\right)_1 = 0.12 \qquad \left(-r_A'\right)_2 = 1.03$$

$$\frac{\tanh (M_T)_1}{\tanh (M_T)_2} = \frac{0.12 \times 0.10}{1.03 \times 0.01}$$

$$= 1.165$$

$$\frac{(M_T)_1}{(M_T)_2} = \frac{R_1}{R_2} = \frac{0.10}{0.01} = 10$$

$$(M_T)_1 = 10\,(M_T)_2$$

As $(M_T)_1$ is large, 10 times as large, $\tanh (M_T)_1 = 1$

$$\frac{\tanh (M_T)_1}{\tanh (M_T)_2} = 1.165$$

$$\therefore \quad \frac{1}{\tanh (M_T)_2} = 1.165$$

$$\tanh (M_T)_2 = \frac{1}{1.165} = 0.85836$$

$$\therefore \quad (M_T)_2 = \tanh^{-1} (0.85836) = 1.2871$$

$$(M_T)_2 = 1.2871$$

$$\eta_2 = \frac{\tanh (M_T)_2}{(M_T)_2}$$

$$= \frac{0.8583}{1.2871}$$

$$= 0.66689 \simeq 0.667$$

$$\left(-r_A'''\right)_2 = k''' C_{As} \eta_2$$

$$k''' = \frac{\left(-r_A'''\right)_2}{C_{As} \eta_2}$$

$$= \frac{1.03}{2 \times 10^{-4} \times 0.667} = 7.721 \times 10^3 \text{ h}^{-1}$$

First order rate constant = $k''' = 7.721 \times 10^3 \text{ h}^{-1}$ ... Ans.

$$(M_T)_2 = L_2 \left(\frac{k'''}{\mathscr{D}_e}\right)^2$$

$$\mathscr{D}_e = \frac{L_2^2 (k''')}{[(M_T)_2]^2}$$

$$L_2 = \frac{R_2}{3} \quad \ldots\ldots \text{ for spherical particles}$$

$$= \frac{0.01}{3} = 3.33 \times 10^{-3} \text{ cm}$$

$$\mathscr{D}_e = \frac{(3.33 \times 10^{-3})^2 \times 7.721 \times 10^3}{(1.2871)^2}$$

$$= 0.05168$$

$$\simeq 0.0517 \text{ cm}^3/\text{h}$$

Effective diffusivity = $\mathscr{D}_e = 0.0517 \text{ cm}^3/\text{h}$ ... Ans

$$\eta_1 = \frac{\tanh (M_T)_1}{(M_T)_1} = \frac{\tanh [10(M_T)_2]}{10(M_T)_2}$$

$$= \frac{\tanh [10 \times 1.2871]}{10 \times 1.2871} = \frac{1}{10 \times 1.2871}$$

$$= 0.07769 \simeq 0.0777$$

Example 6.25 : *What should be the size of particles to virtually eliminate pore diffusion resistance ? Use the data of Example 6.24.*

Solution : We have to determine size of particles for which the effect of pore resistance is negligible, i.e., a size for which pore diffusion offers negligible resistance to reaction.

For pore diffusion to offer negligible resistance : $M_T \leq 0.40$.

The largest particle size free of pore diffusion resistance corresponds to $M_T = 0.40$.

For particles of size R_3,

$$\eta_3 = \frac{\tanh (M_T)_3}{(M_T)_3},$$

$$\eta_3 = \frac{\tanh (0.40)}{0.40} = 0.95$$

i.e., for the largest possible size which is free of pore diffusional resistance, the effectiveness factor is 0.95.

$$(M_T)_3 = 0.4 \text{ for } R_3$$

$$M_T = L \left(\frac{k'''}{D_e}\right)^{0.5}$$

and $$L = \frac{R}{3} \text{ for spherical particles}$$

$$\frac{(M_T)_3}{(M_T)_2} = \frac{R_3}{R_2}$$

$$R_3 = \frac{(M_T)_3 R_2}{(M_T)_2}, \quad (M_T)_2 = 1.2871, \; R_2 = 0.01 \text{ cm} \ldots \text{see Example 6.24.}$$

$$R_3 = \frac{0.4 \times 0.01}{1.2871} = 3.1 \times 10^{-3} \text{ cm}$$

Largest size of spherical particles (which is free of pore diffusion resistance) = R = 3.1×10^3 cm ... **Ans.**

Example 6.26 : *For the catalytic reaction A → 4R, the following rate-concentration data are available :*

C_A, mol/l	0.039	0.0575	0.075	0.092
$-r'_A$, mol A/(h. kg cat)	3.4	5.4	7.6	9.1

Find the size of packed bed, i.e., the amount of catalyst required to treat 2500 mol/h of pure A at 117 °C to achieve 35% conversion, all at 3.2 atm pressure, directly from the data given.

Solution : $A \rightarrow 4R$

For pure A and for this reaction, 1 volume feed on complete conversion yields 4 volumes of product. Thus

$$\varepsilon_A = \frac{4-1}{1} = 3$$

Therefore, we have changing density system.

We have : $C_{Ao} = \frac{p_{Ao}}{RT}$

where p_{Ao} = total pressure = 3.2 atm. ..... since pure A feed

$R = 0.08206\ l.\text{atm/(mol. K)}$

$T = 117\ ^\circ\text{C} = 390\ \text{K}$

$$C_{Ao} = \frac{3.2}{0.08206 \times 390} = 0.0999 \simeq 0.1\ \text{mol}/l$$

$F_{Ao} = 2500\ \text{mol/h}$

For a variable density system, C_A and X_A are related by

$$X_A = \frac{1 - \frac{C_A}{C_{Ao}}}{1 + \frac{\varepsilon_A C_A}{C_{Ao}}}$$

with $\varepsilon_A = 3$ $$X_A = \frac{1 - \frac{C_A}{C_{Ao}}}{1 + \frac{3C_A}{C_{Ao}}}$$

For a packed bed reactor assuming plug flow behaviour, the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r_A'} \qquad \ldots (1)$$

We can determine W knowing RHS of the above equation.

To determine the RHS, we have to make use of the data provided.

Calculate X_A for each C_A.

$$X_A = \frac{1 - \frac{C_A}{C_{Ao}}}{1 + 3\frac{C_A}{C_{Ao}}}$$

$C_{Ao} = 0.10\ \text{mol}/l$

(i) $C_A = 0.039$

$$X_A = \frac{1 - \left(\frac{0.039}{0.10}\right)}{1 + \left(\frac{3 \times 0.039}{0.10}\right)} = 0.281$$

(ii) $C_A = 0.0575$

$$X_A = \frac{1-\left(\frac{0.0575}{0.10}\right)}{1+\left(\frac{3\times 0.0575}{0.10}\right)} = 0.156$$

(iii) $C_A = 0.075$

$$X_A = \frac{1-\left(\frac{0.075}{0.10}\right)}{1+\left(\frac{3\times 0.075}{0.10}\right)} = 0.0769 \approx 0.077$$

(iv) $C_A = 0.092$

$$X_A = \frac{1-\left(\frac{0.092}{0.10}\right)}{1+\left(\frac{3\times 0.092}{0.10}\right)} = 0.0213$$

Now we have $-r_A'$ – X_A data. With this data we can determine the value of integral of Equation (1).

Given : X_A desired $= \frac{35}{100} = 0.35$

$$\frac{W}{F_{Ao}} = \int_0^{0.35} \frac{dX_A}{-r_A'}$$

For a graphical evaluation of the integral on the RHS of the above equation, plot $\frac{1}{(-r_A')}$ v/s X_A and measure the area under the curve between $X_A = 0$ and $X_A = 0.35$.

C_A	0.039	0.0575	0.075	0.092
$-r_A'$	3.4	5.4	7.6	9.1
$\frac{1}{(-r_A')}$	0.294	0.185	0.131	0.11
X_A	0.281	0.156	0.077	0.0213

From the plot, i.e., Fig. Ex 6.26 :

Area under the curve between $X_A = 0$ and $X_A = 0.35 = 2925\ mm^2$

$$\int_0^{0.35} \frac{dX_A}{-r_A'} = \text{Area} \times \text{Scale of x-axis} \times \text{Scale of y-axis}$$

$$= 2925\ mm^2 \times \frac{0.1}{20\ mm} \times \frac{0.1}{20\ mm}$$

$$= 0.0731\ (h.\ kg\ cat.)/mol = \frac{W}{F_{Ao}}$$

$\therefore$ $$W = 0.0731 \times F_{Ao}$$

$$= 0.0731 \times 2500 = 182.75\ kg$$

Amount of catalyst required = 182.75 kg ... Ans.

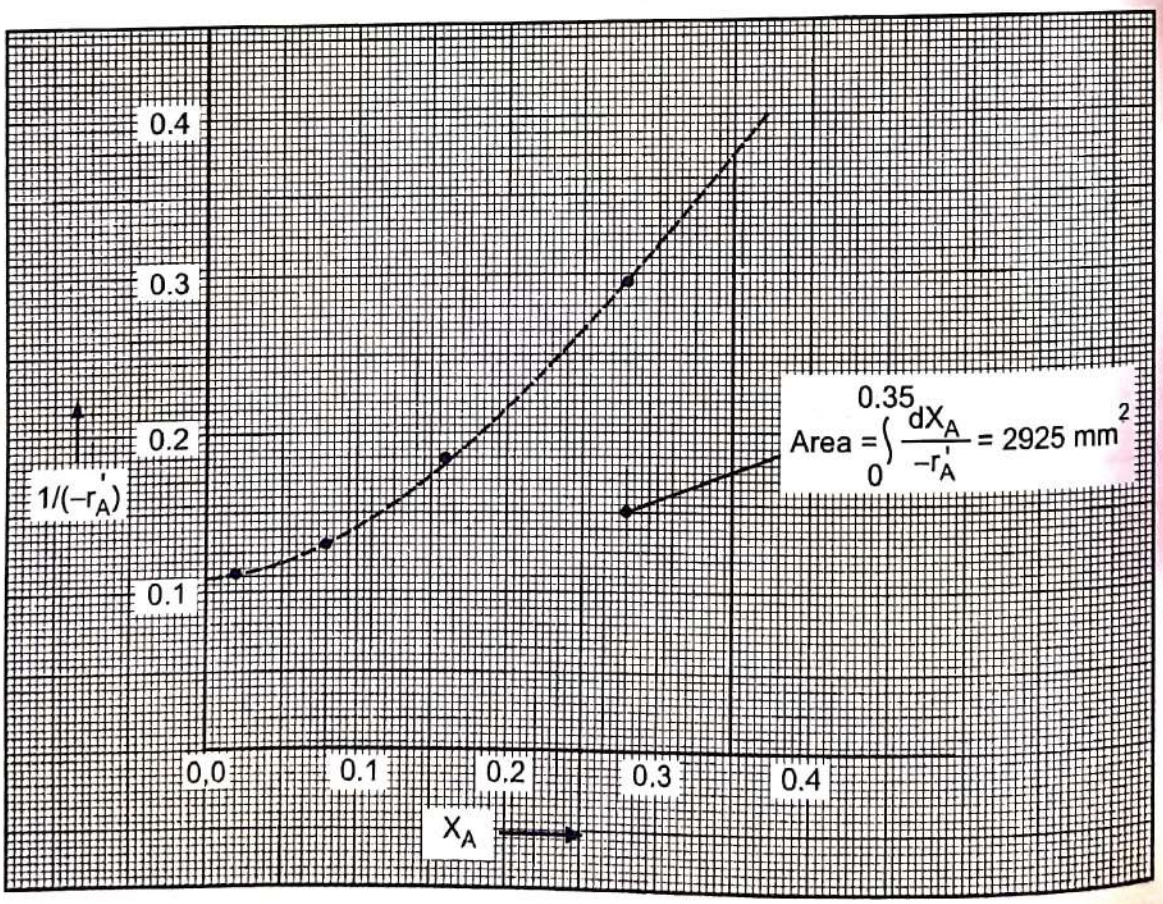


Fig. Ex 6.26

Example 6.27 : *The catalytic reaction A → 3R is run in a packed bed reactor at 3.5 atm and 115 °C. It is desired to treat 1500 mol/h of pure A at 3.5 atm to 32% conversion. The following rate – concentration data are available :*

C_A, mol/l	0.04	0.06	0.075	0.09
$-r_A'$, mol A/(h. kg cat)	3.5	5.7	7.2	8.8

Determine directly from the data, the amount of catalyst needed in the packed bed reactor for this reaction and feed.

Solution : A → 3R

$$\varepsilon_A = \frac{3-1}{1} = 2 \text{ ..... for pure A}$$

$$C_{Ao} = \frac{p_{Ao}}{RT}$$

p_{Ao} = 3.5 atm. ..... pure A feed

R = 0.08206 *l*.atm/(mol. K)

T = 115 °C = 388 K

$$C_{Ao} = \frac{3.5}{0.08206 \times 388}$$

$$= 0.1099$$

$$\simeq 0.11 \text{ mol/}l$$

Conversion to be achieved = $X_A = \frac{32}{100} = 0.32$

F_{Ao} = 1500 mol/h

We have to determine W by making use of the data provided.

The performance equation for a packed bed reactor assuming plug flow is

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r_A'} = \int_0^{0.32} \frac{dX_A}{-r_A'} \quad \text{... (1)}$$

We can determine W knowing RHS for Equation (1).

The integral on the RHS of Equation (1) can be evaluated graphically. Graphical evaluation requires plotting $\frac{1}{\left(-r_A'\right)}$ v/s X_A and measuring the area under the curve between $X_A = 0$ and $X_A = 0.32$ (Given).

Therefore, we have to convert $\left(-r_A'\right) - C_A$ data to $\left(-r_A'\right) - X_A$, i.e., we have to convert C_A to X_A.

For a variable density system, C_A and X_A are related by

$$X_A = \frac{1 - \frac{C_A}{C_{Ao}}}{1 + \frac{\varepsilon_A C_A}{C_{Ao}}}$$

with $\varepsilon_A = 2$

$$X_A = \frac{1 - \frac{C_A}{C_{Ao}}}{1 + \frac{2C_A}{C_{Ao}}}$$

For $C_A = 0.04$ and $C_{Ao} = 0.11$ mol/l

$$X_A = \frac{1 - \left(\frac{0.04}{0.11}\right)}{1 + \left(\frac{2 \times 0.04}{0.11}\right)} = 0.368$$

For $C_A = 0.06$: $X_A = 0.217$, For $C_A = 0.09$: $X_A = 0.069$

For $C_A = 0.075$: $X_A = 0.135$

C_A	0.04	0.06	0.075	0.09
$-r_A'$	3.5	5.7	7.2	8.8
X_A	0.368	0.217	0.135	0.069
$\frac{1}{\left(-r_A'\right)}$	0.2857	0.1754	0.1389	0.1136

Plot $\frac{1}{\left(-r_A'\right)}$ v/s X_A and measure the area under the curve between $X_A = 0$ and $X_A = 0.32$.

From the plot [Fig. Ex 6.27]

Area under the curve = 1950 mm^2

$$\frac{W}{F_{Ao}} = \int_0^{0.32} \frac{dX_A}{-r_A'} = \text{Area} \times \text{Scale of x-axis} \times \text{Scale of y-axis}$$

$$= 1950 \times \frac{0.10}{20} \times \frac{0.10}{20}$$

$$= 0.04875 \ \frac{(\text{h. kg cat})}{\text{mol}}$$

$$\frac{W}{F_{Ao}} = 0.04875$$

$$F_{Ao} = 1500 \text{ mol/h}$$

$$W = 0.04875 \times 1500$$

$$= 73.125 \text{ kg}$$

Amount of catalyst needed = 73.125 kg **... Ans.**

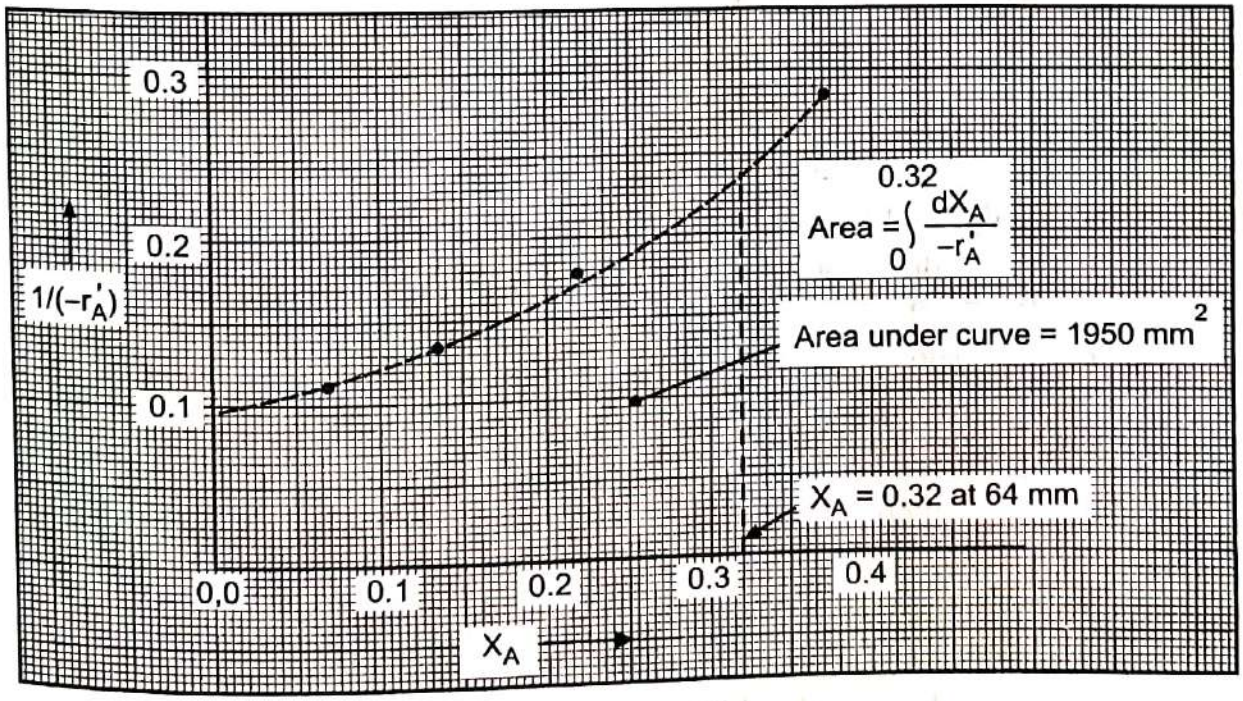


Fig. Ex 6.27

Example 6.28 : *The first order decomposition of A is conducted in an experimental mixed flow reactor. Determine whether the following runs were made under diffusion-free, strong resistance, or intermediate conditions.*

(a)	dp	W	C_{Ao}	ν	X_A	
	8	3	300	60	0.80	$A \rightarrow R$
	4	1	100	160	0.60	

(b)	dp	W	C_{Ao}	ν	X_A	
	2	4	75	10	0.2	$A \rightarrow R$
	1	6	100	5	0.6	

Solution :

(a)	dp	W	C_{Ao}	ν	X_A	
	8	3	300	60	0.80	$A \rightarrow R$
	4	1	100	160	0.60	

For mixed flow reactor :

$$\frac{W}{F_{Ao}} = \frac{W}{\nu\, C_{Ao}} = \frac{X_A}{-r_A'}$$

$$-r_A' = \frac{X_A\, \nu\, C_{Ao}}{W}$$

For particles of size $dp_1 = 8$:

$$R_1 = \frac{dp_1}{2} = \frac{8}{2} = 4$$

$C_{Ao\,1} = 300,\ X_{A1} = 0.80,\ \nu_1 = 60,\ W_1 = 3$

$$-r_{A1}' = \frac{0.80 \times 60 \times 300}{3} = 4800$$

For particles of size $dp_2 = 4$:

$$R_2 = \frac{dp_2}{2} = \frac{4}{2} = 2$$

$X_{A2} = 0.60,\ C_{Ao2} = 100,\ \nu_2 = 160,\ W_2 = 1$

$$-r_{A2}' = \frac{X_{A2}\ C_{Ao2}\ \nu_2}{W_2}$$

$$= \frac{0.60 \times 100 \times 160}{1} = 9600$$

For diffusion-free regime :

$$\frac{-r_{A1}'}{-r_{A2}'} = 1 \quad \ldots\ldots \text{ should be satisfied}$$

For $dp_1 = 8 : -r'_{A1} = 4800$

For $dp_2 = 4 : -r'_{A2} = 9600$

$$\frac{-r'_{A1}}{-r'_{A2}} = \frac{4800}{9600} \neq 1$$

As $$\frac{-r'_{A1}}{-r'_{A2}} \neq 1 \ldots\ldots$$

The runs are not conducted under diffusion-free conditions **... Ans. (a)**

Check for strong pore diffusion resistance :

In the regime of strong pore diffusion :

$$\frac{-r'_{A1}}{-r'_{A2}} = \frac{R_2}{R_1} \ldots\ldots \text{ should be satisfied}$$

$-r'_{A1} = 4800 \ldots\ldots\ldots R_1 = 4 \ldots\ldots$ for $dp_1 = 8$

$-r'_{A2} = 9600 \ldots\ldots\ldots R_2 = 2 \ldots\ldots$ for $dp_2 = 4$

$$\frac{4800}{9600} = 0.50 = \frac{2}{4} = 0.50 \ldots\ldots \text{ satisfied as LHS = RHS}$$

Since $$\frac{-r'_{A1}}{-r'_{A2}} = \frac{R_2}{R_1}$$

i.e., $$-r'_A \propto \frac{1}{R} \ldots\ldots\ldots$$

..... the runs are conducted under conditions of strong pore resistance i.e., runs are conducted in strong pore diffusion resistance regime **... Ans. (a)**

(b)

dp	W	C_{Ao}	ν	X_A	
2	4	75	10	0.20	$A \rightarrow R$
1	6	100	5	0.60	

For $dp_1 = 2 \ldots\ldots R_1 = \frac{2}{2} = 1$:

$W_1 = 4, \quad C_{Ao1} = 75, \quad \nu_1 = 10, \quad X_{A1} = 0.20$

$$-r'_{A1} = \frac{X_{A1}\, \nu_1\, C_{Ao1}}{W_1}$$

$$= \frac{0.2 \times 10 \times 75}{4} = 37.5$$

For $dp_2 = 1$ ..... $R_2 = \frac{1}{2} = 0.50$:

$W_2 = 6, \quad C_{Ao2} = 100, \quad \nu_2 = 5, \quad X_{A2} = 0.60$

$$-r'_{A2} = \frac{X_{A2}\,\nu_2\,C_{Ao2}}{W_2}$$

$$= \frac{0.60 \times 5 \times 100}{6} = 50$$

For diffusion-free regime :

$$\frac{-r'_{A1}}{-r'_{A2}} = 1$$

$-r'_{A1} = 37.5, \qquad -r'_{A2} = 50$

$\frac{37.5}{50} \neq 1$

..... so the runs are not conducted under diffusion-free conditions ... Ans. (b)

In regime of strong pore resistance :

$$\frac{-r'_{A1}}{-r'_{A2}} = \frac{R_2}{R_1} \quad \text{i.e. } -r'_A \propto \frac{1}{R} \quad \text{..... should be satisfied}$$

$-r'_{A1} = 37.5$ ..... $R_1 = 1$

$-r'_{A2} = 50$ ..... $R_2 = 0.50$

$$\frac{-r'_{A1}}{-r'_{A2}} = \frac{37.5}{50} = 0.75 \neq \frac{R_2}{R_1} = \frac{0.5}{1} = 0.5$$

As $-r'_A$ is not inversely proportional to R, the runs are not conducted under strong pore resistance conditions.
So the runs are conducted under intermediate conditions. ... Ans. (b)

Example 6.29 : *The following kinetic data on the catalytic reaction A → 4R are obtained in a plug flow reactor using various amounts of catalyst and 20 l/h of pure A feed at 3.2 atm and 117 °C.*

Run	*1*	*2*	*3*	*4*
W, kg catalyst	*0.02*	*0.04*	*0.08*	*0.16*
$C_{A\,out}$, *mol/l*	*0.074*	*0.06*	*0.044*	*0.029*

Find a rate equation to represent this reaction, using the integral method of analysis.

Solution : A → 4R

For pure A and for this reaction, 1 mol of feed on complete conversion gives 4 mol of product. Therefore, the fractional change in volume of the reaction system is

$$\varepsilon_A = \frac{4-1}{1} = 3$$

Let us find C_{Ao},

$$C_{Ao} = \frac{p_{Ao}}{RT}$$

$p_{Ao} = 3.2$ atm ..... pure A feed

$R = 0.08206$ l.atm/(mol. K)

$T = 117\,^{\circ}C = 390$ K

$$C_{Ao} = \frac{3.2}{0.08206 \times 390}$$

$$= 0.099989 \approx 0.10 \text{ mol/}l$$

20 l/h pure A feed at 3.2 atm and 117 °C

$v_o = 20$ l/h

$F_{Ao} = C_{Ao} \times v_o = 0.10 \times 20 = 2$ mol/h

For a plug flow reactor, the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r_A'}$$

Guess-1 : Assume that the reaction $A \rightarrow 4R$ follows first order kinetics.

$$\therefore \quad -r_A' = k' C_A$$

For changing density systems, C_A and X_A are related by

$$C_A = \frac{C_{Ao}(1-X_A)}{(1+\varepsilon_A X_A)}$$

or

$$X_A = \frac{1-\frac{C_A}{C_{Ao}}}{1+\frac{\varepsilon_A C_A}{C_{Ao}}}$$

$$-r_A' = \frac{k' C_{Ao}(1-X_A)}{(1+\varepsilon_A X_A)} = \frac{k' C_{Ao}(1-X_A)}{(1+3X_A)}$$

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{\frac{k'C_{Ao}(1-X_A)}{(1+3X_A)}}$$

$$\frac{W}{F_{Ao}} = \frac{1}{k'C_{Ao}} \int_0^{X_A} \frac{(1+3X_A)}{(1-X_A)} dX_A$$

$$\frac{W}{F_{Ao}} = \frac{1}{k'C_{Ao}}\left[\int_0^{X_A} \frac{(1+3X_A)}{(1-X_A)}\,dX\right]$$

$$\frac{W}{F_{Ao}} = \frac{1}{k'C_{Ao}}\left[4\,ln\left(\frac{1}{1-X_A}\right) - 3X_A\right]$$

$$\left[4\,ln\left(\frac{1}{1-X_A}\right) - 3X_A\right] = k'\left(\frac{W\,C_{Ao}}{F_{Ao}}\right) \quad \ldots(1)$$

A plot of $\left[4\,ln\left(\frac{1}{1-X_A}\right) - 3X_A\right]$ v/s $\frac{WC_{Ao}}{F_{Ao}}$ should result in a straight line passing through the origin for our guess regarding the order of reaction to be correct.

$$X_A = \frac{1-\frac{C_A}{C_{Ao}}}{1+\frac{\varepsilon_A C_A}{C_{Ao}}} = \frac{1-\frac{C_A}{C_{Ao}}}{1+\frac{3C_A}{C_{Ao}}} \quad \ldots(2)$$

Calculate X_A for each C_A using Equation (2). Using X_A and W, evaluate LHS and RHS $\left(\frac{W\,C_{Ao}}{F_{Ao}}\right)$ of Equation (1) and tabulate.

Run 1 : $C_A = 0.074$ mol/*l*, $C_{Ao} = 0.10$ mol/*l*

$$X_A = \frac{1-\left(\frac{0.074}{0.10}\right)}{1+\left(\frac{3\times 0.074}{0.10}\right)} = 0.0807 \simeq 0.081$$

$$4\,ln\left(\frac{1}{1-X_A}\right) - 3X_A = 4\,ln\left(\frac{1}{1-0.081}\right) - 3\times 0.081$$

$$= 0.09489 = 0.095$$

$$W = 0.02 \text{ kg}$$

$$\left(\frac{W\,C_{Ao}}{F_{Ao}}\right) = \frac{0.02\times 0.10}{2} = 0.001$$

Run	1	2	3	4
$C_{A\,out}$	0.074	0.06	0.044	0.029
X_A	0.081	0.143	0.241	0.380
$4\,ln\left(\frac{1}{1-X_A}\right) - 3X_A$	0.095	0.188	0.38	0.77
$\frac{WC_{Ao}}{F_{Ao}}$	0.001	0.002	0.004	0.008
W	0.02	0.04	0.08	0.16

Refer Fig. Ex 6.29 :

Plot of $\left[4\,ln\left(\frac{1}{1-X_A}\right)-3\,X_A\right]$ v/s $\frac{W\,C_A}{F_{Ao}}$ yields a straight line passing through the origin. Thus, our guess regarding the reaction order is correct, i.e., the decomposition of A, (A → 4R) is first order.

From the plot : slope = k' = 96.30 *l*/(h.kg cat)

The rate equation is

$$-r_A' = -\frac{1}{W}\frac{dN_A}{dt} = \left(96.30\,\frac{l}{\text{h. kg cat}}\right)\left(C_A,\,\frac{\text{mol}}{l}\right)$$ **... Ans.**

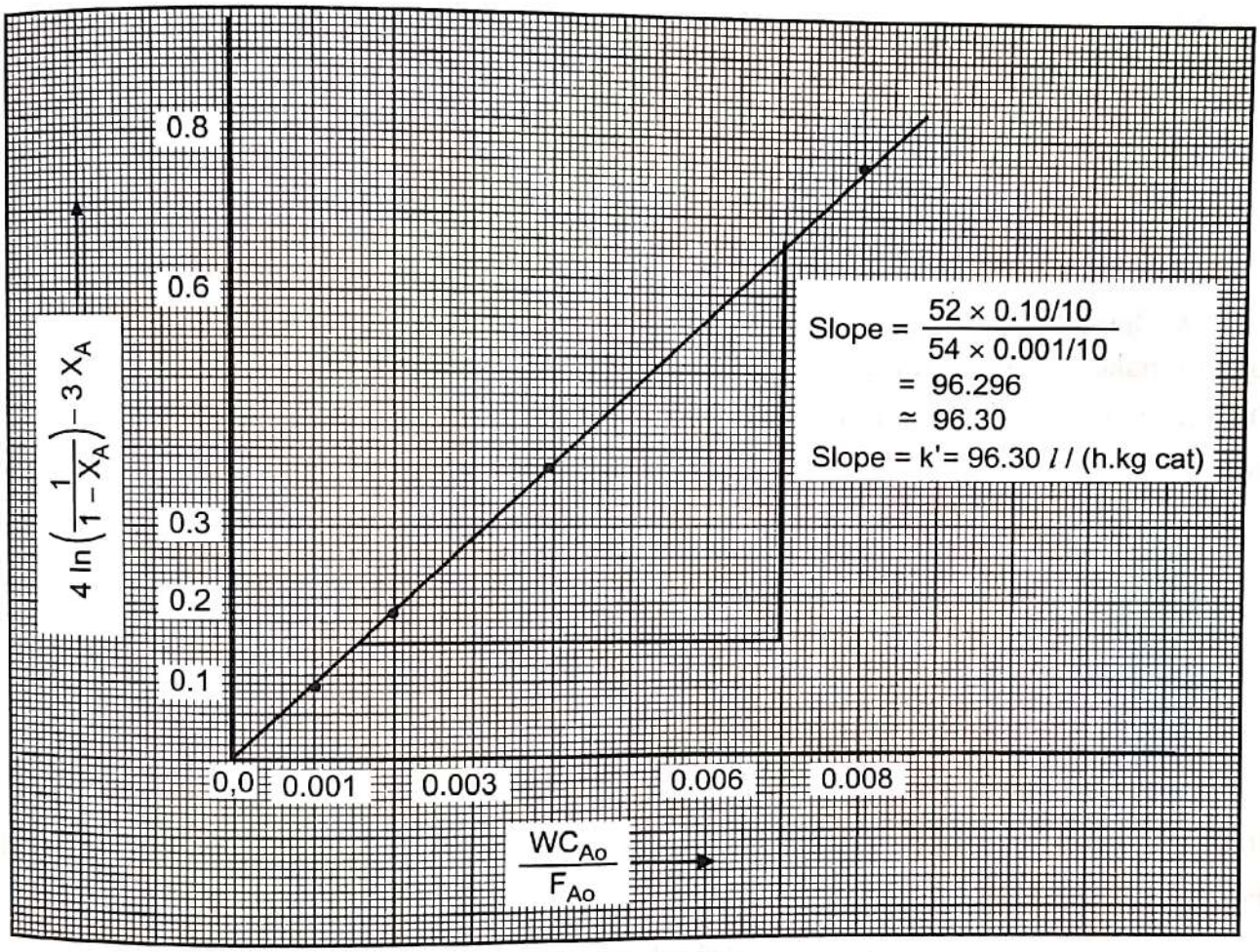


Fig. Ex 6.29

Example 6.30 : *A closed-loop experimental batch G/S reactor, as shown in Fig. Ex 6.30 (a), is used to study the kinetics of a catalytic reaction A → 2 R. Pure A is fed into the system and is circulated at 0 °C and 1 atm at a rate of 10 cm³/s. The results of analysis of the stream from time to time are reported as follows :*

Time, min	*0*	*2*	*4*	*6*	*8*	*10*	*12*	*14*	*(∞)*
Partial pressure of A, mm Hg	*760*	*600*	*475*	*390*	*320*	*275*	*240*	*215*	*150*

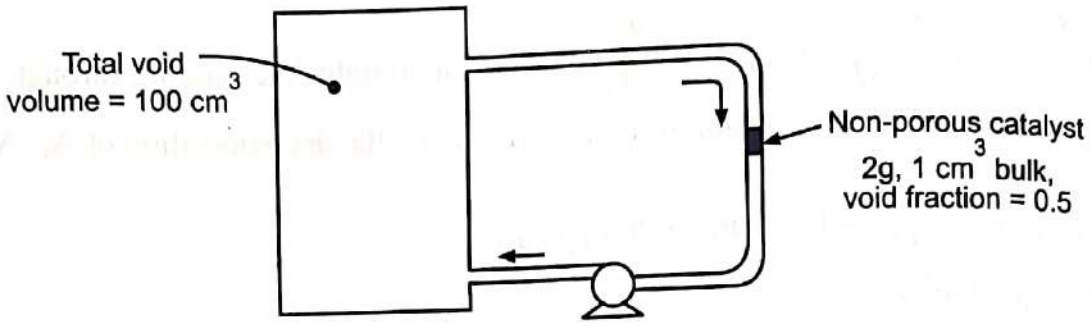


Fig. Ex 6.30 (a)

(i) Find the size of a plug flow reactor operating at 0 °C and 1 atm needed to achieve 50% conversion of A for a feed rate of 100 kmol A/h.

(ii) What would be the size of a plug flow reactor if an inert at a partial pressure of 1 atm is present in the closed loop so that the total pressure at the start is 2 atm. Use the rate equation obtained in part (i).

Solution : (i) To find the size of plug flow reactor for :

$A \rightarrow 2\,R$, $X_A = 0.50$ and $F_{Ao} = 100$ kmol/h

First we have to find the rate equation which will satisfactorily represent the data and then we have to make use of the rate equation to find the size of plug flow reactor.

Data provided are collected in an experimental batch G/S reactor.

For batch reactor for catalytic reaction [shown in Fig. Ex 6.30 (a)]

$$\frac{t}{C_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r_A'} = \frac{V}{W} \int_0^{X_A} \frac{dX_A}{-r_A'}$$

$$\frac{t}{C_{Ao}} = \frac{V}{W} \int_0^{X_A} \frac{dX_A}{-r_A'}$$

In order to use the data provided we have to find the conversion corresponding to each p_A.

For the gas phase reaction

$$A \rightarrow 2\,R \ \ldots \text{ let us find } \varepsilon_A$$

Basis : 1 mol of feed. Initial moles = 1 mol A + 0 mol R = 1 mol

Moles on complete conversion = 0 mol A + (2/1) × 1 = 2 mol R = 0 = 2 = 2 mol

$$\varepsilon_A = \frac{\text{moles on complete conversion} - \text{initial moles}}{\text{initial moles}}$$

$$= \frac{2-1}{1} = 1 \ \ldots \text{ for pure A}$$

For this changing density system, C_A and X_A are related by

$$C_A = \frac{C_{Ao}(1 - X_A)}{(1 + \varepsilon_A X_A)}$$

or
$$X_A = \frac{1 - \frac{C_A}{C_{Ao}}}{1 + \frac{\varepsilon_A C_A}{C_{Ao}}}$$

$\therefore$
$$X_A = \frac{1 - \frac{C_A}{C_{Ao}}}{1 + \frac{C_A}{C_{Ao}}}, \text{ as } \varepsilon_A = 1$$

C_A is related to p_A by

$$C_A = \frac{p_A}{RT} \quad , \quad C_{Ao} = \frac{p_{Ao}}{RT}$$

$\therefore$
$$\frac{C_A}{C_{Ao}} = \frac{p_A}{p_{Ao}}$$

where p_A is the partial pressure of A at any time t

p_{Ao} is the initial partial pressure of A (i.e., at t = 0)

p_{Ao} = 760 mm Hg ..... as pure A is feed.

$$X_A = \frac{1 - \frac{p_A}{760}}{1 + \frac{p_A}{760}}$$ ..... using this get X_A corresponding to each p_A.

t	0	2	4	6	8	10	12	14
p_A	760	600	475	390	320	275	240	215
X_A	0	0.12	0.23	0.32	0.41	0.47	0.47	0.56

For p_A = 600 mm Hg .....

$$X_A = \frac{1 - \left(\frac{600}{760}\right)}{1 + \left(\frac{600}{760}\right)} = 0.1176 \simeq 0.12$$

Let us find the rate equation, i.e., n and k.

Guess 1 : First order reaction, i.e., try a first order rate equation

$$-r_A' = k' C_A = \frac{k' C_{Ao}(1 - X_A)}{(1 + X_A)}, \quad \varepsilon_A = 1$$

$\therefore$
$$\frac{t}{C_{Ao}} = \frac{V}{W} \int_0^{X_A} \frac{dX_A}{\frac{k' C_{Ao}(1 - X_A)}{(1 + X_A)}}$$

$$t = \frac{V}{Wk'} \int_0^{X_A} \frac{(1 + X_A)\, dX_A}{(1 - X_A)}$$

$$t = \frac{V}{Wk'} [-2\, ln\, (1 - X_A) - X_A]$$

$$y = mx$$

$\therefore$ y = t, $x = [-2\, ln\, (1 - X_A) - X_A]$

and $m = \frac{V}{Wk'}$

Thus, a plot of t v/s $[-2\, ln\, (1 - X_A) - X_A]$ should yield a straight line passing through the origin for our guess (the reaction order) to be correct.

t	0	2	4	6	8	10	12	14
X_A	0	0.12	0.23	0.32	0.41	0.47	0.52	0.56
$[-2\, ln\, (1 - X_A) - X_A]$	0	0.136	0.293	0.45	0.645	0.80	0.95	1.082

Plot t v/s $[-2\, ln\, (1 - X_A) - X_A]$

Refer to Fig. Ex 6.30 (b) :

The straight line fit of the data indicates that our guess is correct, i.e., the reaction is first order.

$$\text{Slope} = \frac{V}{Wk'} = 12.31 \text{ min}$$

$$k' = \frac{V}{12.31\, W}$$

where W = 2 g,

V = Volume of gas = total void volume = 100 cm^3

$$k' = \frac{100}{12.31 \times 2} = 4.06 \frac{cm^3}{(g \text{ cat. min})} = 4.06 \frac{l}{(kg \text{ cat.min})}$$

The rate equation is

$$\boxed{-r_A' = \left(4.06 \frac{l}{kg \text{ cat. min}}\right)\left(C_A \frac{mol}{l}\right)}$$

We have to make use of this rate expression to determine the size (i.e., the amount of catalyst) for the plug flow reactor.

For plug flow, the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r_A'} \qquad -r_A' = k' C_{Ao} \frac{(1 - X_A)}{(1 + X_A)}$$

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{(1 + X_A)\, dX_A}{k' C_{Ao} (1 - X_A)}$$

$$\frac{W}{F_{Ao}} = \frac{1}{k' C_{Ao}} [-2\, ln\, (1 - X_A) - X_A]$$

$$C_{Ao} = \frac{p_{Ao}}{RT}, \qquad p_{Ao} = 1 \text{ atm} = 760 \text{ mm Hg} \ldots\ldots \text{ pure A}$$

$$= \frac{1}{0.08206 \times 273}, \quad T = 0 + 273 = 273 \text{ K}$$

$$= 0.0446 \text{ mol/}l$$

$$W = \frac{F_{Ao}}{k' C_{Ao}} [-2\, ln\, (1 - X_A) - X_A]$$

$k' = 4.06\ l$/(kg cat. min), $C_{Ao} = 0.0446$ mol/l

Given : $F_{Ao} = 100$ kmol/h $= 10^5$ mol/h $= 1666.67$ mol/min

$$X_A = \frac{50}{100} = 0.50$$

$$W = \frac{1666.67}{4.06 \times 0.0446} [-2\, ln\, (1 - 0.50) - 0.5]$$

$$= 8157 \text{ kg}$$

Amount of catalyst needed = 8157 kg ..... **... Ans. (i)**

(ii) Let us find the size of plug flow reactor when :

Partial pressure of inert = 1 atm

Total pressure at the start = 2 atm

$\therefore$ Partial pressure of A at the start = 2 – 1 = 1 atm

Therefore, initially we have 50% A + 50% inerts by volume

$A \rightarrow 2R$

At $X_A = 0$ ..... 1 mol A + 1 mol inert ... total moles = 2 mol

At $X_A = 1$ ..... 2 mol R + 1 mol inert + 0 mol A ... total moles = 3 mol

Therefore, $\varepsilon_A = \frac{3 - 2}{2} = 0.50$

Thus, it is a changing density reaction system.

Here also the reaction is first order with $k' = 4.06\ l$/(kg cat.min).

Integrated form of the performance equation of a plug flow reactor for a first order reaction is

$$W = \frac{F_{Ao}}{k' C_{Ao}} [-(1 + \varepsilon_A)\, ln\, (1 - X_A) - \varepsilon_A X_A]$$

F_{Ao} = 1666.67 mol/min [feed rate is the same as in the part (i)]

k' = 4.06 *l*/(kg cat. min)

$$C_{Ao} = \frac{1}{0.08206 \times 273} = 0.0446 \text{ mol/}l, \text{ as } p_{Ao} = 1 \text{ atm}$$

$\varepsilon_A = 0.50, \quad X_A = 0.50$

$$W = \frac{1666.67}{4.06 \times 0.0446} [-(1 + 0.50) \ln (1 - 0.50) - 0.50 \times 0.50]$$

= 7269 kg

Amount of catalyst needed = 7269 kg ... Ans. (ii)

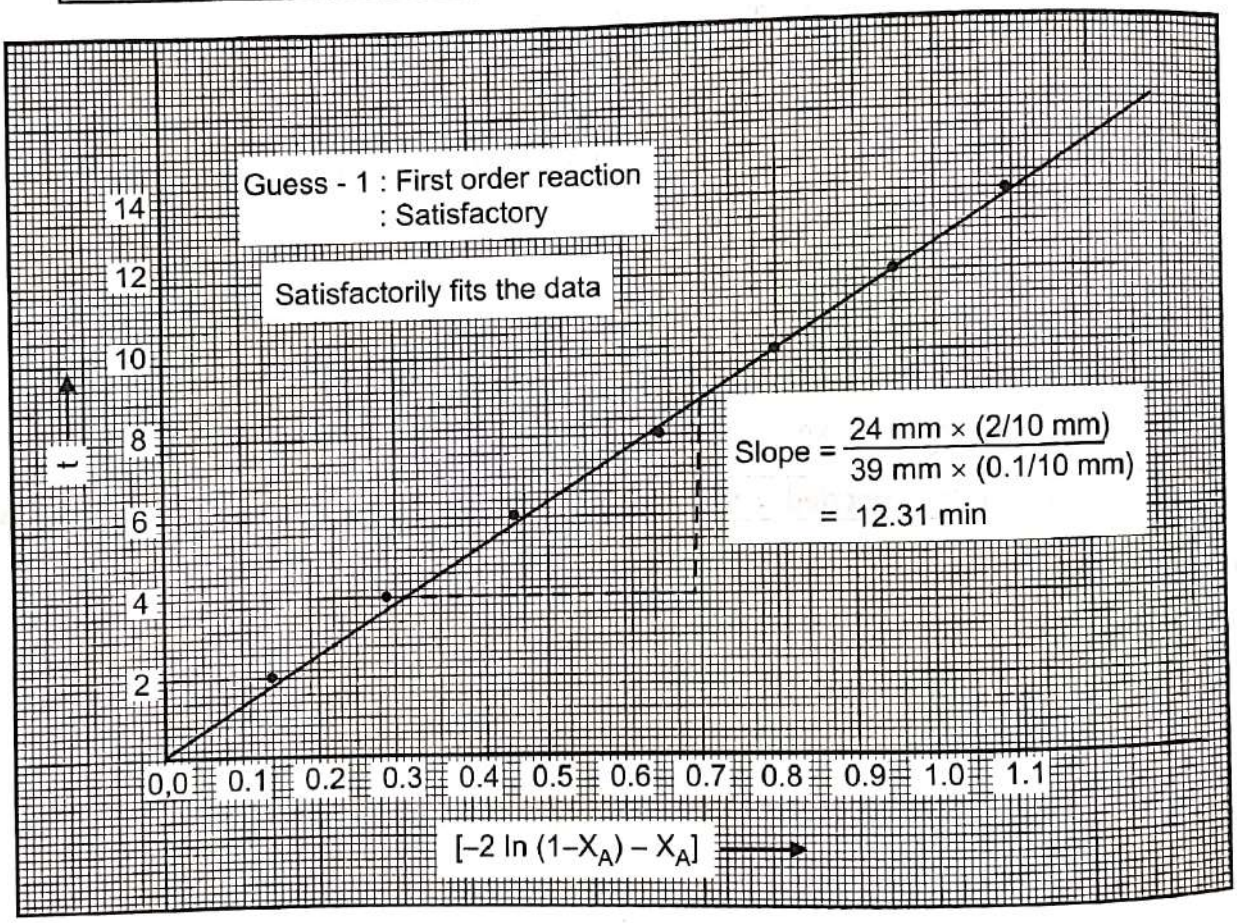


Fig. Ex 6.30 (b)

Example 6.31 : *The catalytic reaction A → R is highly exothermic with rate highl temperature-dependent. Thus, in order to obtain essentially isothermal kinetic data for thi reaction, a long tubular flow reactor immersed in a trough of water [as shown in Fig. Ex 6.31 (a) is used. Pure A at 0 °C and 1 atm flows through this tube at a rate of 10 cm³/s and the strea composition is analysed at various locations and is tabulated below.*

Distance from feed input, m	*0*	*12*	*24*	*36*	*48*	*60*	*72*	*84*	*(∞)*
Partial pressure of A, mm Hg	*760*	*600*	*475*	*390*	*320*	*275*	*240*	*215*	*150*

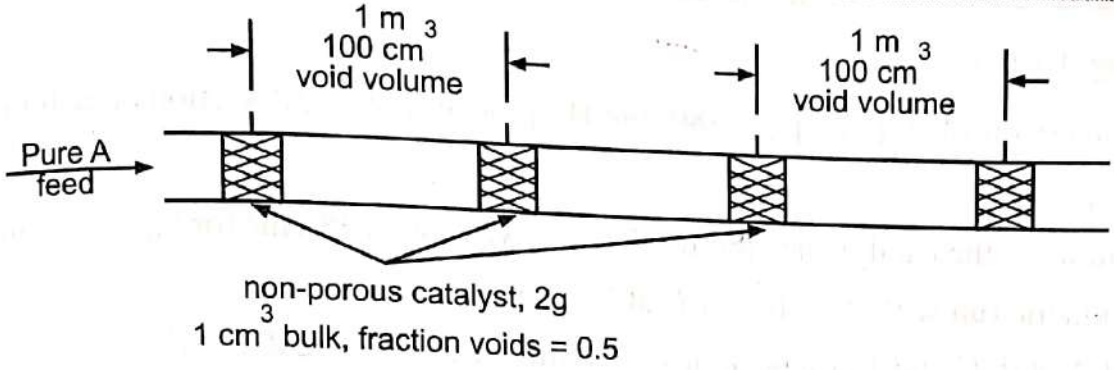


Fig. Ex 6.31 (a)

(i) *Determine the size of a plug flow reactor operating at 0 °C and 1 atm to achieve 50% conversion of A for a feed flow rate of 100 kmol/h of pure A.*

(ii) *If the data given above are valid for a catalytic reaction A → 2.5 R, find the volume of the reactor needed for the same conversion and feed flow rate as given in part (i) ?*

Solution : If we make an appropriate time correction, then the system provided [Fig. Ex 6.31 (a)] is equivalent to

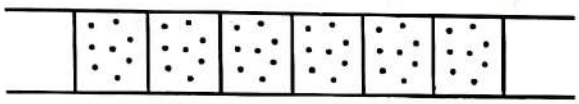


Fig. Ex 6.31 (b)

This means we are provided with the integral data in which we know X_A at various $\frac{V_r}{F_{Ao}}$.

We have to determine the size of the reactor for 50% conversion.

For $A \rightarrow R$

$$\varepsilon_A = 0 \ldots \text{since it is a reaction with unchanging number of moles}$$

$$\therefore \quad C_A = C_{Ao}(1 - X_A)$$

$$\frac{p_A}{RT} = \frac{p_{Ao}}{RT}(1 - X_A)$$

$$p_A = p_{Ao}(1 - X_A)$$

At 50% conversion, i.e., at $X_A = 0.50$

$$p_A = 760\,(1 - 0.50)$$

$$= 380 \text{ mm Hg} \ldots\ldots \text{partial pressure of A}$$

i.e., At 50% conversion, p_A is 380 mm Hg

Refer Fig. Ex 6.31 (c) :

50% conversion of A $\left(\text{i.e., } p_A = 380 \text{ mm Hg}\right)$ occurs after 38 sections (each of 1 cm^3 bulk volume of catalyst).

i.e., Volume of the catalyst needed for 50% conversion is 38 cm^3 for ν_o = 10 cm^3/s.

For the kinetic run with ν_o = 10 cm^3/s at 0 °C and 1 atm :

1 mol of A at 0 °C and 1 atm occupies a volume of 22.4 l = 22400 cm^3

$$F_{Ao} \text{ for the kinetic run} = 10 \text{ cm}^3/\text{s} \times \frac{1 \text{ mol}}{22400 \text{ cm}^3}$$

$$\therefore \quad \frac{V_r}{F_{Ao}} = \frac{38 \text{ cm}^3 \text{ cat}}{10 \text{ cm}^3/\text{s} \times \frac{1 \text{ mol}}{22400 \text{ cm}^3}}$$

$$= 85120 \text{ (cm}^3 \text{ cat.s)/mol}$$

We have to determine the size of plug flow reactor, for 50% conversion, operating at 0°C and 1 atm with F_{Ao} = 100 kmol/h.

$$F_{Ao} = 100 \text{ kmol/h} = \frac{10^5 \text{ mol}}{3600 \text{ s}} = 27.78 \text{ mol/s}$$

$$\therefore \quad V_r = 85120 \left(\frac{\text{cm}^3 \text{ cat. s}}{\text{mol}}\right) \times 27.78 \ \frac{\text{mol}}{\text{s}}$$

$$= 236463 \text{ cm}^3$$

$$= 2.36 \text{ m}^3$$

Volume of reactor = 2.36 m^3 ... Ans. (i)

(ii) $A \rightarrow 2.5\ R$ ... gas phase reaction. Let us find ε_A.

Basis : 1 mol of feed. Initial moles = 1 mol A + 0 mol R = 1 mol.

Moles on complete conversion of A = 0 mol A + [2.5/1) × 1 = 2.5 mol R]

$$= 0 + 2.5 = 2.5 \text{ mol}$$

$$\varepsilon_A = \frac{\text{moles complete conversion} - \text{initial moles}}{\text{initial moles}}$$

$$= \frac{2.5 - 1}{1} = 1.5$$

$$F_{Ao} = 100 \text{ kmol/h}$$

$$= 27.78 \text{ mol/s}$$

$$C_A = \frac{C_{Ao}(1 - X_A)}{(1 + \varepsilon_A X_A)}$$

$$\frac{p_A}{RT} = \frac{p_{Ao}}{RT} \frac{(1 - X_A)}{(1 + \varepsilon_A X_A)}$$

$$p_A = \frac{p_{Ao}(1 - X_A)}{(1 + 1.5\, X_A)}$$

Conversion = 50%, $\therefore X_A = 0.50$

$$\therefore \quad p_A = \frac{760\,(1-0.5)}{(1+1.5\times 0.50)} = 217.14 \text{ mm Hg}$$

i.e., At 50% conversion, $p_A = 217.14$ mm Hg

Refer Fig. Ex 6.31 (c) :

50% conversion of A $\left(\text{i.e., } p_A = 217.14 \text{ mm Hg}\right)$ occurs after 82.5 sections, each of 1 cm^3.

The amount of catalyst needed (catalyst bed needed) is 82.5 m instead of 38 m with all else remain unchanged.

For 50% conversion of A and the feed flow rate of 100 kmol/h, the size of reactor for $A \rightarrow 2.5\ R$ is

$$V_r = 2.36 \times \frac{82.5}{38}, \quad [2.36\ m^3 \text{ volume} - 38 \text{ m} \ldots\ldots \text{ part (i)}]$$

$$= 5.12\ m^3$$

Volume of reactor = 5.12 m^3 **... Ans. (ii)**

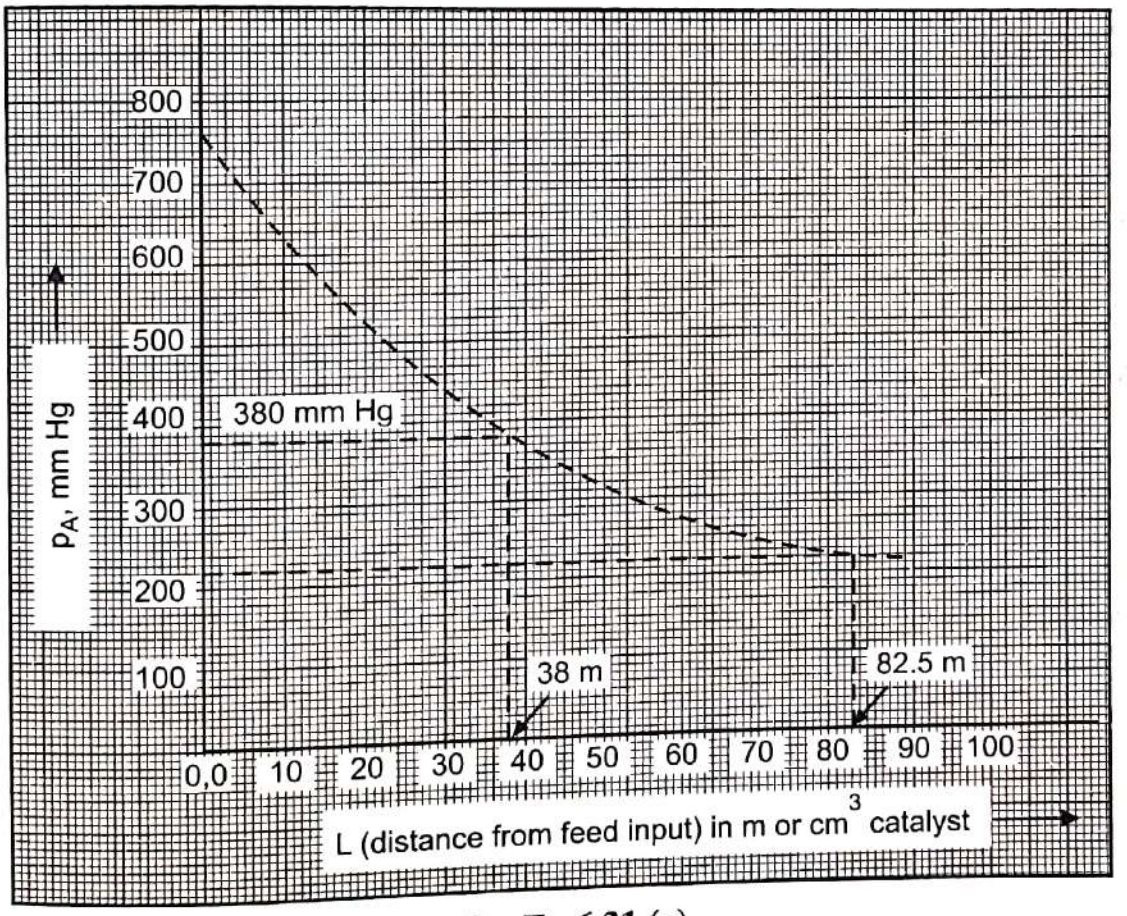


Fig. Ex 6.31 (c)

Example 6.32 : *The mechanism of decomposition of cumene on the catalyst surface is given by*

Reaction : $C_6H_5CH(CH_3)_2 \rightarrow C_6H_6 + C_3H_6$

$C\,(g) \rightarrow B\,(g) + P\,(g)$

Mechanism :

$$C + S \rightleftharpoons C{\cdot}S \quad \ldots\ldots \textit{(adsorption)}$$

$$C{\cdot}S \rightleftharpoons B{\cdot}S + P \quad \ldots\ldots \textit{(surface reaction)}$$

$$B{\cdot}S \rightleftharpoons B + S \quad \ldots\ldots\ldots \textit{(desorption)}$$

Derive the rate expression if surface reaction controls.

Solution : The catalytic decomposition of cumene is given by following overall reaction :

$$C_6H_5CH(CH_3)_2 \rightarrow C_6H_6 + C_3H_6$$

$$C\,(g) \rightarrow B\,(g) + P\,(g)$$

[The catalytic reaction is not diffusion limited.]

The mechanism of cumene decomposition (with no adsorbing inert in the feed) is

$$C + S \underset{k_{-A}}{\overset{k_A}{\rightleftharpoons}} C{\cdot}S \quad \ldots\ldots \text{adsorption of cumene on the surface}$$

$$C{\cdot}S \underset{k_{-S}}{\overset{k_S}{\rightleftharpoons}} B{\cdot}S + P \quad \ldots\ldots \text{surface reaction}$$

[surface reaction forms adsorbed benzene and propylene in the gas phase]

$$B{\cdot}S \underset{k_{-D}}{\overset{k_D}{\rightleftharpoons}} B + S \quad \ldots\ldots \text{desorption of benzene from surface}$$

[while writing rate expressions for these steps we will consider each step as an elementary reaction and we will replace the species concentration in the gas phase by their respective partial pressures].

(1) Adsorption of cumene on the surface :

$$C + S \underset{k_{-A}}{\overset{k_A}{\rightleftharpoons}} C \cdot S, \quad [C\,(g) \ldots \text{use } p_C]$$

The rate of attachment of molecules (C) is proportional to (i) partial pressure of C in the gas phase, p_C and (ii) the concentration of vacant sites.

$$\text{rate of attachment} = k_A p_C C_V$$

The rate of detachment of molecules from the surface is directly proportional to the concentration of sites occupied by the molecules, $C_{C\cdot S}$

$$\text{rate of detachment} = k_{-A} C_{C\cdot S}$$

The net rate of adsorption of cumene (the rate expression for the adsorption of cumene is)

$$r_{AD} = k_A p_C C_V - k_{-A} C_{C\cdot S} \quad \ldots (1)$$

where k_A is the adsorption rate constant (the constant of proportionality for the attachment process). k_{-A} is the desorption constant.

$$K_C = \frac{k_A}{k_{-A}} = \text{the adsorption equilibrium constant for C} \quad \ldots (2)$$

S – represents active site

C·S – means one unit of C is adsorbed on the site S

p_i – partial pressure of species i in the gas phase

$C_{i \cdot s}$ – surface concentration of sites occupied by species i in mol/g cat

$C_{C \cdot s}$ – surface concentration of sites occupied by species C (surface concentration of adsorbed species i) in mol/g cat.

C_V – molar concentration of vacant sites – number of vacant sites per unit mass of catalyst divided by Avogadro's number (mol/g cat)

$$r_{AD} = k_A \left(p_C C_V - \frac{k_{-A}}{k_A} C_{C \cdot S} \right)$$

$$r_{AD} = k_A \left(p_C C_V - \frac{C_{C \cdot S}}{K_C} \right) \quad , \text{ as } K_C = k_A / k_{-A} \quad \ldots (3)$$

If r_{AD} has the units of mol/(g cat.s) and $C_{C \cdot S}$ has units of mol cumene adsorbed/g cat, then

k_A has the units of $(kPa \cdot s)^{-1}$

k_{-A} has the units of s^{-1}

K_C has the units of $(kPa)^{-1}$ or atm^{-1}.

(2) Surface reaction : The rate expression for the surface reaction step yielding adsorbed benzene and propylene in the gas phase (surface reaction is a single site mechanism) is

$$C \cdot S \underset{k_{-S}}{\overset{k_S}{\rightleftharpoons}} B \cdot S + P \quad , [P\,(g) \ldots \text{use } p_P]$$

$$r_S = k_S C_{C \cdot S} - k_{-S} p_P C_{B \cdot s}$$

$$r_S = k_S \left[C_{C \cdot s} - \frac{p_P C_{B \cdot S}}{K_S} \right] \quad \ldots (4)$$

[concentration of species P in the gas phase is replaced by its partial pressure]

$$K_S = \frac{k_S}{k_{-S}} = \text{surface reaction equilibrium constant.}$$

(3) Desorption :

$$B \cdot S \underset{k_{-D}}{\overset{k_D}{\rightleftharpoons}} B + S \quad , [B\,(g) \ldots \text{use } p_B]$$

The rate of benzene desorption is

$$r_D = k_D C_{B\cdot S} - k_{-D} p_B C_V$$

and $$K_{DB} = \frac{k_D}{k_{-D}}$$

K_{DB} is the desorption equilibrium constant for benzene

$$r_D = k_D \left(C_{B\cdot S} - \frac{p_B C_V}{K_{DB}} \right) \quad \ldots (5)$$

Units of k_S (and k_D) and K_S and (K_{DB}) are s^{-1} and atm or kPa, respectively.

The desorption of benzene is just the reverse of the adsorption of benzene.

$$K_B = 1/K_{DB} \quad \ldots (6)$$

Putting value of K_{DB} from Equation (6) into Equation (5), we get

$$r_D = k_D (C_{B\cdot S} - K_B p_B C_V) \quad \ldots (7)$$

Given : The surface reaction is rate limiting or rate controlling step.

The rate of surface reaction is given by

$$r_S = k_S \left(C_{C\cdot S} - \frac{p_P C_{B\cdot S}}{K_S} \right) \quad \ldots (8)$$

As we cannot measure $C_{C\cdot S}$ and $C_{B\cdot S}$ we have to replace these variables in terms of measurable quantities-partial pressures p_C and p_B.

For doing this we have to make use of

(i) Rate expressions for adsorption of C and desorption of B.

(ii) Fact - for surface reaction as rate limiting both k_A and k_D are very large compared with k_S. So $r_{AD}/k_A \approx 0$ and $r_D/k_D \approx 0$.

Evaluation of the surface concentration of adsorbed cumene ($C_{C\cdot S}$) using rate expression for adsorption of cumene :

When surface reaction is controlling k_A is large (and k_S is small) and therefore the ratio r_{AD}/k_A is very small (approximately zero).

$$C + S \underset{k_{-A}}{\overset{k_A}{\rightleftharpoons}} C\cdot S$$

The rate of adsorption of cumene is given by,

$$r_{AD} = k_A \left(p_C C_V - \frac{C_{C\cdot S}}{K_C} \right) \quad \ldots (9)$$

$$\frac{r_{AD}}{k_A} = p_C C_V - \frac{C_{C\cdot S}}{K_C} \approx 0$$

$$\therefore \quad C_{C\cdot S} = K_C p_C C_V \quad \ldots (10)$$

Evaluation of the surface concentration of adsorbed benzene ($C_{B\cdot S}$) using rate expression for desorption of benzene :

When the surface reaction is rate controlling, k_D is very large and therefore the ratio r_D/k_D is very small (approximately zero).

$$B\cdot S \underset{k_{-D}}{\overset{k_D}{\rightleftharpoons}} B + S$$

The rate expression for adsorption or desorption of benzene is

$$r_D = k_D (C_{B\cdot S} - K_B\, p_B\, C_V)$$

$$\frac{r_D}{k_D} = (C_{B\cdot S} - K_B P_B C_V) \approx 0$$

$$\therefore \quad C_{B\cdot S} = K_B P_B C_V \qquad \ldots (11)$$

Substituting $C_{S\cdot S}$ and $C_{B\cdot S}$ from Equations (10) and (11) into Equation (8)

$$r_S = k_S \left(K_C p_C C_V - \frac{p_P K_B p_B C_V}{K_S}\right)$$

$$r_S = k_S \left(p_C K_C - \frac{K_B p_P p_B}{K_S}\right) C_V$$

For the reaction

$$C \rightleftharpoons B + P$$

The overall partial pressure equilibrium constant K_P is

$$K_P = \frac{p_B p_P}{p_C} = \frac{K_S K_C}{K_B}$$

$$r_S = k_S \left(K_C p_C - \frac{K_B p_P p_B}{K_S}\right) C_V$$

$$= k_S K_C \left(p_C - \frac{K_B p_P p_B}{K_S K_C}\right) C_V$$

$$r_S = k_S K_C \left(p_C - \frac{p_P p_B}{K_P}\right) C_V \qquad \ldots (12)$$

We cannot measure C_V so we will eliminate it from the above expression. In the present case, a site balance is/the total concentration of sites is

$$C_t = C_V + C_{B\cdot S} + C_{C\cdot S}$$

$$C_t = C_V + K_C p_C C_V + K_B p_B C_V = C_V (1 + K_C p_C + K_B p_B)$$

$$C_V = \frac{C_t}{1 + K_C p_C + K_B p_B} \qquad \ldots (13)$$

Substituting value of C_v from Equation (13) into Equation (12), we get

$$r_S = \frac{k_S K_C C_t (p_C - p_P p_B/K_P)}{1 + K_C p_C + K_B p_B}$$

Let, $k = k_S K_C C_t$

∴ The rate law for the catalytic decomposition of cumene, assuming the surface reaction as the rate limiting step, is

$$\boxed{-r'_C = r_S = \frac{k (p_C - p_C p_B/K_P)}{1 + K_C p_C + K_B p_B}} \quad \ldots (14)$$

Equation (14) is the desired rate expression.

Where $k = k_S K_C C_t$

and C_t – total molar concentration of active sites per unit mass of catalyst and is equal to the number of active sites per unit mass divided by Avogadro's number (mol/g cat).

Example 6.33 : *The mechanism of decomposition of cumene on the catalyst surface is given by*

Reaction : $C_6H_5CH(CH_3)_2 \longrightarrow C_6H_6 + C_3H_6$

$C(g) \longrightarrow B(g) + P(g)$

Mechanism : $C + S \rightleftharpoons C{\cdot}S$ ..... *(adsorption)*

$C{\cdot}S \rightleftharpoons B{\cdot}S + P(g)$ ..... *(surface reaction)*

$B{\cdot}S \rightleftharpoons B(g) + S$ ..... *(desorption)*

$I + S \rightleftharpoons I{\cdot}S$

where the last step represents adsorption of an inhibitor I present along with reactant in the feed (I is an adsorbing inert in the feed that occupies sites on the catalyst surface). Find the rate expression if surface reaction controls.

Solution : The overall reaction for decomposition of cumene on the catalyst surface is

$$C_6H_5CH(CH_3)_2 \longrightarrow C_6H_6 + C_3H_6$$

$$C(g) \longrightarrow B(g) + P(g)$$

With adsorbing inert (I) in the feed the mechanism of decomposition is

1. $C + S \underset{k_{-A}}{\overset{k_A}{\rightleftharpoons}} C{\cdot}S$ ..... (adsorption of cumene on the surface)

2. $C{\cdot}S \underset{k_{-S}}{\overset{k_S}{\rightleftharpoons}} B{\cdot}S + P$ ..... (surface reaction)

3. $B{\cdot}S \underset{k_{-D}}{\overset{k_D}{\rightleftharpoons}} B + S$ ..... (desorption of benzene from surface)

4. $I + S \underset{k_{-AI}}{\overset{k_{AI}}{\rightleftharpoons}} I{\cdot}S$ ..... (adsorption of inert of the surface)

[While writing the rate expressions for these steps we will consider each step as an elementary reaction and we will replace species concentrations in the gas phase by partial pressures of respective species.]

1. **Adsorption of benzene :**

$$C + S \underset{k_{-A}}{\overset{k_A}{\rightleftharpoons}} C{\cdot}S$$

Rate expression for adsorption of cumene is

$$r_{AD} = k_A p_C C_V - k_{-A} C_{C\cdot S} \quad \ldots (1)$$

The adsorption equilibrium constant for C is

$$K_C = k_A/k_{-A} \quad \ldots (2)$$

Combining Equations (1) and (2), we get

$$r_{AD} = k_A\left(p_C C_V - \frac{C_{C\cdot S}}{K_C}\right) \quad \ldots (3)$$

where r_{AD} = rate of adsorption of species C.

p_C = partial pressure of species C in the gas phase

p_i = partial pressure of species i in the gas phase

$C_{C\cdot S}$ = surface concentration of sites occupied by species C (surface concentration of adsorbed C) in mol/g cat.

C_V = molar concentration of vacant sites – number of vacant sites per unit mass of catalyst divided by Avogadro's number (mol/g cat)

S = active site and C·S – means one unit of C is adsorbed on the site S

k_A = adsorption rate constant and k_{-A} – desorption rate constant

$C_{i\cdot S}$ = surface concentration of sites occupied by species i in mol/g cat

2. Surface reaction to form adsorbed benzene and propylene in the gas phase

$$C{\cdot}S \underset{k_{-S}}{\overset{k_S}{\rightleftharpoons}} B{\cdot}S + P$$

(After adsorption, the adsorbed C decomposes on the site – surface reaction – single-site mechanism.)

The rate of surface reaction is

$$r_S = k_S C_{C\cdot S} - k_{-S}\, p_P C_{B\cdot S} \quad \ldots (4)$$

and

$$K_S = k_S/k_{-S} \quad \ldots (5)$$

K_S is the surface reaction equilibrium constant.

Combining Equations (4) and (5), we get

$$r_S = k_S \left(C_{C\cdot S} - \frac{p_P C_{B\cdot S}}{K_S} \right) \quad \ldots (6)$$

p_P is the partial pressure of species P in the gas phase and $C_{B\cdot S}$ is the surface concentration of sites occupied by species B (surface concentration of adsorbed B) in mol/g cat.

3. Desorption of benzene from the surface :

$$B\cdot S \underset{k_{-D}}{\overset{k_D}{\rightleftharpoons}} B + S$$

The rate of desorption of benzene is

$$r_D = k_D C_{B\cdot S} - k_{-D}\, p_B\, C_V \quad \ldots (7)$$

and

$$K_{DB} = k_D/k_{-D} \quad \ldots (8)$$

K_{DB} is the desorption equilibrium constant for benzene.

Combining Equations (7) and (8), we get

$$r_D = k_D \left(C_{B\cdot S} - \frac{p_B C_V}{K_{DB}} \right) \quad \ldots (9)$$

p_B is the partial pressure of species B in the gas phase.

The desorption of benzene is just the reverse of adsorption of benzene.

$\therefore$ $K_B = 1/K_{DB}$, K_B - adsorption equilibrium constant for benzene

Equation (9) then becomes,

$$r_D = k_D\, (C_{B\cdot S} - K_B\, p_B\, C_V) \quad \ldots (10)$$

4. Adsorption of inert (I) on the surface :

$$I + S \underset{k_{-AI}}{\overset{k_{AI}}{\rightleftharpoons}} I\cdot S$$

The rate of adsorption of I is given by

$$r_{ADI} = k_{AI}\, p_I C_V - k_{-AI} C_{I\cdot S} \quad \ldots (11)$$

and

$$K_I = k_{AI}/k_{-AI} \quad \ldots (12)$$

K_I is the adsorption equilibrium constant for I.

Combining Equations (11) and (12), we get

$$r_{ADI} = k_{AI}\left(p_I C_V - \frac{C_{I\cdot S}}{K_I}\right) \quad \ldots (13)$$

Given : Surface reaction is the rate controlling step.

Now we will obtain the rate equation for the decomposition of cumene when surface reaction controls the rate.

The rate expression for the surface reaction is

$$r_S = k_S\left(C_{C\cdot S} - \frac{p_P C_{B\cdot S}}{K_S}\right) \quad \ldots (14)$$

As we cannot measure $C_{C\cdot S}$ and $C_{B\cdot S}$ we have to replace these variables in terms of measurable quantities – partial pressures p_C and p_B.

For doing this we have to make use of rate expressions for adsorption of C and desorption of B and the fact that for surface reaction as rate limiting step both k_A and k_D are very large compared with k_S.

Evaluation of $C_{C\cdot S}$ – surface concentration of adsorbed cumene :

When surface reaction is the rate controlling step k_A is large (and k_S is small) and therefore the ratio r_{AD}/k_A is very small (approximately zero).

$$r_D/k_A \approx 0 \quad \ldots\ldots \text{ if surface reaction controls.}$$

We have :

$$r_{AD} = k_A\left(p_C C_V - \frac{C_{C\cdot S}}{K_C}\right)$$

$$\frac{r_{AD}}{k_A} = p_C C_V - \frac{C_{C\cdot S}}{K_C} = 0 \quad \ldots\ldots \text{ when surface reaction controls}$$

$$\therefore \quad C_{C\cdot S} = K_C p_C C_V \quad \ldots(15)$$

Evaluation of $C_{B\cdot S}$ – surface concentration of adsorbed benzene :

When the surface reaction is rate controlling, k_D is very large as compared to k_A and therefore, the ratio r_D/k_D is very small (approximately zero).

We have,

$$r_D = k_D\,(C_{B\cdot S} - K_B p_B C_V)$$

$$\frac{r_D}{k_D} = C_{B\cdot S} - K_B\, p_B C_V = 0 \quad \ldots\ldots \text{ when surface reaction controls}$$

$$\therefore \quad C_{B\cdot S} = K_B p_B C_V \quad \ldots (16)$$

Substituting $C_{C\cdot S}$ and $C_{B\cdot S}$ from Equations (15) and (16) into Equation (14), we get

$$r_S = k_S\left(K_C p_C C_V - \frac{p_P K_B p_B C_V}{K_S}\right)$$

$$r_S = k_S\left(p_C K_C - \frac{K_B p_P p_B}{K_S}\right) C_V \quad \ldots (17)$$

For the reaction : $C \rightleftharpoons B + P$

The overall partial pressure equilibrium constant K_P is

$$K_P = \frac{p_B p_P}{p_C} = \frac{K_S K_C}{K_B} \quad \ldots (18)$$

Equation (17) can be written as

$$r_S = k_S K_C\left(p_C - \frac{K_B p_P p_B}{K_S K_C}\right) C_V$$

Using Equation (18) the above equation becomes

$$r_S = k_S K_C\left(p_C - \frac{p_P p_B}{K_P}\right) C_V \quad \ldots (19)$$

We cannot measure C_V so we will eliminate it from the above expression by performing a site balance.

In the present case, the site balance is (the total concentration of sites is)

$$C_t = C_V + C_{C\cdot S} + C_{B\cdot S} + C_{I\cdot S} \quad \ldots (20)$$

where C_t is the total molar concentration of active sites per unit mass of catalyst and is equal to the number of active sites per unit mass divided by Avogadro's number (mol/g cat).

Now we will determine the concentration of sites occupied by the inert ($C_{I\cdot S}$) to make use of Equation (20) to eliminate C_V from Equation (19).

We have, $$r_{ADI} = k_{AI}\left(p_I C_V - \frac{C_{I\cdot S}}{K_I}\right)$$

When surface reaction is rate controlling k_{AI} is very large compared to k_A and therefore the ratio r_{ADI}/k_{AI} is very small (approximately zero).

$$r_{ADI}/k_{AI} = p_I C_V - \frac{C_{I\cdot S}}{K_I} = 0$$

$$\therefore \quad C_{I\cdot S} = K_I p_I C_V \quad \ldots (21)$$

Substituting the values of $C_{C\cdot S}$, $C_{B\cdot S}$ and $C_{I\cdot S}$ from Equations (15), (16) and (21) into Equation (20), we get

$$C_t = C_V + K_C p_C C_V + K_B p_B C_V + K_I p_I C_V$$

$$C_t = (1 + K_C p_C + K_B p_B + K_I p_I) C_V$$

$$C_V = \frac{C_t}{1 + K_C p_C + K_B p_B + K_I p_I} \quad \ldots (22)$$

Substituting the value of C_V from Equation (22) into Equation (19), we get

$$r_S = k_S K_C \left(p_C - \frac{p_P p_B}{K_P}\right) \frac{C_t}{(1 + K_C p_C + K_B p_B + K_I p_I)}$$

$$r_S = \frac{k_S K_C C_t (p_C - p_P p_B/K_P)}{1 + K_C p_C + K_B p_B + K_I p_I}$$

Let, $k = k_S K_C C_t$

∴ The rate law (rate expression) for the catalytic decomposition of cumene (with an inhibitor in the feed), assuming the surface reaction to be rate controlling step, is

$$\boxed{-r'_C = r_S = \frac{k (p_C - p_P p_B/K_P)}{1 + K_C p_C + K_B p_B + K_I p_I}} \quad \ldots (23)$$

where $k = k_S K_C C_t$

Equation (23) is the desired rate expression for the catalytic decomposition of cumene.

Typical units of various constants are :

$r_{AD}, r_S - r'_C$ ..... mol/(g cat·s)

$C_{C\cdot S}$ ..... mol/g cat

k_A ..... (atm. s)$^{-1}$, k_{-A} ..... s^{-1}, k_S ..... s^{-1}

K_S, K_{DB} ..... atm, K_C ..... atm^{-1}, K_P ..... atm.

Example 6.34 : *Toluene is adsorbed on the surface of catalyst and reacts with hydrogen in the gas phase to produce benzene adsorbed on the surface and CH_4 in the gas phase. Benzene is then desorbed from the surface. The proposed mechanism is :*

Adsorption : $T(g) + S \underset{k_{-A}}{\overset{k_A}{\rightleftharpoons}} T\cdot S$

$$r_{AD} = k_A \left(p_T C_V - \frac{C_{T\cdot S}}{K_T}\right)$$

Surface reaction : $T\cdot S + H_2 \underset{k_{-S}}{\overset{k_S}{\rightleftharpoons}} B\cdot S + M(g)$

$$r_S = k_S \left(p_{H_2} C_{T\cdot S} - \frac{C_{B\cdot S} p_M}{K_S}\right)$$

Desorption : $B\cdot S \underset{k_{-D}}{\overset{k_D}{\rightleftharpoons}} B(g) + S$

$$r_D = k_D \left(C_{B\cdot S} - \frac{p_B C_V}{K'_B}\right)$$

When surface reaction controls show that

$$-r_T' = r_S = \frac{C_t K_S K_t [p_{H_2} p_T - p_B p_M/K_P]}{1 + K_T p_T + (p_B/K_B')}$$

Solution : The overall reaction of hydrodemethylation of toluene is

$$C_6H_5CH_3 + H_2 \rightarrow C_6H_6 + CH_4$$
$$T\,(g) + H_2\,(g) \rightarrow B\,(g) + M\,(g)$$

Mechanism :

1. Adsorption of toluene on the surface :

$$T\,(g) + S \underset{k_{-A}}{\overset{k_A}{\rightleftharpoons}} T{\cdot}S$$

where S represents an active site and T·S indicates that one unit of A is adsorbed on the site S.

The rate of adsorption of toluene is given by

$$r_{AD} = k_A\left(p_T C_V - \frac{C_{T\cdot S}}{K_T}\right) \quad \ldots (1)$$

where
k_A = adsorption rate constant
k_{-A} = desorption rate constant.
K_T = k_A/k_{-A} – adsorption equilibrium constant for toluene
P_T = partial pressure of toluene in the gas phase, atm.
C_V = number of vacant sites per unit mass of catalyst divided by Avogadro's number (mol/g cat).
$C_{T\cdot S}$ = surface concentration of sites occupied by toluene (the surface concentration of adsorbed toluene) in mol/g cat

2. Surface reaction to form adsorbed benzene and methane in the gas phase (reaction between an adsorbed molecule and a molecule in the gas phase) :

$$T{\cdot}S + H_2 \underset{k_{-S}}{\overset{k_S}{\rightleftharpoons}} B{\cdot}S + M\,(g)$$

The rate expression for surface reaction step is

$$r_S = k_S\left(p_{H_2} C_{T\cdot S} - \frac{C_{B\cdot S}\, p_M}{K_S}\right) \quad \ldots (2)$$

p_{H_2} and p_M are the partial pressure of hydrogen and methane in the gas phase.

$K_S = k_S/k_{-S}$ = surface reaction equilibrium constant

$C_{T\cdot S}$ and $C_{B\cdot S}$ are the surface concentration of adsorbed toluene and benzene.

3. Desorption of benzene from the surface :

$$B \cdot S \underset{k_{-D}}{\overset{k_D}{\rightleftharpoons}} B\,(g) + S$$

The rate of benzene desorption is

$$r_D = k_D \left(C_{B \cdot S} - \frac{p_B C_V}{K_B'} \right) \quad \ldots (3)$$

$$K_{DB} = K_B' = k_D / k_{-D}$$

$$K_B' = 1/K_B$$

where K_B is the adsorption equilibrium constant for B and K_B' is the desorption equilibrium constant for B (benzene).

Given : Surface reaction is rate controlling step.

Now we will derive the rate expression for hydrodemethylation of toluene when the surface reaction controls the rate.

For the surface reaction, the rate expression is

$$r_S = k_S \left(P_{H_2} C_{T \cdot S} - \frac{C_{B \cdot S} P_M}{K_S} \right) \quad \ldots (4)$$

As we cannot measure $C_{T \cdot S}$ and $C_{B \cdot S}$ we have to replace these variables in terms of measurable quantities – partial pressures P_T and P_B.

For doing this we will adopt following procedure :

(i) Evaluation of $C_{T \cdot S}$ – surface concentration of adsorbed toluene – using the rate expression for adsorption :

When the surface reaction is rate controlling/limiting step, k_A is large compared to k_S and therefore, the ratio r_{AD}/k_A is very small (approximately zero).

$$r_{AD} = k_A \left(p_T C_V - \frac{C_{T \cdot S}}{K_T} \right)$$

$$\frac{r_{AD}}{k_A} = p_T C_V - \frac{C_{T \cdot S}}{K_T} = 0$$

$$C_{T \cdot S} = K_T p_T C_V \quad \ldots (5)$$

(ii) Evaluation of $C_{B \cdot S}$ – surface concentration of adsorbed benzene – using the rate expression for desorption of benzene :

When the surface reaction is rate controlling step, k_D is very large compared to k_S and therefore the ratio r_D/k_D is very small (approximately zero).

$$r_D = k_D \left(C_{B \cdot S} - \frac{p_B C_V}{K_B'} \right)$$

$$\frac{r_D}{k_D} = C_{B \cdot S} - \frac{p_B C_V}{K_B'} = 0$$

$$\therefore \quad C_{B \cdot S} = \frac{p_B C_V}{K_B'} \quad \ldots (6)$$

Substituting $C_{T\cdot S}$ and $C_{B\cdot S}$ from Equations (5) and (6) into Equation (4), we get

$$r_S = k_S \left(p_{H_2} K_T p_T C_V - \frac{p_B C_V p_M}{K_B' K_S} \right)$$

$$\therefore \quad r_S = k_S K_T \left(p_T H_{H_2} - \frac{p_B p_M}{K_T K_S K_B'} \right) C_V \qquad \ldots (7)$$

For the reaction

$$T + H_2 \rightleftharpoons B + M$$

The overall partial pressure equilibrium constant K_P is

$$K_P = \frac{p_B p_M}{p_T p_{H_2}} = \frac{K_S K_T}{K_B}$$

but $\quad K_B' = 1/K_B \quad \therefore \quad K_B = 1/K_B' = 1/K_{DB}$

$$\therefore \quad K_P = K_S K_T K_B' \qquad \ldots (8)$$

With Equation (8), Equation (7) becomes

$$r_S = k_S K_T \left(p_T p_{H_2} - \frac{p_B p_M}{K_P} \right) C_V \qquad \ldots (9)$$

As we cannot measure C_V we will eliminate it from the above equation by performing a site balance.

In present case, the site balance is

$$C_t = C_V + C_{T\cdot S} + C_{B\cdot S} \qquad \ldots (10)$$

where C_t – the total molar concentration of active sites per unit mass of catalyst (mol/g cat) and is equal to the number of active sites per unit mass divided by Avogadro's number.

Putting values of $C_{T\cdot S}$ and $C_{B\cdot S}$ from Equations (5) and (6) into Equation (10), we get

$$C_t = C_V + K_P p_T C_V + \frac{p_B C_V}{K_B'}$$

$$C_t = [1 + K_T p_T + p_B/K_B'] C_V$$

$$\therefore \quad C_V = \frac{C_t}{(1 + K_T p_T + p_B/K_B')} \qquad \ldots (11)$$

Substituting C_V from equation (11) into Equation (9), we get

$$r_S = \frac{k_S K_T C_t \, (p_T p_{H_2} - p_B p_M/K_P)}{(1 + K_T p_T + p_B/K_B')}$$

Therefore, the rate of hydrodemethylation of toluene, when surface reaction controls, is given by

$$-r'_T = r_S = \frac{k_S K_T C_t (p_T p_{H_2} - p_B p_M/K_P)}{(1 + K_T p_T + p_B/K'_B)} \quad \ldots (12)$$

Extra : $K'_B = K_{DB}$ is the desorption equilibrium constant for B (i.e., desorption of B)

$$K_B = 1/K'_B$$

where K_B is the adsorption equilibrium constant for B.

Let, $k = k_S K_T C_t$

Therefore, replacing $1/K'_B$ by K_B and $k_S K_T C_t$ by k, Equation (12) becomes

$$-r'_T = r_S = \frac{k (p_T p_{H_2} - p_B p_M/K_P)}{(1 + K_T p_T + K_B p_B)}$$

Example 6.35 : *Reactant 'A' is adsorbed on the surface of a catalyst and reacts with another component B in the gas phase. The products of the reaction are : C adsorbed on the surface and D in the gas phase. The product C is then desorbed from the surface. The proposed mechanism is*

Adsorption : $A(g) + S \rightleftharpoons A \cdot S$

Surface reaction : $B(g) + A \cdot S \rightleftharpoons C \cdot S + D(g)$

$$r_S = k_S \left(p_B C_{A \cdot S} - \frac{C_{C \cdot S} p_D}{K_S} \right)$$

Desorption : $C \cdot S \rightleftharpoons C(g) + S$

$$r_D = k_D \left(C_{C \cdot S} - \frac{p_C C_v}{K'_C} \right)$$

For the surface reaction controlling, show that

$$-r'_A = r_S = \frac{C_t k_S K_A (p_A p_B - p_C p_D/K_P)}{(1 + p_A K_A + p_C/K'_C)}$$

Solution : [This is the same as Example 6.34]

The overall reaction is

$$A(g) + B(g) \longrightarrow C(g) + D(g)$$

For the surface reaction as a rate controlling step, k_A and k_D are large compared to k_A and the ratios r_{AD}/k_A and r_D/k_D are very small (approximately zero).

The rate of adsorption of A is

$$r_A = k_A\left(C_v p_A - \frac{C_{A\cdot S}}{K_A}\right)$$

$$\frac{r_A}{k_A} = C_v p_A - \frac{C_{A\cdot S}}{K_A} = 0$$

$$\therefore \quad C_{A\cdot S} = K_A p_A C_v \qquad \ldots (1)$$

The rate of desorption of C is

$$r_D = k_D\left(C_{C\cdot S} - \frac{p_C C_v}{K_C'}\right)$$

$$\frac{r_D}{k_D} = C_{C\cdot S} - \frac{p_C C_v}{K_C'} = 0$$

$$\therefore \quad C_{C\cdot S} = \frac{p_C C_v}{K_C'} \qquad \ldots (2)$$

The rate of surface reaction is

$$r_S = k_S\left(p_B C_{A\cdot S} - \frac{C_{C\cdot S} p_D}{K_S}\right) \qquad \ldots (3)$$

Substituting $C_{A\cdot S}$ and $C_{C\cdot S}$ from Equations (1) and (2) into Equation (3), we get

$$r_S = k_S\left(p_B K_A p_A C_v - \frac{p_C C_v p_D}{K_C' K_S}\right)$$

$$r_S = k_S K_A\left(p_A p_B - \frac{p_C p_D}{K_A K_S K_C'}\right) C_v \qquad \ldots (4)$$

For the reaction :

$$A\,(g) + B\,(g) \rightleftharpoons C\,(g) + D\,(g)$$

The overall partial pressure equilibrium constant K_P is given by

$$K_P = \frac{p_C p_D}{p_A p_B} = \frac{K_A K_S}{K_C}$$

where K_C is the adsorption equilibrium constant for C.

$$K_C = 1/K_{DC} = 1/K_C', \quad K_{DC} = K_C'$$

where K_C' is the desorption equilibrium constant for C.

$$\therefore \quad K_P = \frac{K_A K_S}{K_C} = \frac{K_A K_S}{1/K_C'} = K_A K_S K_C' \qquad \ldots (5)$$

Substituting K_P for $K_A K_S K_C'$ Equation (4) becomes

$$r_S = k_S K_A (p_A p_B - p_C p_D/K_P) C_v \quad \ldots (6)$$

Now the site balance is

$$C_t = C_v + C_{A\cdot S} + C_{C\cdot S} \quad \ldots (7)$$

Putting values of $C_{A\cdot S}$ and $C_{C\cdot S}$ in Equation (7) from Equations (1) and (2)

$$C_t = C_v + K_A p_A C_v + p_C C_v/K_C'$$

$$C_t = (1 + K_A p_A + p_C/K_C') C_v$$

$$\therefore \quad C_v = \frac{C_t}{(1 + K_A p_A + p_C/K_C')} \quad \ldots (8)$$

Combining Equations (6) and (8), we get

$$r_S = \frac{k_S K_A C_t (p_A p_B - p_C p_D/K_P)}{(1 + K_A p_A + p_C/K_C')}$$

∴ The rate of reaction of A when surface reaction controls is given by

$$\boxed{-r_A' = r_S = \frac{k_S K_A C_t (p_A p_B - p_C p_D/K_P)}{(1 + K_A p_A + p_C/K_C')}}$$

[Here the solution is given in brief. The solution to this example is the same as Example 6.34. Please refer to Example 6.34 for the detailed solution in which read T as A, H_2 as B, B as C and M as D.]

Example 6.36 : *Ethanol can be produced by catalytic vapour phase hydration of ethylene at 135 atm and 573 K.*

$$C_2H_4 + H_2O \rightleftharpoons C_2H_5OH$$

$$A + B \rightleftharpoons R$$

The rate expression for this reaction is

$$-r' = \frac{z\, k_A k_B (p_A p_B - p_R/K)}{(1 + k_A p_A + k_B p_B)^2}, \frac{mol}{g\ cat.h}$$

where

$$z = 0.018, \quad k_A = k_B = 0.003$$

The equilibrium constant is given by,

$$RT\ lnK = 30\,T - 9730 \quad where\ R = 1.987\ cal/(mol.K)$$

The total feed rate to a plug flow reactor is 10 kg/h with equimolar amounts of ethylene and steam. Find the weight of catalyst needed to achieve 20% conversion of ethylene.

Solution : $C_2H_4 + H_2O \rightleftharpoons C_2H_5OH$

$A + B \rightleftharpoons R$

Conversion of A = 20/100 = 0.20

$P = 135$ atm, $T = 573$ K

$RT \ln K = 30\,T - 9730$

$1.987 \times 573 \ln K = 30 \times 573 - 9730$

$K = 700.8$

Initially let there be 1 mol of CH_4 and 1 mol of water (steam).

Let X_A be the conversion of A at any time t.

Total moles at any time t :

Moles of A $= 1 - 1 \times (X_A) = 1 - X_A$

Moles of B $= 1 - X_A$

Moles of R $= X_A (1/1) = X_A$

Total moles $= (1 - X_A) + (1 - X_A) + X_A$

$= 2 - X_A$

Partial pressure of A $= p_A = \dfrac{1 - X_A}{2 - X_A} \times 135$, atm

Partial pressure of B $= p_B = \dfrac{1 - X_A}{2 - X_A} \times 135$, atm

Partial pressure of R $= p_R = \dfrac{X_A}{2 - X_A} \times 135$, atm

$$-r_A' = \frac{z\,k_A\,k_B\,(p_A\,p_B - p_R/K)}{(1 + k_A\,p_A + k_B\,p_B)^2}, \quad \frac{\text{mol}}{\text{g cat.h}}$$

$z = 0.018$, $k_A = k_B = 0.003$, $K = 700.8$

$$\therefore \quad -r_A' = \frac{0.018 \times 0.003 \times 0.003 \left[\left(\dfrac{1 - X_A}{2 - X_A}\right) \times 135 \times 135 \left(\dfrac{1 - X_A}{2 - X_A}\right) - \left(\dfrac{X_A}{2 - X_A}\right) \times \dfrac{135}{700.8}\right]}{\left[1 + 0.003 \times 135 \left(\dfrac{1 - X_A}{2 - X_A}\right) + 0.003 \times 135 \left(\dfrac{1 - X_A}{2 - X_A}\right)\right]^2}$$

$$-r_A' = \frac{1.67 \times 10^{-7} \left[18225 \left(\dfrac{1 - X_A}{2 - X_A}\right)^2 - 0.193 \left(\dfrac{X_A}{2 - X_A}\right)\right]}{\left[1 + 0.81 \left(\dfrac{1 - X_A}{2 - X_A}\right)\right]^2}$$

Evaluate $-r_A'$ at various X_A values (0 to 0.20).

X_A	0	0.05	0.10	0.15	0.20
$-r_A'$	3.85×10^{-4}	3.714×10^{-4}	3.567×10^{-4}	3.412×10^{-4}	3.25×10^{-4}
$1/-r_A'$	2594	2692	2803	2930	3077

For a plug flow reactor, the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_{Aout}} \frac{dX_A}{-r_A'} = \int_0^{0.20} \left(\frac{1}{-r_A'}\right) dX_A$$

Now we will evaluate the integral on RHS of the above equation by using Simpson's one-third rule (method of numerical evaluation of integrals).

$$\int_{x_0}^{x_2} f(x)\, dx = \frac{h}{3} [f(x_0) + 4 f(x_1) + f(x_2)] \text{ and } h = x_1 = x_0$$

In our case : $h = 0.05 - 0 = 0.05$

$$\int_0^{0.20} \left(\frac{1}{-r_A'}\right) dX_A = \frac{0.05}{3} [\underline{2594 + 4 \times 2692 + 2803} + \underline{2803 + 4 \times 2930 + 3077}]$$

$$= 562.75$$

$$\frac{W}{F_{Ao}} = 562.75$$

For equimolar feed : $M_{avg} = 0.5 \times 28 + 0.5 \times 18 = 23$

Total feed rate = 10 kg/h = 10/23 = 0.435 kmol/h

F_{Ao} = total feed rate/2

$= 0.435/2 = 0.2175$ kmol/h

$W = 562.75 \times F_{Ao}$

$= 562.75 \times 0.2175 = 122.4$ kg

Amount of catalyst needed for 20% conversion = 122.4 kg ... **Ans.**

Example 6.37 : *In a plug flow reactor, carbon monoxide and hydrogen are passed over nickel catalyst to generate methane.*

$$CO + 3\,H_2 \rightleftharpoons CH_4 + H_2O$$

The rate equation is

$$-r' = \frac{1.1\, p_{CO} p_{H_2}^{0.5}}{1 + 1.5\, p_{H_2}}, \; mol/(g\ cat.h)$$

The reaction is carried out under isothermal conditions at one atmosphere. CO and H_2 are fed in stoichiometric proportion with 1 mol/h of CO. Calculate the amount of catalyst needed to achieve 20% conversion of CO.

Data : *Bulk density of catalyst = 480 kg/m³.*

Solution : $CO + 3\,H_2 \rightleftharpoons CH_4 + H_2O$

$A + 3\,B \rightleftharpoons C + D$

Molar flow rate of CO = 1 mol/h)

Molar flow rate of $H_2 = \frac{3}{1} \times 1 = 3$ mol/h (stoichiometric proportion)

Let, X_A be the conversion of A.

At any time t :

A unreacted $= 1 - X_A \times 1 = 1 - X_A$, mol

B unreacted $= 3\,(1 - X_A)$, mol

C produced $= X_A = (1/1 \times X_A)$ mol

D produced $= X_A$, mol

Total moles $= 1 - X_A + 3\,(1 - X_A) + X_A + X_A = 4 - 2\,X_A$, mol

Total pressure = 1 atm

Partial pressure of A = mole fraction of A × total pressure

$$p_A = p_{CO} = \left(\frac{1 - X_A}{4 - 2\,X_A}\right) \times 1 = \frac{1 - X_A}{4 - 2\,X_A}\text{, atm}$$

$$p_B = p_{H_2} = \frac{3\,(1 - X_A)}{4 - 2\,X_A} \times 1 = \frac{3\,(1 - X_A)}{4 - 2\,X_A}\text{, atm}$$

$$p_C = p_{CH_4} = \frac{X_A}{4 - 2\,X_A} \text{ and } p_D = p_{H_2O} = \frac{X_A}{4 - 2\,X_A}\text{, (as P = 1 atm)}$$

$$-r_A' = \frac{1.1\,p_{CO}\,p_{H_2}^{0.5}}{1 + 1.5\,p_{H_2}}$$

$$-r_A' = \frac{1.1\left(\frac{1 - X_A}{4 - 2\,X_A}\right)\left(\frac{3\,(1 - X_A)}{4 - 2\,X_A}\right)^{0.5}}{1 + 1.5\left(\frac{3\,(1 - X_A)}{4 - 2\,X_A}\right)} \quad \ldots (1)$$

For a plug flow reactor, the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_{Aout}} \frac{dX_A}{-r_A'}$$

In our case : $X_{Aout} = 0.20$, $F_{Ao} = 1$ mol/h

$$\frac{W}{F_{Ao}} = \int_0^{0.20} \frac{dX_A}{-r_A'}$$

$-r_A'$ is given by Equation (1).

Now evaluate $-r_A'$ at various X_A using Equation (1) and tabulate.

X_A	0	0.02	0.04	0.06	0.08	0.10	0.12	0.14	0.16	0.18	0.20
$-r_A'$	0.112	0.11	0.1098	0.1087	0.1075	0.1063	0.1051	0.1038	0.1025	0.101	0.0998
$1/-r_A'$	8.92	9.01	9.103	9.20	9.302	9.407	9.515	9.634	9.756	9.901	10.02

$$\frac{W}{F_{Ao}} = \int_0^{0.20} \left(\frac{1}{-r_A'}\right) dX_A$$

Now we will evaluate the integral on RHS of the above equation by Simpson's one-third rule method of numerical evaluation of integrals).

$$\int_{x_0}^{x_2} f(x)\,dx = \frac{h}{3}[f(x_0) + 4f(x_1) + f(x_2)], \quad \text{and} \quad h = x_1 - x_0$$

In our case, $h = 0.02 - 0 = 0.02$ or $0.04 - 0.02 = 0.02$

$$\int_0^{0.2} \left(\frac{1}{-r_A'}\right) dX_A = \frac{0.02}{3}[\underline{8.92 + 4 \times 9.01 + 9.103} + \underline{9.103 + 4 \times 9.20 + 9.302}$$

$$+ \underline{9.302 + 4 \times 9.407 + 9.515}$$

$$+ \underline{9.515 + 4 \times 9.634 + 9.756} + \underline{9.756 + 4 \times 9.901 + 10.02}]$$

$$= \frac{0.02}{3} \times 282.9 = 1.886$$

$$\frac{W}{F_{Ao}} = \int_0^{0.2} \left(\frac{1}{-r_A'}\right) dX_A = 1.886, \quad \frac{\text{g cat·h}}{\text{mol}}$$

$$W = 1.886 \times F_{Ao}$$

$$= 1.886 \times 1, \quad F_{Ao} = 1 \text{ mol/h}$$

$$W = 1.886 \text{ g}$$

Amount of catalyst needed to effect 20% conversion of 1 mol/h CO feed = 1.886 g ... **Ans.**

[You can plot $1/-r_A'$ v/s X_A and evaluate the integral graphically by measuring the area under curve between $X_A = 0$ and $X_A = 0.2$.]

Example 6.38 : *The methane is produced by a catalytic reaction, at 30 atm and 340 °C, between carbon dioxide and hydrogen.*

$$CO + 4\,H_2 \rightleftharpoons CH_4 + 2\,H_2O$$

The rate of reaction is given by

$$-r' = \frac{k\,p_{CO_2}\,p_{H_2}^4}{[1 + k_1 p_{H_2} + k_2 p_{CO_2}]^5}$$

$$k_1 = 1.73\ atm^{-1},\ k_2 = 0.3\ atm^{-1}$$

$$k = 7\ kmol\ CH_4/(kg\ cat.\ h \cdot atm^{-5})$$

Carbon monoxide and hydrogen are fed to a plug flow reactor in the stoichiometric proportion. The feed rate to the reactor is 100 kmol/h. Calculate the amount of catalyst needed to achieve 20% conversion of CO.

Solution : $CO + 4\,H_2 \rightleftharpoons CH_4 + 2\,H_2O$

$$P = 30\ atm,\quad T = 340\ °C$$

$$-r' = \frac{k\,p_{CO_2}\,p_{H_2}^4}{[1 + k_1 p_{H_2} + k_2 p_{CO_2}]^5}$$

$$k_1 = 1.73\ atm^{-1},\ k_2 = 0.3\ atm^{-1}$$

$$k = 7\ kmol/(kg\ cat\ .\ h\ .\ atm^{-5})$$

Fractional conversion of CO = 0.20.

$$A + 4\,B \rightleftharpoons R + 2S \quad \text{(general reaction)}$$

$$X_A = 0.20 \text{ (to be achieved)}$$

Initial moles : 1 kmol A and 4 kmol H_2.

Let X_A be the fractional conversion of A at any time t.

Moles at any time t :

$$\text{Moles of A} = 1 - 1\,(X_A) = 1 - X_A,\ \text{kmol}$$

$$\text{Moles of B} = 4 - 4/1 \cdot (X_A) = 4\,(1 - X_A),\ \text{kmol}$$

$$\text{Moles of R} = (1/1 \times X_A) = X_A,\ \text{kmol}$$

$$\text{Moles of S} = (2/1 \times X_A) = 2\,X_A,\ \text{kmol}$$

[for 1 kmol of A, moles of A reacted = $1 \times X_A = X_A$ kmol]

$$\text{Total moles} = 1 - X_A + 4\,(1 - X_A) + X_A + 2\,X_A = 5 - 2\,X_A,\ \text{kmol}$$

$$p_{CO_2} = p_A = 30\left(\frac{1 - X_A}{5 - 2\,X_A}\right)$$

$$p_{H_2} = p_B = 30\left(\frac{4\,(1-X_A)}{5-2\,X_A}\right) = 120\left(\frac{1-X_A}{5-2\,X_A}\right)$$

$$-r_A' = \frac{k\,p_A\,p_B^4}{[1 + k_1\,p_B + k_2 p_A]^5}$$

$$-r_A' = \frac{7 \times 30\left(\frac{1-X_A}{5-2\,X_A}\right) \times \left[120\left(\frac{1-X_A}{5-2\,X_A}\right)\right]^4}{\left[1 + 1.73 \times 120\left(\frac{1-X_A}{5-2\,X_A}\right) + 0.30 \times 30\left(\frac{1-X_A}{5-2\,X_A}\right)\right]^5}$$

$$= \frac{4.35 \times 10^{10}\,[(1-X_A)/(5-2\,X_A)]^5}{\left[1 + 216.6\left(\frac{1-X_A}{5-2\,X_A}\right)\right]^5}$$

Evaluate $-r_A'$, $1/(-r_A')$ at various X_A and tabulate.

X_A	0	0.05	0.01	0.15	0.20
$-r_A'$	0.0814	0.0811	0.0808	0.08043	0.08
$1/(-r_A')$	12.285	12.33	12.38	12.43	12.5

For a plug flow reactor, the performance equation is

$$\frac{W}{F_{Ao}} = \int_0^{X_{Aout}} \frac{dX_A}{-r_A'} = \int_0^{0.20} \left(\frac{1}{-r_A'}\right) dX_A$$

Use Simpson's one-third rule for evaluation of the integral on RHS of the above equation.

$$h = 0.05 - 0 = 0.05 \quad \text{or} \quad 0.10 - 0.05 = 0.05$$

$$\int_0^{0.20} \left(\frac{1}{-r_A'}\right) dX_A = \frac{0.05}{3}\,[\,\underline{12.285 + 4 \times 12.33 + 12.38} + \underline{12.38 + 4 \times 12.43 + 12.5}\,]$$

$$= \frac{0.05}{3}\,(148.585) = 2.476$$

$$\frac{W}{F_{Ao}} = 2.476$$

$$W = 2.476 \times F_{Ao}, \quad F_{Ao} = 100 \text{ kmol/h}$$

$$= 2.476 \times 100$$

$$= 247.6 \text{ kg}$$

Amount of catalyst needed for 20% conversion of 100 kmol/h CO feed = 247.6 kg

... Ans.

Example 6.39 : *Oxidation of NO is catalysed by an active carbon according to the following rate :*

$$-r' = \frac{p_{NO}^2 \, p_{O_2}}{a + b\, p_{NO}^2 + c\, p_{NO_2}}, \quad \frac{mol\ NO\ converted}{g\ cat.\ h}$$

where $a = 0.0001619,\ b = 4.812,\ c = 0.001352,$

p = *partial pressure in atm*

Find the volume of a reactor for converting 50 metric tonnes per day of NO to NO_2, when using an air - NO mixture containing 15 mole % of NO. The conversion of NO is 90%. The bulk specific gravity of catalyst is 0.48 and the total pressure is 3 atm.

Solution : $NO + \frac{1}{2} O_2 \longrightarrow NO_2$

$A + \frac{1}{2} B \longrightarrow R$

Conversion of NO = 90/100 = 0.90

P = total pressure = 3 atm

Sp. gr. of catalyst = 0.48

Bulk density of catalyst = $0.48 \times 1000 = 480$ kg/m^3

Flow rate of NO = 50×10^3 kg per day = 2083.3 kg/h

Molar flow rate of NO to the reactor = $\frac{2083.3}{30}$ = 69.44 kmol/h

Air - NO mixture to the reactor contains 15 mole % of NO.

Total molar feed rate = 69.44/0.15 = 462.93 kmol/h

Molar feed rate of air = 462.93 – 69.44 = 393.49 kmol/h

O_2 in the air = 0.21 (393.49) = 82.63 kmol/h

N_2 in the air = 0.79 (393.49) = 310.86 kmol/h

At 90% conversion :

NO reacted = 0.90 (69.44) = 62.496 kmol/h

NO_2 produced = $\frac{1}{1} \times 62.496 = 62.496$ kmol/h

O_2 reacted = $\frac{1/2}{1} \times 62.496 = 31.25$ kmol/h

O_2 unreacted = 82.63 – 31.25 = 51.38 kmol/h

NO unreacted = 69.44 – 62.496 = 6.944 kmol/h

Product mixture :

$$NO = 6.944 \text{ kmol/h}$$
$$O_2 = 51.38 \text{ kmol/h}$$
$$N_2 = 310.86 \text{ kmol/h}$$
$$NO_2 = 62.946 \text{ kmol/h}$$
$$\text{Total moles} = 432.13 \text{ kmol/h}$$
$$\text{Partial pressure of NO} = p_{NO} = (6.944/432.13) \times 3 = 0.048 \text{ atm}$$
$$\text{Partial pressure of } O_2 = p_{O_2} = (51.38/432.13) \times 3 = 0.357 \text{ atm}$$
$$\text{Partial pressure of } NO_2 = p_{NO_2} = (62.946/432.13) \times 3 = 0.437 \text{ atm}$$

Now we will find the reaction rate :

$$(-r') = \frac{p_{NO}^2 \, p_{O_2}}{a + b \, p_{NO}^2 + c \, p_{NO_2}}$$

$$a = 0.0001619, \; b = 4.812, \; c = 0.001352$$

$$(-r') = \frac{(0.048)^2 (0.357)}{0.0001619 + 4.812 \times (0.048)^2 + 0.001352 (0.437)}$$

$$= 0.06947 \frac{\text{mol}}{\text{g cat. h}} = 0.06947 \frac{\text{kmol NO converted}}{\text{kg cat} \cdot \text{h}}$$

$$\text{Weight of catalyst} = \frac{\text{molar feed rate of NO}}{\text{reaction rate}}$$

$$\text{Weight of catalyst} = \frac{69.44}{0.06947}$$

$$= 999.568 \approx 999.6 \text{ kg}$$

$$\text{Volume of reactor} = \frac{\text{weight of catalyst}}{\text{bulk density of catalyst}}$$

$$= \frac{999.6}{480}$$

$$= 2.08 \text{ m}^3$$

Volume of reactor = volume of catalyst = 2.08 m³ ... **Ans.**

Catalytic Reactors :

Solid catalysed gas phase reactions are of great industrial importance. These reactions are effected on a commercial scale in fixed-bed and fluidised-bed reactors.

Fixed-bed reactors :

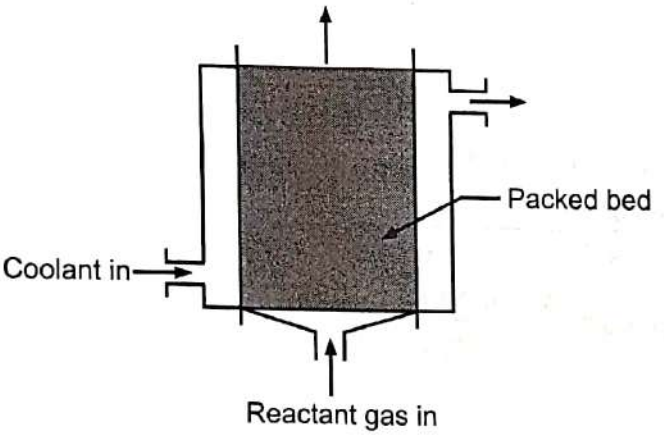
A fixed-bed reactor (also referred to as a packed-bed reactor) consists of a cylinder of large diameter with multiple catalyst beds or many tubes in parallel packed with catalyst and encased in a large shell. In these reactors, reactant gas flows continuously through a stationery bed of catalyst. These reactors make use of relatively large size catalyst particles for low pressure drop. The catalyst particles may be granular, pelleted, cylindrical or spherical.

A multitubular construction is required when large quantities of heat must be transferred to the surroundings as in case of high exothermic reactions.

[Refer Fig. 6.13 (b)]

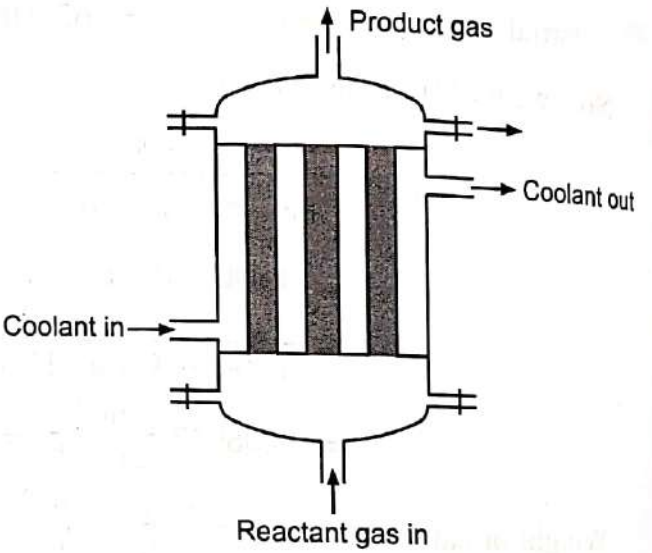
An alternate arrangement that reduces temperature variation in the fixed-bed reactor consists of a multistage catalyst bed with cooling coils placed between the stages. In this arrangement, conversion in the first stage (where the reaction is fastest) is kept low by using less amount of catalyst as compared to that in the other (succeeding) stages.

[Refer Fig. 6.13 (c)]

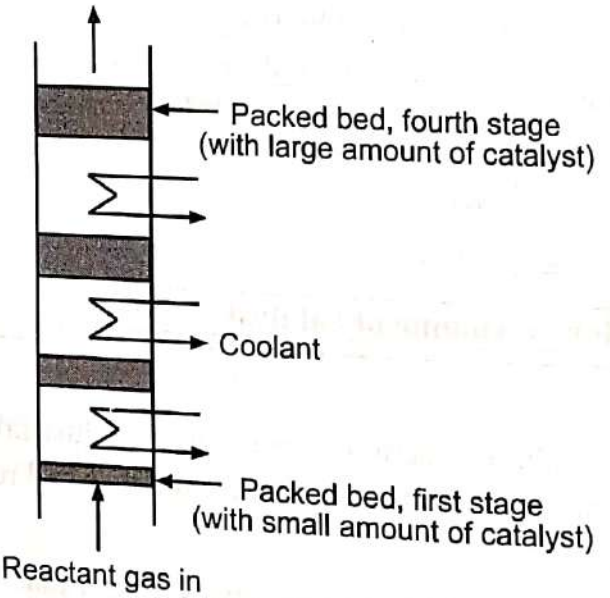


(a)

(a)



(b) Multi-tube fixed bed reactor



(c) Multi-bed reactor

Fig. 6.13 : Fixed bed reactors

Fluidised-bed reactors :

These reactors are especially suitable when frequent catalyst generation is required or for reactions with a very high heat effects.

A fluidised-bed reactor consists of a vertical cylindrical vessel (large diameter) containing fine catalyst particles. A reactant gas is introduced through the bottom of the reactor at a rate such that catalyst particles are suspended in the gas stream without being carried out. Under these conditions the entire bed of catalyst particles behaves like a boiling liquid and this action tends to make the composition of reaction mixture and temperature uniform throughout the bed.

Fig. 6.14 (a) shows a fluidised-bed reactor for a stable catalyst which does not require regeneration continuously.

Fig. 6.14 (b) shows a fluidised-bed reactor for a deactivating catalyst which requires removal and regeneration continuously. Here a portion of the catalyst particles is continuously removed from the reactor and fed to a regenerator. The catalyst particles are regenerated in the regenerator and are then returned to the reactor.

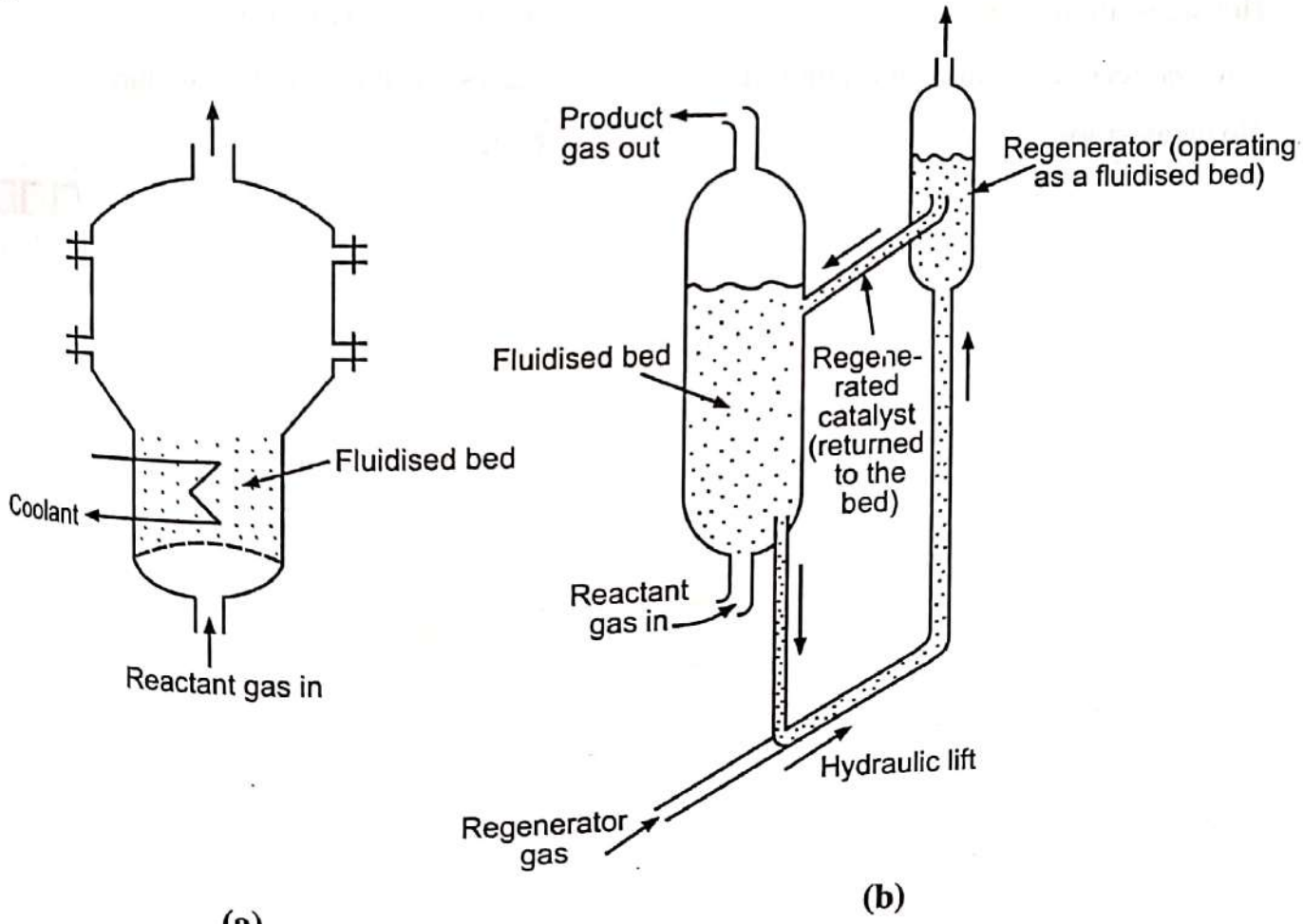


Fig. 6.14 : Fluidised-bed reactors

Fixed-bed reactor	Fluidised-bed reactor
• Catalyst particles are stationery and gas flows through it.	Catalyst particles are suspended in the gas stream. They move about depending upon the gas stream velocity.
• Catalyst regeneration is a serious problem.	Catalyst regeneration is very easy.
• Operation is not practically isothermal.	Practically isothermal operation.
• No carry over of catalyst particle thus does not require recovery units.	Carry over of catalyst particles thus requires recovery units.
• Cannot use very small size catalyst particles due to plugging and high pressure drop. Uses relatively large size particles for low pressure drop.	Uses fine size catalyst particles (which provides large area of contact)
• Gas approximates plug flow.	Gas flow is far from plug flow with considerable bypassing.
• Small diameter multi-tubular units.	Large diameter vessels.
• Hot spots are formed.	No hot spot formation.
• Catalyst recovery unit is not required.	Catalyst recovery unit is required.
• No catalyst loss.	Chances of catalyst loss.

❐❐❐

CHAPTER 7

G/L REACTIONS ON SOLID CATALYSTS

Here we consider the class of heterogeneous reactions in which the gas phase reactant A reacts with the liquid phase reactant B on a solid catalyst to form products. Such multiphase reactions are represented by

$$A\,(g \rightarrow l) + bB\,(l) \xrightarrow[\text{catalyst}]{\text{on solid}} \text{products}$$

These three phase reactions can be carried out in a number of ways, as shown in Fig. 7.1. In all these arrangements, gas and liquid reactants are brought into contact with a solid catalyst. The reaction between them occurs on the surface of the catalyst. Out of these reactors the slurry and trickle bed reactors are the most common.

A multiphase reactor is a reactor in which two or more phases are required to carry out/perform reaction. Most multiphase reactors contain gas and liquid phases which contact a solid. In the multiphase reactors such as slurry and trickle bed reactors, the reaction between the gas and the liquid occurs/takes place on the surface of a solid catalyst.

A slurry reactor is a multiphase flow reactor in which gaseous reactant is bubbled through a solution containing solid catalyst particles (the solid catalyst particles are suspended in the liquid). This reactor may be operated in a semibatch or continuous fashion. In the former case, a fixed charge of liquid is added to the stirred tank initially and gas is bubbled continuously so that the reactor is batch with respect to the liquid phase.

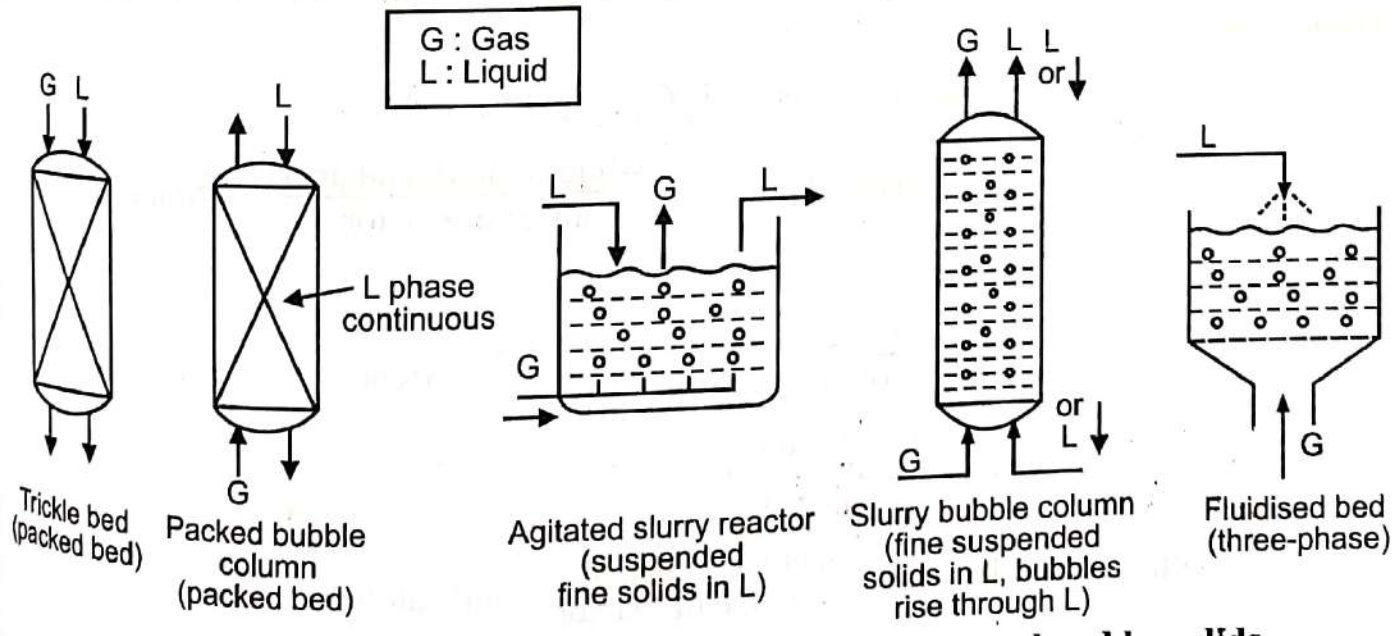


Fig. 7.1 : Various ways of carrying out G/L reactions catalysed by solids

In the trickle bed reactor, gas and liquid flow (trickle) cocurrently downward over a packed/fixed bed of catalyst particles. The trickle bed reator is the three - phase version of the fixed bed arrangement. Trickle bed reactors use large size catalyst particles.

The general rate equation :

Consider the reaction

$$A\,(g \rightarrow l) + bB\,(l) \xrightarrow[\text{surface}]{\text{on catalyst}} \text{products} \qquad \ldots (7.1)$$

(with b = mol B/mol A)

which is first order with respect to A and first order with respect to B.

The rate at which A is consumed is given by

$$-r_A''' = k_A''' C_A C_B, \quad k_A - \text{rate constant referred to A – rate is measured in terms of A}$$

Similarly, for B

$$-r_B''' = k_B''' C_A C_B, \qquad k_B - \text{rate constant referred to B}$$

From the stoichiometry

$$-r_A''' = -\frac{r_B'''}{b}, \quad \text{mol A/(m}^3\,\text{cat·s)}$$

and

$$k_A''' = \frac{k_B'''}{b}, \quad \text{m}^6 l/(\text{mol B·m}^3\text{cat·s})$$

To develop the overall rate equation in terms of the two-film theory, we use the following nomenclature.

H_A = Henry's law constant $= p_A/C_A$, $(P_a \cdot m^3 l)/mol$

a_i = specific interfacial area $= \dfrac{\text{gas-liquid interfacial area}}{\text{volume of reactor}}$, $m^2 i/m^2 r$

a_c = specific surface of particles

$= \dfrac{\text{external surface of catalyst particles}}{\text{volume of reactor}}$, $m^3\,cat/m^3 r$

$\left[\text{for spheres : } a_c = \dfrac{6\,f_s}{dp}\right]$

f_s = solid loading $= \dfrac{\text{volume of particles}}{\text{volume of reactor}}$, $m^3\,cat/m^3 r$

f_l = liquid hold up (fraction) $= \dfrac{\text{volume of liquid}}{\text{volume of reactor}}$, $m^2 l/m^3 r$

Here the gas phase reactant first dissolves in the liquid and then both the reactants – A and B move to the catalyst surface for reaction to take place on the surface.

For reactant A the following steps are involved in the overall reaction process :

(i) Transport from the bulk gas phase to the gas-liquid interface through the gas film.

(ii) Transport from the interface, through the liquid film, to the bulk liquid.

(iii) Transport from the bulk liquid to the external catalyst surface through the liquid film surrounding the catalyst particle.

(iv) Diffusion and reaction within the porous catalyst particle.

A state of equilibrium exists at the interface. At equilibrium, the concentration in the liquid phase is related to that of the gas phase through Henry's law :

$$p_{Ai} = H_A C_{Ai}$$

Thus, the overall rate expression for A will have to account for : the resistance to transfer across the G/L interface (through G and L films), resistance to transfer at the surface of the catalyst particle and the resistance of the chemical reaction step.

These resistances are shown graphically in Fig. 7.2.

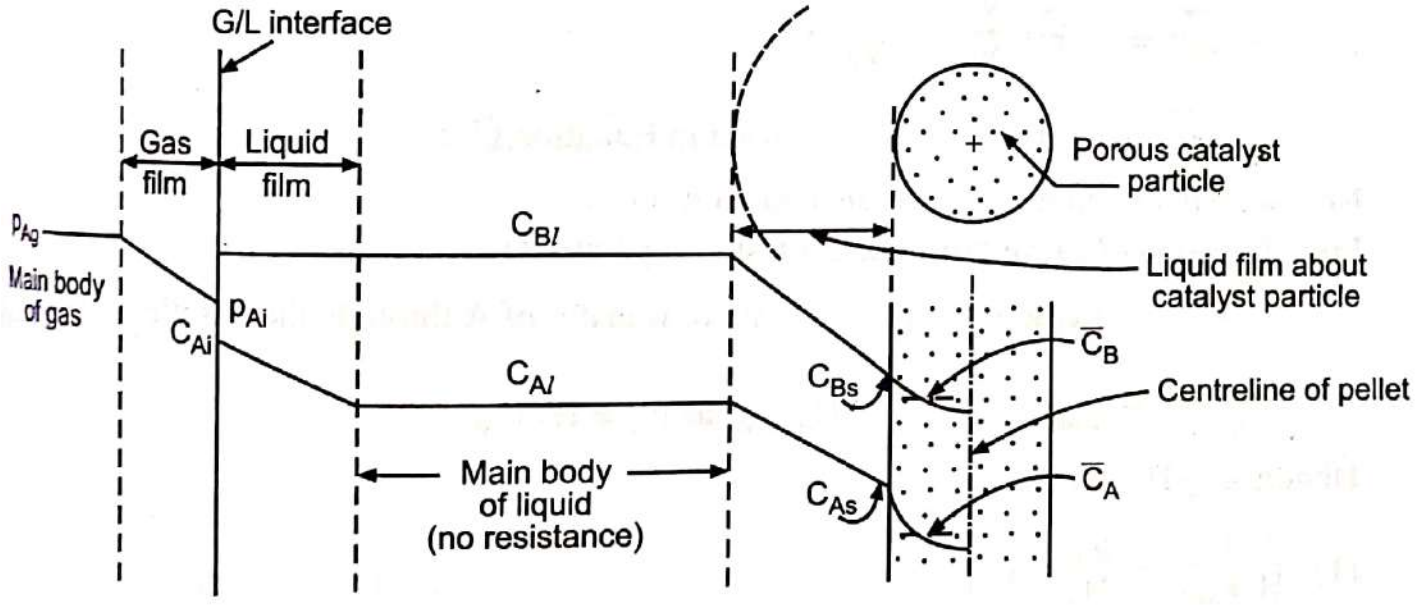


Fig. 7.2 : Resistances in series in G/L reaction on a catalyst surface

k_g – gas film coefficient/gas phase mass transfer coefficient, mol/(Pa·m²·s)

k_l – liquid film coefficient/liquid phase mass transfer coefficient, m³*l*/(m²·s)

k_c – film coefficient about catalyst particle/mass transfer coefficient for particles, m³*l*/(m²cat·s)

$\bar{C}_B, \bar{C}_A$ … in mol/m³*l*.

$\mathscr{D}_{Ae}, \mathscr{D}_{Be}$ – effective diffusion coefficient for A and B, m³*l*/(m cat·s)

The rate of these individual steps in mol A/(m^3 reactor·s) is given by

$$-r_A''' = k_{Ag}\, a_i\, (p_{Ag} - p_{Ai}) = k_{Al}\, a_i\, (C_{Ai} - C_{Al}) = k_{Ac}\, a_c\, (C_{Al} - C_{As})$$

gas film | liquid film | film about catalyst particle surface

$$= (k_A''' \bar{C}_B)\, \eta_A\, f_s\, C_{As} \qquad \ldots (7.2)$$

diffusion and reaction in porous particle

$-r_A'''$ = $(k_A''' \bar{C}_B)\, \bar{C}_A$ – actual rate within pore

$-r_A'''$ = $(k_A''' \bar{C}_B)\, \eta_A\, C_{As}$ – in terms of concentration at the external surface /mouth of pore.

We have

$(-r_A''')\, V_S = (-r_A'''')\, V_r$, $-r_A'''$ → per m^3 catalyst, $-r_A''''$ → per m^3 reactor

$$\therefore \quad (-r_A'''') = (-r_A''') \frac{V_S}{V_r}, \quad \frac{V_S}{V_r} = f_s$$

$-r_A''''$ = $(k_A''' \bar{C}_A)\, \eta_A\, f_s\, C_{As}$ … used in Equation (7.2).

Now we will obtain the overall rate equation for A.

From Equation (7.2) we can write each step separately :

$(-r_A'''')$ = $k_{Ag}\, a_i\, (p_{Ag} - p_{Ai})$ … rate of transfer of A through the gas film

$$\frac{(-r_A'''')}{k_{Ag}\, a_i} = p_{Ag} - p_{Ai} = p_{Ag} - H_A\, C_{Ai} \quad \text{as } p_{Ai} = H_A\, C_{Ai}$$

Dividing by H_A, we get

(1) $$\frac{(-r_A'''')}{H_A k_{Ag} a_i} = \frac{p_{Ag}}{H_A} - C_{Ai}$$

$(-r_A'''')$ = $k_{Al}\, a_i\, (C_{Ai} - C_{Al})$ … rate of transfer of A through the liquid film.

(2) $$\frac{(-r_A'''')}{k_{Al}\, a_i} = C_{Ai} - C_{Al}$$

Similarly, for other steps, we get

(3) $$\frac{(-r_A'''')}{k_{Ac}\, a_c} = C_{Al} - A_{As}$$ … (rate of transfer of A through the film about the particle surface)

(4) $$\frac{(-r_A'''')}{(k_A''' \bar{C}_B)\, \eta_A\, f_s} = C_{As}$$

Adding (1) to (4), we get

$$\frac{(-r_A'''')}{H_A k_{Ag} a_i} + \frac{(-r_A'''')}{k_{Al} a_i} + \frac{(-r_A'''')}{k_{Ac} a_c} + \frac{(-r_A'''')}{(k_A''' \bar{C}_B)\,\eta_A f_s} = \left(\frac{p_{Ag}}{H_A} - C_{Ai}\right) + (C_{Ai} - C_{Al})$$

$$+ (C_{Al} - C_{As}) + C_{As}$$

$$(-r_A'''')\left[\frac{1}{H_A k_{Ag} a_i} + \frac{1}{k_{Al} a_i} + \frac{1}{k_{Ac} a_c} + \frac{1}{(k_A''' \bar{C}_B)\,\eta_A f_s}\right] = \frac{p_{Ag}}{H_A}$$

$$(-r_A'''')\left[\frac{1}{k_{Ag} a_i} + \frac{H_A}{k_{Al} a_i} + \frac{H_A}{k_{Ac} a_c} + \frac{H_A}{(k_A''' \bar{C}_B)\,\eta_A f_s}\right]\frac{1}{H_A} = \frac{p_{Ag}}{H_A}\,.$$

The overall rate of reaction of A is given by

$$(-r_A'''') = \left[\frac{1}{\frac{1}{k_{Ag} a_i} + \frac{H_A}{k_{Al} a_i} + \frac{H_A}{k_{Ac} a_c} + \frac{H_A}{(k_A''' \bar{C}_B)\,\eta_A f_s}}\right] p_{Ag}\,, \quad \frac{\text{mol A}}{\text{m}^3\text{r}\cdot\text{s}} \qquad \ldots (7.3)$$

Units of $k_{Ag} \Rightarrow \frac{\text{mol A}}{\text{Pa}\cdot\text{m}^2\cdot\text{s}}$, $a_i \Rightarrow \text{m}^2/\text{m}^2\text{r}$, $p_{Ag} \Rightarrow \text{Pa}$

Units of $\frac{p_{Ag}}{1/k_{Ag} a_i}$, i.e., $k_{Ag} a_i p_{Ag} \Rightarrow \frac{\text{mol A}}{\text{Pa}\cdot\text{m}^2\cdot\text{s}} \times \frac{\text{m}^2}{\text{m}^3\text{r}} \times \text{Pa} \Rightarrow \frac{\text{mol A}}{\text{m}^3\text{r}\cdot\text{s}}$ ... that of $(-r_A'''')$

Units of $\frac{p_{Ag}}{H_A/k_{Ac} a_c}$, i.e., $\frac{k_{Ac} a_c p_{Ag}}{H_A} \Rightarrow \left(\frac{\text{m}^3 l}{\text{m}^2\,\text{cat}\cdot\text{s}}\right)\left(\frac{\text{m}^2\,\text{cat}}{\text{m}^3\text{r}}\right)\frac{(\text{Pa})}{(\text{Pa}\cdot\text{m}^3 l/\text{mol A})} \Rightarrow \frac{\text{mol A}}{\text{m}^3\text{r}\cdot\text{s}}$

Units of $\frac{p_{Ag}}{H_A/(k_A''' \bar{C}_B)\,\eta_A f_s}$, i.e., $\frac{(k_A''' \bar{C}_B)\,\eta_A f_s\, p_{Ag}}{H_A} \Rightarrow \left(\frac{\text{m}^6 l}{\text{mol B}\cdot\text{m}^3\,\text{cat}\cdot\text{s}}\right)\left(\frac{\text{mol B}}{\text{m}^3 l}\right)\left(\frac{\text{m}^3\,\text{cat}}{\text{m}^3\,\text{r}}\right)$

$$\times\left(\frac{(\text{Pa})}{(\text{Pa}\cdot\text{m}^3 l/\text{mol A})}\right) \Rightarrow \frac{\text{mol A}}{\text{m}^3\text{r}\cdot\text{s}}$$

For reactant B, we have to consider only two resistances (in series).

(1) $(-r_B'''') = k_{Bc} a_c (C_{Bl} - C_{Bs})$

(2) $(-r_B'''') = (k_B''' \bar{C}_B)\,\eta_B f_s C_{Bs}$

Combine (1) and (2) to obtain the overall rate expression for B :

(3) $\frac{(-r_B'''')}{k_{Bc} a_c} = C_{Bl} - C_{Bs}$

(4) $\frac{(-r_B'''')}{(k_B''' \bar{C}_B)\,\eta_B f_s} = C_{Bs}$

Adding (3) and (4), we get

$$(-r_B'''') \left[\frac{1}{k_{Bc}\, a_c} + \frac{1}{(k_B''' \bar{C}_A)\, \eta_B\, f_s}\right] = C_{Bl} - C_{Bs} + C_{Bs} = C_{Bl}$$

$$(-r_B'''') = \left[\frac{1}{\dfrac{1}{k_{Bc}\, a_c} + \dfrac{1}{(k_B''' \bar{C}_A)\, \eta_B\, f_s}}\right] C_{Bl}, \ \frac{\text{mol B}}{\text{m}^3\text{r}\cdot\text{s}} \qquad \ldots(7.4)$$

The rates are related by

$$(-r_A'''') = \frac{(-r_B'''')}{b}$$

η_A – effectiveness factor for the first order reaction of A with rate constant $(k_B''' \bar{C}_B)$

η_B – effectiveness factor for the first order reaction of B with rate constant $(k_B''' \bar{C}_A)$.

$H_A = p_A/C_A$ – Henry's law constant.

The rate of reaction in mol/(m³ reactor·s) is given by either Equation (7.3) or Equation (7.4). It is not possible to find the rate with these expressions as $\bar{C}_A$ and $\bar{C}_B$ are unknown. However, we usually come across either of the following two simplifying extremes :

Extreme 1 : $C_{Bl} >> C_{Al}$

In reaction systems with pure liquid B and slightly soluble gas A (large H_A value), we can take

$C_{Bs} = \bar{C}_{B,\text{ within pellet}} = C_{Bl}$ …… same value of concentration of B everywhere.

With C_B constant everywhere, the rate becomes first order with respect to A and Equation (7.3) becomes

$$\boxed{(-r_A'''') = \left[\frac{1}{\dfrac{1}{k_{Ag}\, a_i} + \dfrac{H_A}{k_{Al}\, a_i} + \dfrac{H_A}{k_{Ac}\, a_c} + \dfrac{H_A}{(k_A''' C_{Bl})\, \eta_A\, f_s}}\right] p_{Ag}, \ \frac{\text{mol A}}{(\text{m}^3\,\text{reactor}\cdot\text{s})}} \qquad \ldots(7.5)$$

where $k_A''' C_{Bl}$ is the first order rate constant for A.

Extreme 2 : $C_{Bl} << C_{Al}$

In reaction systems with dilute reactant B, highly soluble A (low H_A value) and high pressure, we can use

$$C_{Al} = \frac{p_{Ag}}{H_A} \quad \ldots\ldots \text{ throughout the reactor}$$

With C_A constant everywhere, the rate becomes first order with respect to B and Equation (7.4) becomes

$$(-r_B''') = \left[\frac{1}{\frac{1}{k_{Bc}\, a_c} + \frac{1}{\left(\frac{k_B'''\, p_{Ag}}{H_A}\right)\eta_B\, f_s}}\right] C_{Bl}, \quad \frac{\text{mol B}}{(\text{m}^3\,\text{reactor}\cdot\text{s})} \qquad \ldots (7.6)$$

where $k_B'''\, p_{Ag}/H_A$ is the first order rate constant for B.

By the unequal signs >> or << we mean two to three times greater or less than.

Calculate the rate of reaction using Equations (7.5) and (7.6) and use the value whichever is smaller. Thus,

If $(-r_A''')$ by Equation (7.5) << $(-r_B''')$ by Equation (7.6), then C_{Bl} is in excess, i.e., B is in excess and extreme 1 applies.

If $(-r_A''')$ by Equation (7.5) >> $(-r_B''')$ by Equation (7.6), then C_{Al} is in excess, i.e., A is in excess, extreme 2 applies and Equation (7.6) gives the rate of reaction.

Keep in mind that almost always one or other of the extremes applies.

PERFORMANCE EQUATIONS FOR AN EXCESS OF B :

We treat all types of three-phase reactors (trickle bed reactors, slurry reactors, etc.) at the same time. Here it is important to know the flow patterns of the contacting phases and which component, A or B, is in excess.

Let us first consider that the component B is in excess. In such cases, the flow pattern of liquid is not important. We have to consider the flow pattern only of the gas phase. For an excess of B, we have the following cases :

Mixed flow G + any flow L ..... with excess of B ... (7.7)

Here we have :

(i) C_A is uniform throughout ..... as gas is in mixed flow.

(ii) C_B remains constant ..... B is in excess. [Refer to Fig. 7.3].

Therefore the accounting should be made about the reactor as a whole.

A material balance for A and B about the reactor as a whole yields

$$F_{Ao} = \frac{F_{Bo} X_B}{b} = (-r_A'''') V_r \qquad \ldots (7.8)$$

$F_{Ao} X_A$ ..... rate of loss of A

$F_{Bo} X_B$ ..... rate of loss of B

$(-r_A'''') V_r$ ..... rate of disappearance of A by reaction ... (all in mol A/s)

Let us obtain Equation (7.8) :

The material balance of A about the reactor, considering the mixed flow, is

$$F_{Ao} = F_A + (-r_A'''') V_r$$

$$F_{Ao} - F_A = F_{Ao} - F_{Ao}(1 - X_A) = (-r_A'''') V_r$$

$$\therefore \quad F_{Ao} X_A = (-r_A'''') V_r \qquad \ldots (C)$$

Similarly for B

$$F_{Bo} X_B = (-r_B'''') V_r$$

The rates are related by :

$$(-r_A'''') = \frac{(-r_B'''')}{b}$$

$$\therefore \quad (-r_B'''') = b(-r_A'''')$$

$$\therefore \quad F_{Bo} X_B = b(-r_A'''') V_r$$

$$\frac{F_{Bo} X_B}{b} = (-r_A'''') V_r \qquad \ldots (D)$$

Combining Equations (C) and (D) gives

$$\underbrace{F_{Ao} X_A}_{\text{I}} = \underbrace{\frac{F_{Bo} X_B}{b}}_{\text{II}} = \underbrace{(-r_A'''') V_r}_{\text{III}} \qquad \ldots \text{Equation (7.8)}$$

To obtain the solution, either combine terms I and III or II and III.

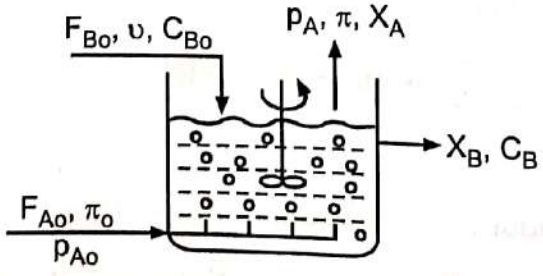


Fig. 7.3 : Gas is in mixed flow as gas bubbles swim around the bed

Plug flow G + any flow L ..... with excess of B :

Since B is in excess, it changes slightly as it goes through the reactor. Hence, C_B remains roughly constant throughout the reactor (at $\bar{C}_B$). As the gas in plug flow, C_A changes along a flow path through the reactor [Refer Fig. 7.4]. Consequently, the accounting for material balance should be made over a differential element of volume dV_r the reactor.

The material balance of A over dV_r is

$$F_A = F_A + dF_A + (-r_A'''')\, dV_r$$

$$-dF_A = -d[F_{Ao}(1 - X_A)] = (-r_A'''')\, dV_r$$

$$F_{Ao}\, dX_A = (-r_A'''')\, dV_r$$

Within dV_r :

$$F_{Ao}\, dX_A = \frac{F_{Bo}\, dX_B}{b}$$

$\therefore$ Integrating about/across the whole reactor, we get

$$F_{Ao} \int \frac{dX_A}{(-r_A'''')} = \int dV_r, \quad F_{Ao} \int dX_A = \frac{F_{Bo}}{b} \int dX_B$$

$$\frac{V_r}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{(-r_A'''')} \qquad \ldots (7.9)$$

and $$F_{Ao} X_A = \frac{F_{Bo} X_B}{b}$$

where $$1 - X_A = \frac{p_A/(\pi - p_A)}{p_{Ao}/(\pi_o - p_{Ao})} = \frac{p_A(\pi_o - p_{Ao})}{p_{Ao}(\pi - p_A)}$$

For dilute systems, it becomes : $1 - X_A = p_A \pi_o / p_{Ao} \pi$

and for dilute systems with π constant : $1 - X_A = p_A / p_{Ao}$.

$$0 - dX_A = \frac{p_{Ao}(\pi - p_A)(\pi_o - p_{Ao})\, dp_A - p_A(\pi_o - p_{Ao}) \cdot p_{Ao}(-dp_A)}{p_{Ao}(\pi - p_A)^2}$$

$$-dX_A = \frac{\pi(\pi_o - p_{Ao})\, dp_A}{p_{Ao}(\pi - p_A)^2} \xrightarrow[\pi = \text{constant}]{\text{dilute}} \frac{dp_A}{p_{Ao}}$$

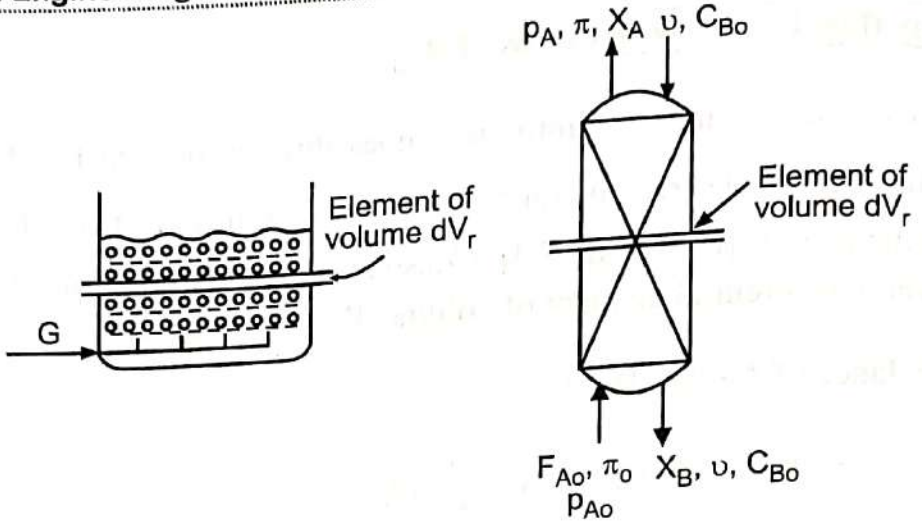


Fig. 7.4 : Gas is in plug flow in these reactors

Mixed Flow G + Batch L ..... with excess of B :

Here gas bubbles through a batch of liquid, which is kept under agitation. Hence, gas is in mixed flow. C_A at the exit is the same as C_A in the reactor. Here we are dealing with a batch of liquid so it is an unsteady-state operation. Here C_B decreases slowly with time as it is being used up by the reaction with A, but at any time t : C_B is roughly constant in the reactor (Refer Fig. 7.5).

The material balance of A and B at any time t over the whole reactor is

$$F_{Ao} X_{A,\,exit} = \frac{V_l}{b}\left(-\frac{dC_B}{dt}\right) = (-r_A'''')\, V_r \qquad \ldots (7.10)$$

I II III

I – loss of A from gas and II – decrease of B with time in L.

Let us obtain Equation (7.10) :

For component A (at any time t) :

$$\text{Input} = \text{output} + \text{disappearance}$$

$$F_{Ao} = F_{Ao}(1 - X_{A,\,exit}) + (-r_A'''')\, V_l$$

$$F_{Ao} X_{A,\,exit} = (-r_A'''')\, V_r \qquad \ldots (C$$

For component B (for a batch of liquid at any time t) :

$$0 = 0 + \frac{dN_B}{dt} + (-r_B)\, V_r$$

input output accumulation of B disappearance of B

$$(-r_B)\, V_l = -\frac{dN_B}{dt}$$

$$(-r_B) = \frac{1}{V_l}\left(-\frac{dN_B}{dt}\right) = -\frac{d\,(N_B/V_l)}{dt} = -\frac{dC_B}{dt}$$

$$(-r_B) = -\frac{dC_B}{dt}$$

The rates of A and B are related by

$$\frac{-r_A}{1} = \frac{-r_B}{b}$$

$$\therefore \quad (-r_B) = b\,(-r_A) = -\frac{dC_B}{dt}$$

$$(-r_A) = \frac{1}{b}\left(-\frac{dC_B}{dt}\right)$$

We have,

$$(-r_A)\, V_l = (-r_A'''')\, V_r$$

and

$$-r_A = \frac{1}{b}\left(-\frac{dC_B}{dt}\right)$$

$$\therefore \quad \frac{V_l}{b}\left(-\frac{dC_B}{dt}\right) = (-r_A'''')\, V_r \qquad \ldots (D)$$

Combining Equations (C) and (D), we get

$$\boxed{F_{Ao}\, X_{A,\,exit} = \frac{V_l}{b}\left(-\frac{dC_B}{dt}\right) = (-r_A'''')\, V_r} \qquad \ldots \text{desired Equation (7.10)}$$

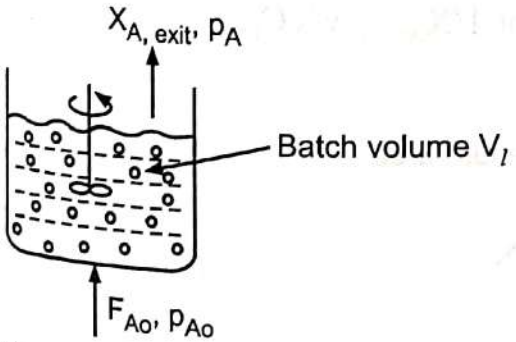


Fig. 7.5 : **Gas is in mixed flow – it bubbles through a batch of liquid**

At $t = 0$: $C_B = C_{Bo}$ and $X_{A,\,exit} = 0$

At $t = t_f$: $C_B = C_{Bf}$ and $X_{A,\,exit} = X_{A,\,exit,\,f}$

With t : $X_{A,\,exit} \downarrow$ and $p_A \uparrow$

$$F_{Ao}\, X_{A,\,exit} = \frac{V_l}{b}\left(-\frac{dC_B}{dt}\right)$$

$$\therefore \quad dt = \frac{V_l}{bF_{Ao}}\,\frac{-dC_B}{X_{A,\,exit}}$$

$$\int_0^t dt = \frac{-V_l}{b F_{Ao}} \int_{C_{Bo}}^{C_{Bf}} \frac{dC_B}{X_{A,\,exit}} = \frac{V_l}{bF_{Ao}} \int_{C_{Bf}}^{C_{Bo}} \frac{dC_B}{X_{A,\,exit}}$$

$$t = \frac{V_l}{bF_{Ao}} \int_{C_{Bf}}^{C_{Bo}} \frac{dC_B}{X_{A,\,exit}} \qquad \ldots (E)$$

$$\frac{V_l}{b}\left(\frac{-dC_B}{dt}\right) = (-r_A'''')\, V_r$$

Integrating gives

$$t = \frac{V_l}{b\,V_r} \int_{C_{Bf}}^{C_{Bo}} \frac{dC_B}{(-r_A'''')} \qquad \ldots (F)$$

Combining Equations (E) and (F) gives

$$\boxed{t = \text{processing time} = \frac{V_l}{bV_r} \int_{C_{Bf}}^{C_{B\,o}} \frac{dC_B}{(-r_A'''')} = \frac{V_l}{bF_{Ao}} \int_{C_{Bf}}^{C_{B\,o}} \frac{dC_B}{X_{A,\,exit}}} \qquad \ldots (7.11)$$

The integral on the RHS of Equation (7.11) can be evaluated graphically by plotting, say, $1/(-r_A'''')$ v/s C_B and measuring the area under the curve between $C_B = C_{Bf}$ and $C_B = C_{Bo}$.

(i) Evaluate $(-r_A'''')$ for each $C_B \rightarrow (-r_A'''')_o, \ldots\ldots (-r_A'''')_f \rightarrow C_{Bo} \ldots C_{Bf}$.

(ii) Plot $1/(-r_A'''')$ v/s C_B.

(iii) Measure the area to get the value of integral.

(iv) Calculate reaction time/processing time using Equation (7.11).

Or evaluate $X_{A,\,exit}$ at $C_{Bo} \ldots C_{Bf}$ and construct a plot of $1/X_{A,\,exit}$ v/s C_B.

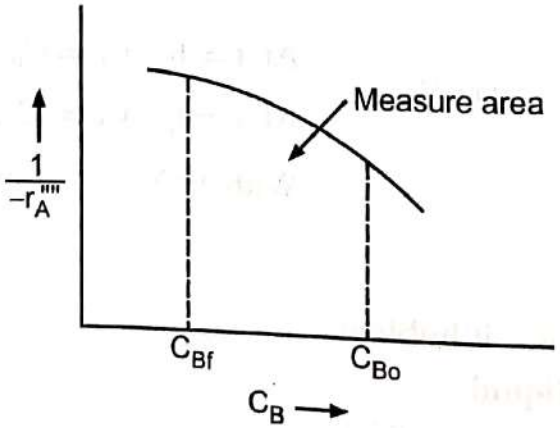


Fig. 7.6 : Determination of the processing time/reaction time graphically for a batch of liquid

$t = \frac{V_l}{bV_r}$ (area under the curve) × (scale of y-axis) (scale of x-axis)

Plug Flow G + Batch L ..... with excess of B :

Here C_B decreases slowly with time as B is used up, but at any instant C_B is the same everywhere and any element of gas contacts with the same C_B as it flows through the reactor (Refer Fig. 7.7).

As gas is in plug flow, the differential balance of A (the material balance of A over a differential element of volume dV_r) is

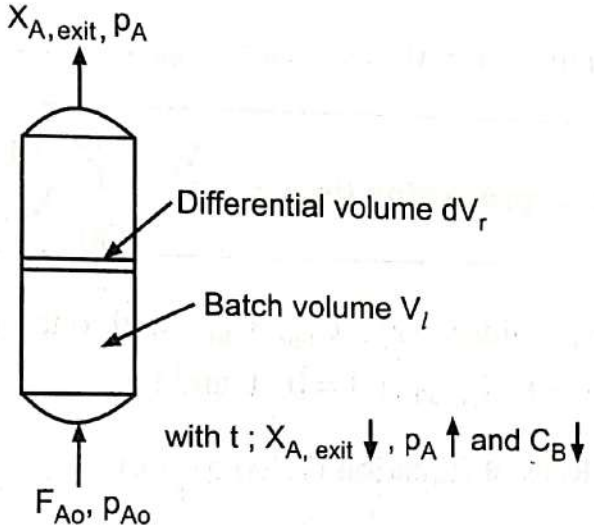


Fig. 7.7 : Gas is in plug flow + batch of liquid

$$F_{Ao}\,dX_A = (-r_A'''')\,dV_r$$

.....	.....
loss of A from gas	disappearance of A by reaction

Integrating over the reactor gives

$$\boxed{\frac{V_r}{F_{Ao}} = \int_0^{X_{A,exit}} \frac{dX_A}{(-r_A'''')}} \qquad \ldots (7.12)$$

Considering B, we can write,

$$\text{loss of A} = \frac{1}{b}\,\text{loss of B}$$

$$F_{Ao}X_{A,exit} = \frac{V_l}{b}\left(-\frac{dC_B}{dt}\right)$$

$$-r_B = -\frac{dC_B}{dt}$$

$$-r_A/1 = -r_B/b$$

$\therefore$

$$-r_B = b\,(-r_A)$$

$$b\,(-r_A) = -\frac{dC_B}{dt}, \qquad -r_A = \frac{1}{b}\left(-\frac{dC_B}{dt}\right)$$

Plug Flow L + Any Flow G – tower and packed bed operations ..... with excess of A :

The performance equation is

$$\boxed{\frac{V_r}{F_{Bo}} = \int_0^{X_B} \frac{dX_B}{-r_B''''}} \quad \ldots (7.17)$$

where $1 - X_B \approx \frac{C_B}{C_{Bo}}$

$-r_B''''$ is given by Equation (7.6).

Mixed Flow L + Any Flow G – tank operations – all types ..... with excess of A :

The performance equation is

$$\boxed{\frac{V_r}{F_{Bo}} = \frac{X_B}{-r_B''''}} \quad \ldots (7.18)$$

$$1 - X_B = \frac{C_B}{C_{Bo}}$$

$-r_B''''$ is given by Equation (7.6).

Batch L + Any Flow G ..... with excess of A :

Since A is in excess, C_A is approximately constant throughout a run.

The performance equation for B is

$$-r_B = \frac{V_r}{V_l}(-r_B'''') = \frac{-dC_B}{dt}$$

$$(-r_B)\, V_l = (-r_A'''')\, V_r$$

or $$\boxed{t = \int_0^t \frac{dC_B}{-r_B}} \quad \ldots (7.19)$$

SOLVED EXAMPLES

Example 7.1 : *Hydrogenation of 2-butyne-1, 4-diol to butenediol is to be carried out in a slurry reactor using a palladium based porous catalyst.*

$$\underset{A}{H_2\,(g \rightarrow l)} + \underset{B}{\text{butynediol}\,(l)} \xrightarrow[\text{catalyst}]{\text{solid}} \text{butenediol}$$

The reaction is first order in hydrogen and in butynediol. The initial concentration of butynediol (C_{Bo}) is 2500 mol/m³. Pure hydrogen is bubbled through the reactor. Unused hydrogen is recompressed and recirculated, and the whole operation takes place at 14.6 atm and 35°C.

Find the time required to achieve 90% conversion of reactant to butenediol.

Data :

Reactor : Mechanically agitated slurry reactor (semibatch)

Volume of reactor = $V_r = 2\ m^3r$, $f_s = 0.0055\ m^3\ cat/m^3r$

Catalyst : $dp = 5 \times 10^{-5}$ m, $\rho_s = 1450\ kg/m^3$

$\mathscr{D}_e = 5 \times 10^{-10}\ m^3l/(m\ cat.s)$

Liquid : $C_{Bo} = 2500\ mol/m^3$, $V_l = V_r$

Gas : Pure H_2 at 14.6 atm, $H_A = 148000\ (Pa.m^3l)/mol$

Kinetics : $(k_{Ai}\ a_i)_{g+l} = 0.277\ m^3l/(m^3r.s)$ ... sum of the gas and liquid film conductances

$k_{Ac} = 4.4 \times 10^{-4}\ m^3l/(m^2\ cat.s)$

Rate constant $k' = 5 \times 10^{-5}\ m^6l/(mol.kg\ cat.s)$ at 35 °C

Solution :

$$\underset{\text{A}}{H_2\ (g \rightarrow l)} + \underset{\text{B}}{\text{butynediol}\ (l)} \xrightarrow[\text{catalyst}]{\text{solid}} \text{butenediol}$$

with rate at 35 °C :

$$-r'_A = k'\ C_A\ C_B \quad , \quad k' = 5 \times 10^{-5}\ m^6l/(mol.kg\ cat.s)$$

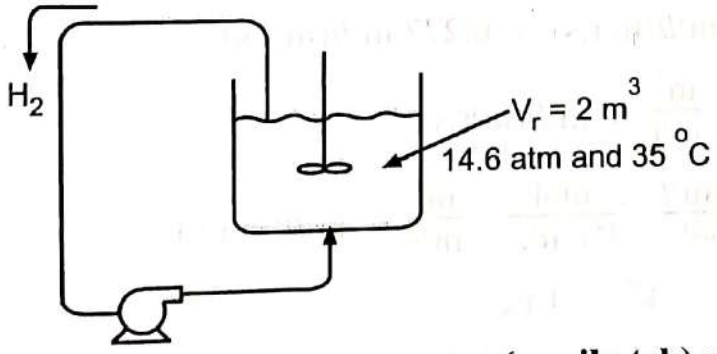


Fig. Ex 7.1 (a) : Mechanically agitated slurry reactor (semibatch) with recompression and recirculation of unused H_2

Check for ..... $C_B >>$ or $<<$ C_A

$C_{Bo} = 2500\ mol/m^3l$ ... initial concentration of B

$X_B = \dfrac{90}{100} = 0.90$ ... conversion of butynediol

At the end of the batch run :

$$C_{Bf} = C_{Bo}(1 - X_B)$$
$$= 2500\,(1 - 0.90) = 250 \text{ mol/m}^3l$$
$$H_A = 148000 \text{ (Pa.m}^3l\text{)/mol}$$
$$p_A = 14.6 \text{ atm} = 14.6 \times \frac{101325}{1} = 1479345 \text{ Pa}$$
$$C_A = \frac{p_A}{H_A} = \frac{1479345}{148000} = 9.995 \approx 10 \text{ mol/m}^3l$$

Therefore, comparing the C_A and C_B values :

Both at the beginning and at the end of the batch run : $C_B >> C_A$

$\therefore$ We are in extreme 1 ..... i.e., excess of B

Here we are dealing with ... pure gaseous reactant A and excess of B.

In our case p_A and C_A remain unchanged throughout the batch run.

Here C_B is changing during the run. Therefore, we are having a changing rate with time and with C_B.

The rate form to be used for this special case of pure gaseous A (excess of B) is

$$-r_A'''' = \left[\frac{1}{\frac{1}{H_A k_{Ag} a_i} + \frac{1}{k_{Al} a_i} + \frac{1}{k_{Ac} a_c} + \frac{1}{(k''' C_B^m C_A^{n-1} \eta_A f_s)}}\right] \frac{p_A}{H_A} \quad \text{... (1)}$$

We have

$$\frac{1}{H_A k_{Ag} a_i} + \frac{1}{k_{Al} a_i} = \frac{1}{(k_{Ai} a_i)_{g+l}}$$

$(k_{Ai}a_i)_{g+l} \longrightarrow m^3l/(m^3r.s) = 0.277\ m^3l/(m^3r.s)$

$k_{Al}a_i \longrightarrow \frac{m^3l}{m^2.s} \cdot \frac{m^2}{m^3r} = m^3l/(m^3r.s)$

$H_A k_{Ag} a_i \longrightarrow \frac{Pa.m^3l}{mol} \cdot \frac{mol}{Pa.m^2.s} \cdot \frac{m^2}{m^3r} = m^3l/(m^3r.s)$

$$k''' = k'\rho_s$$

$k''' \longrightarrow \frac{(m^3l)^{m+n}}{(mol)^{m+n-1}.m^3 \text{ cat.s}} \longrightarrow m + n = 1 + 1 = 2 \longrightarrow \frac{m^6l}{mol.m^3 \text{ cat.s}}$

$k'\rho_s \longrightarrow \frac{m^6l}{mol.kg \text{ cat.s}} \cdot \frac{kg \text{ cat}}{m^3 \text{ cat}} \longrightarrow \frac{m^6l}{mol.m^3 \text{ cat.s}}$

$C_A^{n-1} \longrightarrow n = 1 \longrightarrow C_A^{1-1} = C_A^0 = 1$

$C_B^m \longrightarrow m = 1 \longrightarrow C_B^1 \longrightarrow C_B$

With these Equation (1) becomes

$$-r_A'''' = \left[\frac{1}{\frac{1}{(k_{Ai}a_i)_{g+l}} + \frac{1}{k_{Ac}a_c} + \frac{1}{k'\rho_s C_B f_s \eta_A}}\right]\frac{p_A}{H_A} \quad \ldots (2)$$

For sphere :

$$a_c = \frac{6\, f_s}{dp}$$

Here $dp = 5 \times 10^{-5}$ m, $f_s = 0.0055 \frac{m^3\ cat}{m^3 r}$

$$\therefore \quad a_c = \frac{6 \times 0.0055}{5 \times 10^{-5}} = 660\ m^2\ cat/m^3 r$$

For accounting pore resistance effect on the rate : evaluate ... $M_T \rightarrow \eta_A \rightarrow -r_A''''$

For 1, 1 order reaction, the Thiele modulus is a function of concentration of B and is given by

$$M_T = L\sqrt{\frac{k''' C_B}{\mathscr{D}_e}}$$

For n^{th} order irreversible reaction w.r.t. A and first order w.r.t. B :

$$M_T = L\sqrt{\frac{(n+1)\, k''' C_B C_A^{n-1}}{2\mathscr{D}_e}}$$

n = 1 for first order reaction w.r.t. A. Therefore, the M_T expression reduces to

$$M_T = L\sqrt{\frac{k''' C_B}{\mathscr{D}_e}} \quad \ldots \text{ for first order reaction in A and in B}$$

We have : $k''' = k'\rho_s$

L - characteristic size of catalyst particles

For sphere,

$$L = \frac{R}{3} = \frac{dp}{6}\ m$$

$$M_T = \frac{dp}{6}\sqrt{k'\rho_s C_B/\mathscr{D}_e}$$

$$M_T = \frac{5 \times 10^{-5}}{6}\sqrt{\frac{5 \times 10^{-5} \times 1450\, C_B}{5 \times 10^{-10}}} = 0.10\, C_B^{1/2}$$

Effectiveness factor is given by

$$\eta_A = \frac{1}{M_T} = \frac{1}{0.10\, C_B^{1/2}} = \frac{10}{C_B^{1/2}}$$

$$k_{Ac} = 4.4 \times 10^{-4}\ m^3 l/(m^2\ cat.s)$$

$$-r_A''' = \left[\frac{1}{\frac{1}{0.277} + \frac{1}{4.4 \times 10^{-4} \times 660} + \frac{1}{5 \times 10^{-5} \times 1450 \times C_B \times \frac{10}{C_B^{1/2}} \times 0.0055}}\right] \times \frac{1479345}{148000}$$

$$-r_A''' = \frac{10}{7.054 + 250.8\, C_B^{-0.5}}, \text{ mol/(m}^3\text{r.s)}$$

Evaluate $-r_A'''$ for the various values of C_B and tabulate.

For $C_B = 2500$ mol/m^3 :

$$-r_A''' = \frac{10}{7.054 + 250.8\,(2500)^{-0.5}} = 0.828 \frac{\text{mol}}{(\text{m}^3\text{r.s})}$$

$$\frac{1}{(-r_A''')} = \frac{1}{0.828} = 1.21 \text{ (m}^3\text{r.s)/mol}$$

C_B	$-r_A'''$	$1/(-r_A''')$
2500	0.828	1.21
1500	0.74	1.35
1000	0.667	1.5
500	0.547	1.83
250	0.436	2.29

For the given set up (gas bubbling through a batch of liquid) and excess of B, the material balance at any time t is

$$\frac{V_l}{b}\left(-\frac{dC_B}{dt}\right) = \left(-r_A'''\right) V_r$$

Integrating, we get

$$t = \frac{V_l}{bV_r} \int_{C_{Bf}}^{C_{Bo}} \frac{dC_B}{\left(-r_A'''\right)}$$

where t is the time required to achieve 90% conversion.

The integral on the RHS of the above equation can be evaluated graphically.

Plot $\frac{1}{\left(-r_A'''\right)}$ vs C_B

Area under the curve between $C_B = C_{Bo} = 2500$ mol/m^3 and $C_B = C_{Bf} = 250$ mol/m^3 gives the value of this integral.

Refer to Fig. Ex 7.1 (b).

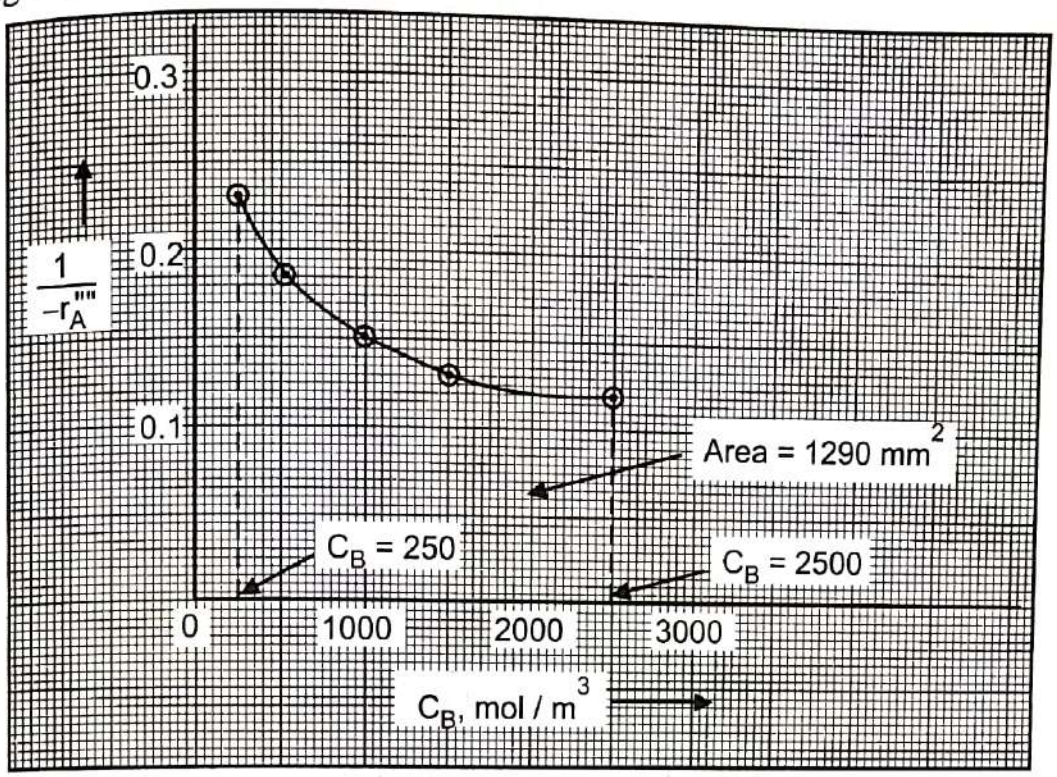


Fig. Ex 7.1 (b)

Area = 1290 mm^2

$$t = \frac{V_l}{bV_r} \int_{C_{Bf}}^{C_{Bo}} \frac{dC_B}{\left(-r_A'''\right)}$$

$$= \frac{V_l}{bV_r} \times \text{Area} \times \text{scale of y-axis} \times \text{scale of x-axis}$$

$$= \frac{1}{1} \times 1290 \text{ mm} \times \frac{1}{20 \text{ mm}} \times \frac{1000}{20 \text{ mm}}, \quad V_l = V_r \therefore \frac{V_l}{V_r} = 1, b = 1$$

$$= 3225 \text{ s or } 53.75 \text{ min}$$

Time required to achieve 90% conversion of butynediol to butenediol = 3225 s or 53.75 min. ... Ans.

$$t = \frac{V_l}{bV_r} \int \frac{dC_B}{\left(-r_A'''\right)} = \frac{V_l}{bV_r} \int \left[\frac{1}{\left(-r_A'''\right)}\right] dC_B$$

Unit of t $\Rightarrow \frac{m^3 l}{m^3 r}\left(\frac{m^3 r.s}{mol}\right)\left(\frac{mol}{m^3 l}\right)$

$\Rightarrow$ s

Unit of the quantities under integral $\Rightarrow \left(\frac{1}{-r_A'''}\right)(C_B)$

$$\Rightarrow \left(\frac{m^3r.s}{mol}\right)\left(\frac{mol}{m^3l}\right) \Rightarrow \frac{m^3r.s}{m^3l} \Rightarrow s$$

Extra :

If you are provided with :

w_c = mass of catalyst per unit volume of reactor ... catalyst charge ... catalyst concentratio in kg cat/m^3r

ρ_s = catalyst density kg cat/m^3 cat

ε = bed porosity

Then, $$\varepsilon = f_l + f_g \Rightarrow \frac{m^3l}{m^3r} + \frac{m^3g}{m^3r} \Rightarrow \frac{m^3 \text{ fluid}}{m^3r}, \; f_g = \frac{V_g}{V_r}, \; f_l = \frac{V_l}{V_r}$$

$$1 - \varepsilon = \frac{m^3r - (m^3l + m^3g)}{m^3r} = \frac{m^3 \text{ cat}}{m^3r}$$

For a sphere :

a_c = external surface area of catalyst particle per unit volume of reactor

$$a_c = \frac{6(1-\varepsilon)}{dp} \Rightarrow \frac{m^3 \text{ cat}}{m^3r.mcat} \Rightarrow \frac{m^2 \text{ cat}}{m^3r}$$

$$a_c = \frac{6w_c}{\rho_s \, dp} \Rightarrow \frac{kg \text{ cat}}{m^3 \, r}\left(\frac{m^3 \text{ cat}}{kg \text{ cat}}\right) \times \frac{1}{m \text{ cat}} \Rightarrow \frac{m^2 \text{ cat}}{m^3r}$$

$$a_c = \frac{6f_s}{dp} \Rightarrow \left(\frac{m^3 \text{ cat}}{m^3r}\right)\left(\frac{1}{m \text{ cat}}\right) \Rightarrow \frac{m^2 \text{ cat}}{m^3r}, \; f_s = \frac{V_s}{V_r}$$

$$w_c = \rho_s(1-\varepsilon) \Rightarrow \frac{kg \text{ cat}}{m^3 \text{ cat}} \times \frac{m^3 \text{ cat}}{m^3r} \Rightarrow \frac{kg \text{ cat}}{m^3r}$$

Example 7.2 : *Acetone is hydrogenated to propanol by the action of pure hydrogen at 1 atm in a long slender column packed with porous Raney nickel catalyst (packed bubble column) and maintained at 14 °C. The reactants, aqueous acetone and hydrogen, are fed at the bottom of the column. The reaction proceeds as follows :*

$$\underset{A}{H_2(g \to l)} + \underset{B}{CH_3COCH_3(l)} \xrightarrow[catalyst]{on} CH_3CHOHCH_3(l)$$

with rate

$$-r_A' = -r_B' = k' C_A^{1/2} C_B^0, \; mol/(kg\ cat.s)$$

$$k' = 2.3 \times 10^{-3} \frac{m^3 l}{kg\ cat.s} \left(\frac{mol}{m^3 l}\right)^{1/2}$$

Find the fractional conversion of acetone to propanol.

Data :

Gas : $\nu_g = 4 \times 10^{-2}$ m³g/s, $H_A = 36850$ (Pa.m³*l*)/mol

Liquid : $\nu_l = 1 \times 10^{-4}$ m³*l*/s, $C_{Bo} = 1000$ mol/m³*l*

Reactor : 0.1 m² cross-section × 5m high, $f_s = 0.60$

Catalyst : $dp = 5 \times 10^{-3}$ m, $\rho_s = 4500$ kg/m³

$\mathscr{D}_e = 8 \times 10^{-10}$ m³*l*/(m cat.s)

Kinetics : $(k_{Ai}\ a_i)_{g+l} = 0.021$ m³*l*/(m³r.s) ... the sum of the gas and liquid film conductances

$k_{Ac} a_c = 0.051$ m³*l*/(m³r.s)

Solution :

$$\underset{A}{H_2\ (g \rightarrow l)} + \underset{B}{CH_3COCH_3\ (l)} \xrightarrow[\text{catalyst}]{\text{on}} CH_3CHOHCH_3\ (l)$$

$$-r_A' = -r_B' = k' C_A^{1/2} C_B^0 = k' C_A^{1/2}$$

$$k' = 2.3 \times 10^{-3} \frac{m^3 l}{kg\ cat.s} \left(\frac{mol}{l}\right)^{1/2}$$

$$C_{Bo} = 1000\ mol/m^3\ l$$

$$H_A = 36850\ (Pa.m^3 l)/mol$$

Evaluate C_A :

$$C_A = \frac{p_A}{H_A}, \; p_A = 1\ atm = 101325\ Pa$$

$$= \frac{101325}{36850}$$

$$= 2.749 \approx 2.75\ mol/m^3 l$$

Comparing the concentration values of A and B :

We see that $C_B >> C_A$. Therefore, we are in extreme 1 (excess of B) and thus in this case we are dealing with pure gaseous A and excess of B.

As we are dealing with pure H_2, p_A is constant throughout the column and as the rate is dependent on C_A and not on C_B (refer the rate equation), the rate of reaction is constant throughout the column.

The material balance with constant rate is

$$F_{Ao} X_A = \frac{F_{Bo} X_B}{b} = \left(-r_A'''\right) V_r$$

$$\frac{F_{Bo} X_B}{b} = \left(-r_A'''\right) V_r \quad \ldots (1)$$

$b = 1, \quad V_r = 0.10 \times 5 = 0.5\ m^3r$

$$F_{Bo} = \nu_l C_{Bo} \ , \ \nu_l = 1 \times 10^{-4}\ m^3l/s, \ C_{Bo} = 1000\ \frac{mol}{m^3l}$$

$$= 1 \times 10^{-4} \times 1000$$

$$= 0.10\ mol/s$$

Evaluate $-r_A'''$ ... considering pure resistance effect on the rate.

For this evaluate $\rightarrow M_T \rightarrow \eta_A \rightarrow -r_A'''$.

The Thiele modulus (M_T) for irreversible n^{th} order reaction with respect to A is given by

$$M_T = L\sqrt{\frac{(n+1)\, k'''\, C_A^{n-1}}{2\mathscr{D}_e}}$$

$$k''' = k'\rho_s$$

Evaluate the characteristic size of a particle :

For sphere particles :

$$L = \frac{R}{3} = \frac{dp}{6}$$

$$dp = 5 \times 10^{-3}\ m$$

$$n = 0.5 \text{ order of reaction}$$

$$k' = 2.3 \times 10^{-3}\ m^3l/(kg\ cat.s).(mol/m^3l)^{1/2}$$

$$\mathscr{D}_e = 8 \times 10^{-10}\ m^3l/(m\ cat.s)$$

$$C_A = 2.75\ mol/m^3l$$

$$\rho_s = 4500\ kg/m^3$$

$$k''' \Rightarrow \frac{(m^3l)^n}{(mol)^{n-1}\ m^3\ cat.s}$$

For n = 1/2,

$$k''' \Rightarrow \frac{(m^2l)^{1/2}}{(mol)^{1/2-1}\ m^3\ cat.s} \Rightarrow \frac{(m^3l.mol)^{1/2}}{m^3\ cat.s}$$

$$k'\rho_s \Rightarrow \frac{m^3l}{kg\ cat.s}\left(\frac{mol}{m^3l}\right)^{1/2} \cdot \frac{kg\ cat}{m^3\ cat} \Rightarrow \frac{(mol.m^3l)^{1/2}}{m^3\ cat.s}$$

$$M_T = L\sqrt{\frac{1.5\,k'\rho_s\,C_A^{-1/2}}{2\mathscr{D}_e}}$$

$$= \frac{dp}{6}\sqrt{\frac{1.5\,k'\rho_s\,C_A^{-1/2}}{2\mathscr{D}_e}}$$

$$= \frac{5\times10^{-3}}{6}\left(\frac{1.5\times2.3\times10^{-3}\times4500\times(2.75)^{-1/2}}{2\times8\times10^{-10}}\right)^{1/2} = 63.74$$

$$\eta_A = \frac{\tanh M_T}{M_T}$$ ... equation of flat plate ... also applicable equally to sphere.

For $M_T > 4$, it can be taken as

$$\eta_A = \frac{1}{M_T} \quad \text{... as } \tanh 4 \approx 1.0$$

$$\eta_A = \frac{1}{M_T} = \frac{1}{63.74} = 0.01566 \approx 0.0157$$ ... effectiveness factor

For our present case ... pure gaseous A ... B in excess, the rate equation to be used is

$$-r_A'''' = \left[\frac{1}{\frac{1}{H_A k_{Ag} a_i} + \frac{1}{k_{Al} a_i} + \frac{1}{k_{Ac} a_c} + \frac{1}{(k''' C_B^m C_A^{n-1}\,\eta_A f_s)}}\right]\frac{p_A}{H_A}$$

With $k''' = k'\rho_s$, $m = 0$ ... $C_B^0 = 1$,

$1/H_A k_{Ag} a_i + 1/k_{Al} a_i = 1/(k_{Ai} a_i)_{g+l}$ and $n = 1/2$, above equation reduces to

$$-r_A'''' = \left[\frac{1}{\frac{1}{(k_{Ai} a_i)_{g+l}} + \frac{1}{k_{Ac} a_c} + \frac{1}{k'\rho_s C_A^{-1/2}\,\eta_A f_s}}\right]\frac{p_A}{H_A} \qquad \text{... (2)}$$

$(k_{Ai} a_i)_{g+l} = 0.021\ s^{-1}$, $k_{Ac} a_c = 0.051\ m^3 l/(m^3 r.s)$

$k' = 2.3\times10^{-3}$, $\eta_A = 0.0157$, $p_A = 101325$ Pa, $f_s = 0.6\ m^3$ cat/m^3r, $H_A = 36850$, $C_A = 2.75$

$$-r_A'''' = \left[\frac{1}{\frac{1}{0.021} + \frac{1}{0.051} + \frac{1}{2.3\times10^{-3}\times4500\times(2.75)^{-1/2}\times1\times0.0157\times0.60}}\right]\times\frac{101325}{36850}$$

$$-r_A'''' = 0.0326\ \text{mol/(m}^3\text{r.s)}$$

From Equation (1), we can write

$$X_B = \frac{(-r_A'''')\,V_r}{F_{Bo}\,b} = \frac{0.0326\times0.50}{0.1\times1}$$

$$= 0.163 \text{ or } 16.3\%$$

The fractional conversion of acetone is 0.163 or 16.3% ... Ans.

Extra :

Units of $k_{Ac}\, a_c \Rightarrow m^3l/(m^3r.s)$

$(k_{Ai}a_i)_{g+l} \Rightarrow m^3l/(m^3r.s)$

$$k'\rho_s C_A^{-1/2}\ \eta_A f_s \Rightarrow \frac{m^3l}{kg\ cat.s}\left(\frac{mol}{m^3l}\right)^{1/2}\cdot\frac{kg\ cat}{m^3\ cat}\left(\frac{mol}{m^3l}\right)^{-1/2}\cdot\frac{m^3\ cat}{m^3r} \Rightarrow \frac{m^3l}{m^3r.s}$$

$$-r_A'''' \Rightarrow \left(\frac{1}{\frac{1}{m^3l/(m^3r.s)}}\right)\times\frac{Pa}{(Pa.m^3l)/mol} \Rightarrow \frac{m^3l}{m^3r.s}\times\frac{Pa\times mol}{Pa.m^3l} \Rightarrow \frac{mol}{m^3r.s}$$

↑ ↑

units of conductance units of p_A/H_A

Example 7.3 : *Dilute aqueous ethyl alcohol is oxidized to acetic acid using pure oxygen at 10 atm in a trickle bed reactor maintained at 30 °C and packed with Pd - Al_2O_3 catalyst pellets. Under these conditions, ethyl alcohol is oxidised according to the reaction :*

$$\underset{A}{O_2\ (g \rightarrow l)} + \underset{B}{CH_3CH_2OH\ (l)} \xrightarrow[catalyst]{on} CH_3COOH\ (l) + H_2O$$

The reaction is first order with respect to hydrogen and zero order with respect to ethyl alcohol with rate constant k' at 30°C = 1.77 × 10^{-5} $m^3l/(kg\ cat.s)$

Calculate the fractional conversion of ethyl alcohol if gas and liquid are fed to the top of the reactor.

Data :

Gas : $v_g = 1 \times 10^{-2}\ m^3g/s$, $H_A = 86000\ (Pa.m^3l)/mol$

Liquid : $v_l = 2 \times 10^{-4}\ m^3l/s$, $C_{Bo} = 400\ mol/m^3l$

Reactor : 0.1 m^2 cross-section × 5m high, $f_s = 0.58$

Catalyst : $dp = 5\ mm$, $\rho_s = 1800\ kg/m^3$

$\mathscr{D}_e = 4.16 \times 10^{-10}\ m^3l/(m\ cat.s)$

Kinetics : $(k_{Ag}\, a_i) = 3 \times 10^{-4}\ mol/(m^3l.Pa.s)$

$k_{Al}a_i = 0.02\ m^3l/(m^3r.s)$, $k_{Ac} = 3.86 \times 10^{-4}\ m^3l/(m^2\ cat.s)$

Solution :

$$\underset{A}{O_2\ (g \rightarrow l)} + \underset{B}{CH_3CH_2OH\ (l)} \xrightarrow[catalyst]{on} CH_3COOH\ (l) + H_2O$$

$$-r'_A = k'C_AC_B^0 = k'C_A \text{ , mol/(kg cat.s)}$$

$$k' = 1.77 \times 10^{-5} \text{ m}^3l\text{/(kg cat.s)} \quad \text{... at } 30\ ^\circ\text{C}$$

Evaluate C_A :

$H_A = 86000$ (Pa.m^3l)/mol

$p_A = 10$ atm $= 1013250$ Pa

$C_{Bo} = 400$ mol/m^3l

$$C_A = \frac{p_A}{H_A} = \frac{1013250}{86000} = 11.78 \approx 11.8 \text{ mol/m}^3l$$

$\therefore$ $C_B >> C_A$ and hence, we are in extreme 1 (excess of B)

Here as we are dealing with pure oxygen, p_A is constant throughout. Also as the rate of reaction depends on C_A and not on C_B, the rate of reaction is constant throughout the reactor.

Evaluate $-r''''_A$ mol/(m^{3}r.s) :

Evaluate $M_T \rightarrow \eta_A \rightarrow -r''''_A \rightarrow$ use $\frac{F_{Bo}X_B}{b} = \left(-r''''_A\right) V_r \rightarrow$ determine X_B

Material balance with constant rate is

$$F_{Ao}X_A = \frac{F_{Bo}X_B}{b} = \left(-r''''_A\right) V_r \qquad \text{... (1)}$$

$b = 1$, $v_l = 2 \times 10^{-4}$ m^3l/s

$C_{Bo} = 400$ mol/m^3l

$F_{Bo} = 2 \times 10^{-4} \times 400 = 0.08$ mol/s

$V_r = 0.1 \text{ m}^2 \times 5 \text{ m} = 0.5 \text{ m}^3\text{r}$

The Thiele modulus for n^{th} order reaction w.r.t. A

$$M_T = L\sqrt{\frac{(n+1)\,k'''C_A^{n-1}}{2\mathcal{D}_e}}$$

L = characteristic size of catalyst particles. $L = \frac{R}{3} = \frac{dp}{6}$ ... for sphere

$n = 1$ ... reaction order w.r.t. A., M_T expression then becomes

$$M_T = \frac{dp}{6}\sqrt{\frac{k'''}{\mathcal{D}_e}}$$

$$k''' = k'\rho_s$$

$$M_T = \frac{dp}{6}\sqrt{\frac{k'\rho_s}{\mathcal{D}_e}}$$

$dp = 5\ mm = 5 \times 10^{-3}\ m$, $k' = 1.77 \times 10^{-3}\ m^3 l/(kg\ cat.s)$

$\mathscr{D}_e = 4.16 \times 10^{-6}\ m^3 l/(m\ cat.s)$, $\rho_s = 1800\ kg/m^3$

$$M_T = \frac{5 \times 10^{-3}}{6}\sqrt{\frac{1.77 \times 10^{-5} \times 1800}{4.16 \times 10^{-10}}}$$

$$= 7.293 \ \ldots\ (> 4)$$

η_A for flat plate is $\eta_A = \dfrac{\tanh M_T}{M_T}$. It is also good approximation for all particle shapes.

$$\therefore \qquad \eta_A = \frac{\tanh (7.293)}{7.293} = 0.137$$

As $\tanh (4) \approx 1.0$, $\quad \eta_A = \dfrac{1}{M_T}$ for $M_T > 4$

$$\therefore \qquad \eta_A = \frac{1}{M_T} = \frac{1}{7.293} = 0.137 \ \ldots \text{effectiveness factor}$$

For sphere : $\quad a_c = \dfrac{6f_s}{dp}$, $f_s = 0.58 = \dfrac{6 \times 0.58}{5 \times 10^{-3}} = 696\ m^2\ cat/m^3 r$

For the present situation, the rate form to be used is

$$-r_A''' = \left[\frac{1}{\frac{1}{H_A k_{Ag} a_i} + \frac{1}{k_{Al} a_i} + \frac{1}{k_{Ac} a_c} + \frac{1}{(k''' C_B^m C_A^{n-1} \eta_A f_s)}}\right] \frac{p_A}{H_A}$$

With $k''' = k'\rho_s$, $m = 0$ $\therefore$ $C_B^m = C_B^0 = 1$, $n = 1$, $\therefore$ $C_A^{n-1} = C_A^{1-1} = 1$, the above equation becomes

$$-r_A''' = \left[\frac{1}{\frac{1}{H_A k_{Ag} a_i} + \frac{1}{k_{Al} a_i} + \frac{1}{k_{Ac} a_c} + \frac{1}{k' \rho_s \eta_A f_s}}\right] \times \frac{p_A}{H_A}$$

$k_{Ag} a_i = 3 \times 10^{-4}\ mol/(m^3 l.Pa.s)$, $k_{Al} a_i = 0.02\ m^3 l/(m^3 r.s)$

$k_{Ac} = 3.86 \times 10^{-4}\ m^3 l/(m^2\ cat.s)$, $a_c = 696\ m^2\ cat/m^3 r$

$\rho_s = 1800\ kg/m^3$, $f_s = 0.58\ m^3\ cat/m^3 r$, $p_A = 1013250\ Pa$

$H_A = 86000\ (Pa.m^3 l)/mol$, $k' = 1.77 \times 10^{-5}\ m^3 l/(kg\ cat.s)$, $\eta_A = 0.137$

$$-r_A''' = \left[\frac{1}{\frac{1}{86000 \times 3 \times 10^{-4}} + \frac{1}{0.02} + \frac{1}{3.86 \times 10^{-4} \times 696} + \frac{1}{1.77 \times 10^{-5} \times 1800 \times 0.137 \times 0.58}}\right] \times \frac{1013250}{86000}$$

$$= 0.02625\ mol/(m^3 r.s)$$

We have : $\quad \dfrac{F_{Bo} X_B}{b} = \left(-r_A'''\right) V_r$

$$X_B = \frac{(-r_A''')\, V_r\, b}{F_{Bo}}$$

$$= \frac{0.02625 \times 0.5 \times 1}{0.08} = 0.164$$

The fractional conversion of ethyl alcohol to acetic acid is 0.164 or 16.4%. ... Ans.

Example 7.4 : *A bubble column slurry reactor is to be used for oxidation of dilute aqueous ethyl alcohol to acetic acid by the action of pure oxygen at 10 atm. The reactor is maintained at 30°C and uses $Pd\text{-}Al_2O_3$ catalyst.*

The reaction proceeds as follows :

$$\underset{A}{O_2\,(g \rightarrow l)} + \underset{B}{CH_3CH_2OH\,(l)} \xrightarrow[\text{catalyst}]{\text{solid}} CH_3COOH\,(l) + H_2O$$

with rate : $-r_A' = k'\, C_A$, $k' = 1.77 \times 10^{-5}\ m^3l/(kg\ cat.s)$

Calculate the fractional conversion of ethyl alcohol to acetic acid.

Data :

Gas : $\nu_g = 1 \times 10^{-2}\ m^3g/s$, $H_A = 86000\ (Pa.m^3l)/mol$

Liquid : $\nu_l = 2 \times 10^{-4}\ m^3l/s$, $C_{Bo} = 400\ mol/m^3l$

Reactor : $0.1\ m^2 \times 5m$ high, $f_l = 0.75$, $f_g = 0.05$ and $f_s = 0.20$

Catalyst : $dp = 1 \times 10^{-4}$ m, $\mathscr{D}_e = 4.16 \times 10^{-10}\ m^3l/(m\ cat.s)$, $\rho_s = 1800\ kg/m^3$

Kinetics : $(k_{Ai}a_i)_{g+l} = 0.052\ m^3l/(m^3r.s)$, $k_{Ac} = 4 \times 10^{-4}\ m^3l/(m^2\ cat.s)$

Solution :

$$\underset{A}{O_2\,(g \rightarrow l)} + \underset{B}{CH_3CH_2OH\,(l)} \xrightarrow[\text{catalyst}]{\text{solid}} CH_3COOH\,(l) + H_2O$$

$$-r_A' = k'C_A = k'C_AC_B^o \ , \quad k' = 1.77 \times 10^{-5}\ m^3l/(kg\ cat.s)$$

The reaction is first order in H_2 and zero order in CH_3CH_2OH.

$C_{Bo} = 400\ mol/m^3l$

$p_A = 10\ atm = 1013250\ Pa$

$H_A = 86000\ (Pa.m^3l)/mol$

$$C_A = \frac{p_A}{H_A} = \frac{1013250}{86000} = 11.78 \approx 11.8\ mol/m^3l$$

We have : $C_B >> C_A$ and hence, we are in extreme 1 (excess of B).

It is a special case of pure gaseous A (excess of B).

p_A is constant throughout this reactor and as the rate depends only on C_A, the rate is constant throughout this reactor.

$$dp = 1 \times 10^{-4} \text{ m}, \quad f_s = 0.20$$

$$a_c = \frac{6f_s}{dp} = \frac{6 \times 0.20}{1 \times 10^{-4}} = 12000 \text{ m}^2 \text{ cat/m}^3\text{r}$$

M_T for first order reaction w.r.t. A and zero order reaction w.r.t. B is given by

$$M_T = L\sqrt{\frac{k'''}{\mathscr{D}_e}}, \quad k''' = k'\rho_s$$

L = characteristic size. $L = R/3 = dp/6$ ... for sphere

$$M_T = \frac{dp}{6}\sqrt{\frac{k'\rho_s}{\mathscr{D}_e}}, \quad \mathscr{D}_e = 4.16 \times 10^{-10} \text{ m}^3\text{l/(m cat.s)}$$

$$\rho_s = 1800 \text{ kg/m}^3 \text{ cat}$$

$$= \frac{1 \times 10^{-4}}{6}\sqrt{\frac{1.77 \times 10^{-5} \times 1800}{4.16 \times 10^{-10}}}$$

$$= 0.1458$$

For $M_T < 0.4$: $\eta_A = 1$ ..... diffusion-free regime.

As $M_T < 0.4$ ... $\eta_A = 1$ ... effectiveness factor

For spherical particle :

$$\eta_A = \frac{1}{M_T}\left[\frac{1}{\tanh(3M_T)} - \frac{1}{3M_T}\right]$$

$$= \frac{1}{0.1458}\left[\frac{1}{\tanh(3 \times 0.1458)} - \frac{1}{3 \times 0.1458}\right]$$

$$= 0.987 \approx 1.0$$

The expression for a flat plate which is also equally applicable to other particle shapes is

$$\eta_A = \frac{\tanh M_T}{M_T} = \frac{\tanh(0.1458)}{0.1458} = 0.993 \approx 1.0$$

Thus, in this case, $\eta_A = 1$ for $M_T = 0.1458$

With constant rate, we have

$$\frac{F_{Bo} X_B}{b} = \left(-r_A'''\right) V_r$$

$$\therefore \quad X_B = \frac{(-r_A''')\, V_r \times b}{F_{Bo}}$$

$\nu_l = 2 \times 10^{-4}$ m³/l/s, $C_{Bo} = 400$ mol/m³l, $F_{Bo} = \nu_l C_{Bo} = 2 \times 10^{-4} \times 400 = 0.08$ mol/s

$b = 1$ mol/mol, $A = 0.1$ m² cross-section, height = 5 m

$$V_r = A \cdot h = 0.1 \times 5 = 0.5 \text{ m}^2$$

$$X_B = \frac{0.067 \times 0.5 \times 1}{0.08} = 0.41875 \approx 0.42$$

The fractional conversion of ethyl alcohol to acetic acid is 0.42 or 42% ... Ans.

The rate form to be used here is

$$-r_A''' = \left[\frac{1}{\frac{1}{H_A k_{Ag} a_i} + \frac{1}{k_{Al} a_i} + \frac{1}{k_{Ac} a_c} + \frac{1}{(k''' C_B^m C_A^{n-1} \eta_A f_s)}}\right] \frac{p_A}{H_A}$$

$$\frac{1}{(k_{Ai} a_i)_{g+l}} = \frac{1}{H_A k_{Ag} a_i} + \frac{1}{k_{Al} a_i}$$

$$k''' = k' \rho_s$$

$m = 0$ ... order w.r.t. B $\quad \therefore \quad C_B^m = 1$

$n = 1$ ... order w.r.t A $\quad \therefore \quad C_A^{n-1} = 1$

$$\therefore \quad -r_A''' = \left[\frac{1}{\frac{1}{(k_{Ai} a_i)_{g+l}} + \frac{1}{k_{Ac} a_c} + \frac{1}{k' \rho_s E_A f_s}}\right] \frac{p_A}{H_A} \qquad \text{... (1)}$$

$(k_{Ai} a_i)_{g+l} = 0.05$ m³l/(m³r.s)

$k_{Ac} = 4 \times 10^{-4}$ m³l/(m² cat.s)

$k' = 1.77 \times 10^{-5}$ m³l/(kg cat.s), $\eta_A = 1$, $f_s = 0.2$ m³ cat/m³r

$\rho_s = 1800$ kg cat/m³ cat

$a_c = 12000$ m² cat/m³r, $H_A = 86000$ (Pa.m³l)/mol, $p_A = 1013250$ Pa

$$\therefore \quad -r_A''' = \left[\frac{1}{\frac{1}{0.052} + \frac{1}{4 \times 10^{-4} \times 12000} + \frac{1}{1.77 \times 10^{-5} \times 1800 \times 1 \times 0.2}}\right] \times \frac{1013250}{86000}$$

$$= 0.0668 \approx 0.067 \text{ mol/(m³r.s)}$$

Example 7.5 : *Aniline is to be hydrogenated in a batch recycle packed column at 1 atm and 135 °C using a nickel based catalyst (dp = 3 mm). The reaction proceeds as follows :*

$$\underset{A}{3\,H_2\,(g \rightarrow l)} + \underset{B}{2\,C_6H_5NH_2\,(l)} \xrightarrow[\text{catalyst}]{Ni} \underset{\text{cyclohexyl aniline}}{C_6H_{11}NHC_6H_5} + NH_3$$

The reaction is first order with respect to hydrogen and zero order with respect to aniline. The specific reaction rate constant k' is 0.05 $m^3l/(kg\ cat.s)$

The batch of liquid aniline is circulated through an external heat exchanger and hydrogen is bubbled through the column. The volume of liquid in the loop outside the reactor is equal to the total volume of the reactor.

Find the time required to achieve 90% conversion of aniline.

Data :

Gas stream : Pure H_2 at 1 atm, $H_A = 28500\ (Pa.m^3l)/mol$

Batch of liquid : $C_{Bo} = 1097\ mol/m^3l$

Reactor : $f_s = 0.40,\ f_l = 0.5,\ f_g = 0.10$

Catalyst : $dp = 3\ mm,\ \rho_s = 750\ kg/m^3$ cat

$D_e = 8.35 \times 10^{-10}\ m^3l/(m\ cat.s)$

Kinetics : $(k_{Ai}a_i)_{g+l} = 0.02\ m^3l/(m^3r.s),\ k_{Ac} = 7 \times 10^{-5}\ m^3l/(m^2\ cat.s)$

Solution :

$$3\,H_2\,(g \rightarrow l) + 2\,C_6H_5NH_2\,(l) \xrightarrow[\text{catalyst}]{Ni} C_6H_{11}NHC_6H_5\,(l) + NH_3$$

$$3\,A\,(g \rightarrow l) + 2\,B\,(l) \longrightarrow \text{Products}$$

This reaction can be written as

$$A\,(g \rightarrow l) + \frac{2}{3}\,B\,(l) \longrightarrow \text{Products}$$

$$\therefore \quad b = \frac{2}{3} = 0.67\ \text{mol B/mol A}$$

Since the reaction is first order in A and zero order in B, the rate is given by

$$-r_A' = k'\,C_A\,C_B^0 = k'C_A,\ k' = 0.05\ m^3l/(kg\ cat.s)$$

$C_{Bo} = 1097\ mol/m^3l$

$p_A = 1\ atm = 101325\ Pa$

$H_A = 28500\ (Pa.m^3l)/mol$

$$C_A = \frac{p_A}{H_A} = \frac{101325}{28500} = 3.56\ mol/m^3l$$

$$X_B = \frac{90}{100} = 0.90$$

$$\therefore \quad C_{Bf} = C_{Bo}\,(1 - X_B) = 1097\,(1 - 0.90) = 109.7\ mol/m^3l$$

Compare the concentration values of C_A and C_B.

We see that both at the beginning and at the end of the batch run : $C_B >> C_A$.

Since $C_B >> C_A$, we are in extreme 1 (excess of B).

Given : Pure H_2 gas is used.

$\therefore$ We have pure gaseous A and an excess of B.

In our present case, C_A stays constant throughout the batch run and C_B is changing with time. As the rate is dependent only on C_A and not on C_B, the rate of reaction is constant throughout the batch run.

$$a_c = \frac{6f_s}{dp}$$

where $f_s = 0.4$, $dp = 3 \text{ mm} = 3 \times 10^{-3} \text{ m}$

$$\therefore \quad a_c = \frac{6 \times 0.4}{3 \times 10^{-3}} = 800 \text{ m}^2 \text{ cat/m}^3\text{r}$$

$$k_{Ac}\, a_c = 7 \times 10^{-5} \times 800 = 0.056 \text{ m}^3 l/(\text{m}^3\text{r.s})$$

$$(k_{Ai}a_i)_{g+l} = 0.02 \text{ m}^3 l/(\text{m}^2\text{r.s})$$

The reaction is first order in A and zero order in B. For this case, the Thiele modulus is given by

$$M_T = L\sqrt{\frac{k'''}{\mathscr{D}_e}} \qquad \left[M_T = L\sqrt{\frac{(n+1)\, k'''\, C_B^m\, C_A^{n-1}}{2\, \mathscr{D}_e}} \right.$$

For sphere, $L = \frac{R}{3} = \frac{dp}{6}$ With n = 1 and m = 0 it reduces to

$k''' = k'\rho_s$ $\left. M_T = L\sqrt{k'''/\mathscr{D}_e} \right]$

$$M_T = \frac{dp}{6}\sqrt{\frac{k'\rho_s}{\mathscr{D}_e}}$$

where, $dp = 3 \times 10^{-3}$ m, $k' = 0.05 \text{ m}^3 l/(\text{kg cat.s})$

$\rho_s = 750 \text{ kg cat/m}^3 \text{ cat}$, $\mathscr{D}_e = 8.35 \times 10^{-10} \text{ m}^3 l/(\text{m cat.s})$

$$M_T = \frac{3 \times 10^{-3}}{6}\sqrt{\frac{0.05 \times 750}{8.35 \times 10^{-10}}}$$

$$= 105.96 \approx 106$$

$$\therefore \quad M_T > 4$$

The catalyst effectiveness factor η_A is given by

$$\eta_A = \frac{1}{M_T} = \frac{1}{106} = 0.0094$$

The expression for a flat plate is

$$\eta_A = \frac{\tanh M_T}{M_T}$$

It is also equally applicable for any particle shape.

For $M_T > 4$, $\tanh (M_T) = 1$

$$\therefore \quad \eta_A = \frac{1}{M_T}$$

The rate form to be used is

$$-r_A'''' = \left[\frac{1}{\frac{1}{H_A k_{Ag} a_i} + \frac{1}{k_{Al} a_i} + \frac{1}{k_{Ac} a_c} + \frac{1}{(k''' C_B^m C_A^{n-1} \eta_A f_s)}}\right] \frac{p_A}{H_A}, \quad \frac{\text{mol}}{(\text{m}^3\text{r.s})}$$

$$\frac{1}{(k_{Ai} a_i)_{g+l}} = \frac{1}{H_A k_{Ag} a_i} + \frac{1}{k_{Al} a_i}$$

$$k''' = k' \rho_s$$

$n = 1 \quad \therefore \quad C_A^{n-1} = 1$

$m = 0 \quad \therefore \quad C_B^0 = 1$

n = 1 ... order w.r.t A and m order w.r.t. B

$$\therefore \quad -r_A'''' = \left[\frac{1}{\frac{1}{(k_{Ai} a_i)_{g+l}} + \frac{1}{k_{Ac} a_c} + \frac{1}{k' \rho_s \eta_A f_s}}\right] \frac{p_A}{H_A} \qquad \ldots (1)$$

$f_s = 0.4 \text{ m}^3 \text{ cat/m}^3\text{r}$

$$\therefore \quad -r_A'''' = \left[\frac{1}{\frac{1}{0.02} + \frac{1}{0.056} + \frac{1}{0.05 \times 750 \times 0.0094 \times 0.4}}\right] \times \frac{101325}{28500}$$

$$= 0.0474 \text{ mol/(m}^3\text{r.s)}$$

$f_l = 0.50$ ... given.

$\therefore \quad V_l$ = volume of liquid in the reactor = $0.5 \times V_r = 0.5 V_r$

Given : Volume of liquid in the loop outside the reactor = $V_l' = V_r$

Total volume of batch of liquid to be handled (processed) is

$$V_{lt} = V_l + V_l' = 0.5 V_r + V_r = 1.5 V_r$$

The reaction time with constant rate is given by

$$t = \frac{V_{lt}}{b} \frac{(C_{Bo} - C_{Bf})}{V_r (-r_A'''')}$$

where, $V_{lt} = 1.5 V_r$, $b = 0.67$, $C_{Bo} = 1097 \text{ mol/m}^3 l$

$C_{Bf} = 109.7 \text{ mol/m}^3\text{r}$, $-r_A'''' = 0.0474 \text{ mol/(m}^3\text{r.s)}$

$$t = \frac{1.5 V_r}{0.67 \times V_r} \times \frac{(1097 - 109.7)}{0.0474}$$

$$= 46632 \text{ s} \approx 12.95 \text{ h}$$

The time required for 90% conversion of aniline is 46632 s, i.e., 12.95 h ... Ans.

❐❐❐

CHAPTER 8

DEACTIVATING CATALYSTS

Most catalysts are subjected to deactivation, which refers to *the continuous decrease in the activity of a catalyst with time*. As the activity of a catalyst decreases with time, the life of the catalyst is usually a major economic consideration in the development of a new catalytic process. In some cases the loss of activity is so rapid that the deactivation period is of the order of second (catalytic cracking of petroleum naphthas), while in other cases it is so slow that the deactivation period is of the order of months or years (synthesis of ammonia). In any case, with deactivating catalysts regeneration or replacement is essential from time to time (i.e., as and when needed).

Types of catalyst deactivation :

Catalyst deactivation can be divided into the following three categories :

(i) Deactivation by sintering (also called as aging),

(ii) Deactivation by fouling or coking,

(iii) Deactivation by poisoning.

(i) **Deactivation by sintering (aging) :** It is the loss of activity of the catalyst due to a loss of active surface area resulting from the prolonged exposure to high temperatures in the reacting atmosphere.

The active surface area of the catalyst may be lost either by crystal agglomeration and growth of metals deposited on the surface or by narrowing or closing of the catalyst pores (i.e., by a change in the surface structure of the catalyst).

(ii) **Deactivation by fouling or coking :** It is caused by a physical deposition of carbonaceous (coke) material on the surface of a catalyst. This deactivation is rapid and common to reactions involving hydrocarbons. Removal of a deposited material from the catalyst surface is called regeneration. Carbon deposition on cracking catalyst is a common example of deactivation by fouling. Catalysts deactivated by coking can be regenerated by burning off the carbon with air and/or steam.

(iii) **Deactivation by poisoning :** It is caused by reversible or irreversible chemisorption of the poisoning material on the active sites. The chemisorption of poisoning material reduces the number of sites available for the main reaction. The poisoning material may be a reactant, a

product or an impurity present in the feed stream. This deactivation is slow and restoration of the activity of a deactivated catalyst is called reactivation. In case of reversible adsorption (temporary poisoning), a change in operating conditions is sufficient to reactivate the catalyst but if the adsorption is irreversible (permanent poisoning) then we need a chemical treatment of the catalyst surface or a replacement of the used/spent catalyst.

In our development that follows we will use the term deactivation for fast as well as slow catalyst decay. Also, we will refer to any material that deposits on the surface to lower its activity as a poison.

Mechanisms of Catalyst Deactivation :

Although many factors affect the deactivation of a porous catalyst, here we consider only two factors : the actual decay reactions and pore diffusion.

Decay Reactions :

Catalyst decay can occur by the following ways :

(i) Parallel deactivation,

(ii) Series deactivation,

(iii) Side-by-side deactivation and

(iv) Independent deactivation.

Parallel Deactivation :

In this case, the reactant may produce a side product which deposits on and deactivates the catalyst surface.

Let P be the material which deposits on and deactivates the surface. Then, this decay reaction can be represented by

$$A \rightarrow R + P\downarrow$$

or

$$A \rightarrow R$$

$$A \rightarrow P\downarrow$$

Here the deposition depends on the concentration of reactant.

Series Deactivation :

In this case, the reaction product may decompose or react further to produce a material which then deposits on and deactivates the surface. This decay reaction can be represented by

$$A \rightarrow R \rightarrow P\downarrow$$

Here the decomposition depends on the concentration of product.

Side-by-Side Deactivation :

In this case, an impurity present in the feed may deposit on and deactivate the catalyst surface. This decay reaction can be represented by

$$A \rightarrow R$$

$$P \rightarrow P\downarrow$$

Here the deposition depends on the concentration of impurity in the feed.

Independent Deactivation :

In this case, decay is caused by sintering of the surface resulting from the prolonged exposure of the catalyst to high temperatures. This type of decay is concentration independent (independent of concentration of any species i present in the gas stream) and it is dependent on the time spent by the catalyst in the high temperature environment.

Pore Diffusion effects :

For a pellet, the progress of catalyst decay may strongly be influenced by the pore diffusion.

In case of parallel deactivation, where A is the source of a poison, the poison will be deposited at the exterior surface for strong pore resistance as in the regime of strong pore resistance ($M_T < 0.40$ and $\eta_A < 1$) the reactant A is found close to the exterior surface and it is uniformly deposited in the pellet for no pore resistance as for no pore resistance ($M_T < 4$ and $\eta_A = 1$) the reactant A is evenly distributed throughout the pellet.

In case of series deactivation, where R is the source of a poison, the poison deposits in high concentration within the pores for strong pore resistance as the concentration of R is higher within the pellet than at the exterior in the regime of strong pore resistance. Thus we have poisoning from the inside-out for series deactivation.

In case of side-by-side deactivation (where an impurity in the feed is the source of the poison), the location of deposition of the poison is determined by the rate at which it reacts with the surface. For a large poison rate constant, poisoning occurs at the exterior surface as soon as it reaches the surface. For a small rate constant, the poisoning occurs uniformly as the poison penetrates the pellet completely.

Thus, it is clear from the above discussion that the progress of deactivation may occur in different ways based on the type of decay reaction taking place and on the value of the pore diffusion factor.

Rate Equations for Main Reaction and Deactivation :

The activity of a catalyst pellet or a catalyst at any time t, **a**, is defined as the *ratio of the rate of reaction on the catalyst pellet at a time t (i.e., on the catalyst pellet which has been used for time t) to the rate of reaction on the same catalyst pellet when it is fresh.*

$$a = \frac{\text{Rate of reaction of A with a pellet at time t}}{\text{Rate of reaction of A with a pellet when fresh}}$$

$$a = \frac{-r'_A}{-r'_{Ao}} \qquad \ldots (8.1)$$

where **a** is the activity of the catalyst/catalyst pellet. It is time dependent, so at t = 0, we have : **a** = 1 ... true for a fresh catalyst.

The rate of disappearance of reactant A on the catalyst pellet, that has been used for time t and soaked by the fluid, depends on the main stream temperature, main stream composition and the activity of the catalyst pellet.

In terms of n^{th} order kinetics, Arrhenius temperature dependency and isothermal conditions, the rate of reaction on a catalyst pellet that has been used for time t is

$$-r'_A = k' C_A^n \cdot a = k'_o e^{-E/RT} \cdot C_A^n \cdot a \qquad \ldots (8.2)$$

Equation (8.2) is the rate expression/law for the main reaction accounting for catalyst activity.

The rate of catalyst decay / the rate at which the catalyst pellet deactivates can be expressed in a rate expression analogous to Equation (8.2). Thus, for deactivation which depends on the concentration of gas phase species, the rate of catalyst decay is

$$r_d = -\frac{da}{dt} = k_d \cdot C_i^m \cdot a^d = k_{do} e^{-E_d/RT} \cdot C_i^m \cdot a^d \qquad \ldots (8.3)$$

where k_d is the specific decay constant, d is the order of deactivation, m is the concentration dependency of the deactivation and E_d is the activation energy or temperature dependency of the deactivation.

Equation (8.3) is the rate expression/law for catalyst decay.

Different forms of Equations (8.2) and (8.3) for different decay reactions are :

For parallel deactivation (A → R and A → P ↓) :

(i) $-r'_A = k' C_A^n \, a$ ... rate of reaction

(ii) $-\frac{da}{dt} = k_d C_A^m \, a^d$ ... rate of deactivation ... (8.4)

For series deactivation (A → R → P ↓) :

(i) $-r'_A = k' C_A^n \, a$

(ii) $-\frac{da}{dt} = k_d C_R^m \, a^d$ ... (8.5)

For side-by-side deactivation (A → R and P → P ↓) :

(i) $-r'_A = k' C_A^n \, a$

(ii) $-\frac{da}{dt} = k_d C_P^m \, a^d$ ... (8.6)

For independent deactivation (Concentration independent) :

(i) $$-r'_A = k' C_A^n \, a$$

(ii) $$-\frac{da}{dt} = k_d \, a^d \qquad \ldots (8.7)$$

In isomerisation and cracking reactions, catalyst deactivation may be caused both by reactant and product. In such cases, the rate law for catalyst decay is

$$\left.\begin{array}{l} A \rightarrow R \\ A \rightarrow P \downarrow \\ R \rightarrow P \downarrow \end{array}\right\} \Rightarrow -\frac{da}{dt} = k_d (C_A + C_R)^m a^d \qquad \ldots (8.8)$$

For a specific feed as $C_A + C_R$ remains constant, this type of deactivation reduces to the independent deactivation represented by Equation (8.7).

The Rate Equation using Experimental Data :

In the experiments to be conducted for studying deactivating catalysts, a decision regarding use of either a batch of solids or a flow of solids is based on whether the process of deactivatior is fast or slow. For slow deactivation we have to use a batch of solids in experimentation and fo fast deactivation we have to use a flow of solids through an experimental device.

Fig. 8.1 shows experimental devices used to study catalyst deactivation, both fast and slow.

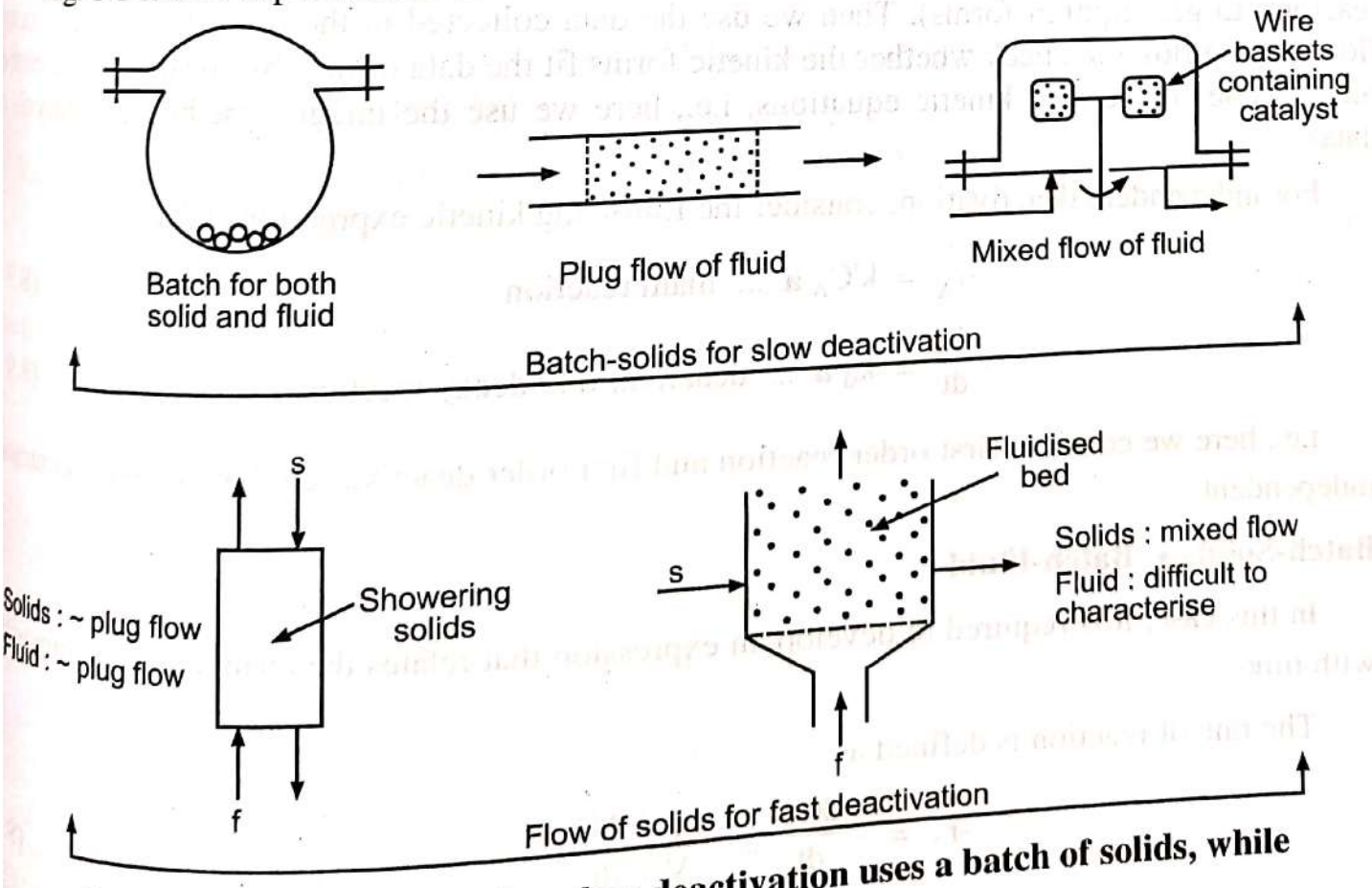


Fig. 8.1 : In experimentation-slow deactivation uses a batch of solids, while fast deactivation uses a flow of solids

Batch-solids devices are usually preferred due to ease in experimentation. However, they can only be used when deactivation is slow (in the order of minutes or longer) so that sufficient fluid composition v/s time data can be collected before the catalyst gets exhausted.

In searching for a rate equation adopt the method which we use in case of homogeneous reactions, irrespective of the presence of an additional factor **a** : start with the simplest rate expression and check if it fits the data. If it does not fit the data satisfactorily, try another rate expression (kinetic expression) and so on.

The type of batch-solids flowing fluid reactor to be used in performing experiments depends on whether the deactivation and hence, the deactivation expression $\frac{da}{dt}$, is concentration independent or not :

When the deactivation is concentration independent, any type of batch-solids system may be used (and can be analyzed easily).

On the other hand, when the deactivation is concentration dependent then use the reactor in which C_A is forced to remain constant/fixed with time.

Batch-Solids : Determination of Rate from Experimental Data for Independent Deactivation :

Here we see how to obtain the kinetic forms - known in all respects (i.e., to obtain the constants of these forms) using the data collected in experimental devices.

We guess the kinetic expression for both main reaction and deactivation reaction (i.e., n and d). For these rate forms we manipulate the performance equations of the various batch-solids reactors (to get required forms). Then we use the data collected in these reactors (batch, mixed flow or plug flow) to check whether the kinetic forms fit the data or not (by constructing a plot of the variables of derived kinetic equations, i.e., here we use the integral method of analysis of data).

For independent deactivation, consider the following kinetic expressions / forms :

$$-r'_A = k'C_A\,\mathbf{a} \quad \text{... main reaction} \qquad \text{... (8.9 a)}$$

$$-\frac{da}{dt} = k_d\,\mathbf{a} \quad \text{... deactivation / decay reaction} \qquad \text{... (8.9 b)}$$

i.e., here we consider first order reaction and first order deactivation that is too concentration independent.

Batch-Solids + Batch-Fluid :

In this case, it is required to develop an expression that relates the changing gas composition with time.

The rate of reaction is defined as

$$-r_A = -\frac{dC_A}{dt} = -\frac{1}{V}\left(\frac{dN_A}{dt}\right) \qquad \text{... (8.10)}$$

Also,
$$-r'_A = -\frac{1}{W}\left(\frac{dN_A}{dt}\right) \qquad \text{... (8.11)}$$

Rearranging, we get

$$-\frac{dN_A}{dt} = (-r'_A)\, W \qquad \ldots (8.12)$$

Replacing $-dN_A/dt$ in Equation (8.10) by the expression of Equation (8.12) gives

$$-\frac{dC_A}{dt} = \frac{W\,(-r'_A)}{V}$$

We have

$$-r'_A = k'C_A\,\mathbf{a}$$

$$\therefore \quad -\frac{dC_A}{dt} = \frac{W}{V}(-r'_A) = \frac{W}{V}\,k'C_A\,\mathbf{a} \qquad \ldots (8.13)$$

The decay rate law is

$$-\frac{d\mathbf{a}}{dt} = k_d\,\mathbf{a} \qquad \ldots (8.14)$$

At $t = t \rightarrow \mathbf{a} = \mathbf{a}$ and at $t = 0 \rightarrow \mathbf{a} = 1\ (= \mathbf{a_o})$

$$-\int_1^{\mathbf{a}} \frac{d\mathbf{a}}{\mathbf{a}} = k_d \int_0^t dt$$

$$-\ln \mathbf{a} = k_d\, t$$

$$\ln \mathbf{a} = -k_d\, t$$

$$\therefore \quad \mathbf{a} = e^{-k_d t} \qquad \ldots (8.15)$$

Replacing **a** in Equation (8.13) by its value in Equation (8.14), we get

$$-\frac{dC_A}{dt} = \frac{W}{V}\,k'\,e^{-k_d t}\,C_A \qquad \ldots (8.16)$$

which on integration gives

$$\ln \frac{C_{Ao}}{C_A} = \frac{Wk'}{Vk_d}\left[1 - e^{-k_d t}\right] \qquad \ldots (8.17)$$

$$-\int_{C_{Ao}}^{C_A} \frac{dC_A}{C_A} = \frac{Wk'}{V}\int_0^t e^{-k_d t}\,dt$$

$$-\ln \frac{C_A}{C_{Ao}} = \frac{Wk'}{V}\left[\frac{1}{-k_d}\,e^{-k_d t}\right]_0^t$$

$$\ln \frac{C_{Ao}}{C_A} = \frac{Wk'}{Vk_d}\left[-e^{-k_d t}\right]_0^t$$

$$\ln \frac{C_{Ao}}{C_A} = \frac{Wk'}{Vk_d}\left[1 - e^{-k_d t}\right]$$

At $t = \infty$, $C_A = C_{A\infty}$ and is not zero (Refer Equation (8.17)).

$\therefore$ Equation (8.17) becomes

$$\ln \frac{C_{Ao}}{C_{A\infty}} = \frac{Wk'}{Vk_d} \qquad \ldots (8.18)$$

$$\ln \frac{C_{Ao}}{C_A} = \frac{Wk'}{Vk_d} - \frac{Wk'}{Vk_d} e^{-k_d t}$$

Using Equation (8.18), the above equation becomes

$$\ln \frac{C_{Ao}}{C_A} = \ln \frac{C_{Ao}}{C_{A\infty}} - \frac{Wk'}{Vk_d} e^{-k_d t}$$

$$\ln \frac{C_{Ao}}{C_A} - \ln \frac{C_{Ao}}{C_{A\infty}} = -\frac{Wk'}{Vk_d} e^{-k_d t}$$

$$\ln \left[\frac{C_{Ao}}{C_A} \cdot \frac{C_{A\times}}{C_{Ao}} \right] = \ln \frac{C_{A\infty}}{C_A} = -\frac{Wk'}{Vk_d} e^{-k_d t}$$

$$-\ln \frac{C_{A\infty}}{C_A} = \frac{Wk'}{Vk_d} e^{-k_d t}$$

$$\ln \frac{C_A}{C_{A\infty}} = \frac{Wk'}{Vk_d} e^{-k_d t}$$

Taking the natural logarithms of both sides gives

$$\ln \ln \left(\frac{C_A}{C_{A\infty}} \right) = \ln \left[\frac{Wk'}{Vk_d} \cdot e^{-k_d t} \right]$$

$$\boxed{\ln \ln \frac{C_A}{C_{A\infty}} = \ln \frac{Wk'}{Vk_d} - k_d t} \qquad \ldots (8.19)$$

The graphical procedure to test for the kinetic expressions of Equation (8.9) and to evaluate the rate constants with data from this type of reactor is shown in Fig. 8.2.

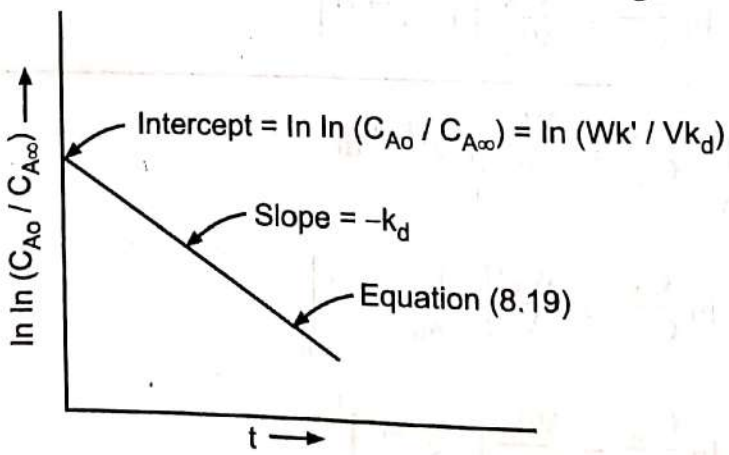


Fig. 8.2 : Test for the kinetic expressions of Equation (8.9) using batch-solids + batch-fluid reactor

Batch-Solids + Mixed Constant Flow of Fluid :

The performance equation for mixed flow is

$$\tau' = \frac{WC_{Ao}}{F_{Ao}} = \frac{W}{\nu} = \frac{C_{Ao} - C_A}{-r'_{A\,out}} \qquad \ldots (8.20)$$

where τ' is the weight time and is analogous to the space time (τ) for homogeneous systems.

$$-r'_A = k'C_A\,\mathbf{a}$$

$$\therefore \quad \tau' = \frac{C_{Ao} - C_A}{k'C_A\,\mathbf{a}}$$

$$k'\mathbf{a}\tau' = \frac{C_{Ao} - C_A}{C_A} = \frac{C_{Ao}}{C_A} - 1$$

$$\therefore \quad \frac{C_{Ao}}{C_A} = 1 + k'\mathbf{a}\,\tau' \qquad \ldots (8.21)$$

We have, $$-\frac{d\mathbf{a}}{dt} = k_d\,\mathbf{a}$$

$$-\int_1^{\mathbf{a}} \frac{d\mathbf{a}}{\mathbf{a}} = k_d \int_0^t dt$$

$$\therefore \quad \mathbf{a} = e^{-k_d t} \qquad \ldots (8.22)$$

Combining Equations (8.21) and (8.22) gives

$$\frac{C_{Ao}}{C_A} = 1 + k' e^{-k_d t}\,\tau' \qquad \ldots (8.22\ a)$$

Rearranging gives

$$\frac{C_{Ao}}{C_A} - 1 = k' e^{-k_d t}\,\tau'$$

Taking the logarithm of both sides, we obtain

$$\ln\left(\frac{C_{Ao}}{C_A} - 1\right) = \ln\left(k' e^{-k_d t}\,\tau'\right)$$

$$\boxed{\ln\left(\frac{C_{Ao}}{C_A} - 1\right) = \ln(k'\tau') - k_d\,t} \qquad \ldots (8.23)$$

A plot of $\ln\left(\frac{C_A}{C_{Ao}} - 1\right)$ v/s t should give a straight line for our guess (regarding the rate forms assumed) to be correct. If the data fall on a straight line, then the slope and intercept of the line yield the values of the rate constants k' and k_d.

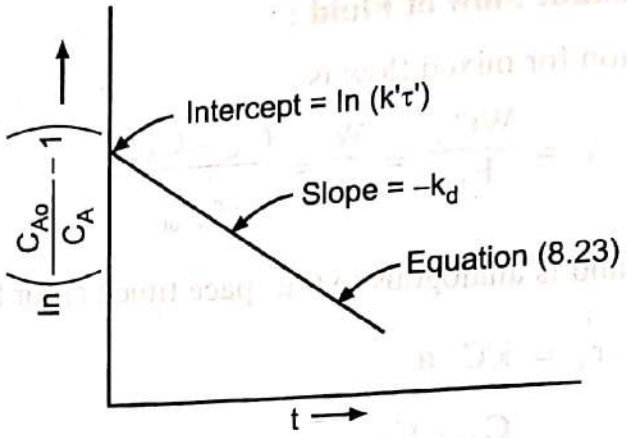


Fig. 8.3 : Test for the kinetic expressions of Equation (8.9) using data obtained in a mixed flow reactor - batch-solids + constant flow of fluid

Batch-Solids + Mixed Changing Flow of Fluid (to keep C_A fixed/constant) :

For steady flow in a mixed flow reactor, we have

$$\frac{C_{Ao}}{C_A} = 1 + k' e^{-k_d t} \cdot \tau' \qquad \ldots (8.24)$$

Since the catalyst is deactivating, for keeping C_A constant, the flow rate must be slowly lowered with time. Hence we have two variables in this situation t and τ'.

Rearranging Equation (8.24) gives

$$\frac{C_{Ao}}{C_A} - 1 = k' e^{-k_d t} \cdot \tau'$$

$$\frac{C_{Ao} - C_A}{C_A} = k' e^{-k_d t} \cdot \tau'$$

$$\frac{C_{Ao} - C_A}{k' C_A} = e^{-k_d t} \cdot \tau'$$

Taking the logarithm of both sides, we obtain

$$\ln\left(\frac{C_{Ao} - C_A}{k' C_A}\right) = \ln\left(e^{-k_d t} \cdot \tau'\right)$$

$$\ln\left(\frac{C_{Ao} - C_A}{k' C_A}\right) = -k_d t + \ln \tau'$$

$$\therefore \quad \boxed{\ln \tau' = k_d t + \ln\left(\frac{C_{Ao} - C_A}{k' C_A}\right)} \qquad \ldots (8.25)$$

Plotting the variables of Equation (8.25) as shown in Fig. 8.4 provides a test for this kinetic equation.

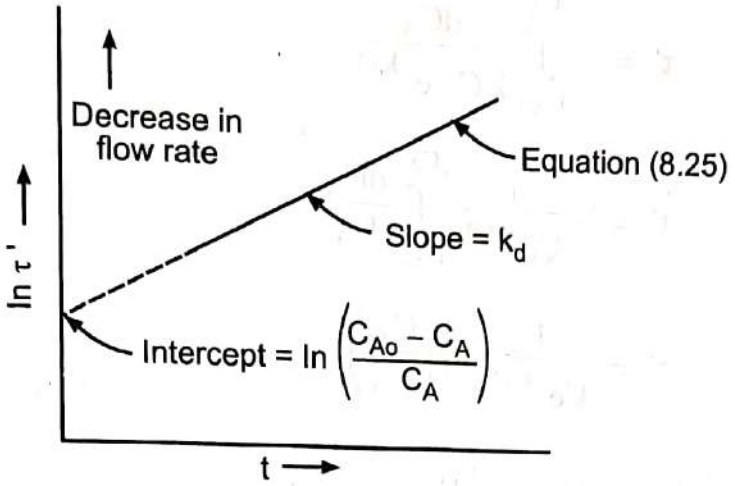


Fig. 8.4 : Test for the guessed kinetic equations of Equation (8.9) using a batch of solids + changing flow of fluid (for keeping C_A fixed) in a mixed flow reactor

Batch-Solids + Plug Constant Flow of Fluid :

The performance equation for a plug flow reactor is

$$\frac{W}{F_{Ao}} = \int_0^{X_A} \frac{dX_A}{-r_A'}$$

For $\varepsilon_A = 0$,

$$\frac{W}{F_{Ao}} = -\frac{1}{C_{Ao}} \int_{C_{Ao}}^{C_A} \frac{dC_A}{-r_A'} \quad \ldots (8.26)$$

$$\frac{WC_{Ao}}{F_{Ao}} = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{-r_A'}$$

But

$$\frac{WC_{Ao}}{F_{Ao}} = \tau'$$

$\therefore$

$$\tau' = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{-r_A'} \quad \ldots (8.27)$$

We have :

$$-r_A' = k'C_A\, a$$

Using $-r_A'$ from the above expression, Equation (8.27) becomes

$$\tau' = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{k'C_A a} \quad \ldots (8.28)$$

$$-\frac{da}{dt} = k_d\, a$$

$$\therefore \quad a = e^{-k_d t}$$

$$\therefore \quad \tau' = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{k'C_A e^{-k_d t}}$$

$$\tau' = -\frac{1}{k'e^{-k_d t}} \int_{C_{Ao}}^{C_A} \frac{dC_A}{C_A}$$

$$\therefore \quad \tau' = \frac{1}{k'e^{-k_d t}} \ln\left(\frac{C_{Ao}}{C_A}\right) \qquad \ldots (8.29)$$

$$\ln\left(\frac{C_{Ao}}{C_A}\right) = k'\tau' \cdot e^{-k_d t}$$

Taking the natural logarithms of both sides gives

$$\ln \ln\left(\frac{C_{Ao}}{C_A}\right) = \ln\left(k'\tau' \cdot e^{-k_d t}\right)$$

$$\boxed{\ln \ln\left(\frac{C_{Ao}}{C_A}\right) = \ln (k'\tau') - k_d t} \qquad \ldots (8.30)$$

A plot of $\ln \ln\left(\frac{C_{Ao}}{C_A}\right)$ v/s t should give a straight line for our guess to be correct and if it is so, then the slope and intercept yield the two rate constants of Equation (8.9).

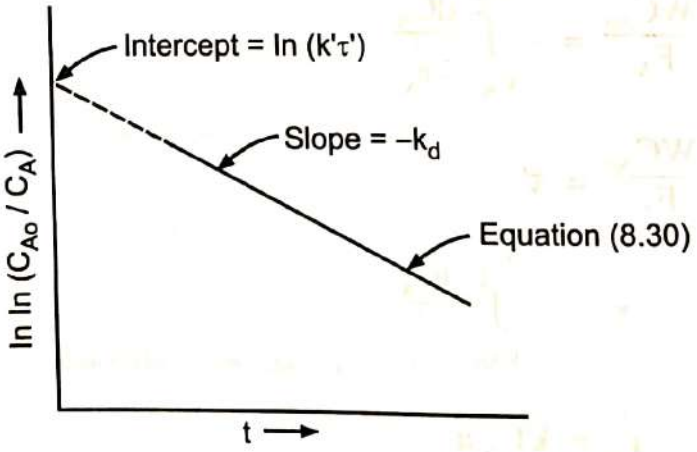


Fig. 8.5 : Integral method to test for the guessed kinetic expressions of Equation (8.9) using a batch of solids and steady plug flow of fluid

Batch-Solids + Changing Plug Flow of Fluid (To keep C_A fixed/constant) :

In the plug flow reactor, Equation (8.29) is applicable. In this case, τ' and t are the two variables.

We have, $$\ln\left(\frac{C_{Ao}}{C_A}\right) = k'\tau' e^{-k_d t}$$

$$\frac{1}{k'} \ln\left(\frac{C_{Ao}}{C_A}\right) = \tau' e^{-k_d t}$$

Taking the natural logarithms of both sides, we obtain

$$\ln\left[\frac{1}{k'} \ln\left(\frac{C_{Ao}}{C_A}\right)\right] = \ln(\tau') - k_d t$$

$$\therefore \quad \boxed{\ln(\tau') = k_d t + \ln\left[\frac{1}{k'} \ln\left(\frac{C_{Ao}}{C_A}\right)\right]} \qquad \ldots (8.31)$$

Constructing a plot of the variables of Equation (8.31) as shown in Fig. 8.6 provides a test for this kinetic equation.

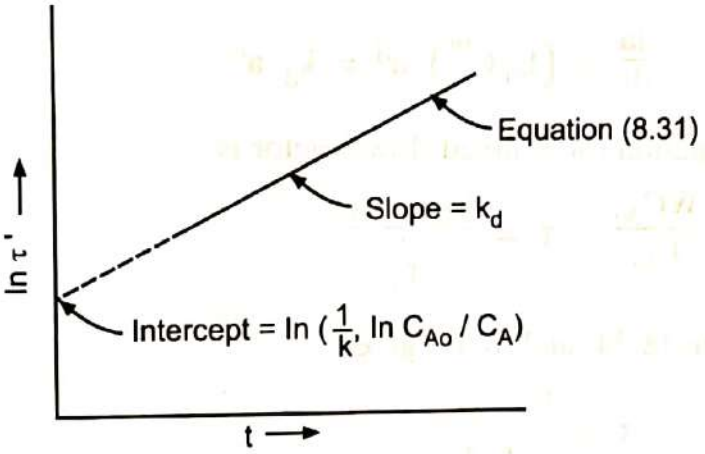


Fig. 8.6 : Test for the guessed kinetic equations of Equation (8.9) using a batch of solids and changing flow of fluid (to keep C_A constant) in a plug flow reactor

Whenever the deactivation is concentration independent, we can extend the above analysis to any reaction order n and any deactivation order d, i.e., in symbols for

$$-r'_A = k'C_A^n \, a$$

$$-\frac{da}{dt} = k_d \, a^d \qquad \ldots (8.32)$$

If the deactivation is **concentration dependent** (i.e., $m \neq 0$), then we have to use reactors which keep C_A fixed/constant by lowering the feed rate to find the rate constants for parallel, series and side-by-side deactivation.

Batch-Solids : Determination of Rate for Parallel, Series or Side-by-Side Deactivation :

To determine the rate constants in such cases from the experiments, we can use a mixed flow reactor with changing flow rate of feed (to keep appropriate concentration constant).

For parallel deactivation maintain C_A constant.

For series deactivation maintain C_R constant.

For side-by-side deactivation maintain C_P constant.

Parallel Deactivation :

For parallel deactivation, the kinetic expressions are :

$$-r_A' = k' C_A^n \, a \quad \text{... for } n^{th} \text{ order reaction}$$

$$-\frac{da}{dt} = k_d C_A^m \quad \text{... for } d^{th} \text{ order decay} \qquad \text{... (8.33)}$$

At constant C_A, the above equations become

$$-r_A' = \left(k' C_A^n\right) a = k'' a \qquad \text{... (8.34)}$$

$$-\frac{da}{dt} = \left(k_d C_A^m\right) a^d = k_d' \, a^d \qquad \text{... (8.35)}$$

The performance equation for a mixed flow reactor is

$$\frac{WC_{Ao}}{F_{Ao}} = \tau' = \frac{C_{Ao} - C_A}{-r_A'} \qquad \text{... (8.36)}$$

Combining Equations (8.34) and (8.36) gives

$$\tau' = \frac{C_{Ao} - C_A}{k'' a} \qquad \text{... (8.37)}$$

We have : $$-\frac{da}{dt} = k_d' \, a^d$$

Assume first order deactivation, i.e., d = 1.

$\therefore$ $$-\frac{da}{dt} = k_d' \, a$$

$$-\frac{da}{a} = k_d' \, dt$$

Integrating, we get

$$a = e^{-k_d' t} \qquad \text{... (8.38)}$$

Combining Equations (8.37) and (8.38) gives

$$\tau' = \frac{C_{Ao} - C_A}{k'' e^{-k_d' t}} \qquad \text{... (8.39)}$$

$$\tau' e^{-k_d' t} = \frac{C_{Ao} - C_A}{k''}$$

Taking the logarithm of both sides, we get

$$\ln (\tau' e^{-k'_d t}) = \ln\left(\frac{C_{Ao} - C_A}{k''}\right)$$

$$\ln \tau' - k'_d t = \ln\left(\frac{C_{Ao} - C_A}{k''}\right)$$

$$\therefore \quad \boxed{\ln \tau' = \ln\left(\frac{C_{Ao} - C_A}{k''}\right) + k'_d t} \qquad \ldots (8.40)$$

Plot ln τ' vs t. If the data fall on a straight line, the guessed order of deactivation is correct. The slope and intercept of this straight line then give the constants k" and k'_d.

After finding the order of deactivation, the concentration dependency can be determined.

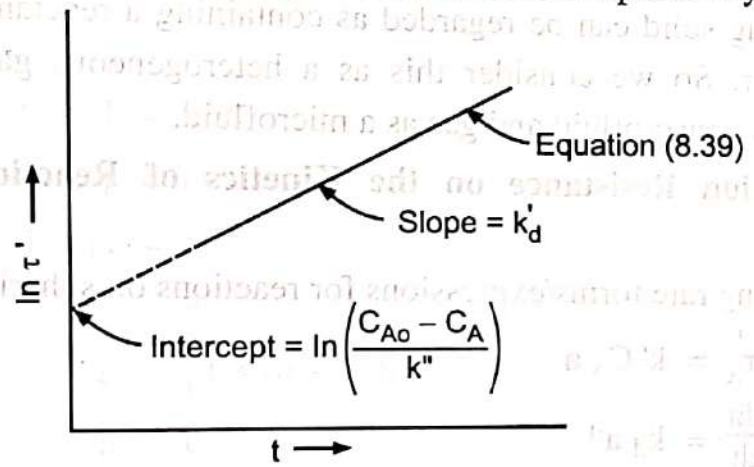


Fig. 8.7 : Test for kinetic expressions of Equation (8.33). Data is collected in a mixed flow reactor with constant C_A

Assume d = 0, d = 2 and d = 3 and obtain expressions similar to that given by Equation (8.40).

For d = 0 ..... zero order deactivation

$$-\frac{da}{dt} = k'_d$$

$$-\int_1^a da = k'_d \int_0^t dt$$

$$1 - a = k'_d t$$

$$a = 1 - k'_d t$$

$$\therefore \quad \tau' = \frac{C_{Ao} - C_A}{k''(1 - k'_d t)}$$

Rearranging gives

$$\frac{k''(1 - k_d' t)}{C_{Ao} - C_A} = \frac{1}{\tau'}$$

$$\frac{k''}{C_{Ao} - C_A} - \frac{k'' k_d' t}{C_{Ao} - C_A} = \frac{1}{\tau'}$$

$$\therefore \quad \boxed{\frac{1}{\tau'} = \frac{k''}{C_{Ao} - C_A} - \frac{k'' k_d' t}{C_{Ao} - C_A}} \qquad \ldots (8.41)$$

Flowing Solids Experimental Reactors :

When deactivation is very rapid, then a reactor with throughflow of both gas and solids must be used. Here the flowing solid can be regarded as containing a reactant (the activity) which is consumed in the reactor. So we consider this as a heterogeneous gas-solid reaction system wherein solid behaves as a macrofluid and gas as a microfluid.

Effect of Pore Diffusion Resistance on the Kinetics of Reactions with Deactivating Catalysts :

Consider the following rate forms/expressions for reactions on spherical catalyst particles :

$$-r_A' = k' C_A a \qquad \ldots (8.42\ a)$$

$$-\frac{da}{dt} = k_d a^d \qquad \ldots (8.42\ b)$$

For no deactivation of catalyst but with or without resistance to pore diffusion, the above rate forms become

$$-r_A' = k' C_A \eta \qquad \ldots (8.43\ a)$$

$$a = 1 \qquad \ldots (8.43\ b)$$

where $\eta = 1$ ... for no pore resistance (i.e., pore diffusion does not offer any resistance to reaction) ... (8.44 a)

$$\eta = \frac{1}{M_T}$$ ... for strong pore diffusion resistance ... (8.44 b)

and $M_T = L\sqrt{k''' / \mathscr{D}_e}$ ... Thiele modulus for first order reaction

$$k''' = k' \rho_s \qquad \ldots (8.44\ c)$$

The effectiveness factor (η) in the rate equation [Equation (8.43)] accounts for the resistance to pore diffusion. It varies from 0 to 1.

With deactivation, Equation (8.43) becomes

$$-r_A' = k' a C_A \eta \qquad \ldots (8.45\ a)$$

$$-\frac{da}{dt} = k_d a^d \qquad \ldots (8.45\ b)$$

where

$$\eta = 1 \quad \text{... for no pore resistance} \qquad \text{... (8.46 a)}$$

$$\eta = \frac{1}{M_{Td}} \quad \text{... for strong pore resistance} \qquad \text{... (8.46 b)}$$

$$M_{Td} = L\sqrt{k'''a / \mathscr{D}_e} = M_T\, a^{1/2} \qquad \text{... (8.46 c)}$$

$$-\frac{da}{dt} = k_d\, a^d$$

$$-\frac{da}{a^d} = k_d\, dt$$

$$-\int_1^a \frac{da}{a^d} = \int_0^t k_d\, dt$$

$$-\left[\frac{a^{-d+1}}{-d+1}\right]_1^a = k_d\, t$$

$$-\frac{1}{1-d}\left[a^{1-d} - 1\right] = k_d\, t$$

$$-a^{1-d} + 1 = (1-d)\, k_d\, t$$

$$1 - (1-d)\, k_d\, t = a^{1-d}$$

$$1 + (d-1)\, k_d\, t = a^{1-d}$$

$\therefore$

$$a = \left[1 + (d-1)\, k_d\, t\right]^{1/(1-d)} \quad \text{... for } d \neq 1$$

and

$$a = e^{-k_d t} \quad \text{... for } d = 1 \qquad \text{... (8.47)}$$

It is clear from Equation (8.47) that **a** decreases with time and in the regime of strong pore diffusion resistance this decrease of **a** causes M_{Td} to decrease [See Equation (8.46 c)]. This means that η increases with time [as it is inversely proportional to M_{Td}, see Equation (8.46 b)], as shown in Fig. 8.8. However, a decrease in **a** is faster than an increase in η and due to this, the reaction rate decreases with time.

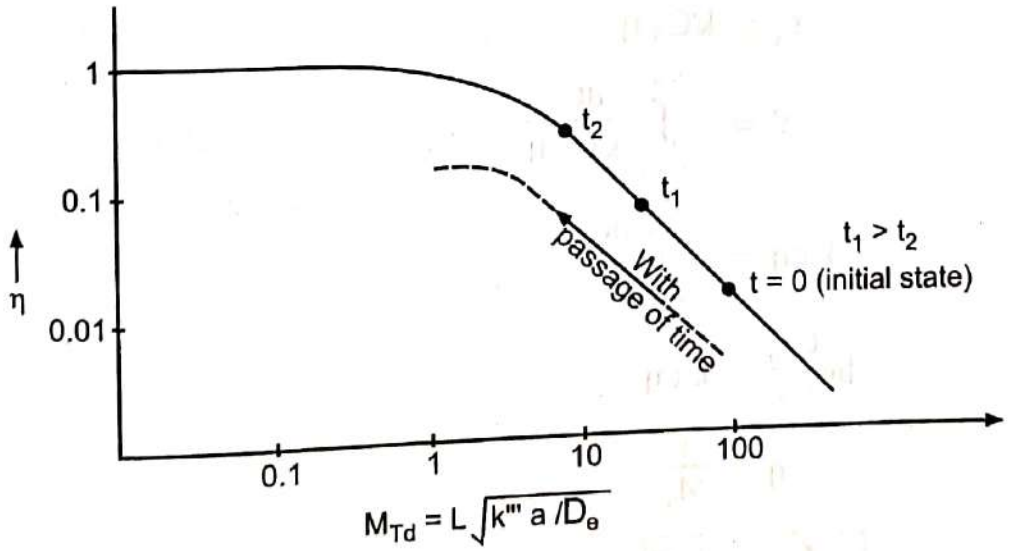


Fig. 8.8 : As catalyst deactivates, η increases with time

Performance Equations in the Regime of Strong Pore Diffusion Resistance :

Now we present the performance equations for plug flow for the simple first order system of Equation (8.9), i.e., for first order reaction + first order deactivation + concentration independent deactivation.

1. No resistance to pore diffusion, i.e., without pore diffusion resistance :

(a) Without deactivation :

$$\ln \frac{C_{Ao}}{C_A} = k'\tau' \quad \text{(easy to obtain)} \qquad \ldots (8.48)$$

(b) With deactivation :

$$\tau' = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{-r_A'} = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{k'C_A\, \mathbf{a}}$$

$$\therefore \quad \ln \frac{C_{Ao}}{C_A} = (k'\mathbf{a})\, \tau'$$

For **first order deactivation**, we have

$$\mathbf{a} = e^{-k_d t}$$

$$\therefore \quad \boxed{\ln \frac{C_{Ao}}{C_A} = (k'\mathbf{a})\, \tau' = (k'\tau')\, e^{-k_d t}} \qquad \ldots (8.49)$$

2. Strong resistance to pore diffusion (strong pore diffusion effects) :

(a) Without deactivation :

$$\tau' = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{-r_A'}$$

$$-r_A' = k'C_A\, \eta$$

$$\therefore \quad \tau' = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{k'C_A\, \eta}$$

$$k'\tau'\eta = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{C_A}$$

$$\ln \frac{C_{Ao}}{C_A} = k'\tau'\eta$$

We have $\eta = \dfrac{1}{M_T}$

$$\therefore \quad \boxed{\ln \frac{C_{Ao}}{C_A} = \frac{k'\tau'}{M_T}} \qquad \ldots (8.50)$$

(b) With deactivation :

$$\tau' = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{-r_A'}$$

$$-r_A' = k'aC_A\,\eta \qquad \ldots\ \eta = \frac{1}{M_{Td}}$$

$$\tau' = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{k'aC_A\,\eta}$$

$$\therefore \quad \ln\frac{dC_A}{C_A} = k'a\tau'\,\eta$$

We have $$\eta = \frac{1}{M_{Td}}$$

$$\therefore \quad \ln\frac{C_{Ao}}{C_A} = \frac{(k'a)\,\tau'}{M_{Td}} \qquad \ldots (8.51)$$

For **first order deactivation**, we have

$$a = e^{-k_d t}$$

$$\therefore \quad a^{1/2} = e^{-k_d t/2}$$

We have, $$M_{Td} = M_T\, a^{1/2}$$

$$\therefore \quad M_{Td} = M_T\,(e^{-k_d t/2})$$

Using the above values of **a** and M_{Td}, Equation (8.51) becomes

$$\ln\frac{C_{Ao}}{C_A} = \frac{(k'a)\,\tau'}{M_{Td}} = \frac{k'\tau'\,(e^{-k_d t})}{M_T\, e^{-k_d t/2}} = \frac{k'\tau'}{M_T}\, e^{-k_d t/2}$$

$$\boxed{\ln\frac{C_{Ao}}{C_A} = \frac{(k'a)\,\tau'}{M_{Td}} = \frac{k'\tau'}{M_T}\, e^{-k_d t/2}} \qquad \ldots (8.52)$$

For a mixed flow do the following change in the equations obtained for the plug flow (... effect of pore diffusion)

$$\ln\frac{C_{Ao}}{C_A} \xrightarrow[\text{to}]{\text{change}} \frac{C_{Ao} - C_A}{C_A}$$

For a given treatment rate, the conversion of a reacting fluid, flowing through a batch of deactivating catalyst, decreases with time during the run.

The average conversion during a run can be obtained by using the following expression :

$$\boxed{\bar{X}_A = \frac{\int_0^{t_{run}} X_A\,(t)\,dt}{t_{run}}} \qquad \ldots (8.53)$$

When the conversion reaches to an unacceptable level, then the run is discontinued / terminated, the catalyst is either discarded or regenerated and the cycle is repeated.

SOLVED EXAMPLES

Example 8.1 : *A recycle reactor with very high recycle ratio is used to study the kinetics of an irreversible catalytic reaction, A → R. The following data are obtained for a constant flow rate of feed [τ' = 2 (kg·s)/l] :*

Time in h after start of operation	*1*	*2*	*4*
X_A , *conversion of A*	*0.889*	*0.865*	*0.804*

From the data given, it is clear that the catalyst deactivates with use (since there is a progressive decrease in conversion). Find the rate equations for the reaction and deactivation which fit the data provided.

Solution : A → R

In case of a recycle reactor with very high recycle ratio, mixed flow is approached. Therefore, in this case, method of mixed flow reactor can be used.

To find the rate equations for the reaction and deactivation, we have to guess the rate equations and see whether they fit the data or not.

As a first guess try first order rate equations for both the reaction and deactivation, i.e., first order reaction and first order deactivation.

$$-r'_A = k' C_A \, a \qquad \text{... reaction}$$

and

$$-\frac{da}{dt} = k_d \, a \qquad \text{... deactivation}$$

For a mixed flow reactor (- recycle reactor with high recycle ratio) with constant flow rate of fluid, we have for the first order rates :

$$\ln\left(\frac{C_{A0}}{C_A} - 1\right) = \ln (k'\tau') - k_d \, t$$

(Equation for a batch of solids, mixed constant flow of fluid)

Test for this kinetic equation is :

A plot of $\ln (C_{A0}/C_A - 1)$ v/s t should yield a straight line, i.e., the data should fall on a straight line. If we get a straight line with intercept on y-axis, then our guess of first-order reaction and of first order deactivation is correct.

$$\text{Slope of this plot} = -k_d$$

$$\text{Intercept of this plot} = \ln (k'\tau')$$

Now convert the data in the required form :

t, h	1	2	4
X_A	0.889	0.865	0.804
$C_A/C_{A0} = (1 - X_A)$	0.111	0.135	0.196
C_{A0}/C_A	9.01	7.41	5.1
$\left(\frac{C_{A0}}{C_A} - 1\right)$	8.01	6.41	4.1
$\ln [(C_{A0}/C_A) - 1]$	2.08	1.86	1.41

$$X_A = \frac{C_{A0} - C_A}{C_{A0}} = 1 - \frac{C_A}{C_{A0}}$$

$$\therefore \quad \frac{C_A}{C_{A0}} = 1 - X_A$$

Plot $\ln\left(\frac{C_{A0}}{C_A} - 1\right)$ v/s t.

Refer Fig. Ex 8.1 :

As the plot of $\ln (C_{A0}/C_A - 1)$ v/s t yields a straight line our guess of first order reaction and first-order deactivation is correct. As the data are falling on a straight line, the rate equations guessed satisfactorily represent the data.

$$\text{Intercept} = \ln (k'\tau') = 2.3$$

$$k'\tau' = 9.974 \approx 10$$

Given : $$\tau' = 2 \text{ (kg·s)}/l$$

$$k' = \frac{10}{2} = 5 \; l/(\text{kg·s})$$

$$\text{Slope} = -0.228 = -k_d$$

$$k_d = 0.228 \text{ h}^{-1}$$

The rate equation for reaction :

and $$\boxed{-r_A' = \left(5 \frac{l}{\text{kg·s}}\right) C_A \, a \, , \frac{\text{mol}}{\text{kg·s}}}$$ **... Ans.**

The rate equation for deactivation :

$$\boxed{-\frac{da}{dt} = (0.228 \text{ h}^{-1}) \, a}$$ **... Ans.**

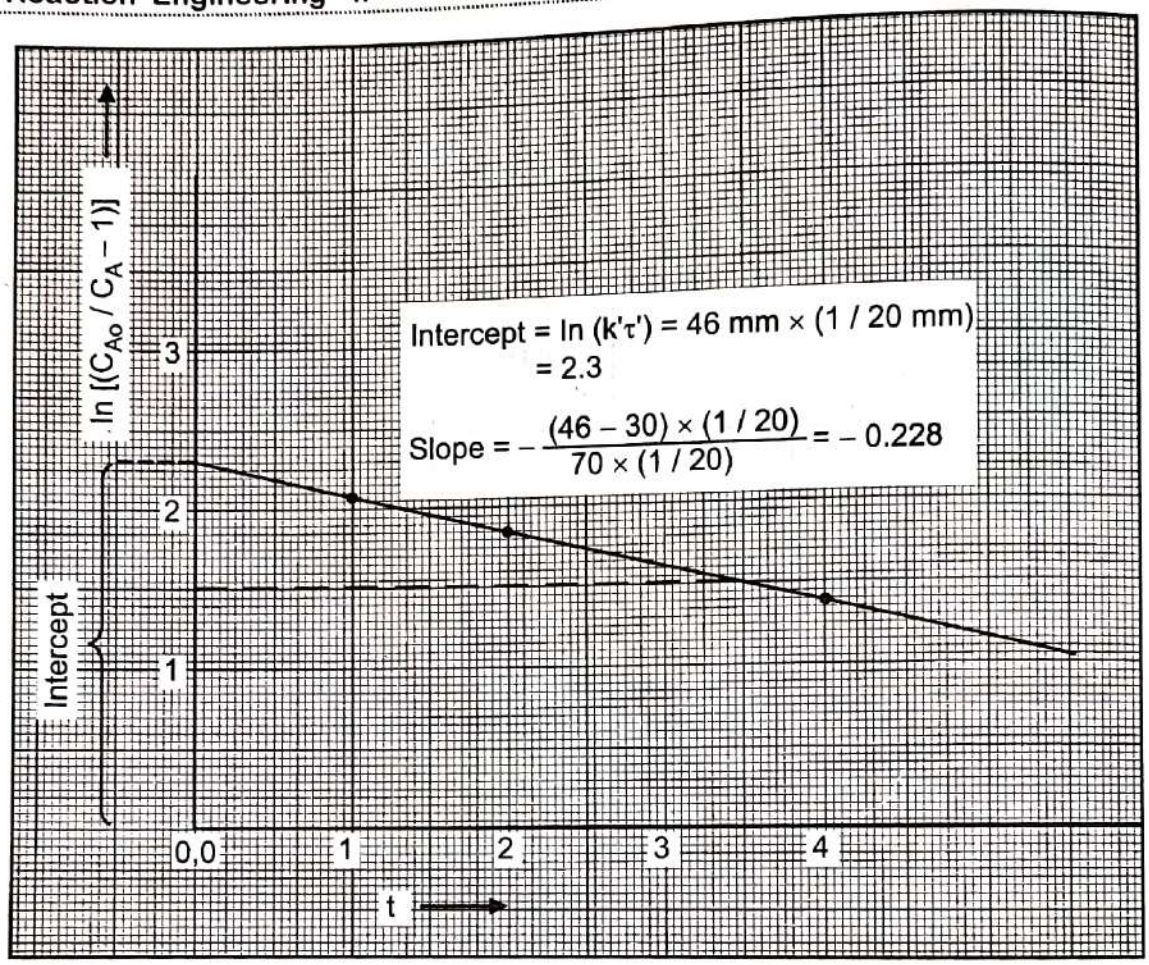


Fig. Ex 8.1

Example 8.2 : *The following data on an irreversible reaction A → R are obtained in a batch reactor (batch solids, batch fluid) using decaying catalyst. What can you say about the kinetics ?*

C_A	*1*	*0.802*	*0.675*	*0.532*	*0.422*	*0.368*
t, h	*0*	*0.25*	*0.5*	*1.0*	*2*	*∞*

Solution : A → R, catalytic reaction

Here we have to guess the rate forms and test the data to fit them.

Let us start by guessing the simplest rate forms.

i.e. guess-1 : First-order reaction and first-order deactivation :

$$-r'_A = k'C_A\, a \qquad \text{... reaction}$$

$$-\frac{da}{dt} = k_d\, a \qquad \text{... deactivation}$$

Now we will see whether the data provided satisfactorily fit these rate forms or not. If not, then we have to guess other rate forms.

The data is collected in a batch reactor.

Therefore, for batch solids, batch-fluid reactor, we have :

$$\ln\left(\ln\frac{C_A}{C_{A\infty}}\right) = \ln\frac{Wk'}{Vk_d} - k_d\,t$$

To construct the plot, we have to convert the data in the form we require.

C_A	t, h	$C_A/C_{A\infty}$	$\ln C_A/C_{A\infty}$	$\ln(\ln C_A/C_{A\infty})$
1	0	2.718	$0.99967 \approx 1.0$	0
0.802	0.25	2.18	0.779	– 0.25
0.675	0.50	1.83	0.604	– 0.50
0.532	1.0	1.446	0.369	– 1.0
0.422	2	1.147	0.137	– 1.988
$C_{A\infty} = 0.368$	∞	1	0	∞

Plot $\ln \ln C_A/C_{A\infty}$ v/s t. For our guess to be correct we should get a straight line.

Refer Fig. Ex 8.2 :

The straight line fit of the data indicates that out guess regarding the forms of rate equation both for the reaction and deactivation is correct. That is, for the reaction A → R, first-order reaction rate equation and first-order deactivation satisfactorily represent the data.

$$\text{Slope} = -k_d = -1$$

$$\therefore \quad k_d = 1$$

$$\text{Intercept} = \ln\frac{Wk'}{Vk_d} = 0$$

$$\therefore \quad \frac{Wk'}{Vk_d} = 1$$

$$k_d = 1\ h^{-1}$$

$$\therefore \quad \frac{Wk'}{V} = 1\ h^{-1}$$

$\therefore$ The rate forms are :

$$-\frac{dC_A}{dt} = \frac{Wk'}{V} C_A\,\mathbf{a} \qquad \text{with } \frac{Wk'}{V} = 1\ h^{-1}$$

$$-\frac{dC_A}{dt} = C_A\,\mathbf{a} = \frac{k'W}{V} C_A\,\mathbf{a} \qquad \text{... for reaction} \qquad \textbf{... Ans.}$$

Or

$$-r_A' = k'C_A\,\mathbf{a}, \qquad \text{with} \qquad k' = \frac{V}{W} \text{ since } \frac{Wk'}{V} = 1, \text{ it becomes}$$

$$\boxed{-r_A' = \frac{V}{W} C_A\,\mathbf{a}} \qquad \textbf{... Ans.}$$

and $-\frac{da}{dt} = k_d\, a$, with $k_d = 1\ h^{-1}$ it becomes

$$\boxed{-\frac{da}{dt} = a}$$ ... for deactivation ... Ans.

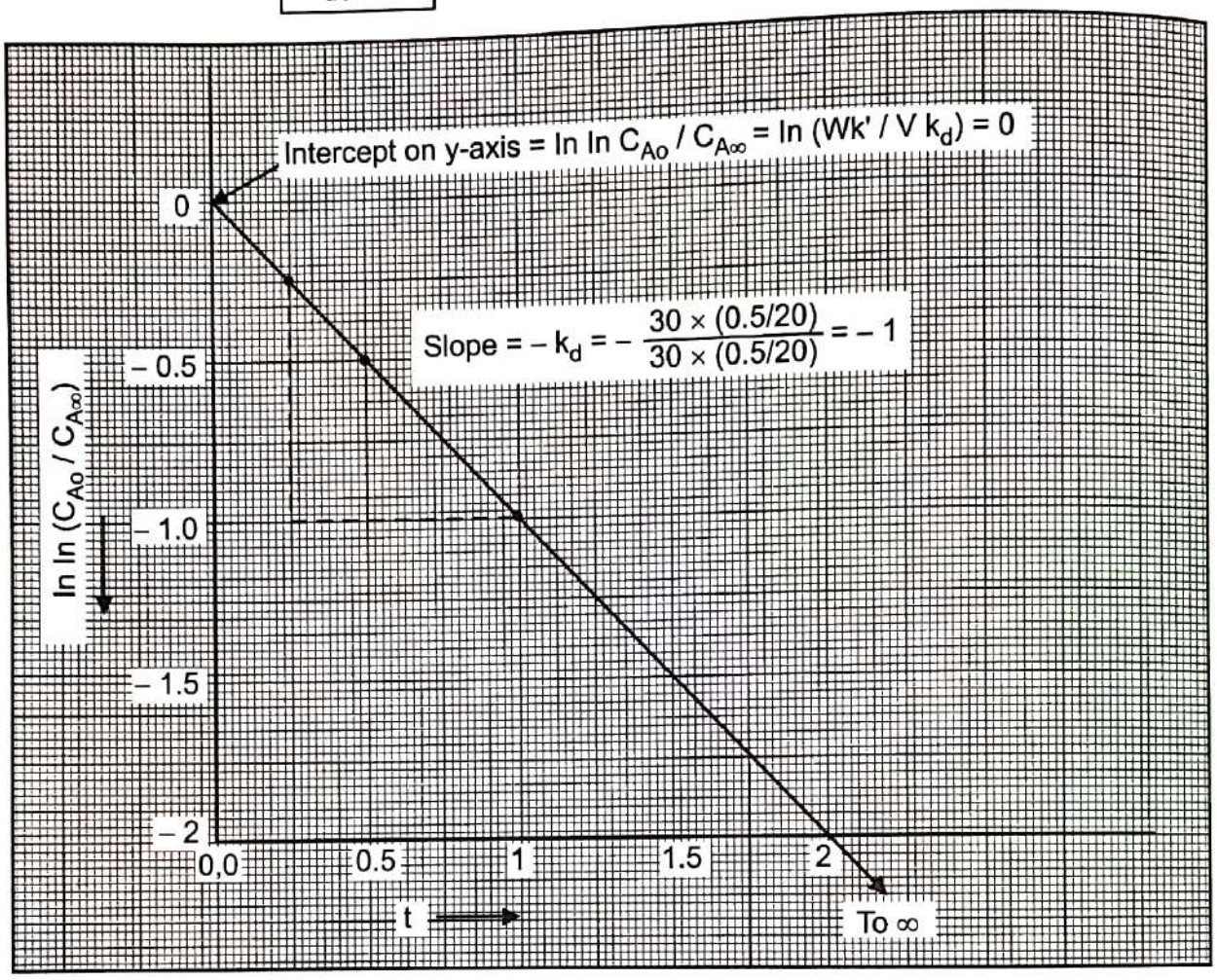


Fig. Ex 8.2

Example 8.3 : *Derive an expression for conversion (X_A) as a function of time in case of an irreversible first-order reaction, A → R, which is carried out isothermally in a batch reactor on a catalyst that is decaying (as a result of aging) as per the following decay law :*

$$-\frac{da}{dt} = k_d\, a^2$$

Solution : A → R

Rate of reaction : $-r'_A = k' C_A\, a$

Rate of deactivation : $-\frac{da}{dt} = k_d\, a^2$

$$-\frac{da}{a^2} = k_d\, dt$$

Limits : t = 0, ... a = 1, t = t ..... a = a

$$-\int_1^a \frac{da}{a^2} = k_d \int_0^t dt$$

$$\left[\frac{1}{\mathbf{a}}\right]_1^{\mathbf{a}} = k_d\, t$$

$$\left[\frac{1}{\mathbf{a}} - 1\right] = k_d\, t$$

$$\therefore \quad \mathbf{a} = \frac{1}{1 + k_d\, t} \qquad \ldots (1)$$

The performance equation for a batch reactor is

$$(-r_A)\, V = N_{A0} \frac{dX_A}{dt} \qquad \ldots \text{homogeneous system}$$

$$-r_A = -\frac{1}{V}\frac{dN_A}{dt}$$

$$-r_A' = -\frac{1}{W}\frac{dN_A}{dt} \qquad \ldots \text{for catalytic reaction}$$

$$\therefore \quad (-r_A) = (-r_A')\, W/V$$

The performance equation becomes

$$\frac{(-r_A')\, W}{V} \cdot V = N_{A0} \frac{dX_A}{dt}$$

$$(-r_A')\, W = N_{A0} \frac{dX_A}{dt} \qquad \ldots (2)$$

For $A \rightarrow R$: $\varepsilon_A = 0$

$$\therefore \quad C_A = C_{A0}(1 - X_A)$$

$$C_A = \frac{N_{A0}}{V}(1 - X_A) \qquad \ldots (3)$$

$$C_{A0} = \frac{N_{A0}}{V}$$

$$-r_A' = k' C_A\, \mathbf{a} \qquad \ldots \text{first-order reaction} \qquad \ldots (4)$$

Combining Equations (2) and (4), we get

$$k' C_A\, \mathbf{a} W = N_{A0} \frac{dX_A}{dt} \qquad \ldots (5)$$

Substituting the value of C_A from Equation (3) in Equation (5), we get

$$k' \frac{N_{A0}}{V}(1 - X_A)\, \mathbf{a} W = N_{A0} \frac{dX_A}{dt}$$

$$\frac{k'W}{V}(1 - X_A)\, \mathbf{a} = \frac{dX_A}{dt}$$

Let, $k'W/V = k$

$$\frac{dX_A}{dt} = k(1 - X_A)\, \mathbf{a} \qquad \ldots (6)$$

Substituting for **a** from Equation (1), Equation (6) becomes

$$\frac{dX_A}{dt} = k(1 - X_A) \times \frac{1}{1 + k_d t}$$

$$\int_0^{X_A} \frac{dX_A}{(1 - X_A)} = k \int_0^t \frac{dt}{1 + k_d t}$$

Integrating, we get

$$\ln \frac{1}{1 - X_A} = \frac{k}{k_d} \ln (1 + k_d t)$$

$$\ln \frac{1}{1 - X_A} = \ln [1 + k_d t]^{k/k_d}$$

$$\frac{1}{1 - X_A} = [1 + k_d t]^{k/k_d}$$

$$\frac{1}{[1 + k_d t]^{k/k_d}} = 1 - X_A$$

$\therefore$

$$\boxed{X_A = 1 - \frac{1}{[1 + k_d t]^{k/k_d}}} \quad \text{where } k = k'W/V \qquad \ldots (7)$$

Equation (7) is the desired relationship.

Extra : $k \int_0^t \frac{dt}{1 + k_d t}$

Let,

$$1 + k_d t = y$$

$$k_d \, dt = dy$$

$$dt = dy/k_d$$

When $t = 0 \ldots \ldots y = 1$

$t = t \ldots \ldots y = 1 + k_d t$

$\therefore$

$$k \int_0^t \frac{dt}{1 + k_d t} = k \int_1^{1 + k_d t} \frac{dy}{k_d \cdot y}$$

$$= \frac{k}{k_d} \int_1^{1 + k_d t} \frac{dy}{y}$$

$$= \frac{k}{k_d} [\ln y]_1^{1 + k_d t}$$

$$= \frac{k}{k_d} [\ln (1 + k_d t) - \ln (1)]$$

$$= \frac{k}{k_d} [\ln (1 + k_d t)]$$

Example 8.4 : *Derive an expression for conversion (X_A) as a function of time for an irreversible first order reaction, $A \rightarrow R$, which is carried out isothermally in a batch reactor on a catalyst that is decaying as per following decay law :*

$$-\frac{da}{dt} = k_d\, a$$

Solution : $A \rightarrow R$ ... irreversible first-order reaction.

Rate of reaction : $$-r_A' = k' C_A\, a$$

Rate of deactivation : $$-\frac{da}{dt} = k_d\, a$$

Limits : $t = 0 \ldots \rightarrow a = 1,\ t = t \ldots \rightarrow a = a$

$$-\int_1^a \frac{da}{a} = k_d \int_0^t dt$$

$$-\ln a = k_d\, t$$

or $$a = e^{-k_d t} \qquad \ldots (1)$$

For batch reactor, we have

$$(-r_A)\, V = N_{A0} \frac{dX_A}{dt} \qquad \text{... for homogeneous system}$$

$$(-r_A')\, W = N_{A0} \frac{dX_A}{dt} \qquad \text{... for heterogeneous system} \qquad \ldots (2)$$

For $A \rightarrow R$: $$\varepsilon_A = 0$$

$\therefore$ $$C_A = C_{A0}\,(1 - X_A)$$

$$C_A = \frac{N_{A0}}{V}(1 - X_A) \qquad \text{as } C_{A0} = \frac{N_{A0}}{V} \qquad \ldots (3)$$

$$-r_A' = k'\, C_A\, a \qquad \text{... first-order catalytic reaction} \qquad \ldots (4)$$

Combining Equations (2) and (4), we get

$$k'\, C_A\, W a = N_{A0} \frac{dX_A}{dt} \qquad \ldots (5)$$

Substituting the value of C_A from Equation (3) in Equation (5), we get

$$k' \frac{N_{A0}}{V}(1 - X_A)\, aW = N_{A0} \frac{dX_A}{dt}$$

$$\frac{k'W}{V}(1 - X_A)\, a = \frac{dX_A}{dt}$$

Let $$\frac{k'W}{V} = k$$

$$\therefore \quad \frac{dX_A}{dt} = k(1 - X_A)\,a$$

But $$a = e^{-k_d t}$$

$$\therefore \quad \frac{dX_A}{dt} = k(1 - X_A)\,e^{-k_d t}$$

$$\int_0^{X_A} \frac{dX_A}{(1 - X_A)} = k \int_0^t e^{-k_d t}\,dt$$

$$\ln \frac{1}{1 - X_A} = k\left[\frac{1}{-k_d}\right]\left[e^{-k_d t}\right]_0^t$$

$$= -\frac{k}{k_d}\left[e^{-k_d t} - 1\right]$$

$$\therefore \quad -\ln(1 - X_A) = -\frac{k}{k_d}\left[e^{-k_d t} - 1\right]$$

$$\ln(1 - X_A) = \frac{k}{k_d}\left[e^{-k_d t} - 1\right]$$

$$1 - X_A = e^{k/k_d\,(e^{-k_d t} - 1)}$$

$$\boxed{X_A = 1 - e^{k/k_d\,(e^{-k_d t} - 1)}} \qquad \text{where } k = k'W/V \qquad \ldots (6)$$

Equation (6) is the desired relationship (conversion as a function of time) for first order reaction and first-order deactivation in a batch reactor.

Example 8.5 : *A particular irreversible catalytic reaction A → R proceeds with decaying catalyst in a batch reactor (batch-solids, batch-fluid). Find the kinetics of reaction and deactivation from the following data.*

C_A	1.0	0.802	0.675	0.532	0.422	0.368
t, h	0	0.25	0.5	1.0	2	∞

Solution : Here we have to determine the order of reaction and order of deactivation by making use of the data provided [this data are collected in the batch solids, batch-fluid (gas) reactor].

As a first guess, try first-order rate expressions, i.e., first-order reaction and first-order deactivation.

i.e., $$-r'_A = k'C_A\,a \qquad \text{with } \varepsilon_A = 0$$

and $$-\frac{da}{dt} = k_d\,a$$

For batch-solids, batch-fluid reactor, we have

$$\ln \ln \frac{C_A}{C_{A\infty}} = \ln \frac{Wk'}{Vk_d} - k_d\, t$$

For our guess to be correct, the data should fit the above expression (i.e., a plot of ln $\ln C_A/C_{A\infty}$ v/s t should yield a straight line with intercept on y-axis).

$$y = \ln \ln (C_A/C_{A\infty}) \quad \text{and } x = t$$

$C_{A\infty}$ is the fluid (gas) concentration at $t = \infty$.

$$C_A = C_{A\infty} = 0.368 \text{ at } t = \infty.$$

t, h	C_A	$C_A/C_{A\infty}$	$\ln C_A/C_{A\infty}$	$\ln (\ln C_A/C_{A\infty})$
0	1	2.717	1	0
0.25	0.802	2.18	0.7793	– 0.25
0.50	0.675	1.834	0.6065	– 0.50
1	0.532	1.446	0.3688	– 0.9975 ≈ – 1.0
2	0.422	1.147	0.1371	– 1.987 ≈ – 1.99
∞	0.368 ($C_{A\infty}$)	1	0	– ∞

Refer Fig. Ex 8.5 :

Since all of the data are falling on a straight line, our guess regarding the kinetic forms is correct.

So we have (i) first-order reaction and (ii) first-order deactivation.

$$-r_A' = k'C_A\, a$$

and

$$-\frac{da}{dt} = k_d\, a$$

$$\text{Slope} = -1 = -k_d \qquad \therefore \quad k_d = 1$$

$$\text{Intercept} = 0 = \ln Wk'/Vk_d$$

as $k_d = 1$, $Wk'/V = 1$ for intercept to be zero (as $\ln 1 = 0$ and $\ln Wk'/Vk_d = \ln 1$ when $Wk'/V = 1$)

$$\therefore \quad \frac{k'W}{V} = 1$$

$$\therefore \quad k' = \frac{V}{W}$$

Substitute $k' = V/W$ and $k_d = 1$ in the rate forms.

$\therefore$ The final rate forms are :

$$\boxed{-r_A' = \frac{V}{W} C_A\, a}\ , \text{ mol/(kg cat·h) … for reaction} \qquad \textbf{... Ans.}$$

and

$$\boxed{-\frac{da}{dt} = a,\ h^{-1}}\ \text{… for deactivation} \qquad \textbf{... Ans.}$$

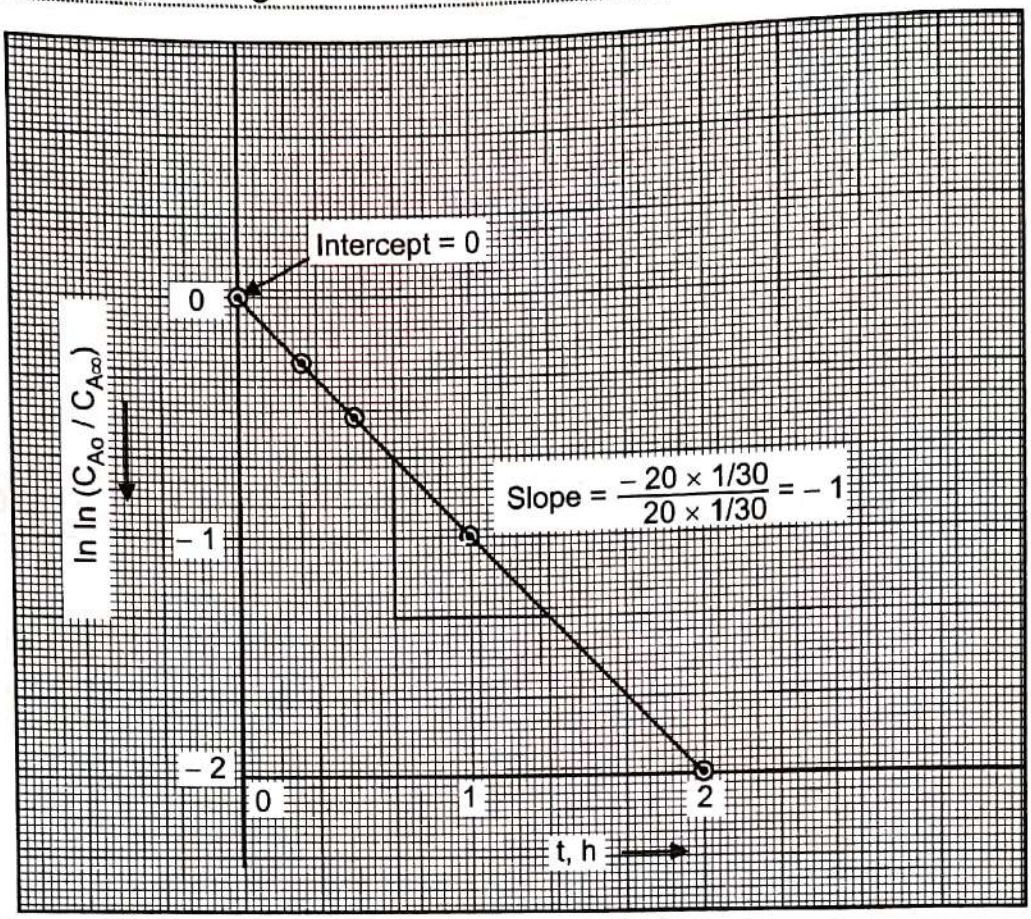


Fig. Ex 8.5

Example 8.6 : *The following data are obtained on a reversible reaction [A* $\rightleftharpoons$ *R,* X_{Ae} *= 0.50] with decaying catalyst in a batch reactor (batch-solids, batch-fluid). What can you say about the kinetics of reaction and deactivation ?*

t, h	*0*	*0.25*	*0.5*	*1*	*2*	*∞*
C_A, *mol/l*	*1.0*	*0.901*	*0.830*	*0.766*	*0.711*	*0.684*

Solution : A $\rightleftharpoons$ R with $X_{Ae} = 0.5$

Here we have to make use of the data to find (i) the order of reaction and (ii) the order of deactivation, i.e., we have to find the rate expressions for both reaction and deactivation.

As a first guess try first order rate expressions.

$$-r'_A = k'(C_A - C_{Ae})\,a \quad \text{... first order reversible reaction}$$

$$-\frac{da}{dt} = k_d\,a \quad \text{... first order deactivation}$$

The data is obtained in a batch-solids, batch-fluid reactor.

For batch solids, batch fluid - for first order reversible reaction and first order deactivation, we have :

$$\ln \ln \frac{C_A - C_{Ae}}{C_{A\infty} - C_{Ae}} = \ln \frac{Wk'}{Vk_d} - k_d t$$

[For irreversible reaction $C_{Ae} = 0$]

For our guess to be correct, plot of ln ln $[C_A - C_{Ae}/C_{A\infty} - C_{Ae}]$ v/s t should give a straight line.

$C_{A\infty}$ is the fluid concentration at $t = \infty$

$C_{A\infty}$ = 0.684 mol/*l* (from data)

Given : $X_{Ae} = 0.50$

$$C_{Ae} = C_{A0}(1 - X_{Ae})$$

C_{A0} is the fluid concentration at $t = 0$

C_{A0} = 1 mol/*l* (from data)

C_{Ae} = 1 (1 – 0.5) = 0.5 mol/*l*

t	C_A	$C_A - C_{Ae}$	$C_{A\infty} - C_{Ae}$	$\frac{C_A - C_{Ae}}{C_{A\infty} - C_{Ae}}$	$\ln\left(\frac{C_A - C_{Ae}}{C_{A\infty} - C_{Ae}}\right)$	$\ln \ln\left(\frac{C_A - C_{Ae}}{C_{A\infty} - C_{Ae}}\right)$
0	1.0	0.5		2.72	1.0	0
0.25	0.901	0.401		2.18	0.7793	– 0.25 (– 0.2493)
0.50	0.830	0.330		1.793	0.584	– 0.54 (– 0.5378)
1	0.766	0.266		1.446	0.369	– 1.0 (– 0.9969)
2	0.711	0.211		1.147	0.1371	– 1.99 (– 1.987)
∞	0.684	0.184	0.184	1	0	– ∞
$C_{Ae} = 0.5$, $C_{A\infty} = 0.684$						

Refer Fig. Ex 8.6 :

We obtained a straight line passing through all the data points. Hence, we conclude that the guessed rate expressions (first order reaction and first order deactivation) satisfactorily fit the data. (i.e., our guess regarding the reaction order and order of deactivation is correct).

$$-r'_A = k'(C_A - C_{Ae})\,a$$

$$-\frac{da}{dt} = k_d\,a$$

From Fig. Ex 8.6,

$$\text{slope} = -k_d = -1 \qquad \therefore \; k_d = 1$$

$$\text{intercept} = 0 = \ln \frac{Wk'}{Vk_d}$$

As $k_d = 1$, Wk'/V (i.e. Wk'/V k_d with $k_d = 1$) has to be equal to one for ln Wk'/Vk_d to be zero.

$$\therefore \quad \frac{k'W}{V} = 1 \qquad \therefore \quad k' = \frac{V}{W}$$

So rate of reaction is given by or the rate expression for the reaction is

$$\boxed{-r_A' = \frac{V}{W}(C_A - C_{Ae})\,a}, \quad \frac{\text{mol}}{\text{g cat}\cdot\text{h}} \qquad \text{... Ans.}$$

The rate of deactivation is given by or the rate form for deactivation is

$$\boxed{-\frac{da}{dt} = a,\ h^{-1}} \qquad \text{... Ans.}$$

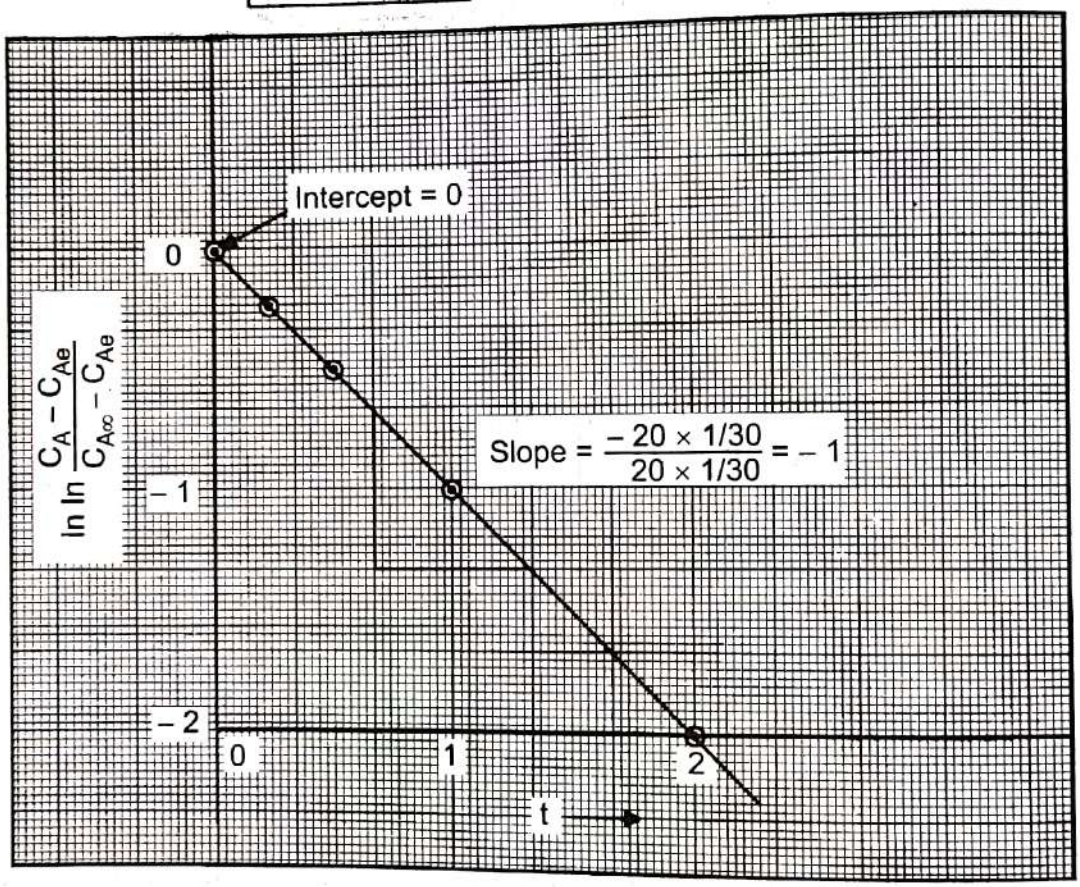


Fig. Ex 8.6

Example 8.7 : *The isomerisation reaction A → R proceeds at 730 K on a slowly deactivating catalyst by a second order rate*

$$-r_A' = k'C_A^2\,a = \left(0.2\frac{m^6}{mol\cdot kg\ cat\cdot h}\right)\left(C_A\frac{mol}{m^3}\right)^2 a, \quad \frac{mol}{(kg\ cat\cdot h)}$$

The rate of deactivation is

$$-\frac{da}{dt} = k_d(C_A + C_R)\,a = \left(10^{-3}\frac{m^3}{mol\cdot day}\right)\left(C_A + C_R\frac{mol}{m^3}\right) a, \quad day^{-1}$$

(This presents poisoning / deactivation by both A and R)

We plan to run the isomerisation reaction in a packed bed reactor (W = 1000 kg) for 120 days using a pure A feed, F_{A0} = 5 kmol/h at 730 K and 3 atm.

Find : (i) the conversion at the start of the run,

(ii) the conversion at the end of the run,

(iii) the average conversion for the 120-days run.

Solution : $A \rightarrow R$

$$-r'_A = 0.2\, C_A^2\, \mathbf{a},\ \text{mol/(kg cat·h)}$$

and $$-\frac{d\mathbf{a}}{dt} = 10^{-3}\,(C_A + C_R)\,\mathbf{a},\ \text{day}^{-1}$$

We have, $C_A + C_R = C_{A0}$ ... (always)

Calculate C_{A0} ... pure A at 730 K and 3 atm.

$$C_{A0} = \frac{p_{A0}}{RT}$$

where $p_{A0} = 3$ atm ... as pure A feed

$T = 730$ K

$R = 0.08206\ l\cdot\text{atm/(mol·K)}$

$= 0.08206 \times 10^{-3}\ \text{m}^3\cdot\text{atm/(mol·K)}$

$$\therefore \quad C_{A0} = \frac{3}{0.08206 \times 10^{-3} \times 730} = 50.08 \approx 50\ \text{mol/m}^3$$

$$-\frac{d\mathbf{a}}{dt} = \left(10^{-3}\,\frac{\text{m}^3}{\text{mol·day}}\right)\left(C_A + C_R\,\frac{\text{mol}}{\text{m}^3}\right)\mathbf{a},\ \text{day}^{-1}$$

$$= 10^{-3} \times 50\,\mathbf{a} = 0.05\,\mathbf{a},\ \text{day}^{-1}$$

$$-\int_{\mathbf{a}}^{1} \frac{d\mathbf{a}}{\mathbf{a}} = 0.05 \int_0^t dt$$

$$-\ln \mathbf{a} = 0.05\, t$$

$$\mathbf{a} = e^{-0.05\,t}$$

$\therefore$ At any time t, the activity is given by

$$\mathbf{a} = e^{-0.05\,t}$$

The activity of the catalyst at the start of run (i.e., at $t = 0$) is

$$\mathbf{a} = e^{-0.05\,t} = e^{-0.05 \times 0} = 1$$

The activity of the catalyst after 120 days is

$$\mathbf{a} = e^{-0.05\,t} = e^{-0.05 \times 120}$$
$$= e^{-6} = 2.478 \times 10^{-3} \approx 2.5 \times 10^{-3}$$

Now consider the isomerisation in the packed bed reactor. The weight-time is given by

$$\tau' = \frac{WC_{A0}}{F_{A0}}$$

where, $W = 1000$ kg

$C_{A0} = 50$ mol/m^3

$F_{A0} = 5$ kmol/h $= 5 \times 10^3$ mol/h

$$\tau' = \frac{1000 \times 50}{5 \times 10^3} = 10 \ \frac{\text{kg cat} \cdot \text{h}}{\text{m}^3 \text{ fluid}}$$

The performance equation for a packed bed reactor assuming plug flow with pseudo steady state assumption is

$$\tau' = C_{A0} \int_0^{X_A} \frac{dX_A}{-r_A'}$$

$$-r_A' = 0.2\, C_A^2\, \mathbf{a} = 0.2\, \mathbf{a}\, C_{A0}^2 (1 - X_A)^2$$

[For $A \rightarrow R$, $\varepsilon_A = 0$ $\therefore$ $C_A = C_{A0}(1 - X_A)$]

$$\tau' = C_{A0} \int_0^{X_A} \frac{dX_A}{0.2\, \mathbf{a}\, C_{A0}^2 (1 - X_A)^2}$$

$$0.2\, \mathbf{a}\, C_{A0}\, \tau' = \int_0^{X_A} \frac{dX_A}{(1 - X_A)^2}$$

$$0.2\, \mathbf{a}\, C_{A0}\, \tau' = \frac{X_A}{1 - X_A}$$

$$\frac{X_A}{1 - X_A} = 0.2\, \mathbf{a}\, C_{A0}\, \tau'$$

(i) To find the conversion at the start of run :

$$\frac{X_A}{1 - X_A} = 0.2\, \mathbf{a}\, C_{A0}\, \tau'$$

where, $C_{A0} = 50$ mol/m^3

$\tau' = 10$ (kg cat · h)/m^3

and $\mathbf{a} = 1$ at $t = 0$

$$\frac{X_A}{1 - X_A} = 0.2 \times 1 \times 50 \times 10 = 100$$

$\therefore$ $X_A = 0.99$

The conversion at the start of the run ($X_{A,\,start}$) is 99 % ... Ans. (i)

(ii) To find the conversion at the end of run, i.e., at the end of 120 days run :

$$\frac{X_A}{1 - X_A} = 0.2\, \mathbf{a}\, C_{A0}\, \tau'$$

where, $\mathbf{a} = 2.5 \times 10^{-3}$ after 120 days

$$\frac{X_A}{1 - X_A} = 0.2 \times 2.5 \times 10^{-3} \times 50 \times 10$$

$\therefore$ $X_A = 0.20$

The conversion at the end of the run, $X_{(A, end)}$ is 20 % ... Ans. (ii)

(iii) To find the average conversion over the 120 days run :

The average conversion over the run is given by

$$\bar{X}_A = \frac{\int_0^{t_{run}} X_A(t)\, dt}{t_{run}}$$

We have, $\dfrac{X_A}{1 - X_A} = 0.2\, \mathbf{a}\, C_{A0}\, \tau'$

and $\mathbf{a} = e^{-0.05\, t}$

$\therefore$ $\dfrac{X_A(t)}{1 - X_A(t)} = 0.2\, e^{-0.05\, t}\, C_{A0}\, \tau'$

$$= 0.2 \times 50 \times 10 \times e^{-0.05\, t} = 100\, e^{-0.05\, t}$$

$$\frac{1}{100\, e^{-0.05\, t}} = \frac{1 - X_A(t)}{X_A(t)} = \frac{1}{X_A(t)} - 1$$

$$\frac{1}{X_A(t)} = \frac{1}{100\, e^{-0.05\, t}} + 1 = \frac{1 + 100\, e^{-0.05\, t}}{100\, e^{-0.05\, t}}$$

$$X_A(t) = \frac{100\, e^{-0.05\, t}}{1 + 100\, e^{-0.05\, t}}$$

The conversion as a function of time is given by the above expression.

$t_{run} = 120$ days

$$\bar{X}_A = \frac{\int_0^{120} [100\, e^{-0.05\, t} / (1 + 100\, e^{-0.05\, t})]\, dt}{120}$$

$$= \frac{1}{120} \int_0^{120} \left[\frac{100\, e^{-0.05\, t}}{1 + 100\, e^{-0.05\, t}}\right] dt$$

Let $1 + 100\, e^{-0.05\, t} = y$

$\therefore$ $(-0.05) \times 100\, e^{-0.05\, t}\, dt = dy$

$\therefore$ $100\, e^{-0.05\, t}\, dt = -dy/0.05$

At t = 0, $y = 101$

At t = 120, $y = 1.25$

$$\therefore \quad \bar{X}_A = \frac{-1}{120 \times 0.05} \int_{101}^{1.25} \frac{dy}{y}$$

$$= \frac{-1}{120 \times 0.05} [\ln y]_{101}^{1.25}$$

$$= \frac{-1}{120 \times 0.05} [\ln 1.25 - \ln 101]$$

$$= \frac{-1}{120 \times 0.05} [-4.392]$$

$$= 0.732$$

The average conversion over the 120 days run ($\bar{X}_A$) is 0.732, i.e., 73.2 % ... **Ans. (iii)**

Extra : The conversion as a function of time is given by

$$X_A(t) = \frac{100\, e^{-0.05\, t}}{1 + 100\, e^{-0.05\, t}}$$

Evaluate X_A at various values of t and tabulate.

t	0	20	40	60	80	100	120
X_A	0.99	0.97	0.93	0.83	0.65	0.40	0.20

The average conversion over the 120 days run is given by

$$\bar{X}_A = \frac{\int_0^t X_A(t)\, dt}{t_{run}} \quad , t_{run} = 120 \text{ days}$$

$$= \frac{1}{120} \int_0^{120} X_A(t)\, dt$$

Now we will use Simpson's one-third rule for the numerical evaluation of the integral on RHS of the above equation.

$$\int_{x_0}^{x_2} f(x)\, dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] \quad , \; h = x_1 - x_0$$

In our case : h = 20 – 0 or 40 – 20 = 20

$$\bar{X}_A = \frac{1}{120} \times \frac{20}{3} [\underline{0.99 + 4 \times 0.97 + 0.93} + \underline{0.93 + 4 \times 0.83 + 0.65} + \underline{0.65 + 4 \times 0.40 + 0.20}]$$

= 0.7305 [approximately equal to that previously obtained]

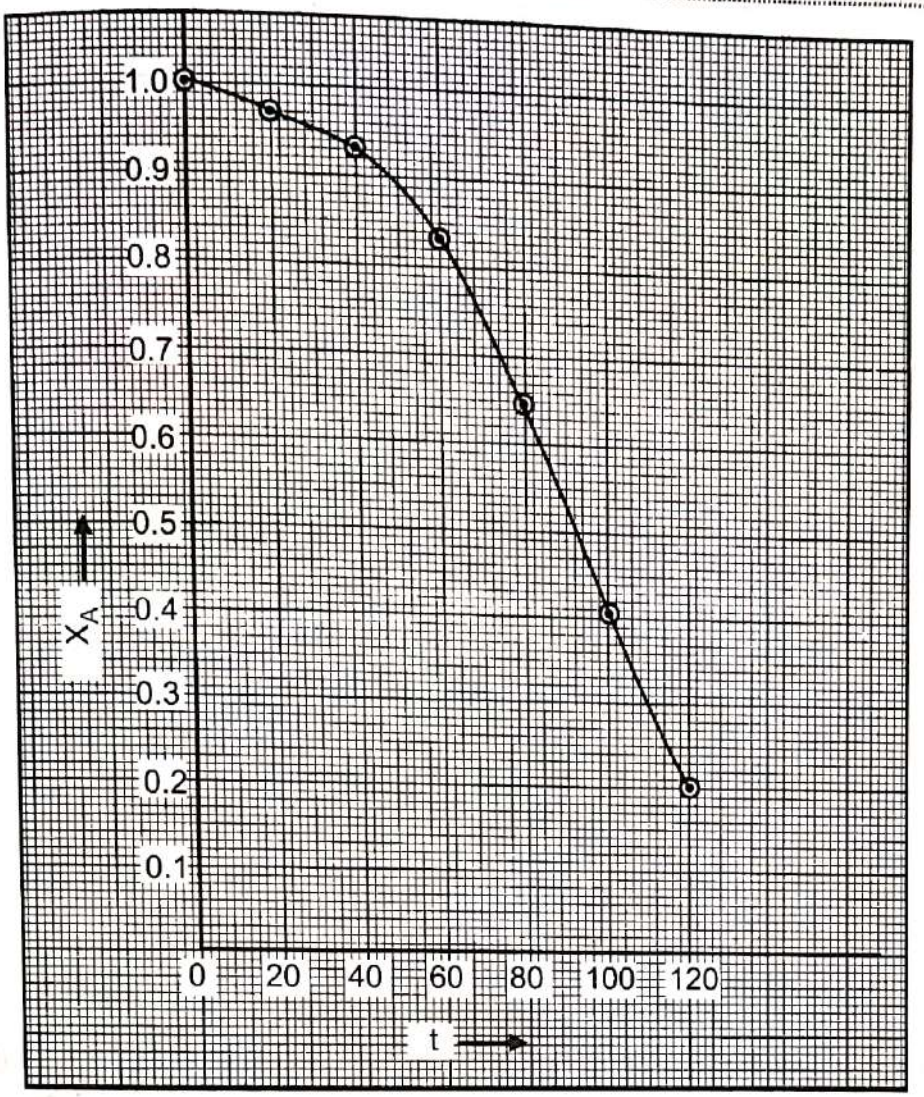


Fig. Ex 8.7

Example 8.8 : *The isomerisation reaction A → R proceeds at 730 K on a slowly deactivating catalyst by a second order rate*

$$-r_A' = k'C_A^2\ a = \left(0.2\frac{m^6}{mol\cdot kg\ (cat\cdot h)}\right)\left(C_A, \frac{mol}{m^3}\right) a, \frac{mol}{(kg\ cat\cdot h)}$$

The rate of deactivation is

$$-\frac{da}{dt} = 3.325\ a^2, day^{-1}\ (indicating\ reasonably\ strong\ pore\ diffusion\ resistance)$$

We plan to run the isomerisation reaction in a packed bed reactor (W = 1000 kg) for 120 days using a pure A feed, F_{A0} = 5 kmol/h at 730 K and 3 atm.

Find the average conversion for 120 days run and construct a plot of X_A v/s t.

Solution : A → R.

$$-r_A' = 0.2\ C_A^2\ a\ ,\ mol/(kg\ cat\cdot h)$$

$$-\frac{da}{dt} = 3.325\ a^2$$

Calculation of C_{A0} :

$$C_{A0} = \frac{p_{A0}}{RT}$$

where $p_{A0} = P = 3$ atm ... since pure A feed

$T = 730$ K

$R = 0.08206 \times 10^{-3}$ m^3·atm/(mol·K)

$$C_{A0} = \frac{3}{0.08206 \times 10^{-3} \times 730} = 50 \text{ mol/m}^3$$

Activity as a function of time :

$$-\frac{da}{dt} = 3.325\, a^2$$

$$-\int_1^a \frac{da}{a^2} = 3.325 \int_0^t dt$$

$$\frac{1}{a} - 1 = 3.325\, t$$

$$a = \frac{1}{1 + 3.325\, t} \quad \text{... (1)}$$

Calculation of τ' and obtaining an expression for the conversion, X_A, as a function of time (t) :

Now consider the isomerisation of A to R in the packed bed reactor. The weight time is given by

$$\tau' = \frac{WC_{A0}}{F_{A0}}$$

where, $W = 1000$ kg

$C_{A0} = 50$ mol/m^3

$F_{A0} = 5$ kmol/h $= 5 \times 10^3$ mol/h

$$\tau' = \frac{1000 \times 50}{5 \times 10^3} = 10 \ \frac{\text{kg cat·h}}{\text{m}^3 \text{ fluid}}$$

The performance equation for a packed bed reactor assuming plug flow with pseudo steady-state assumption is

$$\tau' = C_{A0} \int_0^{X_A} \frac{dX_A}{-r_A'}$$

$$-r_A' = 0.2\, C_A^2\, a = 0.2\, a\, C_{A0}^2 (1 - X_A)^2$$

[For $A \rightarrow R$, $\varepsilon_A = 0$ $\therefore$ $C_A = C_{A0}(1 - X_A)$]

$$\tau' = C_{A0} \int_0^{X_A} \frac{dX_A}{0.2\, a\, C_{A0}^2 (1 - X_A)^2}$$

Integrating and rearranging, we get

$$\frac{X_A}{1 - X_A} = 0.2\, a\, C_{A0}\, \tau' \qquad \ldots (2)$$

Replace **'a'** in Equation (2) by its value from Equation (1). Replace X_A by X_A (t) in Equation (2).

$$\frac{X_A(t)}{1 - X_A(t)} = \frac{0.2\, C_{A0}\, \tau'}{1 + 3.325\, t} = \frac{0.2 \times 50 \times 10}{1 + 3.325\, t} = \frac{100}{1 + 3.325\, t}$$

$$\frac{1 - X_A(t)}{X_A(t)} = \frac{1 + 3.325\, t}{100}$$

$$\frac{1}{X_A(t)} - 1 = \frac{1 + 3.325\, t}{100}$$

$$\frac{1}{X_A(t)} = \frac{1 + 3.325\, t}{100} + 1 = \frac{101 + 3.325\, t}{100}$$

$$X_A(t) = \frac{100}{101 + 3.325\, t} \qquad \ldots (3)$$

Equation (3) is the desired relation for conversion as a function of time.

To find the average conversion over 120 days run :

The average conversion over the run is given by

$$\bar{X}_A = \frac{\int_0^t X_A(t)\, dt}{t_{run}}$$

$$= \frac{\int_0^t [100/(101 + 3.325\, t)]dt}{120} \qquad \text{where } t_{run} = 120 \text{ days}$$

$$= \frac{1}{120} \int_0^{120} \left[\frac{100}{101 + 3.325\, t}\right] dt = \frac{100}{120} \int_0^{120} \frac{dt}{(101 + 3.325\, t)}$$

$$= \frac{100}{120 \times 3.325} \int_{101}^{500} \frac{dy}{y} = \frac{100}{120 \times 3.325} [\ln y]_{101}^{500}$$

$\therefore$ $$\bar{X}_A = \frac{100}{120 \times 3.325} [\ln 500 - \ln 101]$$

$$= 0.40$$

[$y = 101 + 3.325\, t$, $dt = dy/3.325$

At $t = 0$, $y = 101$

At $t = 120$, $y = 500$]

The average conversion over 120 days run ($\bar{X}_A$) is 0.40 or 40 % ... **Ans.**

Now let us construct a plot of X_A v/st.

We have : $$X_A(t) = \frac{100}{101 + 3.325\,t}$$

Evaluate X_A (t) at various values of t (from 0 to 120), tabulate them and construct the desired plot.

At t = 0, $$X_A(t) = \frac{100}{101 + 3.325 \times 0} = 0.99$$

t	0	20	40	60	80	100	120
X_A	0.99	0.60	0.43	0.33	0.27	0.23	0.20

From the above table :

Conversion at the start of the run (i.e., at t = 0) = $X_{A,\,start} = 0.99$

Conversion at the end of the run (i.e., at t = 120 days) = $X_{A,\,end} = 0.20$

$$\bar{X}_A = \frac{\int_0^t X_A(t)\,dt}{t_{run}}$$

$$= \frac{1}{120}\int_0^{120} X_A(t)\,dt$$

Here we will adopt the method of numerical evaluation of integrals - Simpson's one-third rule.

$$\int_{x_0}^{x_2} f(x)\,dx = \frac{h}{3}[f(x_0) + 4f(x_1) + f(x_2)] \quad , \; h = x_1 - x_0$$

In our case, h = 20 – 0 or 40 – 20 = 20

$$\bar{X}_A = \frac{1}{120} \times \frac{20}{3}\,[0.99 + 4 \times 0.60 + 0.43 + 0.43 + 4 \times 0.33 + 0.27 + 0.27 + 4 \times 0.23 + 0.20]$$

$$= 0.4017$$

The average conversion over 120 days run by numerical integration = 0.4017.

This answer is approximately same as that previously obtained, so you can adopt this procedure also to evaluate $\bar{X}_A$.

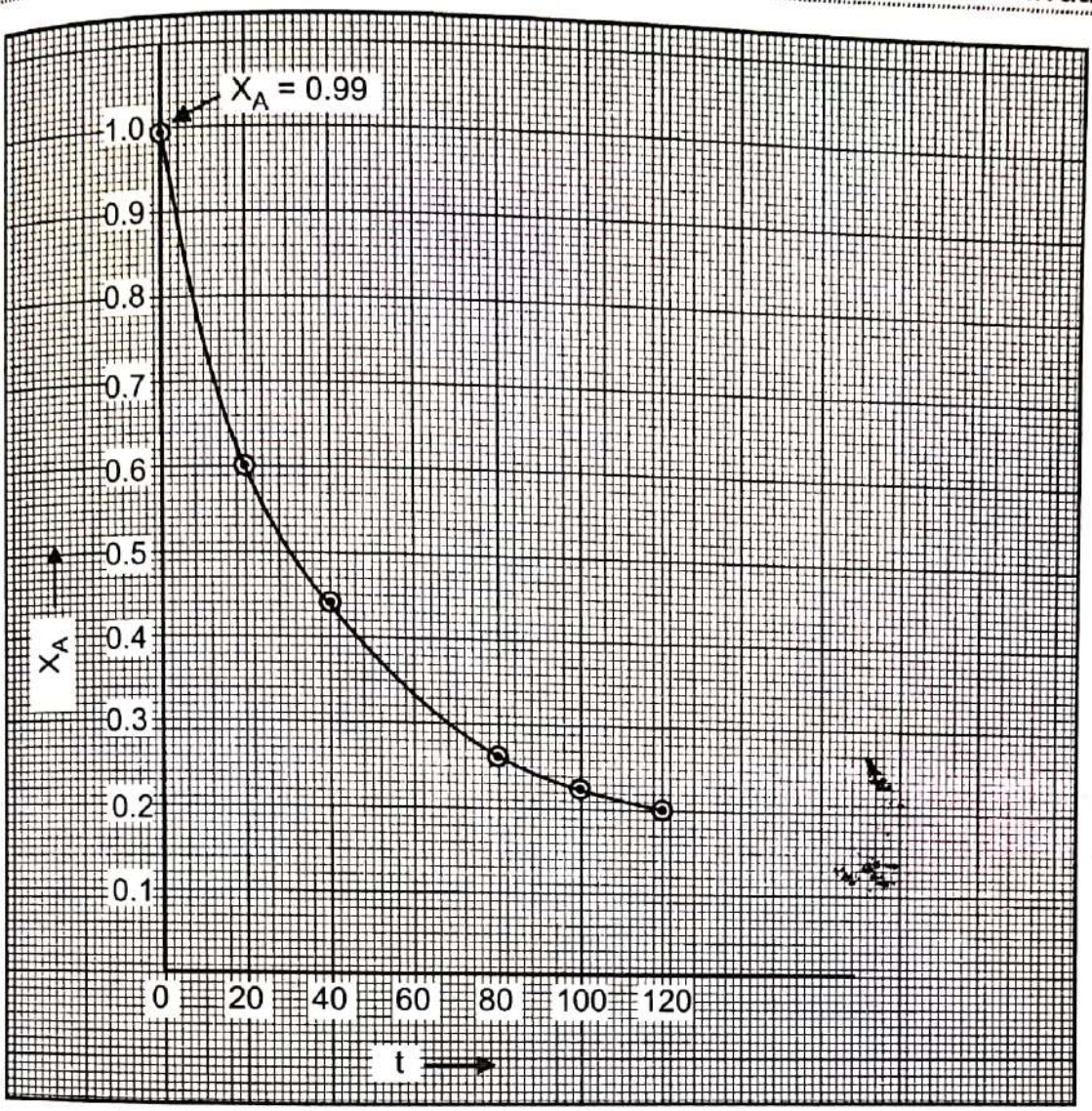


Fig. Ex 8.8

Example 8.9 : *The reaction A → R proceeds at 700 °C on a slowly deactivating catalyst by a first order rate*

$$-r_A' = k'C_A a = 0.03\, C_A a \text{ , mol/(g cat·min), (k' is in l/(g cat·min))}$$

The rate of deactivation (by impurities in the feed) is given by

$$-\frac{da}{dt} = 3a^3 \text{ , day}^{-1}$$

We plan to operate a packed bed reactor (W = 10 kg) with pure A feed, v_o = 100 l/min, at 8 atm and 700 °C until the catalyst activity drops to 10 % of the fresh catalyst, then regenerate it and repeat the cycle.

Find (i) the run time for this operation (i.e., to drop the activity to 10% of the fresh catalyst) and (ii) the mean conversion for the run.

Solution : $A \rightarrow R$, $\varepsilon_A = 0$

$$-r_A' = 0.03\, C_A a, \text{ mol/(g cat·min)}$$

$$-\frac{da}{dt} = 3a^3 \text{ , day}^{-1} \text{ ... so time is in days}$$

$$-\frac{da}{a^3} = 3\,dt$$

$$-\int_1^a \frac{da}{a^3} = 3\int_0^t dt$$

$$-\left[\frac{a^{-3+1}}{-3+1}\right]_1^a = 3t$$

$$\frac{1}{2}\left[\frac{1}{a^2} - 1\right] = 3t$$

$$\frac{1}{a^2} = 1 + 6t$$

$$a = \frac{1}{\sqrt{1+6t}}$$

Let us find the run time when the catalyst activity drops to 10 % of the fresh catalyst :

For fresh catalyst : $a = 1$

Find t for $a = 0.10 \times 1 = 0.1$

$$a = \frac{1}{\sqrt{1+6t}} \quad \text{... (1)}$$

$$0.10 = \frac{1}{\sqrt{1+6t}}$$

$$\sqrt{1+6t} = \frac{1}{0.10} = 10$$

$$1 + 6t = (10)^2$$

$$\therefore \quad t = 16.5 \text{ days}$$

Time required to drop the activity to 10 % that of the fresh catalyst (i.e., the run time for this operation) = 16.5 days ... Ans. (i)

We have : $\tau' = \dfrac{W C_{A0}}{F_{A0}} = \dfrac{W}{\nu_o}$

where $W = 10 \text{ kg} = 10^4 \text{ g}$

$\nu_o = 100\ l/\text{min}$

$$\tau' = \frac{10^4}{100} = 100 \ \frac{\text{g cat} \cdot \text{min}}{l}$$

For a packed bed reactor with pseudo steady-state assumption, the performance equation is

$$\tau' = C_{A0}\int_0^{X_A} \frac{dX_A}{-r_A'}$$

$$-r_A' = 0.03\, C_A\, a = 0.03\, a\, C_{A0}\,(1 - X_A)$$

[For $A \rightarrow R$, $\varepsilon_A = 0$ and $C_A = C_{A0}(1 - X_A)$]

$$\tau' = C_{A0} \int_0^{X_A} \frac{dX_A}{0.03\, a\, C_{A0}(1 - X_A)}$$

$$0.03\, a\, \tau' = \int_0^{X_A} \frac{dX_A}{(1 - X_A)}$$

$$-\ln(1 - X_A) = 0.03\, a\, \tau' = 0.03 \times 100 \times a = 3a$$

$$1 - X_A = e^{-3a}$$

$$X_A = 1 - e^{-3a} \quad \ldots (2)$$

Combining Equations (1) and (2), we get

$$X_A = 1 - e^{-3/\sqrt{1+6t}}$$

i.e., $$X_A(t) = 1 - e^{-3/\sqrt{1+6t}} \quad \ldots (3)$$

This is the required relationship for the conversion as a function of time.

Evaluate X_A at various times using Equation (3) and tabulate, t intervals are taken to suit a Simpson's one-third rule.

t	0	2	4	6	8	12.25	16.5
X_A	0.95	0.565	0.451	0.39	0.35	0.293	0.26

Conversion at the start of the run = 0.95 or 95 %.

Conversion at the end of the run = 0.26 or 26 %.

The mean conversion for the run is given by

$$\bar{X}_A = \frac{\int_0^{X_A} X_A(t)\, dt}{t_{run}}$$

$$t_{run} = 16.5 \text{ days}$$

$$\bar{X}_A = \frac{1}{16.5} \int_0^{16.5} X_A(t)\, dt$$

Use Simpson's one-third rule for the evaluation of the integral on the RHS of the above equation

$$\bar{X}_A = \frac{1}{16.5}\left[\int_0^{8} X_A(t)\, dt + \int_8^{16.5} X_A(t)\, dt\right]$$

For 0 to 8 ..... $h = 2 - 0 = 2$

For 8 to 16.5 ..... $h = 12.25 - 8 = 4.25$

$$\bar{X}_A = \frac{1}{16.5}\left[\frac{2}{3}(0.95 + 4 \times 0.565 + 0.451 + 0.45 + 4 \times 0.39 + 0.35) + \frac{4.25}{3}(0.35 + 4 \times 0.293 + 0.26)\right]$$

$$\bar{X}_A = \frac{1}{16.5}\,[6.5378] = 0.3962 \approx 0.40$$

The mean conversion for the run ($\bar{X}_A$) is 0.40 or 40 %. ... Ans.

Example 8.10 : *The kinetics of an irreversible catalytic reaction, $A \rightarrow R$, are studied in a basket reactor (batch-solids and mixed flow of gas) at temperature T. During the kinetic runs, the gas composition in the reactor is kept unchanged by lowering the flow rate of reactant (to about 5 % of the initial value). Find the kinetics of reaction and deactivation, i.e., find the rate equations/expressions for both the reaction and deactivation from the results of the following runs (The flow rate is lowered as the catalyst is deactivating) :*

Run 1 : $C_{A0} = 1$ *mol/l,* $X_A = 0.5$, $\pi = 1$ *atm.*

t (time from start of run), h	*0*	*1*	*2*	*3*
τ' *(weight time), (g cat·min)/l*	*1*	*e*	e^2	e^3

Run 2 : $C_{A0} = 2$ *mol/l,* $X_A = 0.5$, $\pi = 2$ *atm.*

t (time from start of run), h	*0*	*1*	*2*	*3*
τ' *(weight time), (g cat·min)/l*	*1*	*e*	e^2	e^3

Solution : $A \rightarrow R$, $\varepsilon_A = 0$

Guess-1 : Start by guessing the simplest of rate forms,

i.e., first order reaction,

and first order deactivation

Further assume that the deactivation is concentration independent.

Thus, with this we have :

Rate of reaction : $-r'_A = k'C_A\,\mathbf{a}$, mol/(g cat·min)

Rate of deactivation : $-\frac{da}{dt} = k_d\,\mathbf{a}$, h^{-1}

For mixed flow reactor, the performance equation is

$$\frac{W}{F_{A0}} = \frac{X_A}{-r'_A} = \frac{X_A}{k'C_A\,\mathbf{a}} = \frac{X_A}{k'C_{A0}(1 - X_A)\,\mathbf{a}}$$

$$\frac{WC_{A0}}{F_{A0}} = \frac{X_A}{k'(1-X_A)\,a}$$

$$\tau' = \frac{X_A}{k'(1-X_A)\,a} \quad , \text{ as } \tau' = WC_{Ao}/F_{Ao} \qquad \ldots (1)$$

$$-\frac{da}{dt} = k_d\, a$$

$$-\int_1^a \frac{da}{a} = k_d \int_0^t dt$$

$$a = e^{-k_d t} \qquad \ldots (2)$$

Combining Equations (1) and (2), we get

$$\tau' = \frac{X_A}{k'(1-X_A)\,e^{-k_d t}}$$

$$\tau' = \frac{X_A}{k'(1-X_A)}\, e^{k_d t}$$

$$\ln \tau' = \ln\left[\frac{X_A}{k'(1-X_A)}\right] + k_d\, t$$

$t - \tau'$ data is the same for both the runs.

Plot $\ln \tau'$ v/s t. It should result in a straight line for deactivation to be first order.

t	0	1	2	3
τ'	1	e	e^2	e^3
$\ln \tau'$	0	1	2	3

Refer Fig. Ex 8.10 :

As the plot of $\ln \tau'$ v/s t yields a straight line, the deactivation is first order (i.e., our guess regarding the order of deactivation is correct).

$\therefore$ slope = 1 and intercept = 0

For both runs : $X_A = 0.50$

$$\text{Slope} = 1 = k_d$$

$$\text{Intercept} = \ln\left[\frac{X_A}{k'(1-X_A)}\right] = 0$$

$$\ln\left[\frac{0.5}{k'(1-0.5)}\right] = 0$$

$$\ln\left[\frac{1}{k'}\right] = 0$$

For this to be valid, k' must be 1 as ln 1 = 0

$\therefore \quad k' = 1$

We have, $k' = 1$ and $k_d = 1$

So the rate expressions/kinetic expressions are :

Rate of reaction : $\boxed{-r_A' = C_A \, \mathbf{a}, \ \text{mol/(g cat·min)}}$ ... for reaction

Rate of deactivation : $\boxed{-\frac{da}{dt} = \mathbf{a}, \ h^{-1}}$ ... for deactivation ... Ans.

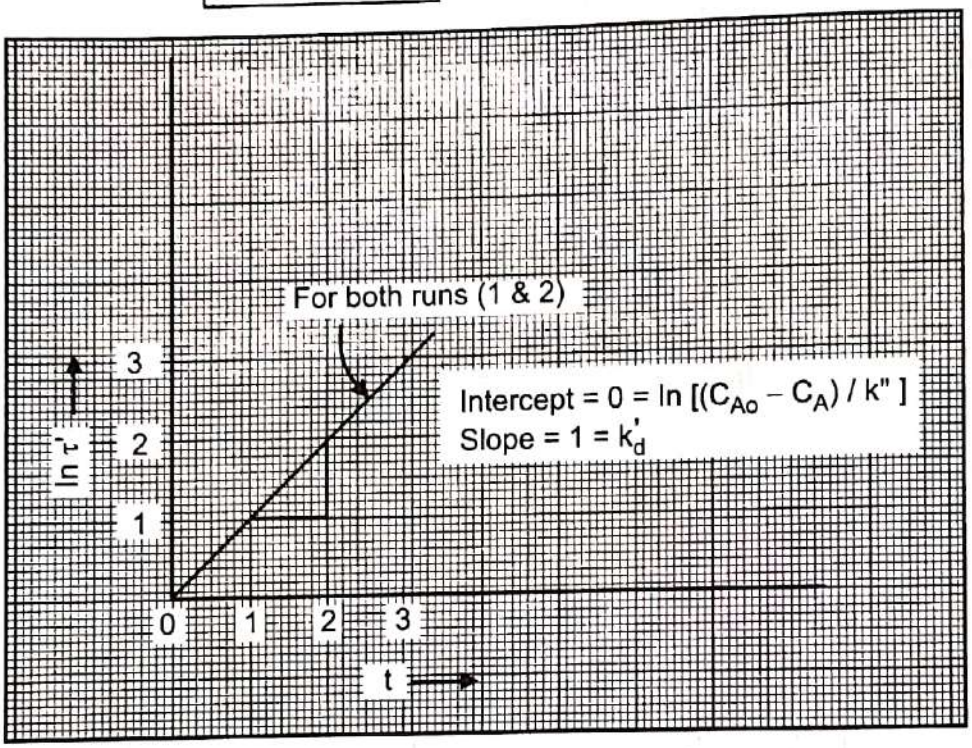


Fig. Ex 8.10

OR :

$A \rightarrow R, \ \varepsilon_A = 0$

General rate forms for constant C_A :

Rate of reaction : $-r_A' = k' C_A^n \, \mathbf{a} = k'' \, \mathbf{a}$ with $k'' = k' C_A^n$

Rate of deactivation : $-\frac{da}{dt} = k_d C_A^m \, \mathbf{a}^d = k_d' \, \mathbf{a}^d$ with $k_d' = k_d C_A^m$

Guess : First order deactivation : $-\frac{da}{dt} = k_d' \, \mathbf{a}$, (d = 1)

For (i) batch-solids and mixed changing flow of gas and

(ii) first order deactivation, we have :

$$\ln \tau' = \ln\left[\frac{C_{A0} - C_A}{k''}\right] + k_d' \, t$$

If the plot of ln τ' v/s t yields a straight line, then the guessed order of deactivation is correct.

$t - \tau'$ data for both the runs is the same, so the plot is the same for both the runs.

Refer Fig. Ex 8.10.

As the data fall on a straight line, the guessed order is correct.

Slope = 1.

Run 1: Slope $= 1 = k'_{d1} = k_d C_{A1}^m$ for $C_{A01} = 1$ mol/l and $X_{A1} = 0.50$

Run 2: Slope $= 1 = k'_{d2} = k_d C_{A2}^m$ for $C_{A02} = 2$ mol/l and $X_{A2} = 0.50$

$$1 = k_d C_{A1}^m$$

$$1 = k_d C_{A2}^m$$

$\therefore$ $$k_d C_{A1}^m = k_d C_{A2}^m$$

$$C_{A1}^m = C_{A2}^m$$

$$[C_{A01}(1 - X_{A1})]^m = [C_{A02}(1 - X_{A2})]^m$$

$$[1(1 - 0.5)]^m = [2(1 - 0.5)]^m$$

$$(0.5)^m = (1)^m$$

$$1 = (1/0.5)^m = (2)^m$$

$$\ln 1 = m \ln 2$$

$$m = \ln 1/\ln 2 = 0/0.693 = 0$$

or $$(0.5)^m = (1)^m$$

For this to be valid, m must be zero.

$\therefore$ $$m = 0$$

$$1 = k_d C_{A2}^m$$

$$1 = k_d [C_{A02}(1 - X_{A2})]^m$$

$$1 = k_d [2(1 - 0.5)]^0$$

$\therefore$ $$k_d = 1$$

$$\text{Intercept} = \ln\left[\frac{C_{A0} - C_A}{k''}\right] = 0 \text{ ... from Fig. Ex 8.10}$$

Run 1 : $$\ln\left[\frac{C_{A01} - C_{A1}}{k_1''}\right] = 0$$

Since ln 1 = 0 and the bracketed term to be one, we must have :

$$k_1'' = C_{A01} - C_{A1} = C_{A01} X_{A1}$$

$$k_1'' = 1 \times 0.5 = 0.5$$

Run 2 : $$\left[\frac{C_{A02} - C_{A2}}{k_2''}\right] = 0$$

As ln 1 = 0, we have :

$$k_2'' = C_{A02} - C_{A2} = C_{A02} X_{A2} = 2 \times 0.50 = 1$$

$$k_1'' = k' C_{A1}^n = 0.5$$

$$k_2'' = k' C_{A2}^n = 1$$

$$\frac{C_{A2}^n}{C_{A1}^n} = \frac{1}{0.5} = 2$$

$$\left[\frac{C_{A2}}{C_{A1}}\right]^n = 2$$

$$\left[\frac{C_{A02} X_{A2}}{C_{A01} X_{A1}}\right]^n = 2$$

$$\left[\frac{2 \times 0.5}{1 \times 0.5}\right]^n = 2$$

$$n \ln 2 = \ln 2$$

$$n = \ln 2 / \ln 2 = 0.693 / 0.693 = 1$$

$\therefore$ $$n = 1$$

$$k_1'' = k' C_{A1}^n = 0.5$$

$$k' (C_{A01} X_{A1})^n = 0.5$$

$$k' (1 \times 0.5)^1 = 0.5$$

$\therefore$ $$k' (0.5) = 0.5$$

$\therefore$ $$k' = 1$$

So we have :

$d = 1,\ n = 1,\ k' = 1,\ k_d = 1$ and $m = 0$

$$-r_A' = k' C_A^n\ a \longrightarrow \boxed{-r_A' = C_A\, a}\ ,\ \text{mol/(g cat·min)}$$

$$-\frac{da}{dt} = k_d C_A^m\ a^d \longrightarrow \boxed{-\frac{da}{dt} = a}\ ,\ h^{-1}$$

... Ans.

Example 8.11 : *The kinetics of an irreversible catalytic reaction, A → R, are studied in a basket reactor (batch solids and mixed flow of gas) at temperature T. During the kinetic runs, the gas composition in the reactor is kept unchanged by lowering the flow rate of reactant (to about 5 % of the initial value). What can you say about the kinetics of reaction and deactivation from the following kinetic runs ? To keep C_A constant the flow rate is reduced because the catalyst is deactivating.*

Run 1 : C_{A0} = 2 *mol/l,* X_A = 0.5

t (time from start of run), h	*0*	*1*	*2*	*3*
τ' (weight time), (g cat·min)/l	*1*	*e*	e^2	e^3

Run 2 : C_{A0} = 20 *mol/l,* X_A = 0.80

t (time from start of run), h	*0*	*0.5*	*1*	*1.5*
τ' (weight time), (g cat·min)/l	*1*	*e*	e^2	e^3

Solution : $A \rightarrow R, \quad \eta_A = 0$

For constant C_A, the general rate forms are :

For reaction : $$-r_A' = k' C_A^n \, a = k'' a \text{ with } k'' = k' C_A^n$$

For deactivation : $$-\frac{da}{dt} = k_d C_A^m \, a^d = k_d' \, a^d \text{ with } k_d' = k_d C_A^m$$

Guess-1 : As a first guess try a first-order deactivation, i.e., first order rate expression for deactivation (i.e., d = 1)

$$-\frac{da}{dt} = k_d' \, a, \; h^{-1}$$

For batch-solids and mixed changing flow of gas and for first order deactivation, we have

$$\ln \tau' = \ln\left[\frac{C_{A0} - C_A}{k''}\right] + k_d' \, t$$

For deactivation to be first order, a plot of ln τ' v/s t should be a straight line.

Run 1 : C_{A01} = 2 mol/*l* and X_{A1} = 0.5

t	0	1	2	3
τ'	1	e	e^2	e^3
ln τ'	0	1	2	3

Run 2 : C_{A02} = 20 mol/*l* and X_{A2} = 0.8

t	0	0.5	1.0	1.5
τ'	1	e	e^2	e^3
ln τ'	0	1	2	3

Plot ln τ' v/s t for both runs.

Refer Fig. Ex 8.11 (a) and (b).

The straight line fit of the data for both runs indicates that the deactivation is first order (i.e., our guess regarding the order of deactivation is correct).

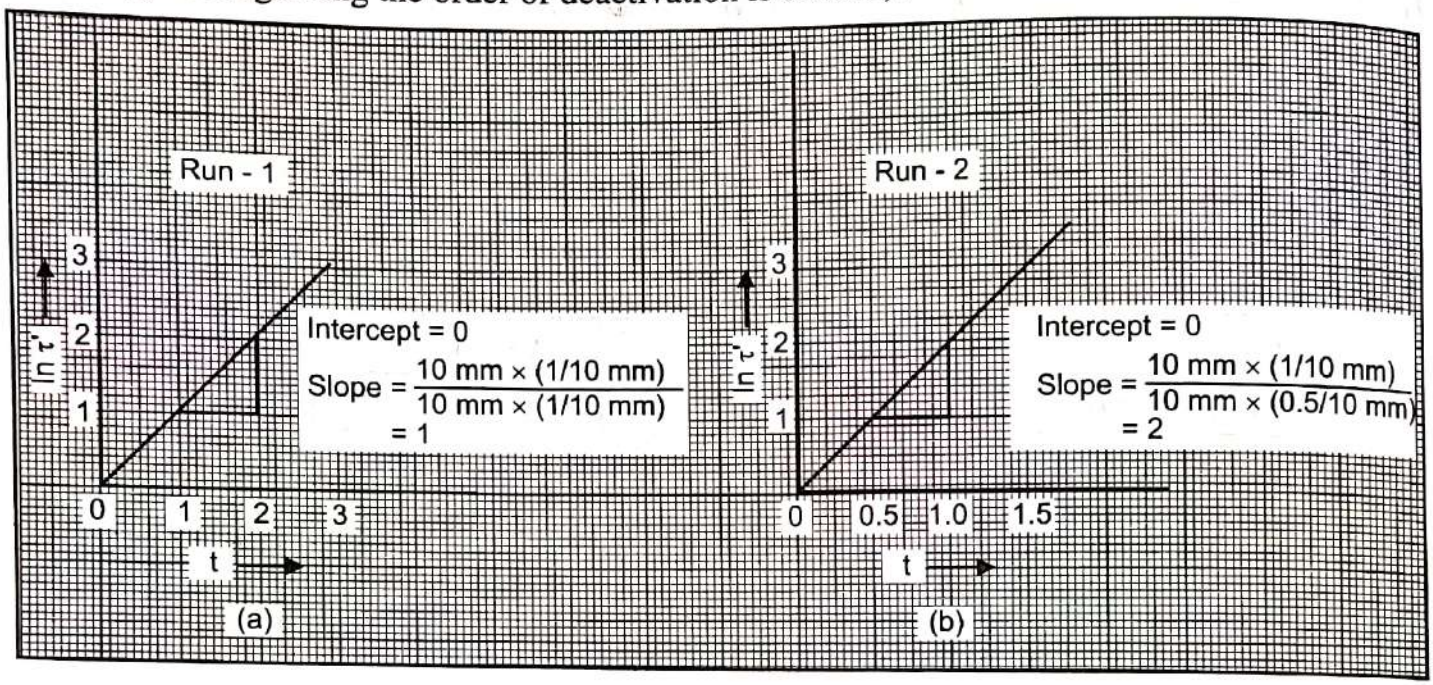


Fig. Ex 8.11 (a) and (b)

From Fig. Ex 8.11 (a), Slope = 1

$$1 = k'_{d1} = k_d\, C_{A1}^m$$

From Fig. Ex 8.11 (b), Slope = 2

$$2 = k'_{d2} = k_d\, C_{A2}^m$$

$$\therefore \quad \frac{1}{2} = \frac{k_d\, C_{A1}^m}{k_d\, C_{A2}^m}$$

$$C_{A2}^m = 2\, C_{A1}^m$$

$$C_{A1} = C_{A01}\,(1 - X_{A1}) = 2\,(1 - 0.5) = 1 \text{ mol/}l$$

$$C_{A2} = C_{A02}\,(1 - X_{A2}) = 20\,(1 - 0.8) = 4 \text{ mol/}l$$

$$\therefore \quad (4)^m = 2\,(1)^m$$

$$\left(\frac{4}{1}\right)^m = 2$$

or $(4)^m = 2$, Taking the logarithms of both sides,

$$m \ln 4 = \ln 2$$

$$m = \ln 2 / \ln 4 = 0.693/1.386 = 1/2$$

We have : $k_d C_{A2}^m = 2$

$$k_d (4)^{1/2} = 2$$

$$\therefore \quad k_d = 1$$

Rate of deactivation : $-\frac{da}{dt} = k_d C_A^m \, a^d$

We have, d = 1, m = 1/2 and $k_d = 1$

$\therefore$ The rate of deactivation is given by

$$-\frac{da}{dt} = C_A^{1/2} \, a \, , \; h^{-1}$$

From Fig. Ex 8.11 (a) :

$$\text{Intercept} = 0$$

$$\text{Intercept} = \ln\left[\frac{C_{A01} - C_{A1}}{k_1''}\right] = 0$$

For this to be valid, the term in bracket must be equal to one as ln 1 = 0.

$$\frac{C_{A01} - C_{A1}}{k_1''} = 1$$

$$k_1'' = C_{A01} - C_{A1}$$

$$= 2 - 1 = 1$$

From Fig. Ex 8.11 (b) :

$$\text{Intercept} = 0$$

$$\ln\left[\frac{C_{A02} - C_{A2}}{k_2''}\right] = 0$$

For this to be valid, the term in bracket must be equal to 1 (possible when $k_2'' = C_{A02} - C_{A2}$ as ln 1 = 0

$$k_2'' = C_{A02} - C_{A2}$$

$$= 20 - 4 = 16$$

$$k_1'' = k' C_{A1}^n = 1$$

$$k_2'' = k' C_{A2}^n = 16$$

$$\frac{1}{16} = \frac{k' C_{A1}^n}{k' C_{A2}^n}$$

$$\left(\frac{C_{A2}}{C_{A1}}\right)^n = 16$$

$$C_{A2} = 20\,(1 - 0.8) = 4$$

$$C_{A1} = 2\,(1 - 0.5) = 1$$

$$(4/1)^n = 16$$

i.e. $$(4)^n = 16$$

For this to be valid, $$n = 2$$

$$n \ln 4 = \ln 16$$

$$n = \ln 16/\ln 4 = 2.7726/1.3863 = 2$$

$$n = 2 \text{ ... second order rate}$$

$$16 = k' C_{A2}^n$$

$$16 = k' (4)^2$$

$\therefore$ $$k' = 1$$

Rate of reaction : $$-r_A' = k' C_A^n \, \mathbf{a}$$

We have, $k' = 1$ and $n = 2$

$\therefore$ Rate of reaction : $$-r_A' = C_A^2 \, \mathbf{a}, \text{ mol/(g cat·min)}$$

The rate forms/expressions are :

$$\boxed{-r_A' = C_A^2 \, \mathbf{a}} \text{ , mol/(g cat·min) ... for reaction}$$

$$\boxed{-\frac{d\mathbf{a}}{dt} = C_A^{1/2} \, \mathbf{a}} \text{ , } h^{-1} \text{ ... for deactivation}$$ ... Ans.

Extra :

First order deactivation

Second order reaction

Half order concentration dependency of deactivation

$\therefore$ $$-r_A' = k' C_A^2 \, \mathbf{a}, \text{ mol/(g cat·min)}$$

$$-\frac{d\mathbf{a}}{dt} = k_d C_A^{1/2} \, \mathbf{a}, \; h^{-1}$$

For mixed flow reactor :

$$\frac{W}{F_{A0}} = \frac{X_A}{-r_A'}$$

$$\frac{W}{F_{A0}} = \frac{X_A}{k' C_A^2 a} = \frac{X_A}{k' C_{A0}^2 (1 - X_A)^2 a}$$

$$\frac{WC_{A0}}{F_{A0}} = \frac{X_A}{k' C_{A0} (1 - X_A)^2 a}$$

$$-\frac{da}{dt} = k_d C_A^{1/2} a$$

$$a = e^{-k_d C_A^{1/2} t}$$

$$\frac{WC_{A0}}{F_{A0}} = \tau' = \frac{X_A}{k' C_{A0} (1 - X_A)^2 e^{-k_d C_A^{1/2} t}}$$

$$\tau' = \frac{X_A}{k' C_{A0} (1 - X_A)^2} e^{k_d C_A^{1/2} t}$$

$$\ln \tau' = \ln \left[\frac{X_A}{k' C_{A0} (1 - X_A)^2} \right] + k_d C_A^{1/2} t$$

For Run 1 :

$$C_{A0} = 2 \text{ mol/}l \text{ and } X_A = 0.5$$

$$C_A = C_{A0} (1 - X_A) = 2 (1 - 0.5) = 1 \text{ mol/}l$$

$$\ln \left[\frac{X_A}{k' C_{A0} (1 - X_A)^2} \right] = \ln \left[\frac{0.50}{k' \times 2 (1 - 0.5)^2} \right] = \ln (1/k')$$

$$\ln (1/k') = \text{intercept} = 0$$

as $\ln 1 = 0 \longrightarrow$ $1/k' = 1$

$\therefore$ $k' = 1$

For Run 2 :

$$C_{A0} = 20 \text{ mol/}l \text{ and } X_A = 0.80$$

$$C_A = C_{A0} (1 - X_A) = 20 (1 - 0.80) = 4 \text{ mol/}l$$

$$\ln \left[\frac{X_A}{k' C_{A0} (1 - X_A)^2} \right] = \ln \left[\frac{0.80}{k' \times 20 (1 - 0.80)^2} \right] = \ln (1/k')$$

$$\text{Intercept} = 0 = \ln (1/k')$$

as $\ln 1 = 0 \longrightarrow$ $1/k' = 1$

$\therefore$ $k' = 1$ ... obtained previously

Slope of plot of $\ln \tau'$ v/s t for Run 1 = 1

$$k_d C_A^{1/2} = 1$$

$$k_d [C_{A0} (1 - X_A)]^{1/2} = 1$$

$\therefore$ $k_d [2 (1 - 0.5)]^{1/2} = 1$

$$k_d = 1$$

Slope of plot of ln τ' v/s t for Run 2 = 2

$$k_d C_A^{1/2} = 2$$

$$k_d [C_{A0} (1 - X_A)]^{1/2} = 2$$

$$k_d [20 (1 - 0.80)]^{1/2} = 2$$

$$\therefore \quad k_d = 1$$

$$\ln \tau' = \ln \left[\frac{X_A}{k' C_{A0} (1 - X_A)^2} \right] + k_d C_A^{1/2} t$$

For Run 2 :

$$\ln \tau' = \ln \left[\frac{0.80}{k' \times 20 (1 - 0.80)^2} \right] + k_d [20 (1 - 0.80)]^{1/2} t$$

$$\ln \tau' = \ln (1/k') + 2k_d t$$

We have, $\tau' = e$ ... $t = 0.5$

$$\ln e = \ln (1/k') + 2k_d \times 0.5$$

$$1 = \ln (1/k') + k_d \quad \text{... (1)}$$

We have, $\tau' = e^3$ ... $t = 1.5$

$$\ln e^3 = \ln (1/k') + 2 \times k_d \times 1.5$$

$$3 = \ln (1/k') + 3k_d \quad \text{... (2)}$$

Subtract Equation (1) from Equation (2),

$$3 - 1 = 3k_d - k_d$$

$$2 = 2k_d$$

$$\therefore \quad k_d = 1$$

Put $k_d = 1$ in Equation (3),

$$3 = \ln (1/k') + 3 \times 1$$

$$\therefore \quad \ln (1/k') = 0$$

as $\ln 1 = 0$ $\therefore$ $1/k' = 1$

$$\therefore \quad k' = 1$$

For Run 1 :

$$\ln \tau' = \ln \left[\frac{X_A}{k' C_{A0} (1 - X_A)^2} \right] + k_d C_A^{1/2} t$$

$$\ln \tau' = \ln \left[\frac{0.50}{k' \times 2 (1 - 0.50)^2} \right] + k_d [2 (1 - 0.50)]^{1/2} t$$

$$\ln \tau' = \ln (1/k') + k_d t$$

We have, $\tau' = e^2$ ... $t = 2$

$$\ln e^2 = \ln (1/k') + 2k_d$$

$$2 = \ln (1/k') + 2k_d \quad \text{... (3)}$$

We have, $\tau' = e^3$ ... $t = 3$

$$\ln e^3 = \ln (1/k') + 3k_d$$

$$3 = \ln (1/k') + 3k_d \quad \text{... (4)}$$

Subtract Equation (3) from Equation (4),

$$3 - 2 = 3k_d - 2k_d$$

$$\therefore \quad k_d = 1$$

Put value of k_d in Equation (4)

$$\therefore \quad 3 = \ln (1/k') + 3 \times 1$$

$$\therefore \quad \ln (1/k') = 0$$

So k' must be equal to 1 as ln 1 = 0

$$\therefore \quad k' = 1 \quad \text{... previously obtained}$$

$\therefore$ $k' = 1$, $k_d = 1$, $n = 2$, $m = 1/2$ and $d = 1$.

Extra :

$$-r_A' = k' C_A^n \, \mathbf{a}$$

Let, $n = 2$

$$-r_A' = k' C_A^2 \, \mathbf{a}$$

$$\frac{W}{F_{A0}} = \frac{X_A}{-r_A'}$$

$$\frac{W}{F_{A0}} = \frac{X_A}{k' C_A^2 \, \mathbf{a}}$$

$A \rightarrow R$, $\varepsilon_A = 0$

$\therefore$

$$C_A = C_{A0} (1 - X_A)$$

$$\frac{W}{F_{A0}} = \frac{X_A}{k' C_{A0}^2 (1 - X_A)^2 \, \mathbf{a}}$$

$$\frac{WC_{A0}}{F_{A0}} = \frac{X_A}{k' C_{A0} (1 - X_A)^2 \, \mathbf{a}}$$

$$\tau' = \frac{X_A}{k' C_{A0} (1 - X_A)^2 \, \mathbf{a}}$$

$$k' = \frac{X_A}{\tau' C_{A0} (1 - X_A)^2 \, \mathbf{a}}$$

For Run 2 : $\tau = 1$ at $t = 0$, $C_{A0} = 20$ and $X_A = 0.80$

For catalyst : At $t = 0$, $\mathbf{a} = 1$

$$\therefore \quad k' = \frac{0.80}{1 \times 20\,(1 - 0.80)^2 \times 1} = 1$$

For Run 1 : $C_{A0} = 2$ and $X_A = 0.5$

For Run 1 : $\tau = 1$ at $t = 0$

For catalyst : At $t = 0$, $\mathbf{a} = 1$

$$k' = \frac{X_A}{\tau'\, C_{A0}\,(1 - X_A)^2\, \mathbf{a}}$$

$$= \frac{0.50}{1 \times 2\,(1 - 0.5)^2 \times 1} = 1$$

For Run 2 : At $t = 1.5$, $\tau' = e^3$

$$-\frac{d\mathbf{a}}{dt} = k_d\, C_A^m\, \mathbf{a}^d$$

$$\mathbf{a} = e^{-k_d\, C_A^m\, t}$$

We have : $k_d = 1$ and $m = 1$

$$\therefore \quad \mathbf{a} = e^{-1 \times C_A^{1/2}\, t} = e^{C_A^{1/2}\, t}$$

$$\mathbf{a} = e^{-[C_{A0}\,(1 - X_A)]^{1/2}\, t}$$

$$\mathbf{a} = e^{-[20\,(1 - 0.80)]^{1/2}\, t}$$

$$\mathbf{a} = e^{-2t}$$

At $t = 1.5$ ..... $\tau' = e^3$

$$\mathbf{a} = e^{-2 \times 1.5} = e^{-3} = 0.04978$$

$$k' = \frac{X_A}{\tau'\, C_{A0}\,(1 - X_A)^2\, \mathbf{a}}$$

$$= \frac{0.80}{e^3 \times 20\,(1 - 0.80)^2 \times 0.04978}$$

$$= 1$$

So $m = 1/2$, $n = 2$ and $d = 1$.

Example 8.12 : *In the catalytic converter of an automobile, CO and hydrocarbons present in the exhaust gases are oxidised. Unfortunately, the effectiveness of this unit decreases with use. The phenomenon was studied by means of an accelerated aging test on a palladium impregnated porous pellet packed bed converter.*

From the following hydrocarbon conversion – time data, obtain an expression to represent the deactivation rate of this catalyst.

t, h	*5*	*10*	*15*	*20*	*25*	*30*	*35*	*40*
$X_{hydrocarbon}$	*0.57*	*0.53*	*0.52*	*0.50*	*0.48*	*0.45*	*0.43*	*0.41*

Solution : Start by guessing the simplest rate forms :

First order reaction, first order deactivation. Further assume that the deactivation is concentration independent.

$$-r_A' = k' C_A a$$

and

$$-\frac{da}{dt} = k_d a$$

For batch-solids and plug constant flow of fluid with the above rate forms, we have

$$\ln \ln (C_{A0}/C_A) = \ln (k' \tau') - k_d t$$

For our guess to be correct, a plot of $\ln \ln (C_{A0}/C_A)$ v/s t should yield a straight line.

Convert the data in the form in which it can be plotted.

$$C_A = C_{A0} (1 - X_A)$$

$$\frac{C_A}{C_{A0}} = 1 - X_A$$

t	X_A	$C_A/C_{A0} = 1 - X_A$	C_{A0}/C_A	$\ln C_{A0}/C_A$	$\ln \ln C_{A0}/C_A$
5	0.57	0.43	2.325	0.844	– 0.17
10	0.53	0.47	2.127	0.755	– 0.28
15	0.52	0.48	2.08	0.734	– 0.31
20	0.50	0.50	2.0	0.693	– 0.366
25	0.48	0.52	1.923	0.654	– 0.425
30	0.45	0.55	1.818	0.598	– 0.514
35	0.43	0.57	1.754	0.562	– 0.576
40	0.41	0.59	1.695	0.527	– 0.64

Plot $\ln \ln C_{A0}/C_A$ v/s t.

Refer Fig. Ex 8.12 :

As all the data points are falling on a straight line our guess is correct.

The deactivation is first order.

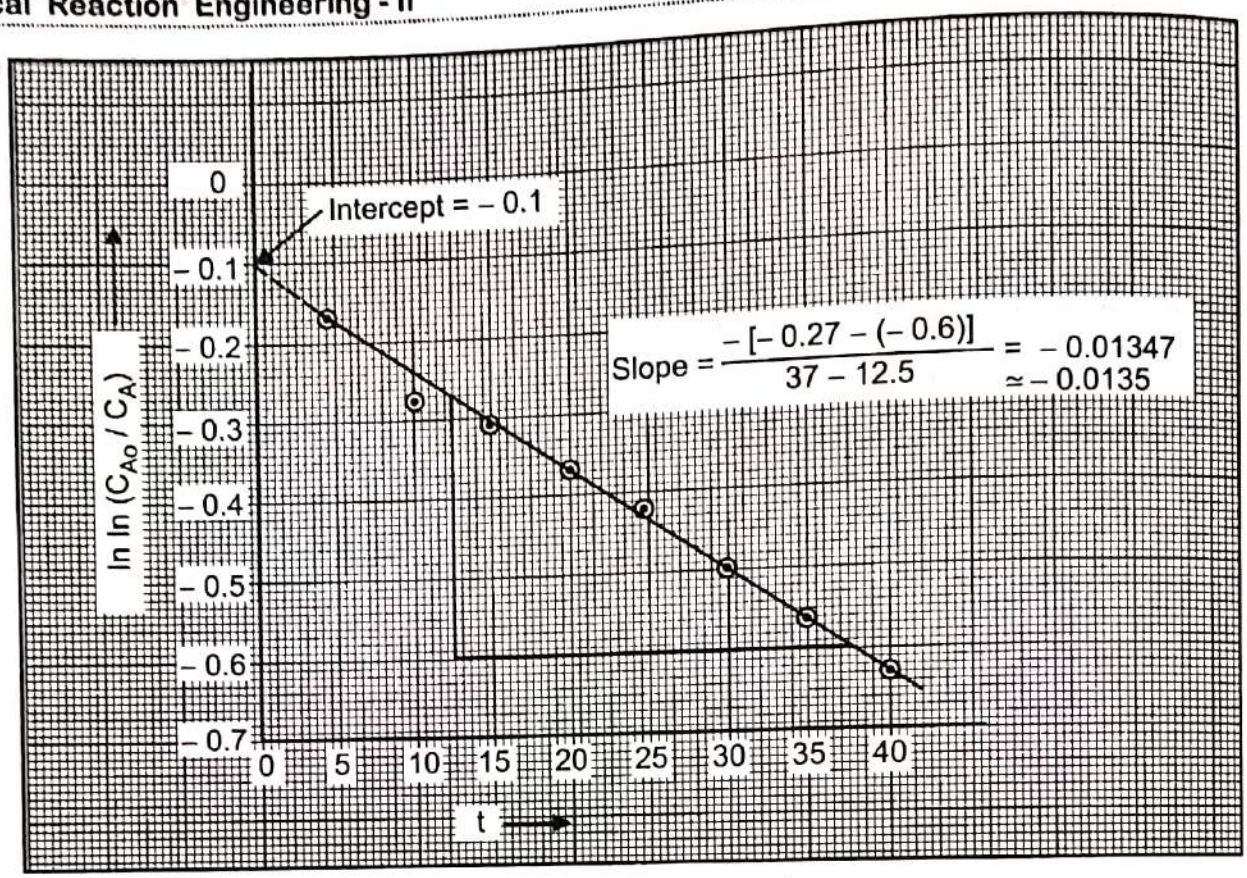


Fig. Ex 8.12

$$\text{Slope} = -0.0135 = -k_d$$

$$\therefore \quad k_d = 0.0135 \text{ h}^{-1}$$

The rate of deactivation is given by or the expression to represent the deactivation rate of the catalyst is

$$\boxed{-\frac{da}{dt} = 0.0135\, a} \text{ , h}^{-1} \qquad \text{... Ans.}$$

Example 8.13 : *The isomerisation of A to R proceeds at 730 K on a slowly deactivating catalyst by a second-order rate.*

$$-r'_A = k' C_A^2 \, a = 200 \, C_A^2 \, a \text{ , mol/(g cat·h)}$$

In this case, deactivation is caused by both A and R as the molecules of reactant and product are similar in structure. The rate of deactivation, with no diffusion effects, is found to be

$$-\frac{da}{dt} = k_d (C_A + C_R) \, a = 10 (C_A + C_R) \, a, \text{ day}^{-1}$$

It is desired to operate a packed bed reactor containing 1000 kg of catalyst for 12 days using a steady feed of pure A (F_{A0} = 5 kmol/h at 730 K and 3 atm)

Find : (i) *the conversion at the start of the run.*

(ii) *the conversion at the end of the run.*

(iii) *the mean/average conversion over 12-days run.*

Solution : $A \rightarrow R, \quad \varepsilon_A = 0$

The rate expressions for the reaction and deactivation are :

Rate of reaction : $-r'_A = 200\, C_A^2\, \mathbf{a}$, mol/(g cat·h)

Rate of deactivation : $-\dfrac{d\mathbf{a}}{dt} = 10\,(C_A + C_R)\,\mathbf{a}$, day^{-1}

Given : W = 1000 kg, F_{A0} = 5 kmol/h, π = 3 atm, T = 730 K, Run time = t = 12 days

Let us obtain C_{Ao}.

We know : $C_{A0} = \dfrac{p_{A0}}{RT}$, p_{Ao} = 3 atm ... as feed is pure A.

$= \dfrac{3}{0.08206 \times 730}$, R = 0.08206 (l·atm)/mol·K

$= 0.05$ mol/l

For a specific feed : $C_A + C_R$ = constant = C_{A0}, $C_A + C_R = 0.05$ mol/l

The rate equation for deactivation at any time t thus becomes

$$-\frac{d\mathbf{a}}{dt} = 10 \times 0.05\, \mathbf{a} \qquad , k_d = 10\ l/(\text{mol}\cdot\text{day})$$

$$= 0.5\, \mathbf{a} \text{ , day}^{-1}$$

$$-\int_1^{\mathbf{a}} \frac{d\mathbf{a}}{\mathbf{a}} = 0.5 \int_0^t dt$$

$\therefore \quad \mathbf{a} = e^{-0.5\,t}$ ... at any time t

For $\varepsilon_A = 0$, we have : $C_A = C_{Ao}\,(1 - X_A) \quad \therefore \quad C_A/C_{Ao} = 1 - X_A$ ... for $\varepsilon_A = 0$

The plug flow equation is

$$\tau' = C_{Ao} \int_0^{X_A} \frac{dX_A}{-r'_A} = C_{Ao} \int_0^{X_A} \frac{dX_A}{200\, \mathbf{a}\, C_{Ao}^2\, (1 - X_A)^2}$$

Integrating gives

$$200\, \mathbf{a}\, \tau'\, C_{Ao} = X_A/(1 - X_A)$$

Rearranging gives

$$X_A = 200\, \mathbf{a}\, \tau'\, C_{Ao}/(1 + 200\, \mathbf{a}\, \tau'\, C_{Ao})$$

Deducting RHS and LHS from 1 gives

$$1 - X_A = \frac{C_A}{C_{A0}} = \frac{1}{1 + 200\, \mathbf{a}\, \tau'\, C_{A0}}$$

Using the above equation, we will evaluate the conversion at the start of the run.

$\tau' = 10$ (g cat·h)/l , $C_{A0} = 0.05$ mol/l

$\mathbf{a} = e^{-0.5\,t}$

At the start of the run, i.e., at t = 0 :

$\mathbf{a} = e^{-0.5 \times 0} = 1$

$$1 - X_A = \frac{C_A}{C_{A0}} = \frac{1}{1 + 200 \times 1 \times 10 \times 0.05}$$

$$= \frac{1}{101} = 0.0099 \approx 0.01$$

Solving, we get $X_A = 1 - 0.01 = 0.99$

∴ **X_A at the start of the run = 0.99 or 99 %** ... Ans. (i)

Let us evaluate the conversion at the end of 12 days run.

$$\tau' = 10 \text{ (g cat·h)}/l$$

$$C_{A0} = 0.05 \text{ mol}/l$$

$$\mathbf{a} = e^{-0.5\,t}$$

$$t = 12 \text{ days}$$

$$\mathbf{a} = e^{-0.5 \times 12} = e^{-6} = 2.478 \times 10^{-3} \approx 2.5 \times 10^{-3}$$

$$1 - X_A = \frac{C_A}{C_{A0}} = \frac{1}{1 + 200\,\mathbf{a}\,\tau'\,C_{A0}}$$

$$= \frac{1}{1 + 200 \times 2.5 \times 10^{-3} \times 10 \times 0.05} = 0.80$$

$$X_A = 1 - 0.80 = 0.20$$

X_A at the end of 12 days run = 0.20 or 20 % ... Ans. (ii)

The average conversion during a run can be obtained by

$$\bar{X}_A = \frac{1}{t_{run}} \int_0^{t_{run}} X_A\,(t)\,dt$$

$t_{run} = 12$ days

$$\bar{X}_A = \frac{1}{12} \int_0^{12} X_A\,(t)\,dt$$

We will adopt a method of numerical evaluation of integrals.

We have, $$1 - X_A = \frac{1}{1 + 200\,\mathbf{a}\,\tau'\,C_{A0}} = \frac{1}{1 + 200 \times 10 \times 0.05 \times e^{-0.5\,t}}$$

$$1 - X_A = \frac{1}{1 + 100\,e^{-0.5\,t}}$$

$$1 - \frac{1}{1 + 100\,e^{-0.5\,t}} = X_A = X_A\,(t)$$

$$X_A\,(t) = \frac{100\,e^{-0.5\,t}}{1 + 100\,e^{-0.5\,t}}$$

Using the above equation, evaluate X_A (t) at various times (from 0 to 12) and tabulate.

t	0	2	4	6	8	10	12
X_A (t)	0.99	0.97	0.93	0.83	0.65	0.40	0.20

We will make use of Simpson's one-third rule for the evaluation of the integral.

$$\int_{x_0}^{x_2} f(x)\,dx = \frac{h}{3}[f(x_0) + 4f(x_1) + f(x_2)] \ , \ h = x_1 - x_0$$

In our case, $h = 2 - 0 = 2$

$$\bar{X}_A = \frac{1}{12}\int_0^{12} X_A(t)\,dt$$

$$= \frac{1}{12}\times\frac{2}{3}\,[\underline{0.99 + 4\times 0.97 + 0.93} + \underline{0.93 + 4\times 0.83 + 0.65} + \underline{0.65 + 4\times 0.40 + 0.20}]$$

$$= \frac{1}{12}\times\frac{2}{3}\times 13.15$$

$$= 0.7305 \approx 0.73$$

Average conversion over 12 days run = 0.73 or 73 % ... **Ans.**

Example 8.14 : *A gaseous feed containing 10 % C_4H_{10} and 90 % inerts (mole basis) at 1 atm and 555 °C flows [$\tau' = 1.1$ (kg·h)/m³] through a packed bed of alumina-chromia catalyst. The decomposition of butane is a first order reaction.*

$$C_4H_{10} \longrightarrow C_4H_8 \longrightarrow Carbon$$
$$\searrow\ other\ gases$$

and the conversion-time data are as follows :

t, h	0	1	2	3	4	5
X_A	0.89	0.78	0.63	0.47	0.34	0.26

The catalyst activity decreases with use because of carbon deposition on the active surfaces. The deactivation is found to be concentration independent. Develop rate equations for the reaction and for the deactivation which fit these data.

Solution : Let $A = C_4H_{10}$

The decomposition of A is a first-order reaction.

$\therefore$
$$-r_A' = k' C_A \ a$$

($\varepsilon_A = 0$ ... neglecting volume changes with reaction)

The deactivation is concentration independent. Therefore,

$$-\frac{da}{dt} = k_d\, a^d$$

Guess-1 : As a first guess try a first order rate expression for deactivation, i.e., try for first-order deactivation (d = 1)

$$\therefore \quad -\frac{da}{dt} = k_d\, a$$

For a packed bed reactor assuming plug flow (batch-solids and plug constant flow of fluid) for first-order reaction and first order deactivation, we have

$$\ln \ln \frac{C_{A0}}{C_A} = \ln (k'\tau') - k_d\, t$$

Construct a plot of $\ln \ln C_{A0}/C_A$ v/s t.

If the plot of $\ln \ln C_{A0}/C_A$ v/s t yields a straight line, then our guess will be correct.

Let us calculate C_{A0}.

$$C_{A0} = \frac{p_{A0}}{RT} \quad p_{A0} = x_A \cdot P = 0.10 \times 1 \text{ atm}, \quad T = 555\ ^\circ\text{C} = 828 \text{ K}$$

$$= \frac{0.10 \times 1}{0.08206 \times 828} = 1.47 \times 10^{-3} \text{ mol}/l = 1.47 \text{ mol/m}^3$$

[mole fraction of A = mole % A/100 = 10/100 = 0.10]

As the feedstream contains a very large proportion of inerts (90 %), i.e., the feedstream is diluted with a large amount of inerts, we can neglect the volume changes with reaction (approximation : $\nu = \nu_o$, $\therefore\ \varepsilon_A = 0$). Then, we have

$$C_A = C_{A0}(1 - X_A)$$

$$\frac{C_A}{C_{A0}} = 1 - X_A$$

$$\therefore \quad \frac{C_{A0}}{C_A} = \frac{1}{1 - X_A}$$

t	X_A	$C_{A0}/C_A = 1/(1 - X_A)$	$\ln C_{A0}/C_A$	$\ln \ln C_{A0}/C_A$
0	0.89	9.091	2.207	0.792
1	0.78	4.545	1.514	0.425
2	0.63	2.703	$0.994 \approx 1$	0
3	0.43	1.754	0.5621	– 0.576
4	0.34	1.515	0.4155	– 0.878
5	0.26	1.351	0.301	– 1.2

Plot $\ln \ln [C_{A0}/C_A]$ v/s t.

Refer Fig. Ex 8.14 :

Since we obtained a straight line, our guess regarding the deactivation order is correct.

So we have : first-order reaction and first-order deactivation.

$$\text{Slope} = -0.396 = -k_d$$

$$k_d = 0.396\ h^{-1}$$

$\therefore$ $$\text{Intercept} = 0.792$$

$$\ln (k'\tau') = 0.792$$

$$k'\tau' = e^{0.792} = 2.2078$$

$$\tau' = 1.1\ (kg \cdot h)/m^3$$

$$k' = \frac{2.2078}{1.1} = 2,\ m^3/(kg \cdot h)$$

The rate equations/expressions for the reaction and deactivation are :

$$\boxed{-r_A' = 2C_A\, a} \quad \text{... for reaction} \qquad \textbf{... Ans.}$$

$$\boxed{-\frac{da}{dt} = 0.396\, a} \quad \text{... for deactivation} \qquad \textbf{... Ans.}$$

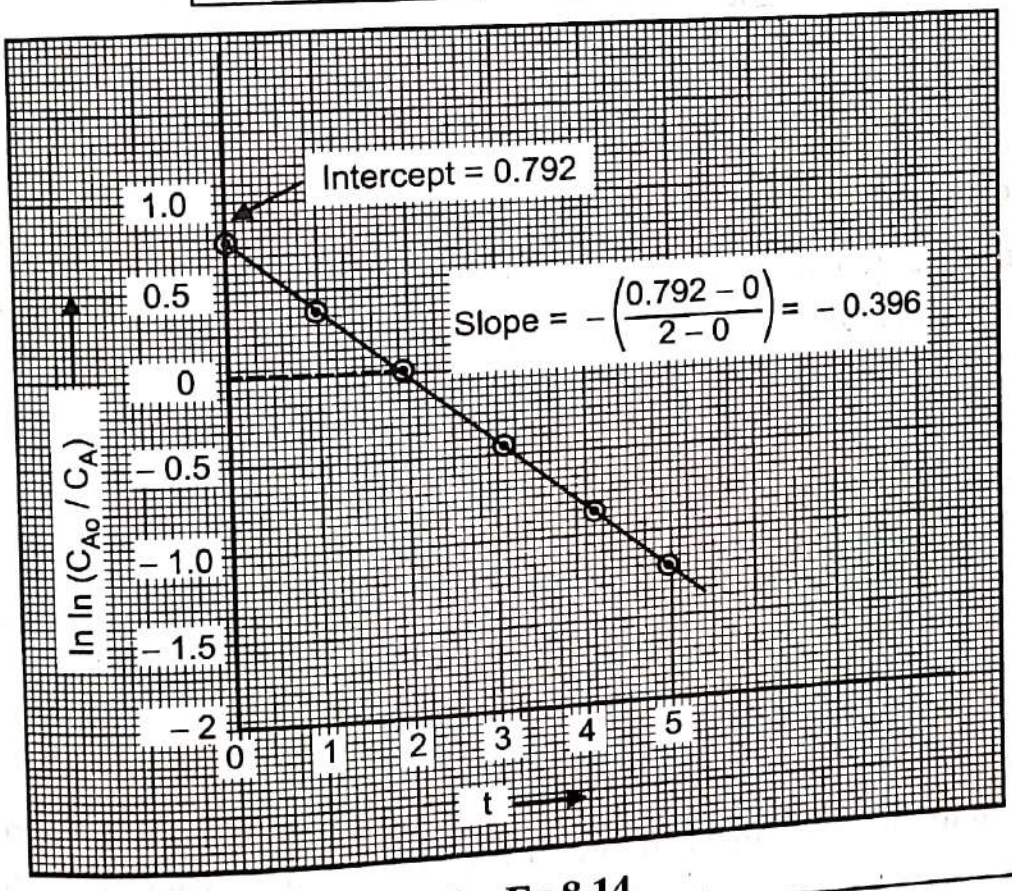


Fig. Ex 8.14

Example 8.15 : *Consider a reactor with mixed flow of catalyst and plug flow of gas where the gas reacts and catalyst deactivates. The kinetics of reaction and deactivation are represented by* $A \rightarrow R$,

$$-r_A' = k'C_A\, a \qquad \text{... reaction}$$

$$-\frac{da}{dt} = k_d\, a \qquad \text{... deactivation}$$

Obtain expressions for the fractional conversion of gas and the activity of the leaving catalyst stream in terms of the variables of the system. The related variables of this steady-state system are shown in Fig. Ex 8.15 (a).

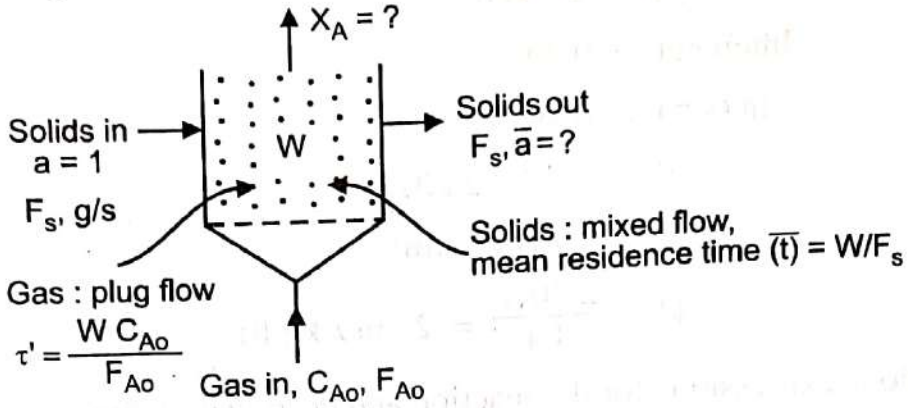


Fig. Ex 8.15 (a)

Solution : $A \rightarrow R$, $\varepsilon_A = 0$

Given : $-r'_A = k'C_A a$ ... rate of reaction

and $-\frac{da}{dt} = k_d a$ ... rate of deactivation

Consider the solids and gas separately - solids (catalyst particles) for obtaining the expressions for mean exit activity ($\bar{a}$) and gas for obtaining the expression for conversion (X_A) and then combine both to obtain the desired expression for X_A.

Refer Fig. Ex 8.15 (a).

Solids : Mixed flow of solids.

Solid stream is treated as a macrofluid (completely segregated). Each solid (catalyst particle) spends a different amount of time in the reactor and thus deactivates to a different extent and thus we have distribution of activities.

To determine the mean activity $\bar{a}$ of solid stream as a whole leaving the reactor (i.e., mean exit activity) we must average the activities of various solids (catalyst particles) in the exit stream.

$$\begin{bmatrix}\text{Mean activity of those particles spending between t and t + dt in the reactor}\end{bmatrix} = \begin{bmatrix}\text{Activity of catalyst particle at time t [spending a time t in the reactor]}\end{bmatrix} \times \begin{bmatrix}\text{Fraction of particles (solids) that spend between t and t + dt in the reactor}\end{bmatrix}$$

$$d\bar{a} = a(t) \times E_s(t)\, dt$$

Summing over all solids, the mean exit activity is

$$\bar{a} = \int_0^\infty a\,(t)\,E_s\,(t)\,dt = \int_0^\infty a\,E_s\,dt \qquad \ldots (1)$$

E (t) for mixed flow reactor is

$$E\,(t) = \frac{1}{\tau}\,e^{-t/\tau}$$

Therefore, on the same line for a mixed flow of solids, the exit age distribution is given by

$$E_s\,(t) = E_s = \frac{1}{\bar{t}}\,e^{-t/\bar{t}} \qquad \ldots (2)$$

where $\bar{t}$ is the mean residence time (contact time).

$$\bar{t} = \frac{\text{Weight of catalyst (kg)}}{\text{Feed rate of catalyst (kg/s)}} = \frac{W}{F_s}$$

The rate of deactivation is given by

$$-\frac{da}{dt} = k_d\,a$$

For any particular catalyst particle integrating the above equation, we get

$$a = e^{-k_d t} \qquad \ldots (3)$$

$$\left[-\frac{da}{a} = k_d\,dt \qquad , \text{At } t = 0 \rightarrow a = 1,\ t = t \rightarrow a = a\right.$$

$$-\int_1^a \frac{da}{a} = k_d \int_0^t dt$$

$$\left. a = e^{-k_d t}\right]$$

Substituting the values of E_s and **a** from Equations (2) and (3) into Equation (1), we get

$$\bar{a} = \int_0^\infty e^{-k_d t} \cdot \frac{1}{\bar{t}} \cdot e^{-t/\bar{t}} \cdot dt$$

$$= \frac{1}{\bar{t}} \int_0^\infty e^{-(k_d t + t/\bar{t})} \cdot dt$$

$$= \frac{1}{\bar{t}} \int_0^\infty e^{-(k_d + 1/\bar{t})\,t} \cdot dt$$

Let $(k_d + 1/\bar{t})\,t = x$

$$\therefore \qquad dt = \frac{dx}{k_d + 1/\bar{t}} = \left(\frac{\bar{t}}{1 + k_d\,\bar{t}}\right) dx$$

Limits : At $t = 0$ ..... $x = 0$, $t = \infty$ ..... $x = \infty$

$$\bar{a} = \frac{1}{\bar{t}} \int_0^\infty e^{-x} \cdot \left(\frac{\bar{t}}{1 + k_d \bar{t}} \right) dx$$

$$= \left(\frac{1}{1 + k_d \bar{t}} \right) \int_0^\infty e^{-x} \, dx$$

$$= \left(\frac{1}{1 + k_d \bar{t}} \right) \left[- e^{-x} \right]_0^\infty$$

$$= \frac{1}{1 + k_d \bar{t}} \left[- e^{-\infty} - (- e^0) \right]$$

$$= \frac{1}{1 + k_d \bar{t}} [0 + 1]$$

$$= \frac{1}{1 + k_d \bar{t}}$$

$$\bar{t} = W/F_s$$

$\therefore$

$$\boxed{\bar{a} = \frac{1}{1 + k_d \bar{t}} = \frac{1}{1 + k_d \, W/F_s}} \quad \text{... (4)}$$

Equation (4) is the desired expression for the exit activity ($\bar{a}$) of the solids. Equation (4) gives the activity of the solids in the reactor and in the exit stream.

Alternatively :

We know that the degree of segregation (complete micromixing or complete segregation) has no effect on the conversion for a first order reaction. So the extent of micromixing does not affect the first order reaction.

Hence, as the deactivation is first order, a macrofluid could be treated as a microfluid. Thus for mixed flow of solids (as a homogeneous fluid), the performance equation

$$\tau = \frac{C_{A0} - C_A}{-r_A}$$

becomes

$$\bar{t} = \frac{1 - \bar{a}}{\text{Rate of deactivation}} = \frac{1 - \bar{a}}{k_d \bar{a}}$$

$$k_d \bar{a} \, \bar{t} = 1 - \bar{a}$$

$$\bar{a} = \frac{1}{1 + k_d \bar{t}} = \frac{1}{1 + k_d \, W/F_s}$$

The above equation is identical to Equation (4).

Gas : Plug flow of gas.

Since the solids are in mixed flow, the up-rising gas (plug flow) comes in contact with solids (catalyst particles) of the same composition (i.e., of the same activity, $\bar{a}$) everywhere as it passes through the bed. Thus, the performance equation for plug flow of gas becomes

$$\tau' = C_{A0} \int \frac{dX_A}{-r_A'}$$

$$= C_{A0} \int \frac{dX_A}{k' C_A \bar{a}}, \quad \text{as } -r_A' = k' C_A \bar{a} \text{ and } a = \bar{a}$$

For $\varepsilon_A = 0$, $\quad C_A = C_{A0}(1 - X_A)$

$$\therefore \quad \tau' = C_{A0} \int_0^{X_A} \frac{dX_A}{k' C_{A0}(1 - X_A)\bar{a}}$$

$$= \frac{1}{k' \bar{a}} \int_0^{X_A} \frac{dX_A}{(1 - X_A)}$$

$$k' \bar{a} \tau' = -\ln(1 - X_A)$$

$$\therefore \quad 1 - X_A = e^{-k'\tau'\bar{a}} \qquad \ldots (5)$$

Combine the expressions for gas and solid :

Substituting the value of $\bar{a}$ from Equation (4) into Equation (5), we get

$$1 - X_A = e^{-k'\tau'[1/(1 + k_d \bar{t})]}$$

$$1 - X_A = e^{-k'\tau'/(1 + k_d \bar{t})}$$

Let $\quad k'\tau' = R$

and $\quad k_d \bar{t} = R_d$

$$\therefore \quad \boxed{1 - X_A = e^{-k'\tau'/(1 + k_d \bar{t})} = e^{-R/(1 + R_d)}} \qquad \ldots (6)$$

where R and R_d are the first order rate groups for reaction and deactivation.

Equation (6) is the desired expression for the fractional conversion of gas in terms of the variables of the system for mixed flow of solids and plug flow of gas.

Example 8.16 : *Consider a reactor with mixed flow of catalyst and mixed flow of gas where the gas reacts and catalyst deactivates. The kinetics of reaction and deactivation are represented by*

$A \rightarrow R,\ \varepsilon_A = 0,$ $\quad -r_A' = k' C_A\, a$ ... *reaction rate*

$$-\frac{da}{dt} = k_d\, a \qquad \text{... deactivation rate}$$

Obtain expressions for the fractional conversion of gas and the activity of the leaving catalyst stream in terms of the variables of the system. The related variables of this steady-state system are shown in Fig. Ex 8.16 (a).

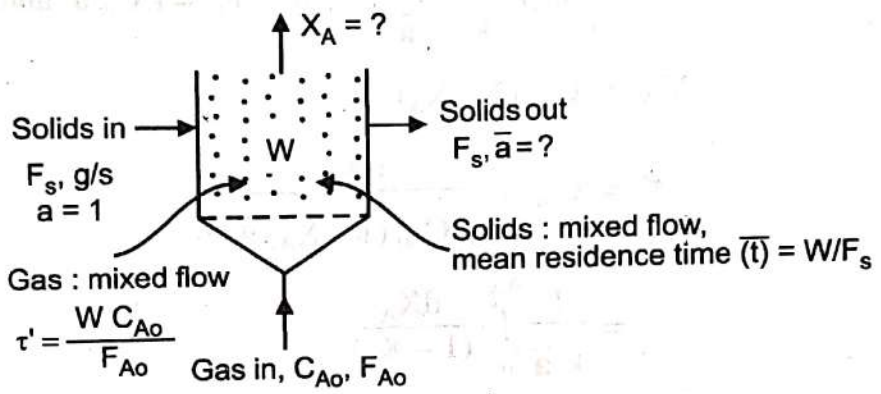


Fig. Ex 8.16 (a)

Solution : $A \rightarrow R,\ \varepsilon_A = 0$

$$-r_A' = k' C_A\, a \qquad \text{... rate of reaction}$$

$$-\frac{da}{dt} = k_d\, a \qquad \text{... rate of deactivation}$$

Procedure to obtain the solution to this example : Consider the solids (in mixed flow) to obtain the expression for $\bar{a}$ ... consider the gas ... (in mixed flow) ... to obtain the expression for X_A in terms of $\bar{a}$ ... and then combine both the expressions to obtain the desired expression for X_A.

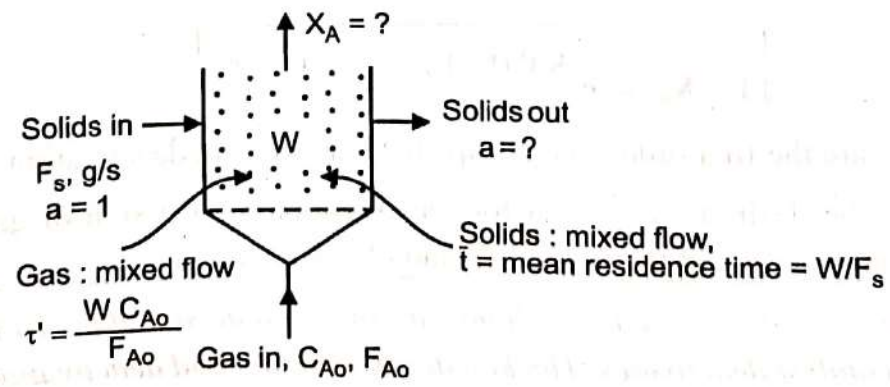


Fig. Ex 8.16 (b)

(i) **Solids :** Solids (catalyst particles) are in mixed flow.

The solid stream is treated as a macrofluid (completely segregated). Each solid (catalyst particle) spends a different amount of time in the reactor and thus deactivates to a different extent and thus we have distribution of activities.

To determine the mean activity $\bar{\mathbf{a}}$ of solid stream (as a whole) leaving the reactor (i.e. mean exit activity of entire solid stream) we must average the activities of various solids (catalyst particles) in the exit stream.

$$\begin{bmatrix}\text{Mean activity of}\\ \text{those particles spending}\\ \text{between t and t + dt}\\ \text{in the reactor}\end{bmatrix} = \begin{bmatrix}\text{Activity of catalyst}\\ \text{particle at time t}\\ \text{[spending a time}\\ \text{t in the reactor]}\end{bmatrix} \times \begin{bmatrix}\text{Fraction of particles}\\ \text{(solids) that spend}\\ \text{between t and t + dt}\\ \text{in the reactor}\end{bmatrix}$$

$$d\bar{\mathbf{a}} = \mathbf{a}\,(t) \times E_s\,(t)\,dt = \mathbf{a}\,E_s\,dt$$

$E_s\,(t)$ is written as E_s for convenience.

Summing over all solids, the mean exit activity is

$$\bar{\mathbf{a}} = \int_0^\infty \mathbf{a}\;E_s\;dt \qquad \ldots (1)$$

E (t) for a mixed flow reactor is

$$E\,(t) = \frac{1}{\tau}\,e^{-t/\tau}$$

∴ On the same line for mixed flow of solids, the exit age distribution is given by

$$E_s\,(t) = E_s = \frac{1}{\bar{t}}\,e^{-t/\bar{t}} \qquad \ldots (2)$$

where $\bar{t}$ is the mean residence time (contact time) of solids.

$$\bar{t} = \frac{\text{Weight of catalyst (g)}}{\text{Feed rate of catalyst (g/s)}} = \frac{W}{F_s}\ ,\ s^{-1}$$

The rate of deactivation is given by

$$-\frac{d\mathbf{a}}{dt} = k_d\,\mathbf{a}$$

For any particular catalyst particle integrating the above equation, we get

$$\mathbf{a} = e^{-k_d t} \qquad \ldots (3)$$

$$\left[-\frac{d\mathbf{a}}{dt} = k_d\,\mathbf{a}\right.$$

$$\int_1^a \frac{d\mathbf{a}}{\mathbf{a}} = -k_d \int_0^t dt$$

$$\ln \mathbf{a} = -k_d\,t$$

∴

$$\left.\mathbf{a} = e^{-k_d t}\right]$$

Substituting the values of E_s and **a** from Equations (2) and (3) into Equation (1), we get

$$\bar{\mathbf{a}} = \int_0^\infty \mathbf{a}\, E_s\, dt = \int_0^\infty e^{-k_d t} \cdot \frac{1}{\bar{t}}\, e^{-t/\bar{t}}\, dt$$

$$= \frac{1}{\bar{t}} \int_0^\infty e^{-(k_d t + t/\bar{t})}\, dt$$

$$= \frac{1}{\bar{t}} \int_0^\infty e^{-(k_d + 1/\bar{t})\, t}\, dt$$

Let $(k_d + 1/\bar{t})\, t = x$

$\therefore$ $$dt = \frac{dx}{k_d + 1/\bar{t}} = \frac{\bar{t}\, dx}{(1 + k_d\, \bar{t})}$$

Limits : At $t = 0$: $x = 0$ and at $t = \infty$: $x = \infty$

$$\bar{\mathbf{a}} = \frac{1}{\bar{t}} \int_0^\infty e^{-x} \left(\frac{\bar{t}}{1 + k_d\, \bar{t}} \right) dx$$

$$= \frac{1}{(1 + k_d\, \bar{t})} \int_0^\infty e^{-x}\, dx = \left(\frac{1}{1 + k_d\, \bar{t}} \right) \left[- e^{-x} \right]_0^\infty$$

$$= \left(\frac{1}{1 + k_d\, \bar{t}} \right) \left[- e^{-\infty} - (- e^0) \right]$$

$$= \left(\frac{1}{1 + k_d\, \bar{t}} \right) (0 + 1)$$

$$= \frac{1}{1 + k_d\, \bar{t}}$$

$\therefore$ $$\bar{t} = W/F_s$$

$\therefore$ $$\bar{\mathbf{a}} = \frac{1}{1 + k_d\, \bar{t}} = \frac{1}{1 + k_d W/F_s} \qquad \ldots (4)$$

Equation (4) is the desired expression for the mean exit activity ($\bar{\mathbf{a}}$) of the solids (all catalysts particles).

Equation (4) gives the activity of the solids in the reactor and in the exit stream (as composition of the solids in the exit stream is identical to the composition of the solids anywhere in the reactor as it is perfectly mixed).

Gas : Mixed flow of gas.

The gas and solids are in mixed flow so the gas meets the solids of activity $\bar{a}$ everywhere in the reactor. Thus, the performance equation for mixed flow of gas becomes

$$\tau' = \frac{C_{A0} X_A}{-r_A'}$$

$$= \frac{C_{A0} X_A}{k' C_A \bar{a}} \quad , (\text{as } -r_A' = k' C_A \bar{a} \text{ and } a = \bar{a})$$

For $\varepsilon_A = 0$, $C_A = C_{A0}(1 - X_A)$

$$\tau' = \frac{C_{A0} X_A}{k' C_{A0} \bar{a} (1 - X_A)}$$

$$k' \bar{a} \tau' = \frac{X_A}{1 - X_A}$$

Adding 1 on both sides, we get

$$1 + k' \bar{a} \tau' = \frac{X_A}{1 - X_A} + 1 = \frac{X_A + 1 - X_A}{1 - X_A} = \frac{1}{1 - X_A}$$

$$\therefore \quad 1 - X_A = \frac{1}{1 + k' \tau' \bar{a}} \qquad \ldots (5)$$

Combine the expressions for gas and solid :

Substituting the value of $\bar{a}$ from Equation (4) into Equation (5), we get

$$1 - X_A = \frac{1}{1 + k'\tau' \times 1/(1 + k_d \bar{t})} = \frac{1}{1 + k'\tau'/(1 + k_d \bar{t})}$$

Let, $R = k'\tau'$ ... first order rate group for reaction

$R_d = k_d \bar{t}$ ... first order rate group for deactivation

$$\boxed{1 - X_A = \frac{1}{1 + R/(1 + R_d)} = \frac{1 + R_d}{1 + R_d + R}} \qquad \ldots (6)$$

Equation (6) is the desired expression for the fractional conversion of gas in terms of the variables of the system for mixed flow of solids and mixed flow of gas.

Example 8.17 : *Consider a reactor - regenerator system with a steady state circulation of solids between the units. Reactant gas flows through the reactor, reacts to form product and deactivates the catalyst. The spent catalyst is continuously fed to the regenerator to restore the activity of catalyst partially by contact with regenerator gas and the catalyst is then returned to the reactor.*

The kinetics of reaction, deactivation and regeneration are given by

$$A \rightarrow R, \quad \varepsilon_A = 0$$

$$-r_A' = k' C_A a \qquad \text{...for reacting gas ... rate expression for reaction}$$

$$-\frac{da}{dt} = k_d a \qquad \text{...for deactivation ... rate expression for deactivation}$$

$$+\frac{da}{dt} = k_r (1 - a) \qquad \text{...for regeneration ... rate expression for regeneration}$$

Obtain expressions for the fractional conversion of gas and the activity of catalyst streams in terms of the variables of the system for mixed flow of gas in two units assuming that solids are in mixed flow in two units. Fig. Ex 8.17 shows the related variables of this steady-state system.

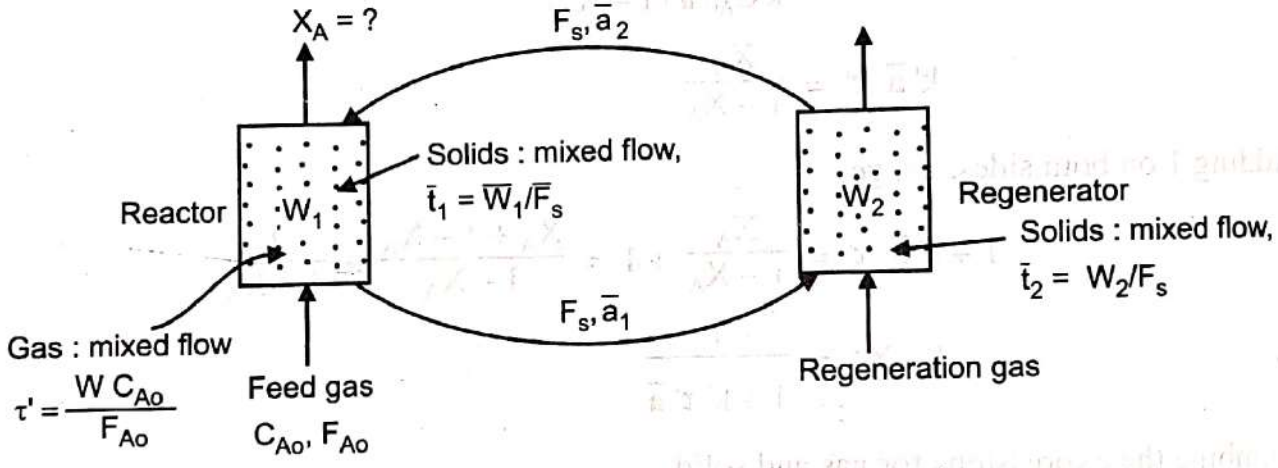


Fig. Ex 8.17 (a) : Reactor-Regenerator set up

Solution :

$$A \rightarrow R, \quad \varepsilon_A = 0$$

$$-r_A' = k' C_A a \qquad \text{... rate of reaction}$$

$$-\frac{da}{dt} = k_d t \qquad \text{... rate of deactivation}$$

$$+\frac{da}{dt} = k_r (1 - a) \qquad \text{... rate of regeneration}$$

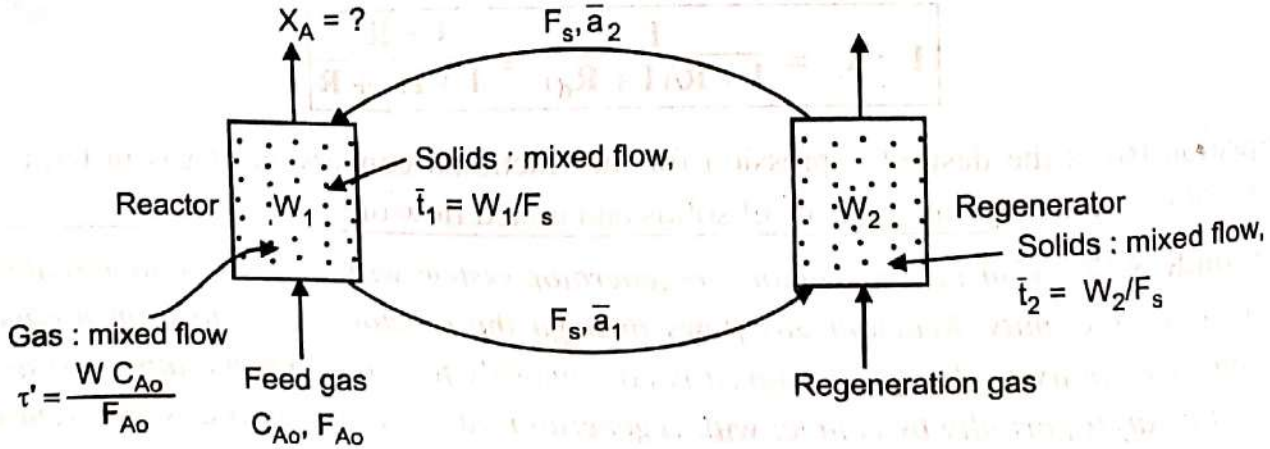


Fig. Ex 8.17 (b)

Procedure : Obtain expressions ... for the activity of the catalyst ... for the conversion of gas ... and then combine the expressions obtained.

Solids : Mixed flow of solids in both units.

Both deactivation and regeneration are first order processes.

We know that the extent of micromixing (any degree of mixing between complete mixing and complete segregation) does not affect a first order reaction (mixing has no effect on conversion).

Hence, as the deactivation and regeneration are first order processes, a macrofluid can be treated as a microfluid. So treating the solids as a microfluid, the performance equation for mixed flow is

$$\tau = \frac{C_{A0} - C_A}{-r_A'}$$

For solids (mixed flow of solids) becomes

$$\bar{t}_1 = \frac{\bar{a}_2 - \bar{a}_1}{\text{Rate of deactivation}}$$

$$= \frac{\bar{a}_2 - \bar{a}_1}{k_d \bar{a}_1} \quad \text{... for reactor}$$

and

$$\bar{t}_2 = \frac{\bar{a}_2 - \bar{a}_1}{\text{Rate of regeneration}}$$

$$= \frac{\bar{a}_2 - \bar{a}_1}{k_r (1 - \bar{a}_2)} \quad \text{... for regenerator}$$

Let $k_d \bar{t}_1 = R_d$ and $k_r \bar{t}_2 = R_r$

$$\bar{t}\,\bar{t}_1 = \frac{\bar{a}_2 - \bar{a}_1}{k_d \bar{a}_1}$$

$$k_d \bar{t}_1 \bar{a}_1 = \bar{a}_2 - \bar{a}_1$$

$$\bar{a}_2 = \bar{a}_1 + R_d \bar{a}_1 \quad \text{... (1)}$$

$$\bar{t}_2 = \frac{\bar{a}_2 - \bar{a}_1}{k_r (1 - \bar{a}_2)}$$

$$k_r \bar{t}_2 (1 - \bar{a}_2) = \bar{a}_2 - \bar{a}_1$$

$$R_r (1 - \bar{a}_2) = \bar{a}_2 - \bar{a}_1$$

$$R_r - R_r \bar{a}_2 = \bar{a}_2 - \bar{a}_1 \quad \ldots (2)$$

Substituting the value of $\bar{a}_2$ from Equation (1) into Equation (2), we get

$$R_r - R_r (\bar{a}_1 + R_d \bar{a}_1) = \bar{a}_1 + R_d \bar{a}_1 - \bar{a}_1$$

$$R_r - R_r \bar{a}_1 - R_r R_d \bar{a}_1 = R_d \bar{a}_1$$

$$\bar{a}_1 (R_d + R_r R_d + R_r) = R_r$$

$$\bar{a}_1 = \frac{R_r}{R_d + R_r R_d + R_r} \quad \ldots (3)$$

Equation (3) gives the activity of catalyst in the reactor as well as the activity of catalyst in the exit stream of solids from the reactor in terms of the variables of the system.

Gas : Mixed flow of gas.

For mixed flow of gas, X_A is given by

$$1 - X_A = \frac{1}{1 + k'\tau'\bar{a}_1} \quad \ldots (4)$$

(For this expression please refer to Example 8.16)

Substituting the value of $\bar{a}_1$ from Equation (3) into Equation (4), we get

$$1 - X_A = \frac{1}{1 + k'\tau' [R_r/(R_d + R_r R_d + R_r)]}$$

Let $R = k'\tau'$

$$\boxed{1 - X_A = \frac{1}{1 + RR_r/(R_d + R_r R_d + R_r)}} \quad \ldots (5$$

Equation (5) is the desired expression for the fractional conversion of gas in terms of th variables of the system for mixed flow of solids and mixed flow of gas in the two units.

Extra : Obtain an expression for the fractional conversion for plug flow of gas in the tw units ... all else remaining unchanged.

We have for solids :

$$\bar{a}_1 = \frac{R_r}{R_r + R_r R_d + R_d} \quad \ldots (A)$$

(Obtained previously)

Gas : Plug flow of gas.

For plug flow of gas X_A is given by

$$1 - X_A = e^{-k'\tau'\bar{a}_1}$$

(Please refer to the solution of Example 8.15 for derivation of the above expression.)

Let $R = k'\tau'$

$$1 - X_A = e^{-R\bar{a}_1} \quad \ldots (B)$$

Putting value of $\bar{a}_1$ from Equation (A) into Equation (B), we get

$$\boxed{1 - X_A = e^{-RR_r/(R_r + R_r R_d + R_d)}} \quad \ldots (C)$$

Equation (C) is the desired expression for the fractional conversion of gas in terms of the variables of the system for - mixed flow of solids and plug flow of gas - in the two units (reactor and regenerator).

Example 8.18 : *A packed bed reactor is run with fresh catalyst at 600 K. Five weeks later when the temperature reaches 800 K, the reactor is shut down for reactivation of the catalyst. In addition, at any time/moment the reactor is isothermal. Determine the activity of the catalyst at the time of reactivation, assuming optimal operations.*

The rate of reaction with fresh catalyst is given by

$$-r_A = kC_A^2 \qquad , \; k = k_0 e^{-7200/T}$$

The rate of deactivation is unknown.

Solution : The optimal operation is the one in which the temperature is raised with time in order to keep the conversion from the reactor unchanged. The optimal operation criterion can be written as :

$$k'a = \text{constant during the run}$$

$\therefore$

$$ka = \text{constant during the run}$$

$\therefore$

$$k_1 a_1 = k_2 a_2$$

a_1 = activity of catalyst at 600 K (at the start of the run) $T_1 = 600$ K

a_2 = activity of catalyst at 800 K (at the end of the run) $T_2 = 800$ K

a_1 = activity of the fresh catalyst

a_2 = activity of the catalyst at the time of reactivation

$$\frac{a_2}{a_1} = \frac{k_1}{k_2}$$

$$a_2 = \frac{k_2}{k_1} a_1$$

Given : $k = k_0 e^{-7200/T}$

$$a_2 = \frac{k_0 e^{-7200/T_1}}{k_0 e^{-7200/T_2}} a_1$$

$$a_2 = e^{-7200/T_1 + 7200/T_2} a_1$$

$$a_2 = e^{-7200/600 + 7200/800} a_1$$

$$a_2 = 0.04978\, a_1 \approx 0.05\, a_1$$

Activity of catalyst at the time of reactivation = 0.05 a_1 = 5% of a_1 = 5% of the activity of the fresh catalyst, i.e., 5% of the activity of catalyst at the start of the run ... Ans.

❑❑❑

CHAPTER 9

ENZYME FERMENTATION

Fermentations are reactions in which a raw organic feed is converted into product by the action of microbes or by the action of enzymes.

1. Enzyme Fermentations :

These can be represented by

$$\text{(Organic feed, A)} \xrightarrow[\text{acting as a catalyst}]{\text{enzyme E}} \text{(Product chemical, R)}$$

2. Microbial Fermentations :

These can be represented by

$$\text{(Organic feed, A)} \xrightarrow[\text{which acts as a catalyst}]{\text{microbe C}} \text{(Product, R)} + \text{(More cells, C)}$$

(i) In enzyme fermentation, the catalytic agent - the enzyme does not reproduce itself.

(ii) In microbial fermentation, the catalytic agent - the cell or microbe reproduces itself.

(iii) Enzyme fermentation is not autocatalytic.

(iv) Microbial fermentation is autocatalytic.

Fermentations are used to produce alcoholic beverages, antibiotics, vitamins, yeast, tea, etc. and also to treat waste water.

Enzyme :

An enzyme (E) is a protein or high molecular weight (molecular weight > 10000) protein-like substance with catalytic properties. Enzymes are highly specific in nature - in that one enzyme can catalyze only one particular reaction or one group of reactions. Enzymes are produced by micro-organisms and they usually work, i.e., catalyze reactions under mild conditions (pH : 4 to 9 and temperature : 24 to 70°C).

In deriving the names of most enzymes it is a usual practice to add the suffix '– ase' to a major part of the name of the substrate on which the enzyme acts. For example, the enzyme catalyzing the decomposition of urea is urease (urease is derived as : ure from urea (substrate) + ase = urease).

Substrate (A) :

It is the substance that is chemically converted into product at an accelerated rate by the action of an enzyme.

We consider soluble enzyme - soluble substrate type enzyme reactions in our development to follow.

Michaelis – Menten Kinetics ⇒ M - M Kinetics :

Consider the enzyme - substrate reaction

$$A \xrightarrow{\text{enzyme E}} R$$

It is observed that the said reaction exhibits the following rate characteristics :

(i) At all C_A (the reactant concentration) : The rate is proportional to the total enzyme concentration (C_{Eo}).

(ii) At low C_A : The rate is proportional to the reactant concentration (C_A). Hence, the rate is first order in A at low (C_A).

(iii) At high C_A : The rate is independent of the reactant concentration (C_A).

Fig. 9.1 shows this behaviour of the rate.

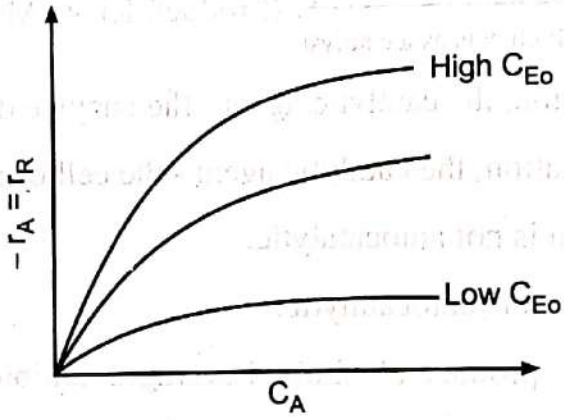


Fig. 9.1 : Rate - concentration curves for enzyme catalyzed reactions

The Michaelis - Menten equation that accounts for this rate behaviour is

$$-r_A = r_R = k\frac{C_{Eo}\,C_A}{C_M + C_A} \qquad \ldots (9.1)$$

where C_M is called the Michaelis constant and C_{Eo} is the total enzyme concentration.

Michaelis and Menten explained this behaviour and the rate form with the following two-step elementary reaction mechanism :

1. $A + E \underset{k_2}{\overset{k_1}{\rightleftharpoons}} X \quad , \quad (X = E \cdot A)$

2. $X \xrightarrow{k_3} R + E$

1. Reversible reaction between the enzyme (E) and the substrate / reactant (A) to form intermediate enzyme - substrate complex (X).
2. Irreversible decomposition of complex (X) to give product (R) and enzyme (E).

M - M considered that the reaction proceeded as per the mechanism given above.

The rate of formation of the product R is

$$r_R = k_3 C_X \quad \ldots (9.2)$$

The net rate of formation of the complex is

$$r_X = \underbrace{k_1 C_E C_A}_{\text{formation of X}} - \underbrace{k_2 C_X - k_3 C_X}_{\text{disappearance of X}} \quad \ldots (9.3)$$

From the elementary steps of reaction it is clear that the enzyme is not consumed by the reaction. So the total enzyme concentration in the reaction system C_{Eo} is constant and equal to the sum of the concentrations of free enzyme E and enzyme-substrate complex X.

$$\underbrace{C_{Eo}}_{\text{total enzyme}} = \underbrace{C_E}_{\text{free enzyme}} + \underbrace{C_X}_{\text{enzyme attached to reactant/substrate}} \quad \ldots (9.4)$$

$$\therefore \quad C_E = C_{Eo} - C_X \quad \ldots (9.5)$$

According to the pseudo-steady-state hypothesis (PSSH), the net rate of formation of the intermediate complex is zero. Thus,

$$r_X = k_1 C_E C_A - k_2 C_X - k_3 C_X = 0$$

$$\therefore \quad k_1 C_E C_A = k_2 C_X + k_3 C_X$$

$$C_X (k_2 + k_3) = k_1 C_E C_A$$

Substituting the value of C_E from Equation (9.5) in the above equation, we get

$$C_X (k_1 + k_3) = k_1 (C_{Eo} - C_X) C_A$$

$$C_X (k_2 + k_3) = k_1 C_{Eo} C_A - k_1 C_A C_X$$

$$C_X (k_2 + k_3) + k_1 C_A C_X = k_1 C_{Eo} C_A$$

$$C_X [(k_2 + k_3) + k_1 C_A] = k_1 C_{Eo} C_A$$

$$\therefore \quad C_X = \frac{k_1 C_{Eo} C_A}{(k_2 + k_3) + k_1 C_A} \quad \ldots (9.6)$$

Replacing C_X in Equation (9.2) by its value given by Equation (9.6) gives

$$r_R = \frac{k_3 k_1 C_{Eo} C_A}{(k_2 + k_3) + k_1 C_A}$$

Rearranging, we get

$$r_R = \frac{k_3 C_{Eo} C_A}{\left(\frac{k_2 + k_3}{k_1}\right) + C_A}$$

Let, $C_M = \frac{k_2 + k_3}{k_1}$, a constant, called the Michaelis constant

$$\therefore \quad r_R = \frac{k_3 C_{Eo} C_A}{C_M + C_A} \quad \ldots (9.7)$$

As $-r_A = r_R$, we can write

$$\boxed{-r_A = r_R = \frac{k_3 C_{Eo} C_A}{C_M + C_A}} \quad \ldots (9.8)$$

Equation (9.8) is the Michaelis - Menten Equation.

[Comparing Equation (9.8) with Equation (9.1) ... $k = k_3$]

where k_3 is the reaction rate constant for irreversible decomposition of enzyme-substrate complex.

Let r_{max} represents the maximum rate of reaction for a given total enzyme concentration (C_{Eo}). The rate of formation of product is maximum when all the enzyme is associated with the enzyme-substrate complex.

$$\therefore \quad r_{max} = k_3 C_{Eo} \quad \ldots (9.9)$$

With this the Michaelis - Menten Equation [Equation (9.8)] takes the form

$$-r_A = r_R = \frac{r_{max} C_A}{C_M + C_A} \quad \ldots (9.10)$$

At low C_A (at low substrate reactant concentration) : $C_A << C_M$

$\therefore$ Equation (9.10) becomes

$$-r_A = \frac{r_{max} C_A}{C_M} \quad \ldots \text{first order behaviour}$$

At high C_A : $C_A >> C_M$

$\therefore$ Equation (9.10) becomes

$$-r_A = r_R = r = r_{max} \quad \ldots \text{ zero order behaviour}$$

Let us now consider the case when C_A is such that the rate of reaction is equal to one-half the maximum rate.

$$-r_A = r = \frac{r_{max}}{2}$$

Using this value of rate in Equation (9.10), we get

$$\frac{r_{max}}{2} = \frac{r_{max} C_A}{C_M + C_A}$$

$$C_M + C_A = 2 C_A$$

$$\therefore \quad C_M = C_A$$

i.e., the Michaelis constant is equal to the substrate concentration at which the reaction rate is equal to one-half the maximum rate.

Special features of M-M equation :

(i) When $C_A = C_M$ ... half the enzyme is in free/unbounded form and the other half is in bound form/combined form with A.

(ii) When $C_A << C_M$ ... most of the enzyme is in free form.

(iii) When $C_A >> C_M$ ... most of the enzyme is in bound form in ~~X~~ (A·E).

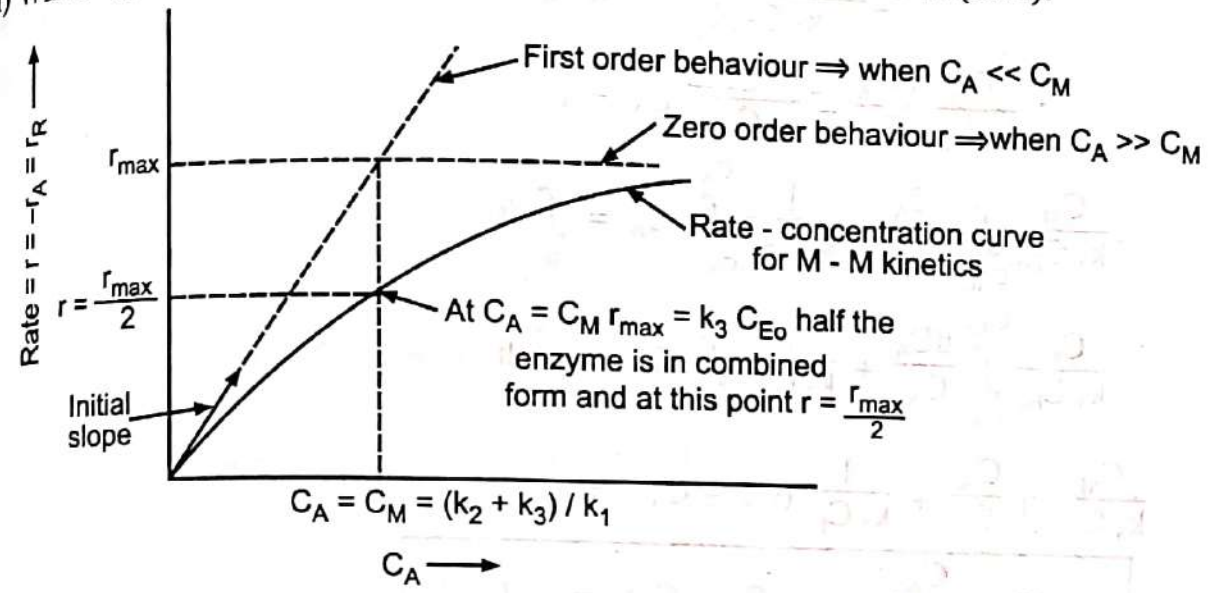


Fig. 9.2 : Special features of the Michaelis - Menten equation (Eq. 9.7)

$$-r_A = \frac{k_3 C_{Eo} C_A}{C_M + C_A}$$

$$-r_A = \frac{k_3 C_{Eo} C_A}{\frac{k_2 + k_3}{k_1} + C_A} = \frac{k_1 k_3 C_{Eo} C_A}{(k_2 + k_3) + k_1 C_A}$$

At very low C_A, it reduces to

$$-r_A = \frac{k_1 k_3 C_{Eo} C_A}{(k_2 + k_3)}$$

$$\therefore \quad \text{Initial slope} = \frac{k_1 k_3 C_{Eo}}{(k_2 + k_3)}$$

Fermentors :

Batch Fermentor or Plug Flow Fermentor :

[Here we are dealing with homogeneous liquid phase reactions → constant density reaction systems.]

For constant density systems :

$$t = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{(-r_A)} \qquad \text{... batch reactor}$$

$$\tau = -\int_{C_{Ao}}^{C_A} \frac{dC_A}{(-r_A)} \qquad \text{... plug flow reactor}$$

τ for plug flow is equivalent to t for the batch reactor. For a given duty : $\tau = t$

For this system let us integrate the Michaelis - Menten Equation.

$$-r_A = -\frac{dC_A}{dt} = \frac{k_3 C_{Eo} C_A}{C_M + C_A}$$

Rearranging, we get

$$-\frac{(C_M + C_A)\, dC_A}{k_3 C_{Eo} C_A} = dt$$

$$-\frac{C_M}{k_3 C_{Eo}} \frac{dC_A}{C_A} - \frac{1}{k_3 C_{Eo}} dC_A = dt$$

$$-\frac{C_M}{k_3 C_{Eo}} \int_{C_{Ao}}^{C_A} \frac{dC_A}{C_A} - \frac{1}{k_3 C_{Eo}} \int_{C_{Ao}}^{C_A} dC_A = \int_0^t dt$$

$$\frac{C_M}{k_3 C_{Eo}} \int_{C_A}^{C_{Ao}} \frac{dC_A}{C_A} + \frac{1}{k_3 C_{Eo}} \int_{C_A}^{C_{Ao}} dC_A = \int_0^t dt$$

$$\frac{C_M}{k_3 C_{Eo}} \ln \frac{C_{Ao}}{C_A} + \frac{1}{k_3 C_{Eo}} (C_{Ao} - C_A) = t$$

$$\underbrace{C_M \ln \frac{C_{Ao}}{C_A}}_{\text{first order term}} + \underbrace{(C_{Ao} - C_A)}_{\text{zero order term}} = k_3 C_{Eo}\, t \qquad \text{... (9.11)}$$

The concentration-time behaviour given by Equation (9.11) is graphically shown in Fig. 9.3.

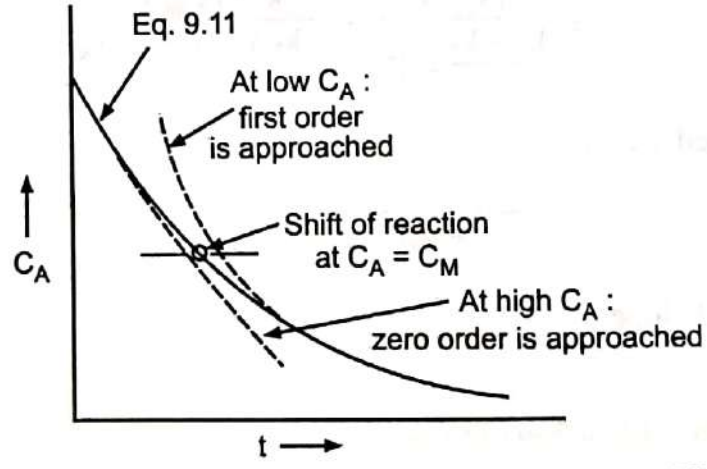


Fig. 9.3 : Concentration-time behaviour of the integrated M-M equation

Equation (9.11) cannot be plotted directly to get the values of the constants k_3 and C_M. To linearise, rearrange Equation (9.11) to give

$$C_M \ln \frac{C_{Ao}}{C_A} + (C_{Ao} - C_A) = k_3 C_{Eo}\, t$$

$$C_M + \frac{(C_{Ao} - C_A)}{\ln (C_{Ao} / C_A)} = k_3 C_{Eo} \cdot \frac{t}{\ln (C_{Ao} / C_A)}$$

$$\boxed{\frac{(C_{Ao} - C_A)}{\ln (C_{Ao} / C_A)} = -C_M + k_3 C_{Eo} \frac{t}{\ln (C_{Ao} / C_A)}} \qquad \text{... (9.12)}$$

Equation (9.12) can be plotted, as shown in Fig. 9.4, to give the constants k_3 and C_M.

Equation (9.11) can be expressed in terms of X_A.

$$C_M \ln\left(\frac{C_{Ao}}{C_A}\right) + (C_{Ao} - C_A) = k_3 C_{Eo} t \quad \text{... [Equation (9.11)]}$$

We have,

$$X_A = \frac{C_{Ao} - C_A}{C_{Ao}}$$

$$\therefore \quad C_{Ao} - C_A = C_{Ao} X_A$$

$$C_A = C_{Ao}(1 - X_A)$$

$$\frac{C_{Ao}}{C_A} = \frac{1}{(1 - X_A)}$$

In terms of X_A Equation (9.11) becomes

$$C_M \ln\left(\frac{1}{1 - X_A}\right) + C_{Ao} X_A = k_3 C_{Eo} t$$

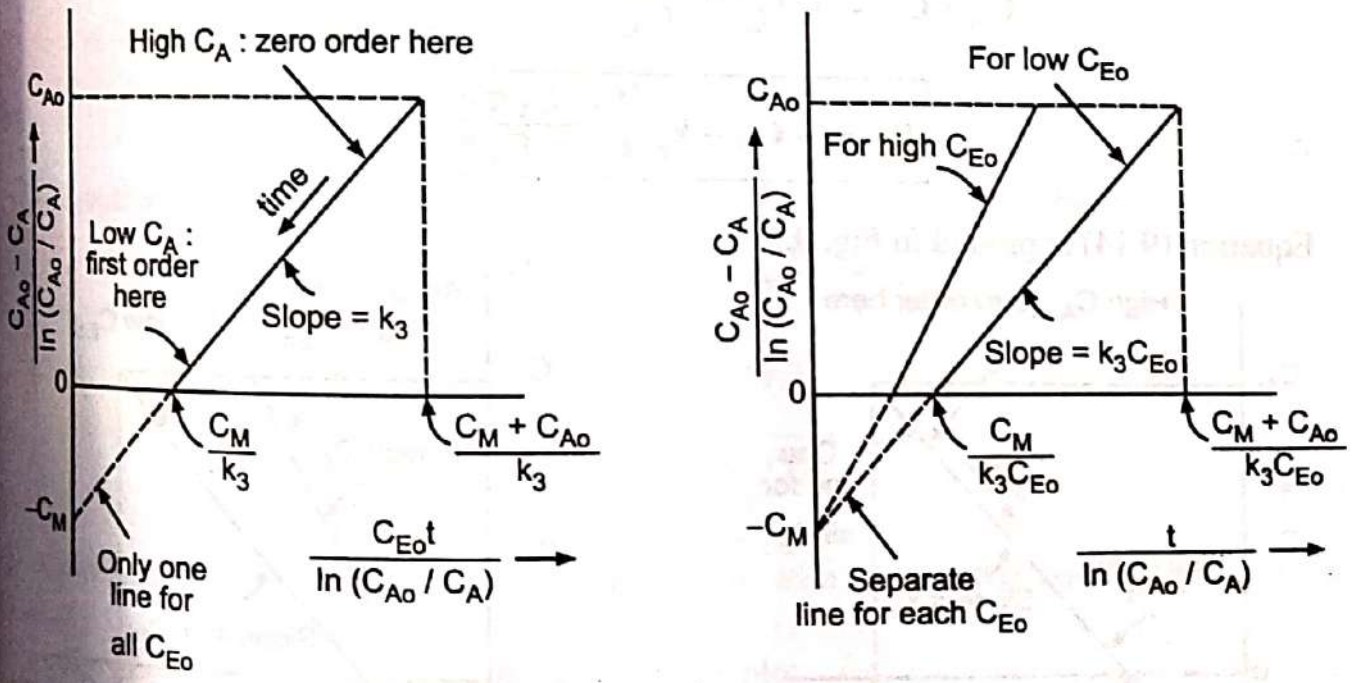


Fig. 9.4 : Either of these plots can be used to test and fit the M-M Equation [Equation (9.12)] using data taken in a batch reactor

Mixed Flow Fermentor :

The performance equation for a mixed flow reactor is

$$\tau = \frac{C_{Ao} - C_A}{-r_A}$$

The M-M equation is

$$-r_A = \frac{k_3 C_{Eo} C_A}{C_M + C_A}$$

Combining these two equations, we get

$$\tau = \frac{(C_{Ao} - C_A)(C_M + C_A)}{k_3 C_{Eo} C_A}$$

or
$$k_3 C_{Eo} \tau = \frac{(C_{Ao} - C_A)(C_M + C_A)}{C_A} \quad \text{... (9.13)}$$

It is not possible to get k_3 and C_M by plotting Equation (9.13). Therefore, rearranging Equation (9.13) to a form which when plotted gives k_3 and C_M.

$$\tau = \frac{(C_{Ao} - C_A)(C_M + C_A)}{k_3 C_{Eo} C_A}$$

$$\frac{(C_{Ao} - C_A)(C_M + C_A)}{k_3} = C_{Eo} C_A \tau$$

$$\frac{(C_M + C_A)}{k_3} = \frac{C_{Eo} C_A \tau}{C_{Ao} - C_A}$$

$$\therefore \quad C_M + C_A = k_3 \left(\frac{C_{Eo} C_A \tau}{C_{Ao} - C_A} \right)$$

$$\therefore \quad \boxed{C_A = -C_M + k_3 \left(\frac{C_{Eo} C_A \tau}{C_{Ao} - C_A} \right)} \quad \text{... (9.14)}$$

Equation (9.14) is plotted in Fig. 9.5.

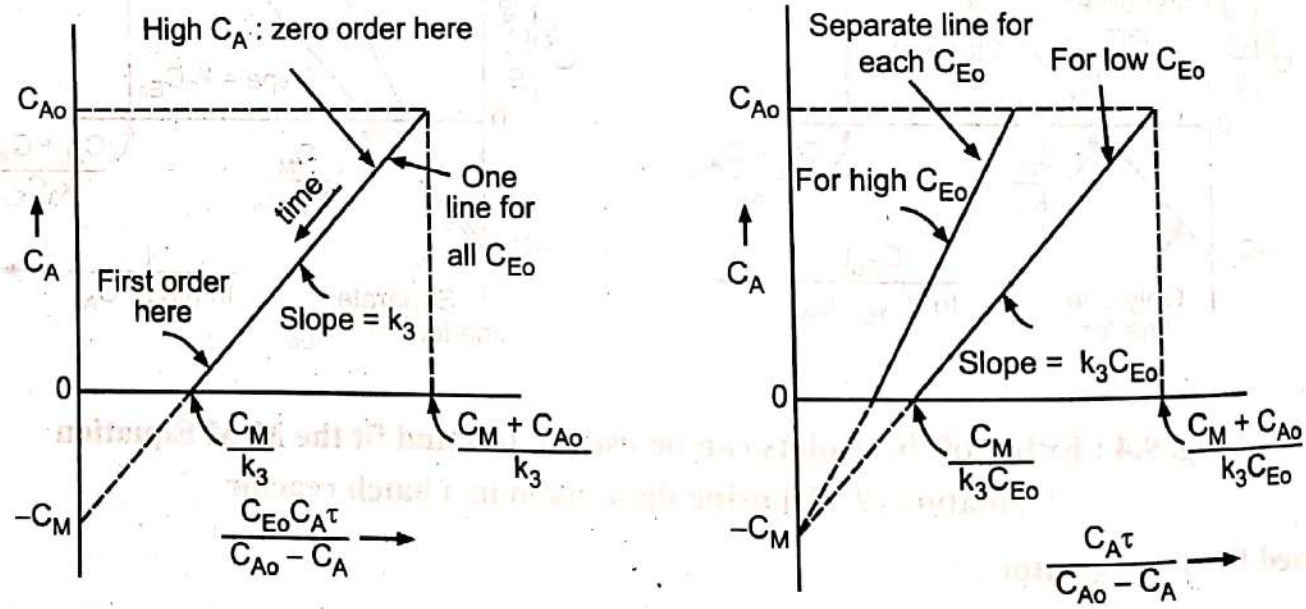


Fig. 9.5 : Either of these plots can be used to test and fit the M-M equation [Equation (9.14)] using mixed flow reactor data

Alternate approach to evaluate k, i.e., k_3 and C_M :

The Michaelis - Menten equation is

$$-r_A = \frac{k_3 C_{Eo} C_A}{C_M + C_A}$$

With $-r_a$ v/s C_A data in hand to obtain a straight line plot of the M-M equation to evaluate the constants k_3 and C_M, let us manipulate the M-M equation.

$\therefore$ Taking the reciprocal of both sides of the M-M Equation or inverting the M-M equation, we get

$$\frac{1}{(-r_A)} = \frac{C_M + C_A}{k_3 C_{Eo} C_A}$$

$$\boxed{\frac{1}{(-r_A)} = \frac{C_M}{k_3 C_{Eo}}\left(\frac{1}{C_A}\right) + \frac{1}{k_3 C_{Eo}}} \quad \ldots (9.15)$$

A plot of $\frac{1}{(-r_A)}$ v/s $\frac{1}{C_A}$ gives a straight line with an intercept of $\frac{1}{k_3 C_{Eo}}$ and having a slope of $\frac{C_M}{k_3 C_{Eo}}$. A plot of $\frac{1}{(-r_A)}$ v/s $\frac{1}{C_A}$ is called the Lineweaver plot.

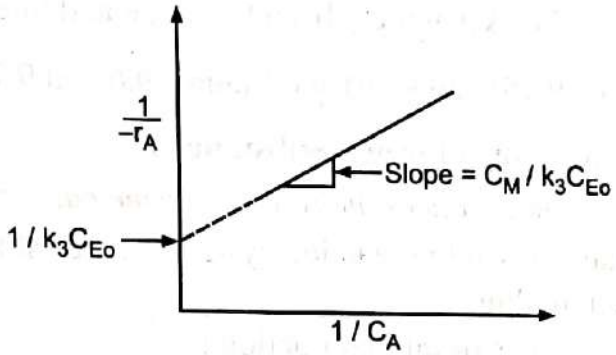


Fig. 9.6 : Lineweaver plot

Alternatively, multiplying Equation (9.15) by $k_3 C_{Eo} (-r_A)$ gives another form

$$\frac{k_3 C_{Eo} (-r_A)}{(-r_A)} = \frac{C_M k_3 C_{Eo} (-r_A)}{k_3 C_{Eo} C_A} + \frac{k_3 C_{Eo} (-r_A)}{k_3 C_{Eo}}$$

$$k_3 C_{Eo} = C_M\left[\frac{(-r_A)}{C_A}\right] + (-r_A)$$

$$\boxed{(-r_A) = k_3 C_{Eo} - C_M\left[\frac{(-r_A)}{C_A}\right]} \quad \ldots (9.16)$$

A plot of $(-r_A)$ v/s $(-r_A)/C_A$ gives a straight line with an intercept equal to $k_3 C_{Eo}$ and a slope equal to $-C_M$.

A plot of $(-r_A)$ v/s $(-r_A)/C_A$ is called the Eadie plot.

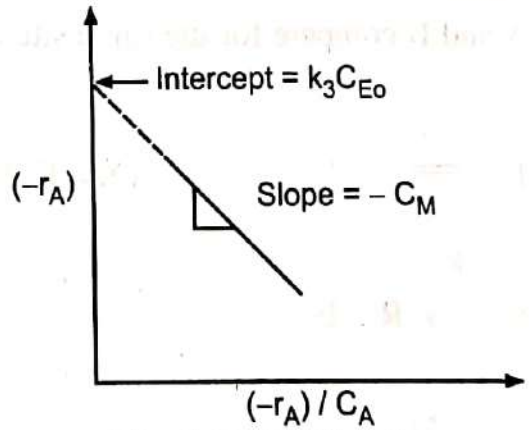


Fig. 9.7 : Eadie plot

1. Measure $C_{A\,out}$ v/s τ (or t) data in any one of the three ideal reactors - batch, mixed, or plug flow.
2. Evaluate $(-r_A)$ at various C_A.

For mixed flow data : use $-r_A = (C_{Ao} - C_A)/\tau$ to get $-r_A$.

For plug flow or batch data : plot C_A v/s τ (or t), draw tangents to the curve at various C_A and measure slopes. These slopes are the rates of reaction at these C_A.

3. Plot $-r_A$ v/s C_A in one of the following ways :

   $(-r_A)$ v/s $(-r_A)/C_A$ $\rightarrow$ the Eadie plot.

   $1/(-r_A)$ v/s $1/C_A$ $\rightarrow$ the Lineweaver plot.

and obtain the constants k (i.e., k_3) and C_M from the slope and intercept of these plots.

[Please refer to Equations (9.15) and 9.16) and Figures 9.6 and 9.7.]

Inhibition of Enzyme Reactions by a foreign Substance :

An inhibitor is a substance that reduces the rate of enzyme catalyzed reaction of A to R.

If the presence of substance B (in the reaction system) causes slowdown of enzyme reaction of A to R, then B is called an inhibitor.

We consider the following types of inhibitor action :

1. Competitive inhibition.
2. Noncompetitive inhibition ..... both are reversible.

The enzyme molecule contains active sites and in this respect it is analogous to the surface of solid catalyst.

When the substrate (A) and inhibitor (B) attack the same site on the enzyme then the inhibition is called as competitive. Here the resulting inhibitor - enzyme complex is inactive.

Noncompetitive inhibition occurs with enzymes containing at least two different types of sites.

When the inhibitor (B) attacks different type of site on the enzyme than the substrate, and in doing so it stops the action of A, then the inhibition is called as noncompetitive.

Competitive Inhibition – Kinetics :

In competitive inhibition, A and B compete for the same site on the enzyme. The mechanism of this action is given by

$$A + E \underset{k_2}{\overset{k_1}{\rightleftharpoons}} X \qquad (X = E \cdot A)$$

$$X \xrightarrow{k_3} R + E$$

$$B + E \underset{k_5}{\overset{k_4}{\rightleftharpoons}} Y \qquad (Y = E \cdot B)$$

For this mechanism the final rate form is given by

$$\boxed{r_R = \frac{k_3 C_{Eo} C_A}{C_M + C_A + NC_{Bo} C_M} = \frac{k_3 C_{Eo} C_A}{C_M (1 + NC_{Bo}) + C_A}} \quad \ldots (9.17)$$

where, $C_M = \frac{k_2 + k_3}{k_1}$, mol/m^3

and $N = \frac{k_4}{k_5}$, m^3/mol

Equation (9.17) is the kinetic expression for competitive inhibition.

Comparing Equation (9.17) ... of systems with inhibition with Equation (9.7) ... of systems without inhibition, we come to know that to get the rate equation for systems with competitive inhibition we have to only replace C_M in Equation (9.7) (in the M-M equation) by $C_M (1 + NC_{Bo})$

Non-competitive Inhibition – Kinetics :

In non-competitive inhibition, A attacks one type of site on the enzyme, B attacks the other and in doing so it stops the action of A.

For inhibition of this type with mechanism

$$A + E \underset{k_2}{\overset{k_1}{\rightleftharpoons}} X \qquad , X = E \cdot A$$

$$X \xrightarrow{k_3} R + E$$

$$B + E \underset{k_5}{\overset{k_4}{\rightleftharpoons}} Y \qquad , Y = E \cdot B$$

$$B + X \underset{k_7}{\overset{k_6}{\rightleftharpoons}} Z \qquad , Z = B \cdot X$$

the rate is given by

$$r_R = \frac{k_3 C_{Eo} C_A}{C_M + C_A + NC_{Bo} C_M + LC_A C_{Bo}}$$

$$\boxed{r_R = \frac{\frac{k_3}{(1 + LC_{Bo})} \cdot C_{Eo} C_A}{C_M \left(\frac{1 + NC_{Bo}}{1 + LC_{Bo}}\right) + C_A}} \quad \ldots (9.18)$$

where, $C_M = \frac{k_2 + k_3}{k_1}$, $N = \frac{k_4}{k_5}$ and $L = \frac{k_6}{k_7}$

Comparison of Equation (9.18) with Equation (9.7) reveals that both k_3 and C_M of the M-M equation are modified here. To obtain the rate equation for non-competitive inhibition using Equation (9.7) replace k_3 by $\frac{k_3}{(1 + LC_{Bo})}$ and C_M by $C_M \left[\frac{(1 + NC_{Bo})}{(1 + LC_{Bo})}\right]$

Method to find Inhibition Type from Experiment :

From C v/s t (or τ) data collected by conducting either batch, mixed flow or plug flow runs, construct one of the recommended plots for inhibition-free systems as shown in Fig. 9.4 or Fig. 9.5. For the systems with inhibitions, these plots are modified as shown in Fig. 9.8 and Fig. 9.9.

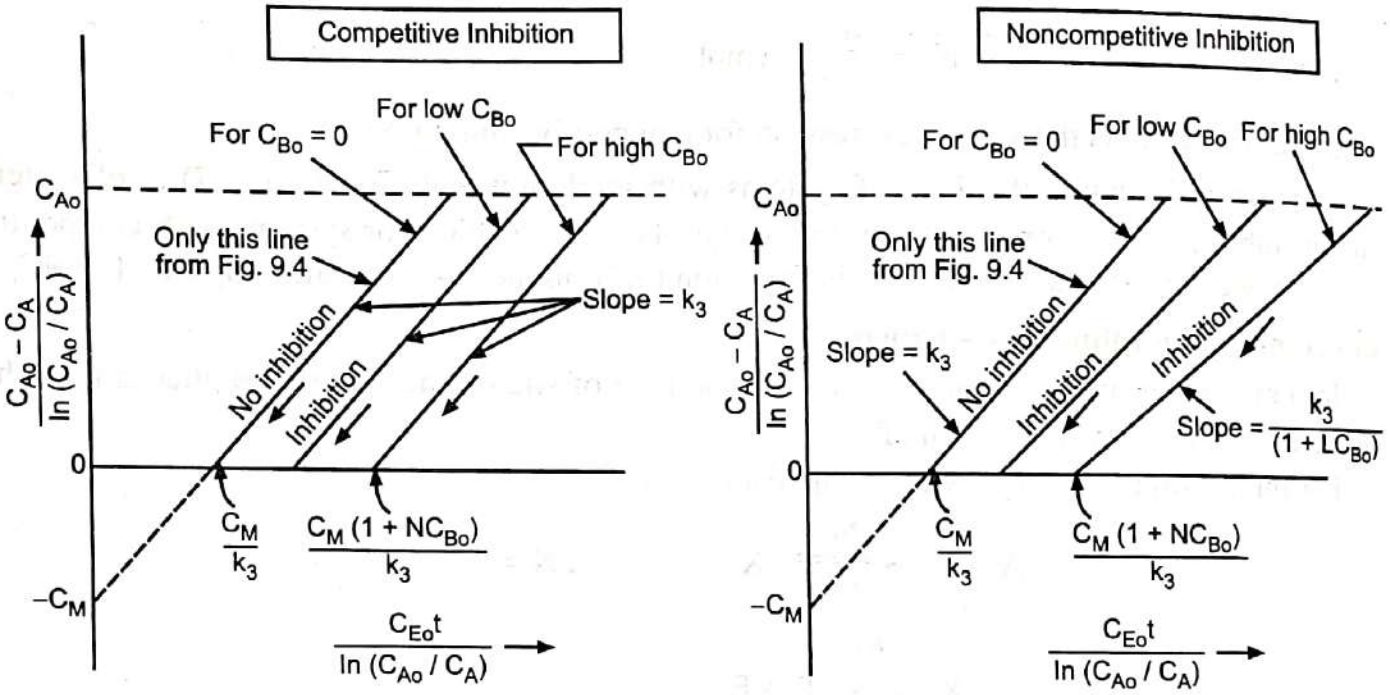


Fig. 9.8 : Effect of inhibition on batch or plug flow reactor data (Parallel lines for competitive inhibition. Lines are not parallel for non-competitive inhibition)

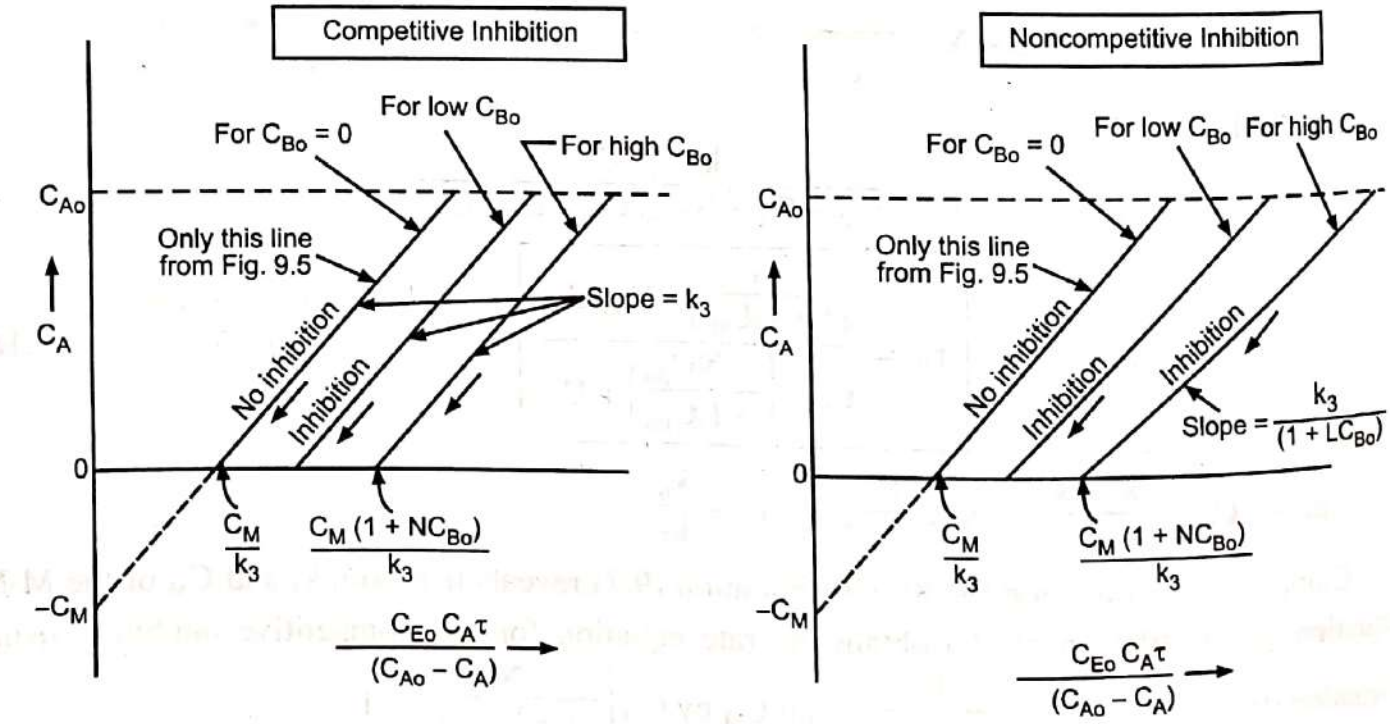


Fig. 9.9 : Effect of inhibition (i.e., presence of B) on mixed flow reactor data

SOLVED EXAMPLES

Example 9.1 : *Substrate A and enzyme E flow through a mixed flow reactor of olume (V) 6l. Find a rate equation to represent the action of enzyme on the substrate using the llowing data :*

C_{Eo}, mol/l	C_{Ao}, mol/l	C_A, mol/l	v, l/h
0.02	0.20	0.04	3
0.01	0.30	0.15	4
0.001	0.69	0.60	1.2

Solution : Let us try the M - M equation.

The Michaelis-Menten equation is

$$-r_A = \frac{k_3 C_{Eo} C_A}{C_M + C_A}$$

where C_M is the Michaelis constant.

Coupling M – M equation and the performance equation for a mixed flow reactor, we get

$$C_A = -C_M + k_3\left(\frac{C_{Eo} C_A \tau}{C_{Ao} - C_A}\right)$$

Plot C_A v/s $\frac{C_{Eo} C_A \tau}{(C_{Ao} - C_A)}$ for all C_{Eo}. It should result in a straight line for data to fit reasonably he M-M equation.

$\tau = V/v$ ... calculate it for each feed rate and then $\frac{C_{Eo} C_A \tau}{C_{Ao} - C_A}$ and tabulate

$V = 6l$

C_{Eo}, mol/l	0.02	0.01	0.001
C_{Ao}, mol/l	0.20	0.30	0.69
C_A, mol/l	0.04	0.15	0.60
v, l/h	3	4	1.2
V, l	6	6	6
τ, h^{-1}	2	1.5	5
$\frac{C_{Eo} C_A \tau}{C_{Ao} - C_A}$, mol/(l·h)	0.01	0.015	0.033

Plot C_A v/s $\frac{C_{Eo} C_A \tau}{(C_{Ao} - C_A)}$.

Refer to Fig. Ex 9.1.

Since we get a straight line, the data satisfactorily fit the M - M equation.

From the plot : slope = k_3 = 24 h

C_M = 0.20 mol/l

Thus, the rate equation to represent the action of enzyme on the substrate is

$$\boxed{-r_A = \frac{24\, C_{Eo} C_A}{0.20 + C_A}}$$ ... Ans.

Extra :

From Fig. Ex 9.1 : $\frac{C_M}{k_3} = 8.5 \times 10^{-3}$

$C_M = 8.5 \times 10^{-3} k_3 = 8.5 \times 10^{-3} \times 24 = 0.204 \approx 0.20$

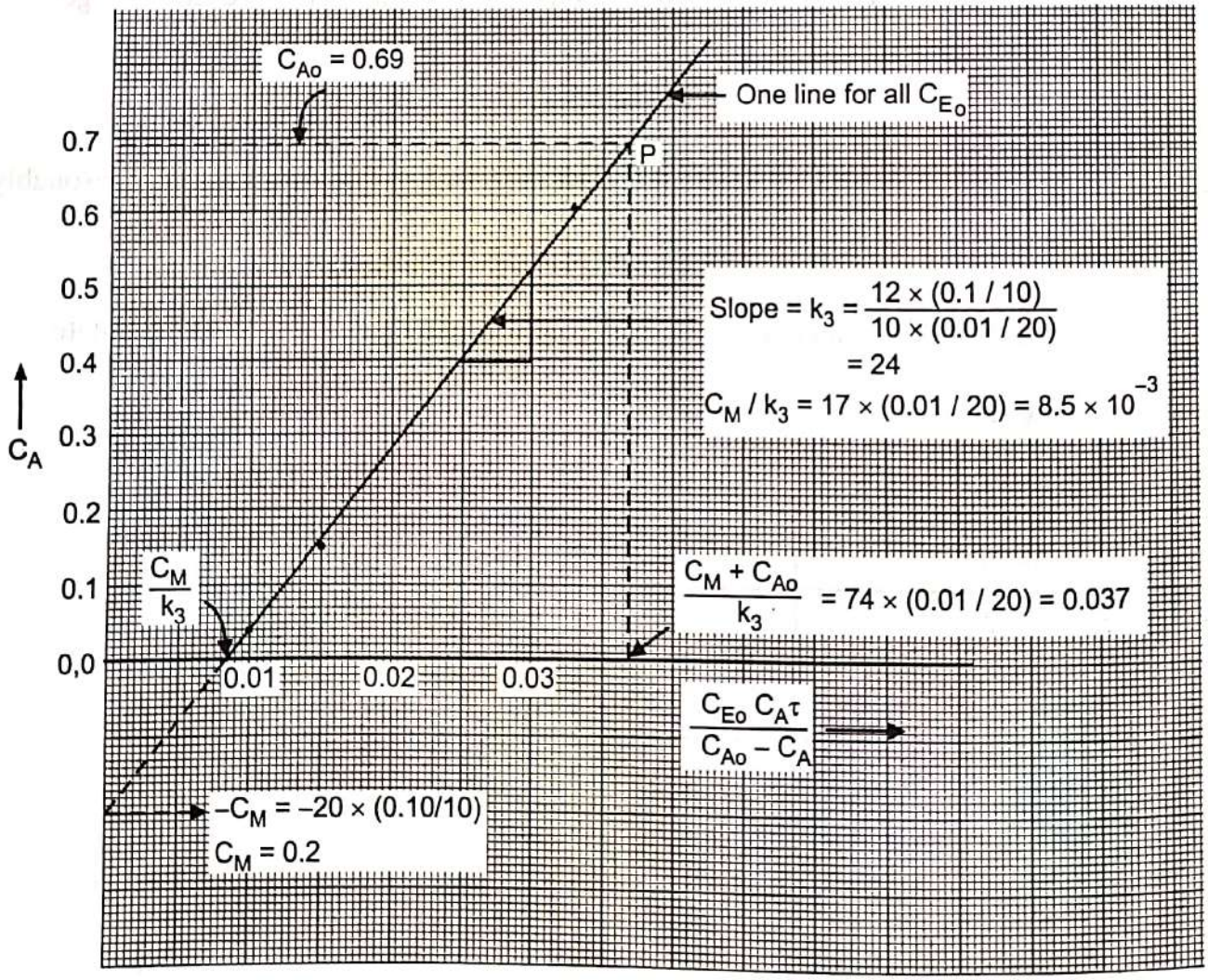


Fig. Ex 9.1

Through $C_{Ao} = 0.69$ on y-axis, draw a horizontal line to cut the straight line at point P and through the point P drop a vertical on x-axis and read it.

$$\text{Abscissa} = \frac{C_M + C_{Ao}}{k_3} = 0.037 \quad \text{... from Fig. E 9.1}$$

$$\frac{C_M + C_{Ao}}{k_3} = 0.037$$

$$k_3 = \frac{C_M + C_{Ao}}{0.037}$$

$$= \frac{0.20 + 0.69}{0037}$$

$$= 24.05 \approx 24 \text{ h} \Leftarrow \text{check.}$$

Example 9.2 : *Sucrose is hydrolysed at ambient temperature by the enzyme sucrase as follows :*

$$\textit{sucrose} \xrightarrow{\textit{sucrase}} \textit{products.}$$

Starting with a sucrose concentration $C_{Ao} = 1$ mol/m³ and a sucrase (enzyme) concentration $C_{Eo} = 0.01$ mol/m³, the following kinetic data are obtained in a batch reactor.

t, h	*2*	*6*	*10*
C_A, mol/m³	*0.68*	*0.16*	*0.006*

Find the rate equation for this hydrolysis reaction.

Solution : $\text{sucrose} \xrightarrow{\text{sucrase}} \text{products.}$

Assume that the Michaelis - Menten equation represents the kinetics of this reaction

$$-r_A = \frac{k_3 C_{Eo} C_A}{C_M + C_A}$$

where C_M is the Michaelis constant.

For a batch reactor with the M - M equation, we have

$$\frac{C_{Ao} - C_A}{\ln (C_{Ao}/C_A)} = -C_M + \frac{k_3 C_{Eo} t}{\ln (C_{Ao}/C_A)}$$

For our assumption to be correct, a plot of $(C_{Ao} - C_A)/\ln (C_{Ao}/C_A)$ v/s $t / \ln (C_{Ao}/C_A)$ should yield a straight line.

$C_{Ao} = 1$ mol/m³ and $C_{Eo} = 0.01$ mol/m³

t, h	2	6	10
C_A, mol/m³	0.68	0.16	0.006
$C_{Ao} - C_A$, mol/m³	0.32	0.84	0.994
$\ln (C_{Ao}/C_A)$	0.3857	1.83	5.11
$(C_{Ao} - C_A)/\ln (C_{Ao}/C_A)$, mol/m³	0.83	0.46	0.194
$\frac{t}{\ln (C_{Ao}/C_A)}$, h	5.2	3.3	1.96

The runs are carried out in a batch reactor at different C_{Eo}.

For a batch reactor with M - M equation, we have

$$(C_{Ao} - C_A)/\ln (C_{Ao}/C_A) = -C_M + k_3 C_{Eo} t / \ln (C_{Ao}/C_A)$$

For our guess to be correct, a plot of $(C_{Ao} - C_A)/\ln (C_{Ao}/C_A)$ v/s $C_{Eo}t / \ln (C_{Ao}/C_A)$ should yield a straight line. This plot gives one line for all C_{Eo}.

Run	1	2	3
t	1	1	1
C_{Eo}	3	2	1
C_{Ao}	400	200	20
C_A	10	5	1
$C_{Ao} - C_A$	390	195	19
$\ln (C_{Ao}/C_A)$	3.7	3.7	3
$\frac{C_{Ao} - C_A}{\ln C_{Ao}/C_A}$	105.7	52.84	6.33
$\frac{C_{Eo}\, t}{\ln C_{Ao}/C_A}$	0.813	0.54	0.333

Plot $(C_{Ao} - C_A)/\ln (C_{Ao}/C_A)$ v/s $C_{Eo}t / \ln (C_{Ao}/C_A)$
(as the runs are taken at different C_{Eo}).

Refer to Fig Ex 9.3 :

The plot results in a straight line. A straight line fit of the data indicates that the M - M equation satisfactorily represents the action of enzyme on the substrate (kinetics of the reaction), i.e., our guess is correct.

$$\text{slope} = k_3 = 206 \text{ h}^{-1}$$

$$C_M = 60 \text{ mol/m}^3$$

$\therefore$ The rate equation is

$$\boxed{-r_A = \frac{206\, C_{Eo}\, C_A}{60 + C_A}} \quad \text{... Ans.}$$

Extra :

From Fig. Ex 9.2, we have

$$\frac{C_M}{k_3} = 0.29$$

$$k_3 = \frac{C_M}{0.29} = \frac{60}{0.29} = 206.9 \ (\approx \text{ same as slope value})$$

For $C_{Ao} = 20 \text{ mol/m}^3$ we have,

$$\frac{C_M + C_{Ao}}{k_3} = 0.39 \ \ldots \text{ from Fig. E 9.2}$$

$$k_3 = \frac{C_M + C_{Ao}}{0.39} = \frac{60 + 20}{0.39} = 205.13 \text{ h}^{-1}$$

[k values obtained by different approaches are the almost same.]

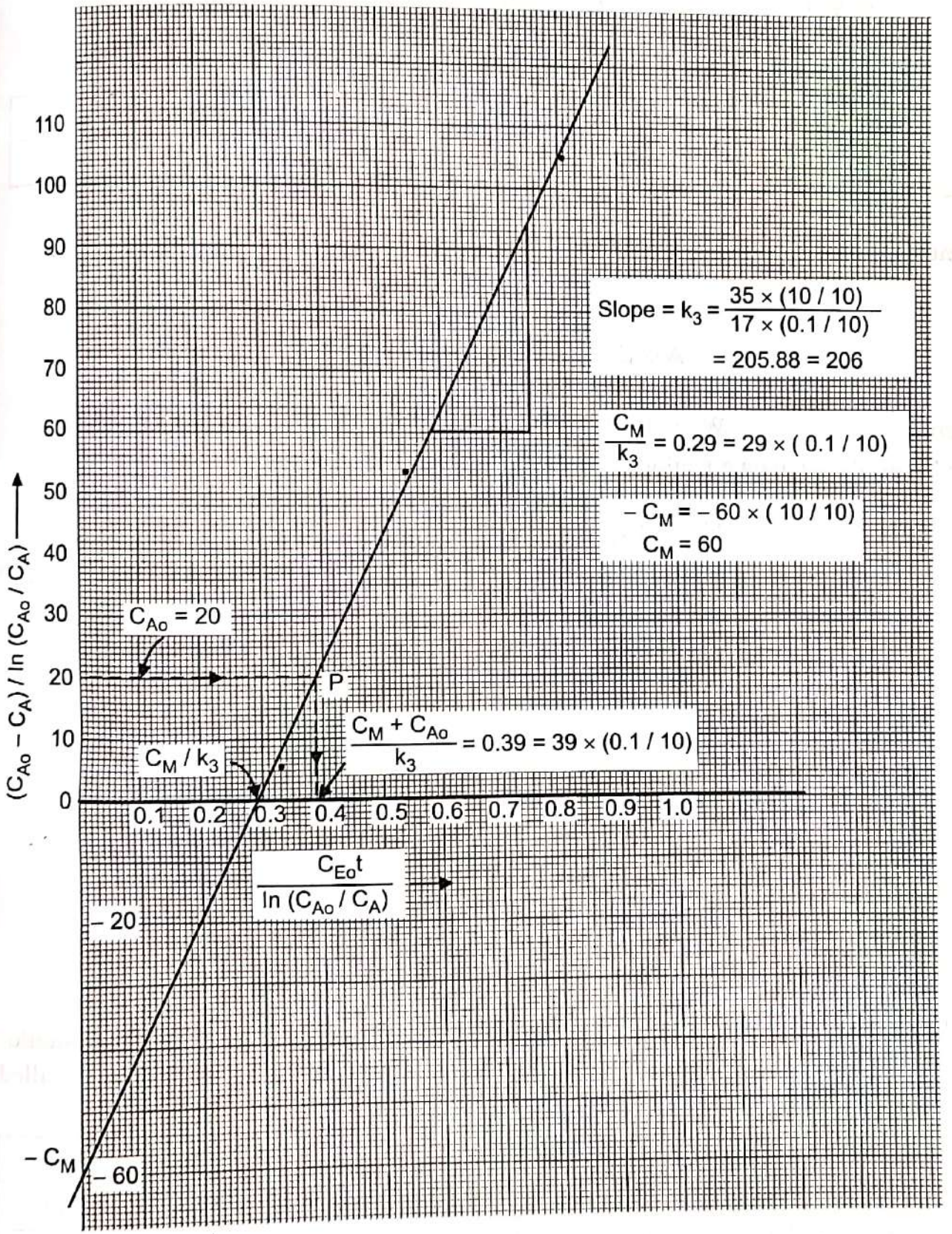

110
100
90
80
70
60
50
40
30
20
10
0
– 20
– 60
– C_M
$(C_{Ao} - C_A) / \ln (C_{Ao} / C_A)$
$C_{Ao} = 20$
P
C_M / k_3
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
$\frac{C_{Eo}t}{\ln (C_{Ao} / C_A)}$
Slope = $k_3 = \frac{35 \times (10 / 10)}{17 \times (0.1 / 10)}$
$= 205.88 = 206$
$\frac{C_M}{k_3} = 0.29 = 29 \times (0.1 / 10)$
$- C_M = - 60 \times (10 / 10)$
$C_M = 60$
$\frac{C_M + C_{Ao}}{k_3} = 0.39 = 39 \times (0.1 / 10)$

Fig. Ex 9.3

Example 9.4 : *The following data (the rate of reaction as a function of urea concentration) is available for the reaction*

$$urea + urease \underset{k_2}{\overset{k_1}{\rightleftharpoons}} urea \cdot urease \xrightarrow[-H_2O]{k_3} 2NH_3 + CO_2 + urease$$

Determine the Michaelis-Menten parameters for this reaction.

C_{urea}, kmol/m³	0.20	0.02	0.01	0.005	0.002
$-r_{urea}$, kmol/(m³·s)	1.08	0.55	0.38	0.20	0.09

Solution : $\text{urea} + \text{urease} \underset{k_2}{\overset{k_1}{\rightleftharpoons}} \text{urea} \cdot \text{urease} \xrightarrow[-H_2O]{k_3} 2NH_3 + CO_2 + \text{urease}$

$$A + E \underset{k_2}{\overset{k_1}{\rightleftharpoons}} A \cdot E \xrightarrow[-W]{k_3} P + E$$

where $W = H_2O$, $P = 2NH_3 + CO_2$, $A = \text{urea } [NH_2CONH_2]$

The Michaelis-Menten equation is

$$-r_A = \frac{k_3 C_{Eo} C_A}{C_M + C_A}$$

Let, $r_{max} = k_3 C_{Eo}$ ... the maximum rate of reaction for a given total enzyme concentration

$$\therefore \quad -r_A = \frac{r_{max} C_A}{C_M + C_A}$$

Taking the reciprocal of both sides, we get

$$\frac{1}{-r_A} = \frac{C_M + C_A}{r_{max} C_A}$$

$$\frac{1}{-r_A} = \left(\frac{C_M}{r_{max}}\right)\frac{1}{C_A} + \frac{1}{r_{max}}$$

where r_{max} and C_M are the Michaelis-Menten parameters.

Now we will evaluate the M - M parameters by plotting $1/(-r_A)$ v/s $1/C_A$.

A plot of the reciprocal of rate of reaction versus the concentration of urea should be a straight line with a slope = C_M/r_{max} and an intercept = $1/r_{max}$. Such a type of plot is called as a Lineweaver plot.

C_A, kmol/m³	0.20	0.02	0.01	0.005	0.002
$-r_A$, kmol/(m³·s)	1.08	0.55	0.38	0.20	0.09
$1/C_A$, m³/kmol	5	50	100	200	500
$1/(-r_A)$, $\frac{(m^3 \cdot s)}{kmol}$	0.926	1.82	2.63	5	11.11

Plot $1/(-r_A)$ v/s $1/C_A$.

Refer to Fig. Ex 9.4 :

As the plot results in a straight line, the M-M equation satisfactorily represents the kinetics of this reaction.

From Fig. Ex 9.4 :

$$\text{slope} = 0.0208 = \frac{C_M}{r_{max}}$$

$$\text{intercept} = 0.80 = \frac{1}{r_{max}}, \ (m^3 \cdot s)/mol$$

$$r_{max} = 1/0.80 = \mathbf{1.25 \ kmol/(m^3 \cdot s)} \quad \text{... Ans.}$$

$$C_M = 0.0208 \, r_{max}$$

$$= 0.0208 \times 1.25 = \mathbf{0.026 \ kmol/m^3} \quad \text{... Ans.}$$

$\therefore$ The rate equation is

$$\boxed{-r_A = \frac{1.25 \, C_A}{0.026 + C_A}}, \ \frac{\text{kmol}}{(m^3 \cdot s)} \quad \text{... Ans.}$$

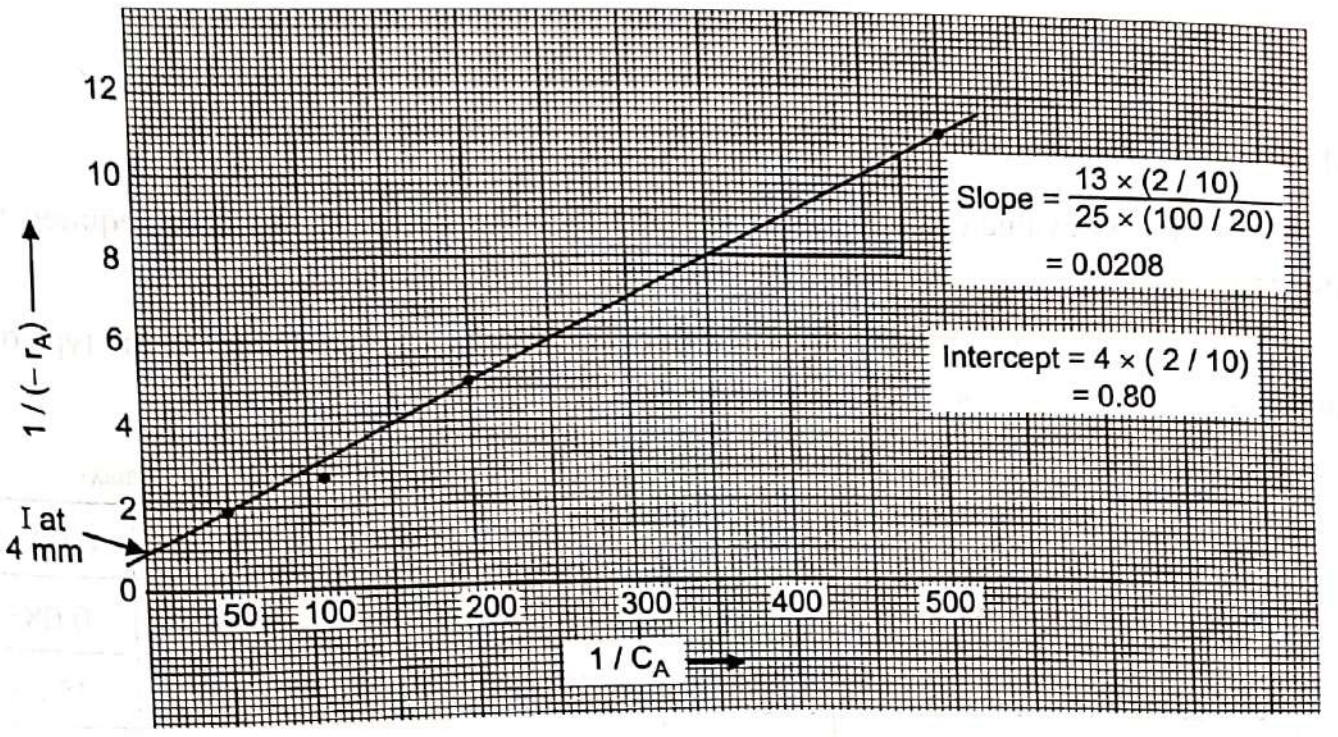


Fig. Ex 9.4

Plot $1/(-r_A)$ v/s $1/C_A$.

Refer to Fig. Ex 9.4 :

As the plot results in a straight line, the M-M equation satisfactorily represents the kinetics of this reaction.

From Fig. Ex 9.4 :

$$\text{slope} = 0.0208 = \frac{C_M}{r_{max}}$$

$$\text{intercept} = 0.80 = \frac{1}{r_{max}}, \ (m^3 \cdot s)/mol$$

$$r_{max} = 1/0.80 = \mathbf{1.25 \ kmol/(m^3 \cdot s)} \quad \text{... Ans.}$$

$$C_M = 0.0208 \ r_{max}$$

$$= 0.0208 \times 1.25 = \mathbf{0.026 \ kmol/m^3} \quad \text{... Ans.}$$

$\therefore$ The rate equation is

$$\boxed{-r_A = \frac{1.25 \ C_A}{0.026 + C_A}}, \ \frac{\text{kmol}}{(m^3 \cdot s)} \quad \text{... Ans.}$$

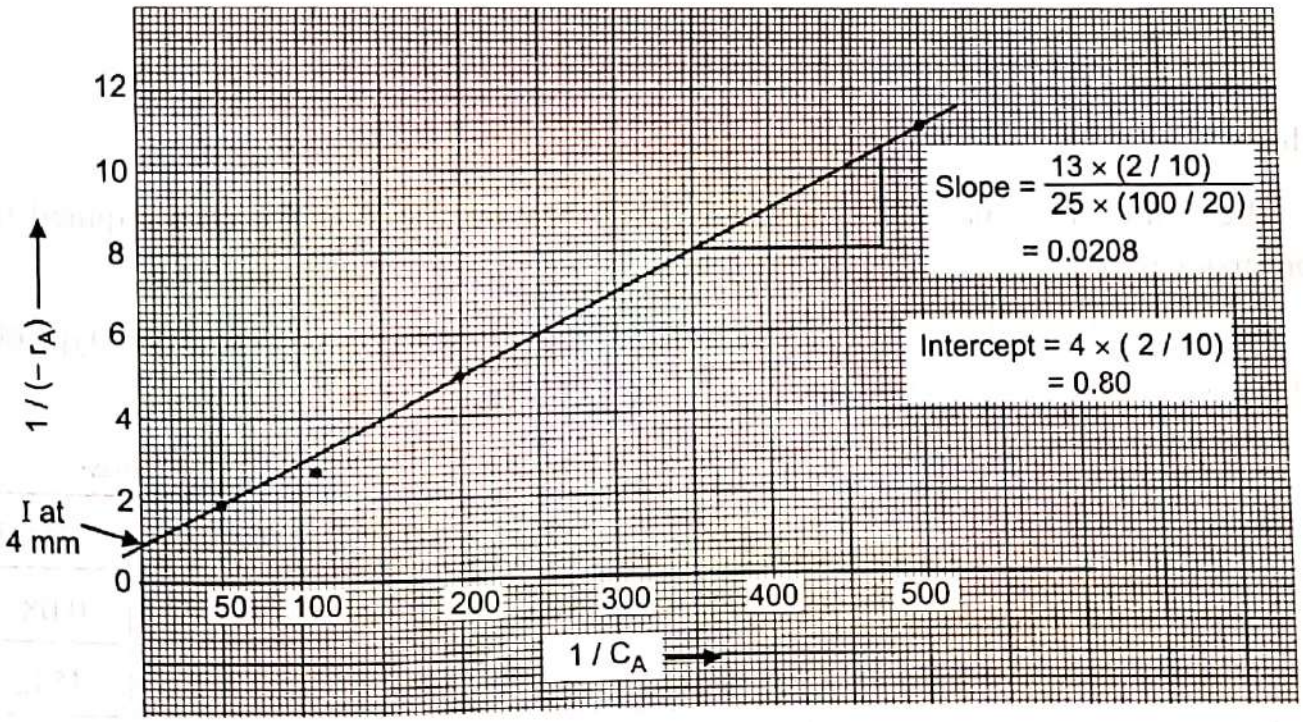


Fig. Ex 9.4

Example 9.5 : *Calculate the time needed to convert 80% of urea to ammonia and carbon dioxide in a batch reactor. The initial concentration of urea is 0.10 mol/l and urease concentration is 0.001 g/l. The reaction is carried out isothermally at temperature T. The following data obtained at temperature T with urease concentration of 5 g/l are available :*

C_{urea}, mol/l	0.04	0.01	0.0067	0.0033	0.0022
$-r_{urea}$, mol/(l·s)	0.5	0.27	0.204	0.119	0.083

Solution : Let A = urea, $P = 2\,NH_3 + CO_2$, $W = H_2O$.

$$A + E \underset{k_2}{\overset{k_1}{\rightleftharpoons}} A\cdot E\,(X) \xrightarrow[-W]{k_3} P + E$$

The Michaelis-Menten equation is

$$-r_A = \frac{k_3 C_{E_0} C_A}{C_M + C_A}$$

Let $r_{max} = k_3 C_{E_0}$ ... the maximum rate of reaction for a given total enzyme concentration.

$$\therefore \qquad -r_A = \frac{r_{max} C_A}{C_M + C_A}$$

Inverting the above equation, we get

$$\frac{1}{-r_A} = \frac{C_M + C_A}{r_{max} C_A}$$

$$\frac{1}{-r_A} = \left(\frac{C_M}{r_{max}}\right)\frac{1}{C_A} + \frac{1}{r_{max}}$$

where r_{max} and C_M are the M-M parameters.

We will first evaluate these parameters and then we will find the time required for 85% conversion of urea.

Now we will evaluate the M - M parameters by plotting $1/(-r_A)$ v/s $1/C_A$ (this type of plot is called as a Lineweaver plot).

A plot should be a straight line with a slope = C_M/r_{max} and an intercept = $1/r_{max}$.

C_A, mol/l	0.04	0.01	0.0067	0.0033	0.0022
$-r_A$, mol/(l·s)	0.5	0.27	0.204	0.119	0.083
$1/C_A$, l/mol	25	100	149.25	303	454.5
$1/(-r_A)$, (l·s)/mol	2	3.7	4.9	8.4	12.05

Plot $1/(-r_A)$ v/s $1/C_A$.

Refer Fig. Ex 9.5 :

We get a straight line ..... through the data points.

Hence, the M - M equation represents the kinetics of this reaction.

$$\text{slope} = \frac{C_M}{r_{max}} = 0.023$$

$$\text{intercept} = 1/r_{max} = 1.5$$

$$\therefore \quad r_{max} = 1/1.5 = 0.67, \quad \text{mol}/(l \cdot \text{s})$$

$$C_M = 0.023\, r_{max}$$

$$= 0.023 \times 0.67 = 0.0154 \text{ mol}/l$$

$\therefore$ The M - M equation is

$$\boxed{-r_A = \frac{0.67\, C_A}{0.0154 + C_A}} \quad , \text{ mol}/(l \cdot \text{s})$$

Now we will make use of this to find the time needed to convert 85% urea in the batch reactor.

We have for the data provided :

$$C_M = 0.0154 \text{ mol}/l$$

$$r_{max} = 0.67 \text{ mol}/(l \cdot \text{s}) \ldots \text{ for the enzyme concentration of } C_{Eo} = 5 \text{ g}/l$$

For a batch reactor with the M – M equation, we have

$$C_M \ln\left(\frac{C_{Ao}}{C_A}\right) + (C_{Ao} - C_A) = k_3\, C_{Eo}\, t$$

$$C_A = C_{Ao}(1 - X_A)$$

$$\frac{C_{Ao}}{C_A} = \frac{1}{1 - X_A}, \quad C_{Ao} - C_A = C_{Ao}\, X_A$$

$k_3\, C_{Eo} = r_{max}$ ..... for conditions in the batch reactor i.e. for enzyme concentration of 0.001 g/l.

$$C_M \ln\left(\frac{1}{1 - X_A}\right) + C_{Ao}\, X_A = r_{max}\, t$$

$$t = \frac{C_M}{r_{max}} \ln\left(\frac{1}{1 - X_A}\right) + \frac{C_{Ao}\, X_A}{r_{max}} \qquad \ldots (1)$$

Note : Important : In the above equation, r_{max} is to be used at an enzyme concentration of 0.001 g/l . Let it be $r_{max\,2}$.

From the data provided we obtain $r_{max} = 0.67$ at the enzyme concentration of 5 g/l. Let it be $r_{max\,1}$.

$$r_{max\,1} = 0.67 \quad \ldots\ldots \text{ for } C_{Eo\,1} = 5 \text{ g/}l$$

$$r_{max\,2} = ? \quad \ldots\ldots \text{ for } C_{Eo\,2} = 0.001 \text{ g/}l$$

$$r_{max\,2} = \frac{C_{Eo\,2}}{C_{Eo\,1}} r_{max\,1}$$

$$= \frac{0.001}{5} \times 0.67$$

$$= 1.34 \times 10^{-4} \text{ mol/}(l \cdot \text{s}) \ldots\ldots \text{ in a batch reactor}$$

Equation (1) for the conditions in a batch reactor becomes

$$t = \frac{C_M}{r_{max\,2}} \ln\left(\frac{1}{1 - X_A}\right) + \frac{C_{Ao} X_A}{r_{max\,2}}$$

where $r_{max\,2} = 1.34 \times 10^{-4}$, mol/($l \cdot$s)

$C_M = 0.0154$, $X_A = 80/100 = 0.80$

$C_{Ao} = 0.10$ mol/l

$$t = \frac{0.0154}{1.34 \times 10^{-4}} \ln\left(\frac{1}{1 - 0.80}\right) + \frac{0.10 \times 0.80}{1.34 \times 10^{-4}}$$

$$= 782 \text{ s (13 min)}$$

Time required to convert 80% urea = 782 s or 13 min ... Ans.

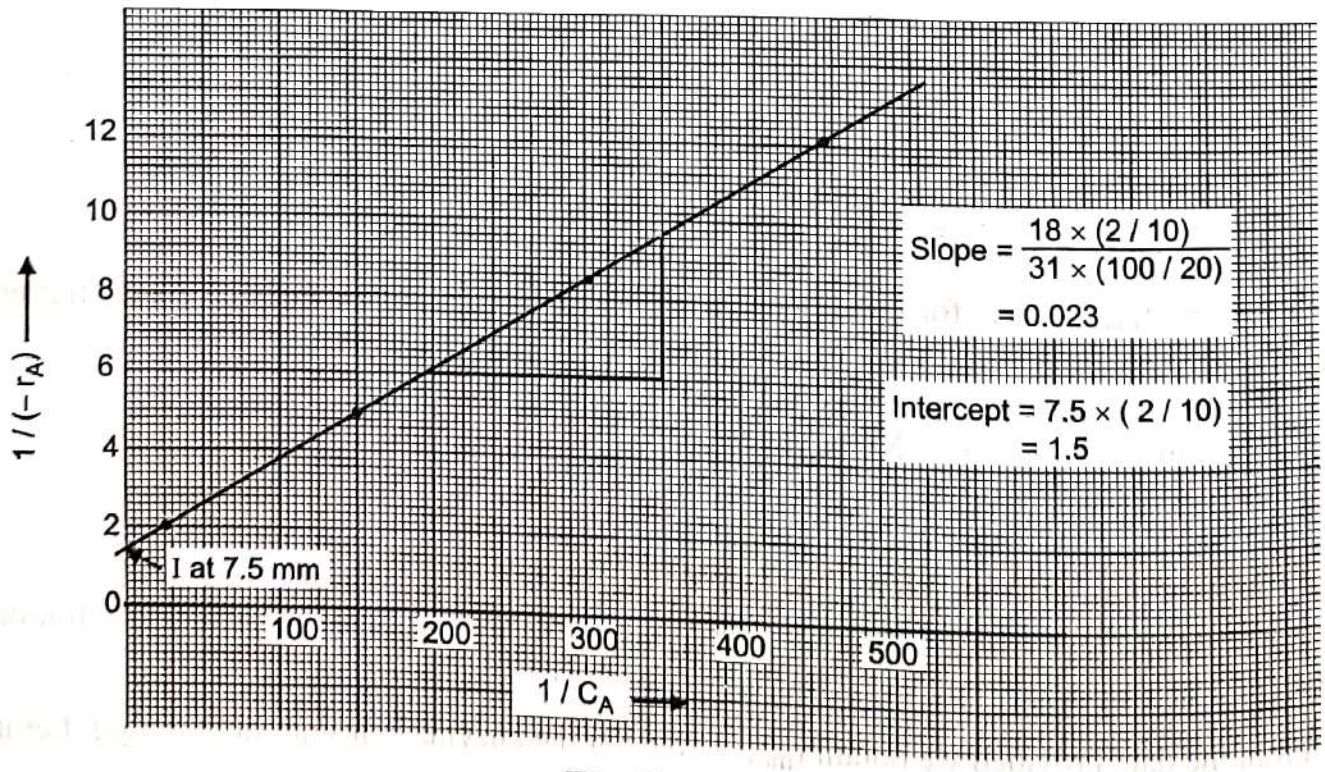


Fig. Ex 9.5

Example 9.6 : *Carbohydrate A decomposes in the presence of enzyme E. It is suspected that bohydrate B in some way influences the decomposition of A. In order to study this nomenon various concentrations of A, B and E flow into and out of a mixed flow reactor. The ume (V) of the mixed flow reactor is 240 cm³. From the following data :*

(i) Find the rate equation for the decomposition of A.

(ii) What is the role of B in this decomposition ?

(iii) Suggest a mechanism for this decomposition reaction.

C_{Ao}, mol/m³	C_A, mol/m³	C_{Bo}, mol/m³	C_{Eo}, mol/m³	ν, cm³/min
200	50	0	12.5	80
900	300	0	5	24
1200	800	0	5	48
700	33.3	33.3	33.3	24
200	80	33.3	10	80
900	500	33.3	20	120

Solution : It is clear from the data that :

In the first three data $\Rightarrow C_{Bo} = 0$ and in the remaining three $\Rightarrow C_{Bo} = 33.3$ mol/m³, i.e., have provided with the data – (i) without B (inhibition free) and (ii) with B (inhibition).

The data are obtained in a mixed flow reactor (V = 240 cm³).

In order to find the rate equation and to find the inhibitor (B) action-competitive or ncompetitive we have to construct the following plots for – (i) no inhibition and (ii) with hibition.

Construct a plot of C_A v/s $C_{Eo} C_A \tau/(C_{Ao} - C_A)$.

C_{Ao}	200	900	1200	700	200	900
C_A	50	300	800	33.3	80	500
C_{Eo}	12.5	5	5	33.3	10	20
C_{Bo}	0	0	0	33.3	33.3	33.3
$C_{Ao} - C_A$	150	600	400	666.7	120	400
ν	80	24	48	24	80	120
$\tau = V/\nu$	3	10	5	10	3	2
$\frac{C_A C_{Eo} \tau}{(C_{Ao} - C_A)}$	12.5	25	50	16.63	20	50
	No inhibition by B			Inhibition by B		

Refer Fig. Ex 9.6 :

1. For no inhibition :

$$-C_M = -20 \times (100/20)$$

$$C_M = 200 \text{ mol/m}^3$$

$$\text{slope} = 20$$

$$\text{slope} = k_3 = 20 \text{ min}^{-1}$$

The rate equation for the decomposition of A with no B present is

$$\boxed{-r_A = \frac{20\, C_{Eo}\, C_A}{200 + C_A}} \text{ , mol/(m}^3\text{·min)} \quad \text{... Ans. (i)}$$

$$\frac{C_M}{k_3} = 10.2$$

$$k_3 = C_M/10.2 = 200/10.2 = 19.16 \text{ min}^{-1} \text{ (} \approx \text{ly equal to slope value)}$$

2. For inhibition ..... with B present :

$$\text{slope} = 13.64$$

$$\frac{k_3}{1 + L\, C_{Bo}} = 13.64$$

and

$$\frac{C_M (1 + N\, C_{Bo})}{k_3} = 14$$

$$C_M \left(\frac{1 + N\, C_{Bo}}{1 + L\, C_{Bo}}\right) = C_M \left(\frac{1 + N\, C_{Bo}}{k_3}\right) \left(\frac{k_3}{1 + L\, C_{Bo}}\right)$$

$$= 14 \times 13.64$$

$$= 190.96 \approx 191$$

The rate equation when B influences the decomposition of A is

$$-r_A = r_R = \frac{[k_3/(1 + L\, C_{Bo})]\, C_{Eo}\, C_A}{C_M \left(\frac{1 + N\, C_{Bo}}{1 + L\, C_{Bo}}\right) + C_A}$$

$$\boxed{-r_A = r_R = \frac{13.64\, C_{Eo}\, C_A}{191 + C_A}} \quad \text{... Ans. (i)}$$

The role of B in the decomposition of A :

Refer to Fig. Ex 96 :

The plot of C_A v/s $C_{Eo}\, C_A\, \tau/(C_{Ao} - C_A)$, for – (i) no inhibition and (ii) with inhibition, yields straight lines but these lines are not parallel to each other (they have different slopes). Therefore, the action of inhibitor (B) is noncompetitive. B attacks a different site on the enzyme but in doing so stops the action of A (noncompetitive inhibition) .... **Ans.** (ii)

The mechanism for – inhibitor action-noncompetitive is

$$A + B \underset{k_2}{\overset{k_1}{\rightleftharpoons}} X \xrightarrow{k_3} R + E$$

$$B + E \underset{k_5}{\overset{k_4}{\rightleftharpoons}} Y$$

$$B + X \underset{k_7}{\overset{k_6}{\rightleftharpoons}} Z$$

... **Ans.** (iii)

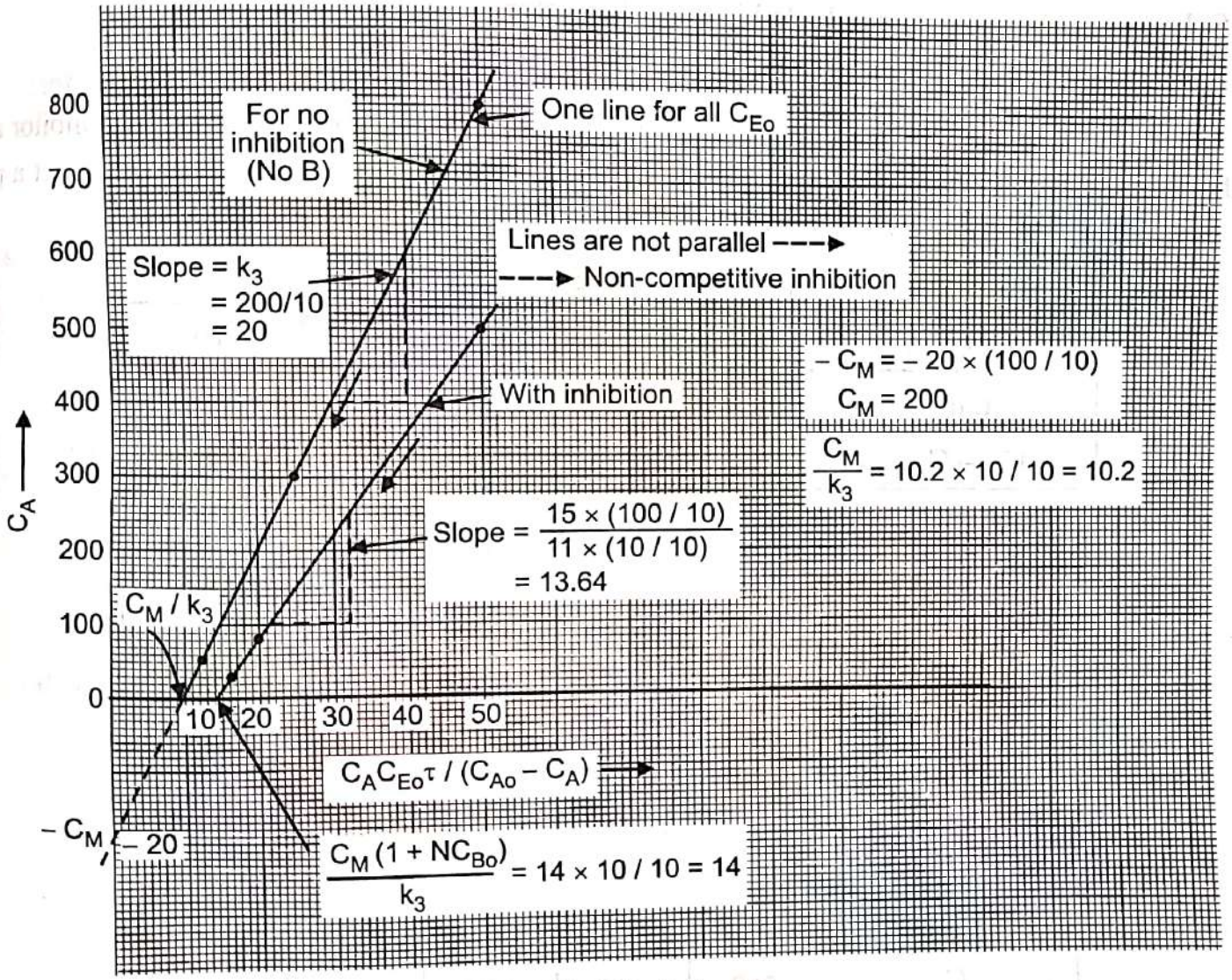


Fig. Ex 9.6

Example 9.7 : *Substrate A decomposes in the presence of enzyme E. To check whether substance B acts as an inhibitor, two kinetic runs are conducted in a batch reactor, one with B present and the other without B.*

(i) From the following data find the rate equation for the decomposition of A.

(ii) What is the role of B in the decomposition of A ?

(iii) Suggest a mechanism for this reaction.

Run 1 : $C_{Ao} = 600$ mol/m³, $C_{Eo} = 8$ g/m³, No B present.

C_A, mol/m³	350	160	40	10
t, h	1	2	3	4

Run 2 : $C_{Ao} = 800$ mol/m³, $C_{Eo} = 8$ g/m³, $C_B = C_{Bo} = 100$ mol/m³.

C_A, mol/m³	560	340	180	80	30
t, h	1	2	3	4	5

Solution : The runs are conducted in a batch reactor

(i) With no B and (ii) with B (inhibitor).

In order to find the rate equation for the decomposition of A in the presence of inhibitor and in the absence of inhibitor and to find a particular type of inhibition, we have to construct a plot of $(C_{Ao} - C_A)/\ln (C_{Ao}/C_A)$ v/s $C_{Eo}t /\ln (C_{Ao}/C_A)$.

Run 1 : $C_{Ao} = 600$ mol/m³, $C_{Eo} = 8$ g/m³, No B present.

C_A, mol/m³	350	160	40	10
t, h	1	2	3	4
$C_{Ao} - C_A$	250	440	560	590
$\ln C_{Ao}/C_A$	0.54	1.32	2.71	4.1
$\frac{C_{Ao} - C_A}{\ln C_{Ao}/C_A}$	463	333	206.6	143.9
$\frac{C_{Eo}\, t}{\ln C_{Ao}/C_A}$	14.8	12.12	8.86	7.8

$C_{Eo}\, t/\ln C_{Ao}/C_A$ has the units of (g·h)/m³.

$\frac{C_{Ao} - C_A}{\ln (C_{Ao}/C_A)}$ has the units of mol/m³.

Run 2 : $C_{Ao} = 800$ mol/m³, $C_{Eo} = 8$ g/m³, $C_B = C_{Bo} = 100$ mol/m³.

C_A	560	340	180	80	30
t	1	2	3	4	5
$C_{Ao} - C_A$	240	460	620	720	770
$\ln C_{Ao}/C_A$	0.357	0.86	1.49	2.3	3.28
$\frac{C_{Ao} - C_A}{\ln C_{Ao}/C_A}$	672	535	416	313	235
$\frac{C_{Eo}\, t}{\ln C_{Ao}/C_A}$	22.2	18.6	16.1	13.9	12.2

Construct a plot of $(C_{Ao} - C_A)/\ln (C_{Ao}/C_A)$ v/s $C_{Eo}\, t/\ln (C_{Ao}/C_A)$ for both runs (1 and 2).

Refer to Fig. Ex 9.7 :

The plot results in a straight line for the Run 1 as well as for the Run 2. As the straight lines parallel to each other (i.e., they have the same slope), we have competitive inhibition.

(1) **No B present, i.e., in the absence of inhibitor (no inhibition) :** Please refer to g. Ex 9.7 ... straight line for run 1.

$$\text{slope} = k_3 = 45.45 \text{ mol/(g·h)}$$

$$C_M = 210 \text{ mol/m}^3$$

The rate equation for the decomposition of A is

$$\boxed{-r_A = \frac{45.45\, C_{Eo}\, C_A}{210 + C_A}}, \quad \text{mol/(m}^3\text{·h)} \qquad \text{... Ans. (i)}$$

This rate equation is for no inhibition.

(2) **With inhibition – in the presence of inhibitor (with B present) :** Please refer to g. Ex 9.7 ..... straight line for run 2.

$$\text{slope} = k_3 = 45.45 \text{ mol/(g·h)}$$

$$\frac{C_M (1 + N C_{Bo})}{k_3} = 7$$

$$C_M (1 + N C_{Bo}) = \frac{C_M (1 + N C_{Bo})}{k_3} \times k_3$$

$$= 7 \times 45.45 = 318.15 \approx 318$$

The rate equation for the decomposition of A with B present (i.e., with inhibition) is

$$-r_A = r_R = \frac{k_3\, C_{Eo}\, C_A}{C_M (1 + N C_{Bo}) + C_A}$$

$$C_M = \frac{k_2 + k_3}{k_1}, \text{ mol/m}^3 \text{ and } N = k_4/k_5, \text{ mol/m}^3$$

$$\boxed{-r_A = r_R = \frac{45.45\, C_{Eo}\, C_A}{318 + C_A}}, \quad \text{mol/(m}^3\text{·h)} \qquad \text{... Ans. (i)}$$

As the straight lines are parallel to each other, the kind of inhibitor action is competitive. erefore, we have competitive inhibition (A and B attack the same site on the enzyme)

... Ans. (ii)

The suggested mechanism is

$$A + E \underset{k_2}{\overset{k_1}{\rightleftharpoons}} X \xrightarrow{k_3} R + E$$

$$B + E \underset{k_5}{\overset{k_4}{\rightleftharpoons}} Y$$

... Ans. (iii)

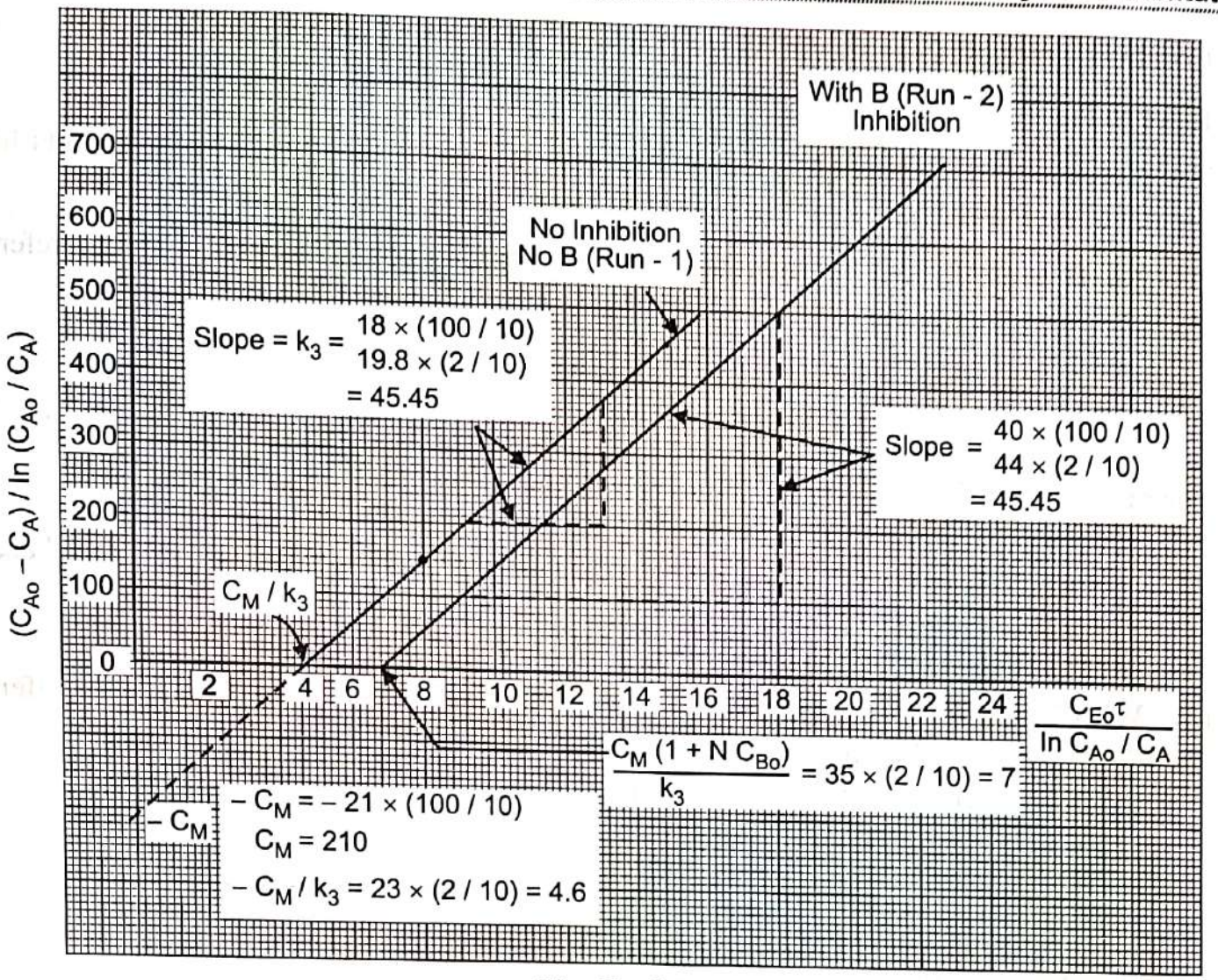


Fig. Ex 9.7

Example 9.8 : *Cellulose can be converted to sugar, by the catalytic action of enzyme-cellulase, as follows :*

$$\text{cellulose} \xrightarrow{\text{cellulase}} \text{sugar}$$

Both cellubiose and glucose act to inhibit the breakdown of cellulase. For studying the kinetics of this reaction a number of runs are conducted in a mixed flow reactor, maintained at 50 °C using a feed of finely shreaded cellulose (C_{Ao} = 15 kg/m³), enzyme (C_{Eo} = 0.01 kg/m³, same for all runs) and various inhibitors. The results obtained are tabulated below :

Run	Exit stream C_A, kg/m³	Series - 1 No inhibitor τ, min -	Series - 2 With cellubiose C_{Bo} = 5 kg/m³ τ, min	Series - 3 With glucose C_{Go} = 10 kg/m³ τ, min
1	1.5	587	940	1020
2	4.5	279	387	433
3	9	171	213	250
4	21	36	40	50

(i) Find the rate equation for the breakdown (decomposition) of cellulose by cellulase in the absence of inhibitor.

(ii) Find the type of inhibition and the rate equation in presence of cellubiose (i.e., the role of cellubiose in the breakdown of cellulose).

(iii) Find the type of inhibition and the rate equation for the breakdown of cellulose in the presence of glucose (inhibitor).

Solution : cellulose $\xrightarrow{\text{cellulase}}$ sugar

Here we have to obtain :

(i) rate equation in the absence of inhibitor

(ii) rate equation in the presence of cellubiose (inhibitor)

(iii) type of inhibition in the presence of cellubiose

(iv) rate equation in the presence of glucose (inhibitor)

(v) type of inhibition in the presence of glucose (inhibitor).

The results provided are obtained in a mixed flow reactor with C_{Ao} = 25 kg/m^3 and C_{Eo} = 0.01 kg/m^3 (same for all the runs).

In order to find the rate equation and type of inhibition we have to construct plots for no inhibition and with inhibition by making use of the data provided.

Plot C_A v/s $C_{Eo} C_A \tau/(C_{Ao} - C_A)$ by making use of the mixed flow reactor data.

(1) With no inhibitor : C_{Eo} = 0.01 kg/m^3.

Run	C_A, kg/m^3	No inhibitor τ, min	$(C_{Ao} - C_A)$, kg/m^3	$C_{Eo} C_A \tau/(C_{Ao} - C_A)$, (kg.min)/m^3
1	1.5	587	25 – 1.5 = 23.5	0.375
2	4.5	279	20.5	0.61
3	9	171	16	0.96
4	21	36	25 – 21 = 4	1.89

(2) With cellubiose : C_{Eo} = 0.01 kg/m^3 and $C_{Bo} = C_{Co}$ = 5 kg/m^3.

Run	C_A	τ	$C_{Ao} - C_A$	$C_{Eo} C_A \tau/(C_{Ao} - C_A)$
1	1.5	940	25 – 1.5 = 23.5	0.60
2	4.5	387	20.5	0.85
3	9	213	16	1.198 ≈ 1.2
4	21	40	4	2.1

(3) With glucose (inhibitor) : $C_{Eo} = 0.01$ kg/m^3 and $C_{Bo} = C_{Go} = 10$ kg/m^3.

Run	C_A	τ	$C_{Ao} - C_A$	$C_{Eo} C_A \tau/(C_{Ao} - C_A)$
1	1.5	1020	23.5	0.65
2	4.5	433	20.5	0.95
3	9	250	16	1.41
4	21	50	25 – 21 = 4	2.625

Plot C_A v/s $C_{Eo} C_A \tau/(C_{Ao} - C_A)$.

For ..... no inhibition, inhibition with cellubiose and inhibition with glucose.

We obtain straight lines for no inhibition and inhibition.

(1) The rate equation in the absence of inhibitor.

Please refer to Fig. Ex 9.8 ..... straight line-1.

$$\text{slope} = k_3 = 12.8 \text{ min}^{-1}$$

$$C_M = 3.32 \text{ kg/m}^3$$

$\therefore$ The rate equation for the breakdown of cellulose is

$$-r_A = \frac{k_3 C_{Eo} C_A}{C_M + C_A}$$

$$\boxed{-r_A = \frac{12.8\, C_{Eo} C_A}{3.32 + C_A}} \text{ , kg/(m}^3\text{·min)} \quad \text{... Ans. (i}$$

Also from Fig. Ex 9.8 :

$$\frac{C_M}{k_3} = 0.26$$

$$k_3 = C_M/0.26 = 3.32/0.26 = 12.77 \text{ (almost same as the slope value)}$$

For $C_{Ao} = 25$ kg/m^3, we have

$$\frac{C_M + C_{Ao}}{k_3} = 2.22$$

$$k_3 = (C_M + C_{Ao})/2.22 = (3.32 + 25)/2.22 = 12.77$$

(matches with the slope valu

[Please note that a small error in a graphical construction results in a large deviation in t answer. So precision is required in the graphical construction.]

(2) The rate equation for the breakdown of cellulose in the presence of inhibitor - cellubiose : As the plots for no inhibition and with inhibition are having the same slope – lin are parallel to each other - we have competitive inhibition by cellubiose.

Type of inhibition ..... with cellubiose ..... competitive ... Ans. (i

[A and B (cellubiose) will attack the same site on the enzyme].

Refer Fig. Ex 9.8 :

$$\text{slope} = k_3 = 12.8 \text{ min}^{-1} \text{ ... straight line - 2}$$

$$\frac{C_M (1 + N C_{Bo})}{k_3} = 4.8 \quad \text{..... straight line - 2.}$$

The rate equation for the breakdown of cellulose for competitive inhibition by cellubiose is

$$r_R = \frac{k_3 C_{Eo} C_A}{C_M(1 + N C_{Bo}) + C_A}$$

$$C_M(1 + N C_{Bo}) = \frac{C_M(1 + N C_B)}{k_3} \cdot k_3$$

$$= 4.8 \times 12.8 = 61.44$$

$$\therefore \quad \boxed{-r_A = r_R = \frac{12.8\, C_{Eo} C_A}{61.44 + C_A}}, \ kg/(m^3 \cdot min) \qquad \text{... Ans. (ii)}$$

(3) The rate equation for the breakdown of cellulose in the presence of inhibitor ..... glucose :

Please refer to Fig. Ex 9.8 ..... straight line - 3.

As the straight line-3 is not parallel to the straight line-1, i.e., slopes of line 3 and line 1 are different, we have noncompetitive inhibition by glucose.

Type of inhibition ..... with glucose ..... noncompetitive ... **Ans.** (iii)

[B (glucose) attacks a different site on the enzyme and in doing so stops the action of A.]

Here, $C_{Bo} = C_{Go} = 10\ kg/m^3$

See straight line-3 (Fig. Ex 9.8) :

$$\text{slope} = \frac{k_3}{1 + L C_{Bo}} = 9.7$$

and

$$\frac{C_M(1 + N C_{Bo})}{k_3} = 5$$

$$\frac{C_M(1 + N C_{Bo})}{(1 + L C_{Bo})} = \frac{C_M(1 + N C_{Bo})}{k_3} \times \frac{k_3}{(1 + L C_{Bo})}, \quad [C_{Bo} = C_{Go}]$$

$$= 5 \times 9.7$$

$$= 48.5$$

The rate equation for the breakdown of cellulose with noncompetitive inhibition by glucose is

$$-r_A = r_R = \left(\frac{k_3}{1 + L C_{Bo}}\right) C_{Eo} C_A \Big/ \left[C_M\left(\frac{1 + N C_{Bo}}{1 + L C_{Bo}}\right) + C_A\right]$$

$$-r_A = r_R = \frac{\left(\frac{k_3}{1 + L C_{Bo}}\right) C_{Eo} C_A}{C_M\left(\frac{1 + N C_{Bo}}{1 + L C_{Bo}}\right) + C_A}, \quad C_{Bo} = C_{Go}$$

$$\therefore \quad \boxed{-r_A = \frac{9.7\, C_{Eo} C_A}{48.5 + C_A}}, \ kg/(m^3 \cdot min) \qquad \text{... Ans. (iii)}$$

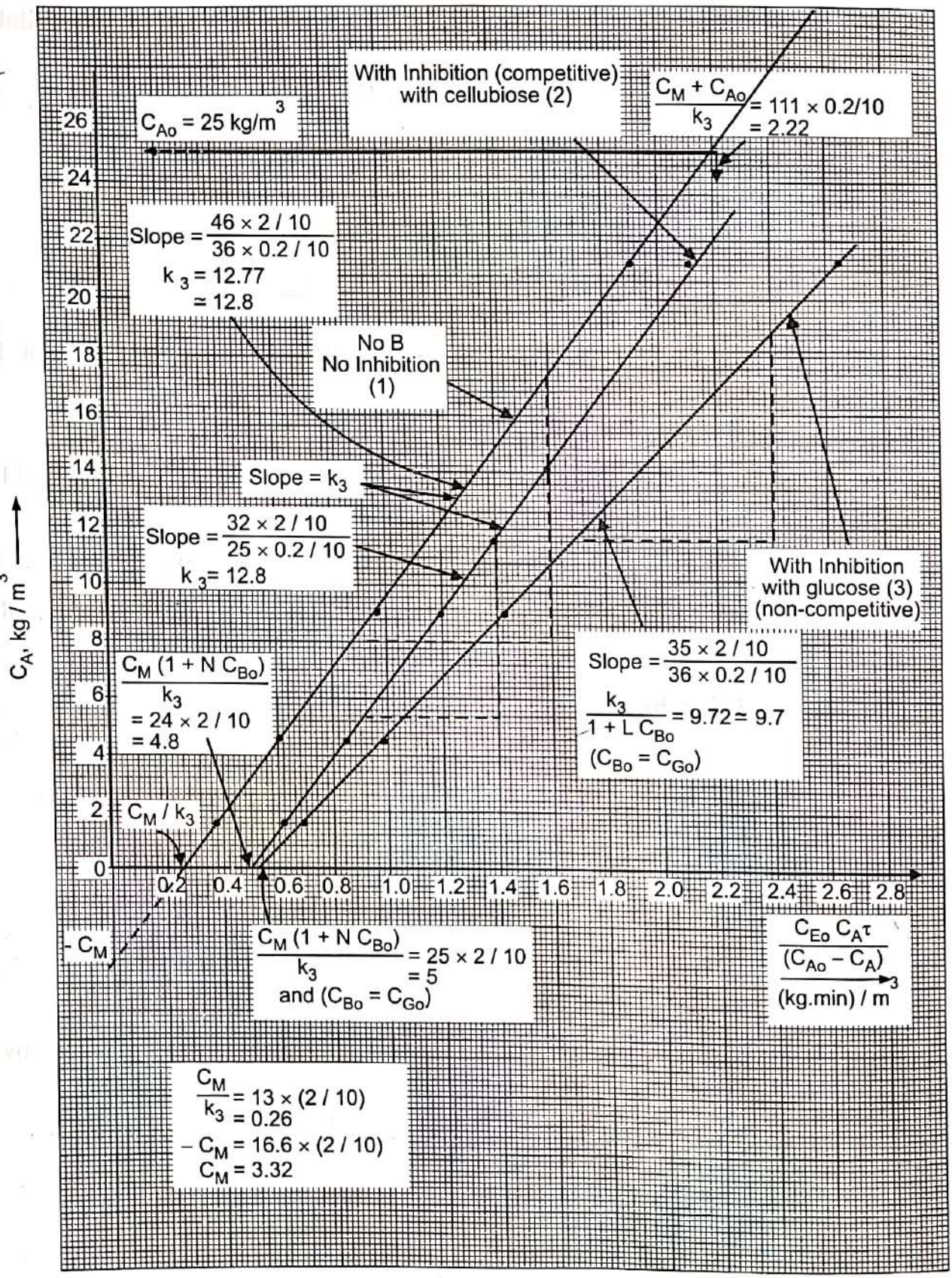
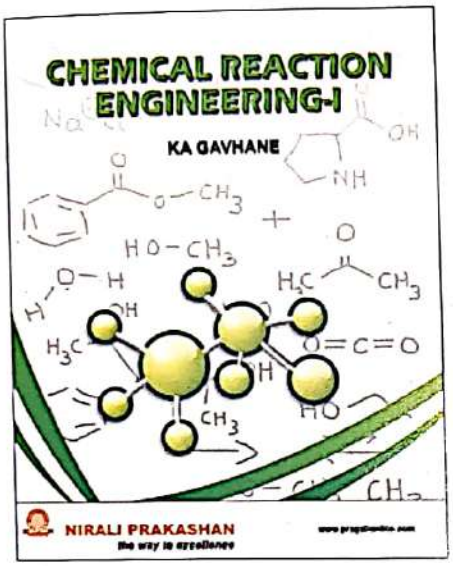
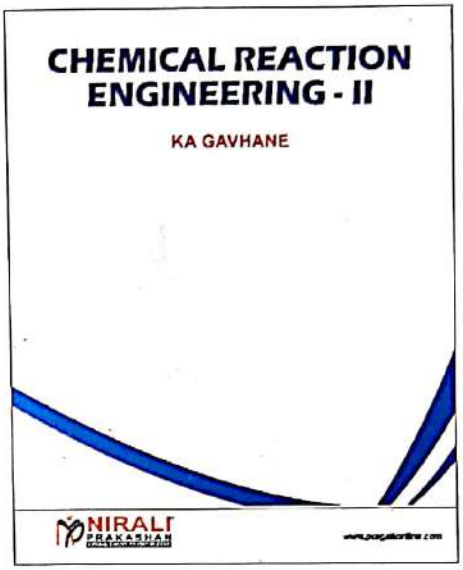

With Inhibition (competitive) with cellubiose (2)
$\frac{C_M + C_{Ao}}{k_3} = 111 \times 0.2/10 = 2.22$
$C_{Ao} = 25$ kg/m^3
Slope $= \frac{46 \times 2/10}{36 \times 0.2/10}$
$k_3 = 12.77 \simeq 12.8$
No B
No Inhibition
(1)
Slope $= k_3$
Slope $= \frac{32 \times 2/10}{25 \times 0.2/10}$
$k_3 = 12.8$
With Inhibition with glucose (3) (non-competitive)
Slope $= \frac{35 \times 2/10}{36 \times 0.2/10}$
$\frac{k_3}{1 + L\,C_{Bo}} = 9.72 \simeq 9.7$
$(C_{Bo} = C_{Go})$
$\frac{C_M (1 + N\,C_{Bo})}{k_3} = 24 \times 2/10 = 4.8$
C_M / k_3
$\frac{C_M (1 + N\,C_{Bo})}{k_3} = 25 \times 2/10 = 5$
and $(C_{Bo} = C_{Go})$
$\frac{C_M}{k_3} = 13 \times (2/10) = 0.26$
$-C_M = 16.6 \times (2/10)$
$C_M = 3.32$
$-C_M$
C_A, kg / m^3
$\frac{C_{Eo} C_A \tau}{(C_{Ao} - C_A)}$ m^3 / (kg.min)
26
24
22
20
18
16
14
12
10
8
6
4
2
0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
2.4
2.6
2.8

Fig. Ex 9.8

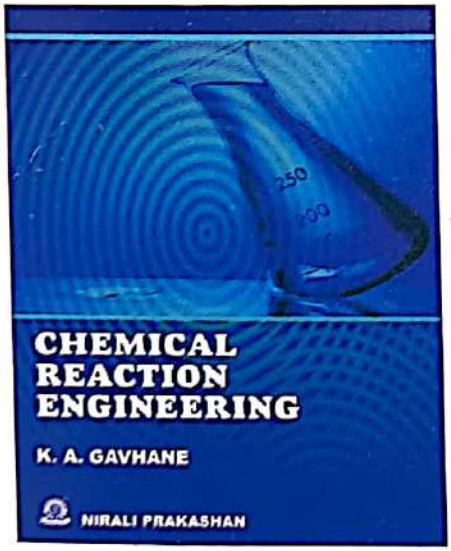
Chemical Engineering



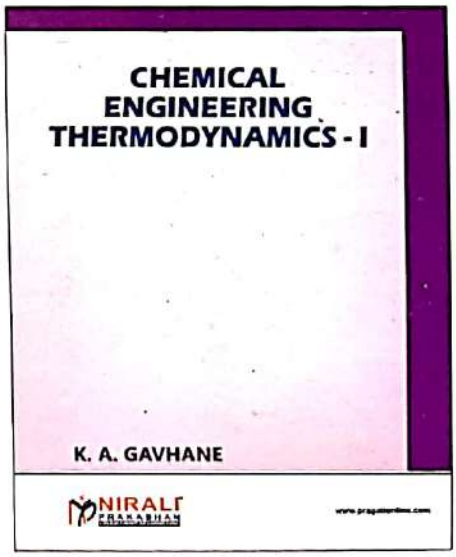
CHEMICAL REACTION ENGINEERING-I



CHEMICAL REACTION ENGINEERING-II



CHEMICAL REACTION ENGINEERING



CHEMICAL ENGINEERING THERMODYNAMICS - I